

Proyecto_Space_Invaders_v0

July 4, 2026

1 Actividad - Proyecto práctico

La actividad se desarrollará en grupos pre-definidos de 4 alumnos. Se debe indicar los nombres en orden alfabético (de apellidos). Recordad que esta actividad se corresponde con un 30% de la nota final de la asignatura. Se debe entregar el trabajo en la presente notebook. * Alumno 1: Daniel Alfonso Correa Angarita * Alumno 2: Cesar Daniel Meneses Guevara * Alumno 3: America Garrido Sánchez

Colab: [Proyecto Space Invaders v0](#)

Git Hub: [Link del Repositorio](#)

Evidencias de entrenamiento: En esta [carpeta compartida de google drive](#) se guardan los pesos de cada modelo, videos, logs, y el ipynb de este proyecto.

Videos Agente Jugando:

- [Video Modelo DQN Clasico](#)
- [Video Modelo DQN Spatial](#)
- [Video Modelo DQN Dueling](#)

Nota. Se hicieron varias sesiones de trabajo (Peer Programming) actualizando el colab, puede verificar el historial de revisiones en colab para lo mencionado.

1.1 PARTE 1 - Instalación y requisitos previos

Las prácticas han sido preparadas para poder realizarse en el entorno de trabajo de Google Colab. Sin embargo, esta plataforma presenta ciertas incompatibilidades a la hora de visualizar la renderización en gym. Por ello, para obtener estas visualizaciones, se deberá trasladar el entorno de trabajo a local. Por ello, el presente dossier presenta instrucciones para poder trabajar en ambos entornos. Siga los siguientes pasos para un correcto funcionamiento: 1. **LOCAL:** Preparar el enviroment, siguiendo las intrucciones detalladas en la sección *1.1.Preparar enviroment*. 2. **AMBOS:** Modificar las variables “mount” y “drive_mount” a la carpeta de trabajo en drive en el caso de estar en Colab, y ejecturar la celda *1.2.Localizar entorno de trabajo*. 3. **COLAB:** se deberá ejecutar las celdas correspondientes al montaje de la carpeta de trabajo en Drive. Esta corresponde a la sección *1.3.Montar carpeta de datos local*. 4. **AMBOS:** Instalar las librerías necesarias, siguiendo la sección *1.4.Instalar librerías necesarias*.

1.1.1 1.1. Preparar enviroment (solo local)

Para preparar el entorno de trabajo en local, se han seguido los siguientes pasos: 1. En Windows, puede ser necesario instalar las C++ Build Tools. Para ello, siga los siguientes pasos. Alternativamente puedes utilizar WSL2: <https://towardsdatascience.com/how-to-install-openai-gym-in-a-windows-environment-338969e24d30>. 2. Instalar Anaconda 3. Siguiendo el código que se presenta comentado en la próxima celda: Crear un enviroment, cambiar la ruta de trabajo, e instalar librerías básicas.

```
conda create --name miar_rl python=3.8
conda activate miar_rl
cd "PATH_TO_FOLDER"
conda install git
pip install jupyter
```

4. Abrir la notebook con *jupyter-notebook*.

```
jupyter-notebook
```

1.1.2 1.2. Localizar entorno de trabajo: Google colab o local

```
[ ]: # ATENCIÓN!! Modificar ruta relativa a la práctica si es distinta (drive_root)
mount='/content/gdrive'
drive_root = mount + "/My Drive/08_MIAR/actividades/proyecto practico"

try:
    from google.colab import drive
    IN_COLAB=True
except:
    IN_COLAB=False
```

1.1.3 1.3. Montar carpeta de datos local (solo Colab)

```
[ ]: import os
if IN_COLAB:
    print("We're running Colab")

if IN_COLAB:
    print("Colab: mounting Google drive on ", mount)

    drive.mount(mount)

    create_drive_root = True
    if create_drive_root:
        print("\nColab: making sure ", drive_root, " exists.")
        os.makedirs(drive_root, exist_ok=True)
```

```

print("\nColab: Changing directory to ", drive_root)
%cd $drive_root
%pwd
print("Archivos en el directorio: ")
print(os.listdir())

```

We're running Colab

Colab: mounting Google drive on /content/gdrive

Mounted at /content/gdrive

Colab: making sure /content/gdrive/My Drive/08_MIAR/actividades/proyecto practico exists.

Colab: Changing directory to /content/gdrive/My

Drive/08_MIAR/actividades/proyecto practico

/content/gdrive/My Drive/08_MIAR/actividades/proyecto practico

Archivos en el directorio:

['viejo entrenamiento', 'dqn clasico', 'dqn spartial', 'Copia entrenamiento',
'dqn dueling', 'Proyecto_práctico.ipynb']

1.1.4 1.4. Instalar librerías necesarias

```

[ ]: if IN_COLAB:
    !pip uninstall jax jaxlib ml-dtypes -y
    %pip install --upgrade ml_dtypes==0.5.0

    %pip install tensorflow==2.18.0
    %pip install tf-keras==2.18.0
    %pip install gym==0.17.3
    %pip install git+https://github.com/Kojoley/atari-py.git
    %pip install keras-rl2==1.0.5

    print(f"Ecosistema clásico cargado en la nube de Google.")

else:
    """
    %pip install gym==0.17.3
    %pip install git+https://github.com/Kojoley/atari-py.git
    %pip install pygame==1.5.0
    %pip install h5py==3.1.0
    %pip install Pillow==9.5.0
    %pip install keras-rl2==1.0.5
    %pip install Keras==2.2.4
    %pip install tensorflow==2.5.3
    %pip install torch==2.0.1
    """

```

```
%pip install agents==1.4.0
"""
%pip install numpy pandas matplotlib seaborn tensorflow==2.18.0
```

```
Found existing installation: jax 0.7.2
Uninstalling jax-0.7.2:
  Successfully uninstalled jax-0.7.2
Found existing installation: jaxlib 0.7.2
Uninstalling jaxlib-0.7.2:
  Successfully uninstalled jaxlib-0.7.2
Found existing installation: ml_dtypes 0.5.4
Uninstalling ml_dtypes-0.5.4:
  Successfully uninstalled ml_dtypes-0.5.4
Collecting ml_dtypes==0.5.0
  Downloading ml_dtypes-0.5.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (21 kB)
Requirement already satisfied: numpy>=1.21 in /usr/local/lib/python3.12/dist-packages (from ml_dtypes==0.5.0) (2.0.2)
Downloading
ml_dtypes-0.5.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (4.5 MB)

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Installing collected packages: ml_dtypes
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

dopamine-rl 4.1.2 requires jax>=0.1.72, which is not installed.
dopamine-rl 4.1.2 requires jaxlib>=0.1.51, which is not installed.
tensorflow 2.20.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml-dtypes 0.5.0
which is incompatible.

Successfully installed ml_dtypes-0.5.0
Collecting tensorflow==2.18.0
  Downloading tensorflow-2.18.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (0.7.0)
Requirement already satisfied: google-pasta>=0.1.1 in
```

```

/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (18.1.1)
Requirement already satisfied: opt-einsum>=2.3.2 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (26.2)
Requirement already satisfied:
protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3
in /usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (3.3.0)
Requirement already satisfied: typing-extensions>=3.6.6 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (4.15.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (2.2.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (1.81.1)
Collecting tensorboard<2.19,>=2.18 (from tensorflow==2.18.0)
  Downloading tensorboard-2.18.0-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (3.13.2)
Requirement already satisfied: numpy<2.1.0,>=1.26.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.18.0) (2.0.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow==2.18.0) (3.16.0)
Collecting ml-dtypes<0.5.0,>=0.4.0 (from tensorflow==2.18.0)
  Downloading ml_dtypes-0.4.1-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from
astunparse>=1.6.0->tensorflow==2.18.0) (0.47.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow==2.18.0) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow==2.18.0) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow==2.18.0) (0.19.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow==2.18.0) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-

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packages (from requests<3,>=2.21.0->tensorflow==2.18.0) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow==2.18.0) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow==2.18.0) (2026.6.17)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow==2.18.0) (3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow==2.18.0) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow==2.18.0) (3.1.8)
Requirement already satisfied: markupsafe>=2.1.1 in
/usr/local/lib/python3.12/dist-packages (from
werkzeug>=1.0.1->tensorboard<2.19,>=2.18->tensorflow==2.18.0) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow==2.18.0) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow==2.18.0) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow==2.18.0)
(0.1.2)
Downloading
tensorflow-2.18.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
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ml_dtypes-0.4.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.2
MB)

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Downloading tensorboard-2.18.0-py3-none-any.whl (5.5 MB)

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Installing collected packages: ml-dtypes, tensorboard, tensorflow

Attempting uninstall: ml-dtypes

Found existing installation: ml_dtypes 0.5.0

Uninstalling ml_dtypes-0.5.0:

Successfully uninstalled ml_dtypes-0.5.0

Attempting uninstall: tensorboard

Found existing installation: tensorboard 2.20.0

```

Uninstalling tensorboard-2.20.0:
  Successfully uninstalled tensorboard-2.20.0
Attempting uninstall: tensorflow
  Found existing installation: tensorflow 2.20.0
  Uninstalling tensorflow-2.20.0:
    Successfully uninstalled tensorflow-2.20.0
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

dopamine-rl 4.1.2 requires jax>=0.1.72, which is not installed.
dopamine-rl 4.1.2 requires jaxlib>=0.1.51, which is not installed.
tensorflow-text 2.20.1 requires tensorflow<2.21,>=2.20.0, but you have
tensorflow 2.18.0 which is incompatible.
tf-keras 2.20.0 requires tensorflow<2.21,>=2.20, but you have tensorflow 2.18.0
which is incompatible.
tensorstore 0.1.84 requires ml_dtypes>=0.5.0, but you have ml-dtypes 0.4.1 which
is incompatible.
ydf-tf 2.20.0 requires tensorflow==2.20.0, but you have tensorflow 2.18.0 which
is incompatible.

Successfully installed ml-dtypes-0.4.1 tensorboard-2.18.0 tensorflow-2.18.0
Collecting tf-keras==2.18.0
  Downloading tf_keras-2.18.0-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: tensorflow<2.19,>=2.18 in
/usr/local/lib/python3.12/dist-packages (from tf-keras==2.18.0) (2.18.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-
packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in
/usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (0.7.0)
Requirement already satisfied: google-pasta>=0.1.1 in
/usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (18.1.1)

```

Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.4.0)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (26.2)

Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (5.29.6)

Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.32.4)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (75.2.0)

Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (1.17.0)

Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.3.0)

Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (4.15.0)

Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.2.2)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (1.81.1)

Requirement already satisfied: tensorboard<2.19,>=2.18 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.18.0)

Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.13.2)

Requirement already satisfied: numpy<2.1.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.0.2)

Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.16.0)

Requirement already satisfied: ml-dtypes<0.5.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.4.1)

Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.47.0)

Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (13.9.4)

Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.1.0)

Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages


```

(from keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.19.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests<3,>=2.21.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0)
(3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2026.6.17)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (3.1.8)
Requirement already satisfied: markupsafe>=2.1.1 in
/usr/local/lib/python3.12/dist-packages (from
werkzeug>=1.0.1->tensorboard<2.19,>=2.18->tensorflow<2.19,>=2.18->tf-
keras==2.18.0) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-
py>=2.2.0->rich->keras>=3.5.0->tensorflow<2.19,>=2.18->tf-keras==2.18.0) (0.1.2)
Downloading tf_keras-2.18.0-py3-none-any.whl (1.7 MB)

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Installing collected packages: tf-keras

Attempting uninstall: tf-keras

Found existing installation: tf_keras 2.20.0

Uninstalling tf_keras-2.20.0:

Successfully uninstalled tf_keras-2.20.0

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

dopamine-rl 4.1.2 requires jax>=0.1.72, which is not installed.

dopamine-rl 4.1.2 requires jaxlib>=0.1.51, which is not installed.

Successfully installed tf-keras-2.18.0

Collecting gym==0.17.3

Downloading gym-0.17.3.tar.gz (1.6 MB)

1.6/1.6 MB

76.8 MB/s eta 0:00:00

Preparing metadata (setup.py) ... done

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (1.16.3)

Requirement already satisfied: numpy>=1.10.4 in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (2.0.2)

Collecting pygame<=1.5.0,>=1.4.0 (from gym==0.17.3)

Downloading pygame-1.5.0-py2.py3-none-any.whl.metadata (7.6 kB)

Collecting cloudpickle<1.7.0,>=1.2.0 (from gym==0.17.3)

Downloading cloudpickle-1.6.0-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages (from pygame<=1.5.0,>=1.4.0->gym==0.17.3) (1.0.0)

Downloading cloudpickle-1.6.0-py3-none-any.whl (23 kB)

Downloading pygame-1.5.0-py2.py3-none-any.whl (1.0 MB)

1.0/1.0 MB

83.3 MB/s eta 0:00:00

Building wheels for collected packages: gym

Building wheel for gym (setup.py) ... done

Created wheel for gym: filename=gym-0.17.3-py3-none-any.whl size=1654617 sha256=bec55492d7e13cabb3a9604ff7f3bdc7fb2b2c69422a968eb96fc72ca2a0b4a9

Stored in directory: /root/.cache/pip/wheels/99/7e/03/69cdeb432380e05bc9a56ff68451d8063e41aa06db508f4be5

Successfully built gym

Installing collected packages: pygame, cloudpickle, gym

Attempting uninstall: cloudpickle

Found existing installation: cloudpickle 3.1.2

Uninstalling cloudpickle-3.1.2:

Successfully uninstalled cloudpickle-3.1.2

Attempting uninstall: gym

Found existing installation: gym 0.25.2

Uninstalling gym-0.25.2:

Successfully uninstalled gym-0.25.2

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

dopamine-rl 4.1.2 requires jax>=0.1.72, which is not installed.

dopamine-rl 4.1.2 requires jaxlib>=0.1.51, which is not installed.

bigframes 2.42.0 requires cloudpickle>=2.0.0, but you have cloudpickle 1.6.0 which is incompatible.

distributed 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

dask 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.

Successfully installed cloudpickle-1.6.0 gym-0.17.3 pygame-1.5.0

Collecting git+https://github.com/Kojoley/atari-py.git

Cloning https://github.com/Kojoley/atari-py.git to /tmp/pip-req-build-upuobtb3

Running command git clone --filter=blob:none --quiet https://github.com/Kojoley/atari-py.git /tmp/pip-req-build-upuobtb3

Resolved https://github.com/Kojoley/atari-py.git to commit 86a1e05c0a95e9e6233c3a413521fdb34ca8a089

Preparing metadata (setup.py) ... done

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from atari-py==1.2.2) (2.0.2)

Building wheels for collected packages: atari-py

Building wheel for atari-py (setup.py) ... done

Created wheel for atari-py:

filename=atari_py-1.2.2-cp312-cp312-linux_x86_64.whl size=4739147 sha256=a93bce60398d3bf1cc1062a2875ad37cd275ddebbaee654eb843614f058bcd9e2

Stored in directory: /tmp/pip-ephem-wheel-cache-x95b920b/wheels/cb/1f/56/35b393898f74ae3d978063738e8eeb4b3d7357d0d7eeec6b9f

Successfully built atari-py

Installing collected packages: atari-py

Successfully installed atari-py-1.2.2

Collecting keras-rl2==1.0.5

Downloading keras_rl2-1.0.5-py3-none-any.whl.metadata (304 bytes)

Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (from keras-rl2==1.0.5) (2.18.0)

Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (1.4.0)

Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (1.6.3)

Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5)

(25.12.19)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (0.7.0)

Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (0.2.0)

Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (18.1.1)

Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (3.4.0)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (26.2)

Requirement already satisfied: protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (5.29.6)

Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (2.32.4)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (75.2.0)

Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (1.17.0)

Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (3.3.0)

Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (4.15.0)

Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (2.2.2)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (1.81.1)

Requirement already satisfied: tensorboard<2.19,>=2.18 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (2.18.0)

Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (3.13.2)

Requirement already satisfied: numpy<2.1.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (2.0.2)

Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5) (3.16.0)

Requirement already satisfied: ml-dtypes<0.5.0,>=0.4.0 in

```

/usr/local/lib/python3.12/dist-packages (from tensorflow->keras-rl2==1.0.5)
(0.4.1)
Requirement already satisfied: wheel<1.0,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from
astunparse>=1.6.0->tensorflow->keras-rl2==1.0.5) (0.47.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow->keras-rl2==1.0.5) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow->keras-rl2==1.0.5) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages
(from keras>=3.5.0->tensorflow->keras-rl2==1.0.5) (0.19.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow->keras-rl2==1.0.5) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests<3,>=2.21.0->tensorflow->keras-rl2==1.0.5) (3.18)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow->keras-rl2==1.0.5) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests<3,>=2.21.0->tensorflow->keras-rl2==1.0.5) (2026.6.17)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow->keras-rl2==1.0.5) (3.10.2)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow->keras-rl2==1.0.5) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.19,>=2.18->tensorflow->keras-rl2==1.0.5) (3.1.8)
Requirement already satisfied: markupsafe>=2.1.1 in
/usr/local/lib/python3.12/dist-packages (from
werkzeug>=1.0.1->tensorboard<2.19,>=2.18->tensorflow->keras-rl2==1.0.5) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow->keras-rl2==1.0.5) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from
rich->keras>=3.5.0->tensorflow->keras-rl2==1.0.5) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0->tensorflow->keras-
rl2==1.0.5) (0.1.2)
Downloading keras_rl2-1.0.5-py3-none-any.whl (52 kB)
52.1/52.1 kB
5.9 MB/s eta 0:00:00
Installing collected packages: keras-rl2
Successfully installed keras-rl2-1.0.5

```

Ecosistema clásico cargado en la nube de Google.

1.2 PARTE 2. Enunciado

Consideraciones a tener en cuenta:

- El entorno sobre el que trabajaremos será el indicado en el listado correspondien de cada grupo y el algoritmo que usaremos será *DQN*.
- Para nuestro ejercicio, el requisito mínimo será alcanzado cuando el agente consiga una **media de recompensa por encima de los puntos indicados en el listado por grupos en modo test**. Por ello, esta media de la recompensa se calculará a partir del código de test en la última celda del notebook.

Este proyecto práctico consta de tres partes:

1. Implementar la red neuronal que se usará en la solución
2. Implementar las distintas piezas de la solución DQN y probar al menos 3 propuestas diferentes de mejora.
3. Justificar la respuesta en relación a los resultados obtenidos e incluir al menos 3 gráficas relevantes comparando las 3 propuestas.

Rúbrica: Se valorará la originalidad en la solución aportada, así como la capacidad de discutir los resultados de forma detallada. El requisito mínimo servirá para aprobar la actividad, bajo premisa de que la discusión del resultado sera apropiada.

IMPORTANTE:

- Si no se consigue una puntuación óptima, responder sobre la mejor puntuación obtenida.
- Para entrenamientos largos, recordad que podéis usar checkpoints de vuestros modelos para retomar los entrenamientos. En este caso, recordad cambiar los parámetros adecuadamente (sobre todo los relacionados con el proceso de exploración).
- Se deberá entregar unicamente el notebook y los pesos del mejor modelo en un fichero .zip, de forma organizada.
- Cada alumno deberá de subir la solución de forma individual.

1.2.1 SpaceInvaders-v0

En nuestro caso es necesario realizar el entrenamiento por refuerzo en el ambiente en el entorno de **SpaceInvaders-v0** la documentación se encuentra en https://ale.farama.org/environments/space_invaders/.

Le objetivo en nuestro caso es: - Alcanzar más de 20 puntos con reward clipping durante más de 100 episodios consecutivos

1.3 PARTE 3. Desarrollo y preguntas

Importar librerías

```
[ ]: import os
os.environ['TF_USE_LEGACY_KERAS']="1"

import tensorflow.keras as tf
import tensorflow as tf1

from keras import __version__
tf.__version__ = __version__

print(tf1.__version__)
print(tf1.config.list_physical_devices('GPU'))
print(tf1.config.list_logical_devices('GPU'))
```

2.18.0

```
[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
[LogicalDevice(name='/device:GPU:0', device_type='GPU')]
```

```
[ ]: from __future__ import division

# -----

import sys
import os

# PARCHE DE COMPATIBILIDAD K2 PARA TENSORFLOW 2.18 + KERAS-RL2
import tensorflow.keras.models as keras_models
import tf_keras
from tf_keras.models import model_from_config
keras_models.model_from_config = model_from_config
sys.modules['tensorflow.keras.models'].model_from_config = model_from_config

if 'keras.models' in sys.modules:
    sys.modules['keras.models'].model_from_config = model_from_config

print("Parche 'model_from_config' inyectado con éxito usando tf-keras.")

# -----

from PIL import Image
import numpy as np
import gym
import csv
import os

from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense, Activation, Flatten, Conv2D,
↳ Permute, SpatialDropout2D, Input, Lambda
```

```

from tensorflow.keras.optimizers.legacy import Adam
import tensorflow.keras.backend as K

from rl.agents.dqn import DQNAgent
from rl.policy import LinearAnnealedPolicy, BoltzmannQPolicy, EpsGreedyQPolicy
from rl.memory import SequentialMemory
from rl.core import Processor
from rl.callbacks import FileLogger, ModelIntervalCheckpoint, Callback

print("Todos los módulos y agentes de Keras-RL2 cargados correctamente")

```

Parche 'model_from_config' inyectado con éxito usando tf-keras.
 Todos los módulos y agentes de Keras-RL2 cargados correctamente

```

[ ]: import os
import json
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def graph_training(logs_path, model_name, variables=['mean_q']):
    """
    Lee los logs en JSON y genera una cuadrícula 1x3 con las variables_
    solicitadas.
    """
    with open(logs_path, 'r') as f:
        data = json.load(f)

    # Crear una figura con 1 fila y 3 columnas, ajustando el tamaño
    fig, axes = plt.subplots(1, 3, figsize=(18, 5))
    fig.suptitle(f"Rendimiento de Entrenamiento: {model_name}", fontsize=16,
    ↪y=1.05)

    for i, var in enumerate(variables):
        if var in data:
            # Extraemos los datos para la variable actual
            df_var = pd.DataFrame(data[var]).dropna()

            # Graficamos en el sub-gráfico correspondiente (axes[i])
            df_var.plot(ax=axes[i], legend=False, color='#1f77b4') # Azul_
            ↪clásico

            # Configuramos títulos y etiquetas
            axes[i].set_title(f"Evolución de {var}")
            axes[i].set_xlabel('Steps')
            axes[i].set_ylabel(var)

```



```

        axes[i].grid(True, linestyle='--', alpha=0.7) # Cuadrícula para
↪mejor lectura
    else:
        print(f"Advertencia: La variable '{var}' no se encontró en
↪{logs_path}")

    # Ajusta el espaciado para que no se superpongan los textos
    plt.tight_layout()

    # Manejo de rutas y guardado
    # os.path.dirname(logs_path) obtiene la carpeta donde está el JSON
    path_imgs = os.path.join(os.path.dirname(logs_path), "imgs")
    os.makedirs(path_imgs, exist_ok=True)

    save_path = os.path.join(path_imgs, f"{model_name}_metricas_cuadrícula.png")
    plt.savefig(save_path, bbox_inches='tight')
    plt.show() # Muestra la gráfica en el notebook antes de continuar

def graph_test_results(test_metrics, model_name):
    """
    Recibe el diccionario de resultados del test y genera una cuadrícula 1x3.
    """
    rewards = test_metrics['episode_reward']
    steps = test_metrics['nb_steps']
    episodios = range(1, len(rewards) + 1)

    fig, axes = plt.subplots(1, 3, figsize=(18, 5))
    fig.suptitle(f"Resultados de Evaluación (Test): {model_name}", fontsize=16,
↪y=1.05)

    # 1. Gráfica de Recompensas a lo largo de los episodios
    axes[0].plot(episodios, rewards, marker='o', linestyle='-', color='#2ca02c')
    axes[0].set_title("Recompensas por Episodio")
    axes[0].set_xlabel("Número de Episodio")
    axes[0].set_ylabel("Reward")
    axes[0].grid(True, linestyle='--', alpha=0.7)

    # 2. Gráfica de Supervivencia (Pasos por episodio)
    axes[1].plot(episodios, steps, marker='s', linestyle='-', color='#ff7f0e')
    axes[1].set_title("Supervivencia (Pasos por Episodio)")
    axes[1].set_xlabel("Número de Episodio")
    axes[1].set_ylabel("Cantidad de Steps")
    axes[1].grid(True, linestyle='--', alpha=0.7)

    # 3. Diagrama de Caja (Boxplot) para ver la estabilidad
    caja = axes[2].boxplot(rewards, vert=True, patch_artist=True)

```

```

caja['boxes'][0].set_facecolor('#1f77b4') # Color azul de relleno
axes[2].set_title("Distribución de Rewards")
axes[2].set_ylabel("Reward")
axes[2].set_xticks([1])
axes[2].set_xticklabels([f'{len(rewards)} Episodios'])
axes[2].grid(True, linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()

```

Configuración base

```

[ ]: BASE_CONFIG = {
    # Entorno y preprocesamiento
    'ENV_NAME': 'SpaceInvaders-v0',
    'WINDOW_LENGTH': 4,
    'INPUT_SHAPE': (84, 84),
    'SEED': 123,

    # Hiperparámetros del agente
    'learning_rate': 0.00025,
    'discount_factor': 0.99,
    'batch_size': 32,
    'target_update_freq': 5000,
    'train_frequency': 4,
    'learning_starts': 10000,

    # Exploración
    'epsilon_start': 1.0,
    'epsilon_end': 0.1,
    'epsilon_decay_steps': 750000,
    'epsilon_test': 0.05,

    # Memoria
    'memory_size': 800000,

    # Entrenamiento
    'max_steps': 1000000,
    'optimizer': 'adam',
    'gradient_clip': 1.0,
}

```

```

[ ]: from gym import Wrapper
import numpy as np

class FixedFrameSkip(Wrapper):
    def __init__(self, env, skip=BASE_CONFIG['WINDOW_LENGTH']):

```

```

super(FixedFrameSkip, self).__init__(env)
self._skip = skip
if hasattr(self.env.unwrapped, 'ale'):
    self.env.unwrapped.frameskip = 1

def step(self, action):
    total_reward = 0.0
    done = False
    info = {}

    for i in range(self._skip):
        obs, reward, done, info = self.env.step(action)
        total_reward += reward
        if done:
            break

    return obs, total_reward, done, info

```

```

[ ]: env_base = gym.make(BASE_CONFIG['ENV_NAME'])
env = FixedFrameSkip(env_base, skip=4)

np.random.seed(BASE_CONFIG["SEED"])
env.seed(BASE_CONFIG["SEED"])
nb_actions = env.action_space.n
print(f"Acciones disponibles: {str(nb_actions)}" )
print(f"Formato de las observaciones: {env.observation_space}")

```

Acciones disponibles: 6

Formato de las observaciones: Box(0, 255, (210, 160, 3), uint8)

```

[ ]: class AtariProcessor(Processor):
    def process_observation(self, observation):
        assert observation.ndim == 3
        img = Image.fromarray(observation)
        img = img.resize(BASE_CONFIG["INPUT_SHAPE"], Image.LANCZOS).convert('L')
        processed_observation = np.array(img)
        assert processed_observation.shape == BASE_CONFIG["INPUT_SHAPE"]
        return processed_observation.astype('uint8')

    def process_state_batch(self, batch):
        processed_batch = batch.astype('float32') / 255.
        return processed_batch

    def process_reward(self, reward):
        return np.clip(reward, -1., 1.)

```

```
[ ]: def create_memory():
    return SequentialMemory(
        limit=BASE_CONFIG['memory_size'],
        window_length=BASE_CONFIG['WINDOW_LENGTH']
    )
```

```
[ ]: def create_policy():
    return LinearAnnealedPolicy(
        EpsGreedyQPolicy(),
        attr='eps',
        value_max=BASE_CONFIG['epsilon_start'],
        value_min=BASE_CONFIG['epsilon_end'],
        value_test=BASE_CONFIG['epsilon_test'],
        nb_steps=BASE_CONFIG['epsilon_decay_steps']
    )
```

```
[ ]: def create_agent(model, nb_actions, enable_double=False):
    return DQNAgent(
        model=model,
        nb_actions=nb_actions,
        memory=create_memory(),
        nb_steps_warmup=BASE_CONFIG['learning_starts'],
        target_model_update=BASE_CONFIG['target_update_freq'],
        policy=create_policy(),
        processor=AtariProcessor(),
        batch_size=BASE_CONFIG['batch_size'],
        gamma=BASE_CONFIG['discount_factor'],
        train_interval=BASE_CONFIG['train_frequency'],
        enable_double_dqn=enable_double,
    )
```

```
[ ]: def compile_agent(agent):
    agent.compile(
        optimizer=Adam(
            learning_rate=BASE_CONFIG['learning_rate'],
            clipnorm=BASE_CONFIG['gradient_clip']
        ),
        metrics=['mae']
    )
    return agent
```

```
[ ]: import csv
from rl.callbacks import Callback
import os

class MetricLogger(Callback):
    def __init__(self, version, base_path):
```

```

self.version = version
self.folder = os.path.join(base_path, f"version_{version}")
os.makedirs(self.folder, exist_ok=True)
self.file_path = os.path.join(self.folder, "metrics.csv")
self.fields = ['episode', 'steps', 'reward']

archivo_existe = os.path.exists(self.file_path) and os.path.
↳getsize(self.file_path) > 0
modo = 'a' if archivo_existe else 'w'

with open(self.file_path, modo, newline='') as f:
    if not archivo_existe:
        writer = csv.DictWriter(f, fieldnames=self.fields)
        writer.writeheader()

def on_episode_end(self, episode, logs):
    row = {
        'episode': episode,
        'steps': logs.get('nb_steps'),
        'reward': logs.get('episode_reward')
    }

    with open(self.file_path, 'a', newline='') as f:
        writer = csv.DictWriter(f, fieldnames=self.fields)
        writer.writerow(row)

```

```

[ ]: class SaveWeightsInterval(Callback):
    def __init__(self, agent, version, interval, base_path="."):
        self.agent = agent
        self.version = version
        self.interval = interval
        self.folder = os.path.join(base_path, f"version_{version}", "pesos")
        os.makedirs(self.folder, exist_ok=True)

    def on_episode_end(self, episode, logs={}):
        if (episode % self.interval == 0):
            filename = os.path.join(self.folder, f"dqn_v_{self.
↳version}_weights_{episode}_episodes.h5f")
            self.agent.save_weights(filename, overwrite=True)
            print(f"\n[Callback] Pesos guardados en: {filename}")

```

```

[ ]: from rl.callbacks import Callback
import os

class SaveBestWeights(Callback):
    def __init__(self, agent, version, base_path):

```

```

        self.agent = agent
        self.version = version
        self.best_reward = -float('inf')
        self.folder = os.path.join(base_path, f"version_{version}",
↪"mejor_modelo")
        os.makedirs(self.folder, exist_ok=True)

    def on_episode_end(self, episode, logs={}):
        current_reward = logs.get('episode_reward')

        if current_reward is not None and current_reward > self.best_reward:
            self.best_reward = current_reward
            filename = os.path.join(self.folder, "best_weights.h5f")
            self.agent.save_weights(filename, overwrite=True)
            print(f"\nNuevo récord: {current_reward}. Pesos guardados en
↪{filename}")

```

```

[ ]: class TotalStepStopper(Callback):
    def __init__(self, max_steps):
        super(TotalStepStopper, self).__init__()
        self.max_steps = max_steps

    def on_step_end(self, step, logs={}):
        if self.model.step >= self.max_steps:
            print(f"\n[Stopper] ¡Meta global alcanzada ({self.model.step} >=
↪{self.max_steps})! Deteniendo...")
            self.model.stop_training = True

```

```

[ ]: import os
import glob
import re
import pandas as pd

def entrenando_modelo_con_recuperacion(agent, env, version, max_steps,
↪carpeta_guardado, callbacks_lista, ruta_csv):
    """
    Busca los pesos más recientes, lee el CSV para saber el progreso exacto y
↪continúa el entrenamiento.
    """
    os.makedirs(carpeta_guardado, exist_ok=True)

    # 1. Buscar los últimos pesos
    patron_busqueda = os.path.join(carpeta_guardado, f"version_{version}",
↪"pesos", f"dqn_v_{version}_weights*_episodes.h5f.index")
    archivos = glob.glob(patron_busqueda)

```

```

archivo_reciente_base = None
max_episodio = -1

for archivo in archivos:

    match = re.search(r'weights_(\d+)_episodes', archivo)

    if match:
        episodio = int(match.group(1))
        if episodio > max_episodio:
            max_episodio = episodio
            archivo_reciente_base = archivo[:-6]

steps_reales_completados = 0

if archivo_reciente_base:
    print(f";Guardado encontrado! Restaurando pesos desde el episodio_{max_episodio}: {archivo_reciente_base}")
    agent.load_weights(archivo_reciente_base)

    try:
        if os.path.exists(ruta_csv):
            df = pd.read_csv(ruta_csv)
            steps_reales_completados = int(df['steps'].max())
            print(f"CSV leído correctamente. El modelo ya ha entrenado_{steps_reales_completados} steps.")
        else:
            print("No se encontró el CSV. Asumiendo 0 steps (Peligro de_sobreentrenamiento).")
            except Exception as e:
                print(f"Error al leer el CSV: {e}. Asumiendo 0 steps.")

    else:
        print("No se encontraron pesos previos. Iniciando desde cero.")

steps_restantes = max_steps - steps_reales_completados

if steps_restantes <= 0:
    print(f"El agente ya cumplió o superó la meta de los {max_steps} steps.")
    return None

    print(f"Retomando entrenamiento. Faltan exactamente: {steps_restantes} steps.")

historial = agent.fit(

```

```

        env,
        nb_steps=steps_restantes,
        visualize=False,
        verbose=2,
        callbacks=callbacks_lista
    )

    return historial

```

```

[ ]: import os
import shutil
import glob
import numpy as np
from gym import wrappers

def evaluar_agente(agent, env, folder_path, video_dir=None, nb_episodes=5):
    weights_filename = os.path.join(folder_path, 'best_weights.h5f')
    agent.load_weights(weights_filename)
    agent.test_policy.eps = 0.0

    grabar_video = video_dir is not None
    local_video_folder = "/content/video_eval_temporal" if grabar_video else None

    if grabar_video:
        env_ejecucion = wrappers.Monitor(
            env,
            local_video_folder,
            force=True,
            video_callable=lambda episode_id: True
        )
    else:
        env_ejecucion = env

    print(f"-> Iniciando evaluación ({'Modo Video Activo' if grabar_video else 'Modo Métricas Puro'})...")
    results = agent.test(env_ejecucion, nb_episodes=nb_episodes, visualize=False, verbose=0)

    if grabar_video:
        env_ejecucion.close()
        os.makedirs(video_dir, exist_ok=True)
        for archivo in glob.glob(os.path.join(local_video_folder, "*")):
            shutil.copy(archivo, video_dir)
        print(f"Video exportado exitosamente a: {video_dir}")

    return results

```



```
def print_results(agent_name, results):
    rewards = results.history['episode_reward']

    print(f"Resultados del Agente: {agent_name}")
    print(f"Rendimiento promedio: {np.mean(rewards)}")
    print(f"Rendimiento mínimo: {np.min(rewards)}")
    print(f"Rendimiento máximo: {np.max(rewards)}")
    print(f"Desviación estándar: {np.std(rewards)}")
```

1. Implementación de la red neuronal

```
[ ]: def create_dqn_model(input_shape, window_length, nb_actions,
    ↪with_spatial_dropout = False):
    input_layer = Input(shape=(window_length,) + input_shape)

    print(K.image_data_format())
    if K.image_data_format() == 'channels_last':
        x = Permute((2, 3, 1))(input_layer)
    elif K.image_data_format() == 'channels_first':
        x = Permute((1, 2, 3))(input_layer)
    else:
        raise RuntimeError('Unknown image_dim_ordering.')

    x = Conv2D(32, (8, 8), strides=(4, 4), padding='valid')(x)
    x = Activation('relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Conv2D(64, (4, 4), strides=(2, 2), padding='valid')(x)
    x = Activation('relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Conv2D(64, (3, 3), strides=(1, 1), padding='valid')(x)
    x = Activation('relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Flatten()(x)
    x = Dense(512)(x)
    x = Activation('relu')(x)

    output_layer = Dense(nb_actions)(x)
    output_layer = Activation('linear')(output_layer)

    model = Model(inputs=input_layer, outputs=output_layer)
```

```
return model
```

```
[ ]: import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense, Flatten, Conv2D, Activation,
    ↳ SpatialDropout2D, Layer
import tensorflow.keras.backend as K

# Definimos la operación matemática de Dueling como una Capa segura
@tf.keras.utils.register_keras_serializable()
class DuelingOutputLayer(Layer):
    def __init__(self, **kwargs):
        super(DuelingOutputLayer, self).__init__(**kwargs)

    def call(self, inputs):
        value_stream, advantage_stream = inputs
        # Aplicamos la fórmula:  $Q(s,a) = V(s) + (A(s,a) - \text{mean}(A(s,a)))$ 
        mean_advantage = tf.reduce_mean(advantage_stream, axis=1, keepdims=True)
        return value_stream + (advantage_stream - mean_advantage)

def create_dueling_dqn_model(input_shape, window_length, nb_actions,
    ↳ with_spatial_dropout=False):
    input_layer = Input(shape=(window_length,) + input_shape)

    if K.image_data_format() == 'channels_last':
        x = tf.keras.layers.Permute((2, 3, 1))(input_layer)
    else:
        x = tf.keras.layers.Permute((1, 2, 3))(input_layer)

    x = Conv2D(32, (8, 8), strides=(4, 4), padding='valid',
    ↳ activation='relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Conv2D(64, (4, 4), strides=(2, 2), padding='valid',
    ↳ activation='relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Conv2D(64, (3, 3), strides=(1, 1), padding='valid',
    ↳ activation='relu')(x)
    if with_spatial_dropout:
        x = SpatialDropout2D(0.1)(x)

    x = Flatten()(x)

    value_stream = Dense(512, activation='relu')(x)
```

```

value_stream = Dense(1, activation='linear')(value_stream)

advantage_stream = Dense(512, activation='relu')(x)
advantage_stream = Dense(nb_actions, activation='linear')(advantage_stream)

outputs = DuelingOutputLayer(name='dueling_output')([value_stream,
↪advantage_stream])

model = Model(inputs=input_layer, outputs=outputs,
↪name='Dueling_DQN_SpaceInvaders')

return model

```

2. Implementación de la solución DQN A continuación se realizan las 3 implementaciones solicitadas: - Implementación 1: **DQN Clásico (Baseline)** - Implementación 2: **DQN con SpatialDropout** - Implementación 3: **Dueling DQN**

Implementación 1: DQN Clásico (Baseline)

```

[ ]: dqn_model_without_spatial_dropout =
↪create_dqn_model(input_shape=BASE_CONFIG["INPUT_SHAPE"],
↪window_length=BASE_CONFIG["WINDOW_LENGTH"],
↪nb_actions=nb_actions)

print(dqn_model_without_spatial_dropout.summary())

baseline_agent = create_agent(model=dqn_model_without_spatial_dropout,
↪nb_actions=nb_actions)
baseline_agent = compile_agent(baseline_agent)

DQN_CLASICO_VERSION_ACTUAL = "1.0"
DQN_CLASICO_CARPETA = '/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
↪practico/dqn clasico/'
DQN_CLASICO_METRICS =
↪f'{DQN_CLASICO_CARPETA}version_{DQN_CLASICO_VERSION_ACTUAL}/metricas.csv'
DQN_CLASICO_LOGS = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
↪practico/dqn clasico/version_{DQN_CLASICO_VERSION_ACTUAL}/logs.json'

```

channels_last
Model: "model_5"

Layer (type)	Output Shape	Param #
input_4 (InputLayer)	[(None, 4, 84, 84)]	0

permute_3 (Permute)	(None, 84, 84, 4)	0
conv2d_9 (Conv2D)	(None, 20, 20, 32)	8224
activation_10 (Activation)	(None, 20, 20, 32)	0
conv2d_10 (Conv2D)	(None, 9, 9, 64)	32832
activation_11 (Activation)	(None, 9, 9, 64)	0
conv2d_11 (Conv2D)	(None, 7, 7, 64)	36928
activation_12 (Activation)	(None, 7, 7, 64)	0
flatten_3 (Flatten)	(None, 3136)	0
dense_8 (Dense)	(None, 512)	1606144
activation_13 (Activation)	(None, 512)	0
dense_9 (Dense)	(None, 6)	3078
activation_14 (Activation)	(None, 6)	0

=====

Total params: 1687206 (6.44 MB)

Trainable params: 1687206 (6.44 MB)

Non-trainable params: 0 (0.00 Byte)

None

```
[ ]: dqn_clasico_guardar_pesos = SaveWeightsInterval(
    agent=baseline_agent,
    version=DQN_CLASICO_VERSION_ACTUAL,
    interval=160,
    base_path=DQN_CLASICO_CARPETA
)
dqn_clasico_guardar_mejor_modelo = SaveBestWeights(
    agent=baseline_agent,
    version=DQN_CLASICO_VERSION_ACTUAL,
    base_path=DQN_CLASICO_CARPETA)

dqn_clasico_metricas = MetricLogger(
    version=DQN_CLASICO_VERSION_ACTUAL,
    base_path=DQN_CLASICO_CARPETA)

dqn_clasico_lista_callbacks = [dqn_clasico_guardar_pesos,
```

```

        dqn_clasico_guardar_mejor_modelo,
        dqn_clasico_metricas,
        FileLogger(DQN_CLASICO_LOGS, interval=160),
        ↵
        ↵TotalStepStopper(max_steps=BASE_CONFIG['max_steps']))]

history_1 = entrenando_modelo_con_recuperacion(
    agent=baseline_agent,
    env=env,
    version=DQN_CLASICO_VERSION_ACTUAL,
    max_steps=BASE_CONFIG['max_steps'],
    carpeta_guardado=DQN_CLASICO_CARPETA,
    callbacks_lista=dqn_clasico_lista_callbacks,
    ruta_csv=DQN_CLASICO_METRICS
)

```

¡Guardado encontrado! Restaurando pesos desde el episodio 3200:
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
clasico/version_1.0/pesos/dqn_v_1.0_weights_3200_episodes.h5f
CSV leído correctamente. El modelo ya ha entrenado 32718 steps.
Retomando entrenamiento. Faltan exactamente: 967282 steps.
Training for 967282 steps ...
done, took 0.549 seconds

```

[ ]: # Pasamos la carpeta raíz donde están guardados los archivos de la versión ↵
      ↵clásica
DQN_CLASICO_VERSION_ACTUAL = "1.0"
CARPETA_CHECKPOINT_CLASICO = f'/content/gdrive/MyDrive/08_MIAR/actividades/
      ↵proyecto practico/dqn clasico/version_{DQN_CLASICO_VERSION_ACTUAL}/
      ↵mejor_modelo'
CARPETA_VIDEO_CLASICO = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto ↵
      ↵practico/dqn clasico/version_{DQN_CLASICO_VERSION_ACTUAL}/video'

# Ejecucion del test con 100 episodios para metricas.
clasico_results = evaluar_agente(
    agent=baseline_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_CLASICO,
    nb_episodes=100
)

# Generación de video.
clasico_video = evaluar_agente(
    agent=baseline_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_CLASICO,
    video_dir=CARPETA_VIDEO_CLASICO,

```

```

        nb_episodes=2
    )

print_results("DQN Clásico", clasico_results)

```

-> Iniciando evaluación (Modo Métricas Puro)...

```

/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:
UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied
automatically.

```

```

    updates=self.state_updates,

```

-> Iniciando evaluación (Modo Video Activo)...

Video exportado exitosamente a:

```

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
clasico/version_1.0/video

```

Resultados del Agente: DQN Clásico

Rendimiento promedio: 17.24

Rendimiento mínimo: 6.0

Rendimiento máximo: 31.0

Desviación estándar: 5.729083696368906

Implementación 2: DQN con SpatialDropout

```

[ ]: dqn_model_with_spatial_dropout = create_dqn_model(input_shape=BASE_CONFIG["INPUT_SHAPE"],
    window_length=BASE_CONFIG["WINDOW_LENGTH"],
    nb_actions=nb_actions,
    with_spatial_dropout=True)

print(dqn_model_with_spatial_dropout.summary())

spatial_dropout_agent = create_agent(model=dqn_model_with_spatial_dropout,
    nb_actions=nb_actions)
spatial_dropout_agent = compile_agent(spatial_dropout_agent)

DQN_SPATIAL_VERSION_ACTUAL = "1.0"
DQN_SPATIAL_CARPETA = '/content/gdrive/MyDrive/08_MIAR/actividades/proyecto
    practico/dqn spatial/'
DQN_SPATIAL_METRICS = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto
    practico/dqn spatial/version_{DQN_SPATIAL_VERSION_ACTUAL}/metricas.csv'
DQN_SPATIAL_LOGS = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto
    practico/dqn spatial/version_{DQN_SPATIAL_VERSION_ACTUAL}/logs.json'

```

channels_last

Model: "model_2"

Layer (type)	Output Shape	Param #
--------------	--------------	---------

```

=====
input_2 (InputLayer)      [(None, 4, 84, 84)]      0

permute_1 (Permute)       (None, 84, 84, 4)        0

conv2d_3 (Conv2D)         (None, 20, 20, 32)       8224

activation_5 (Activation)  (None, 20, 20, 32)       0

spatial_dropout2d_3 (Spat (None, 20, 20, 32)       0
ialDropout2D)

conv2d_4 (Conv2D)         (None, 9, 9, 64)         32832

activation_6 (Activation)  (None, 9, 9, 64)         0

spatial_dropout2d_4 (Spat (None, 9, 9, 64)         0
ialDropout2D)

conv2d_5 (Conv2D)         (None, 7, 7, 64)         36928

activation_7 (Activation)  (None, 7, 7, 64)         0

spatial_dropout2d_5 (Spat (None, 7, 7, 64)         0
ialDropout2D)

flatten_1 (Flatten)       (None, 3136)             0

dense_2 (Dense)           (None, 512)              1606144

activation_8 (Activation)  (None, 512)              0

dense_3 (Dense)           (None, 6)                3078

activation_9 (Activation)  (None, 6)                0

=====
Total params: 1687206 (6.44 MB)
Trainable params: 1687206 (6.44 MB)
Non-trainable params: 0 (0.00 Byte)
-----
None

```

```

[ ]: # Entrenamiento
      # Prepararemos los callbacks contruidos
      dqn_spatial_guardar_pesos = SaveWeightsInterval(
          agent=spatial_dropout_agent,

```

```

        version=DQN_SPATIAL_VERSION_ACTUAL,
        interval=160,
        base_path = DQN_SPATIAL_CARPETA)

dqn_spatial_guardar_mejor_modelo = SaveBestWeights(
    agent=spatial_dropout_agent,
    version=DQN_SPATIAL_VERSION_ACTUAL,
    base_path=DQN_SPATIAL_CARPETA)

dqn_spatial_metricas = MetricLogger(
    version=DQN_SPATIAL_VERSION_ACTUAL,
    base_path=DQN_SPATIAL_CARPETA)

dqn_spatial_lista_callbacks = [dqn_spatial_guardar_pesos,
                                dqn_spatial_guardar_mejor_modelo,
                                dqn_spatial_metricas,
                                FileLogger(DQN_SPATIAL_LOGS, interval=160),
                                ↵
                                ↪TotalStepStopper(max_steps=BASE_CONFIG['max_steps'])]

history_2 = entrenando_modelo_con_recuperacion(
    agent=spatial_dropout_agent,
    env=env,
    version=DQN_SPATIAL_VERSION_ACTUAL,
    max_steps=BASE_CONFIG['max_steps'],
    carpeta_guardado=DQN_SPATIAL_CARPETA,
    callbacks_lista=dqn_spatial_lista_callbacks,
    ruta_csv=DQN_SPATIAL_METRICS
)

```

No se encontraron pesos previos. Iniciando desde cero.
 Retomando entrenamiento. Faltan exactamente: 1000000 steps.
 Training for 1000000 steps ...

```

/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:
UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied
automatically.
    updates=self.state_updates,

```

[Callback] Pesos guardados en:
 /content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_0_episodes.h5f

Nuevo récord: 22.0. Pesos guardados en
 /content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/mejor_modelo/best_weights.h5f

920/1000000: episode: 1, duration: 5.287s, episode steps: 920, steps per second: 174, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1599/1000000: episode: 2, duration: 2.827s, episode steps: 679, steps per second: 240, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

2136/1000000: episode: 3, duration: 2.261s, episode steps: 537, steps per second: 237, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

2751/1000000: episode: 4, duration: 2.569s, episode steps: 615, steps per second: 239, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.353 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

3233/1000000: episode: 5, duration: 2.017s, episode steps: 482, steps per second: 239, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

3638/1000000: episode: 6, duration: 1.738s, episode steps: 405, steps per second: 233, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

4011/1000000: episode: 7, duration: 1.576s, episode steps: 373, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

4582/1000000: episode: 8, duration: 2.422s, episode steps: 571, steps per second: 236, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

5301/1000000: episode: 9, duration: 3.029s, episode steps: 719, steps per second: 237, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

Nuevo récord: 23.0. Pesos guardados en
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/mejor_modelo/best_weights.h5f

6068/1000000: episode: 10, duration: 3.237s, episode steps: 767, steps per second: 237, episode reward: 23.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

6534/1000000: episode: 11, duration: 1.942s, episode steps: 466, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

7027/1000000: episode: 12, duration: 2.050s, episode steps: 493, steps per second: 241, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

7441/1000000: episode: 13, duration: 1.755s, episode steps: 414, steps per second: 236, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

8057/1000000: episode: 14, duration: 2.633s, episode steps: 616, steps per second: 234, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

8348/1000000: episode: 15, duration: 1.224s, episode steps: 291, steps per second: 238, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean

```

action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
9114/1000000: episode: 16, duration: 3.174s, episode steps: 766, steps per
second: 241, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
9525/1000000: episode: 17, duration: 1.715s, episode steps: 411, steps per
second: 240, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
9839/1000000: episode: 18, duration: 1.324s, episode steps: 314, steps per
second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:
UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied
automatically.
updates=self.state_updates,

10504/1000000: episode: 19, duration: 18.376s, episode steps: 665, steps per
second: 36, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.448 [0.000, 5.000], loss: 0.010455, mae: 0.046785, mean_q: 0.076358,
mean_eps: 0.987698
10876/1000000: episode: 20, duration: 10.668s, episode steps: 372, steps per
second: 35, episode reward: 12.000, mean reward: 0.032 [0.000, 1.000], mean
action: 2.438 [0.000, 5.000], loss: 0.010648, mae: 0.048388, mean_q: 0.073615,
mean_eps: 0.987174
11460/1000000: episode: 21, duration: 17.267s, episode steps: 584, steps per
second: 34, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.411 [0.000, 5.000], loss: 0.010640, mae: 0.049341, mean_q: 0.069348,
mean_eps: 0.986601
12016/1000000: episode: 22, duration: 17.544s, episode steps: 556, steps per
second: 32, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.448 [0.000, 5.000], loss: 0.008906, mae: 0.046135, mean_q: 0.065770,
mean_eps: 0.985917
12539/1000000: episode: 23, duration: 16.425s, episode steps: 523, steps per
second: 32, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.577 [0.000, 5.000], loss: 0.010632, mae: 0.050978, mean_q: 0.070966,
mean_eps: 0.985269
12927/1000000: episode: 24, duration: 12.233s, episode steps: 388, steps per
second: 32, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.497 [0.000, 5.000], loss: 0.009416, mae: 0.048892, mean_q: 0.072945,
mean_eps: 0.984722
13431/1000000: episode: 25, duration: 15.874s, episode steps: 504, steps per
second: 32, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.552 [0.000, 5.000], loss: 0.010382, mae: 0.049589, mean_q: 0.076718,
mean_eps: 0.984186
14011/1000000: episode: 26, duration: 18.171s, episode steps: 580, steps per
second: 32, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.578 [0.000, 5.000], loss: 0.009500, mae: 0.049370, mean_q: 0.071733,
mean_eps: 0.983536

```

14544/1000000: episode: 27, duration: 16.893s, episode steps: 533, steps per second: 32, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.009647, mae: 0.050440, mean_q: 0.074583, mean_eps: 0.982869

14820/1000000: episode: 28, duration: 8.741s, episode steps: 276, steps per second: 32, episode reward: 8.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.008996, mae: 0.048338, mean_q: 0.072802, mean_eps: 0.982384

15558/1000000: episode: 29, duration: 21.852s, episode steps: 738, steps per second: 34, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.008553, mae: 0.059572, mean_q: 0.088977, mean_eps: 0.981774

15901/1000000: episode: 30, duration: 10.760s, episode steps: 343, steps per second: 32, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: 0.010103, mae: 0.068622, mean_q: 0.105188, mean_eps: 0.981124

16277/1000000: episode: 31, duration: 11.816s, episode steps: 376, steps per second: 32, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: 0.011750, mae: 0.071745, mean_q: 0.105248, mean_eps: 0.980692

17129/1000000: episode: 32, duration: 26.700s, episode steps: 852, steps per second: 32, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.010689, mae: 0.070978, mean_q: 0.104438, mean_eps: 0.979955

17758/1000000: episode: 33, duration: 19.711s, episode steps: 629, steps per second: 32, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: 0.010850, mae: 0.072156, mean_q: 0.107606, mean_eps: 0.979067

18702/1000000: episode: 34, duration: 29.649s, episode steps: 944, steps per second: 32, episode reward: 13.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.009385, mae: 0.068044, mean_q: 0.101259, mean_eps: 0.978124

19253/1000000: episode: 35, duration: 17.302s, episode steps: 551, steps per second: 32, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.010624, mae: 0.070559, mean_q: 0.104911, mean_eps: 0.977226

19732/1000000: episode: 36, duration: 15.098s, episode steps: 479, steps per second: 32, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: 0.008421, mae: 0.066613, mean_q: 0.099474, mean_eps: 0.976610

20005/1000000: episode: 37, duration: 8.691s, episode steps: 273, steps per second: 31, episode reward: 7.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.008804, mae: 0.066432, mean_q: 0.091882, mean_eps: 0.976158

20478/1000000: episode: 38, duration: 13.528s, episode steps: 473, steps per second: 35, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.008625, mae: 0.088886, mean_q: 0.123293, mean_eps: 0.975710

20917/1000000: episode: 39, duration: 13.885s, episode steps: 439, steps per second: 32, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.010831, mae: 0.097250, mean_q: 0.140420, mean_eps: 0.975162

21201/1000000: episode: 40, duration: 8.916s, episode steps: 284, steps per second: 32, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: 0.009849, mae: 0.091810, mean_q: 0.130927, mean_eps: 0.974728

21581/1000000: episode: 41, duration: 11.993s, episode steps: 380, steps per second: 32, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.371 [0.000, 5.000], loss: 0.009554, mae: 0.093865, mean_q: 0.129649, mean_eps: 0.974330

22015/1000000: episode: 42, duration: 13.641s, episode steps: 434, steps per second: 32, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: 0.008318, mae: 0.091143, mean_q: 0.127541, mean_eps: 0.973842

22548/1000000: episode: 43, duration: 16.826s, episode steps: 533, steps per second: 32, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.006976, mae: 0.089925, mean_q: 0.126133, mean_eps: 0.973264

22923/1000000: episode: 44, duration: 11.922s, episode steps: 375, steps per second: 31, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.007969, mae: 0.092462, mean_q: 0.128980, mean_eps: 0.972719

23352/1000000: episode: 45, duration: 13.591s, episode steps: 429, steps per second: 32, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.008301, mae: 0.092562, mean_q: 0.125869, mean_eps: 0.972237

24149/1000000: episode: 46, duration: 25.195s, episode steps: 797, steps per second: 32, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: 0.007984, mae: 0.090391, mean_q: 0.125023, mean_eps: 0.971500

24500/1000000: episode: 47, duration: 11.134s, episode steps: 351, steps per second: 32, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.007580, mae: 0.088722, mean_q: 0.125738, mean_eps: 0.970811

24804/1000000: episode: 48, duration: 9.689s, episode steps: 304, steps per second: 31, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: 0.008197, mae: 0.095388, mean_q: 0.131228, mean_eps: 0.970420

25196/1000000: episode: 49, duration: 11.896s, episode steps: 392, steps per second: 33, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.005417, mae: 0.090477, mean_q: 0.126448, mean_eps: 0.970002

25591/1000000: episode: 50, duration: 11.708s, episode steps: 395, steps per second: 34, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.008953, mae: 0.103468, mean_q: 0.144332, mean_eps: 0.969530

26168/1000000: episode: 51, duration: 18.235s, episode steps: 577, steps per second: 32, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.009012, mae: 0.103606, mean_q: 0.139713, mean_eps: 0.968946

26466/1000000: episode: 52, duration: 9.444s, episode steps: 298, steps per second: 32, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: 0.010552, mae: 0.113176, mean_q: 0.151749, mean_eps: 0.968421

26960/1000000: episode: 53, duration: 15.623s, episode steps: 494, steps per second: 32, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.007898, mae: 0.102995, mean_q: 0.143286, mean_eps: 0.967946

27398/1000000: episode: 54, duration: 13.868s, episode steps: 438, steps per second: 32, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: 0.008826, mae: 0.104895, mean_q: 0.140513, mean_eps: 0.967386

27694/1000000: episode: 55, duration: 9.321s, episode steps: 296, steps per second: 32, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: 0.006958, mae: 0.100599, mean_q: 0.136350, mean_eps: 0.966945

28173/1000000: episode: 56, duration: 15.141s, episode steps: 479, steps per second: 32, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.007087, mae: 0.101354, mean_q: 0.140407, mean_eps: 0.966479

28715/1000000: episode: 57, duration: 17.110s, episode steps: 542, steps per second: 32, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: 0.007905, mae: 0.104650, mean_q: 0.146458, mean_eps: 0.965867

29353/1000000: episode: 58, duration: 20.127s, episode steps: 638, steps per second: 32, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.007975, mae: 0.102915, mean_q: 0.140912, mean_eps: 0.965159

29809/1000000: episode: 59, duration: 14.393s, episode steps: 456, steps per second: 32, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: 0.007204, mae: 0.102014, mean_q: 0.139696, mean_eps: 0.964502

30377/1000000: episode: 60, duration: 16.785s, episode steps: 568, steps per second: 34, episode reward: 3.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.008499, mae: 0.119561, mean_q: 0.166947, mean_eps: 0.963887

30842/1000000: episode: 61, duration: 14.429s, episode steps: 465, steps per second: 32, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: 0.008380, mae: 0.127210, mean_q: 0.172571, mean_eps: 0.963268

31299/1000000: episode: 62, duration: 14.385s, episode steps: 457, steps per second: 32, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.007861, mae: 0.124296, mean_q: 0.167868, mean_eps: 0.962716

31661/1000000: episode: 63, duration: 11.480s, episode steps: 362, steps per second: 32, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.007311, mae: 0.122526, mean_q: 0.162906, mean_eps: 0.962224

32550/1000000: episode: 64, duration: 27.971s, episode steps: 889, steps per second: 32, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.007460, mae: 0.125186, mean_q: 0.164689, mean_eps: 0.961473

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

```
self.step += 1
```

33219/1000000: episode: 65, duration: 8.763s, episode steps: 669, steps per second: 76, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.008214, mae: 0.134625, mean_q: 0.175752, mean_eps: 0.960810

33681/1000000: episode: 66, duration: 1.952s, episode steps: 462, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

34055/1000000: episode: 67, duration: 1.569s, episode steps: 374, steps per second: 238, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

34555/1000000: episode: 68, duration: 2.095s, episode steps: 500, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

35048/1000000: episode: 69, duration: 2.079s, episode steps: 493, steps per second: 237, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

35343/1000000: episode: 70, duration: 1.271s, episode steps: 295, steps per second: 232, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

35756/1000000: episode: 71, duration: 1.760s, episode steps: 413, steps per second: 235, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

36271/1000000: episode: 72, duration: 2.155s, episode steps: 515, steps per second: 239, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

36583/1000000: episode: 73, duration: 1.313s, episode steps: 312, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37116/1000000: episode: 74, duration: 2.229s, episode steps: 533, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37575/1000000: episode: 75, duration: 1.932s, episode steps: 459, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37968/1000000: episode: 76, duration: 1.648s, episode steps: 393, steps per second: 238, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 38439/1000000: episode: 77, duration: 1.984s, episode steps: 471, steps per second: 237, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 38953/1000000: episode: 78, duration: 2.173s, episode steps: 514, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 39384/1000000: episode: 79, duration: 1.821s, episode steps: 431, steps per second: 237, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 39785/1000000: episode: 80, duration: 1.683s, episode steps: 401, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 40168/1000000: episode: 81, duration: 1.608s, episode steps: 383, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 40660/1000000: episode: 82, duration: 2.086s, episode steps: 492, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 41096/1000000: episode: 83, duration: 1.840s, episode steps: 436, steps per second: 237, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 41813/1000000: episode: 84, duration: 3.004s, episode steps: 717, steps per second: 239, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 42496/1000000: episode: 85, duration: 2.849s, episode steps: 683, steps per second: 240, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 43033/1000000: episode: 86, duration: 2.258s, episode steps: 537, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 43619/1000000: episode: 87, duration: 2.464s, episode steps: 586, steps per second: 238, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 44312/1000000: episode: 88, duration: 2.921s, episode steps: 693, steps per second: 237, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 44619/1000000: episode: 89, duration: 1.300s, episode steps: 307, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 45247/1000000: episode: 90, duration: 2.625s, episode steps: 628, steps per second: 239, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 45842/1000000: episode: 91, duration: 2.487s, episode steps: 595, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46204/1000000: episode: 92, duration: 1.514s, episode steps: 362, steps per second: 239, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean

action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46493/1000000: episode: 93, duration: 1.209s, episode steps: 289, steps per second: 239, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.256 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46875/1000000: episode: 94, duration: 1.609s, episode steps: 382, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 47271/1000000: episode: 95, duration: 1.674s, episode steps: 396, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 47556/1000000: episode: 96, duration: 1.205s, episode steps: 285, steps per second: 237, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 47940/1000000: episode: 97, duration: 1.604s, episode steps: 384, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 48436/1000000: episode: 98, duration: 2.070s, episode steps: 496, steps per second: 240, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 48979/1000000: episode: 99, duration: 2.277s, episode steps: 543, steps per second: 238, episode reward: 15.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 49371/1000000: episode: 100, duration: 1.644s, episode steps: 392, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 49762/1000000: episode: 101, duration: 1.640s, episode steps: 391, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 50080/1000000: episode: 102, duration: 1.381s, episode steps: 318, steps per second: 230, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.704 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 50759/1000000: episode: 103, duration: 2.856s, episode steps: 679, steps per second: 238, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 51173/1000000: episode: 104, duration: 1.723s, episode steps: 414, steps per second: 240, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 51548/1000000: episode: 105, duration: 1.577s, episode steps: 375, steps per second: 238, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 51941/1000000: episode: 106, duration: 1.648s, episode steps: 393, steps per second: 239, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 52338/1000000: episode: 107, duration: 1.664s, episode steps: 397, steps per second: 239, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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spartial/version_1.0/mejor_modelo/best_weights.h5f

53425/1000000: episode: 108, duration: 4.578s, episode steps: 1087, steps per second: 237, episode reward: 25.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

53713/1000000: episode: 109, duration: 1.219s, episode steps: 288, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

54264/1000000: episode: 110, duration: 2.309s, episode steps: 551, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

54785/1000000: episode: 111, duration: 2.188s, episode steps: 521, steps per second: 238, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

55325/1000000: episode: 112, duration: 2.264s, episode steps: 540, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

55640/1000000: episode: 113, duration: 1.339s, episode steps: 315, steps per second: 235, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

56340/1000000: episode: 114, duration: 2.939s, episode steps: 700, steps per second: 238, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

57077/1000000: episode: 115, duration: 3.081s, episode steps: 737, steps per second: 239, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

57606/1000000: episode: 116, duration: 2.218s, episode steps: 529, steps per second: 238, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

58373/1000000: episode: 117, duration: 3.207s, episode steps: 767, steps per second: 239, episode reward: 14.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

58966/1000000: episode: 118, duration: 2.495s, episode steps: 593, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

59396/1000000: episode: 119, duration: 1.822s, episode steps: 430, steps per second: 236, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

59842/1000000: episode: 120, duration: 1.868s, episode steps: 446, steps per second: 239, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

60416/1000000: episode: 121, duration: 2.399s, episode steps: 574, steps per second: 239, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

60823/1000000: episode: 122, duration: 1.713s, episode steps: 407, steps per second: 238, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

61402/1000000: episode: 123, duration: 2.418s, episode steps: 579, steps per

second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

62290/1000000: episode: 124, duration: 3.721s, episode steps: 888, steps per second: 239, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

62768/1000000: episode: 125, duration: 2.025s, episode steps: 478, steps per second: 236, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

63214/1000000: episode: 126, duration: 1.881s, episode steps: 446, steps per second: 237, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

63492/1000000: episode: 127, duration: 1.173s, episode steps: 278, steps per second: 237, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

64166/1000000: episode: 128, duration: 2.821s, episode steps: 674, steps per second: 239, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

64806/1000000: episode: 129, duration: 2.669s, episode steps: 640, steps per second: 240, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.380 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

65091/1000000: episode: 130, duration: 1.201s, episode steps: 285, steps per second: 237, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.333 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

65584/1000000: episode: 131, duration: 2.074s, episode steps: 493, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

66104/1000000: episode: 132, duration: 2.163s, episode steps: 520, steps per second: 240, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

66922/1000000: episode: 133, duration: 3.399s, episode steps: 818, steps per second: 241, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

67688/1000000: episode: 134, duration: 3.187s, episode steps: 766, steps per second: 240, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

67970/1000000: episode: 135, duration: 1.193s, episode steps: 282, steps per second: 236, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

68288/1000000: episode: 136, duration: 1.346s, episode steps: 318, steps per second: 236, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

68619/1000000: episode: 137, duration: 1.402s, episode steps: 331, steps per second: 236, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.644 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

69093/1000000: episode: 138, duration: 1.991s, episode steps: 474, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

69391/1000000: episode: 139, duration: 1.258s, episode steps: 298, steps per

second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

69770/1000000: episode: 140, duration: 1.599s, episode steps: 379, steps per second: 237, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

70296/1000000: episode: 141, duration: 2.193s, episode steps: 526, steps per second: 240, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

70763/1000000: episode: 142, duration: 1.971s, episode steps: 467, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

71305/1000000: episode: 143, duration: 2.277s, episode steps: 542, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

72071/1000000: episode: 144, duration: 3.201s, episode steps: 766, steps per second: 239, episode reward: 22.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

72663/1000000: episode: 145, duration: 2.465s, episode steps: 592, steps per second: 240, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

73283/1000000: episode: 146, duration: 2.582s, episode steps: 620, steps per second: 240, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

73751/1000000: episode: 147, duration: 1.953s, episode steps: 468, steps per second: 240, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

74046/1000000: episode: 148, duration: 1.258s, episode steps: 295, steps per second: 235, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

74414/1000000: episode: 149, duration: 1.549s, episode steps: 368, steps per second: 238, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

74906/1000000: episode: 150, duration: 2.057s, episode steps: 492, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

75709/1000000: episode: 151, duration: 8.038s, episode steps: 803, steps per second: 100, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.010300, mae: 0.168494, mean_q: 0.221307, mean_eps: 0.987894

76539/1000000: episode: 152, duration: 26.244s, episode steps: 830, steps per second: 32, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: 0.011350, mae: 0.168812, mean_q: 0.216407, mean_eps: 0.987294

76848/1000000: episode: 153, duration: 9.861s, episode steps: 309, steps per second: 31, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.288 [0.000, 5.000], loss: 0.009670, mae: 0.164511, mean_q: 0.210010, mean_eps: 0.986613

77368/1000000: episode: 154, duration: 16.649s, episode steps: 520, steps per

second: 31, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.010567, mae: 0.169057, mean_q: 0.215122, mean_eps: 0.986116

78034/1000000: episode: 155, duration: 21.165s, episode steps: 666, steps per second: 31, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: 0.011088, mae: 0.175766, mean_q: 0.221456, mean_eps: 0.985403

78816/1000000: episode: 156, duration: 24.824s, episode steps: 782, steps per second: 32, episode reward: 6.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.008379, mae: 0.166023, mean_q: 0.207543, mean_eps: 0.984534

79522/1000000: episode: 157, duration: 22.519s, episode steps: 706, steps per second: 31, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.007866, mae: 0.166990, mean_q: 0.207751, mean_eps: 0.983642

79984/1000000: episode: 158, duration: 14.809s, episode steps: 462, steps per second: 31, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.007582, mae: 0.161058, mean_q: 0.199917, mean_eps: 0.982941

80285/1000000: episode: 159, duration: 9.663s, episode steps: 301, steps per second: 31, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.007490, mae: 0.163694, mean_q: 0.204207, mean_eps: 0.982482

80914/1000000: episode: 160, duration: 19.939s, episode steps: 629, steps per second: 32, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.008653, mae: 0.173154, mean_q: 0.216432, mean_eps: 0.981923

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spartial/version_1.0/pesos/dqn_v_1.0_weights_160_episodes.h5f

81456/1000000: episode: 161, duration: 17.377s, episode steps: 542, steps per second: 31, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.010812, mae: 0.186091, mean_q: 0.231646, mean_eps: 0.981222

82050/1000000: episode: 162, duration: 18.934s, episode steps: 594, steps per second: 31, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: 0.009099, mae: 0.187366, mean_q: 0.233445, mean_eps: 0.980541

82497/1000000: episode: 163, duration: 14.252s, episode steps: 447, steps per second: 31, episode reward: 13.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.007520, mae: 0.171640, mean_q: 0.213132, mean_eps: 0.979914

82959/1000000: episode: 164, duration: 14.640s, episode steps: 462, steps per second: 32, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.008366, mae: 0.181074, mean_q: 0.225118, mean_eps: 0.979370

83359/1000000: episode: 165, duration: 12.692s, episode steps: 400, steps per

second: 32, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: 0.008734, mae: 0.186782, mean_q: 0.230157, mean_eps: 0.978854

83861/1000000: episode: 166, duration: 15.978s, episode steps: 502, steps per second: 31, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.007800, mae: 0.179465, mean_q: 0.225229, mean_eps: 0.978311

84345/1000000: episode: 167, duration: 15.338s, episode steps: 484, steps per second: 32, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: 0.008458, mae: 0.180433, mean_q: 0.223372, mean_eps: 0.977718

84794/1000000: episode: 168, duration: 14.226s, episode steps: 449, steps per second: 32, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.008376, mae: 0.182701, mean_q: 0.227340, mean_eps: 0.977159

85504/1000000: episode: 169, duration: 22.585s, episode steps: 710, steps per second: 31, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: 0.007242, mae: 0.179672, mean_q: 0.223151, mean_eps: 0.976466

85854/1000000: episode: 170, duration: 11.201s, episode steps: 350, steps per second: 31, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.009541, mae: 0.201026, mean_q: 0.248456, mean_eps: 0.975830

86302/1000000: episode: 171, duration: 14.219s, episode steps: 448, steps per second: 32, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: 0.010260, mae: 0.209983, mean_q: 0.258330, mean_eps: 0.975350

86595/1000000: episode: 172, duration: 9.330s, episode steps: 293, steps per second: 31, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.009744, mae: 0.213267, mean_q: 0.262725, mean_eps: 0.974906

87114/1000000: episode: 173, duration: 16.520s, episode steps: 519, steps per second: 31, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.010426, mae: 0.210980, mean_q: 0.260385, mean_eps: 0.974418

87566/1000000: episode: 174, duration: 14.385s, episode steps: 452, steps per second: 31, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: 0.009135, mae: 0.209635, mean_q: 0.258686, mean_eps: 0.973835

87906/1000000: episode: 175, duration: 10.819s, episode steps: 340, steps per second: 31, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.671 [0.000, 5.000], loss: 0.007961, mae: 0.210510, mean_q: 0.260670, mean_eps: 0.973360

88522/1000000: episode: 176, duration: 19.617s, episode steps: 616, steps per second: 31, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: 0.008858, mae: 0.208732, mean_q: 0.256610, mean_eps: 0.972786

88938/1000000: episode: 177, duration: 13.261s, episode steps: 416, steps per

second: 31, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.008150, mae: 0.206698, mean_q: 0.255445, mean_eps: 0.972167

89350/1000000: episode: 178, duration: 13.136s, episode steps: 412, steps per second: 31, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.008296, mae: 0.208252, mean_q: 0.257471, mean_eps: 0.971670

89825/1000000: episode: 179, duration: 15.161s, episode steps: 475, steps per second: 31, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.008817, mae: 0.211739, mean_q: 0.262058, mean_eps: 0.971138

90549/1000000: episode: 180, duration: 23.040s, episode steps: 724, steps per second: 31, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.008828, mae: 0.206184, mean_q: 0.254139, mean_eps: 0.970418

91047/1000000: episode: 181, duration: 15.794s, episode steps: 498, steps per second: 32, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: 0.010165, mae: 0.229662, mean_q: 0.280420, mean_eps: 0.969686

91568/1000000: episode: 182, duration: 16.694s, episode steps: 521, steps per second: 31, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.009050, mae: 0.221078, mean_q: 0.269492, mean_eps: 0.969076

91862/1000000: episode: 183, duration: 9.424s, episode steps: 294, steps per second: 31, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.344 [0.000, 5.000], loss: 0.008043, mae: 0.212893, mean_q: 0.263526, mean_eps: 0.968586

92228/1000000: episode: 184, duration: 11.750s, episode steps: 366, steps per second: 31, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.008743, mae: 0.221711, mean_q: 0.273605, mean_eps: 0.968190

92806/1000000: episode: 185, duration: 18.463s, episode steps: 578, steps per second: 31, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.009424, mae: 0.226571, mean_q: 0.279799, mean_eps: 0.967624

93304/1000000: episode: 186, duration: 15.909s, episode steps: 498, steps per second: 31, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.007972, mae: 0.224058, mean_q: 0.275606, mean_eps: 0.966978

93801/1000000: episode: 187, duration: 15.926s, episode steps: 497, steps per second: 31, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.009012, mae: 0.224639, mean_q: 0.275788, mean_eps: 0.966381

94662/1000000: episode: 188, duration: 27.401s, episode steps: 861, steps per second: 31, episode reward: 14.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.009121, mae: 0.226091, mean_q: 0.278042, mean_eps: 0.965565

95237/1000000: episode: 189, duration: 18.370s, episode steps: 575, steps per

second: 31, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.009085, mae: 0.223092, mean_q: 0.274516, mean_eps: 0.964703

96125/1000000: episode: 190, duration: 26.812s, episode steps: 888, steps per second: 33, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: 0.008846, mae: 0.231170, mean_q: 0.283811, mean_eps: 0.963825

96749/1000000: episode: 191, duration: 19.826s, episode steps: 624, steps per second: 31, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: 0.009995, mae: 0.240162, mean_q: 0.293786, mean_eps: 0.962918

97264/1000000: episode: 192, duration: 16.506s, episode steps: 515, steps per second: 31, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.009358, mae: 0.241823, mean_q: 0.298552, mean_eps: 0.962236

97978/1000000: episode: 193, duration: 22.828s, episode steps: 714, steps per second: 31, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: 0.008988, mae: 0.237856, mean_q: 0.296299, mean_eps: 0.961499

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

98492/1000000: episode: 194, duration: 11.154s, episode steps: 514, steps per second: 46, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: 0.009653, mae: 0.248579, mean_q: 0.304768, mean_eps: 0.960875

98988/1000000: episode: 195, duration: 2.100s, episode steps: 496, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

99603/1000000: episode: 196, duration: 2.570s, episode steps: 615, steps per second: 239, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

100111/1000000: episode: 197, duration: 2.118s, episode steps: 508, steps per second: 240, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

100803/1000000: episode: 198, duration: 2.874s, episode steps: 692, steps per second: 241, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

101775/1000000: episode: 199, duration: 4.067s, episode steps: 972, steps per second: 239, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

102083/1000000: episode: 200, duration: 1.310s, episode steps: 308, steps per second: 235, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

102650/1000000: episode: 201, duration: 2.382s, episode steps: 567, steps per second: 238, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.256 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

103128/1000000: episode: 202, duration: 2.009s, episode steps: 478, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.364 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

103535/1000000: episode: 203, duration: 1.738s, episode steps: 407, steps per second: 234, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

103837/1000000: episode: 204, duration: 1.291s, episode steps: 302, steps per second: 234, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

104457/1000000: episode: 205, duration: 2.616s, episode steps: 620, steps per second: 237, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

104964/1000000: episode: 206, duration: 2.156s, episode steps: 507, steps per second: 235, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

105570/1000000: episode: 207, duration: 2.558s, episode steps: 606, steps per second: 237, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

106262/1000000: episode: 208, duration: 2.930s, episode steps: 692, steps per second: 236, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

106550/1000000: episode: 209, duration: 1.209s, episode steps: 288, steps per second: 238, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

107023/1000000: episode: 210, duration: 1.975s, episode steps: 473, steps per second: 239, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

107514/1000000: episode: 211, duration: 2.058s, episode steps: 491, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

107847/1000000: episode: 212, duration: 1.408s, episode steps: 333, steps per second: 236, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

108558/1000000: episode: 213, duration: 2.956s, episode steps: 711, steps per second: 241, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

109336/1000000: episode: 214, duration: 3.235s, episode steps: 778, steps per second: 241, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

109961/1000000: episode: 215, duration: 2.594s, episode steps: 625, steps per second: 241, episode reward: 17.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

110407/1000000: episode: 216, duration: 1.886s, episode steps: 446, steps per second: 236, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

110993/1000000: episode: 217, duration: 2.484s, episode steps: 586, steps per second: 236, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

111645/1000000: episode: 218, duration: 2.722s, episode steps: 652, steps per second: 239, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

112195/1000000: episode: 219, duration: 2.289s, episode steps: 550, steps per second: 240, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

113046/1000000: episode: 220, duration: 3.532s, episode steps: 851, steps per second: 241, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

113339/1000000: episode: 221, duration: 1.235s, episode steps: 293, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

113862/1000000: episode: 222, duration: 2.201s, episode steps: 523, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

114134/1000000: episode: 223, duration: 1.155s, episode steps: 272, steps per second: 235, episode reward: 8.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

114742/1000000: episode: 224, duration: 2.545s, episode steps: 608, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

115082/1000000: episode: 225, duration: 1.430s, episode steps: 340, steps per second: 238, episode reward: 10.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

115580/1000000: episode: 226, duration: 2.082s, episode steps: 498, steps per second: 239, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.293 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

115952/1000000: episode: 227, duration: 1.564s, episode steps: 372, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

116394/1000000: episode: 228, duration: 1.876s, episode steps: 442, steps per second: 236, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

116759/1000000: episode: 229, duration: 1.539s, episode steps: 365, steps per second: 237, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

117312/1000000: episode: 230, duration: 2.313s, episode steps: 553, steps per second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

117929/1000000: episode: 231, duration: 2.569s, episode steps: 617, steps per second: 240, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

118447/1000000: episode: 232, duration: 2.166s, episode steps: 518, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

118827/1000000: episode: 233, duration: 1.596s, episode steps: 380, steps per second: 238, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

119399/1000000: episode: 234, duration: 2.405s, episode steps: 572, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

119874/1000000: episode: 235, duration: 2.017s, episode steps: 475, steps per second: 235, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

120542/1000000: episode: 236, duration: 2.808s, episode steps: 668, steps per second: 238, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

120925/1000000: episode: 237, duration: 1.605s, episode steps: 383, steps per second: 239, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

121576/1000000: episode: 238, duration: 2.732s, episode steps: 651, steps per second: 238, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

122049/1000000: episode: 239, duration: 1.980s, episode steps: 473, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

122776/1000000: episode: 240, duration: 3.070s, episode steps: 727, steps per second: 237, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

123718/1000000: episode: 241, duration: 3.942s, episode steps: 942, steps per second: 239, episode reward: 15.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

124164/1000000: episode: 242, duration: 1.866s, episode steps: 446, steps per second: 239, episode reward: 11.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

124765/1000000: episode: 243, duration: 2.525s, episode steps: 601, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

125539/1000000: episode: 244, duration: 3.254s, episode steps: 774, steps per second: 238, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

126232/1000000: episode: 245, duration: 2.929s, episode steps: 693, steps per second: 237, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

126624/1000000: episode: 246, duration: 1.640s, episode steps: 392, steps per second: 239, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

127008/1000000: episode: 247, duration: 1.605s, episode steps: 384, steps per second: 239, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

127982/1000000: episode: 248, duration: 4.033s, episode steps: 974, steps per second: 242, episode reward: 15.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

128532/1000000: episode: 249, duration: 2.313s, episode steps: 550, steps per second: 238, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

129033/1000000: episode: 250, duration: 2.122s, episode steps: 501, steps per second: 236, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

129760/1000000: episode: 251, duration: 3.041s, episode steps: 727, steps per second: 239, episode reward: 19.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

130365/1000000: episode: 252, duration: 2.533s, episode steps: 605, steps per second: 239, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

131014/1000000: episode: 253, duration: 2.702s, episode steps: 649, steps per second: 240, episode reward: 18.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

131729/1000000: episode: 254, duration: 3.000s, episode steps: 715, steps per second: 238, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

132356/1000000: episode: 255, duration: 2.640s, episode steps: 627, steps per second: 237, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

132899/1000000: episode: 256, duration: 2.261s, episode steps: 543, steps per second: 240, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

133273/1000000: episode: 257, duration: 1.564s, episode steps: 374, steps per second: 239, episode reward: 11.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

133840/1000000: episode: 258, duration: 2.370s, episode steps: 567, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

134308/1000000: episode: 259, duration: 1.974s, episode steps: 468, steps per second: 237, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

134627/1000000: episode: 260, duration: 1.358s, episode steps: 319, steps per second: 235, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

135223/1000000: episode: 261, duration: 2.507s, episode steps: 596, steps per second: 238, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

135867/1000000: episode: 262, duration: 2.690s, episode steps: 644, steps per second: 239, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

136369/1000000: episode: 263, duration: 2.111s, episode steps: 502, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

136737/1000000: episode: 264, duration: 1.549s, episode steps: 368, steps per second: 238, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

137113/1000000: episode: 265, duration: 1.589s, episode steps: 376, steps per second: 237, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

137844/1000000: episode: 266, duration: 3.072s, episode steps: 731, steps per second: 238, episode reward: 19.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

138525/1000000: episode: 267, duration: 2.859s, episode steps: 681, steps per second: 238, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

139306/1000000: episode: 268, duration: 3.249s, episode steps: 781, steps per second: 240, episode reward: 9.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

139713/1000000: episode: 269, duration: 1.701s, episode steps: 407, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

140195/1000000: episode: 270, duration: 2.015s, episode steps: 482, steps per second: 239, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

140798/1000000: episode: 271, duration: 2.519s, episode steps: 603, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

141265/1000000: episode: 272, duration: 7.224s, episode steps: 467, steps per second: 65, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.010049, mae: 0.250697, mean_q: 0.309257, mean_eps: 0.987882

141611/1000000: episode: 273, duration: 11.083s, episode steps: 346, steps per second: 31, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.010598, mae: 0.262675, mean_q: 0.320858, mean_eps: 0.987561

142156/1000000: episode: 274, duration: 17.611s, episode steps: 545, steps per second: 31, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.010623, mae: 0.255224, mean_q: 0.312292, mean_eps: 0.987028

142745/1000000: episode: 275, duration: 19.043s, episode steps: 589, steps per second: 31, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.010806, mae: 0.260271, mean_q: 0.320531, mean_eps: 0.986346

143326/1000000: episode: 276, duration: 18.600s, episode steps: 581, steps per second: 31, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: 0.010676, mae: 0.261599, mean_q: 0.319558, mean_eps: 0.985643

143753/1000000: episode: 277, duration: 13.802s, episode steps: 427, steps per second: 31, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.008972, mae: 0.264589, mean_q: 0.326021, mean_eps: 0.985038

144047/1000000: episode: 278, duration: 9.390s, episode steps: 294, steps per second: 31, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.010714, mae: 0.255555, mean_q: 0.312879, mean_eps: 0.984606

144784/1000000: episode: 279, duration: 23.872s, episode steps: 737, steps per second: 31, episode reward: 9.000, mean reward: 0.012 [0.000, 1.000], mean

action: 2.449 [0.000, 5.000], loss: 0.009466, mae: 0.257909, mean_q: 0.318186, mean_eps: 0.983990

145596/1000000: episode: 280, duration: 26.256s, episode steps: 812, steps per second: 31, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.008983, mae: 0.260524, mean_q: 0.321841, mean_eps: 0.983061

146065/1000000: episode: 281, duration: 15.241s, episode steps: 469, steps per second: 31, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.009606, mae: 0.256229, mean_q: 0.316017, mean_eps: 0.982290

146994/1000000: episode: 282, duration: 28.456s, episode steps: 929, steps per second: 33, episode reward: 16.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.011135, mae: 0.305755, mean_q: 0.376637, mean_eps: 0.981450

147522/1000000: episode: 283, duration: 16.988s, episode steps: 528, steps per second: 31, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.629 [0.000, 5.000], loss: 0.012029, mae: 0.301396, mean_q: 0.367896, mean_eps: 0.980577

147904/1000000: episode: 284, duration: 12.379s, episode steps: 382, steps per second: 31, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: 0.009039, mae: 0.301300, mean_q: 0.369702, mean_eps: 0.980032

148426/1000000: episode: 285, duration: 16.891s, episode steps: 522, steps per second: 31, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: 0.010475, mae: 0.304713, mean_q: 0.371363, mean_eps: 0.979490

149126/1000000: episode: 286, duration: 22.574s, episode steps: 700, steps per second: 31, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.010775, mae: 0.307395, mean_q: 0.376381, mean_eps: 0.978755

149434/1000000: episode: 287, duration: 9.960s, episode steps: 308, steps per second: 31, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.010669, mae: 0.309236, mean_q: 0.377736, mean_eps: 0.978150

149913/1000000: episode: 288, duration: 15.571s, episode steps: 479, steps per second: 31, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.010403, mae: 0.303808, mean_q: 0.374899, mean_eps: 0.977678

150368/1000000: episode: 289, duration: 14.713s, episode steps: 455, steps per second: 31, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.010307, mae: 0.292421, mean_q: 0.360545, mean_eps: 0.977118

150956/1000000: episode: 290, duration: 19.108s, episode steps: 588, steps per second: 31, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: 0.011316, mae: 0.311789, mean_q: 0.381940, mean_eps: 0.976494

151352/1000000: episode: 291, duration: 12.909s, episode steps: 396, steps per second: 31, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.581 [0.000, 5.000], loss: 0.010436, mae: 0.304714, mean_q: 0.374484, mean_eps: 0.975904

151922/1000000: episode: 292, duration: 18.442s, episode steps: 570, steps per second: 31, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: 0.010709, mae: 0.318575, mean_q: 0.389918, mean_eps: 0.975323

152409/1000000: episode: 293, duration: 15.785s, episode steps: 487, steps per second: 31, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: 0.012306, mae: 0.321000, mean_q: 0.396248, mean_eps: 0.974687

152734/1000000: episode: 294, duration: 10.437s, episode steps: 325, steps per second: 31, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: 0.008876, mae: 0.312130, mean_q: 0.384914, mean_eps: 0.974200

153206/1000000: episode: 295, duration: 15.314s, episode steps: 472, steps per second: 31, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.653 [0.000, 5.000], loss: 0.008898, mae: 0.309744, mean_q: 0.381978, mean_eps: 0.973722

153465/1000000: episode: 296, duration: 8.375s, episode steps: 259, steps per second: 31, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.008119, mae: 0.306782, mean_q: 0.379023, mean_eps: 0.973283

153942/1000000: episode: 297, duration: 15.468s, episode steps: 477, steps per second: 31, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.009232, mae: 0.320680, mean_q: 0.395151, mean_eps: 0.972842

154375/1000000: episode: 298, duration: 14.003s, episode steps: 433, steps per second: 31, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: 0.011982, mae: 0.324719, mean_q: 0.397473, mean_eps: 0.972297

154991/1000000: episode: 299, duration: 19.932s, episode steps: 616, steps per second: 31, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.008807, mae: 0.305403, mean_q: 0.374201, mean_eps: 0.971668

155595/1000000: episode: 300, duration: 19.599s, episode steps: 604, steps per second: 31, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: 0.009901, mae: 0.319589, mean_q: 0.395304, mean_eps: 0.970936

156075/1000000: episode: 301, duration: 15.591s, episode steps: 480, steps per second: 31, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.009571, mae: 0.317692, mean_q: 0.391473, mean_eps: 0.970286

156384/1000000: episode: 302, duration: 9.170s, episode steps: 309, steps per second: 34, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.340 [0.000, 5.000], loss: 0.012469, mae: 0.334648, mean_q: 0.408622, mean_eps: 0.969813

156908/1000000: episode: 303, duration: 16.692s, episode steps: 524, steps per second: 31, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean

action: 2.611 [0.000, 5.000], loss: 0.011257, mae: 0.342446, mean_q: 0.423352, mean_eps: 0.969314

157511/1000000: episode: 304, duration: 19.565s, episode steps: 603, steps per second: 31, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.012899, mae: 0.355301, mean_q: 0.440702, mean_eps: 0.968637

157876/1000000: episode: 305, duration: 11.956s, episode steps: 365, steps per second: 31, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.011630, mae: 0.353576, mean_q: 0.436719, mean_eps: 0.968056

158467/1000000: episode: 306, duration: 19.181s, episode steps: 591, steps per second: 31, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.360 [0.000, 5.000], loss: 0.010316, mae: 0.334897, mean_q: 0.415514, mean_eps: 0.967482

158920/1000000: episode: 307, duration: 14.741s, episode steps: 453, steps per second: 31, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.012099, mae: 0.352584, mean_q: 0.438528, mean_eps: 0.966856

159515/1000000: episode: 308, duration: 19.354s, episode steps: 595, steps per second: 31, episode reward: 17.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.011954, mae: 0.337936, mean_q: 0.421047, mean_eps: 0.966227

159937/1000000: episode: 309, duration: 13.729s, episode steps: 422, steps per second: 31, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.009729, mae: 0.330310, mean_q: 0.413286, mean_eps: 0.965615

160390/1000000: episode: 310, duration: 14.676s, episode steps: 453, steps per second: 31, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.011416, mae: 0.346170, mean_q: 0.427124, mean_eps: 0.965090

161491/1000000: episode: 311, duration: 34.332s, episode steps: 1101, steps per second: 32, episode reward: 17.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.011701, mae: 0.347828, mean_q: 0.432957, mean_eps: 0.964158

162009/1000000: episode: 312, duration: 16.618s, episode steps: 518, steps per second: 31, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.012031, mae: 0.364649, mean_q: 0.455331, mean_eps: 0.963186

162828/1000000: episode: 313, duration: 26.686s, episode steps: 819, steps per second: 31, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.012638, mae: 0.370561, mean_q: 0.462924, mean_eps: 0.962385

163272/1000000: episode: 314, duration: 14.738s, episode steps: 444, steps per second: 30, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.014230, mae: 0.378328, mean_q: 0.472575, mean_eps: 0.961629

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow

encountered in scalar add

self.step += 1

163991/1000000: episode: 315, duration: 19.201s, episode steps: 719, steps per second: 37, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: 0.012076, mae: 0.368827, mean_q: 0.461569, mean_eps: 0.961022

164805/1000000: episode: 316, duration: 3.393s, episode steps: 814, steps per second: 240, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

165278/1000000: episode: 317, duration: 1.976s, episode steps: 473, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

165799/1000000: episode: 318, duration: 2.208s, episode steps: 521, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.647 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

166136/1000000: episode: 319, duration: 1.433s, episode steps: 337, steps per second: 235, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.288 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

166663/1000000: episode: 320, duration: 2.231s, episode steps: 527, steps per second: 236, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_320_episodes.h5f

167165/1000000: episode: 321, duration: 2.167s, episode steps: 502, steps per second: 232, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

167567/1000000: episode: 322, duration: 1.701s, episode steps: 402, steps per second: 236, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

168037/1000000: episode: 323, duration: 1.977s, episode steps: 470, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.351 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

168901/1000000: episode: 324, duration: 3.652s, episode steps: 864, steps per second: 237, episode reward: 19.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

169493/1000000: episode: 325, duration: 2.512s, episode steps: 592, steps per second: 236, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

170099/1000000: episode: 326, duration: 2.529s, episode steps: 606, steps per second: 240, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

170407/1000000: episode: 327, duration: 1.293s, episode steps: 308, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

171118/1000000: episode: 328, duration: 2.961s, episode steps: 711, steps per

second: 240, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

172016/1000000: episode: 329, duration: 3.777s, episode steps: 898, steps per second: 238, episode reward: 14.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

172816/1000000: episode: 330, duration: 3.338s, episode steps: 800, steps per second: 240, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

173108/1000000: episode: 331, duration: 1.231s, episode steps: 292, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

173398/1000000: episode: 332, duration: 1.230s, episode steps: 290, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

173754/1000000: episode: 333, duration: 1.514s, episode steps: 356, steps per second: 235, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

174058/1000000: episode: 334, duration: 1.295s, episode steps: 304, steps per second: 235, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

174341/1000000: episode: 335, duration: 1.212s, episode steps: 283, steps per second: 234, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

175265/1000000: episode: 336, duration: 3.913s, episode steps: 924, steps per second: 236, episode reward: 20.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

175671/1000000: episode: 337, duration: 1.712s, episode steps: 406, steps per second: 237, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.330 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

176279/1000000: episode: 338, duration: 2.544s, episode steps: 608, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177029/1000000: episode: 339, duration: 3.149s, episode steps: 750, steps per second: 238, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177508/1000000: episode: 340, duration: 2.014s, episode steps: 479, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177907/1000000: episode: 341, duration: 1.700s, episode steps: 399, steps per second: 235, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

178289/1000000: episode: 342, duration: 1.600s, episode steps: 382, steps per second: 239, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

178809/1000000: episode: 343, duration: 2.169s, episode steps: 520, steps per second: 240, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

179353/1000000: episode: 344, duration: 2.270s, episode steps: 544, steps per

second: 240, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

179866/1000000: episode: 345, duration: 2.143s, episode steps: 513, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

180491/1000000: episode: 346, duration: 2.615s, episode steps: 625, steps per second: 239, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

181261/1000000: episode: 347, duration: 3.244s, episode steps: 770, steps per second: 237, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

181772/1000000: episode: 348, duration: 2.140s, episode steps: 511, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

182258/1000000: episode: 349, duration: 2.044s, episode steps: 486, steps per second: 238, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

182761/1000000: episode: 350, duration: 2.097s, episode steps: 503, steps per second: 240, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

183409/1000000: episode: 351, duration: 2.699s, episode steps: 648, steps per second: 240, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

183906/1000000: episode: 352, duration: 2.119s, episode steps: 497, steps per second: 235, episode reward: 3.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

184418/1000000: episode: 353, duration: 2.167s, episode steps: 512, steps per second: 236, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

184698/1000000: episode: 354, duration: 1.181s, episode steps: 280, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

185438/1000000: episode: 355, duration: 3.094s, episode steps: 740, steps per second: 239, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

185972/1000000: episode: 356, duration: 2.230s, episode steps: 534, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.341 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

186236/1000000: episode: 357, duration: 1.110s, episode steps: 264, steps per second: 238, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

186712/1000000: episode: 358, duration: 2.033s, episode steps: 476, steps per second: 234, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

187355/1000000: episode: 359, duration: 2.686s, episode steps: 643, steps per second: 239, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

187860/1000000: episode: 360, duration: 2.108s, episode steps: 505, steps per

second: 240, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
188304/1000000: episode: 361, duration: 1.853s, episode steps: 444, steps per second: 240, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
188985/1000000: episode: 362, duration: 2.833s, episode steps: 681, steps per second: 240, episode reward: 8.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
189446/1000000: episode: 363, duration: 1.942s, episode steps: 461, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
190287/1000000: episode: 364, duration: 3.511s, episode steps: 841, steps per second: 240, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
190789/1000000: episode: 365, duration: 2.105s, episode steps: 502, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
191182/1000000: episode: 366, duration: 1.635s, episode steps: 393, steps per second: 240, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
191953/1000000: episode: 367, duration: 3.206s, episode steps: 771, steps per second: 241, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
192565/1000000: episode: 368, duration: 2.549s, episode steps: 612, steps per second: 240, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
193048/1000000: episode: 369, duration: 2.038s, episode steps: 483, steps per second: 237, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.362 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
193469/1000000: episode: 370, duration: 1.763s, episode steps: 421, steps per second: 239, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.371 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
194013/1000000: episode: 371, duration: 2.272s, episode steps: 544, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
194649/1000000: episode: 372, duration: 2.648s, episode steps: 636, steps per second: 240, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
194950/1000000: episode: 373, duration: 1.270s, episode steps: 301, steps per second: 237, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
195263/1000000: episode: 374, duration: 1.323s, episode steps: 313, steps per second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
195828/1000000: episode: 375, duration: 2.402s, episode steps: 565, steps per second: 235, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.356 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
196329/1000000: episode: 376, duration: 2.107s, episode steps: 501, steps per

second: 238, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 196921/1000000: episode: 377, duration: 2.490s, episode steps: 592, steps per second: 238, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 197504/1000000: episode: 378, duration: 2.444s, episode steps: 583, steps per second: 239, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 197959/1000000: episode: 379, duration: 1.913s, episode steps: 455, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 198499/1000000: episode: 380, duration: 2.270s, episode steps: 540, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 198786/1000000: episode: 381, duration: 1.224s, episode steps: 287, steps per second: 235, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 199243/1000000: episode: 382, duration: 1.919s, episode steps: 457, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 199701/1000000: episode: 383, duration: 1.910s, episode steps: 458, steps per second: 240, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 200131/1000000: episode: 384, duration: 1.797s, episode steps: 430, steps per second: 239, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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201025/1000000: episode: 385, duration: 3.786s, episode steps: 894, steps per second: 236, episode reward: 27.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 201317/1000000: episode: 386, duration: 1.227s, episode steps: 292, steps per second: 238, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 202055/1000000: episode: 387, duration: 3.095s, episode steps: 738, steps per second: 238, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.386 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 202508/1000000: episode: 388, duration: 1.902s, episode steps: 453, steps per second: 238, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 203084/1000000: episode: 389, duration: 2.417s, episode steps: 576, steps per second: 238, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 203755/1000000: episode: 390, duration: 2.822s, episode steps: 671, steps per second: 238, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

204285/1000000: episode: 391, duration: 2.212s, episode steps: 530, steps per second: 240, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

204583/1000000: episode: 392, duration: 1.263s, episode steps: 298, steps per second: 236, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

205050/1000000: episode: 393, duration: 1.983s, episode steps: 467, steps per second: 236, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

205669/1000000: episode: 394, duration: 2.589s, episode steps: 619, steps per second: 239, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

206037/1000000: episode: 395, duration: 1.545s, episode steps: 368, steps per second: 238, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

206543/1000000: episode: 396, duration: 2.119s, episode steps: 506, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

206917/1000000: episode: 397, duration: 10.294s, episode steps: 374, steps per second: 36, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: 0.012254, mae: 0.393677, mean_q: 0.490893, mean_eps: 0.987813

207212/1000000: episode: 398, duration: 9.712s, episode steps: 295, steps per second: 30, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.012684, mae: 0.404618, mean_q: 0.507543, mean_eps: 0.987453

207616/1000000: episode: 399, duration: 13.393s, episode steps: 404, steps per second: 30, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.012048, mae: 0.382400, mean_q: 0.478937, mean_eps: 0.987035

207950/1000000: episode: 400, duration: 11.041s, episode steps: 334, steps per second: 30, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.012519, mae: 0.416770, mean_q: 0.520983, mean_eps: 0.986591

208514/1000000: episode: 401, duration: 18.534s, episode steps: 564, steps per second: 30, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.011583, mae: 0.408040, mean_q: 0.511520, mean_eps: 0.986051

208958/1000000: episode: 402, duration: 14.598s, episode steps: 444, steps per second: 30, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.010574, mae: 0.398117, mean_q: 0.502324, mean_eps: 0.985446

209440/1000000: episode: 403, duration: 15.924s, episode steps: 482, steps per second: 30, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.013524, mae: 0.396625, mean_q: 0.497259, mean_eps: 0.984892

209729/1000000: episode: 404, duration: 9.577s, episode steps: 289, steps per second: 30, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.512 [0.000, 5.000], loss: 0.011525, mae: 0.388928, mean_q: 0.489976, mean_eps: 0.984429

210210/1000000: episode: 405, duration: 15.825s, episode steps: 481, steps per second: 30, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: 0.013310, mae: 0.393244, mean_q: 0.493008, mean_eps: 0.983966

211032/1000000: episode: 406, duration: 27.111s, episode steps: 822, steps per second: 30, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: 0.012429, mae: 0.401568, mean_q: 0.510227, mean_eps: 0.983186

211545/1000000: episode: 407, duration: 17.002s, episode steps: 513, steps per second: 30, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.011016, mae: 0.393118, mean_q: 0.495502, mean_eps: 0.982384

211827/1000000: episode: 408, duration: 8.617s, episode steps: 282, steps per second: 33, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.011908, mae: 0.428298, mean_q: 0.538592, mean_eps: 0.981906

212129/1000000: episode: 409, duration: 9.370s, episode steps: 302, steps per second: 32, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: 0.013123, mae: 0.450675, mean_q: 0.568057, mean_eps: 0.981556

212579/1000000: episode: 410, duration: 14.770s, episode steps: 450, steps per second: 30, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.013940, mae: 0.447031, mean_q: 0.562760, mean_eps: 0.981105

213437/1000000: episode: 411, duration: 28.233s, episode steps: 858, steps per second: 30, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: 0.011404, mae: 0.439758, mean_q: 0.554536, mean_eps: 0.980320

213721/1000000: episode: 412, duration: 9.373s, episode steps: 284, steps per second: 30, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.011293, mae: 0.436701, mean_q: 0.551173, mean_eps: 0.979634

214224/1000000: episode: 413, duration: 16.616s, episode steps: 503, steps per second: 30, episode reward: 14.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: 0.011515, mae: 0.449881, mean_q: 0.566009, mean_eps: 0.979163

214606/1000000: episode: 414, duration: 12.644s, episode steps: 382, steps per second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.011681, mae: 0.437237, mean_q: 0.549679, mean_eps: 0.978633

214894/1000000: episode: 415, duration: 9.498s, episode steps: 288, steps per second: 30, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: 0.012326, mae: 0.431545, mean_q: 0.548112, mean_eps: 0.978230

215276/1000000: episode: 416, duration: 12.634s, episode steps: 382, steps per second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean

action: 2.524 [0.000, 5.000], loss: 0.012295, mae: 0.453500, mean_q: 0.573082, mean_eps: 0.977829

215672/1000000: episode: 417, duration: 13.236s, episode steps: 396, steps per second: 30, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.011356, mae: 0.428196, mean_q: 0.542192, mean_eps: 0.977363

216177/1000000: episode: 418, duration: 16.788s, episode steps: 505, steps per second: 30, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.011999, mae: 0.438668, mean_q: 0.551265, mean_eps: 0.976821

216902/1000000: episode: 419, duration: 23.996s, episode steps: 725, steps per second: 30, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.011774, mae: 0.461211, mean_q: 0.580471, mean_eps: 0.976082

217366/1000000: episode: 420, duration: 15.315s, episode steps: 464, steps per second: 30, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: 0.011685, mae: 0.486850, mean_q: 0.610364, mean_eps: 0.975369

218187/1000000: episode: 421, duration: 27.028s, episode steps: 821, steps per second: 30, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.011838, mae: 0.494148, mean_q: 0.624240, mean_eps: 0.974598

218961/1000000: episode: 422, duration: 25.674s, episode steps: 774, steps per second: 30, episode reward: 14.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.012093, mae: 0.491052, mean_q: 0.616028, mean_eps: 0.973641

219443/1000000: episode: 423, duration: 15.919s, episode steps: 482, steps per second: 30, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.011421, mae: 0.499232, mean_q: 0.628130, mean_eps: 0.972887

219860/1000000: episode: 424, duration: 13.881s, episode steps: 417, steps per second: 30, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.012846, mae: 0.493255, mean_q: 0.624716, mean_eps: 0.972350

220382/1000000: episode: 425, duration: 17.301s, episode steps: 522, steps per second: 30, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: 0.010388, mae: 0.500808, mean_q: 0.631241, mean_eps: 0.971786

221269/1000000: episode: 426, duration: 29.368s, episode steps: 887, steps per second: 30, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.011927, mae: 0.493485, mean_q: 0.621103, mean_eps: 0.970938

221564/1000000: episode: 427, duration: 9.836s, episode steps: 295, steps per second: 30, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.654 [0.000, 5.000], loss: 0.011573, mae: 0.491081, mean_q: 0.619797, mean_eps: 0.970230

222113/1000000: episode: 428, duration: 16.836s, episode steps: 549, steps per second: 33, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.587 [0.000, 5.000], loss: 0.012890, mae: 0.514283, mean_q: 0.647506,
 mean_eps: 0.969724
 223206/1000000: episode: 429, duration: 36.060s, episode steps: 1093, steps per
 second: 30, episode reward: 24.000, mean reward: 0.022 [0.000, 1.000], mean
 action: 2.538 [0.000, 5.000], loss: 0.011482, mae: 0.519096, mean_q: 0.654537,
 mean_eps: 0.968738
 223586/1000000: episode: 430, duration: 12.601s, episode steps: 380, steps per
 second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.413 [0.000, 5.000], loss: 0.010956, mae: 0.512676, mean_q: 0.647388,
 mean_eps: 0.967854
 224237/1000000: episode: 431, duration: 21.620s, episode steps: 651, steps per
 second: 30, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.550 [0.000, 5.000], loss: 0.012629, mae: 0.529288, mean_q: 0.668427,
 mean_eps: 0.967235
 224528/1000000: episode: 432, duration: 9.661s, episode steps: 291, steps per
 second: 30, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.515 [0.000, 5.000], loss: 0.010182, mae: 0.521335, mean_q: 0.663377,
 mean_eps: 0.966671
 225103/1000000: episode: 433, duration: 19.123s, episode steps: 575, steps per
 second: 30, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.555 [0.000, 5.000], loss: 0.012845, mae: 0.516355, mean_q: 0.655364,
 mean_eps: 0.966153
 225558/1000000: episode: 434, duration: 15.165s, episode steps: 455, steps per
 second: 30, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.459 [0.000, 5.000], loss: 0.011080, mae: 0.520584, mean_q: 0.658378,
 mean_eps: 0.965534
 226072/1000000: episode: 435, duration: 17.111s, episode steps: 514, steps per
 second: 30, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.609 [0.000, 5.000], loss: 0.011908, mae: 0.523243, mean_q: 0.658749,
 mean_eps: 0.964953
 226416/1000000: episode: 436, duration: 11.533s, episode steps: 344, steps per
 second: 30, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.535 [0.000, 5.000], loss: 0.010874, mae: 0.506636, mean_q: 0.634189,
 mean_eps: 0.964439
 227178/1000000: episode: 437, duration: 24.033s, episode steps: 762, steps per
 second: 32, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.445 [0.000, 5.000], loss: 0.011858, mae: 0.550346, mean_q: 0.695584,
 mean_eps: 0.963774
 227697/1000000: episode: 438, duration: 17.253s, episode steps: 519, steps per
 second: 30, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.593 [0.000, 5.000], loss: 0.013381, mae: 0.555193, mean_q: 0.703697,
 mean_eps: 0.963004
 227988/1000000: episode: 439, duration: 9.704s, episode steps: 291, steps per
 second: 30, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.364 [0.000, 5.000], loss: 0.011497, mae: 0.566325, mean_q: 0.716379,
 mean_eps: 0.962519
 228373/1000000: episode: 440, duration: 12.858s, episode steps: 385, steps per
 second: 30, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean

action: 2.595 [0.000, 5.000], loss: 0.012187, mae: 0.561783, mean_q: 0.710910, mean_eps: 0.962114

228773/1000000: episode: 441, duration: 13.328s, episode steps: 400, steps per second: 30, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.013080, mae: 0.570154, mean_q: 0.719480, mean_eps: 0.961641

229246/1000000: episode: 442, duration: 15.691s, episode steps: 473, steps per second: 30, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.011059, mae: 0.558386, mean_q: 0.707013, mean_eps: 0.961118

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

229898/1000000: episode: 443, duration: 6.503s, episode steps: 652, steps per second: 100, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: 0.012173, mae: 0.563947, mean_q: 0.718990, mean_eps: 0.960758

230475/1000000: episode: 444, duration: 2.409s, episode steps: 577, steps per second: 240, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

230894/1000000: episode: 445, duration: 1.751s, episode steps: 419, steps per second: 239, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

231231/1000000: episode: 446, duration: 1.423s, episode steps: 337, steps per second: 237, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

231672/1000000: episode: 447, duration: 1.858s, episode steps: 441, steps per second: 237, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

232095/1000000: episode: 448, duration: 1.784s, episode steps: 423, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

232453/1000000: episode: 449, duration: 1.518s, episode steps: 358, steps per second: 236, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

232931/1000000: episode: 450, duration: 1.997s, episode steps: 478, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

233456/1000000: episode: 451, duration: 2.205s, episode steps: 525, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

233761/1000000: episode: 452, duration: 1.288s, episode steps: 305, steps per second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.334 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

234696/1000000: episode: 453, duration: 3.902s, episode steps: 935, steps per second: 240, episode reward: 18.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

235284/1000000: episode: 454, duration: 2.505s, episode steps: 588, steps per second: 235, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

235804/1000000: episode: 455, duration: 2.181s, episode steps: 520, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

236089/1000000: episode: 456, duration: 1.208s, episode steps: 285, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

236574/1000000: episode: 457, duration: 2.029s, episode steps: 485, steps per second: 239, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

237234/1000000: episode: 458, duration: 2.776s, episode steps: 660, steps per second: 238, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

237647/1000000: episode: 459, duration: 1.750s, episode steps: 413, steps per second: 236, episode reward: 1.000, mean reward: 0.002 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

238161/1000000: episode: 460, duration: 2.190s, episode steps: 514, steps per second: 235, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

238659/1000000: episode: 461, duration: 2.092s, episode steps: 498, steps per second: 238, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

239217/1000000: episode: 462, duration: 2.332s, episode steps: 558, steps per second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

240196/1000000: episode: 463, duration: 4.094s, episode steps: 979, steps per second: 239, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

240562/1000000: episode: 464, duration: 1.549s, episode steps: 366, steps per second: 236, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

241359/1000000: episode: 465, duration: 3.396s, episode steps: 797, steps per second: 235, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

241823/1000000: episode: 466, duration: 1.980s, episode steps: 464, steps per second: 234, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.638 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

242600/1000000: episode: 467, duration: 3.257s, episode steps: 777, steps per second: 239, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

243052/1000000: episode: 468, duration: 1.889s, episode steps: 452, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

243421/1000000: episode: 469, duration: 1.550s, episode steps: 369, steps per second: 238, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

243810/1000000: episode: 470, duration: 1.655s, episode steps: 389, steps per second: 235, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

244542/1000000: episode: 471, duration: 3.066s, episode steps: 732, steps per second: 239, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

244991/1000000: episode: 472, duration: 1.884s, episode steps: 449, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

245359/1000000: episode: 473, duration: 1.543s, episode steps: 368, steps per second: 238, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

245955/1000000: episode: 474, duration: 2.501s, episode steps: 596, steps per second: 238, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

246451/1000000: episode: 475, duration: 2.080s, episode steps: 496, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

246732/1000000: episode: 476, duration: 1.189s, episode steps: 281, steps per second: 236, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

247443/1000000: episode: 477, duration: 2.984s, episode steps: 711, steps per second: 238, episode reward: 24.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.319 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

247921/1000000: episode: 478, duration: 2.000s, episode steps: 478, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

248207/1000000: episode: 479, duration: 1.200s, episode steps: 286, steps per second: 238, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.273 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

248843/1000000: episode: 480, duration: 2.644s, episode steps: 636, steps per second: 241, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v1.0_weights_480_episodes.h5f

249142/1000000: episode: 481, duration: 1.329s, episode steps: 299, steps per second: 225, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

249577/1000000: episode: 482, duration: 1.820s, episode steps: 435, steps per second: 239, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

250424/1000000: episode: 483, duration: 3.570s, episode steps: 847, steps per second: 237, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

250979/1000000: episode: 484, duration: 2.324s, episode steps: 555, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 251557/1000000: episode: 485, duration: 2.402s, episode steps: 578, steps per second: 241, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 251999/1000000: episode: 486, duration: 1.841s, episode steps: 442, steps per second: 240, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 252596/1000000: episode: 487, duration: 2.505s, episode steps: 597, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 253084/1000000: episode: 488, duration: 2.066s, episode steps: 488, steps per second: 236, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 253481/1000000: episode: 489, duration: 1.684s, episode steps: 397, steps per second: 236, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 253993/1000000: episode: 490, duration: 2.152s, episode steps: 512, steps per second: 238, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 254512/1000000: episode: 491, duration: 2.186s, episode steps: 519, steps per second: 237, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 255047/1000000: episode: 492, duration: 2.244s, episode steps: 535, steps per second: 238, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 255464/1000000: episode: 493, duration: 1.750s, episode steps: 417, steps per second: 238, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 255951/1000000: episode: 494, duration: 2.050s, episode steps: 487, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 256562/1000000: episode: 495, duration: 2.575s, episode steps: 611, steps per second: 237, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 257044/1000000: episode: 496, duration: 2.014s, episode steps: 482, steps per second: 239, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 257778/1000000: episode: 497, duration: 3.051s, episode steps: 734, steps per second: 241, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 258534/1000000: episode: 498, duration: 3.133s, episode steps: 756, steps per second: 241, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 259015/1000000: episode: 499, duration: 2.029s, episode steps: 481, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 259458/1000000: episode: 500, duration: 1.864s, episode steps: 443, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 259950/1000000: episode: 501, duration: 2.062s, episode steps: 492, steps per second: 239, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 260336/1000000: episode: 502, duration: 1.628s, episode steps: 386, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 260629/1000000: episode: 503, duration: 1.228s, episode steps: 293, steps per second: 239, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261082/1000000: episode: 504, duration: 1.895s, episode steps: 453, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261410/1000000: episode: 505, duration: 1.378s, episode steps: 328, steps per second: 238, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261881/1000000: episode: 506, duration: 1.992s, episode steps: 471, steps per second: 236, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.327 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 262383/1000000: episode: 507, duration: 2.143s, episode steps: 502, steps per second: 234, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 262801/1000000: episode: 508, duration: 1.742s, episode steps: 418, steps per second: 240, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 263096/1000000: episode: 509, duration: 1.244s, episode steps: 295, steps per second: 237, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 263678/1000000: episode: 510, duration: 2.434s, episode steps: 582, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 264783/1000000: episode: 511, duration: 4.579s, episode steps: 1105, steps per second: 241, episode reward: 16.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 265276/1000000: episode: 512, duration: 2.096s, episode steps: 493, steps per second: 235, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 266020/1000000: episode: 513, duration: 3.100s, episode steps: 744, steps per second: 240, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 266336/1000000: episode: 514, duration: 1.323s, episode steps: 316, steps per second: 239, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 266638/1000000: episode: 515, duration: 1.269s, episode steps: 302, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.338 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 267202/1000000: episode: 516, duration: 2.363s, episode steps: 564, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 267716/1000000: episode: 517, duration: 2.146s, episode steps: 514, steps per second: 240, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 268250/1000000: episode: 518, duration: 2.259s, episode steps: 534, steps per second: 236, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 268820/1000000: episode: 519, duration: 2.400s, episode steps: 570, steps per second: 238, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 269181/1000000: episode: 520, duration: 1.530s, episode steps: 361, steps per second: 236, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 269894/1000000: episode: 521, duration: 2.982s, episode steps: 713, steps per second: 239, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 270407/1000000: episode: 522, duration: 2.147s, episode steps: 513, steps per second: 239, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 271029/1000000: episode: 523, duration: 2.610s, episode steps: 622, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 271588/1000000: episode: 524, duration: 2.350s, episode steps: 559, steps per second: 238, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 272161/1000000: episode: 525, duration: 2.845s, episode steps: 573, steps per second: 201, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.024446, mae: 0.614236, mean_q: 0.768138, mean_eps: 0.987988
 272654/1000000: episode: 526, duration: 16.557s, episode steps: 493, steps per second: 30, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.625 [0.000, 5.000], loss: 0.014602, mae: 0.623519, mean_q: 0.788023, mean_eps: 0.987683
 273060/1000000: episode: 527, duration: 13.753s, episode steps: 406, steps per second: 30, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.013019, mae: 0.601496, mean_q: 0.758846, mean_eps: 0.987146
 273473/1000000: episode: 528, duration: 14.022s, episode steps: 413, steps per second: 29, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: 0.011337, mae: 0.607612, mean_q: 0.771266, mean_eps: 0.986654
 273752/1000000: episode: 529, duration: 9.459s, episode steps: 279, steps per second: 29, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.013266, mae: 0.627091, mean_q: 0.791472, mean_eps: 0.986238

Nuevo récord: 35.0. Pesos guardados en
 /content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn

spartial/version_1.0/mejor_modelo/best_weights.h5f

274801/1000000: episode: 530, duration: 35.598s, episode steps: 1049, steps per second: 29, episode reward: 35.000, mean reward: 0.033 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.012817, mae: 0.608367, mean_q: 0.771398, mean_eps: 0.985442

275119/1000000: episode: 531, duration: 10.719s, episode steps: 318, steps per second: 30, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.011748, mae: 0.603946, mean_q: 0.763320, mean_eps: 0.984621

275406/1000000: episode: 532, duration: 9.754s, episode steps: 287, steps per second: 29, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: 0.012396, mae: 0.609541, mean_q: 0.767942, mean_eps: 0.984258

275901/1000000: episode: 533, duration: 16.797s, episode steps: 495, steps per second: 29, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.011930, mae: 0.607752, mean_q: 0.770509, mean_eps: 0.983788

276293/1000000: episode: 534, duration: 13.244s, episode steps: 392, steps per second: 30, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.011742, mae: 0.614955, mean_q: 0.776290, mean_eps: 0.983255

276566/1000000: episode: 535, duration: 9.236s, episode steps: 273, steps per second: 30, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: 0.014053, mae: 0.626967, mean_q: 0.792600, mean_eps: 0.982857

277380/1000000: episode: 536, duration: 27.593s, episode steps: 814, steps per second: 30, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: 0.012666, mae: 0.634839, mean_q: 0.805209, mean_eps: 0.982206

277846/1000000: episode: 537, duration: 15.895s, episode steps: 466, steps per second: 29, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: 0.013378, mae: 0.655204, mean_q: 0.829328, mean_eps: 0.981438

278316/1000000: episode: 538, duration: 16.016s, episode steps: 470, steps per second: 29, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: 0.014921, mae: 0.664837, mean_q: 0.837821, mean_eps: 0.980877

278863/1000000: episode: 539, duration: 18.615s, episode steps: 547, steps per second: 29, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.013085, mae: 0.656810, mean_q: 0.831034, mean_eps: 0.980267

279403/1000000: episode: 540, duration: 18.219s, episode steps: 540, steps per second: 30, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.013735, mae: 0.647865, mean_q: 0.822554, mean_eps: 0.979614

280006/1000000: episode: 541, duration: 20.494s, episode steps: 603, steps per second: 29, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.629 [0.000, 5.000], loss: 0.012905, mae: 0.657261, mean_q: 0.826818,

mean_eps: 0.978928

280370/1000000: episode: 542, duration: 12.349s, episode steps: 364, steps per second: 29, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.013680, mae: 0.652817, mean_q: 0.824636, mean_eps: 0.978347

280937/1000000: episode: 543, duration: 19.260s, episode steps: 567, steps per second: 29, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.012252, mae: 0.643332, mean_q: 0.814999, mean_eps: 0.977788

281558/1000000: episode: 544, duration: 21.040s, episode steps: 621, steps per second: 30, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: 0.012031, mae: 0.657120, mean_q: 0.827149, mean_eps: 0.977075

281966/1000000: episode: 545, duration: 13.845s, episode steps: 408, steps per second: 29, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: 0.012611, mae: 0.657825, mean_q: 0.829228, mean_eps: 0.976458

282272/1000000: episode: 546, duration: 10.091s, episode steps: 306, steps per second: 30, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.284 [0.000, 5.000], loss: 0.013499, mae: 0.662563, mean_q: 0.833966, mean_eps: 0.976031

282740/1000000: episode: 547, duration: 14.960s, episode steps: 468, steps per second: 31, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: 0.014172, mae: 0.710053, mean_q: 0.889049, mean_eps: 0.975568

283225/1000000: episode: 548, duration: 16.493s, episode steps: 485, steps per second: 29, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.012934, mae: 0.698354, mean_q: 0.871413, mean_eps: 0.974994

283484/1000000: episode: 549, duration: 8.816s, episode steps: 259, steps per second: 29, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.013906, mae: 0.720361, mean_q: 0.897036, mean_eps: 0.974548

283793/1000000: episode: 550, duration: 10.587s, episode steps: 309, steps per second: 29, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.014131, mae: 0.705394, mean_q: 0.876230, mean_eps: 0.974207

284270/1000000: episode: 551, duration: 16.100s, episode steps: 477, steps per second: 30, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.013953, mae: 0.711209, mean_q: 0.884891, mean_eps: 0.973734

284553/1000000: episode: 552, duration: 9.677s, episode steps: 283, steps per second: 29, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.728 [0.000, 5.000], loss: 0.014389, mae: 0.701352, mean_q: 0.870164, mean_eps: 0.973278

285211/1000000: episode: 553, duration: 22.259s, episode steps: 658, steps per second: 30, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.013402, mae: 0.704681, mean_q: 0.875619,

mean_eps: 0.972714

285610/1000000: episode: 554, duration: 13.594s, episode steps: 399, steps per second: 29, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.624 [0.000, 5.000], loss: 0.011306, mae: 0.677400, mean_q: 0.841555, mean_eps: 0.972081

286211/1000000: episode: 555, duration: 20.400s, episode steps: 601, steps per second: 29, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.012940, mae: 0.692829, mean_q: 0.864381, mean_eps: 0.971481

286729/1000000: episode: 556, duration: 17.700s, episode steps: 518, steps per second: 29, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.666 [0.000, 5.000], loss: 0.014017, mae: 0.702261, mean_q: 0.876382, mean_eps: 0.970809

287230/1000000: episode: 557, duration: 16.747s, episode steps: 501, steps per second: 30, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.013660, mae: 0.715905, mean_q: 0.888185, mean_eps: 0.970197

287545/1000000: episode: 558, duration: 9.746s, episode steps: 315, steps per second: 32, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.014037, mae: 0.753590, mean_q: 0.932789, mean_eps: 0.969707

288149/1000000: episode: 559, duration: 20.393s, episode steps: 604, steps per second: 30, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.015482, mae: 0.760164, mean_q: 0.939045, mean_eps: 0.969155

288678/1000000: episode: 560, duration: 17.940s, episode steps: 529, steps per second: 29, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.015816, mae: 0.778845, mean_q: 0.961984, mean_eps: 0.968476

289123/1000000: episode: 561, duration: 15.104s, episode steps: 445, steps per second: 29, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.013712, mae: 0.745048, mean_q: 0.920865, mean_eps: 0.967893

289470/1000000: episode: 562, duration: 11.865s, episode steps: 347, steps per second: 29, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.326 [0.000, 5.000], loss: 0.013053, mae: 0.764570, mean_q: 0.942232, mean_eps: 0.967418

289867/1000000: episode: 563, duration: 13.477s, episode steps: 397, steps per second: 29, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.014232, mae: 0.750552, mean_q: 0.928501, mean_eps: 0.966971

290670/1000000: episode: 564, duration: 27.353s, episode steps: 803, steps per second: 29, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.013888, mae: 0.765633, mean_q: 0.944590, mean_eps: 0.966251

290965/1000000: episode: 565, duration: 10.082s, episode steps: 295, steps per second: 29, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.678 [0.000, 5.000], loss: 0.013816, mae: 0.756003, mean_q: 0.933833,

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mean_eps: 0.965591
291669/1000000: episode: 566, duration: 24.021s, episode steps: 704, steps per
second: 29, episode reward: 16.000, mean reward: 0.023 [ 0.000, 1.000], mean
action: 2.612 [0.000, 5.000], loss: 0.012934, mae: 0.752261, mean_q: 0.928743,
mean_eps: 0.964991
292217/1000000: episode: 567, duration: 18.477s, episode steps: 548, steps per
second: 30, episode reward: 8.000, mean reward: 0.015 [ 0.000, 1.000], mean
action: 2.589 [0.000, 5.000], loss: 0.013120, mae: 0.769897, mean_q: 0.952437,
mean_eps: 0.964240
292684/1000000: episode: 568, duration: 14.769s, episode steps: 467, steps per
second: 32, episode reward: 11.000, mean reward: 0.024 [ 0.000, 1.000], mean
action: 2.617 [0.000, 5.000], loss: 0.014854, mae: 0.777959, mean_q: 0.961805,
mean_eps: 0.963633
293125/1000000: episode: 569, duration: 15.067s, episode steps: 441, steps per
second: 29, episode reward: 8.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.590 [0.000, 5.000], loss: 0.015722, mae: 0.793231, mean_q: 0.981815,
mean_eps: 0.963088
293575/1000000: episode: 570, duration: 15.315s, episode steps: 450, steps per
second: 29, episode reward: 9.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.480 [0.000, 5.000], loss: 0.014581, mae: 0.795826, mean_q: 0.985682,
mean_eps: 0.962553
294303/1000000: episode: 571, duration: 24.843s, episode steps: 728, steps per
second: 29, episode reward: 20.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.523 [0.000, 5.000], loss: 0.014860, mae: 0.799773, mean_q: 0.988963,
mean_eps: 0.961847
294776/1000000: episode: 572, duration: 16.340s, episode steps: 473, steps per
second: 29, episode reward: 10.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.666 [0.000, 5.000], loss: 0.014033, mae: 0.788995, mean_q: 0.975020,
mean_eps: 0.961127

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

295311/1000000: episode: 573, duration: 6.300s, episode steps: 535, steps per
second: 85, episode reward: 6.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.527 [0.000, 5.000], loss: 0.014277, mae: 0.809271, mean_q: 0.994650,
mean_eps: 0.960762
295609/1000000: episode: 574, duration: 1.251s, episode steps: 298, steps per
second: 238, episode reward: 2.000, mean reward: 0.007 [ 0.000, 1.000], mean
action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
296149/1000000: episode: 575, duration: 2.271s, episode steps: 540, steps per
second: 238, episode reward: 9.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
296627/1000000: episode: 576, duration: 2.022s, episode steps: 478, steps per
second: 236, episode reward: 10.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
297010/1000000: episode: 577, duration: 1.636s, episode steps: 383, steps per
second: 234, episode reward: 7.000, mean reward: 0.018 [ 0.000, 1.000], mean

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action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 297462/1000000: episode: 578, duration: 1.916s, episode steps: 452, steps per second: 236, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 298038/1000000: episode: 579, duration: 2.425s, episode steps: 576, steps per second: 237, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 298657/1000000: episode: 580, duration: 2.576s, episode steps: 619, steps per second: 240, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 299163/1000000: episode: 581, duration: 2.142s, episode steps: 506, steps per second: 236, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 300152/1000000: episode: 582, duration: 4.100s, episode steps: 989, steps per second: 241, episode reward: 22.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 300987/1000000: episode: 583, duration: 3.493s, episode steps: 835, steps per second: 239, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 301675/1000000: episode: 584, duration: 2.910s, episode steps: 688, steps per second: 236, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 302789/1000000: episode: 585, duration: 4.703s, episode steps: 1114, steps per second: 237, episode reward: 14.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 303483/1000000: episode: 586, duration: 2.891s, episode steps: 694, steps per second: 240, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 303776/1000000: episode: 587, duration: 1.240s, episode steps: 293, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 304526/1000000: episode: 588, duration: 3.158s, episode steps: 750, steps per second: 238, episode reward: 8.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 305180/1000000: episode: 589, duration: 2.749s, episode steps: 654, steps per second: 238, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 305698/1000000: episode: 590, duration: 2.203s, episode steps: 518, steps per second: 235, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 306168/1000000: episode: 591, duration: 1.992s, episode steps: 470, steps per second: 236, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 306631/1000000: episode: 592, duration: 1.937s, episode steps: 463, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 307164/1000000: episode: 593, duration: 2.229s, episode steps: 533, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 307444/1000000: episode: 594, duration: 1.183s, episode steps: 280, steps per second: 237, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 307949/1000000: episode: 595, duration: 2.128s, episode steps: 505, steps per second: 237, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 308418/1000000: episode: 596, duration: 1.975s, episode steps: 469, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 309104/1000000: episode: 597, duration: 2.892s, episode steps: 686, steps per second: 237, episode reward: 7.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 309575/1000000: episode: 598, duration: 1.973s, episode steps: 471, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 310020/1000000: episode: 599, duration: 1.864s, episode steps: 445, steps per second: 239, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 310483/1000000: episode: 600, duration: 1.943s, episode steps: 463, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 310952/1000000: episode: 601, duration: 1.976s, episode steps: 469, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 311403/1000000: episode: 602, duration: 1.918s, episode steps: 451, steps per second: 235, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 312162/1000000: episode: 603, duration: 3.210s, episode steps: 759, steps per second: 236, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 312739/1000000: episode: 604, duration: 2.436s, episode steps: 577, steps per second: 237, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.362 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 313416/1000000: episode: 605, duration: 2.821s, episode steps: 677, steps per second: 240, episode reward: 20.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 313768/1000000: episode: 606, duration: 1.482s, episode steps: 352, steps per second: 237, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 314918/1000000: episode: 607, duration: 4.810s, episode steps: 1150, steps per second: 239, episode reward: 15.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 315523/1000000: episode: 608, duration: 2.529s, episode steps: 605, steps per second: 239, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 316030/1000000: episode: 609, duration: 2.118s, episode steps: 507, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 316552/1000000: episode: 610, duration: 2.184s, episode steps: 522, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 316959/1000000: episode: 611, duration: 1.694s, episode steps: 407, steps per second: 240, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 317334/1000000: episode: 612, duration: 1.573s, episode steps: 375, steps per second: 238, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.701 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 317716/1000000: episode: 613, duration: 1.625s, episode steps: 382, steps per second: 235, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 317999/1000000: episode: 614, duration: 1.190s, episode steps: 283, steps per second: 238, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.717 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 318506/1000000: episode: 615, duration: 2.114s, episode steps: 507, steps per second: 240, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 319028/1000000: episode: 616, duration: 2.193s, episode steps: 522, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 319518/1000000: episode: 617, duration: 2.057s, episode steps: 490, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 319968/1000000: episode: 618, duration: 1.893s, episode steps: 450, steps per second: 238, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 320250/1000000: episode: 619, duration: 1.200s, episode steps: 282, steps per second: 235, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 320559/1000000: episode: 620, duration: 1.309s, episode steps: 309, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 321156/1000000: episode: 621, duration: 2.529s, episode steps: 597, steps per second: 236, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 321536/1000000: episode: 622, duration: 1.598s, episode steps: 380, steps per second: 238, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 322049/1000000: episode: 623, duration: 2.151s, episode steps: 513, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 322453/1000000: episode: 624, duration: 1.684s, episode steps: 404, steps per second: 240, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 323248/1000000: episode: 625, duration: 3.348s, episode steps: 795, steps per second: 237, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 324142/1000000: episode: 626, duration: 3.745s, episode steps: 894, steps per second: 239, episode reward: 12.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 324724/1000000: episode: 627, duration: 2.434s, episode steps: 582, steps per second: 239, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 325262/1000000: episode: 628, duration: 2.256s, episode steps: 538, steps per second: 239, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 325740/1000000: episode: 629, duration: 2.009s, episode steps: 478, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 326354/1000000: episode: 630, duration: 2.595s, episode steps: 614, steps per second: 237, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 327015/1000000: episode: 631, duration: 2.782s, episode steps: 661, steps per second: 238, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 327630/1000000: episode: 632, duration: 2.573s, episode steps: 615, steps per second: 239, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 328057/1000000: episode: 633, duration: 1.798s, episode steps: 427, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 328663/1000000: episode: 634, duration: 2.519s, episode steps: 606, steps per second: 241, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 329167/1000000: episode: 635, duration: 2.124s, episode steps: 504, steps per second: 237, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 330014/1000000: episode: 636, duration: 3.538s, episode steps: 847, steps per second: 239, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 330504/1000000: episode: 637, duration: 2.040s, episode steps: 490, steps per second: 240, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 330971/1000000: episode: 638, duration: 1.949s, episode steps: 467, steps per second: 240, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 331350/1000000: episode: 639, duration: 1.586s, episode steps: 379, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 331810/1000000: episode: 640, duration: 1.916s, episode steps: 460, steps per second: 240, episode reward: 13.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

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spartial/version_1.0/pesos/dqn_v_1.0_weights_640_episodes.h5f
332444/1000000: episode: 641, duration: 2.754s, episode steps: 634, steps per
second: 230, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
332777/1000000: episode: 642, duration: 1.425s, episode steps: 333, steps per
second: 234, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
333559/1000000: episode: 643, duration: 3.260s, episode steps: 782, steps per
second: 240, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
333850/1000000: episode: 644, duration: 1.220s, episode steps: 291, steps per
second: 238, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
334451/1000000: episode: 645, duration: 2.510s, episode steps: 601, steps per
second: 239, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
334890/1000000: episode: 646, duration: 1.847s, episode steps: 439, steps per
second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
335177/1000000: episode: 647, duration: 1.212s, episode steps: 287, steps per
second: 237, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
335483/1000000: episode: 648, duration: 1.294s, episode steps: 306, steps per
second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
336487/1000000: episode: 649, duration: 4.174s, episode steps: 1004, steps per
second: 241, episode reward: 21.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
336901/1000000: episode: 650, duration: 1.736s, episode steps: 414, steps per
second: 239, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
337339/1000000: episode: 651, duration: 1.842s, episode steps: 438, steps per
second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
337942/1000000: episode: 652, duration: 10.465s, episode steps: 603, steps per
second: 58, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.551 [0.000, 5.000], loss: 0.016710, mae: 0.838327, mean_q: 1.025493,
mean_eps: 0.987842
338580/1000000: episode: 653, duration: 22.175s, episode steps: 638, steps per
second: 29, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.549 [0.000, 5.000], loss: 0.013836, mae: 0.835052, mean_q: 1.025181,
mean_eps: 0.987304
338988/1000000: episode: 654, duration: 14.315s, episode steps: 408, steps per
second: 29, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.468 [0.000, 5.000], loss: 0.015231, mae: 0.840002, mean_q: 1.034094,
mean_eps: 0.986678
339588/1000000: episode: 655, duration: 20.943s, episode steps: 600, steps per

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second: 29, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.015406, mae: 0.829071, mean_q: 1.022549, mean_eps: 0.986073

339976/1000000: episode: 656, duration: 13.639s, episode steps: 388, steps per second: 28, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.014940, mae: 0.830510, mean_q: 1.021239, mean_eps: 0.985480

340411/1000000: episode: 657, duration: 15.081s, episode steps: 435, steps per second: 29, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.015842, mae: 0.838122, mean_q: 1.030234, mean_eps: 0.984986

340943/1000000: episode: 658, duration: 18.459s, episode steps: 532, steps per second: 29, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.015396, mae: 0.835449, mean_q: 1.027332, mean_eps: 0.984405

341489/1000000: episode: 659, duration: 19.000s, episode steps: 546, steps per second: 29, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: 0.013104, mae: 0.823627, mean_q: 1.014013, mean_eps: 0.983757

341992/1000000: episode: 660, duration: 17.488s, episode steps: 503, steps per second: 29, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.013777, mae: 0.838108, mean_q: 1.029085, mean_eps: 0.983128

342601/1000000: episode: 661, duration: 21.318s, episode steps: 609, steps per second: 29, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.015188, mae: 0.828566, mean_q: 1.015647, mean_eps: 0.982461

342985/1000000: episode: 662, duration: 12.484s, episode steps: 384, steps per second: 31, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.016814, mae: 0.856625, mean_q: 1.051308, mean_eps: 0.981863

343478/1000000: episode: 663, duration: 16.693s, episode steps: 493, steps per second: 30, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: 0.014248, mae: 0.875954, mean_q: 1.077441, mean_eps: 0.981338

343764/1000000: episode: 664, duration: 10.008s, episode steps: 286, steps per second: 29, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.016180, mae: 0.867861, mean_q: 1.064268, mean_eps: 0.980872

344757/1000000: episode: 665, duration: 34.654s, episode steps: 993, steps per second: 29, episode reward: 14.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.014593, mae: 0.873248, mean_q: 1.069827, mean_eps: 0.980104

345525/1000000: episode: 666, duration: 26.744s, episode steps: 768, steps per second: 29, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: 0.014813, mae: 0.876820, mean_q: 1.076838, mean_eps: 0.979046

345975/1000000: episode: 667, duration: 15.635s, episode steps: 450, steps per

second: 29, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: 0.013992, mae: 0.860283, mean_q: 1.057698, mean_eps: 0.978316

346563/1000000: episode: 668, duration: 20.544s, episode steps: 588, steps per second: 29, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.013619, mae: 0.871136, mean_q: 1.075337, mean_eps: 0.977694

347175/1000000: episode: 669, duration: 21.360s, episode steps: 612, steps per second: 29, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.014400, mae: 0.864985, mean_q: 1.060288, mean_eps: 0.976974

347853/1000000: episode: 670, duration: 23.188s, episode steps: 678, steps per second: 29, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.014353, mae: 0.879267, mean_q: 1.077958, mean_eps: 0.976199

348674/1000000: episode: 671, duration: 27.807s, episode steps: 821, steps per second: 30, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.015065, mae: 0.918194, mean_q: 1.126690, mean_eps: 0.975299

349092/1000000: episode: 672, duration: 14.685s, episode steps: 418, steps per second: 28, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.014453, mae: 0.883455, mean_q: 1.090226, mean_eps: 0.974558

349773/1000000: episode: 673, duration: 23.888s, episode steps: 681, steps per second: 29, episode reward: 8.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.014434, mae: 0.900391, mean_q: 1.105400, mean_eps: 0.973898

350158/1000000: episode: 674, duration: 13.482s, episode steps: 385, steps per second: 29, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.015201, mae: 0.881455, mean_q: 1.083037, mean_eps: 0.973257

350456/1000000: episode: 675, duration: 10.453s, episode steps: 298, steps per second: 29, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.014522, mae: 0.907402, mean_q: 1.113366, mean_eps: 0.972849

351001/1000000: episode: 676, duration: 19.229s, episode steps: 545, steps per second: 28, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.013583, mae: 0.894199, mean_q: 1.102241, mean_eps: 0.972342

351784/1000000: episode: 677, duration: 27.544s, episode steps: 783, steps per second: 28, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.013418, mae: 0.904333, mean_q: 1.108739, mean_eps: 0.971546

352138/1000000: episode: 678, duration: 12.478s, episode steps: 354, steps per second: 28, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.013406, mae: 0.917235, mean_q: 1.125786, mean_eps: 0.970864

352784/1000000: episode: 679, duration: 22.687s, episode steps: 646, steps per

second: 28, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.014838, mae: 0.912999, mean_q: 1.121295, mean_eps: 0.970264

353484/1000000: episode: 680, duration: 24.608s, episode steps: 700, steps per second: 28, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.015092, mae: 0.949760, mean_q: 1.170951, mean_eps: 0.969458

354411/1000000: episode: 681, duration: 32.416s, episode steps: 927, steps per second: 29, episode reward: 21.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.014780, mae: 0.944785, mean_q: 1.159368, mean_eps: 0.968481

354896/1000000: episode: 682, duration: 17.179s, episode steps: 485, steps per second: 28, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.013349, mae: 0.953915, mean_q: 1.172267, mean_eps: 0.967634

355769/1000000: episode: 683, duration: 30.661s, episode steps: 873, steps per second: 28, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.015773, mae: 0.927529, mean_q: 1.138871, mean_eps: 0.966818

356051/1000000: episode: 684, duration: 9.831s, episode steps: 282, steps per second: 29, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.012667, mae: 0.944013, mean_q: 1.160637, mean_eps: 0.966124

356470/1000000: episode: 685, duration: 14.776s, episode steps: 419, steps per second: 28, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.013483, mae: 0.942153, mean_q: 1.157311, mean_eps: 0.965704

356875/1000000: episode: 686, duration: 14.279s, episode steps: 405, steps per second: 28, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.015432, mae: 0.938678, mean_q: 1.153840, mean_eps: 0.965210

357186/1000000: episode: 687, duration: 10.989s, episode steps: 311, steps per second: 28, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.012747, mae: 0.918892, mean_q: 1.127884, mean_eps: 0.964780

357662/1000000: episode: 688, duration: 16.809s, episode steps: 476, steps per second: 28, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.015404, mae: 0.935063, mean_q: 1.148076, mean_eps: 0.964307

358419/1000000: episode: 689, duration: 27.097s, episode steps: 757, steps per second: 28, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.014583, mae: 0.992077, mean_q: 1.216134, mean_eps: 0.963568

358915/1000000: episode: 690, duration: 17.625s, episode steps: 496, steps per second: 28, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: 0.013629, mae: 0.975057, mean_q: 1.197883, mean_eps: 0.962817

359224/1000000: episode: 691, duration: 10.988s, episode steps: 309, steps per

second: 28, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.741 [0.000, 5.000], loss: 0.015283, mae: 1.010775, mean_q: 1.236984,
mean_eps: 0.962334

359653/1000000: episode: 692, duration: 15.268s, episode steps: 429, steps per
second: 28, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean
action: 2.485 [0.000, 5.000], loss: 0.014556, mae: 1.011341, mean_q: 1.240525,
mean_eps: 0.961890

360241/1000000: episode: 693, duration: 20.821s, episode steps: 588, steps per
second: 28, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.512 [0.000, 5.000], loss: 0.015309, mae: 0.981947, mean_q: 1.208264,
mean_eps: 0.961278

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

360588/1000000: episode: 694, duration: 7.778s, episode steps: 347, steps per
second: 45, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.565 [0.000, 5.000], loss: 0.012644, mae: 0.958441, mean_q: 1.180054,
mean_eps: 0.960803

361285/1000000: episode: 695, duration: 2.929s, episode steps: 697, steps per
second: 238, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

362051/1000000: episode: 696, duration: 3.213s, episode steps: 766, steps per
second: 238, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean
action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

362769/1000000: episode: 697, duration: 3.014s, episode steps: 718, steps per
second: 238, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

363275/1000000: episode: 698, duration: 2.135s, episode steps: 506, steps per
second: 237, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

363686/1000000: episode: 699, duration: 1.728s, episode steps: 411, steps per
second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

364416/1000000: episode: 700, duration: 3.061s, episode steps: 730, steps per
second: 239, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

364859/1000000: episode: 701, duration: 1.869s, episode steps: 443, steps per
second: 237, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

365258/1000000: episode: 702, duration: 1.684s, episode steps: 399, steps per
second: 237, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

365746/1000000: episode: 703, duration: 2.051s, episode steps: 488, steps per
second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

366225/1000000: episode: 704, duration: 2.011s, episode steps: 479, steps per
second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean

action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 366697/1000000: episode: 705, duration: 1.989s, episode steps: 472, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 367381/1000000: episode: 706, duration: 2.884s, episode steps: 684, steps per second: 237, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.658 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 367971/1000000: episode: 707, duration: 2.473s, episode steps: 590, steps per second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 368331/1000000: episode: 708, duration: 1.511s, episode steps: 360, steps per second: 238, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 368855/1000000: episode: 709, duration: 2.193s, episode steps: 524, steps per second: 239, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 369231/1000000: episode: 710, duration: 1.581s, episode steps: 376, steps per second: 238, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 369730/1000000: episode: 711, duration: 2.103s, episode steps: 499, steps per second: 237, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 370013/1000000: episode: 712, duration: 1.201s, episode steps: 283, steps per second: 236, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 370499/1000000: episode: 713, duration: 2.040s, episode steps: 486, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 370880/1000000: episode: 714, duration: 1.596s, episode steps: 381, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 371455/1000000: episode: 715, duration: 2.407s, episode steps: 575, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 371762/1000000: episode: 716, duration: 1.293s, episode steps: 307, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 372050/1000000: episode: 717, duration: 1.205s, episode steps: 288, steps per second: 239, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 372513/1000000: episode: 718, duration: 1.939s, episode steps: 463, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 372958/1000000: episode: 719, duration: 1.878s, episode steps: 445, steps per second: 237, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 373659/1000000: episode: 720, duration: 2.931s, episode steps: 701, steps per second: 239, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 373958/1000000: episode: 721, duration: 1.251s, episode steps: 299, steps per second: 239, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 374446/1000000: episode: 722, duration: 2.041s, episode steps: 488, steps per second: 239, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 374925/1000000: episode: 723, duration: 2.004s, episode steps: 479, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 375346/1000000: episode: 724, duration: 1.755s, episode steps: 421, steps per second: 240, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.257 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 375677/1000000: episode: 725, duration: 1.415s, episode steps: 331, steps per second: 234, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 376063/1000000: episode: 726, duration: 1.660s, episode steps: 386, steps per second: 232, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 376511/1000000: episode: 727, duration: 1.894s, episode steps: 448, steps per second: 236, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 376959/1000000: episode: 728, duration: 1.890s, episode steps: 448, steps per second: 237, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 377323/1000000: episode: 729, duration: 1.547s, episode steps: 364, steps per second: 235, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 377790/1000000: episode: 730, duration: 1.970s, episode steps: 467, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 378156/1000000: episode: 731, duration: 1.547s, episode steps: 366, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 378690/1000000: episode: 732, duration: 2.248s, episode steps: 534, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 379430/1000000: episode: 733, duration: 3.094s, episode steps: 740, steps per second: 239, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 379912/1000000: episode: 734, duration: 2.025s, episode steps: 482, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 380404/1000000: episode: 735, duration: 2.059s, episode steps: 492, steps per second: 239, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 380790/1000000: episode: 736, duration: 1.616s, episode steps: 386, steps per second: 239, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 381159/1000000: episode: 737, duration: 1.551s, episode steps: 369, steps per second: 238, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 381691/1000000: episode: 738, duration: 2.229s, episode steps: 532, steps per second: 239, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 381990/1000000: episode: 739, duration: 1.266s, episode steps: 299, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 382638/1000000: episode: 740, duration: 2.716s, episode steps: 648, steps per second: 239, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 383160/1000000: episode: 741, duration: 2.181s, episode steps: 522, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 383639/1000000: episode: 742, duration: 2.011s, episode steps: 479, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 384357/1000000: episode: 743, duration: 2.990s, episode steps: 718, steps per second: 240, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.728 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 385110/1000000: episode: 744, duration: 3.184s, episode steps: 753, steps per second: 236, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 385719/1000000: episode: 745, duration: 2.560s, episode steps: 609, steps per second: 238, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 386395/1000000: episode: 746, duration: 2.824s, episode steps: 676, steps per second: 239, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 386962/1000000: episode: 747, duration: 2.361s, episode steps: 567, steps per second: 240, episode reward: 16.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 387578/1000000: episode: 748, duration: 2.594s, episode steps: 616, steps per second: 238, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 388074/1000000: episode: 749, duration: 2.120s, episode steps: 496, steps per second: 234, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 388755/1000000: episode: 750, duration: 2.833s, episode steps: 681, steps per second: 240, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 389439/1000000: episode: 751, duration: 2.834s, episode steps: 684, steps per second: 241, episode reward: 19.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 389862/1000000: episode: 752, duration: 1.766s, episode steps: 423, steps per second: 240, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean

action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 390476/1000000: episode: 753, duration: 2.573s, episode steps: 614, steps per second: 239, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 390781/1000000: episode: 754, duration: 1.304s, episode steps: 305, steps per second: 234, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 391321/1000000: episode: 755, duration: 2.268s, episode steps: 540, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 391889/1000000: episode: 756, duration: 2.362s, episode steps: 568, steps per second: 240, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 392340/1000000: episode: 757, duration: 1.889s, episode steps: 451, steps per second: 239, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 392823/1000000: episode: 758, duration: 2.026s, episode steps: 483, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.315 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 393293/1000000: episode: 759, duration: 1.981s, episode steps: 470, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 394070/1000000: episode: 760, duration: 3.262s, episode steps: 777, steps per second: 238, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 394364/1000000: episode: 761, duration: 1.235s, episode steps: 294, steps per second: 238, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 395018/1000000: episode: 762, duration: 2.726s, episode steps: 654, steps per second: 240, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 395788/1000000: episode: 763, duration: 3.213s, episode steps: 770, steps per second: 240, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 396236/1000000: episode: 764, duration: 1.876s, episode steps: 448, steps per second: 239, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 396537/1000000: episode: 765, duration: 1.274s, episode steps: 301, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.701 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 397197/1000000: episode: 766, duration: 2.779s, episode steps: 660, steps per second: 237, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 397694/1000000: episode: 767, duration: 2.084s, episode steps: 497, steps per second: 238, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 398179/1000000: episode: 768, duration: 2.021s, episode steps: 485, steps per second: 240, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 398719/1000000: episode: 769, duration: 2.249s, episode steps: 540, steps per second: 240, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 399360/1000000: episode: 770, duration: 2.669s, episode steps: 641, steps per second: 240, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 399972/1000000: episode: 771, duration: 2.573s, episode steps: 612, steps per second: 238, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 400267/1000000: episode: 772, duration: 1.246s, episode steps: 295, steps per second: 237, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 400572/1000000: episode: 773, duration: 1.281s, episode steps: 305, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 401145/1000000: episode: 774, duration: 2.391s, episode steps: 573, steps per second: 240, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 401579/1000000: episode: 775, duration: 1.816s, episode steps: 434, steps per second: 239, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 401864/1000000: episode: 776, duration: 1.201s, episode steps: 285, steps per second: 237, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 402411/1000000: episode: 777, duration: 2.289s, episode steps: 547, steps per second: 239, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 403222/1000000: episode: 778, duration: 3.529s, episode steps: 811, steps per second: 230, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.010217, mae: 0.928664, mean_q: 1.149186, mean_eps: 0.987995
 403907/1000000: episode: 779, duration: 24.306s, episode steps: 685, steps per second: 28, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: 0.016528, mae: 1.050627, mean_q: 1.290568, mean_eps: 0.987582
 404320/1000000: episode: 780, duration: 14.777s, episode steps: 413, steps per second: 28, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.351 [0.000, 5.000], loss: 0.015949, mae: 1.061978, mean_q: 1.302792, mean_eps: 0.986925
 404703/1000000: episode: 781, duration: 13.674s, episode steps: 383, steps per second: 28, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.014052, mae: 1.019459, mean_q: 1.252363, mean_eps: 0.986447
 405268/1000000: episode: 782, duration: 20.239s, episode steps: 565, steps per second: 28, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.015240, mae: 1.037889, mean_q: 1.276748, mean_eps: 0.985878

406294/1000000: episode: 783, duration: 36.625s, episode steps: 1026, steps per second: 28, episode reward: 14.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.017001, mae: 1.039861, mean_q: 1.276926, mean_eps: 0.984923

406793/1000000: episode: 784, duration: 17.985s, episode steps: 499, steps per second: 28, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.014847, mae: 1.042717, mean_q: 1.281489, mean_eps: 0.984006

407275/1000000: episode: 785, duration: 17.173s, episode steps: 482, steps per second: 28, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: 0.014370, mae: 1.049973, mean_q: 1.290976, mean_eps: 0.983418

407587/1000000: episode: 786, duration: 11.156s, episode steps: 312, steps per second: 28, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.017472, mae: 1.037048, mean_q: 1.272794, mean_eps: 0.982943

407962/1000000: episode: 787, duration: 13.345s, episode steps: 375, steps per second: 28, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.014807, mae: 1.042046, mean_q: 1.281922, mean_eps: 0.982530

408407/1000000: episode: 788, duration: 15.335s, episode steps: 445, steps per second: 29, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.017532, mae: 1.071249, mean_q: 1.316834, mean_eps: 0.982038

408920/1000000: episode: 789, duration: 17.885s, episode steps: 513, steps per second: 29, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: 0.015041, mae: 1.082184, mean_q: 1.330542, mean_eps: 0.981465

409279/1000000: episode: 790, duration: 12.854s, episode steps: 359, steps per second: 28, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.016123, mae: 1.095271, mean_q: 1.341140, mean_eps: 0.980942

409871/1000000: episode: 791, duration: 21.174s, episode steps: 592, steps per second: 28, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.016026, mae: 1.105251, mean_q: 1.354286, mean_eps: 0.980370

410288/1000000: episode: 792, duration: 14.989s, episode steps: 417, steps per second: 28, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.015958, mae: 1.058605, mean_q: 1.297456, mean_eps: 0.979766

410690/1000000: episode: 793, duration: 14.417s, episode steps: 402, steps per second: 28, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.016261, mae: 1.091645, mean_q: 1.338836, mean_eps: 0.979274

411631/1000000: episode: 794, duration: 33.630s, episode steps: 941, steps per second: 28, episode reward: 23.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.016381, mae: 1.098083, mean_q: 1.343856, mean_eps: 0.978467

412267/1000000: episode: 795, duration: 22.818s, episode steps: 636, steps per second: 28, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.017367, mae: 1.096824, mean_q: 1.346104, mean_eps: 0.977522

412626/1000000: episode: 796, duration: 12.927s, episode steps: 359, steps per second: 28, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.017741, mae: 1.100970, mean_q: 1.353929, mean_eps: 0.976924

413064/1000000: episode: 797, duration: 15.812s, episode steps: 438, steps per second: 28, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.016332, mae: 1.106956, mean_q: 1.358219, mean_eps: 0.976446

413908/1000000: episode: 798, duration: 29.125s, episode steps: 844, steps per second: 29, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.018828, mae: 1.103313, mean_q: 1.353782, mean_eps: 0.975678

414211/1000000: episode: 799, duration: 10.945s, episode steps: 303, steps per second: 28, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.017133, mae: 1.103920, mean_q: 1.358340, mean_eps: 0.974990

414879/1000000: episode: 800, duration: 23.959s, episode steps: 668, steps per second: 28, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.015229, mae: 1.116772, mean_q: 1.373617, mean_eps: 0.974406

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_800_episodes.h5f

415515/1000000: episode: 801, duration: 22.948s, episode steps: 636, steps per second: 28, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.016040, mae: 1.114744, mean_q: 1.366243, mean_eps: 0.973624

415920/1000000: episode: 802, duration: 14.719s, episode steps: 405, steps per second: 28, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.013464, mae: 1.115617, mean_q: 1.368647, mean_eps: 0.973000

416433/1000000: episode: 803, duration: 18.521s, episode steps: 513, steps per second: 28, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.015635, mae: 1.099818, mean_q: 1.349195, mean_eps: 0.972448

417027/1000000: episode: 804, duration: 21.278s, episode steps: 594, steps per second: 28, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.016351, mae: 1.103880, mean_q: 1.353544, mean_eps: 0.971783

417624/1000000: episode: 805, duration: 21.458s, episode steps: 597, steps per second: 28, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.016448, mae: 1.111308, mean_q: 1.362209, mean_eps: 0.971070

417923/1000000: episode: 806, duration: 10.817s, episode steps: 299, steps per second: 28, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: 0.013600, mae: 1.121483, mean_q: 1.376324, mean_eps: 0.970533

418561/1000000: episode: 807, duration: 23.058s, episode steps: 638, steps per second: 28, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.014653, mae: 1.118384, mean_q: 1.370491, mean_eps: 0.969969

419351/1000000: episode: 808, duration: 28.298s, episode steps: 790, steps per second: 28, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.016317, mae: 1.157580, mean_q: 1.423124, mean_eps: 0.969112

419959/1000000: episode: 809, duration: 21.925s, episode steps: 608, steps per second: 28, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: 0.015739, mae: 1.155754, mean_q: 1.421670, mean_eps: 0.968274

420588/1000000: episode: 810, duration: 22.786s, episode steps: 629, steps per second: 28, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.017095, mae: 1.160882, mean_q: 1.428217, mean_eps: 0.967533

421090/1000000: episode: 811, duration: 18.136s, episode steps: 502, steps per second: 28, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.015845, mae: 1.158603, mean_q: 1.426251, mean_eps: 0.966854

421435/1000000: episode: 812, duration: 12.452s, episode steps: 345, steps per second: 28, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.371 [0.000, 5.000], loss: 0.016576, mae: 1.153572, mean_q: 1.419664, mean_eps: 0.966345

421834/1000000: episode: 813, duration: 14.440s, episode steps: 399, steps per second: 28, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: 0.015215, mae: 1.132121, mean_q: 1.394149, mean_eps: 0.965898

422240/1000000: episode: 814, duration: 14.687s, episode steps: 406, steps per second: 28, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.015434, mae: 1.129298, mean_q: 1.386453, mean_eps: 0.965416

422697/1000000: episode: 815, duration: 16.565s, episode steps: 457, steps per second: 28, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.014389, mae: 1.133857, mean_q: 1.392711, mean_eps: 0.964898

423086/1000000: episode: 816, duration: 13.998s, episode steps: 389, steps per second: 28, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.013737, mae: 1.135194, mean_q: 1.393315, mean_eps: 0.964389

423582/1000000: episode: 817, duration: 16.777s, episode steps: 496, steps per second: 30, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.015977, mae: 1.171819, mean_q: 1.443287, mean_eps: 0.963858

423965/1000000: episode: 818, duration: 13.681s, episode steps: 383, steps per second: 28, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.329 [0.000, 5.000], loss: 0.015622, mae: 1.173039, mean_q: 1.440104, mean_eps: 0.963330

424427/1000000: episode: 819, duration: 16.627s, episode steps: 462, steps per second: 28, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.015410, mae: 1.165808, mean_q: 1.432003, mean_eps: 0.962824

425008/1000000: episode: 820, duration: 21.083s, episode steps: 581, steps per second: 28, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: 0.015294, mae: 1.173475, mean_q: 1.439838, mean_eps: 0.962200

425545/1000000: episode: 821, duration: 19.469s, episode steps: 537, steps per second: 28, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: 0.014889, mae: 1.175544, mean_q: 1.442479, mean_eps: 0.961528

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

426020/1000000: episode: 822, duration: 15.946s, episode steps: 475, steps per second: 30, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: 0.016585, mae: 1.180489, mean_q: 1.448233, mean_eps: 0.960942

426473/1000000: episode: 823, duration: 1.924s, episode steps: 453, steps per second: 235, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

427023/1000000: episode: 824, duration: 2.331s, episode steps: 550, steps per second: 236, episode reward: 21.000, mean reward: 0.038 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

427452/1000000: episode: 825, duration: 1.796s, episode steps: 429, steps per second: 239, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.716 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

427716/1000000: episode: 826, duration: 1.113s, episode steps: 264, steps per second: 237, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

428165/1000000: episode: 827, duration: 1.877s, episode steps: 449, steps per second: 239, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

428565/1000000: episode: 828, duration: 1.684s, episode steps: 400, steps per second: 237, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.277 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

429068/1000000: episode: 829, duration: 2.122s, episode steps: 503, steps per second: 237, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

429491/1000000: episode: 830, duration: 1.797s, episode steps: 423, steps per second: 235, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

429976/1000000: episode: 831, duration: 2.045s, episode steps: 485, steps per second: 237, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

430462/1000000: episode: 832, duration: 2.041s, episode steps: 486, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.387 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

430980/1000000: episode: 833, duration: 2.173s, episode steps: 518, steps per second: 238, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

431812/1000000: episode: 834, duration: 3.458s, episode steps: 832, steps per second: 241, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

432523/1000000: episode: 835, duration: 2.979s, episode steps: 711, steps per second: 239, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

433000/1000000: episode: 836, duration: 2.007s, episode steps: 477, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

433439/1000000: episode: 837, duration: 1.838s, episode steps: 439, steps per second: 239, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

434002/1000000: episode: 838, duration: 2.357s, episode steps: 563, steps per second: 239, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

434688/1000000: episode: 839, duration: 2.844s, episode steps: 686, steps per second: 241, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

435260/1000000: episode: 840, duration: 2.389s, episode steps: 572, steps per second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

435870/1000000: episode: 841, duration: 2.586s, episode steps: 610, steps per second: 236, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

436383/1000000: episode: 842, duration: 2.157s, episode steps: 513, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

437061/1000000: episode: 843, duration: 2.835s, episode steps: 678, steps per second: 239, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

437800/1000000: episode: 844, duration: 3.085s, episode steps: 739, steps per second: 240, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

438143/1000000: episode: 845, duration: 1.451s, episode steps: 343, steps per second: 236, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

438610/1000000: episode: 846, duration: 1.997s, episode steps: 467, steps per second: 234, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

439366/1000000: episode: 847, duration: 3.162s, episode steps: 756, steps per second: 239, episode reward: 9.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

439833/1000000: episode: 848, duration: 1.948s, episode steps: 467, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

440289/1000000: episode: 849, duration: 1.900s, episode steps: 456, steps per second: 240, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.693 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

440923/1000000: episode: 850, duration: 2.635s, episode steps: 634, steps per second: 241, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

441469/1000000: episode: 851, duration: 2.284s, episode steps: 546, steps per second: 239, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

442166/1000000: episode: 852, duration: 2.914s, episode steps: 697, steps per second: 239, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

442639/1000000: episode: 853, duration: 1.971s, episode steps: 473, steps per second: 240, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

443170/1000000: episode: 854, duration: 2.222s, episode steps: 531, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

443464/1000000: episode: 855, duration: 1.240s, episode steps: 294, steps per second: 237, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

443928/1000000: episode: 856, duration: 1.951s, episode steps: 464, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.375 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

444465/1000000: episode: 857, duration: 2.247s, episode steps: 537, steps per second: 239, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

445381/1000000: episode: 858, duration: 3.819s, episode steps: 916, steps per second: 240, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

445925/1000000: episode: 859, duration: 2.268s, episode steps: 544, steps per second: 240, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

446585/1000000: episode: 860, duration: 2.763s, episode steps: 660, steps per second: 239, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

447036/1000000: episode: 861, duration: 1.896s, episode steps: 451, steps per second: 238, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

447636/1000000: episode: 862, duration: 2.543s, episode steps: 600, steps per second: 236, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

447958/1000000: episode: 863, duration: 1.377s, episode steps: 322, steps per second: 234, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

448531/1000000: episode: 864, duration: 2.388s, episode steps: 573, steps per second: 240, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

449140/1000000: episode: 865, duration: 2.553s, episode steps: 609, steps per second: 239, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

449750/1000000: episode: 866, duration: 2.548s, episode steps: 610, steps per second: 239, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

450064/1000000: episode: 867, duration: 1.321s, episode steps: 314, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

450351/1000000: episode: 868, duration: 1.220s, episode steps: 287, steps per second: 235, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.672 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

450651/1000000: episode: 869, duration: 1.269s, episode steps: 300, steps per second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

451136/1000000: episode: 870, duration: 2.036s, episode steps: 485, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

451426/1000000: episode: 871, duration: 1.221s, episode steps: 290, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.272 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

452310/1000000: episode: 872, duration: 3.671s, episode steps: 884, steps per second: 241, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

452723/1000000: episode: 873, duration: 1.723s, episode steps: 413, steps per second: 240, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

453075/1000000: episode: 874, duration: 1.472s, episode steps: 352, steps per second: 239, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.384 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

453389/1000000: episode: 875, duration: 1.330s, episode steps: 314, steps per second: 236, episode reward: 8.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

454074/1000000: episode: 876, duration: 2.906s, episode steps: 685, steps per second: 236, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

454551/1000000: episode: 877, duration: 1.990s, episode steps: 477, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

454951/1000000: episode: 878, duration: 1.676s, episode steps: 400, steps per second: 239, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

455439/1000000: episode: 879, duration: 2.051s, episode steps: 488, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

455936/1000000: episode: 880, duration: 2.078s, episode steps: 497, steps per second: 239, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

456623/1000000: episode: 881, duration: 2.875s, episode steps: 687, steps per second: 239, episode reward: 19.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

457024/1000000: episode: 882, duration: 1.690s, episode steps: 401, steps per second: 237, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

457739/1000000: episode: 883, duration: 2.960s, episode steps: 715, steps per second: 242, episode reward: 23.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

458562/1000000: episode: 884, duration: 3.424s, episode steps: 823, steps per second: 240, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

458862/1000000: episode: 885, duration: 1.275s, episode steps: 300, steps per second: 235, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

459368/1000000: episode: 886, duration: 2.126s, episode steps: 506, steps per second: 238, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

459731/1000000: episode: 887, duration: 1.544s, episode steps: 363, steps per second: 235, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

460027/1000000: episode: 888, duration: 1.262s, episode steps: 296, steps per second: 235, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

460364/1000000: episode: 889, duration: 1.410s, episode steps: 337, steps per second: 239, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

461064/1000000: episode: 890, duration: 2.906s, episode steps: 700, steps per second: 241, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

461629/1000000: episode: 891, duration: 2.358s, episode steps: 565, steps per second: 240, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

462091/1000000: episode: 892, duration: 1.934s, episode steps: 462, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

462598/1000000: episode: 893, duration: 2.150s, episode steps: 507, steps per second: 236, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

463070/1000000: episode: 894, duration: 1.993s, episode steps: 472, steps per second: 237, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.684 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

463728/1000000: episode: 895, duration: 2.738s, episode steps: 658, steps per second: 240, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

464344/1000000: episode: 896, duration: 2.579s, episode steps: 616, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

465245/1000000: episode: 897, duration: 3.734s, episode steps: 901, steps per second: 241, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

465882/1000000: episode: 898, duration: 2.707s, episode steps: 637, steps per second: 235, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

466425/1000000: episode: 899, duration: 2.262s, episode steps: 543, steps per second: 240, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

466778/1000000: episode: 900, duration: 1.482s, episode steps: 353, steps per second: 238, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

467533/1000000: episode: 901, duration: 3.154s, episode steps: 755, steps per second: 239, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

468001/1000000: episode: 902, duration: 1.969s, episode steps: 468, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

468521/1000000: episode: 903, duration: 2.213s, episode steps: 520, steps per second: 235, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

469044/1000000: episode: 904, duration: 11.672s, episode steps: 523, steps per second: 45, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.016480, mae: 1.246308, mean_q: 1.531347, mean_eps: 0.987825

469514/1000000: episode: 905, duration: 17.327s, episode steps: 470, steps per second: 27, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.016237, mae: 1.194737, mean_q: 1.465554, mean_eps: 0.987369

470302/1000000: episode: 906, duration: 29.016s, episode steps: 788, steps per second: 27, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.016422, mae: 1.237237, mean_q: 1.516709, mean_eps: 0.986613

470669/1000000: episode: 907, duration: 13.537s, episode steps: 367, steps per second: 27, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.360 [0.000, 5.000], loss: 0.017430, mae: 1.238633, mean_q: 1.518599, mean_eps: 0.985919

471049/1000000: episode: 908, duration: 13.996s, episode steps: 380, steps per second: 27, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.016664, mae: 1.234978, mean_q: 1.515562, mean_eps: 0.985470

471631/1000000: episode: 909, duration: 21.336s, episode steps: 582, steps per

second: 27, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.017094, mae: 1.239806, mean_q: 1.524059, mean_eps: 0.984894

472225/1000000: episode: 910, duration: 21.986s, episode steps: 594, steps per second: 27, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.016157, mae: 1.248279, mean_q: 1.532793, mean_eps: 0.984189

472507/1000000: episode: 911, duration: 10.304s, episode steps: 282, steps per second: 27, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.015853, mae: 1.212254, mean_q: 1.489112, mean_eps: 0.983663

473001/1000000: episode: 912, duration: 18.266s, episode steps: 494, steps per second: 27, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.014154, mae: 1.233350, mean_q: 1.516608, mean_eps: 0.983198

473283/1000000: episode: 913, duration: 10.325s, episode steps: 282, steps per second: 27, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: 0.018348, mae: 1.245364, mean_q: 1.533541, mean_eps: 0.982732

473640/1000000: episode: 914, duration: 13.284s, episode steps: 357, steps per second: 27, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.016430, mae: 1.241606, mean_q: 1.523991, mean_eps: 0.982350

474051/1000000: episode: 915, duration: 14.320s, episode steps: 411, steps per second: 29, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.014400, mae: 1.275112, mean_q: 1.566453, mean_eps: 0.981890

474544/1000000: episode: 916, duration: 17.795s, episode steps: 493, steps per second: 28, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: 0.016994, mae: 1.270178, mean_q: 1.555664, mean_eps: 0.981347

475006/1000000: episode: 917, duration: 17.177s, episode steps: 462, steps per second: 27, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.015472, mae: 1.315029, mean_q: 1.612627, mean_eps: 0.980774

475439/1000000: episode: 918, duration: 15.976s, episode steps: 433, steps per second: 27, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: 0.015851, mae: 1.298318, mean_q: 1.590227, mean_eps: 0.980236

476020/1000000: episode: 919, duration: 21.491s, episode steps: 581, steps per second: 27, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.015653, mae: 1.291070, mean_q: 1.582150, mean_eps: 0.979629

476870/1000000: episode: 920, duration: 31.304s, episode steps: 850, steps per second: 27, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.015038, mae: 1.282594, mean_q: 1.573469, mean_eps: 0.978770

477244/1000000: episode: 921, duration: 13.932s, episode steps: 374, steps per

second: 27, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean
action: 2.471 [0.000, 5.000], loss: 0.015756, mae: 1.290013, mean_q: 1.581720,
mean_eps: 0.978035

477693/1000000: episode: 922, duration: 16.714s, episode steps: 449, steps per
second: 27, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
action: 2.606 [0.000, 5.000], loss: 0.016423, mae: 1.289748, mean_q: 1.579106,
mean_eps: 0.977541

478156/1000000: episode: 923, duration: 17.056s, episode steps: 463, steps per
second: 27, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean
action: 2.417 [0.000, 5.000], loss: 0.014438, mae: 1.293062, mean_q: 1.585196,
mean_eps: 0.976994

478707/1000000: episode: 924, duration: 20.435s, episode steps: 551, steps per
second: 27, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.481 [0.000, 5.000], loss: 0.016032, mae: 1.287234, mean_q: 1.576271,
mean_eps: 0.976386

479247/1000000: episode: 925, duration: 18.570s, episode steps: 540, steps per
second: 29, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.557 [0.000, 5.000], loss: 0.016387, mae: 1.318638, mean_q: 1.618865,
mean_eps: 0.975731

479766/1000000: episode: 926, duration: 19.362s, episode steps: 519, steps per
second: 27, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.518 [0.000, 5.000], loss: 0.015642, mae: 1.338362, mean_q: 1.637927,
mean_eps: 0.975095

480241/1000000: episode: 927, duration: 17.617s, episode steps: 475, steps per
second: 27, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.520 [0.000, 5.000], loss: 0.015667, mae: 1.345146, mean_q: 1.646916,
mean_eps: 0.974498

480763/1000000: episode: 928, duration: 19.218s, episode steps: 522, steps per
second: 27, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.534 [0.000, 5.000], loss: 0.014672, mae: 1.316883, mean_q: 1.615426,
mean_eps: 0.973900

481051/1000000: episode: 929, duration: 10.641s, episode steps: 288, steps per
second: 27, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.476 [0.000, 5.000], loss: 0.015892, mae: 1.330478, mean_q: 1.628154,
mean_eps: 0.973415

481563/1000000: episode: 930, duration: 18.972s, episode steps: 512, steps per
second: 27, episode reward: 14.000, mean reward: 0.027 [0.000, 1.000], mean
action: 2.684 [0.000, 5.000], loss: 0.014727, mae: 1.348186, mean_q: 1.651301,
mean_eps: 0.972935

481863/1000000: episode: 931, duration: 11.128s, episode steps: 300, steps per
second: 27, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.480 [0.000, 5.000], loss: 0.014543, mae: 1.331905, mean_q: 1.628193,
mean_eps: 0.972448

482410/1000000: episode: 932, duration: 20.312s, episode steps: 547, steps per
second: 27, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.448 [0.000, 5.000], loss: 0.014472, mae: 1.315552, mean_q: 1.606716,
mean_eps: 0.971939

482914/1000000: episode: 933, duration: 18.725s, episode steps: 504, steps per

second: 27, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: 0.015072, mae: 1.323442, mean_q: 1.617305, mean_eps: 0.971308

483318/1000000: episode: 934, duration: 15.003s, episode steps: 404, steps per second: 27, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.728 [0.000, 5.000], loss: 0.015873, mae: 1.329143, mean_q: 1.626989, mean_eps: 0.970763

483623/1000000: episode: 935, duration: 11.289s, episode steps: 305, steps per second: 27, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.014981, mae: 1.318391, mean_q: 1.614896, mean_eps: 0.970338

484092/1000000: episode: 936, duration: 16.480s, episode steps: 469, steps per second: 28, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.014677, mae: 1.348847, mean_q: 1.649905, mean_eps: 0.969875

484500/1000000: episode: 937, duration: 14.778s, episode steps: 408, steps per second: 28, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: 0.017636, mae: 1.372171, mean_q: 1.677213, mean_eps: 0.969350

485504/1000000: episode: 938, duration: 37.282s, episode steps: 1004, steps per second: 27, episode reward: 16.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.016587, mae: 1.368181, mean_q: 1.670305, mean_eps: 0.968502

486297/1000000: episode: 939, duration: 29.489s, episode steps: 793, steps per second: 27, episode reward: 20.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.015637, mae: 1.361980, mean_q: 1.662382, mean_eps: 0.967422

486776/1000000: episode: 940, duration: 17.818s, episode steps: 479, steps per second: 27, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: 0.015961, mae: 1.373379, mean_q: 1.678331, mean_eps: 0.966659

487310/1000000: episode: 941, duration: 19.909s, episode steps: 534, steps per second: 27, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.365 [0.000, 5.000], loss: 0.015744, mae: 1.373318, mean_q: 1.679580, mean_eps: 0.966052

488084/1000000: episode: 942, duration: 28.757s, episode steps: 774, steps per second: 27, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.015164, mae: 1.361152, mean_q: 1.661322, mean_eps: 0.965267

488522/1000000: episode: 943, duration: 16.279s, episode steps: 438, steps per second: 27, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: 0.015387, mae: 1.360044, mean_q: 1.660904, mean_eps: 0.964540

489017/1000000: episode: 944, duration: 18.431s, episode steps: 495, steps per second: 27, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: 0.014482, mae: 1.381697, mean_q: 1.685943, mean_eps: 0.963978

489506/1000000: episode: 945, duration: 18.054s, episode steps: 489, steps per

second: 27, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.521 [0.000, 5.000], loss: 0.014695, mae: 1.387306, mean_q: 1.696421,
mean_eps: 0.963388

489970/1000000: episode: 946, duration: 17.306s, episode steps: 464, steps per
second: 27, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.489 [0.000, 5.000], loss: 0.015320, mae: 1.378288, mean_q: 1.682993,
mean_eps: 0.962817

490268/1000000: episode: 947, duration: 11.131s, episode steps: 298, steps per
second: 27, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.389 [0.000, 5.000], loss: 0.014466, mae: 1.402728, mean_q: 1.712969,
mean_eps: 0.962361

490737/1000000: episode: 948, duration: 17.573s, episode steps: 469, steps per
second: 27, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.623 [0.000, 5.000], loss: 0.016362, mae: 1.417917, mean_q: 1.733074,
mean_eps: 0.961900

491199/1000000: episode: 949, duration: 17.093s, episode steps: 462, steps per
second: 27, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.502 [0.000, 5.000], loss: 0.016517, mae: 1.386732, mean_q: 1.690862,
mean_eps: 0.961341

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

491707/1000000: episode: 950, duration: 12.691s, episode steps: 508, steps per
second: 40, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.506 [0.000, 5.000], loss: 0.016285, mae: 1.379022, mean_q: 1.682306,
mean_eps: 0.960873

491974/1000000: episode: 951, duration: 1.126s, episode steps: 267, steps per
second: 237, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean
action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

492609/1000000: episode: 952, duration: 2.682s, episode steps: 635, steps per
second: 237, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

493055/1000000: episode: 953, duration: 1.883s, episode steps: 446, steps per
second: 237, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

493833/1000000: episode: 954, duration: 3.227s, episode steps: 778, steps per
second: 241, episode reward: 26.000, mean reward: 0.033 [0.000, 1.000], mean
action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

494435/1000000: episode: 955, duration: 2.536s, episode steps: 602, steps per
second: 237, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

494958/1000000: episode: 956, duration: 2.179s, episode steps: 523, steps per
second: 240, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

495435/1000000: episode: 957, duration: 2.026s, episode steps: 477, steps per
second: 235, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

495911/1000000: episode: 958, duration: 2.013s, episode steps: 476, steps per second: 236, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 496200/1000000: episode: 959, duration: 1.216s, episode steps: 289, steps per second: 238, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.349 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 496706/1000000: episode: 960, duration: 2.112s, episode steps: 506, steps per second: 240, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_960_episodes.h5f

497273/1000000: episode: 961, duration: 2.444s, episode steps: 567, steps per second: 232, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 497922/1000000: episode: 962, duration: 2.704s, episode steps: 649, steps per second: 240, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 498765/1000000: episode: 963, duration: 3.552s, episode steps: 843, steps per second: 237, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 499274/1000000: episode: 964, duration: 2.142s, episode steps: 509, steps per second: 238, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 499589/1000000: episode: 965, duration: 1.328s, episode steps: 315, steps per second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.714 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 499963/1000000: episode: 966, duration: 1.581s, episode steps: 374, steps per second: 237, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 500706/1000000: episode: 967, duration: 3.077s, episode steps: 743, steps per second: 242, episode reward: 28.000, mean reward: 0.038 [0.000, 1.000], mean action: 2.328 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 501301/1000000: episode: 968, duration: 2.491s, episode steps: 595, steps per second: 239, episode reward: 16.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 501876/1000000: episode: 969, duration: 2.453s, episode steps: 575, steps per second: 234, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 502217/1000000: episode: 970, duration: 1.454s, episode steps: 341, steps per second: 234, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 502509/1000000: episode: 971, duration: 1.239s, episode steps: 292, steps per second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 502824/1000000: episode: 972, duration: 1.332s, episode steps: 315, steps per second: 237, episode reward: 8.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 503290/1000000: episode: 973, duration: 1.966s, episode steps: 466, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 503586/1000000: episode: 974, duration: 1.257s, episode steps: 296, steps per second: 235, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 504171/1000000: episode: 975, duration: 2.476s, episode steps: 585, steps per second: 236, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 504647/1000000: episode: 976, duration: 2.009s, episode steps: 476, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 504951/1000000: episode: 977, duration: 1.287s, episode steps: 304, steps per second: 236, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 505339/1000000: episode: 978, duration: 1.630s, episode steps: 388, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 505672/1000000: episode: 979, duration: 1.403s, episode steps: 333, steps per second: 237, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 506635/1000000: episode: 980, duration: 4.003s, episode steps: 963, steps per second: 241, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 506922/1000000: episode: 981, duration: 1.211s, episode steps: 287, steps per second: 237, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 507541/1000000: episode: 982, duration: 2.602s, episode steps: 619, steps per second: 238, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 507995/1000000: episode: 983, duration: 1.937s, episode steps: 454, steps per second: 234, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 508394/1000000: episode: 984, duration: 1.682s, episode steps: 399, steps per second: 237, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 508842/1000000: episode: 985, duration: 1.874s, episode steps: 448, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 509316/1000000: episode: 986, duration: 1.991s, episode steps: 474, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 510365/1000000: episode: 987, duration: 4.343s, episode steps: 1049, steps per second: 242, episode reward: 17.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 510994/1000000: episode: 988, duration: 2.661s, episode steps: 629, steps per second: 236, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 511504/1000000: episode: 989, duration: 2.142s, episode steps: 510, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 511797/1000000: episode: 990, duration: 1.235s, episode steps: 293, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 512178/1000000: episode: 991, duration: 1.598s, episode steps: 381, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 512561/1000000: episode: 992, duration: 1.614s, episode steps: 383, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 512858/1000000: episode: 993, duration: 1.252s, episode steps: 297, steps per second: 237, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 513141/1000000: episode: 994, duration: 1.190s, episode steps: 283, steps per second: 238, episode reward: 10.000, mean reward: 0.035 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 513821/1000000: episode: 995, duration: 2.879s, episode steps: 680, steps per second: 236, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 514125/1000000: episode: 996, duration: 1.282s, episode steps: 304, steps per second: 237, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 514646/1000000: episode: 997, duration: 2.193s, episode steps: 521, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 515117/1000000: episode: 998, duration: 1.962s, episode steps: 471, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 515483/1000000: episode: 999, duration: 1.537s, episode steps: 366, steps per second: 238, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 515857/1000000: episode: 1000, duration: 1.562s, episode steps: 374, steps per second: 240, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 516268/1000000: episode: 1001, duration: 1.732s, episode steps: 411, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 516596/1000000: episode: 1002, duration: 1.388s, episode steps: 328, steps per second: 236, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.341 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 517015/1000000: episode: 1003, duration: 1.775s, episode steps: 419, steps per second: 236, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.334 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 517760/1000000: episode: 1004, duration: 3.086s, episode steps: 745, steps per second: 241, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 518239/1000000: episode: 1005, duration: 2.007s, episode steps: 479, steps per second: 239, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 518681/1000000: episode: 1006, duration: 1.862s, episode steps: 442, steps per second: 237, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 518984/1000000: episode: 1007, duration: 1.282s, episode steps: 303, steps per second: 236, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 519313/1000000: episode: 1008, duration: 1.403s, episode steps: 329, steps per second: 235, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 519890/1000000: episode: 1009, duration: 2.430s, episode steps: 577, steps per second: 237, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.340 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 520368/1000000: episode: 1010, duration: 2.008s, episode steps: 478, steps per second: 238, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 520856/1000000: episode: 1011, duration: 2.049s, episode steps: 488, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 521430/1000000: episode: 1012, duration: 2.395s, episode steps: 574, steps per second: 240, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 521851/1000000: episode: 1013, duration: 1.761s, episode steps: 421, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 522399/1000000: episode: 1014, duration: 2.309s, episode steps: 548, steps per second: 237, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 523288/1000000: episode: 1015, duration: 3.709s, episode steps: 889, steps per second: 240, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 523777/1000000: episode: 1016, duration: 2.042s, episode steps: 489, steps per second: 240, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 524473/1000000: episode: 1017, duration: 2.909s, episode steps: 696, steps per second: 239, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 525046/1000000: episode: 1018, duration: 2.395s, episode steps: 573, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 525549/1000000: episode: 1019, duration: 2.144s, episode steps: 503, steps per second: 235, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 526239/1000000: episode: 1020, duration: 2.919s, episode steps: 690, steps per second: 236, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 526926/1000000: episode: 1021, duration: 2.888s, episode steps: 687, steps per second: 238, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 527350/1000000: episode: 1022, duration: 1.786s, episode steps: 424, steps per second: 237, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 527643/1000000: episode: 1023, duration: 1.249s, episode steps: 293, steps per second: 235, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.700 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 528370/1000000: episode: 1024, duration: 3.067s, episode steps: 727, steps per second: 237, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 528774/1000000: episode: 1025, duration: 1.731s, episode steps: 404, steps per second: 233, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 529366/1000000: episode: 1026, duration: 2.502s, episode steps: 592, steps per second: 237, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 529790/1000000: episode: 1027, duration: 1.778s, episode steps: 424, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 530525/1000000: episode: 1028, duration: 3.067s, episode steps: 735, steps per second: 240, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 530808/1000000: episode: 1029, duration: 1.196s, episode steps: 283, steps per second: 237, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 531204/1000000: episode: 1030, duration: 1.670s, episode steps: 396, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 531907/1000000: episode: 1031, duration: 2.961s, episode steps: 703, steps per second: 237, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 532226/1000000: episode: 1032, duration: 1.342s, episode steps: 319, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 533074/1000000: episode: 1033, duration: 3.517s, episode steps: 848, steps per second: 241, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 533495/1000000: episode: 1034, duration: 1.778s, episode steps: 421, steps per second: 237, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 534616/1000000: episode: 1035, duration: 15.748s, episode steps: 1121, steps per second: 71, episode reward: 16.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: 0.014678, mae: 1.395102, mean_q: 1.703875, mean_eps: 0.987803
 534998/1000000: episode: 1036, duration: 14.522s, episode steps: 382, steps per

second: 26, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: 0.017282, mae: 1.432324, mean_q: 1.743095, mean_eps: 0.987378

535851/1000000: episode: 1037, duration: 32.280s, episode steps: 853, steps per second: 26, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: 0.015564, mae: 1.443812, mean_q: 1.757183, mean_eps: 0.986637

536845/1000000: episode: 1038, duration: 37.669s, episode steps: 994, steps per second: 26, episode reward: 29.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: 0.016298, mae: 1.425642, mean_q: 1.738195, mean_eps: 0.985528

537346/1000000: episode: 1039, duration: 18.969s, episode steps: 501, steps per second: 26, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.015298, mae: 1.431704, mean_q: 1.743601, mean_eps: 0.984630

537869/1000000: episode: 1040, duration: 19.826s, episode steps: 523, steps per second: 26, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.016492, mae: 1.415530, mean_q: 1.723198, mean_eps: 0.984016

538308/1000000: episode: 1041, duration: 16.736s, episode steps: 439, steps per second: 26, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.016373, mae: 1.402992, mean_q: 1.711271, mean_eps: 0.983440

538696/1000000: episode: 1042, duration: 14.838s, episode steps: 388, steps per second: 26, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: 0.017766, mae: 1.441556, mean_q: 1.754972, mean_eps: 0.982946

539336/1000000: episode: 1043, duration: 24.418s, episode steps: 640, steps per second: 26, episode reward: 20.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: 0.015370, mae: 1.418302, mean_q: 1.728319, mean_eps: 0.982329

539911/1000000: episode: 1044, duration: 21.867s, episode steps: 575, steps per second: 26, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.016549, mae: 1.449743, mean_q: 1.769251, mean_eps: 0.981599

540304/1000000: episode: 1045, duration: 15.108s, episode steps: 393, steps per second: 26, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.018320, mae: 1.451239, mean_q: 1.770701, mean_eps: 0.981018

540602/1000000: episode: 1046, duration: 11.386s, episode steps: 298, steps per second: 26, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.017957, mae: 1.467940, mean_q: 1.791941, mean_eps: 0.980603

541176/1000000: episode: 1047, duration: 21.840s, episode steps: 574, steps per second: 26, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.015953, mae: 1.463639, mean_q: 1.785175, mean_eps: 0.980080

541580/1000000: episode: 1048, duration: 15.506s, episode steps: 404, steps per

second: 26, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.015985, mae: 1.459099, mean_q: 1.777392, mean_eps: 0.979494

541981/1000000: episode: 1049, duration: 15.358s, episode steps: 401, steps per second: 26, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: 0.016267, mae: 1.459117, mean_q: 1.774010, mean_eps: 0.979010

542883/1000000: episode: 1050, duration: 34.169s, episode steps: 902, steps per second: 26, episode reward: 21.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.016454, mae: 1.446729, mean_q: 1.764812, mean_eps: 0.978227

543474/1000000: episode: 1051, duration: 22.504s, episode steps: 591, steps per second: 26, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.016292, mae: 1.453294, mean_q: 1.769485, mean_eps: 0.977332

543868/1000000: episode: 1052, duration: 15.018s, episode steps: 394, steps per second: 26, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: 0.016687, mae: 1.449377, mean_q: 1.764788, mean_eps: 0.976742

544369/1000000: episode: 1053, duration: 19.210s, episode steps: 501, steps per second: 26, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: 0.015556, mae: 1.437693, mean_q: 1.751459, mean_eps: 0.976204

544779/1000000: episode: 1054, duration: 15.560s, episode steps: 410, steps per second: 26, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: 0.018576, mae: 1.484040, mean_q: 1.809220, mean_eps: 0.975657

545394/1000000: episode: 1055, duration: 23.465s, episode steps: 615, steps per second: 26, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.015806, mae: 1.494597, mean_q: 1.825279, mean_eps: 0.975042

545768/1000000: episode: 1056, duration: 14.311s, episode steps: 374, steps per second: 26, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: 0.017080, mae: 1.493918, mean_q: 1.820808, mean_eps: 0.974450

546157/1000000: episode: 1057, duration: 14.990s, episode steps: 389, steps per second: 26, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.017704, mae: 1.510251, mean_q: 1.841935, mean_eps: 0.973991

546752/1000000: episode: 1058, duration: 22.778s, episode steps: 595, steps per second: 26, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.017170, mae: 1.484323, mean_q: 1.810859, mean_eps: 0.973401

547116/1000000: episode: 1059, duration: 13.987s, episode steps: 364, steps per second: 26, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.015628, mae: 1.455512, mean_q: 1.773775, mean_eps: 0.972827

547590/1000000: episode: 1060, duration: 18.246s, episode steps: 474, steps per

second: 26, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.017035, mae: 1.494430, mean_q: 1.822750, mean_eps: 0.972323

548077/1000000: episode: 1061, duration: 18.595s, episode steps: 487, steps per second: 26, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.345 [0.000, 5.000], loss: 0.016843, mae: 1.481871, mean_q: 1.807259, mean_eps: 0.971745

548544/1000000: episode: 1062, duration: 17.908s, episode steps: 467, steps per second: 26, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.017123, mae: 1.497730, mean_q: 1.830700, mean_eps: 0.971174

548844/1000000: episode: 1063, duration: 11.578s, episode steps: 300, steps per second: 26, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.740 [0.000, 5.000], loss: 0.015203, mae: 1.466486, mean_q: 1.789053, mean_eps: 0.970715

549245/1000000: episode: 1064, duration: 15.448s, episode steps: 401, steps per second: 26, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.015177, mae: 1.506973, mean_q: 1.839544, mean_eps: 0.970293

549589/1000000: episode: 1065, duration: 12.245s, episode steps: 344, steps per second: 28, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.017331, mae: 1.553174, mean_q: 1.894030, mean_eps: 0.969844

550042/1000000: episode: 1066, duration: 16.965s, episode steps: 453, steps per second: 27, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: 0.016867, mae: 1.549436, mean_q: 1.887707, mean_eps: 0.969366

550693/1000000: episode: 1067, duration: 24.949s, episode steps: 651, steps per second: 26, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.016105, mae: 1.523521, mean_q: 1.857081, mean_eps: 0.968704

551240/1000000: episode: 1068, duration: 21.009s, episode steps: 547, steps per second: 26, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.017501, mae: 1.511536, mean_q: 1.842386, mean_eps: 0.967986

551858/1000000: episode: 1069, duration: 23.753s, episode steps: 618, steps per second: 26, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.018340, mae: 1.531992, mean_q: 1.868425, mean_eps: 0.967288

552370/1000000: episode: 1070, duration: 19.492s, episode steps: 512, steps per second: 26, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.015115, mae: 1.535462, mean_q: 1.874328, mean_eps: 0.966609

552755/1000000: episode: 1071, duration: 14.764s, episode steps: 385, steps per second: 26, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.017509, mae: 1.552344, mean_q: 1.893887, mean_eps: 0.966071

553398/1000000: episode: 1072, duration: 24.579s, episode steps: 643, steps per

second: 26, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.585 [0.000, 5.000], loss: 0.017462, mae: 1.540805, mean_q: 1.882750,
mean_eps: 0.965454

554019/1000000: episode: 1073, duration: 23.784s, episode steps: 621, steps per
second: 26, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.593 [0.000, 5.000], loss: 0.016056, mae: 1.537163, mean_q: 1.872015,
mean_eps: 0.964696

554364/1000000: episode: 1074, duration: 13.333s, episode steps: 345, steps per
second: 26, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.478 [0.000, 5.000], loss: 0.017467, mae: 1.550378, mean_q: 1.889848,
mean_eps: 0.964118

554653/1000000: episode: 1075, duration: 11.213s, episode steps: 289, steps per
second: 26, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean
action: 2.505 [0.000, 5.000], loss: 0.018483, mae: 1.569019, mean_q: 1.913539,
mean_eps: 0.963736

555192/1000000: episode: 1076, duration: 20.736s, episode steps: 539, steps per
second: 26, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.499 [0.000, 5.000], loss: 0.017178, mae: 1.574233, mean_q: 1.918507,
mean_eps: 0.963239

555472/1000000: episode: 1077, duration: 10.859s, episode steps: 280, steps per
second: 26, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.479 [0.000, 5.000], loss: 0.018288, mae: 1.589232, mean_q: 1.937294,
mean_eps: 0.962750

555864/1000000: episode: 1078, duration: 15.138s, episode steps: 392, steps per
second: 26, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.431 [0.000, 5.000], loss: 0.018798, mae: 1.541731, mean_q: 1.877899,
mean_eps: 0.962346

556358/1000000: episode: 1079, duration: 19.054s, episode steps: 494, steps per
second: 26, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.504 [0.000, 5.000], loss: 0.018262, mae: 1.573637, mean_q: 1.918902,
mean_eps: 0.961814

556750/1000000: episode: 1080, duration: 15.056s, episode steps: 392, steps per
second: 26, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.452 [0.000, 5.000], loss: 0.017591, mae: 1.555827, mean_q: 1.894635,
mean_eps: 0.961281

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

557294/1000000: episode: 1081, duration: 12.635s, episode steps: 544, steps per
second: 43, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.629 [0.000, 5.000], loss: 0.020524, mae: 1.563037, mean_q: 1.905726,
mean_eps: 0.960863

557746/1000000: episode: 1082, duration: 1.910s, episode steps: 452, steps per
second: 237, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean
action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

558263/1000000: episode: 1083, duration: 2.166s, episode steps: 517, steps per
second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 558543/1000000: episode: 1084, duration: 1.183s, episode steps: 280, steps per second: 237, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.804 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 558841/1000000: episode: 1085, duration: 1.259s, episode steps: 298, steps per second: 237, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 559137/1000000: episode: 1086, duration: 1.254s, episode steps: 296, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 559944/1000000: episode: 1087, duration: 3.383s, episode steps: 807, steps per second: 239, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 560378/1000000: episode: 1088, duration: 1.827s, episode steps: 434, steps per second: 238, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 560888/1000000: episode: 1089, duration: 2.144s, episode steps: 510, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 561179/1000000: episode: 1090, duration: 1.223s, episode steps: 291, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 561676/1000000: episode: 1091, duration: 2.077s, episode steps: 497, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 562768/1000000: episode: 1092, duration: 4.498s, episode steps: 1092, steps per second: 243, episode reward: 30.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 563176/1000000: episode: 1093, duration: 1.750s, episode steps: 408, steps per second: 233, episode reward: 1.000, mean reward: 0.002 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 563667/1000000: episode: 1094, duration: 2.125s, episode steps: 491, steps per second: 231, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.664 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 564077/1000000: episode: 1095, duration: 1.733s, episode steps: 410, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 564695/1000000: episode: 1096, duration: 2.611s, episode steps: 618, steps per second: 237, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 565204/1000000: episode: 1097, duration: 2.169s, episode steps: 509, steps per second: 235, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 565669/1000000: episode: 1098, duration: 1.970s, episode steps: 465, steps per second: 236, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 565990/1000000: episode: 1099, duration: 1.366s, episode steps: 321, steps per second: 235, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean

action: 2.343 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 566318/1000000: episode: 1100, duration: 1.400s, episode steps: 328, steps per second: 234, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 566793/1000000: episode: 1101, duration: 2.027s, episode steps: 475, steps per second: 234, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 567183/1000000: episode: 1102, duration: 1.642s, episode steps: 390, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 567489/1000000: episode: 1103, duration: 1.292s, episode steps: 306, steps per second: 237, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 568053/1000000: episode: 1104, duration: 2.367s, episode steps: 564, steps per second: 238, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 568622/1000000: episode: 1105, duration: 2.388s, episode steps: 569, steps per second: 238, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 569092/1000000: episode: 1106, duration: 1.986s, episode steps: 470, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.291 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 569545/1000000: episode: 1107, duration: 1.913s, episode steps: 453, steps per second: 237, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 570232/1000000: episode: 1108, duration: 2.879s, episode steps: 687, steps per second: 239, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 570602/1000000: episode: 1109, duration: 1.555s, episode steps: 370, steps per second: 238, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 571230/1000000: episode: 1110, duration: 2.635s, episode steps: 628, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 571527/1000000: episode: 1111, duration: 1.250s, episode steps: 297, steps per second: 238, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 572084/1000000: episode: 1112, duration: 2.345s, episode steps: 557, steps per second: 238, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 572468/1000000: episode: 1113, duration: 1.622s, episode steps: 384, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 573436/1000000: episode: 1114, duration: 4.042s, episode steps: 968, steps per second: 239, episode reward: 16.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 573881/1000000: episode: 1115, duration: 1.875s, episode steps: 445, steps per second: 237, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 574401/1000000: episode: 1116, duration: 2.204s, episode steps: 520, steps per second: 236, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 574771/1000000: episode: 1117, duration: 1.570s, episode steps: 370, steps per second: 236, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 575308/1000000: episode: 1118, duration: 2.271s, episode steps: 537, steps per second: 236, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 575763/1000000: episode: 1119, duration: 1.918s, episode steps: 455, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 576170/1000000: episode: 1120, duration: 1.709s, episode steps: 407, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.686 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

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 spartial/version_1.0/pesos/dqn_v_1.0_weights_1120_episodes.h5f

576619/1000000: episode: 1121, duration: 1.963s, episode steps: 449, steps per second: 229, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 577030/1000000: episode: 1122, duration: 1.726s, episode steps: 411, steps per second: 238, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 577720/1000000: episode: 1123, duration: 2.871s, episode steps: 690, steps per second: 240, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 578262/1000000: episode: 1124, duration: 2.273s, episode steps: 542, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 578806/1000000: episode: 1125, duration: 2.319s, episode steps: 544, steps per second: 235, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 579393/1000000: episode: 1126, duration: 2.463s, episode steps: 587, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 579954/1000000: episode: 1127, duration: 2.372s, episode steps: 561, steps per second: 236, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 580609/1000000: episode: 1128, duration: 2.719s, episode steps: 655, steps per second: 241, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 581133/1000000: episode: 1129, duration: 2.207s, episode steps: 524, steps per second: 237, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 581469/1000000: episode: 1130, duration: 1.438s, episode steps: 336, steps per

second: 234, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

582283/1000000: episode: 1131, duration: 3.396s, episode steps: 814, steps per second: 240, episode reward: 9.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

582582/1000000: episode: 1132, duration: 1.252s, episode steps: 299, steps per second: 239, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

583054/1000000: episode: 1133, duration: 1.969s, episode steps: 472, steps per second: 240, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

583619/1000000: episode: 1134, duration: 2.363s, episode steps: 565, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

584555/1000000: episode: 1135, duration: 3.923s, episode steps: 936, steps per second: 239, episode reward: 19.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

585108/1000000: episode: 1136, duration: 2.307s, episode steps: 553, steps per second: 240, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

585392/1000000: episode: 1137, duration: 1.196s, episode steps: 284, steps per second: 238, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

585701/1000000: episode: 1138, duration: 1.308s, episode steps: 309, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

586593/1000000: episode: 1139, duration: 3.720s, episode steps: 892, steps per second: 240, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

586987/1000000: episode: 1140, duration: 1.674s, episode steps: 394, steps per second: 235, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

587470/1000000: episode: 1141, duration: 2.030s, episode steps: 483, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

588313/1000000: episode: 1142, duration: 3.512s, episode steps: 843, steps per second: 240, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

589191/1000000: episode: 1143, duration: 3.635s, episode steps: 878, steps per second: 242, episode reward: 24.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

589818/1000000: episode: 1144, duration: 2.625s, episode steps: 627, steps per second: 239, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

590233/1000000: episode: 1145, duration: 1.771s, episode steps: 415, steps per second: 234, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

590700/1000000: episode: 1146, duration: 1.976s, episode steps: 467, steps per

second: 236, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

591082/1000000: episode: 1147, duration: 1.605s, episode steps: 382, steps per second: 238, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

591376/1000000: episode: 1148, duration: 1.236s, episode steps: 294, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

591891/1000000: episode: 1149, duration: 2.157s, episode steps: 515, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

592272/1000000: episode: 1150, duration: 1.592s, episode steps: 381, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

592807/1000000: episode: 1151, duration: 2.236s, episode steps: 535, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

593554/1000000: episode: 1152, duration: 3.126s, episode steps: 747, steps per second: 239, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

593918/1000000: episode: 1153, duration: 1.523s, episode steps: 364, steps per second: 239, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

594426/1000000: episode: 1154, duration: 2.122s, episode steps: 508, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

594792/1000000: episode: 1155, duration: 1.539s, episode steps: 366, steps per second: 238, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

595369/1000000: episode: 1156, duration: 2.419s, episode steps: 577, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

596043/1000000: episode: 1157, duration: 2.813s, episode steps: 674, steps per second: 240, episode reward: 7.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

596951/1000000: episode: 1158, duration: 3.781s, episode steps: 908, steps per second: 240, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

597287/1000000: episode: 1159, duration: 1.404s, episode steps: 336, steps per second: 239, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

597903/1000000: episode: 1160, duration: 2.580s, episode steps: 616, steps per second: 239, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

598373/1000000: episode: 1161, duration: 1.987s, episode steps: 470, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

598789/1000000: episode: 1162, duration: 1.758s, episode steps: 416, steps per

second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

599253/1000000: episode: 1163, duration: 1.957s, episode steps: 464, steps per second: 237, episode reward: 3.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

599894/1000000: episode: 1164, duration: 5.065s, episode steps: 641, steps per second: 127, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.024416, mae: 1.601154, mean_q: 1.951313, mean_eps: 0.987957

600566/1000000: episode: 1165, duration: 26.298s, episode steps: 672, steps per second: 26, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: 0.019378, mae: 1.616165, mean_q: 1.970883, mean_eps: 0.987513

600972/1000000: episode: 1166, duration: 15.910s, episode steps: 406, steps per second: 26, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.017784, mae: 1.618595, mean_q: 1.977463, mean_eps: 0.986867

601369/1000000: episode: 1167, duration: 15.666s, episode steps: 397, steps per second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.019270, mae: 1.647097, mean_q: 2.004945, mean_eps: 0.986385

601790/1000000: episode: 1168, duration: 16.430s, episode steps: 421, steps per second: 26, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.018869, mae: 1.630652, mean_q: 1.990085, mean_eps: 0.985893

602583/1000000: episode: 1169, duration: 29.280s, episode steps: 793, steps per second: 27, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: 0.018178, mae: 1.608438, mean_q: 1.961805, mean_eps: 0.985166

603362/1000000: episode: 1170, duration: 30.392s, episode steps: 779, steps per second: 26, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.019030, mae: 1.616311, mean_q: 1.969662, mean_eps: 0.984222

603656/1000000: episode: 1171, duration: 11.616s, episode steps: 294, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.017033, mae: 1.631505, mean_q: 1.988170, mean_eps: 0.983579

604146/1000000: episode: 1172, duration: 19.293s, episode steps: 490, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.017168, mae: 1.646673, mean_q: 2.008912, mean_eps: 0.983109

604705/1000000: episode: 1173, duration: 21.898s, episode steps: 559, steps per second: 26, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: 0.018036, mae: 1.645945, mean_q: 2.008974, mean_eps: 0.982478

605005/1000000: episode: 1174, duration: 11.642s, episode steps: 300, steps per second: 26, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: 0.017614, mae: 1.659312, mean_q: 2.025346,

mean_eps: 0.981962

605469/1000000: episode: 1175, duration: 18.088s, episode steps: 464, steps per second: 26, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: 0.016370, mae: 1.650584, mean_q: 2.012924, mean_eps: 0.981503

606011/1000000: episode: 1176, duration: 21.149s, episode steps: 542, steps per second: 26, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.018244, mae: 1.663480, mean_q: 2.026426, mean_eps: 0.980901

606701/1000000: episode: 1177, duration: 27.141s, episode steps: 690, steps per second: 25, episode reward: 19.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.017813, mae: 1.646852, mean_q: 2.004410, mean_eps: 0.980162

607301/1000000: episode: 1178, duration: 23.646s, episode steps: 600, steps per second: 25, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.019799, mae: 1.661208, mean_q: 2.021462, mean_eps: 0.979386

607815/1000000: episode: 1179, duration: 20.024s, episode steps: 514, steps per second: 26, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.360 [0.000, 5.000], loss: 0.018570, mae: 1.638673, mean_q: 1.996894, mean_eps: 0.978719

608283/1000000: episode: 1180, duration: 18.408s, episode steps: 468, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.018257, mae: 1.662627, mean_q: 2.024540, mean_eps: 0.978131

608633/1000000: episode: 1181, duration: 13.830s, episode steps: 350, steps per second: 25, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.021494, mae: 1.678988, mean_q: 2.041995, mean_eps: 0.977639

609113/1000000: episode: 1182, duration: 18.864s, episode steps: 480, steps per second: 25, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.017275, mae: 1.666488, mean_q: 2.031256, mean_eps: 0.977140

609490/1000000: episode: 1183, duration: 14.851s, episode steps: 377, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.023737, mae: 1.626584, mean_q: 1.980315, mean_eps: 0.976626

609789/1000000: episode: 1184, duration: 11.796s, episode steps: 299, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.656 [0.000, 5.000], loss: 0.018044, mae: 1.669108, mean_q: 2.036740, mean_eps: 0.976221

610532/1000000: episode: 1185, duration: 29.223s, episode steps: 743, steps per second: 25, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: 0.021123, mae: 1.699793, mean_q: 2.068971, mean_eps: 0.975597

611158/1000000: episode: 1186, duration: 24.607s, episode steps: 626, steps per second: 25, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: 0.020330, mae: 1.691789, mean_q: 2.060405,

mean_eps: 0.974776

611713/1000000: episode: 1187, duration: 21.853s, episode steps: 555, steps per second: 25, episode reward: 15.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: 0.019635, mae: 1.684879, mean_q: 2.050169, mean_eps: 0.974066

612235/1000000: episode: 1188, duration: 20.452s, episode steps: 522, steps per second: 26, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: 0.020759, mae: 1.703461, mean_q: 2.072794, mean_eps: 0.973420

612727/1000000: episode: 1189, duration: 19.396s, episode steps: 492, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.019352, mae: 1.677808, mean_q: 2.040500, mean_eps: 0.972813

613450/1000000: episode: 1190, duration: 28.468s, episode steps: 723, steps per second: 25, episode reward: 22.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.711 [0.000, 5.000], loss: 0.017275, mae: 1.693177, mean_q: 2.062281, mean_eps: 0.972083

614280/1000000: episode: 1191, duration: 32.836s, episode steps: 830, steps per second: 25, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.020129, mae: 1.693984, mean_q: 2.062891, mean_eps: 0.971152

614583/1000000: episode: 1192, duration: 11.946s, episode steps: 303, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: 0.016993, mae: 1.682331, mean_q: 2.048980, mean_eps: 0.970473

615453/1000000: episode: 1193, duration: 34.180s, episode steps: 870, steps per second: 25, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.021145, mae: 1.730753, mean_q: 2.107591, mean_eps: 0.969767

615756/1000000: episode: 1194, duration: 11.963s, episode steps: 303, steps per second: 25, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: 0.021616, mae: 1.736085, mean_q: 2.117472, mean_eps: 0.969064

616562/1000000: episode: 1195, duration: 31.893s, episode steps: 806, steps per second: 25, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.020119, mae: 1.736057, mean_q: 2.112274, mean_eps: 0.968399

617165/1000000: episode: 1196, duration: 23.858s, episode steps: 603, steps per second: 25, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.020902, mae: 1.734342, mean_q: 2.112819, mean_eps: 0.967552

617697/1000000: episode: 1197, duration: 21.019s, episode steps: 532, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.019388, mae: 1.743590, mean_q: 2.121255, mean_eps: 0.966870

618000/1000000: episode: 1198, duration: 12.052s, episode steps: 303, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.019828, mae: 1.735819, mean_q: 2.112718,

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mean_eps: 0.966371
618410/1000000: episode: 1199, duration: 16.309s, episode steps: 410, steps per
second: 25, episode reward: 7.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.593 [0.000, 5.000], loss: 0.023897, mae: 1.723577, mean_q: 2.095805,
mean_eps: 0.965944
618709/1000000: episode: 1200, duration: 11.823s, episode steps: 299, steps per
second: 25, episode reward: 5.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.381 [0.000, 5.000], loss: 0.019692, mae: 1.719053, mean_q: 2.092216,
mean_eps: 0.965517
619169/1000000: episode: 1201, duration: 18.121s, episode steps: 460, steps per
second: 25, episode reward: 11.000, mean reward: 0.024 [ 0.000, 1.000], mean
action: 2.407 [0.000, 5.000], loss: 0.019983, mae: 1.729907, mean_q: 2.106901,
mean_eps: 0.965061
619572/1000000: episode: 1202, duration: 15.974s, episode steps: 403, steps per
second: 25, episode reward: 9.000, mean reward: 0.022 [ 0.000, 1.000], mean
action: 2.506 [0.000, 5.000], loss: 0.020173, mae: 1.708255, mean_q: 2.078666,
mean_eps: 0.964545
619952/1000000: episode: 1203, duration: 15.212s, episode steps: 380, steps per
second: 25, episode reward: 7.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.603 [0.000, 5.000], loss: 0.020428, mae: 1.769223, mean_q: 2.155339,
mean_eps: 0.964077
620535/1000000: episode: 1204, duration: 23.085s, episode steps: 583, steps per
second: 25, episode reward: 11.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.602 [0.000, 5.000], loss: 0.019658, mae: 1.755703, mean_q: 2.141312,
mean_eps: 0.963498
621095/1000000: episode: 1205, duration: 22.198s, episode steps: 560, steps per
second: 25, episode reward: 15.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.520 [0.000, 5.000], loss: 0.019946, mae: 1.769493, mean_q: 2.157650,
mean_eps: 0.962812
621379/1000000: episode: 1206, duration: 11.196s, episode steps: 284, steps per
second: 25, episode reward: 7.000, mean reward: 0.025 [ 0.000, 1.000], mean
action: 2.718 [0.000, 5.000], loss: 0.017189, mae: 1.758032, mean_q: 2.144694,
mean_eps: 0.962306
621876/1000000: episode: 1207, duration: 19.823s, episode steps: 497, steps per
second: 25, episode reward: 10.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.503 [0.000, 5.000], loss: 0.021988, mae: 1.744707, mean_q: 2.125184,
mean_eps: 0.961838
622347/1000000: episode: 1208, duration: 18.705s, episode steps: 471, steps per
second: 25, episode reward: 6.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.550 [0.000, 5.000], loss: 0.020260, mae: 1.720403, mean_q: 2.093658,
mean_eps: 0.961257

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

622729/1000000: episode: 1209, duration: 10.239s, episode steps: 382, steps per
second: 37, episode reward: 4.000, mean reward: 0.010 [ 0.000, 1.000], mean
action: 2.495 [0.000, 5.000], loss: 0.022354, mae: 1.752638, mean_q: 2.134804,

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mean_eps: 0.960827

623264/1000000: episode: 1210, duration: 2.243s, episode steps: 535, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

623716/1000000: episode: 1211, duration: 1.889s, episode steps: 452, steps per second: 239, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

623995/1000000: episode: 1212, duration: 1.171s, episode steps: 279, steps per second: 238, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

624450/1000000: episode: 1213, duration: 1.906s, episode steps: 455, steps per second: 239, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

625348/1000000: episode: 1214, duration: 3.777s, episode steps: 898, steps per second: 238, episode reward: 12.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

625936/1000000: episode: 1215, duration: 2.484s, episode steps: 588, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

626284/1000000: episode: 1216, duration: 1.474s, episode steps: 348, steps per second: 236, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

626767/1000000: episode: 1217, duration: 2.032s, episode steps: 483, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

627235/1000000: episode: 1218, duration: 1.977s, episode steps: 468, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

627754/1000000: episode: 1219, duration: 2.202s, episode steps: 519, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

628370/1000000: episode: 1220, duration: 2.605s, episode steps: 616, steps per second: 236, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

628749/1000000: episode: 1221, duration: 1.588s, episode steps: 379, steps per second: 239, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

629286/1000000: episode: 1222, duration: 2.243s, episode steps: 537, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.669 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

629897/1000000: episode: 1223, duration: 2.547s, episode steps: 611, steps per second: 240, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

630492/1000000: episode: 1224, duration: 2.477s, episode steps: 595, steps per second: 240, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

631107/1000000: episode: 1225, duration: 2.581s, episode steps: 615, steps per second: 238, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean

action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 631583/1000000: episode: 1226, duration: 2.013s, episode steps: 476, steps per second: 236, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 631979/1000000: episode: 1227, duration: 1.664s, episode steps: 396, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 632252/1000000: episode: 1228, duration: 1.157s, episode steps: 273, steps per second: 236, episode reward: 7.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 632560/1000000: episode: 1229, duration: 1.301s, episode steps: 308, steps per second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 633000/1000000: episode: 1230, duration: 1.847s, episode steps: 440, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 633602/1000000: episode: 1231, duration: 2.523s, episode steps: 602, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 634080/1000000: episode: 1232, duration: 2.017s, episode steps: 478, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 634671/1000000: episode: 1233, duration: 2.493s, episode steps: 591, steps per second: 237, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 635098/1000000: episode: 1234, duration: 1.787s, episode steps: 427, steps per second: 239, episode reward: 13.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 635428/1000000: episode: 1235, duration: 1.401s, episode steps: 330, steps per second: 235, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 636195/1000000: episode: 1236, duration: 3.175s, episode steps: 767, steps per second: 242, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 636576/1000000: episode: 1237, duration: 1.594s, episode steps: 381, steps per second: 239, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 637252/1000000: episode: 1238, duration: 2.853s, episode steps: 676, steps per second: 237, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 637767/1000000: episode: 1239, duration: 2.158s, episode steps: 515, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 638283/1000000: episode: 1240, duration: 2.169s, episode steps: 516, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 638672/1000000: episode: 1241, duration: 1.645s, episode steps: 389, steps per second: 237, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean

action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 639364/1000000: episode: 1242, duration: 2.888s, episode steps: 692, steps per second: 240, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 639806/1000000: episode: 1243, duration: 1.863s, episode steps: 442, steps per second: 237, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 640341/1000000: episode: 1244, duration: 2.278s, episode steps: 535, steps per second: 235, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 640942/1000000: episode: 1245, duration: 2.535s, episode steps: 601, steps per second: 237, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 641344/1000000: episode: 1246, duration: 1.688s, episode steps: 402, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 641910/1000000: episode: 1247, duration: 2.362s, episode steps: 566, steps per second: 240, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 642476/1000000: episode: 1248, duration: 2.372s, episode steps: 566, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 643605/1000000: episode: 1249, duration: 4.683s, episode steps: 1129, steps per second: 241, episode reward: 29.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 644351/1000000: episode: 1250, duration: 3.111s, episode steps: 746, steps per second: 240, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 644721/1000000: episode: 1251, duration: 1.554s, episode steps: 370, steps per second: 238, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 645322/1000000: episode: 1252, duration: 2.511s, episode steps: 601, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 645907/1000000: episode: 1253, duration: 2.463s, episode steps: 585, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 646323/1000000: episode: 1254, duration: 1.759s, episode steps: 416, steps per second: 237, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 646821/1000000: episode: 1255, duration: 2.076s, episode steps: 498, steps per second: 240, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 647455/1000000: episode: 1256, duration: 2.630s, episode steps: 634, steps per second: 241, episode reward: 18.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 647757/1000000: episode: 1257, duration: 1.273s, episode steps: 302, steps per second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean

action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 648049/1000000: episode: 1258, duration: 1.226s, episode steps: 292, steps per second: 238, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 648414/1000000: episode: 1259, duration: 1.528s, episode steps: 365, steps per second: 239, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 648869/1000000: episode: 1260, duration: 1.922s, episode steps: 455, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 649623/1000000: episode: 1261, duration: 3.182s, episode steps: 754, steps per second: 237, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 650231/1000000: episode: 1262, duration: 2.556s, episode steps: 608, steps per second: 238, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 650619/1000000: episode: 1263, duration: 1.639s, episode steps: 388, steps per second: 237, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 651196/1000000: episode: 1264, duration: 2.419s, episode steps: 577, steps per second: 239, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 651783/1000000: episode: 1265, duration: 2.464s, episode steps: 587, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 652311/1000000: episode: 1266, duration: 2.245s, episode steps: 528, steps per second: 235, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 652784/1000000: episode: 1267, duration: 1.979s, episode steps: 473, steps per second: 239, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 653408/1000000: episode: 1268, duration: 2.608s, episode steps: 624, steps per second: 239, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 653736/1000000: episode: 1269, duration: 1.373s, episode steps: 328, steps per second: 239, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 654263/1000000: episode: 1270, duration: 2.197s, episode steps: 527, steps per second: 240, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 654722/1000000: episode: 1271, duration: 1.930s, episode steps: 459, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 655255/1000000: episode: 1272, duration: 2.252s, episode steps: 533, steps per second: 237, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 655708/1000000: episode: 1273, duration: 1.908s, episode steps: 453, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean

action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 655992/1000000: episode: 1274, duration: 1.196s, episode steps: 284, steps per second: 237, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.669 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 656707/1000000: episode: 1275, duration: 3.010s, episode steps: 715, steps per second: 238, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 657165/1000000: episode: 1276, duration: 1.929s, episode steps: 458, steps per second: 237, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 657622/1000000: episode: 1277, duration: 1.920s, episode steps: 457, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 658258/1000000: episode: 1278, duration: 2.680s, episode steps: 636, steps per second: 237, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 658736/1000000: episode: 1279, duration: 2.021s, episode steps: 478, steps per second: 236, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 659356/1000000: episode: 1280, duration: 2.578s, episode steps: 620, steps per second: 240, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_1280_episodes.h5f

660125/1000000: episode: 1281, duration: 3.278s, episode steps: 769, steps per second: 235, episode reward: 14.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 660535/1000000: episode: 1282, duration: 1.732s, episode steps: 410, steps per second: 237, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 660807/1000000: episode: 1283, duration: 1.165s, episode steps: 272, steps per second: 234, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 661381/1000000: episode: 1284, duration: 2.421s, episode steps: 574, steps per second: 237, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 661977/1000000: episode: 1285, duration: 2.480s, episode steps: 596, steps per second: 240, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 662375/1000000: episode: 1286, duration: 1.662s, episode steps: 398, steps per second: 239, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 663297/1000000: episode: 1287, duration: 3.827s, episode steps: 922, steps per second: 241, episode reward: 19.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 663886/1000000: episode: 1288, duration: 2.513s, episode steps: 589, steps per

second: 234, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

664380/1000000: episode: 1289, duration: 2.081s, episode steps: 494, steps per second: 237, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

664878/1000000: episode: 1290, duration: 2.089s, episode steps: 498, steps per second: 238, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

665278/1000000: episode: 1291, duration: 1.680s, episode steps: 400, steps per second: 238, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

665925/1000000: episode: 1292, duration: 23.012s, episode steps: 647, steps per second: 28, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.021985, mae: 1.773559, mean_q: 2.161170, mean_eps: 0.987659

666219/1000000: episode: 1293, duration: 11.798s, episode steps: 294, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.025702, mae: 1.771112, mean_q: 2.159189, mean_eps: 0.987146

666594/1000000: episode: 1294, duration: 15.078s, episode steps: 375, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.018768, mae: 1.760006, mean_q: 2.142382, mean_eps: 0.986745

667057/1000000: episode: 1295, duration: 18.638s, episode steps: 463, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.019331, mae: 1.761422, mean_q: 2.142468, mean_eps: 0.986241

667336/1000000: episode: 1296, duration: 11.258s, episode steps: 279, steps per second: 25, episode reward: 8.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.019631, mae: 1.799645, mean_q: 2.186535, mean_eps: 0.985797

667715/1000000: episode: 1297, duration: 15.252s, episode steps: 379, steps per second: 25, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.019806, mae: 1.757724, mean_q: 2.136802, mean_eps: 0.985403

668212/1000000: episode: 1298, duration: 20.182s, episode steps: 497, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.020363, mae: 1.749080, mean_q: 2.126144, mean_eps: 0.984878

668736/1000000: episode: 1299, duration: 21.282s, episode steps: 524, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.018371, mae: 1.750707, mean_q: 2.128113, mean_eps: 0.984266

669123/1000000: episode: 1300, duration: 15.538s, episode steps: 387, steps per second: 25, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.020719, mae: 1.763400, mean_q: 2.150120, mean_eps: 0.983718

669565/1000000: episode: 1301, duration: 17.828s, episode steps: 442, steps per

second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: 0.019687, mae: 1.774462, mean_q: 2.162816, mean_eps: 0.983219

669859/1000000: episode: 1302, duration: 11.748s, episode steps: 294, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.022058, mae: 1.738207, mean_q: 2.113961, mean_eps: 0.982778

670250/1000000: episode: 1303, duration: 15.908s, episode steps: 391, steps per second: 25, episode reward: 11.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: 0.022734, mae: 1.777678, mean_q: 2.164806, mean_eps: 0.982367

670867/1000000: episode: 1304, duration: 24.783s, episode steps: 617, steps per second: 25, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.019714, mae: 1.760777, mean_q: 2.140916, mean_eps: 0.981762

671795/1000000: episode: 1305, duration: 37.322s, episode steps: 928, steps per second: 25, episode reward: 20.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.018702, mae: 1.777981, mean_q: 2.166160, mean_eps: 0.980836

672462/1000000: episode: 1306, duration: 27.031s, episode steps: 667, steps per second: 25, episode reward: 18.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: 0.022762, mae: 1.777151, mean_q: 2.160615, mean_eps: 0.979878

673226/1000000: episode: 1307, duration: 30.912s, episode steps: 764, steps per second: 25, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.021900, mae: 1.771327, mean_q: 2.156619, mean_eps: 0.979019

673585/1000000: episode: 1308, duration: 14.568s, episode steps: 359, steps per second: 25, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: 0.019772, mae: 1.762007, mean_q: 2.141286, mean_eps: 0.978345

674053/1000000: episode: 1309, duration: 18.806s, episode steps: 468, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: 0.020088, mae: 1.795437, mean_q: 2.184025, mean_eps: 0.977848

674567/1000000: episode: 1310, duration: 20.707s, episode steps: 514, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: 0.023338, mae: 1.752473, mean_q: 2.133418, mean_eps: 0.977260

674901/1000000: episode: 1311, duration: 13.718s, episode steps: 334, steps per second: 24, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: 0.021550, mae: 1.814717, mean_q: 2.209452, mean_eps: 0.976751

675411/1000000: episode: 1312, duration: 20.399s, episode steps: 510, steps per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: 0.019914, mae: 1.783086, mean_q: 2.167792, mean_eps: 0.976245

675791/1000000: episode: 1313, duration: 14.064s, episode steps: 380, steps per

second: 27, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.020277, mae: 1.823496, mean_q: 2.224538, mean_eps: 0.975712

676122/1000000: episode: 1314, duration: 13.327s, episode steps: 331, steps per second: 25, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.023109, mae: 1.800235, mean_q: 2.194691, mean_eps: 0.975285

676547/1000000: episode: 1315, duration: 17.131s, episode steps: 425, steps per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.021367, mae: 1.813600, mean_q: 2.208649, mean_eps: 0.974831

677249/1000000: episode: 1316, duration: 28.456s, episode steps: 702, steps per second: 25, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: 0.019987, mae: 1.802758, mean_q: 2.196528, mean_eps: 0.974154

677575/1000000: episode: 1317, duration: 13.109s, episode steps: 326, steps per second: 25, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: 0.021898, mae: 1.818070, mean_q: 2.215918, mean_eps: 0.973538

677856/1000000: episode: 1318, duration: 11.557s, episode steps: 281, steps per second: 24, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.022202, mae: 1.811138, mean_q: 2.210043, mean_eps: 0.973175

678150/1000000: episode: 1319, duration: 11.990s, episode steps: 294, steps per second: 25, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.018229, mae: 1.844140, mean_q: 2.249453, mean_eps: 0.972830

678536/1000000: episode: 1320, duration: 15.632s, episode steps: 386, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.020894, mae: 1.810652, mean_q: 2.207972, mean_eps: 0.972422

678897/1000000: episode: 1321, duration: 14.686s, episode steps: 361, steps per second: 25, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: 0.022609, mae: 1.835177, mean_q: 2.235125, mean_eps: 0.971973

679581/1000000: episode: 1322, duration: 27.844s, episode steps: 684, steps per second: 25, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.629 [0.000, 5.000], loss: 0.019109, mae: 1.793290, mean_q: 2.186195, mean_eps: 0.971344

679884/1000000: episode: 1323, duration: 12.315s, episode steps: 303, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: 0.018830, mae: 1.830560, mean_q: 2.229567, mean_eps: 0.970754

680760/1000000: episode: 1324, duration: 34.397s, episode steps: 876, steps per second: 25, episode reward: 14.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.021085, mae: 1.834241, mean_q: 2.235387, mean_eps: 0.970048

681308/1000000: episode: 1325, duration: 22.045s, episode steps: 548, steps per

second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.022134, mae: 1.859553, mean_q: 2.269789, mean_eps: 0.969194

681810/1000000: episode: 1326, duration: 20.380s, episode steps: 502, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.017288, mae: 1.833187, mean_q: 2.236612, mean_eps: 0.968562

682106/1000000: episode: 1327, duration: 12.033s, episode steps: 296, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.018477, mae: 1.836723, mean_q: 2.240998, mean_eps: 0.968082

682592/1000000: episode: 1328, duration: 19.843s, episode steps: 486, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.019130, mae: 1.837999, mean_q: 2.242217, mean_eps: 0.967614

682975/1000000: episode: 1329, duration: 15.562s, episode steps: 383, steps per second: 25, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: 0.022471, mae: 1.900856, mean_q: 2.321212, mean_eps: 0.967094

683354/1000000: episode: 1330, duration: 15.387s, episode steps: 379, steps per second: 25, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: 0.024634, mae: 1.871788, mean_q: 2.285355, mean_eps: 0.966635

683657/1000000: episode: 1331, duration: 12.297s, episode steps: 303, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.021488, mae: 1.857116, mean_q: 2.271732, mean_eps: 0.966225

684135/1000000: episode: 1332, duration: 19.372s, episode steps: 478, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.020326, mae: 1.858422, mean_q: 2.266585, mean_eps: 0.965757

684789/1000000: episode: 1333, duration: 26.655s, episode steps: 654, steps per second: 25, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.021900, mae: 1.863501, mean_q: 2.274429, mean_eps: 0.965078

685263/1000000: episode: 1334, duration: 19.267s, episode steps: 474, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: 0.021012, mae: 1.862915, mean_q: 2.269800, mean_eps: 0.964401

686357/1000000: episode: 1335, duration: 42.885s, episode steps: 1094, steps per second: 26, episode reward: 22.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.020324, mae: 1.912205, mean_q: 2.333660, mean_eps: 0.963460

687089/1000000: episode: 1336, duration: 29.690s, episode steps: 732, steps per second: 25, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: 0.022701, mae: 1.907457, mean_q: 2.330005, mean_eps: 0.962363

687347/1000000: episode: 1337, duration: 10.465s, episode steps: 258, steps per

second: 25, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.415 [0.000, 5.000], loss: 0.017572, mae: 1.925241, mean_q: 2.354147,
mean_eps: 0.961770

687849/1000000: episode: 1338, duration: 20.558s, episode steps: 502, steps per
second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.468 [0.000, 5.000], loss: 0.023039, mae: 1.919506, mean_q: 2.341436,
mean_eps: 0.961314

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

688304/1000000: episode: 1339, duration: 12.047s, episode steps: 455, steps per
second: 38, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.495 [0.000, 5.000], loss: 0.019081, mae: 1.901048, mean_q: 2.315076,
mean_eps: 0.960846

688688/1000000: episode: 1340, duration: 1.620s, episode steps: 384, steps per
second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

689001/1000000: episode: 1341, duration: 1.319s, episode steps: 313, steps per
second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

689291/1000000: episode: 1342, duration: 1.227s, episode steps: 290, steps per
second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

689719/1000000: episode: 1343, duration: 1.797s, episode steps: 428, steps per
second: 238, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean
action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

690120/1000000: episode: 1344, duration: 1.724s, episode steps: 401, steps per
second: 233, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

690713/1000000: episode: 1345, duration: 2.493s, episode steps: 593, steps per
second: 238, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean
action: 2.307 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

691289/1000000: episode: 1346, duration: 2.438s, episode steps: 576, steps per
second: 236, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

691938/1000000: episode: 1347, duration: 2.708s, episode steps: 649, steps per
second: 240, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

692490/1000000: episode: 1348, duration: 2.309s, episode steps: 552, steps per
second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

693361/1000000: episode: 1349, duration: 3.667s, episode steps: 871, steps per
second: 238, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

693859/1000000: episode: 1350, duration: 2.091s, episode steps: 498, steps per
second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.317 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

694218/1000000: episode: 1351, duration: 1.507s, episode steps: 359, steps per second: 238, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.694 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

694593/1000000: episode: 1352, duration: 1.590s, episode steps: 375, steps per second: 236, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

695517/1000000: episode: 1353, duration: 3.867s, episode steps: 924, steps per second: 239, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

695883/1000000: episode: 1354, duration: 1.564s, episode steps: 366, steps per second: 234, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

696386/1000000: episode: 1355, duration: 2.145s, episode steps: 503, steps per second: 235, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

697148/1000000: episode: 1356, duration: 3.217s, episode steps: 762, steps per second: 237, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

697740/1000000: episode: 1357, duration: 2.499s, episode steps: 592, steps per second: 237, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

698181/1000000: episode: 1358, duration: 1.871s, episode steps: 441, steps per second: 236, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

698525/1000000: episode: 1359, duration: 1.439s, episode steps: 344, steps per second: 239, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

699027/1000000: episode: 1360, duration: 2.115s, episode steps: 502, steps per second: 237, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

699399/1000000: episode: 1361, duration: 1.581s, episode steps: 372, steps per second: 235, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

699771/1000000: episode: 1362, duration: 1.568s, episode steps: 372, steps per second: 237, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

700081/1000000: episode: 1363, duration: 1.302s, episode steps: 310, steps per second: 238, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

700512/1000000: episode: 1364, duration: 1.801s, episode steps: 431, steps per second: 239, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

701012/1000000: episode: 1365, duration: 2.117s, episode steps: 500, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

701526/1000000: episode: 1366, duration: 2.147s, episode steps: 514, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

701967/1000000: episode: 1367, duration: 1.857s, episode steps: 441, steps per second: 237, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

702379/1000000: episode: 1368, duration: 1.746s, episode steps: 412, steps per second: 236, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

703481/1000000: episode: 1369, duration: 4.555s, episode steps: 1102, steps per second: 242, episode reward: 21.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

703915/1000000: episode: 1370, duration: 1.820s, episode steps: 434, steps per second: 239, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

704380/1000000: episode: 1371, duration: 1.946s, episode steps: 465, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

704840/1000000: episode: 1372, duration: 1.928s, episode steps: 460, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

705501/1000000: episode: 1373, duration: 2.792s, episode steps: 661, steps per second: 237, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

706046/1000000: episode: 1374, duration: 2.286s, episode steps: 545, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

706619/1000000: episode: 1375, duration: 2.388s, episode steps: 573, steps per second: 240, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.658 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

707269/1000000: episode: 1376, duration: 2.716s, episode steps: 650, steps per second: 239, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

707899/1000000: episode: 1377, duration: 2.632s, episode steps: 630, steps per second: 239, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

708846/1000000: episode: 1378, duration: 3.992s, episode steps: 947, steps per second: 237, episode reward: 10.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

709376/1000000: episode: 1379, duration: 2.218s, episode steps: 530, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

710066/1000000: episode: 1380, duration: 2.865s, episode steps: 690, steps per second: 241, episode reward: 24.000, mean reward: 0.035 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

710871/1000000: episode: 1381, duration: 3.352s, episode steps: 805, steps per second: 240, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

711397/1000000: episode: 1382, duration: 2.256s, episode steps: 526, steps per second: 233, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

712210/1000000: episode: 1383, duration: 3.387s, episode steps: 813, steps per second: 240, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

712797/1000000: episode: 1384, duration: 2.451s, episode steps: 587, steps per second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

713132/1000000: episode: 1385, duration: 1.398s, episode steps: 335, steps per second: 240, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

713630/1000000: episode: 1386, duration: 2.074s, episode steps: 498, steps per second: 240, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

714174/1000000: episode: 1387, duration: 2.302s, episode steps: 544, steps per second: 236, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

714767/1000000: episode: 1388, duration: 2.507s, episode steps: 593, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

715063/1000000: episode: 1389, duration: 1.252s, episode steps: 296, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

715572/1000000: episode: 1390, duration: 2.139s, episode steps: 509, steps per second: 238, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

715971/1000000: episode: 1391, duration: 1.680s, episode steps: 399, steps per second: 237, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

716480/1000000: episode: 1392, duration: 2.142s, episode steps: 509, steps per second: 238, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

716942/1000000: episode: 1393, duration: 1.947s, episode steps: 462, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

717414/1000000: episode: 1394, duration: 1.999s, episode steps: 472, steps per second: 236, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

717816/1000000: episode: 1395, duration: 1.696s, episode steps: 402, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

718105/1000000: episode: 1396, duration: 1.225s, episode steps: 289, steps per second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

718633/1000000: episode: 1397, duration: 2.215s, episode steps: 528, steps per second: 238, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

719139/1000000: episode: 1398, duration: 2.118s, episode steps: 506, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

719689/1000000: episode: 1399, duration: 2.309s, episode steps: 550, steps per second: 238, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

720129/1000000: episode: 1400, duration: 1.879s, episode steps: 440, steps per second: 234, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.330 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

720424/1000000: episode: 1401, duration: 1.260s, episode steps: 295, steps per second: 234, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

720766/1000000: episode: 1402, duration: 1.467s, episode steps: 342, steps per second: 233, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

721140/1000000: episode: 1403, duration: 1.594s, episode steps: 374, steps per second: 235, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

721621/1000000: episode: 1404, duration: 2.037s, episode steps: 481, steps per second: 236, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

722098/1000000: episode: 1405, duration: 2.006s, episode steps: 477, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

722603/1000000: episode: 1406, duration: 2.115s, episode steps: 505, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

723438/1000000: episode: 1407, duration: 3.497s, episode steps: 835, steps per second: 239, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

723716/1000000: episode: 1408, duration: 1.178s, episode steps: 278, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

724472/1000000: episode: 1409, duration: 3.131s, episode steps: 756, steps per second: 241, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

725102/1000000: episode: 1410, duration: 2.642s, episode steps: 630, steps per second: 238, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

725796/1000000: episode: 1411, duration: 2.915s, episode steps: 694, steps per second: 238, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

726269/1000000: episode: 1412, duration: 2.036s, episode steps: 473, steps per second: 232, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

727063/1000000: episode: 1413, duration: 3.327s, episode steps: 794, steps per second: 239, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

727445/1000000: episode: 1414, duration: 1.605s, episode steps: 382, steps per second: 238, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

727860/1000000: episode: 1415, duration: 1.746s, episode steps: 415, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.737 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

728525/1000000: episode: 1416, duration: 2.776s, episode steps: 665, steps per second: 240, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

728878/1000000: episode: 1417, duration: 1.491s, episode steps: 353, steps per second: 237, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

729521/1000000: episode: 1418, duration: 2.740s, episode steps: 643, steps per second: 235, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

730047/1000000: episode: 1419, duration: 2.221s, episode steps: 526, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.380 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

730334/1000000: episode: 1420, duration: 1.218s, episode steps: 287, steps per second: 236, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

730935/1000000: episode: 1421, duration: 3.776s, episode steps: 601, steps per second: 159, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.026675, mae: 1.911555, mean_q: 2.338112, mean_eps: 0.987976

731309/1000000: episode: 1422, duration: 14.418s, episode steps: 374, steps per second: 26, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.021531, mae: 1.971487, mean_q: 2.410979, mean_eps: 0.987729

731765/1000000: episode: 1423, duration: 18.915s, episode steps: 456, steps per second: 24, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.704 [0.000, 5.000], loss: 0.024195, mae: 1.953096, mean_q: 2.380988, mean_eps: 0.987230

732316/1000000: episode: 1424, duration: 22.942s, episode steps: 551, steps per second: 24, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: 0.025340, mae: 1.945378, mean_q: 2.372325, mean_eps: 0.986627

732890/1000000: episode: 1425, duration: 23.784s, episode steps: 574, steps per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.023349, mae: 1.935257, mean_q: 2.360989, mean_eps: 0.985953

733348/1000000: episode: 1426, duration: 19.063s, episode steps: 458, steps per second: 24, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.019544, mae: 1.986221, mean_q: 2.427586, mean_eps: 0.985334

733821/1000000: episode: 1427, duration: 19.761s, episode steps: 473, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: 0.022275, mae: 1.963344, mean_q: 2.397719, mean_eps: 0.984774

734415/1000000: episode: 1428, duration: 24.603s, episode steps: 594, steps per second: 24, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.411 [0.000, 5.000], loss: 0.023017, mae: 1.935193, mean_q: 2.360362,
 mean_eps: 0.984134
 734735/1000000: episode: 1429, duration: 13.352s, episode steps: 320, steps per
 second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.413 [0.000, 5.000], loss: 0.027841, mae: 1.967213, mean_q: 2.399745,
 mean_eps: 0.983586
 735291/1000000: episode: 1430, duration: 23.193s, episode steps: 556, steps per
 second: 24, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean
 action: 2.482 [0.000, 5.000], loss: 0.025601, mae: 1.948078, mean_q: 2.373005,
 mean_eps: 0.983061
 735877/1000000: episode: 1431, duration: 24.361s, episode steps: 586, steps per
 second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.531 [0.000, 5.000], loss: 0.021579, mae: 1.959016, mean_q: 2.392092,
 mean_eps: 0.982374
 736336/1000000: episode: 1432, duration: 17.736s, episode steps: 459, steps per
 second: 26, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.403 [0.000, 5.000], loss: 0.023344, mae: 2.005738, mean_q: 2.451926,
 mean_eps: 0.981748
 736625/1000000: episode: 1433, duration: 12.087s, episode steps: 289, steps per
 second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.519 [0.000, 5.000], loss: 0.016604, mae: 1.999011, mean_q: 2.448030,
 mean_eps: 0.981299
 737107/1000000: episode: 1434, duration: 20.048s, episode steps: 482, steps per
 second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.500 [0.000, 5.000], loss: 0.025473, mae: 1.994311, mean_q: 2.435897,
 mean_eps: 0.980836
 737573/1000000: episode: 1435, duration: 19.442s, episode steps: 466, steps per
 second: 24, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.485 [0.000, 5.000], loss: 0.029972, mae: 2.006006, mean_q: 2.449860,
 mean_eps: 0.980267
 738178/1000000: episode: 1436, duration: 25.187s, episode steps: 605, steps per
 second: 24, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.572 [0.000, 5.000], loss: 0.024827, mae: 2.017822, mean_q: 2.462266,
 mean_eps: 0.979624
 738704/1000000: episode: 1437, duration: 22.075s, episode steps: 526, steps per
 second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.490 [0.000, 5.000], loss: 0.024407, mae: 2.000584, mean_q: 2.444228,
 mean_eps: 0.978947
 739701/1000000: episode: 1438, duration: 41.669s, episode steps: 997, steps per
 second: 24, episode reward: 21.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.478 [0.000, 5.000], loss: 0.023798, mae: 2.004152, mean_q: 2.452810,
 mean_eps: 0.978033
 740298/1000000: episode: 1439, duration: 24.960s, episode steps: 597, steps per
 second: 24, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.461 [0.000, 5.000], loss: 0.024164, mae: 2.012882, mean_q: 2.457908,
 mean_eps: 0.977075
 740573/1000000: episode: 1440, duration: 11.567s, episode steps: 275, steps per
 second: 24, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean

action: 2.611 [0.000, 5.000], loss: 0.025630, mae: 2.024004, mean_q: 2.470674,
mean_eps: 0.976552

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_1440_episodes.h5f

741053/1000000: episode: 1441, duration: 19.578s, episode steps: 480, steps per
second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.708 [0.000, 5.000], loss: 0.023926, mae: 2.031395, mean_q: 2.481680,
mean_eps: 0.976098

741632/1000000: episode: 1442, duration: 23.280s, episode steps: 579, steps per
second: 25, episode reward: 16.000, mean reward: 0.028 [0.000, 1.000], mean
action: 2.487 [0.000, 5.000], loss: 0.023483, mae: 2.030797, mean_q: 2.486972,
mean_eps: 0.975465

742052/1000000: episode: 1443, duration: 17.677s, episode steps: 420, steps per
second: 24, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.479 [0.000, 5.000], loss: 0.027595, mae: 2.010191, mean_q: 2.454641,
mean_eps: 0.974867

742545/1000000: episode: 1444, duration: 20.648s, episode steps: 493, steps per
second: 24, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean
action: 2.535 [0.000, 5.000], loss: 0.025923, mae: 2.055642, mean_q: 2.513905,
mean_eps: 0.974318

743038/1000000: episode: 1445, duration: 20.538s, episode steps: 493, steps per
second: 24, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.465 [0.000, 5.000], loss: 0.026032, mae: 2.070061, mean_q: 2.530527,
mean_eps: 0.973725

743516/1000000: episode: 1446, duration: 20.154s, episode steps: 478, steps per
second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.525 [0.000, 5.000], loss: 0.022740, mae: 2.031573, mean_q: 2.483167,
mean_eps: 0.973144

744034/1000000: episode: 1447, duration: 21.665s, episode steps: 518, steps per
second: 24, episode reward: 15.000, mean reward: 0.029 [0.000, 1.000], mean
action: 2.496 [0.000, 5.000], loss: 0.024563, mae: 2.022222, mean_q: 2.468764,
mean_eps: 0.972546

744411/1000000: episode: 1448, duration: 15.716s, episode steps: 377, steps per
second: 24, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.607 [0.000, 5.000], loss: 0.027388, mae: 2.033381, mean_q: 2.486873,
mean_eps: 0.972009

745014/1000000: episode: 1449, duration: 25.345s, episode steps: 603, steps per
second: 24, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.687 [0.000, 5.000], loss: 0.027099, mae: 2.020945, mean_q: 2.468495,
mean_eps: 0.971421

745536/1000000: episode: 1450, duration: 21.867s, episode steps: 522, steps per
second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.467 [0.000, 5.000], loss: 0.030412, mae: 2.043546, mean_q: 2.504597,
mean_eps: 0.970746

745841/1000000: episode: 1451, duration: 12.941s, episode steps: 305, steps per
second: 24, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.751 [0.000, 5.000], loss: 0.027348, mae: 2.020182, mean_q: 2.473627,
 mean_eps: 0.970250
 746336/1000000: episode: 1452, duration: 19.269s, episode steps: 495, steps per
 second: 26, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.531 [0.000, 5.000], loss: 0.029523, mae: 2.072002, mean_q: 2.533132,
 mean_eps: 0.969770
 746706/1000000: episode: 1453, duration: 15.462s, episode steps: 370, steps per
 second: 24, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.403 [0.000, 5.000], loss: 0.020642, mae: 2.060782, mean_q: 2.525744,
 mean_eps: 0.969251
 747236/1000000: episode: 1454, duration: 22.317s, episode steps: 530, steps per
 second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.485 [0.000, 5.000], loss: 0.026141, mae: 2.062969, mean_q: 2.524989,
 mean_eps: 0.968711
 747715/1000000: episode: 1455, duration: 20.221s, episode steps: 479, steps per
 second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.491 [0.000, 5.000], loss: 0.027472, mae: 2.067136, mean_q: 2.527253,
 mean_eps: 0.968106
 748011/1000000: episode: 1456, duration: 12.350s, episode steps: 296, steps per
 second: 24, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.429 [0.000, 5.000], loss: 0.025156, mae: 2.084516, mean_q: 2.548420,
 mean_eps: 0.967641
 748522/1000000: episode: 1457, duration: 21.368s, episode steps: 511, steps per
 second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.603 [0.000, 5.000], loss: 0.026955, mae: 2.037118, mean_q: 2.490762,
 mean_eps: 0.967156
 748925/1000000: episode: 1458, duration: 16.882s, episode steps: 403, steps per
 second: 24, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean
 action: 2.464 [0.000, 5.000], loss: 0.027072, mae: 2.049620, mean_q: 2.502182,
 mean_eps: 0.966606
 749259/1000000: episode: 1459, duration: 14.029s, episode steps: 334, steps per
 second: 24, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.473 [0.000, 5.000], loss: 0.021945, mae: 2.097097, mean_q: 2.562957,
 mean_eps: 0.966165
 750049/1000000: episode: 1460, duration: 33.294s, episode steps: 790, steps per
 second: 24, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.518 [0.000, 5.000], loss: 0.027070, mae: 2.055253, mean_q: 2.513026,
 mean_eps: 0.965490
 750505/1000000: episode: 1461, duration: 19.084s, episode steps: 456, steps per
 second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.640 [0.000, 5.000], loss: 0.026780, mae: 2.053508, mean_q: 2.508325,
 mean_eps: 0.964742
 750989/1000000: episode: 1462, duration: 20.016s, episode steps: 484, steps per
 second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.463 [0.000, 5.000], loss: 0.024046, mae: 2.062327, mean_q: 2.523030,
 mean_eps: 0.964178
 751430/1000000: episode: 1463, duration: 17.458s, episode steps: 441, steps per
 second: 25, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean

action: 2.544 [0.000, 5.000], loss: 0.025094, mae: 2.131703, mean_q: 2.606024, mean_eps: 0.963623

751797/1000000: episode: 1464, duration: 15.369s, episode steps: 367, steps per second: 24, episode reward: 11.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.029899, mae: 2.135877, mean_q: 2.608546, mean_eps: 0.963138

752399/1000000: episode: 1465, duration: 25.004s, episode steps: 602, steps per second: 24, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.025119, mae: 2.107172, mean_q: 2.579356, mean_eps: 0.962558

752791/1000000: episode: 1466, duration: 16.419s, episode steps: 392, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.022605, mae: 2.108976, mean_q: 2.576026, mean_eps: 0.961962

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

753665/1000000: episode: 1467, duration: 36.514s, episode steps: 874, steps per second: 24, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.024505, mae: 2.105299, mean_q: 2.570436, mean_eps: 0.961204

754317/1000000: episode: 1468, duration: 2.724s, episode steps: 652, steps per second: 239, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

754982/1000000: episode: 1469, duration: 2.780s, episode steps: 665, steps per second: 239, episode reward: 18.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

755513/1000000: episode: 1470, duration: 2.232s, episode steps: 531, steps per second: 238, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.652 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

755939/1000000: episode: 1471, duration: 1.781s, episode steps: 426, steps per second: 239, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

756232/1000000: episode: 1472, duration: 1.234s, episode steps: 293, steps per second: 237, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

756806/1000000: episode: 1473, duration: 2.418s, episode steps: 574, steps per second: 237, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

757622/1000000: episode: 1474, duration: 3.414s, episode steps: 816, steps per second: 239, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

758138/1000000: episode: 1475, duration: 2.181s, episode steps: 516, steps per second: 237, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

758790/1000000: episode: 1476, duration: 2.727s, episode steps: 652, steps per second: 239, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean

action: 2.356 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 759274/1000000: episode: 1477, duration: 2.035s, episode steps: 484, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 759775/1000000: episode: 1478, duration: 2.091s, episode steps: 501, steps per second: 240, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 760204/1000000: episode: 1479, duration: 1.806s, episode steps: 429, steps per second: 238, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 760685/1000000: episode: 1480, duration: 2.024s, episode steps: 481, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 761167/1000000: episode: 1481, duration: 2.055s, episode steps: 482, steps per second: 235, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.307 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 761572/1000000: episode: 1482, duration: 1.711s, episode steps: 405, steps per second: 237, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.368 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 761863/1000000: episode: 1483, duration: 1.221s, episode steps: 291, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.656 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 762697/1000000: episode: 1484, duration: 3.470s, episode steps: 834, steps per second: 240, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 763552/1000000: episode: 1485, duration: 3.560s, episode steps: 855, steps per second: 240, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 764414/1000000: episode: 1486, duration: 3.619s, episode steps: 862, steps per second: 238, episode reward: 10.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 764909/1000000: episode: 1487, duration: 2.079s, episode steps: 495, steps per second: 238, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.384 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 765375/1000000: episode: 1488, duration: 1.949s, episode steps: 466, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 766337/1000000: episode: 1489, duration: 4.005s, episode steps: 962, steps per second: 240, episode reward: 14.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 766843/1000000: episode: 1490, duration: 2.156s, episode steps: 506, steps per second: 235, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.666 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 767287/1000000: episode: 1491, duration: 1.883s, episode steps: 444, steps per second: 236, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 768380/1000000: episode: 1492, duration: 4.544s, episode steps: 1093, steps per second: 241, episode reward: 20.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 769222/1000000: episode: 1493, duration: 3.500s, episode steps: 842, steps per second: 241, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 769782/1000000: episode: 1494, duration: 2.350s, episode steps: 560, steps per second: 238, episode reward: 4.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 770189/1000000: episode: 1495, duration: 1.718s, episode steps: 407, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 770752/1000000: episode: 1496, duration: 2.358s, episode steps: 563, steps per second: 239, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 771080/1000000: episode: 1497, duration: 1.373s, episode steps: 328, steps per second: 239, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.345 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 771592/1000000: episode: 1498, duration: 2.147s, episode steps: 512, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 772299/1000000: episode: 1499, duration: 2.952s, episode steps: 707, steps per second: 239, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 772717/1000000: episode: 1500, duration: 1.755s, episode steps: 418, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 773165/1000000: episode: 1501, duration: 1.893s, episode steps: 448, steps per second: 237, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 773698/1000000: episode: 1502, duration: 2.243s, episode steps: 533, steps per second: 238, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 774195/1000000: episode: 1503, duration: 2.087s, episode steps: 497, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 774859/1000000: episode: 1504, duration: 2.771s, episode steps: 664, steps per second: 240, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 775273/1000000: episode: 1505, duration: 1.736s, episode steps: 414, steps per second: 238, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 775713/1000000: episode: 1506, duration: 1.860s, episode steps: 440, steps per second: 237, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 776258/1000000: episode: 1507, duration: 2.316s, episode steps: 545, steps per second: 235, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 777015/1000000: episode: 1508, duration: 3.164s, episode steps: 757, steps per second: 239, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean

action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 777314/1000000: episode: 1509, duration: 1.264s, episode steps: 299, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.314 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 777827/1000000: episode: 1510, duration: 2.157s, episode steps: 513, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 778710/1000000: episode: 1511, duration: 3.681s, episode steps: 883, steps per second: 240, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 779465/1000000: episode: 1512, duration: 3.160s, episode steps: 755, steps per second: 239, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 779820/1000000: episode: 1513, duration: 1.487s, episode steps: 355, steps per second: 239, episode reward: 11.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 780316/1000000: episode: 1514, duration: 2.063s, episode steps: 496, steps per second: 240, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.673 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 781047/1000000: episode: 1515, duration: 3.050s, episode steps: 731, steps per second: 240, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 781509/1000000: episode: 1516, duration: 1.945s, episode steps: 462, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 782367/1000000: episode: 1517, duration: 3.610s, episode steps: 858, steps per second: 238, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 782777/1000000: episode: 1518, duration: 1.715s, episode steps: 410, steps per second: 239, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 783451/1000000: episode: 1519, duration: 2.813s, episode steps: 674, steps per second: 240, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 783923/1000000: episode: 1520, duration: 1.967s, episode steps: 472, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 784407/1000000: episode: 1521, duration: 2.040s, episode steps: 484, steps per second: 237, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 784739/1000000: episode: 1522, duration: 1.409s, episode steps: 332, steps per second: 236, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 785096/1000000: episode: 1523, duration: 1.515s, episode steps: 357, steps per second: 236, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.653 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 785548/1000000: episode: 1524, duration: 1.944s, episode steps: 452, steps per second: 233, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean

action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 786188/1000000: episode: 1525, duration: 2.687s, episode steps: 640, steps per second: 238, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 786542/1000000: episode: 1526, duration: 1.502s, episode steps: 354, steps per second: 236, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.387 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 787040/1000000: episode: 1527, duration: 2.090s, episode steps: 498, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 787588/1000000: episode: 1528, duration: 2.291s, episode steps: 548, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 787891/1000000: episode: 1529, duration: 1.299s, episode steps: 303, steps per second: 233, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 788730/1000000: episode: 1530, duration: 3.549s, episode steps: 839, steps per second: 236, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 789216/1000000: episode: 1531, duration: 2.052s, episode steps: 486, steps per second: 237, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 789760/1000000: episode: 1532, duration: 2.310s, episode steps: 544, steps per second: 236, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 790303/1000000: episode: 1533, duration: 2.299s, episode steps: 543, steps per second: 236, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.355 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 790688/1000000: episode: 1534, duration: 1.634s, episode steps: 385, steps per second: 236, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 791148/1000000: episode: 1535, duration: 1.949s, episode steps: 460, steps per second: 236, episode reward: 1.000, mean reward: 0.002 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 791438/1000000: episode: 1536, duration: 1.240s, episode steps: 290, steps per second: 234, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.700 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 791728/1000000: episode: 1537, duration: 1.220s, episode steps: 290, steps per second: 238, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 792444/1000000: episode: 1538, duration: 2.979s, episode steps: 716, steps per second: 240, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 793006/1000000: episode: 1539, duration: 2.352s, episode steps: 562, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 793489/1000000: episode: 1540, duration: 2.014s, episode steps: 483, steps per second: 240, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean

action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 793962/1000000: episode: 1541, duration: 1.998s, episode steps: 473, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 794447/1000000: episode: 1542, duration: 2.066s, episode steps: 485, steps per second: 235, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 794839/1000000: episode: 1543, duration: 1.673s, episode steps: 392, steps per second: 234, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 795216/1000000: episode: 1544, duration: 1.595s, episode steps: 377, steps per second: 236, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 795918/1000000: episode: 1545, duration: 2.931s, episode steps: 702, steps per second: 240, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 796473/1000000: episode: 1546, duration: 3.912s, episode steps: 555, steps per second: 142, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.019435, mae: 2.058297, mean_q: 2.517077, mean_eps: 0.987974
 797020/1000000: episode: 1547, duration: 23.404s, episode steps: 547, steps per second: 23, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.024730, mae: 2.142101, mean_q: 2.621402, mean_eps: 0.987623
 797549/1000000: episode: 1548, duration: 22.703s, episode steps: 529, steps per second: 23, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: 0.026505, mae: 2.148517, mean_q: 2.625291, mean_eps: 0.986978
 797960/1000000: episode: 1549, duration: 17.599s, episode steps: 411, steps per second: 23, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: 0.023793, mae: 2.195541, mean_q: 2.678797, mean_eps: 0.986414
 798580/1000000: episode: 1550, duration: 26.708s, episode steps: 620, steps per second: 23, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.030000, mae: 2.184141, mean_q: 2.662192, mean_eps: 0.985797
 799638/1000000: episode: 1551, duration: 45.297s, episode steps: 1058, steps per second: 23, episode reward: 24.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.029845, mae: 2.171291, mean_q: 2.645115, mean_eps: 0.984789
 800277/1000000: episode: 1552, duration: 27.387s, episode steps: 639, steps per second: 23, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.024981, mae: 2.185291, mean_q: 2.659344, mean_eps: 0.983769
 800772/1000000: episode: 1553, duration: 21.221s, episode steps: 495, steps per second: 23, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: 0.022896, mae: 2.110136, mean_q: 2.571251, mean_eps: 0.983090

801193/1000000: episode: 1554, duration: 18.266s, episode steps: 421, steps per second: 23, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.025493, mae: 2.155092, mean_q: 2.621641, mean_eps: 0.982540

801919/1000000: episode: 1555, duration: 30.937s, episode steps: 726, steps per second: 23, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.027952, mae: 2.213725, mean_q: 2.691747, mean_eps: 0.981851

802657/1000000: episode: 1556, duration: 31.617s, episode steps: 738, steps per second: 23, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.027148, mae: 2.208131, mean_q: 2.685982, mean_eps: 0.980973

803169/1000000: episode: 1557, duration: 22.064s, episode steps: 512, steps per second: 23, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.029991, mae: 2.255019, mean_q: 2.747807, mean_eps: 0.980222

803550/1000000: episode: 1558, duration: 16.411s, episode steps: 381, steps per second: 23, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.031288, mae: 2.218128, mean_q: 2.696299, mean_eps: 0.979686

803913/1000000: episode: 1559, duration: 15.607s, episode steps: 363, steps per second: 23, episode reward: 0.000, mean reward: 0.000 [0.000, 0.000], mean action: 2.317 [0.000, 5.000], loss: 0.027293, mae: 2.226845, mean_q: 2.709932, mean_eps: 0.979240

804356/1000000: episode: 1560, duration: 19.045s, episode steps: 443, steps per second: 23, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.024597, mae: 2.201860, mean_q: 2.671333, mean_eps: 0.978758

804888/1000000: episode: 1561, duration: 23.060s, episode steps: 532, steps per second: 23, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.022872, mae: 2.221091, mean_q: 2.702163, mean_eps: 0.978174

805734/1000000: episode: 1562, duration: 36.279s, episode steps: 846, steps per second: 23, episode reward: 11.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.028876, mae: 2.225531, mean_q: 2.709767, mean_eps: 0.977346

806211/1000000: episode: 1563, duration: 20.490s, episode steps: 477, steps per second: 23, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: 0.027727, mae: 2.238731, mean_q: 2.722139, mean_eps: 0.976552

807196/1000000: episode: 1564, duration: 42.269s, episode steps: 985, steps per second: 23, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.028995, mae: 2.259203, mean_q: 2.745298, mean_eps: 0.975676

807926/1000000: episode: 1565, duration: 31.266s, episode steps: 730, steps per second: 23, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.028614, mae: 2.260135, mean_q: 2.750527, mean_eps: 0.974646

808461/1000000: episode: 1566, duration: 22.890s, episode steps: 535, steps per second: 23, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.024697, mae: 2.297113, mean_q: 2.793440, mean_eps: 0.973886

808766/1000000: episode: 1567, duration: 12.947s, episode steps: 305, steps per second: 24, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.028657, mae: 2.305921, mean_q: 2.801691, mean_eps: 0.973382

809310/1000000: episode: 1568, duration: 23.192s, episode steps: 544, steps per second: 23, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.030443, mae: 2.265300, mean_q: 2.748282, mean_eps: 0.972873

809872/1000000: episode: 1569, duration: 24.259s, episode steps: 562, steps per second: 23, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.029946, mae: 2.301506, mean_q: 2.796754, mean_eps: 0.972210

810354/1000000: episode: 1570, duration: 20.837s, episode steps: 482, steps per second: 23, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.026004, mae: 2.279176, mean_q: 2.773530, mean_eps: 0.971584

810939/1000000: episode: 1571, duration: 25.014s, episode steps: 585, steps per second: 23, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.027846, mae: 2.269686, mean_q: 2.762442, mean_eps: 0.970943

811656/1000000: episode: 1572, duration: 30.132s, episode steps: 717, steps per second: 24, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.028776, mae: 2.279190, mean_q: 2.768600, mean_eps: 0.970163

812210/1000000: episode: 1573, duration: 21.977s, episode steps: 554, steps per second: 25, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.026269, mae: 2.320575, mean_q: 2.819381, mean_eps: 0.969400

812898/1000000: episode: 1574, duration: 27.257s, episode steps: 688, steps per second: 25, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.719 [0.000, 5.000], loss: 0.028069, mae: 2.309786, mean_q: 2.807083, mean_eps: 0.968654

813513/1000000: episode: 1575, duration: 24.407s, episode steps: 615, steps per second: 25, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.027334, mae: 2.317968, mean_q: 2.816556, mean_eps: 0.967871

813978/1000000: episode: 1576, duration: 18.354s, episode steps: 465, steps per second: 25, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.028160, mae: 2.345346, mean_q: 2.851533, mean_eps: 0.967223

814466/1000000: episode: 1577, duration: 19.440s, episode steps: 488, steps per second: 25, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: 0.029664, mae: 2.338311, mean_q: 2.837607, mean_eps: 0.966652

814962/1000000: episode: 1578, duration: 19.673s, episode steps: 496, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.032060, mae: 2.326846, mean_q: 2.827454, mean_eps: 0.966062

815567/1000000: episode: 1579, duration: 23.977s, episode steps: 605, steps per second: 25, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: 0.030920, mae: 2.325264, mean_q: 2.823904, mean_eps: 0.965402

816134/1000000: episode: 1580, duration: 22.415s, episode steps: 567, steps per second: 25, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.027942, mae: 2.314409, mean_q: 2.810121, mean_eps: 0.964698

816580/1000000: episode: 1581, duration: 17.757s, episode steps: 446, steps per second: 25, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: 0.026107, mae: 2.337908, mean_q: 2.838602, mean_eps: 0.964091

816973/1000000: episode: 1582, duration: 15.635s, episode steps: 393, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: 0.032056, mae: 2.363448, mean_q: 2.870501, mean_eps: 0.963587

817297/1000000: episode: 1583, duration: 12.825s, episode steps: 324, steps per second: 25, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: 0.031060, mae: 2.321041, mean_q: 2.821017, mean_eps: 0.963155

817822/1000000: episode: 1584, duration: 20.792s, episode steps: 525, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.027789, mae: 2.350774, mean_q: 2.856920, mean_eps: 0.962646

818346/1000000: episode: 1585, duration: 20.744s, episode steps: 524, steps per second: 25, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: 0.031254, mae: 2.362607, mean_q: 2.871619, mean_eps: 0.962018

818647/1000000: episode: 1586, duration: 11.949s, episode steps: 301, steps per second: 25, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.651 [0.000, 5.000], loss: 0.023607, mae: 2.350559, mean_q: 2.858536, mean_eps: 0.961523

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

819283/1000000: episode: 1587, duration: 22.172s, episode steps: 636, steps per second: 29, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.032474, mae: 2.351406, mean_q: 2.855255, mean_eps: 0.961012

820088/1000000: episode: 1588, duration: 3.351s, episode steps: 805, steps per second: 240, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

820860/1000000: episode: 1589, duration: 3.209s, episode steps: 772, steps per

second: 241, episode reward: 21.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
821231/1000000: episode: 1590, duration: 1.577s, episode steps: 371, steps per second: 235, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
821635/1000000: episode: 1591, duration: 1.704s, episode steps: 404, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
822067/1000000: episode: 1592, duration: 1.816s, episode steps: 432, steps per second: 238, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
822567/1000000: episode: 1593, duration: 2.100s, episode steps: 500, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
823138/1000000: episode: 1594, duration: 2.376s, episode steps: 571, steps per second: 240, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
823822/1000000: episode: 1595, duration: 2.832s, episode steps: 684, steps per second: 242, episode reward: 23.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
824107/1000000: episode: 1596, duration: 1.202s, episode steps: 285, steps per second: 237, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
824540/1000000: episode: 1597, duration: 1.834s, episode steps: 433, steps per second: 236, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
824845/1000000: episode: 1598, duration: 1.290s, episode steps: 305, steps per second: 236, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
825357/1000000: episode: 1599, duration: 2.144s, episode steps: 512, steps per second: 239, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
825821/1000000: episode: 1600, duration: 1.949s, episode steps: 464, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_1600_episodes.h5f

826499/1000000: episode: 1601, duration: 2.916s, episode steps: 678, steps per second: 232, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
827327/1000000: episode: 1602, duration: 3.483s, episode steps: 828, steps per second: 238, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
827931/1000000: episode: 1603, duration: 2.506s, episode steps: 604, steps per second: 241, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

828313/1000000: episode: 1604, duration: 1.601s, episode steps: 382, steps per second: 239, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

828688/1000000: episode: 1605, duration: 1.567s, episode steps: 375, steps per second: 239, episode reward: 11.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

829340/1000000: episode: 1606, duration: 2.722s, episode steps: 652, steps per second: 240, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.325 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

830056/1000000: episode: 1607, duration: 3.257s, episode steps: 716, steps per second: 220, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

830546/1000000: episode: 1608, duration: 2.068s, episode steps: 490, steps per second: 237, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

830849/1000000: episode: 1609, duration: 1.281s, episode steps: 303, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

831253/1000000: episode: 1610, duration: 1.708s, episode steps: 404, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.319 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

831964/1000000: episode: 1611, duration: 2.956s, episode steps: 711, steps per second: 241, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

832554/1000000: episode: 1612, duration: 2.477s, episode steps: 590, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

833056/1000000: episode: 1613, duration: 2.121s, episode steps: 502, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

833500/1000000: episode: 1614, duration: 1.869s, episode steps: 444, steps per second: 238, episode reward: 11.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

834073/1000000: episode: 1615, duration: 2.394s, episode steps: 573, steps per second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

834820/1000000: episode: 1616, duration: 3.112s, episode steps: 747, steps per second: 240, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

835633/1000000: episode: 1617, duration: 3.398s, episode steps: 813, steps per second: 239, episode reward: 11.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

836134/1000000: episode: 1618, duration: 2.116s, episode steps: 501, steps per second: 237, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

836400/1000000: episode: 1619, duration: 1.126s, episode steps: 266, steps per second: 236, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

836859/1000000: episode: 1620, duration: 1.927s, episode steps: 459, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

837883/1000000: episode: 1621, duration: 4.229s, episode steps: 1024, steps per second: 242, episode reward: 18.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

838200/1000000: episode: 1622, duration: 1.323s, episode steps: 317, steps per second: 240, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

838699/1000000: episode: 1623, duration: 2.085s, episode steps: 499, steps per second: 239, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

839157/1000000: episode: 1624, duration: 1.931s, episode steps: 458, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.666 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

839738/1000000: episode: 1625, duration: 2.443s, episode steps: 581, steps per second: 238, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

840193/1000000: episode: 1626, duration: 1.906s, episode steps: 455, steps per second: 239, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

840989/1000000: episode: 1627, duration: 3.330s, episode steps: 796, steps per second: 239, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

841349/1000000: episode: 1628, duration: 1.518s, episode steps: 360, steps per second: 237, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

841638/1000000: episode: 1629, duration: 1.224s, episode steps: 289, steps per second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

842238/1000000: episode: 1630, duration: 2.543s, episode steps: 600, steps per second: 236, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

842595/1000000: episode: 1631, duration: 1.500s, episode steps: 357, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

843033/1000000: episode: 1632, duration: 1.841s, episode steps: 438, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

844088/1000000: episode: 1633, duration: 4.393s, episode steps: 1055, steps per second: 240, episode reward: 16.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

844540/1000000: episode: 1634, duration: 1.891s, episode steps: 452, steps per second: 239, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

844910/1000000: episode: 1635, duration: 1.562s, episode steps: 370, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

845524/1000000: episode: 1636, duration: 2.607s, episode steps: 614, steps per second: 236, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

845950/1000000: episode: 1637, duration: 1.777s, episode steps: 426, steps per second: 240, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

846315/1000000: episode: 1638, duration: 1.529s, episode steps: 365, steps per second: 239, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

846889/1000000: episode: 1639, duration: 2.380s, episode steps: 574, steps per second: 241, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

847315/1000000: episode: 1640, duration: 1.790s, episode steps: 426, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

847616/1000000: episode: 1641, duration: 1.267s, episode steps: 301, steps per second: 238, episode reward: 9.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

848253/1000000: episode: 1642, duration: 2.717s, episode steps: 637, steps per second: 234, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

848639/1000000: episode: 1643, duration: 1.626s, episode steps: 386, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

848944/1000000: episode: 1644, duration: 1.287s, episode steps: 305, steps per second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

849600/1000000: episode: 1645, duration: 2.742s, episode steps: 656, steps per second: 239, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

850059/1000000: episode: 1646, duration: 1.917s, episode steps: 459, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

850358/1000000: episode: 1647, duration: 1.265s, episode steps: 299, steps per second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

851117/1000000: episode: 1648, duration: 3.180s, episode steps: 759, steps per second: 239, episode reward: 24.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

851705/1000000: episode: 1649, duration: 2.484s, episode steps: 588, steps per second: 237, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

852196/1000000: episode: 1650, duration: 2.054s, episode steps: 491, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

852964/1000000: episode: 1651, duration: 3.197s, episode steps: 768, steps per second: 240, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

853281/1000000: episode: 1652, duration: 1.328s, episode steps: 317, steps per second: 239, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

853680/1000000: episode: 1653, duration: 1.677s, episode steps: 399, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

854211/1000000: episode: 1654, duration: 2.245s, episode steps: 531, steps per second: 237, episode reward: 5.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

854780/1000000: episode: 1655, duration: 2.389s, episode steps: 569, steps per second: 238, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

855104/1000000: episode: 1656, duration: 1.357s, episode steps: 324, steps per second: 239, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

855641/1000000: episode: 1657, duration: 2.215s, episode steps: 537, steps per second: 242, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

855974/1000000: episode: 1658, duration: 1.390s, episode steps: 333, steps per second: 240, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

856544/1000000: episode: 1659, duration: 2.388s, episode steps: 570, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

857125/1000000: episode: 1660, duration: 2.461s, episode steps: 581, steps per second: 236, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

857519/1000000: episode: 1661, duration: 1.659s, episode steps: 394, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

858176/1000000: episode: 1662, duration: 2.752s, episode steps: 657, steps per second: 239, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

858668/1000000: episode: 1663, duration: 2.060s, episode steps: 492, steps per second: 239, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

859414/1000000: episode: 1664, duration: 3.097s, episode steps: 746, steps per second: 241, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.626 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

860009/1000000: episode: 1665, duration: 2.492s, episode steps: 595, steps per second: 239, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

860365/1000000: episode: 1666, duration: 1.533s, episode steps: 356, steps per second: 232, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

860782/1000000: episode: 1667, duration: 1.743s, episode steps: 417, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

861275/1000000: episode: 1668, duration: 2.060s, episode steps: 493, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

861645/1000000: episode: 1669, duration: 1.550s, episode steps: 370, steps per second: 239, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

861928/1000000: episode: 1670, duration: 1.186s, episode steps: 283, steps per second: 239, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

862511/1000000: episode: 1671, duration: 21.703s, episode steps: 583, steps per second: 27, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.024638, mae: 2.367678, mean_q: 2.875622, mean_eps: 0.987674

862981/1000000: episode: 1672, duration: 18.656s, episode steps: 470, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.032940, mae: 2.352989, mean_q: 2.856249, mean_eps: 0.987066

863502/1000000: episode: 1673, duration: 20.728s, episode steps: 521, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.032174, mae: 2.371380, mean_q: 2.877318, mean_eps: 0.986471

863973/1000000: episode: 1674, duration: 18.841s, episode steps: 471, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.028499, mae: 2.359678, mean_q: 2.865886, mean_eps: 0.985876

864626/1000000: episode: 1675, duration: 26.182s, episode steps: 653, steps per second: 25, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.031403, mae: 2.379361, mean_q: 2.887573, mean_eps: 0.985202

865110/1000000: episode: 1676, duration: 19.366s, episode steps: 484, steps per second: 25, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.028006, mae: 2.359962, mean_q: 2.867978, mean_eps: 0.984520

865603/1000000: episode: 1677, duration: 19.578s, episode steps: 493, steps per second: 25, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.031894, mae: 2.368241, mean_q: 2.880613, mean_eps: 0.983934

866124/1000000: episode: 1678, duration: 20.836s, episode steps: 521, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.028923, mae: 2.370860, mean_q: 2.878645, mean_eps: 0.983327

866599/1000000: episode: 1679, duration: 18.925s, episode steps: 475, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: 0.030538, mae: 2.384559, mean_q: 2.899879, mean_eps: 0.982730

867103/1000000: episode: 1680, duration: 19.981s, episode steps: 504, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.030621, mae: 2.388435, mean_q: 2.903073,

mean_eps: 0.982142

867609/1000000: episode: 1681, duration: 20.217s, episode steps: 506, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: 0.025696, mae: 2.424623, mean_q: 2.947576, mean_eps: 0.981534

868691/1000000: episode: 1682, duration: 43.017s, episode steps: 1082, steps per second: 25, episode reward: 17.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: 0.030112, mae: 2.378872, mean_q: 2.888927, mean_eps: 0.980582

869171/1000000: episode: 1683, duration: 19.080s, episode steps: 480, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: 0.022982, mae: 2.405674, mean_q: 2.922312, mean_eps: 0.979646

869693/1000000: episode: 1684, duration: 20.887s, episode steps: 522, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.031665, mae: 2.381303, mean_q: 2.900065, mean_eps: 0.979043

870086/1000000: episode: 1685, duration: 15.602s, episode steps: 393, steps per second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.031730, mae: 2.418969, mean_q: 2.936041, mean_eps: 0.978494

870765/1000000: episode: 1686, duration: 27.113s, episode steps: 679, steps per second: 25, episode reward: 18.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.028704, mae: 2.389339, mean_q: 2.899531, mean_eps: 0.977850

871172/1000000: episode: 1687, duration: 16.345s, episode steps: 407, steps per second: 25, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.032200, mae: 2.393294, mean_q: 2.912296, mean_eps: 0.977200

871996/1000000: episode: 1688, duration: 32.934s, episode steps: 824, steps per second: 25, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.034957, mae: 2.412894, mean_q: 2.936041, mean_eps: 0.976463

872366/1000000: episode: 1689, duration: 14.934s, episode steps: 370, steps per second: 25, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.031554, mae: 2.485223, mean_q: 3.019750, mean_eps: 0.975746

872770/1000000: episode: 1690, duration: 16.069s, episode steps: 404, steps per second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: 0.026824, mae: 2.490871, mean_q: 3.025102, mean_eps: 0.975280

873148/1000000: episode: 1691, duration: 15.207s, episode steps: 378, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: 0.023877, mae: 2.501623, mean_q: 3.035197, mean_eps: 0.974812

873622/1000000: episode: 1692, duration: 18.986s, episode steps: 474, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.026658, mae: 2.456022, mean_q: 2.985950,

mean_eps: 0.974301

874376/1000000: episode: 1693, duration: 30.148s, episode steps: 754, steps per second: 25, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: 0.028959, mae: 2.487351, mean_q: 3.022436, mean_eps: 0.973564

875207/1000000: episode: 1694, duration: 33.045s, episode steps: 831, steps per second: 25, episode reward: 21.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: 0.026998, mae: 2.498362, mean_q: 3.034772, mean_eps: 0.972614

875719/1000000: episode: 1695, duration: 20.526s, episode steps: 512, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: 0.028495, mae: 2.464678, mean_q: 2.999568, mean_eps: 0.971807

876351/1000000: episode: 1696, duration: 25.249s, episode steps: 632, steps per second: 25, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.025020, mae: 2.471926, mean_q: 3.008549, mean_eps: 0.971121

876630/1000000: episode: 1697, duration: 11.140s, episode steps: 279, steps per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.032749, mae: 2.435076, mean_q: 2.957136, mean_eps: 0.970574

877072/1000000: episode: 1698, duration: 17.590s, episode steps: 442, steps per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: 0.026794, mae: 2.451769, mean_q: 2.981740, mean_eps: 0.970142

877498/1000000: episode: 1699, duration: 17.103s, episode steps: 426, steps per second: 25, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.035923, mae: 2.511771, mean_q: 3.049358, mean_eps: 0.969621

877992/1000000: episode: 1700, duration: 19.910s, episode steps: 494, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.644 [0.000, 5.000], loss: 0.028645, mae: 2.534401, mean_q: 3.082785, mean_eps: 0.969069

878541/1000000: episode: 1701, duration: 21.987s, episode steps: 549, steps per second: 25, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.028268, mae: 2.530011, mean_q: 3.070534, mean_eps: 0.968442

879247/1000000: episode: 1702, duration: 28.174s, episode steps: 706, steps per second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.028336, mae: 2.531093, mean_q: 3.077350, mean_eps: 0.967689

879721/1000000: episode: 1703, duration: 19.049s, episode steps: 474, steps per second: 25, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: 0.030931, mae: 2.526956, mean_q: 3.074990, mean_eps: 0.966981

880047/1000000: episode: 1704, duration: 12.995s, episode steps: 326, steps per second: 25, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: 0.036778, mae: 2.499577, mean_q: 3.039693,

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mean_eps: 0.966501
880438/1000000: episode: 1705, duration: 15.726s, episode steps: 391, steps per
second: 25, episode reward: 3.000, mean reward: 0.008 [ 0.000, 1.000], mean
action: 2.488 [0.000, 5.000], loss: 0.028450, mae: 2.520780, mean_q: 3.068740,
mean_eps: 0.966071
880732/1000000: episode: 1706, duration: 11.817s, episode steps: 294, steps per
second: 25, episode reward: 5.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.578 [0.000, 5.000], loss: 0.027897, mae: 2.522265, mean_q: 3.067612,
mean_eps: 0.965661
881143/1000000: episode: 1707, duration: 16.376s, episode steps: 411, steps per
second: 25, episode reward: 11.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.521 [0.000, 5.000], loss: 0.030857, mae: 2.558550, mean_q: 3.108142,
mean_eps: 0.965238
881675/1000000: episode: 1708, duration: 21.332s, episode steps: 532, steps per
second: 25, episode reward: 16.000, mean reward: 0.030 [ 0.000, 1.000], mean
action: 2.577 [0.000, 5.000], loss: 0.033050, mae: 2.503645, mean_q: 3.043342,
mean_eps: 0.964672
882196/1000000: episode: 1709, duration: 20.833s, episode steps: 521, steps per
second: 25, episode reward: 10.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.468 [0.000, 5.000], loss: 0.027911, mae: 2.532320, mean_q: 3.079725,
mean_eps: 0.964041
882747/1000000: episode: 1710, duration: 21.963s, episode steps: 551, steps per
second: 25, episode reward: 8.000, mean reward: 0.015 [ 0.000, 1.000], mean
action: 2.612 [0.000, 5.000], loss: 0.033907, mae: 2.553805, mean_q: 3.104955,
mean_eps: 0.963398
883385/1000000: episode: 1711, duration: 25.626s, episode steps: 638, steps per
second: 25, episode reward: 13.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.619 [0.000, 5.000], loss: 0.028875, mae: 2.555080, mean_q: 3.097677,
mean_eps: 0.962682
883881/1000000: episode: 1712, duration: 19.716s, episode steps: 496, steps per
second: 25, episode reward: 6.000, mean reward: 0.012 [ 0.000, 1.000], mean
action: 2.468 [0.000, 5.000], loss: 0.031632, mae: 2.568915, mean_q: 3.120682,
mean_eps: 0.962001
884348/1000000: episode: 1713, duration: 18.724s, episode steps: 467, steps per
second: 25, episode reward: 6.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.510 [0.000, 5.000], loss: 0.031178, mae: 2.521317, mean_q: 3.063879,
mean_eps: 0.961425

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

884952/1000000: episode: 1714, duration: 16.487s, episode steps: 604, steps per
second: 37, episode reward: 15.000, mean reward: 0.025 [ 0.000, 1.000], mean
action: 2.474 [0.000, 5.000], loss: 0.029652, mae: 2.536381, mean_q: 3.082983,
mean_eps: 0.960914
885259/1000000: episode: 1715, duration: 1.304s, episode steps: 307, steps per
second: 235, episode reward: 4.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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885884/1000000: episode: 1716, duration: 2.625s, episode steps: 625, steps per second: 238, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

886363/1000000: episode: 1717, duration: 2.002s, episode steps: 479, steps per second: 239, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

886756/1000000: episode: 1718, duration: 1.644s, episode steps: 393, steps per second: 239, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

887202/1000000: episode: 1719, duration: 1.869s, episode steps: 446, steps per second: 239, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

887692/1000000: episode: 1720, duration: 2.066s, episode steps: 490, steps per second: 237, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

888230/1000000: episode: 1721, duration: 2.251s, episode steps: 538, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

888752/1000000: episode: 1722, duration: 2.202s, episode steps: 522, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

889338/1000000: episode: 1723, duration: 2.456s, episode steps: 586, steps per second: 239, episode reward: 6.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

889914/1000000: episode: 1724, duration: 2.412s, episode steps: 576, steps per second: 239, episode reward: 6.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

890404/1000000: episode: 1725, duration: 2.061s, episode steps: 490, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

890861/1000000: episode: 1726, duration: 1.913s, episode steps: 457, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

891220/1000000: episode: 1727, duration: 1.525s, episode steps: 359, steps per second: 235, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

891702/1000000: episode: 1728, duration: 2.079s, episode steps: 482, steps per second: 232, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

892007/1000000: episode: 1729, duration: 1.285s, episode steps: 305, steps per second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

892591/1000000: episode: 1730, duration: 2.455s, episode steps: 584, steps per second: 238, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

892888/1000000: episode: 1731, duration: 1.247s, episode steps: 297, steps per second: 238, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.646 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

893572/1000000: episode: 1732, duration: 2.854s, episode steps: 684, steps per second: 240, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.670 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

894254/1000000: episode: 1733, duration: 2.869s, episode steps: 682, steps per second: 238, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

894735/1000000: episode: 1734, duration: 2.030s, episode steps: 481, steps per second: 237, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

895220/1000000: episode: 1735, duration: 2.039s, episode steps: 485, steps per second: 238, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

895682/1000000: episode: 1736, duration: 1.932s, episode steps: 462, steps per second: 239, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

896091/1000000: episode: 1737, duration: 1.714s, episode steps: 409, steps per second: 239, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

896592/1000000: episode: 1738, duration: 2.090s, episode steps: 501, steps per second: 240, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

896899/1000000: episode: 1739, duration: 1.297s, episode steps: 307, steps per second: 237, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

897585/1000000: episode: 1740, duration: 2.896s, episode steps: 686, steps per second: 237, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

897985/1000000: episode: 1741, duration: 1.703s, episode steps: 400, steps per second: 235, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

898518/1000000: episode: 1742, duration: 2.232s, episode steps: 533, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

899037/1000000: episode: 1743, duration: 2.178s, episode steps: 519, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

899505/1000000: episode: 1744, duration: 1.965s, episode steps: 468, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

899814/1000000: episode: 1745, duration: 1.300s, episode steps: 309, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

900283/1000000: episode: 1746, duration: 1.991s, episode steps: 469, steps per second: 236, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

900684/1000000: episode: 1747, duration: 1.701s, episode steps: 401, steps per second: 236, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.242 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

901072/1000000: episode: 1748, duration: 1.622s, episode steps: 388, steps per second: 239, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

901385/1000000: episode: 1749, duration: 1.321s, episode steps: 313, steps per second: 237, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

901912/1000000: episode: 1750, duration: 2.188s, episode steps: 527, steps per second: 241, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

902303/1000000: episode: 1751, duration: 1.641s, episode steps: 391, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

902770/1000000: episode: 1752, duration: 1.961s, episode steps: 467, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

903363/1000000: episode: 1753, duration: 2.489s, episode steps: 593, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

903776/1000000: episode: 1754, duration: 1.744s, episode steps: 413, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

904281/1000000: episode: 1755, duration: 2.111s, episode steps: 505, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

904769/1000000: episode: 1756, duration: 2.044s, episode steps: 488, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

905057/1000000: episode: 1757, duration: 1.216s, episode steps: 288, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

905563/1000000: episode: 1758, duration: 2.121s, episode steps: 506, steps per second: 239, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

906273/1000000: episode: 1759, duration: 2.975s, episode steps: 710, steps per second: 239, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

906886/1000000: episode: 1760, duration: 2.591s, episode steps: 613, steps per second: 237, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_1760_episodes.h5f

907361/1000000: episode: 1761, duration: 2.088s, episode steps: 475, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

908033/1000000: episode: 1762, duration: 2.829s, episode steps: 672, steps per second: 238, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 908472/1000000: episode: 1763, duration: 1.848s, episode steps: 439, steps per second: 238, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 909183/1000000: episode: 1764, duration: 3.014s, episode steps: 711, steps per second: 236, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 909910/1000000: episode: 1765, duration: 3.069s, episode steps: 727, steps per second: 237, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 910486/1000000: episode: 1766, duration: 2.426s, episode steps: 576, steps per second: 237, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 911108/1000000: episode: 1767, duration: 2.599s, episode steps: 622, steps per second: 239, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 912125/1000000: episode: 1768, duration: 4.224s, episode steps: 1017, steps per second: 241, episode reward: 25.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 912450/1000000: episode: 1769, duration: 1.368s, episode steps: 325, steps per second: 238, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 913167/1000000: episode: 1770, duration: 3.019s, episode steps: 717, steps per second: 238, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 913543/1000000: episode: 1771, duration: 1.588s, episode steps: 376, steps per second: 237, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 913931/1000000: episode: 1772, duration: 1.641s, episode steps: 388, steps per second: 236, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 914432/1000000: episode: 1773, duration: 2.108s, episode steps: 501, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 914715/1000000: episode: 1774, duration: 1.203s, episode steps: 283, steps per second: 235, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 915293/1000000: episode: 1775, duration: 2.438s, episode steps: 578, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 915999/1000000: episode: 1776, duration: 2.976s, episode steps: 706, steps per second: 237, episode reward: 9.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 916485/1000000: episode: 1777, duration: 2.030s, episode steps: 486, steps per second: 239, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 917238/1000000: episode: 1778, duration: 3.135s, episode steps: 753, steps per second: 240, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean

action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 917993/1000000: episode: 1779, duration: 3.145s, episode steps: 755, steps per second: 240, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 918603/1000000: episode: 1780, duration: 2.561s, episode steps: 610, steps per second: 238, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 919193/1000000: episode: 1781, duration: 2.464s, episode steps: 590, steps per second: 239, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 919576/1000000: episode: 1782, duration: 1.615s, episode steps: 383, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 920107/1000000: episode: 1783, duration: 2.227s, episode steps: 531, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 920948/1000000: episode: 1784, duration: 3.497s, episode steps: 841, steps per second: 241, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 921245/1000000: episode: 1785, duration: 1.262s, episode steps: 297, steps per second: 235, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 921904/1000000: episode: 1786, duration: 2.779s, episode steps: 659, steps per second: 237, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 922400/1000000: episode: 1787, duration: 2.076s, episode steps: 496, steps per second: 239, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 923079/1000000: episode: 1788, duration: 2.827s, episode steps: 679, steps per second: 240, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 923544/1000000: episode: 1789, duration: 1.950s, episode steps: 465, steps per second: 238, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 924063/1000000: episode: 1790, duration: 2.185s, episode steps: 519, steps per second: 238, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 924546/1000000: episode: 1791, duration: 2.059s, episode steps: 483, steps per second: 235, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 924941/1000000: episode: 1792, duration: 1.665s, episode steps: 395, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.691 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 925596/1000000: episode: 1793, duration: 2.736s, episode steps: 655, steps per second: 239, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 925889/1000000: episode: 1794, duration: 1.242s, episode steps: 293, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.662 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 926281/1000000: episode: 1795, duration: 1.652s, episode steps: 392, steps per second: 237, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 926875/1000000: episode: 1796, duration: 2.470s, episode steps: 594, steps per second: 241, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 927351/1000000: episode: 1797, duration: 2.002s, episode steps: 476, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 927693/1000000: episode: 1798, duration: 8.118s, episode steps: 342, steps per second: 42, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: 0.026236, mae: 2.588832, mean_q: 3.142486, mean_eps: 0.987885
 928037/1000000: episode: 1799, duration: 13.720s, episode steps: 344, steps per second: 25, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: 0.031042, mae: 2.632769, mean_q: 3.190758, mean_eps: 0.987566
 928417/1000000: episode: 1800, duration: 15.160s, episode steps: 380, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.684 [0.000, 5.000], loss: 0.029353, mae: 2.592091, mean_q: 3.143693, mean_eps: 0.987131
 928942/1000000: episode: 1801, duration: 20.983s, episode steps: 525, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.027047, mae: 2.597342, mean_q: 3.154057, mean_eps: 0.986589
 929357/1000000: episode: 1802, duration: 16.529s, episode steps: 415, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: 0.031530, mae: 2.606710, mean_q: 3.167260, mean_eps: 0.986025
 929746/1000000: episode: 1803, duration: 15.492s, episode steps: 389, steps per second: 25, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.036225, mae: 2.594894, mean_q: 3.150830, mean_eps: 0.985542
 930229/1000000: episode: 1804, duration: 19.246s, episode steps: 483, steps per second: 25, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.037300, mae: 2.615924, mean_q: 3.171536, mean_eps: 0.985019
 930665/1000000: episode: 1805, duration: 17.507s, episode steps: 436, steps per second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.030409, mae: 2.563647, mean_q: 3.111957, mean_eps: 0.984467
 931070/1000000: episode: 1806, duration: 16.092s, episode steps: 405, steps per second: 25, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.033041, mae: 2.613423, mean_q: 3.173047, mean_eps: 0.983963
 931557/1000000: episode: 1807, duration: 19.504s, episode steps: 487, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.526 [0.000, 5.000], loss: 0.037072, mae: 2.614703, mean_q: 3.173022,
 mean_eps: 0.983428
 932164/1000000: episode: 1808, duration: 24.248s, episode steps: 607, steps per
 second: 25, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.578 [0.000, 5.000], loss: 0.029522, mae: 2.607356, mean_q: 3.165601,
 mean_eps: 0.982773
 932907/1000000: episode: 1809, duration: 29.588s, episode steps: 743, steps per
 second: 25, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean
 action: 2.515 [0.000, 5.000], loss: 0.034767, mae: 2.625888, mean_q: 3.186381,
 mean_eps: 0.981964
 933284/1000000: episode: 1810, duration: 15.134s, episode steps: 377, steps per
 second: 25, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean
 action: 2.594 [0.000, 5.000], loss: 0.037029, mae: 2.637791, mean_q: 3.198753,
 mean_eps: 0.981292
 933755/1000000: episode: 1811, duration: 18.734s, episode steps: 471, steps per
 second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.569 [0.000, 5.000], loss: 0.034123, mae: 2.601415, mean_q: 3.154165,
 mean_eps: 0.980783
 934492/1000000: episode: 1812, duration: 29.396s, episode steps: 737, steps per
 second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.374 [0.000, 5.000], loss: 0.031084, mae: 2.618426, mean_q: 3.176074,
 mean_eps: 0.980058
 934758/1000000: episode: 1813, duration: 10.668s, episode steps: 266, steps per
 second: 25, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.556 [0.000, 5.000], loss: 0.031131, mae: 2.641169, mean_q: 3.206902,
 mean_eps: 0.979456
 935250/1000000: episode: 1814, duration: 19.666s, episode steps: 492, steps per
 second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.569 [0.000, 5.000], loss: 0.030127, mae: 2.608839, mean_q: 3.166588,
 mean_eps: 0.979000
 935723/1000000: episode: 1815, duration: 18.963s, episode steps: 473, steps per
 second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.518 [0.000, 5.000], loss: 0.029508, mae: 2.581993, mean_q: 3.132432,
 mean_eps: 0.978422
 936037/1000000: episode: 1816, duration: 12.619s, episode steps: 314, steps per
 second: 25, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.729 [0.000, 5.000], loss: 0.029084, mae: 2.672750, mean_q: 3.243301,
 mean_eps: 0.977949
 936432/1000000: episode: 1817, duration: 15.787s, episode steps: 395, steps per
 second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean
 action: 2.572 [0.000, 5.000], loss: 0.045608, mae: 2.626623, mean_q: 3.180732,
 mean_eps: 0.977524
 937019/1000000: episode: 1818, duration: 23.448s, episode steps: 587, steps per
 second: 25, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean
 action: 2.574 [0.000, 5.000], loss: 0.036763, mae: 2.615379, mean_q: 3.173115,
 mean_eps: 0.976936
 937481/1000000: episode: 1819, duration: 18.448s, episode steps: 462, steps per
 second: 25, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.548 [0.000, 5.000], loss: 0.034693, mae: 2.633063, mean_q: 3.198423,
 mean_eps: 0.976305
 937846/1000000: episode: 1820, duration: 14.472s, episode steps: 365, steps per
 second: 25, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean
 action: 2.512 [0.000, 5.000], loss: 0.032763, mae: 2.630181, mean_q: 3.184297,
 mean_eps: 0.975808
 938368/1000000: episode: 1821, duration: 20.662s, episode steps: 522, steps per
 second: 25, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.504 [0.000, 5.000], loss: 0.037122, mae: 2.631434, mean_q: 3.193569,
 mean_eps: 0.975278
 938887/1000000: episode: 1822, duration: 20.867s, episode steps: 519, steps per
 second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.495 [0.000, 5.000], loss: 0.026589, mae: 2.660866, mean_q: 3.230697,
 mean_eps: 0.974654
 939359/1000000: episode: 1823, duration: 18.704s, episode steps: 472, steps per
 second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.551 [0.000, 5.000], loss: 0.036947, mae: 2.658336, mean_q: 3.225245,
 mean_eps: 0.974058
 940280/1000000: episode: 1824, duration: 36.865s, episode steps: 921, steps per
 second: 25, episode reward: 15.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.522 [0.000, 5.000], loss: 0.029373, mae: 2.666419, mean_q: 3.237045,
 mean_eps: 0.973223
 940586/1000000: episode: 1825, duration: 12.222s, episode steps: 306, steps per
 second: 25, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.480 [0.000, 5.000], loss: 0.025105, mae: 2.589927, mean_q: 3.144688,
 mean_eps: 0.972486
 940981/1000000: episode: 1826, duration: 15.817s, episode steps: 395, steps per
 second: 25, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean
 action: 2.471 [0.000, 5.000], loss: 0.028576, mae: 2.694160, mean_q: 3.268367,
 mean_eps: 0.972064
 941986/1000000: episode: 1827, duration: 40.091s, episode steps: 1005, steps
 per second: 25, episode reward: 25.000, mean reward: 0.025 [0.000, 1.000],
 mean action: 2.444 [0.000, 5.000], loss: 0.030801, mae: 2.629828, mean_q:
 3.188491, mean_eps: 0.971224
 942289/1000000: episode: 1828, duration: 12.103s, episode steps: 303, steps per
 second: 25, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.703 [0.000, 5.000], loss: 0.025821, mae: 2.626415, mean_q: 3.183846,
 mean_eps: 0.970439
 942821/1000000: episode: 1829, duration: 21.125s, episode steps: 532, steps per
 second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.566 [0.000, 5.000], loss: 0.033484, mae: 2.684724, mean_q: 3.257978,
 mean_eps: 0.969938
 943119/1000000: episode: 1830, duration: 11.735s, episode steps: 298, steps per
 second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.430 [0.000, 5.000], loss: 0.035727, mae: 2.785740, mean_q: 3.379146,
 mean_eps: 0.969441
 943741/1000000: episode: 1831, duration: 24.820s, episode steps: 622, steps per
 second: 25, episode reward: 18.000, mean reward: 0.029 [0.000, 1.000], mean

action: 2.537 [0.000, 5.000], loss: 0.038145, mae: 2.684370, mean_q: 3.259453,
 mean_eps: 0.968889
 944229/1000000: episode: 1832, duration: 19.412s, episode steps: 488, steps per
 second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.516 [0.000, 5.000], loss: 0.032031, mae: 2.674052, mean_q: 3.242701,
 mean_eps: 0.968222
 944696/1000000: episode: 1833, duration: 18.621s, episode steps: 467, steps per
 second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.619 [0.000, 5.000], loss: 0.032901, mae: 2.691663, mean_q: 3.261576,
 mean_eps: 0.967650
 945219/1000000: episode: 1834, duration: 20.945s, episode steps: 523, steps per
 second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.438 [0.000, 5.000], loss: 0.039697, mae: 2.681783, mean_q: 3.250667,
 mean_eps: 0.967058
 945669/1000000: episode: 1835, duration: 18.017s, episode steps: 450, steps per
 second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.596 [0.000, 5.000], loss: 0.033134, mae: 2.716936, mean_q: 3.299347,
 mean_eps: 0.966472
 946038/1000000: episode: 1836, duration: 14.522s, episode steps: 369, steps per
 second: 25, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.566 [0.000, 5.000], loss: 0.034666, mae: 2.697634, mean_q: 3.270600,
 mean_eps: 0.965980
 946347/1000000: episode: 1837, duration: 12.340s, episode steps: 309, steps per
 second: 25, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.592 [0.000, 5.000], loss: 0.034937, mae: 2.653117, mean_q: 3.212497,
 mean_eps: 0.965574
 946840/1000000: episode: 1838, duration: 19.871s, episode steps: 493, steps per
 second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.491 [0.000, 5.000], loss: 0.039715, mae: 2.687055, mean_q: 3.258219,
 mean_eps: 0.965094
 947246/1000000: episode: 1839, duration: 16.307s, episode steps: 406, steps per
 second: 25, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean
 action: 2.594 [0.000, 5.000], loss: 0.029371, mae: 2.683792, mean_q: 3.250011,
 mean_eps: 0.964554
 947559/1000000: episode: 1840, duration: 12.444s, episode steps: 313, steps per
 second: 25, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean
 action: 2.585 [0.000, 5.000], loss: 0.032817, mae: 2.723913, mean_q: 3.307746,
 mean_eps: 0.964122
 948449/1000000: episode: 1841, duration: 35.651s, episode steps: 890, steps per
 second: 25, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.554 [0.000, 5.000], loss: 0.033862, mae: 2.720090, mean_q: 3.299732,
 mean_eps: 0.963400
 948788/1000000: episode: 1842, duration: 13.574s, episode steps: 339, steps per
 second: 25, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean
 action: 2.519 [0.000, 5.000], loss: 0.038925, mae: 2.747161, mean_q: 3.330263,
 mean_eps: 0.962663
 949185/1000000: episode: 1843, duration: 15.934s, episode steps: 397, steps per
 second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean

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action: 2.416 [0.000, 5.000], loss: 0.035317, mae: 2.733995, mean_q: 3.317640,
mean_eps: 0.962222
949646/1000000: episode: 1844, duration: 18.278s, episode steps: 461, steps per
second: 25, episode reward: 8.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.508 [0.000, 5.000], loss: 0.025496, mae: 2.732088, mean_q: 3.310684,
mean_eps: 0.961706
950130/1000000: episode: 1845, duration: 19.262s, episode steps: 484, steps per
second: 25, episode reward: 10.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.492 [0.000, 5.000], loss: 0.031474, mae: 2.729275, mean_q: 3.305814,
mean_eps: 0.961139

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

950593/1000000: episode: 1846, duration: 7.001s, episode steps: 463, steps per
second: 66, episode reward: 6.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.652 [0.000, 5.000], loss: 0.024017, mae: 2.738099, mean_q: 3.310758,
mean_eps: 0.960765
951038/1000000: episode: 1847, duration: 1.858s, episode steps: 445, steps per
second: 240, episode reward: 8.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
951653/1000000: episode: 1848, duration: 2.569s, episode steps: 615, steps per
second: 239, episode reward: 12.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
952161/1000000: episode: 1849, duration: 2.146s, episode steps: 508, steps per
second: 237, episode reward: 7.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
952455/1000000: episode: 1850, duration: 1.248s, episode steps: 294, steps per
second: 236, episode reward: 9.000, mean reward: 0.031 [ 0.000, 1.000], mean
action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
952837/1000000: episode: 1851, duration: 1.604s, episode steps: 382, steps per
second: 238, episode reward: 8.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
953212/1000000: episode: 1852, duration: 1.578s, episode steps: 375, steps per
second: 238, episode reward: 3.000, mean reward: 0.008 [ 0.000, 1.000], mean
action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
953653/1000000: episode: 1853, duration: 1.843s, episode steps: 441, steps per
second: 239, episode reward: 5.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
954136/1000000: episode: 1854, duration: 2.018s, episode steps: 483, steps per
second: 239, episode reward: 7.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
954580/1000000: episode: 1855, duration: 1.872s, episode steps: 444, steps per
second: 237, episode reward: 5.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
954962/1000000: episode: 1856, duration: 1.611s, episode steps: 382, steps per
second: 237, episode reward: 9.000, mean reward: 0.024 [ 0.000, 1.000], mean
action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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955677/1000000: episode: 1857, duration: 3.008s, episode steps: 715, steps per second: 238, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

956091/1000000: episode: 1858, duration: 1.737s, episode steps: 414, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.662 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

957036/1000000: episode: 1859, duration: 3.917s, episode steps: 945, steps per second: 241, episode reward: 26.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

957853/1000000: episode: 1860, duration: 3.394s, episode steps: 817, steps per second: 241, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

958165/1000000: episode: 1861, duration: 1.322s, episode steps: 312, steps per second: 236, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.263 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

958602/1000000: episode: 1862, duration: 1.848s, episode steps: 437, steps per second: 236, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

958975/1000000: episode: 1863, duration: 1.574s, episode steps: 373, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

959471/1000000: episode: 1864, duration: 2.074s, episode steps: 496, steps per second: 239, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

960044/1000000: episode: 1865, duration: 2.390s, episode steps: 573, steps per second: 240, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

960429/1000000: episode: 1866, duration: 1.619s, episode steps: 385, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

960726/1000000: episode: 1867, duration: 1.246s, episode steps: 297, steps per second: 238, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

961053/1000000: episode: 1868, duration: 1.379s, episode steps: 327, steps per second: 237, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

961539/1000000: episode: 1869, duration: 2.050s, episode steps: 486, steps per second: 237, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

962052/1000000: episode: 1870, duration: 2.138s, episode steps: 513, steps per second: 240, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

962584/1000000: episode: 1871, duration: 2.218s, episode steps: 532, steps per second: 240, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

963253/1000000: episode: 1872, duration: 2.798s, episode steps: 669, steps per second: 239, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

963546/1000000: episode: 1873, duration: 1.235s, episode steps: 293, steps per second: 237, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

964057/1000000: episode: 1874, duration: 2.153s, episode steps: 511, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

964446/1000000: episode: 1875, duration: 1.662s, episode steps: 389, steps per second: 234, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

965260/1000000: episode: 1876, duration: 3.421s, episode steps: 814, steps per second: 238, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

965654/1000000: episode: 1877, duration: 1.649s, episode steps: 394, steps per second: 239, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

966356/1000000: episode: 1878, duration: 2.922s, episode steps: 702, steps per second: 240, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

966797/1000000: episode: 1879, duration: 1.868s, episode steps: 441, steps per second: 236, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

967094/1000000: episode: 1880, duration: 1.260s, episode steps: 297, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

967381/1000000: episode: 1881, duration: 1.216s, episode steps: 287, steps per second: 236, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

967756/1000000: episode: 1882, duration: 1.613s, episode steps: 375, steps per second: 233, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

968284/1000000: episode: 1883, duration: 2.211s, episode steps: 528, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

968821/1000000: episode: 1884, duration: 2.259s, episode steps: 537, steps per second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

969428/1000000: episode: 1885, duration: 2.555s, episode steps: 607, steps per second: 238, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

970108/1000000: episode: 1886, duration: 2.847s, episode steps: 680, steps per second: 239, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

970389/1000000: episode: 1887, duration: 1.198s, episode steps: 281, steps per second: 235, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

970927/1000000: episode: 1888, duration: 2.275s, episode steps: 538, steps per second: 236, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

971607/1000000: episode: 1889, duration: 2.835s, episode steps: 680, steps per second: 240, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

972636/1000000: episode: 1890, duration: 4.266s, episode steps: 1029, steps per second: 241, episode reward: 20.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

973156/1000000: episode: 1891, duration: 2.199s, episode steps: 520, steps per second: 237, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

973624/1000000: episode: 1892, duration: 2.004s, episode steps: 468, steps per second: 234, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

974031/1000000: episode: 1893, duration: 1.716s, episode steps: 407, steps per second: 237, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

974446/1000000: episode: 1894, duration: 1.732s, episode steps: 415, steps per second: 240, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

974937/1000000: episode: 1895, duration: 2.057s, episode steps: 491, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

975351/1000000: episode: 1896, duration: 1.747s, episode steps: 414, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

975826/1000000: episode: 1897, duration: 2.007s, episode steps: 475, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

976315/1000000: episode: 1898, duration: 2.088s, episode steps: 489, steps per second: 234, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.628 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

976926/1000000: episode: 1899, duration: 2.572s, episode steps: 611, steps per second: 238, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

977760/1000000: episode: 1900, duration: 3.483s, episode steps: 834, steps per second: 239, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

978237/1000000: episode: 1901, duration: 2.009s, episode steps: 477, steps per second: 237, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

978763/1000000: episode: 1902, duration: 2.220s, episode steps: 526, steps per second: 237, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

979036/1000000: episode: 1903, duration: 1.168s, episode steps: 273, steps per second: 234, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.330 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

979700/1000000: episode: 1904, duration: 2.783s, episode steps: 664, steps per second: 239, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

980240/1000000: episode: 1905, duration: 2.250s, episode steps: 540, steps per second: 240, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

980966/1000000: episode: 1906, duration: 3.024s, episode steps: 726, steps per second: 240, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

981294/1000000: episode: 1907, duration: 1.381s, episode steps: 328, steps per second: 237, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

981688/1000000: episode: 1908, duration: 1.665s, episode steps: 394, steps per second: 237, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

982023/1000000: episode: 1909, duration: 1.426s, episode steps: 335, steps per second: 235, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

982471/1000000: episode: 1910, duration: 1.891s, episode steps: 448, steps per second: 237, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

983009/1000000: episode: 1911, duration: 2.249s, episode steps: 538, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

983410/1000000: episode: 1912, duration: 1.689s, episode steps: 401, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

983967/1000000: episode: 1913, duration: 2.325s, episode steps: 557, steps per second: 240, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

984520/1000000: episode: 1914, duration: 2.317s, episode steps: 553, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

985049/1000000: episode: 1915, duration: 2.236s, episode steps: 529, steps per second: 237, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

985435/1000000: episode: 1916, duration: 1.640s, episode steps: 386, steps per second: 235, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

985806/1000000: episode: 1917, duration: 1.561s, episode steps: 371, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

986318/1000000: episode: 1918, duration: 2.151s, episode steps: 512, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

986643/1000000: episode: 1919, duration: 1.375s, episode steps: 325, steps per second: 236, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

986902/1000000: episode: 1920, duration: 1.102s, episode steps: 259, steps per second: 235, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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987265/1000000: episode: 1921, duration: 1.629s, episode steps: 363, steps per second: 223, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

987565/1000000: episode: 1922, duration: 1.260s, episode steps: 300, steps per second: 238, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

988109/1000000: episode: 1923, duration: 2.307s, episode steps: 544, steps per second: 236, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

988687/1000000: episode: 1924, duration: 2.435s, episode steps: 578, steps per second: 237, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

989510/1000000: episode: 1925, duration: 3.427s, episode steps: 823, steps per second: 240, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

989807/1000000: episode: 1926, duration: 1.254s, episode steps: 297, steps per second: 237, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

990126/1000000: episode: 1927, duration: 1.336s, episode steps: 319, steps per second: 239, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

990609/1000000: episode: 1928, duration: 2.028s, episode steps: 483, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.646 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

990994/1000000: episode: 1929, duration: 1.645s, episode steps: 385, steps per second: 234, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

991412/1000000: episode: 1930, duration: 1.769s, episode steps: 418, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

991780/1000000: episode: 1931, duration: 1.554s, episode steps: 368, steps per second: 237, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

992230/1000000: episode: 1932, duration: 1.894s, episode steps: 450, steps per second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

992668/1000000: episode: 1933, duration: 1.838s, episode steps: 438, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

993206/1000000: episode: 1934, duration: 8.152s, episode steps: 538, steps per second: 66, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: 0.029998, mae: 2.682211, mean_q: 3.246766, mean_eps: 0.987899

993621/1000000: episode: 1935, duration: 16.479s, episode steps: 415, steps per

second: 25, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.027645, mae: 2.705666, mean_q: 3.279305, mean_eps: 0.987551

994116/1000000: episode: 1936, duration: 19.731s, episode steps: 495, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: 0.030436, mae: 2.747762, mean_q: 3.333210, mean_eps: 0.987006

994397/1000000: episode: 1937, duration: 11.284s, episode steps: 281, steps per second: 25, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.036675, mae: 2.770050, mean_q: 3.354561, mean_eps: 0.986541

995003/1000000: episode: 1938, duration: 24.012s, episode steps: 606, steps per second: 25, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.033226, mae: 2.726672, mean_q: 3.310352, mean_eps: 0.986008

995642/1000000: episode: 1939, duration: 25.512s, episode steps: 639, steps per second: 25, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.029678, mae: 2.702707, mean_q: 3.282345, mean_eps: 0.985262

996116/1000000: episode: 1940, duration: 18.830s, episode steps: 474, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.029523, mae: 2.727746, mean_q: 3.309342, mean_eps: 0.984594

996590/1000000: episode: 1941, duration: 18.988s, episode steps: 474, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.037206, mae: 2.705356, mean_q: 3.280985, mean_eps: 0.984026

997082/1000000: episode: 1942, duration: 19.618s, episode steps: 492, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: 0.029953, mae: 2.702841, mean_q: 3.277502, mean_eps: 0.983445

997484/1000000: episode: 1943, duration: 16.099s, episode steps: 402, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: 0.034849, mae: 2.693280, mean_q: 3.265865, mean_eps: 0.982910

997963/1000000: episode: 1944, duration: 19.016s, episode steps: 479, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.034316, mae: 2.697785, mean_q: 3.266952, mean_eps: 0.982382

998292/1000000: episode: 1945, duration: 13.159s, episode steps: 329, steps per second: 25, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.034020, mae: 2.779386, mean_q: 3.365916, mean_eps: 0.981897

998955/1000000: episode: 1946, duration: 26.377s, episode steps: 663, steps per second: 25, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.039316, mae: 2.719329, mean_q: 3.294207, mean_eps: 0.981302

999364/1000000: episode: 1947, duration: 16.371s, episode steps: 409, steps per

second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.029507, mae: 2.734533, mean_q: 3.320224, mean_eps: 0.980658

999809/1000000: episode: 1948, duration: 17.696s, episode steps: 445, steps per second: 25, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: 0.031745, mae: 2.729419, mean_q: 3.306451, mean_eps: 0.980145

1000328/1000000: episode: 1949, duration: 20.616s, episode steps: 519, steps per second: 25, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.033851, mae: 2.755431, mean_q: 3.343063, mean_eps: 0.979566

1000956/1000000: episode: 1950, duration: 25.251s, episode steps: 628, steps per second: 25, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.028841, mae: 2.758083, mean_q: 3.341479, mean_eps: 0.978880

1001366/1000000: episode: 1951, duration: 16.380s, episode steps: 410, steps per second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: 0.032617, mae: 2.746948, mean_q: 3.327381, mean_eps: 0.978256

1001897/1000000: episode: 1952, duration: 21.197s, episode steps: 531, steps per second: 25, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.038484, mae: 2.735297, mean_q: 3.314948, mean_eps: 0.977690

1002983/1000000: episode: 1953, duration: 43.112s, episode steps: 1086, steps per second: 25, episode reward: 18.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.034196, mae: 2.742816, mean_q: 3.326281, mean_eps: 0.976720

1003423/1000000: episode: 1954, duration: 17.591s, episode steps: 440, steps per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.033035, mae: 2.799776, mean_q: 3.391058, mean_eps: 0.975806

1004035/1000000: episode: 1955, duration: 24.399s, episode steps: 612, steps per second: 25, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.034363, mae: 2.776814, mean_q: 3.369249, mean_eps: 0.975174

1004718/1000000: episode: 1956, duration: 27.255s, episode steps: 683, steps per second: 25, episode reward: 23.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.035993, mae: 2.801955, mean_q: 3.396689, mean_eps: 0.974397

1005237/1000000: episode: 1957, duration: 20.710s, episode steps: 519, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.034131, mae: 2.772481, mean_q: 3.360196, mean_eps: 0.973674

1006024/1000000: episode: 1958, duration: 31.573s, episode steps: 787, steps per second: 25, episode reward: 22.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.037205, mae: 2.770087, mean_q: 3.353272, mean_eps: 0.972892

1006450/1000000: episode: 1959, duration: 17.057s, episode steps: 426, steps

per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.486 [0.000, 5.000], loss: 0.040940, mae: 2.820316, mean_q:
3.415624, mean_eps: 0.972165

1006918/1000000: episode: 1960, duration: 18.658s, episode steps: 468, steps
per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000],
mean action: 2.543 [0.000, 5.000], loss: 0.031448, mae: 2.780878, mean_q:
3.367367, mean_eps: 0.971627

1007190/1000000: episode: 1961, duration: 10.817s, episode steps: 272, steps
per second: 25, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.438 [0.000, 5.000], loss: 0.033649, mae: 2.769003, mean_q:
3.350542, mean_eps: 0.971183

1007767/1000000: episode: 1962, duration: 22.934s, episode steps: 577, steps
per second: 25, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.548 [0.000, 5.000], loss: 0.039835, mae: 2.795195, mean_q:
3.389214, mean_eps: 0.970674

1008202/1000000: episode: 1963, duration: 17.475s, episode steps: 435, steps
per second: 25, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.591 [0.000, 5.000], loss: 0.032499, mae: 2.790190, mean_q:
3.381080, mean_eps: 0.970067

1008650/1000000: episode: 1964, duration: 17.712s, episode steps: 448, steps
per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.520 [0.000, 5.000], loss: 0.027126, mae: 2.821323, mean_q:
3.422229, mean_eps: 0.969537

1008942/1000000: episode: 1965, duration: 11.577s, episode steps: 292, steps
per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.377 [0.000, 5.000], loss: 0.033929, mae: 2.784762, mean_q:
3.376333, mean_eps: 0.969093

1009326/1000000: episode: 1966, duration: 15.306s, episode steps: 384, steps
per second: 25, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000],
mean action: 2.305 [0.000, 5.000], loss: 0.031382, mae: 2.784260, mean_q:
3.383657, mean_eps: 0.968687

1009820/1000000: episode: 1967, duration: 19.730s, episode steps: 494, steps
per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.474 [0.000, 5.000], loss: 0.035219, mae: 2.807229, mean_q:
3.405575, mean_eps: 0.968162

1010309/1000000: episode: 1968, duration: 19.626s, episode steps: 489, steps
per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.519 [0.000, 5.000], loss: 0.029528, mae: 2.829017, mean_q:
3.430005, mean_eps: 0.967571

1010681/1000000: episode: 1969, duration: 14.848s, episode steps: 372, steps
per second: 25, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.414 [0.000, 5.000], loss: 0.036963, mae: 2.821224, mean_q:
3.416327, mean_eps: 0.967053

1011181/1000000: episode: 1970, duration: 19.873s, episode steps: 500, steps
per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.476 [0.000, 5.000], loss: 0.037684, mae: 2.846747, mean_q:
3.449389, mean_eps: 0.966530

1011532/1000000: episode: 1971, duration: 14.091s, episode steps: 351, steps

per second: 25, episode reward: 9.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.436 [0.000, 5.000], loss: 0.033302, mae: 2.852207, mean_q:
3.459921, mean_eps: 0.966021

1012055/1000000: episode: 1972, duration: 20.824s, episode steps: 523, steps
per second: 25, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.468 [0.000, 5.000], loss: 0.029451, mae: 2.819205, mean_q:
3.413743, mean_eps: 0.965498

1012555/1000000: episode: 1973, duration: 19.953s, episode steps: 500, steps
per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.514 [0.000, 5.000], loss: 0.036689, mae: 2.792576, mean_q:
3.390243, mean_eps: 0.964883

1013325/1000000: episode: 1974, duration: 30.624s, episode steps: 770, steps
per second: 25, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.477 [0.000, 5.000], loss: 0.033386, mae: 2.827308, mean_q:
3.425230, mean_eps: 0.964120

1013666/1000000: episode: 1975, duration: 13.537s, episode steps: 341, steps
per second: 25, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.463 [0.000, 5.000], loss: 0.035545, mae: 2.811832, mean_q:
3.410134, mean_eps: 0.963453

1013966/1000000: episode: 1976, duration: 12.044s, episode steps: 300, steps
per second: 25, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000],
mean action: 2.533 [0.000, 5.000], loss: 0.036148, mae: 2.785122, mean_q:
3.377423, mean_eps: 0.963069

1014796/1000000: episode: 1977, duration: 33.146s, episode steps: 830, steps
per second: 25, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.531 [0.000, 5.000], loss: 0.033753, mae: 2.834061, mean_q:
3.438224, mean_eps: 0.962392

1015335/1000000: episode: 1978, duration: 21.543s, episode steps: 539, steps
per second: 25, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.449 [0.000, 5.000], loss: 0.038228, mae: 2.827711, mean_q:
3.426479, mean_eps: 0.961571

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

1015856/1000000: episode: 1979, duration: 19.064s, episode steps: 521, steps
per second: 27, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.526 [0.000, 5.000], loss: 0.041447, mae: 2.851671, mean_q:
3.455989, mean_eps: 0.960964

1016572/1000000: episode: 1980, duration: 2.974s, episode steps: 716, steps per
second: 241, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1016959/1000000: episode: 1981, duration: 1.634s, episode steps: 387, steps per
second: 237, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1017335/1000000: episode: 1982, duration: 1.600s, episode steps: 376, steps per
second: 235, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean
action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1017941/1000000: episode: 1983, duration: 2.543s, episode steps: 606, steps per second: 238, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1018623/1000000: episode: 1984, duration: 2.868s, episode steps: 682, steps per second: 238, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1019191/1000000: episode: 1985, duration: 2.378s, episode steps: 568, steps per second: 239, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1019902/1000000: episode: 1986, duration: 3.015s, episode steps: 711, steps per second: 236, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1020519/1000000: episode: 1987, duration: 2.632s, episode steps: 617, steps per second: 234, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1021016/1000000: episode: 1988, duration: 2.096s, episode steps: 497, steps per second: 237, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1021758/1000000: episode: 1989, duration: 3.110s, episode steps: 742, steps per second: 239, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.674 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1022343/1000000: episode: 1990, duration: 2.467s, episode steps: 585, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1022820/1000000: episode: 1991, duration: 2.019s, episode steps: 477, steps per second: 236, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1023364/1000000: episode: 1992, duration: 2.319s, episode steps: 544, steps per second: 235, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1024050/1000000: episode: 1993, duration: 2.881s, episode steps: 686, steps per second: 238, episode reward: 7.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1024514/1000000: episode: 1994, duration: 1.952s, episode steps: 464, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.353 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1025132/1000000: episode: 1995, duration: 2.578s, episode steps: 618, steps per second: 240, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1026073/1000000: episode: 1996, duration: 3.960s, episode steps: 941, steps per second: 238, episode reward: 26.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1026877/1000000: episode: 1997, duration: 3.349s, episode steps: 804, steps per second: 240, episode reward: 20.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1027989/1000000: episode: 1998, duration: 4.616s, episode steps: 1112, steps per second: 241, episode reward: 16.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1028632/1000000: episode: 1999, duration: 2.736s, episode steps: 643, steps per second: 235, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1029138/1000000: episode: 2000, duration: 2.141s, episode steps: 506, steps per second: 236, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1029866/1000000: episode: 2001, duration: 3.044s, episode steps: 728, steps per second: 239, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1030526/1000000: episode: 2002, duration: 2.731s, episode steps: 660, steps per second: 242, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1030975/1000000: episode: 2003, duration: 1.887s, episode steps: 449, steps per second: 238, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1031372/1000000: episode: 2004, duration: 1.667s, episode steps: 397, steps per second: 238, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1031766/1000000: episode: 2005, duration: 1.676s, episode steps: 394, steps per second: 235, episode reward: 11.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1032191/1000000: episode: 2006, duration: 1.800s, episode steps: 425, steps per second: 236, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1032856/1000000: episode: 2007, duration: 2.764s, episode steps: 665, steps per second: 241, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1033355/1000000: episode: 2008, duration: 2.100s, episode steps: 499, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1033736/1000000: episode: 2009, duration: 1.614s, episode steps: 381, steps per second: 236, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.675 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1034275/1000000: episode: 2010, duration: 2.257s, episode steps: 539, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1034598/1000000: episode: 2011, duration: 1.367s, episode steps: 323, steps per second: 236, episode reward: 9.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1034918/1000000: episode: 2012, duration: 1.364s, episode steps: 320, steps per second: 235, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1035423/1000000: episode: 2013, duration: 2.125s, episode steps: 505, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1035888/1000000: episode: 2014, duration: 1.944s, episode steps: 465, steps per second: 239, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1036445/1000000: episode: 2015, duration: 2.337s, episode steps: 557, steps per second: 238, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1037285/1000000: episode: 2016, duration: 3.487s, episode steps: 840, steps per second: 241, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1038410/1000000: episode: 2017, duration: 4.697s, episode steps: 1125, steps per second: 240, episode reward: 15.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1038980/1000000: episode: 2018, duration: 2.415s, episode steps: 570, steps per second: 236, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1039484/1000000: episode: 2019, duration: 2.134s, episode steps: 504, steps per second: 236, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1039777/1000000: episode: 2020, duration: 1.244s, episode steps: 293, steps per second: 236, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1040284/1000000: episode: 2021, duration: 2.145s, episode steps: 507, steps per second: 236, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1040950/1000000: episode: 2022, duration: 2.831s, episode steps: 666, steps per second: 235, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1041565/1000000: episode: 2023, duration: 2.581s, episode steps: 615, steps per second: 238, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1042448/1000000: episode: 2024, duration: 3.644s, episode steps: 883, steps per second: 242, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1042822/1000000: episode: 2025, duration: 1.562s, episode steps: 374, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1043351/1000000: episode: 2026, duration: 2.225s, episode steps: 529, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1044116/1000000: episode: 2027, duration: 3.218s, episode steps: 765, steps per second: 238, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1044579/1000000: episode: 2028, duration: 1.947s, episode steps: 463, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1045015/1000000: episode: 2029, duration: 1.822s, episode steps: 436, steps per second: 239, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1045649/1000000: episode: 2030, duration: 2.651s, episode steps: 634, steps per second: 239, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1046124/1000000: episode: 2031, duration: 1.992s, episode steps: 475, steps per second: 238, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.217 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1046768/1000000: episode: 2032, duration: 2.705s, episode steps: 644, steps per second: 238, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1047290/1000000: episode: 2033, duration: 2.194s, episode steps: 522, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1047799/1000000: episode: 2034, duration: 2.130s, episode steps: 509, steps per second: 239, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1048197/1000000: episode: 2035, duration: 1.674s, episode steps: 398, steps per second: 238, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.332 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1048712/1000000: episode: 2036, duration: 2.169s, episode steps: 515, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049075/1000000: episode: 2037, duration: 1.523s, episode steps: 363, steps per second: 238, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049633/1000000: episode: 2038, duration: 2.351s, episode steps: 558, steps per second: 237, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049994/1000000: episode: 2039, duration: 1.545s, episode steps: 361, steps per second: 234, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1050511/1000000: episode: 2040, duration: 2.155s, episode steps: 517, steps per second: 240, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1051033/1000000: episode: 2041, duration: 2.179s, episode steps: 522, steps per second: 240, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1051564/1000000: episode: 2042, duration: 2.210s, episode steps: 531, steps per second: 240, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052184/1000000: episode: 2043, duration: 2.586s, episode steps: 620, steps per second: 240, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.342 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052561/1000000: episode: 2044, duration: 1.585s, episode steps: 377, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052978/1000000: episode: 2045, duration: 1.758s, episode steps: 417, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1053509/1000000: episode: 2046, duration: 2.221s, episode steps: 531, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1054009/1000000: episode: 2047, duration: 2.095s, episode steps: 500, steps per second: 239, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1054412/1000000: episode: 2048, duration: 1.691s, episode steps: 403, steps per second: 238, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1054781/1000000: episode: 2049, duration: 1.551s, episode steps: 369, steps per second: 238, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1055173/1000000: episode: 2050, duration: 1.646s, episode steps: 392, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1055593/1000000: episode: 2051, duration: 1.758s, episode steps: 420, steps per second: 239, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1056426/1000000: episode: 2052, duration: 3.441s, episode steps: 833, steps per second: 242, episode reward: 25.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1057004/1000000: episode: 2053, duration: 2.399s, episode steps: 578, steps per second: 241, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1057385/1000000: episode: 2054, duration: 1.585s, episode steps: 381, steps per second: 240, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1057959/1000000: episode: 2055, duration: 2.406s, episode steps: 574, steps per second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1058544/1000000: episode: 2056, duration: 2.449s, episode steps: 585, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1058848/1000000: episode: 2057, duration: 11.064s, episode steps: 304, steps per second: 27, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.041491, mae: 2.794304, mean_q: 3.391244, mean_eps: 0.987837

1059411/1000000: episode: 2058, duration: 22.406s, episode steps: 563, steps per second: 25, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.044318, mae: 2.851339, mean_q: 3.457834, mean_eps: 0.987338

1059795/1000000: episode: 2059, duration: 15.300s, episode steps: 384, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.036862, mae: 2.828921, mean_q: 3.427459, mean_eps: 0.986769

1060254/1000000: episode: 2060, duration: 18.353s, episode steps: 459, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.029565, mae: 2.843861, mean_q: 3.447619, mean_eps: 0.986262

1060666/1000000: episode: 2061, duration: 16.389s, episode steps: 412, steps per second: 25, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000],

mean action: 2.410 [0.000, 5.000], loss: 0.036913, mae: 2.814846, mean_q:
 3.407349, mean_eps: 0.985739
 1061110/1000000: episode: 2062, duration: 17.752s, episode steps: 444, steps
 per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000],
 mean action: 2.486 [0.000, 5.000], loss: 0.039469, mae: 2.876901, mean_q:
 3.485472, mean_eps: 0.985226
 1061779/1000000: episode: 2063, duration: 26.657s, episode steps: 669, steps
 per second: 25, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000],
 mean action: 2.414 [0.000, 5.000], loss: 0.039217, mae: 2.858745, mean_q:
 3.460907, mean_eps: 0.984558
 1062835/1000000: episode: 2064, duration: 42.049s, episode steps: 1056, steps
 per second: 25, episode reward: 12.000, mean reward: 0.011 [0.000, 1.000],
 mean action: 2.569 [0.000, 5.000], loss: 0.036068, mae: 2.852734, mean_q:
 3.456320, mean_eps: 0.983524
 1063351/1000000: episode: 2065, duration: 20.622s, episode steps: 516, steps
 per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000],
 mean action: 2.500 [0.000, 5.000], loss: 0.035429, mae: 2.855215, mean_q:
 3.461723, mean_eps: 0.982581
 1064164/1000000: episode: 2066, duration: 32.400s, episode steps: 813, steps
 per second: 25, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000],
 mean action: 2.526 [0.000, 5.000], loss: 0.037766, mae: 2.875971, mean_q:
 3.487952, mean_eps: 0.981784
 1064549/1000000: episode: 2067, duration: 15.630s, episode steps: 385, steps
 per second: 25, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000],
 mean action: 2.374 [0.000, 5.000], loss: 0.039674, mae: 2.865714, mean_q:
 3.473147, mean_eps: 0.981064
 1064929/1000000: episode: 2068, duration: 15.138s, episode steps: 380, steps
 per second: 25, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000],
 mean action: 2.545 [0.000, 5.000], loss: 0.038741, mae: 2.867673, mean_q:
 3.475220, mean_eps: 0.980603
 1065412/1000000: episode: 2069, duration: 19.186s, episode steps: 483, steps
 per second: 25, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000],
 mean action: 2.414 [0.000, 5.000], loss: 0.040608, mae: 2.868202, mean_q:
 3.478637, mean_eps: 0.980087
 1065989/1000000: episode: 2070, duration: 23.169s, episode steps: 577, steps
 per second: 25, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000],
 mean action: 2.459 [0.000, 5.000], loss: 0.044262, mae: 2.874084, mean_q:
 3.483241, mean_eps: 0.979451
 1066680/1000000: episode: 2071, duration: 27.445s, episode steps: 691, steps
 per second: 25, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000],
 mean action: 2.575 [0.000, 5.000], loss: 0.040078, mae: 2.896659, mean_q:
 3.512580, mean_eps: 0.978690
 1067258/1000000: episode: 2072, duration: 23.049s, episode steps: 578, steps
 per second: 25, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000],
 mean action: 2.557 [0.000, 5.000], loss: 0.043864, mae: 2.883430, mean_q:
 3.496088, mean_eps: 0.977930
 1067548/1000000: episode: 2073, duration: 11.637s, episode steps: 290, steps
 per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000],

mean action: 2.659 [0.000, 5.000], loss: 0.036861, mae: 2.891609, mean_q: 3.502162, mean_eps: 0.977409

1068305/1000000: episode: 2074, duration: 30.297s, episode steps: 757, steps per second: 25, episode reward: 23.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.038435, mae: 2.876453, mean_q: 3.485105, mean_eps: 0.976780

1068718/1000000: episode: 2075, duration: 16.393s, episode steps: 413, steps per second: 25, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.339 [0.000, 5.000], loss: 0.043551, mae: 2.908069, mean_q: 3.523642, mean_eps: 0.976077

1069255/1000000: episode: 2076, duration: 21.231s, episode steps: 537, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.035198, mae: 2.903613, mean_q: 3.519181, mean_eps: 0.975508

1069808/1000000: episode: 2077, duration: 22.193s, episode steps: 553, steps per second: 25, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.037664, mae: 2.897463, mean_q: 3.521178, mean_eps: 0.974855

1070536/1000000: episode: 2078, duration: 28.962s, episode steps: 728, steps per second: 25, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.036415, mae: 2.927576, mean_q: 3.545545, mean_eps: 0.974087

1070828/1000000: episode: 2079, duration: 11.775s, episode steps: 292, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.037157, mae: 2.923763, mean_q: 3.540962, mean_eps: 0.973475

1071489/1000000: episode: 2080, duration: 26.400s, episode steps: 661, steps per second: 25, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.033516, mae: 2.915952, mean_q: 3.533395, mean_eps: 0.972902

[Callback] Pesos guardados en:
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_2080_episodes.h5f

1071766/1000000: episode: 2081, duration: 11.062s, episode steps: 277, steps per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.038377, mae: 2.881275, mean_q: 3.498442, mean_eps: 0.972338

1072422/1000000: episode: 2082, duration: 26.209s, episode steps: 656, steps per second: 25, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.040935, mae: 2.895548, mean_q: 3.509367, mean_eps: 0.971778

1072928/1000000: episode: 2083, duration: 20.306s, episode steps: 506, steps per second: 25, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: 0.038461, mae: 2.903391, mean_q: 3.519488, mean_eps: 0.971082

1074070/1000000: episode: 2084, duration: 45.727s, episode steps: 1142, steps per second: 25, episode reward: 16.000, mean reward: 0.014 [0.000, 1.000],

mean action: 2.467 [0.000, 5.000], loss: 0.037804, mae: 2.920325, mean_q: 3.545672, mean_eps: 0.970094

1074723/1000000: episode: 2085, duration: 26.456s, episode steps: 653, steps per second: 25, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: 0.035881, mae: 2.960002, mean_q: 3.591002, mean_eps: 0.969016

1075510/1000000: episode: 2086, duration: 31.413s, episode steps: 787, steps per second: 25, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.037687, mae: 2.944805, mean_q: 3.569966, mean_eps: 0.968152

1076051/1000000: episode: 2087, duration: 21.586s, episode steps: 541, steps per second: 25, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.032564, mae: 2.953081, mean_q: 3.585729, mean_eps: 0.967355

1076341/1000000: episode: 2088, duration: 11.601s, episode steps: 290, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.040282, mae: 2.916865, mean_q: 3.536853, mean_eps: 0.966856

1076916/1000000: episode: 2089, duration: 22.958s, episode steps: 575, steps per second: 25, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: 0.039395, mae: 2.928001, mean_q: 3.549165, mean_eps: 0.966338

1077280/1000000: episode: 2090, duration: 14.787s, episode steps: 364, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.037990, mae: 2.971223, mean_q: 3.605388, mean_eps: 0.965776

1077816/1000000: episode: 2091, duration: 21.568s, episode steps: 536, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.038235, mae: 2.957398, mean_q: 3.588417, mean_eps: 0.965236

1078366/1000000: episode: 2092, duration: 22.105s, episode steps: 550, steps per second: 25, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: 0.036257, mae: 2.924938, mean_q: 3.547086, mean_eps: 0.964583

1078759/1000000: episode: 2093, duration: 15.586s, episode steps: 393, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.035746, mae: 2.995226, mean_q: 3.635098, mean_eps: 0.964017

1079079/1000000: episode: 2094, duration: 12.717s, episode steps: 320, steps per second: 25, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.362 [0.000, 5.000], loss: 0.039686, mae: 2.981720, mean_q: 3.617164, mean_eps: 0.963590

1079596/1000000: episode: 2095, duration: 20.709s, episode steps: 517, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.034048, mae: 2.961718, mean_q: 3.591613, mean_eps: 0.963088

1080163/1000000: episode: 2096, duration: 22.703s, episode steps: 567, steps per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000],

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mean action: 2.533 [0.000, 5.000], loss: 0.038940, mae: 2.998672, mean_q:
3.639822, mean_eps: 0.962438
1080984/1000000: episode: 2097, duration: 32.759s, episode steps: 821, steps
per second: 25, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.497 [0.000, 5.000], loss: 0.040367, mae: 2.979968, mean_q:
3.611327, mean_eps: 0.961605

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

1081649/1000000: episode: 2098, duration: 15.621s, episode steps: 665, steps
per second: 43, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.469 [0.000, 5.000], loss: 0.047394, mae: 2.937224, mean_q:
3.558984, mean_eps: 0.960897
1081947/1000000: episode: 2099, duration: 1.252s, episode steps: 298, steps per
second: 238, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1082658/1000000: episode: 2100, duration: 2.973s, episode steps: 711, steps per
second: 239, episode reward: 21.000, mean reward: 0.030 [0.000, 1.000], mean
action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1083165/1000000: episode: 2101, duration: 2.123s, episode steps: 507, steps per
second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1083613/1000000: episode: 2102, duration: 1.873s, episode steps: 448, steps per
second: 239, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1084222/1000000: episode: 2103, duration: 2.576s, episode steps: 609, steps per
second: 236, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1084905/1000000: episode: 2104, duration: 2.897s, episode steps: 683, steps per
second: 236, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1085411/1000000: episode: 2105, duration: 2.116s, episode steps: 506, steps per
second: 239, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1085844/1000000: episode: 2106, duration: 1.816s, episode steps: 433, steps per
second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1086347/1000000: episode: 2107, duration: 2.109s, episode steps: 503, steps per
second: 239, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1086886/1000000: episode: 2108, duration: 2.258s, episode steps: 539, steps per
second: 239, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean
action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1087541/1000000: episode: 2109, duration: 2.746s, episode steps: 655, steps per
second: 239, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1087890/1000000: episode: 2110, duration: 1.460s, episode steps: 349, steps per

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second: 239, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1088601/1000000: episode: 2111, duration: 2.958s, episode steps: 711, steps per second: 240, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1089722/1000000: episode: 2112, duration: 4.658s, episode steps: 1121, steps per second: 241, episode reward: 15.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1090237/1000000: episode: 2113, duration: 2.179s, episode steps: 515, steps per second: 236, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1090695/1000000: episode: 2114, duration: 1.918s, episode steps: 458, steps per second: 239, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.697 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1091107/1000000: episode: 2115, duration: 1.734s, episode steps: 412, steps per second: 238, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1091560/1000000: episode: 2116, duration: 1.904s, episode steps: 453, steps per second: 238, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1092064/1000000: episode: 2117, duration: 2.107s, episode steps: 504, steps per second: 239, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1092565/1000000: episode: 2118, duration: 2.089s, episode steps: 501, steps per second: 240, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1093427/1000000: episode: 2119, duration: 3.623s, episode steps: 862, steps per second: 238, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1093808/1000000: episode: 2120, duration: 1.599s, episode steps: 381, steps per second: 238, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1094705/1000000: episode: 2121, duration: 3.745s, episode steps: 897, steps per second: 240, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1095582/1000000: episode: 2122, duration: 3.641s, episode steps: 877, steps per second: 241, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1096200/1000000: episode: 2123, duration: 2.635s, episode steps: 618, steps per second: 234, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1096761/1000000: episode: 2124, duration: 2.342s, episode steps: 561, steps per second: 240, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1097245/1000000: episode: 2125, duration: 2.036s, episode steps: 484, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1097767/1000000: episode: 2126, duration: 2.196s, episode steps: 522, steps per

second: 238, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1098067/1000000: episode: 2127, duration: 1.274s, episode steps: 300, steps per second: 235, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1098441/1000000: episode: 2128, duration: 1.567s, episode steps: 374, steps per second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1098923/1000000: episode: 2129, duration: 2.037s, episode steps: 482, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1099432/1000000: episode: 2130, duration: 2.171s, episode steps: 509, steps per second: 234, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1099813/1000000: episode: 2131, duration: 1.607s, episode steps: 381, steps per second: 237, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1101007/1000000: episode: 2132, duration: 4.948s, episode steps: 1194, steps per second: 241, episode reward: 15.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1101392/1000000: episode: 2133, duration: 1.620s, episode steps: 385, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1102170/1000000: episode: 2134, duration: 3.270s, episode steps: 778, steps per second: 238, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1102611/1000000: episode: 2135, duration: 1.853s, episode steps: 441, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.270 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1102878/1000000: episode: 2136, duration: 1.125s, episode steps: 267, steps per second: 237, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1103358/1000000: episode: 2137, duration: 2.008s, episode steps: 480, steps per second: 239, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1103851/1000000: episode: 2138, duration: 2.072s, episode steps: 493, steps per second: 238, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1104284/1000000: episode: 2139, duration: 1.824s, episode steps: 433, steps per second: 237, episode reward: 11.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1104573/1000000: episode: 2140, duration: 1.224s, episode steps: 289, steps per second: 236, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1104981/1000000: episode: 2141, duration: 1.725s, episode steps: 408, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.360 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1105472/1000000: episode: 2142, duration: 2.060s, episode steps: 491, steps per

second: 238, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1106058/1000000: episode: 2143, duration: 2.442s, episode steps: 586, steps per second: 240, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.650 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1106557/1000000: episode: 2144, duration: 2.077s, episode steps: 499, steps per second: 240, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1107018/1000000: episode: 2145, duration: 1.940s, episode steps: 461, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1108204/1000000: episode: 2146, duration: 4.945s, episode steps: 1186, steps per second: 240, episode reward: 18.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1108932/1000000: episode: 2147, duration: 3.041s, episode steps: 728, steps per second: 239, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1109387/1000000: episode: 2148, duration: 1.914s, episode steps: 455, steps per second: 238, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1110445/1000000: episode: 2149, duration: 4.401s, episode steps: 1058, steps per second: 240, episode reward: 16.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.370 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1110965/1000000: episode: 2150, duration: 2.206s, episode steps: 520, steps per second: 236, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1111437/1000000: episode: 2151, duration: 2.013s, episode steps: 472, steps per second: 234, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1111917/1000000: episode: 2152, duration: 2.023s, episode steps: 480, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1112394/1000000: episode: 2153, duration: 1.998s, episode steps: 477, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1112882/1000000: episode: 2154, duration: 2.054s, episode steps: 488, steps per second: 238, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1113188/1000000: episode: 2155, duration: 1.309s, episode steps: 306, steps per second: 234, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1113671/1000000: episode: 2156, duration: 2.049s, episode steps: 483, steps per second: 236, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1114654/1000000: episode: 2157, duration: 4.112s, episode steps: 983, steps per second: 239, episode reward: 25.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1115169/1000000: episode: 2158, duration: 2.150s, episode steps: 515, steps per

second: 240, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1115713/1000000: episode: 2159, duration: 2.273s, episode steps: 544, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1116208/1000000: episode: 2160, duration: 2.069s, episode steps: 495, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1116684/1000000: episode: 2161, duration: 2.011s, episode steps: 476, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1117239/1000000: episode: 2162, duration: 2.339s, episode steps: 555, steps per second: 237, episode reward: 17.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.241 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1117685/1000000: episode: 2163, duration: 1.887s, episode steps: 446, steps per second: 236, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1118231/1000000: episode: 2164, duration: 2.285s, episode steps: 546, steps per second: 239, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1118873/1000000: episode: 2165, duration: 2.679s, episode steps: 642, steps per second: 240, episode reward: 18.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1119280/1000000: episode: 2166, duration: 1.723s, episode steps: 407, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1120148/1000000: episode: 2167, duration: 3.638s, episode steps: 868, steps per second: 239, episode reward: 23.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1120455/1000000: episode: 2168, duration: 1.296s, episode steps: 307, steps per second: 237, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.684 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1120921/1000000: episode: 2169, duration: 1.952s, episode steps: 466, steps per second: 239, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1121245/1000000: episode: 2170, duration: 1.380s, episode steps: 324, steps per second: 235, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1121834/1000000: episode: 2171, duration: 2.458s, episode steps: 589, steps per second: 240, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1122288/1000000: episode: 2172, duration: 1.904s, episode steps: 454, steps per second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1123189/1000000: episode: 2173, duration: 3.780s, episode steps: 901, steps per second: 238, episode reward: 10.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1123596/1000000: episode: 2174, duration: 1.713s, episode steps: 407, steps per

second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1124054/1000000: episode: 2175, duration: 1.921s, episode steps: 458, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1124452/1000000: episode: 2176, duration: 13.802s, episode steps: 398, steps per second: 29, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.036599, mae: 2.972016, mean_q: 3.607439, mean_eps: 0.987796

1124857/1000000: episode: 2177, duration: 16.197s, episode steps: 405, steps per second: 25, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.043714, mae: 3.026872, mean_q: 3.672023, mean_eps: 0.987350

1125228/1000000: episode: 2178, duration: 14.948s, episode steps: 371, steps per second: 25, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: 0.037746, mae: 3.036552, mean_q: 3.683045, mean_eps: 0.986884

1125741/1000000: episode: 2179, duration: 20.622s, episode steps: 513, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.037421, mae: 3.004720, mean_q: 3.646477, mean_eps: 0.986354

1126251/1000000: episode: 2180, duration: 20.397s, episode steps: 510, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.035036, mae: 3.013787, mean_q: 3.656419, mean_eps: 0.985739

1126900/1000000: episode: 2181, duration: 26.063s, episode steps: 649, steps per second: 25, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.035319, mae: 3.042380, mean_q: 3.690793, mean_eps: 0.985046

1127188/1000000: episode: 2182, duration: 11.695s, episode steps: 288, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.029488, mae: 2.993759, mean_q: 3.630642, mean_eps: 0.984484

1127683/1000000: episode: 2183, duration: 19.787s, episode steps: 495, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: 0.033204, mae: 3.012066, mean_q: 3.656795, mean_eps: 0.984014

1128034/1000000: episode: 2184, duration: 14.096s, episode steps: 351, steps per second: 25, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.041409, mae: 3.014393, mean_q: 3.665544, mean_eps: 0.983505

1128496/1000000: episode: 2185, duration: 18.559s, episode steps: 462, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.029835, mae: 2.988070, mean_q: 3.624495, mean_eps: 0.983018

1129326/1000000: episode: 2186, duration: 33.192s, episode steps: 830, steps per second: 25, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.036060, mae: 3.022232, mean_q:

3.663928, mean_eps: 0.982242
1129611/1000000: episode: 2187, duration: 11.342s, episode steps: 285, steps per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: 0.032198, mae: 3.010409, mean_q: 3.656664, mean_eps: 0.981573
1130417/1000000: episode: 2188, duration: 32.172s, episode steps: 806, steps per second: 25, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.033136, mae: 3.040819, mean_q: 3.688390, mean_eps: 0.980918
1130996/1000000: episode: 2189, duration: 23.160s, episode steps: 579, steps per second: 25, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.038224, mae: 3.047608, mean_q: 3.691349, mean_eps: 0.980087
1131280/1000000: episode: 2190, duration: 11.569s, episode steps: 284, steps per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.034053, mae: 3.028917, mean_q: 3.676956, mean_eps: 0.979571
1131796/1000000: episode: 2191, duration: 20.751s, episode steps: 516, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.032479, mae: 3.055361, mean_q: 3.711056, mean_eps: 0.979091
1132189/1000000: episode: 2192, duration: 15.715s, episode steps: 393, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.047321, mae: 2.989819, mean_q: 3.620763, mean_eps: 0.978544
1132764/1000000: episode: 2193, duration: 22.979s, episode steps: 575, steps per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.038710, mae: 3.012415, mean_q: 3.653255, mean_eps: 0.977963
1133273/1000000: episode: 2194, duration: 20.339s, episode steps: 509, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.042522, mae: 3.044161, mean_q: 3.686270, mean_eps: 0.977313
1134151/1000000: episode: 2195, duration: 35.015s, episode steps: 878, steps per second: 25, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.037214, mae: 3.032635, mean_q: 3.676475, mean_eps: 0.976480
1134634/1000000: episode: 2196, duration: 19.337s, episode steps: 483, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.037663, mae: 3.048628, mean_q: 3.698664, mean_eps: 0.975664
1135259/1000000: episode: 2197, duration: 24.892s, episode steps: 625, steps per second: 25, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.334 [0.000, 5.000], loss: 0.038283, mae: 3.100187, mean_q: 3.758062, mean_eps: 0.974999
1135654/1000000: episode: 2198, duration: 15.843s, episode steps: 395, steps per second: 25, episode reward: 11.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: 0.048116, mae: 3.111610, mean_q:

3.773002, mean_eps: 0.974387
 1135910/1000000: episode: 2199, duration: 10.282s, episode steps: 256, steps per second: 25, episode reward: 4.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.332 [0.000, 5.000], loss: 0.042458, mae: 3.087859, mean_q: 3.742776, mean_eps: 0.973996
 1136384/1000000: episode: 2200, duration: 19.078s, episode steps: 474, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: 0.034115, mae: 3.056050, mean_q: 3.708431, mean_eps: 0.973559
 1136754/1000000: episode: 2201, duration: 14.878s, episode steps: 370, steps per second: 25, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: 0.036212, mae: 3.090191, mean_q: 3.744127, mean_eps: 0.973053
 1137254/1000000: episode: 2202, duration: 19.910s, episode steps: 500, steps per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.039152, mae: 3.074708, mean_q: 3.730864, mean_eps: 0.972530
 1137797/1000000: episode: 2203, duration: 21.787s, episode steps: 543, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.036022, mae: 3.058133, mean_q: 3.710843, mean_eps: 0.971903
 1138294/1000000: episode: 2204, duration: 19.839s, episode steps: 497, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.035112, mae: 3.092560, mean_q: 3.751301, mean_eps: 0.971279
 1138893/1000000: episode: 2205, duration: 23.975s, episode steps: 599, steps per second: 25, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: 0.037768, mae: 3.070064, mean_q: 3.728536, mean_eps: 0.970622
 1139424/1000000: episode: 2206, duration: 21.335s, episode steps: 531, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.033015, mae: 3.103352, mean_q: 3.764072, mean_eps: 0.969945
 1139963/1000000: episode: 2207, duration: 21.699s, episode steps: 539, steps per second: 25, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.325 [0.000, 5.000], loss: 0.030797, mae: 3.087819, mean_q: 3.742116, mean_eps: 0.969304
 1140632/1000000: episode: 2208, duration: 26.742s, episode steps: 669, steps per second: 25, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.038789, mae: 3.092991, mean_q: 3.750450, mean_eps: 0.968579
 1141205/1000000: episode: 2209, duration: 23.063s, episode steps: 573, steps per second: 25, episode reward: 15.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.032423, mae: 3.158374, mean_q: 3.827533, mean_eps: 0.967833
 1141948/1000000: episode: 2210, duration: 29.791s, episode steps: 743, steps per second: 25, episode reward: 10.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.700 [0.000, 5.000], loss: 0.043316, mae: 3.095573, mean_q:

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3.754013, mean_eps: 0.967043
1142411/1000000: episode: 2211, duration: 18.539s, episode steps: 463, steps
per second: 25, episode reward: 12.000, mean reward: 0.026 [ 0.000, 1.000],
mean action: 2.523 [0.000, 5.000], loss: 0.034169, mae: 3.061312, mean_q:
3.711048, mean_eps: 0.966321
1143562/1000000: episode: 2212, duration: 45.900s, episode steps: 1151, steps
per second: 25, episode reward: 27.000, mean reward: 0.023 [ 0.000, 1.000],
mean action: 2.469 [0.000, 5.000], loss: 0.036535, mae: 3.101042, mean_q:
3.759692, mean_eps: 0.965351
1144124/1000000: episode: 2213, duration: 22.540s, episode steps: 562, steps
per second: 25, episode reward: 8.000, mean reward: 0.014 [ 0.000, 1.000],
mean action: 2.571 [0.000, 5.000], loss: 0.040434, mae: 3.080319, mean_q:
3.733651, mean_eps: 0.964324
1144622/1000000: episode: 2214, duration: 20.071s, episode steps: 498, steps
per second: 25, episode reward: 13.000, mean reward: 0.026 [ 0.000, 1.000],
mean action: 2.448 [0.000, 5.000], loss: 0.034256, mae: 3.158120, mean_q:
3.829807, mean_eps: 0.963688
1144994/1000000: episode: 2215, duration: 14.817s, episode steps: 372, steps
per second: 25, episode reward: 6.000, mean reward: 0.016 [ 0.000, 1.000],
mean action: 2.610 [0.000, 5.000], loss: 0.038666, mae: 3.133337, mean_q:
3.799112, mean_eps: 0.963165
1145259/1000000: episode: 2216, duration: 10.565s, episode steps: 265, steps
per second: 25, episode reward: 4.000, mean reward: 0.015 [ 0.000, 1.000],
mean action: 2.377 [0.000, 5.000], loss: 0.034201, mae: 3.137010, mean_q:
3.805003, mean_eps: 0.962783
1145771/1000000: episode: 2217, duration: 20.492s, episode steps: 512, steps
per second: 25, episode reward: 10.000, mean reward: 0.020 [ 0.000, 1.000],
mean action: 2.438 [0.000, 5.000], loss: 0.035745, mae: 3.148060, mean_q:
3.815523, mean_eps: 0.962318
1146150/1000000: episode: 2218, duration: 15.232s, episode steps: 379, steps
per second: 25, episode reward: 0.000, mean reward: 0.000 [ 0.000, 0.000],
mean action: 2.512 [0.000, 5.000], loss: 0.041477, mae: 3.084369, mean_q:
3.738275, mean_eps: 0.961782

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

1146956/1000000: episode: 2219, duration: 29.360s, episode steps: 806, steps
per second: 27, episode reward: 21.000, mean reward: 0.026 [ 0.000, 1.000],
mean action: 2.525 [0.000, 5.000], loss: 0.037703, mae: 3.087232, mean_q:
3.742196, mean_eps: 0.961118
1147691/1000000: episode: 2220, duration: 3.068s, episode steps: 735, steps per
second: 240, episode reward: 10.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1148142/1000000: episode: 2221, duration: 1.897s, episode steps: 451, steps per
second: 238, episode reward: 10.000, mean reward: 0.022 [ 0.000, 1.000], mean
action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1148443/1000000: episode: 2222, duration: 1.275s, episode steps: 301, steps per

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second: 236, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1148837/1000000: episode: 2223, duration: 1.653s, episode steps: 394, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1149366/1000000: episode: 2224, duration: 2.195s, episode steps: 529, steps per second: 241, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1149751/1000000: episode: 2225, duration: 1.622s, episode steps: 385, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1150431/1000000: episode: 2226, duration: 2.841s, episode steps: 680, steps per second: 239, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1150925/1000000: episode: 2227, duration: 2.092s, episode steps: 494, steps per second: 236, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1151425/1000000: episode: 2228, duration: 2.116s, episode steps: 500, steps per second: 236, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1151836/1000000: episode: 2229, duration: 1.725s, episode steps: 411, steps per second: 238, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1152261/1000000: episode: 2230, duration: 1.788s, episode steps: 425, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1152739/1000000: episode: 2231, duration: 2.009s, episode steps: 478, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1153598/1000000: episode: 2232, duration: 3.566s, episode steps: 859, steps per second: 241, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1154023/1000000: episode: 2233, duration: 1.802s, episode steps: 425, steps per second: 236, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1154431/1000000: episode: 2234, duration: 1.724s, episode steps: 408, steps per second: 237, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1154769/1000000: episode: 2235, duration: 1.429s, episode steps: 338, steps per second: 237, episode reward: 9.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1155087/1000000: episode: 2236, duration: 1.340s, episode steps: 318, steps per second: 237, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1155484/1000000: episode: 2237, duration: 1.666s, episode steps: 397, steps per second: 238, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1156025/1000000: episode: 2238, duration: 2.264s, episode steps: 541, steps per

second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1156917/1000000: episode: 2239, duration: 3.737s, episode steps: 892, steps per
second: 239, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1157435/1000000: episode: 2240, duration: 2.195s, episode steps: 518, steps per
second: 236, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/pesos/dqn_v_1.0_weights_2240_episodes.h5f

1157750/1000000: episode: 2241, duration: 1.432s, episode steps: 315, steps per
second: 220, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1158386/1000000: episode: 2242, duration: 2.657s, episode steps: 636, steps per
second: 239, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1158772/1000000: episode: 2243, duration: 1.615s, episode steps: 386, steps per
second: 239, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1159256/1000000: episode: 2244, duration: 2.034s, episode steps: 484, steps per
second: 238, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1159822/1000000: episode: 2245, duration: 2.381s, episode steps: 566, steps per
second: 238, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1160478/1000000: episode: 2246, duration: 2.760s, episode steps: 656, steps per
second: 238, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1160948/1000000: episode: 2247, duration: 1.972s, episode steps: 470, steps per
second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1161346/1000000: episode: 2248, duration: 1.674s, episode steps: 398, steps per
second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean
action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1161961/1000000: episode: 2249, duration: 2.571s, episode steps: 615, steps per
second: 239, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1162496/1000000: episode: 2250, duration: 2.249s, episode steps: 535, steps per
second: 238, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1163043/1000000: episode: 2251, duration: 2.318s, episode steps: 547, steps per
second: 236, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1163335/1000000: episode: 2252, duration: 1.230s, episode steps: 292, steps per
second: 237, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean
action: 2.716 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1163764/1000000: episode: 2253, duration: 1.801s, episode steps: 429, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1164263/1000000: episode: 2254, duration: 2.088s, episode steps: 499, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1164877/1000000: episode: 2255, duration: 2.570s, episode steps: 614, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1165369/1000000: episode: 2256, duration: 2.061s, episode steps: 492, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1165999/1000000: episode: 2257, duration: 2.664s, episode steps: 630, steps per second: 236, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1166608/1000000: episode: 2258, duration: 2.554s, episode steps: 609, steps per second: 238, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1167190/1000000: episode: 2259, duration: 2.446s, episode steps: 582, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1167614/1000000: episode: 2260, duration: 1.780s, episode steps: 424, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1168140/1000000: episode: 2261, duration: 2.228s, episode steps: 526, steps per second: 236, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1168591/1000000: episode: 2262, duration: 1.924s, episode steps: 451, steps per second: 234, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1169335/1000000: episode: 2263, duration: 3.142s, episode steps: 744, steps per second: 237, episode reward: 10.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1169721/1000000: episode: 2264, duration: 1.642s, episode steps: 386, steps per second: 235, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1170272/1000000: episode: 2265, duration: 2.329s, episode steps: 551, steps per second: 237, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1170687/1000000: episode: 2266, duration: 1.753s, episode steps: 415, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.706 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1171288/1000000: episode: 2267, duration: 2.525s, episode steps: 601, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1171978/1000000: episode: 2268, duration: 2.961s, episode steps: 690, steps per second: 233, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1172629/1000000: episode: 2269, duration: 2.746s, episode steps: 651, steps per second: 237, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1173064/1000000: episode: 2270, duration: 1.842s, episode steps: 435, steps per second: 236, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1173692/1000000: episode: 2271, duration: 2.627s, episode steps: 628, steps per second: 239, episode reward: 21.000, mean reward: 0.033 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1173968/1000000: episode: 2272, duration: 1.168s, episode steps: 276, steps per second: 236, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1174437/1000000: episode: 2273, duration: 1.991s, episode steps: 469, steps per second: 236, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1175066/1000000: episode: 2274, duration: 2.673s, episode steps: 629, steps per second: 235, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1175917/1000000: episode: 2275, duration: 3.567s, episode steps: 851, steps per second: 239, episode reward: 22.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1176370/1000000: episode: 2276, duration: 1.911s, episode steps: 453, steps per second: 237, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1177075/1000000: episode: 2277, duration: 2.970s, episode steps: 705, steps per second: 237, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1177862/1000000: episode: 2278, duration: 3.318s, episode steps: 787, steps per second: 237, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1178310/1000000: episode: 2279, duration: 1.882s, episode steps: 448, steps per second: 238, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1178723/1000000: episode: 2280, duration: 1.742s, episode steps: 413, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1179022/1000000: episode: 2281, duration: 1.265s, episode steps: 299, steps per second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1179429/1000000: episode: 2282, duration: 1.721s, episode steps: 407, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1179909/1000000: episode: 2283, duration: 2.027s, episode steps: 480, steps per second: 237, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1180382/1000000: episode: 2284, duration: 2.002s, episode steps: 473, steps per second: 236, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1180909/1000000: episode: 2285, duration: 2.227s, episode steps: 527, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1181727/1000000: episode: 2286, duration: 3.413s, episode steps: 818, steps per second: 240, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1182116/1000000: episode: 2287, duration: 1.632s, episode steps: 389, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.365 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1182637/1000000: episode: 2288, duration: 2.192s, episode steps: 521, steps per second: 238, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1183051/1000000: episode: 2289, duration: 1.746s, episode steps: 414, steps per second: 237, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1183551/1000000: episode: 2290, duration: 2.100s, episode steps: 500, steps per second: 238, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1183992/1000000: episode: 2291, duration: 1.861s, episode steps: 441, steps per second: 237, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1184362/1000000: episode: 2292, duration: 1.557s, episode steps: 370, steps per second: 238, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1185062/1000000: episode: 2293, duration: 2.935s, episode steps: 700, steps per second: 238, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1185527/1000000: episode: 2294, duration: 1.968s, episode steps: 465, steps per second: 236, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1186009/1000000: episode: 2295, duration: 2.043s, episode steps: 482, steps per second: 236, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1186509/1000000: episode: 2296, duration: 2.111s, episode steps: 500, steps per second: 237, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1187035/1000000: episode: 2297, duration: 2.228s, episode steps: 526, steps per second: 236, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1187444/1000000: episode: 2298, duration: 1.720s, episode steps: 409, steps per second: 238, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1188162/1000000: episode: 2299, duration: 3.001s, episode steps: 718, steps per second: 239, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1188550/1000000: episode: 2300, duration: 1.627s, episode steps: 388, steps per second: 238, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1188852/1000000: episode: 2301, duration: 1.269s, episode steps: 302, steps per second: 238, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1189567/1000000: episode: 2302, duration: 3.001s, episode steps: 715, steps per second: 238, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1189927/1000000: episode: 2303, duration: 11.388s, episode steps: 360, steps per second: 32, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.319 [0.000, 5.000], loss: 0.027558, mae: 3.213715, mean_q: 3.894178, mean_eps: 0.987832

1190411/1000000: episode: 2304, duration: 19.359s, episode steps: 484, steps per second: 25, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: 0.038225, mae: 3.174966, mean_q: 3.845138, mean_eps: 0.987376

1190923/1000000: episode: 2305, duration: 20.619s, episode steps: 512, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.037841, mae: 3.167467, mean_q: 3.839103, mean_eps: 0.986778

1191272/1000000: episode: 2306, duration: 14.030s, episode steps: 349, steps per second: 25, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.037673, mae: 3.163658, mean_q: 3.832822, mean_eps: 0.986262

1191941/1000000: episode: 2307, duration: 26.849s, episode steps: 669, steps per second: 25, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.036202, mae: 3.177531, mean_q: 3.848501, mean_eps: 0.985650

1192482/1000000: episode: 2308, duration: 21.484s, episode steps: 541, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.035239, mae: 3.179312, mean_q: 3.852472, mean_eps: 0.984923

1192956/1000000: episode: 2309, duration: 19.059s, episode steps: 474, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.036344, mae: 3.204141, mean_q: 3.882075, mean_eps: 0.984316

1193266/1000000: episode: 2310, duration: 12.490s, episode steps: 310, steps per second: 25, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.035086, mae: 3.160163, mean_q: 3.821662, mean_eps: 0.983846

1193926/1000000: episode: 2311, duration: 26.450s, episode steps: 660, steps per second: 25, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.042284, mae: 3.180509, mean_q: 3.853171, mean_eps: 0.983262

1194444/1000000: episode: 2312, duration: 20.760s, episode steps: 518, steps per second: 25, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.042371, mae: 3.202427, mean_q: 3.882518, mean_eps: 0.982557

1195391/1000000: episode: 2313, duration: 38.073s, episode steps: 947, steps per second: 25, episode reward: 22.000, mean reward: 0.023 [0.000, 1.000],

mean action: 2.603 [0.000, 5.000], loss: 0.043944, mae: 3.200558, mean_q: 3.878489, mean_eps: 0.981678

1195770/1000000: episode: 2314, duration: 15.295s, episode steps: 379, steps per second: 25, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: 0.032078, mae: 3.227229, mean_q: 3.909863, mean_eps: 0.980882

1196345/1000000: episode: 2315, duration: 23.013s, episode steps: 575, steps per second: 25, episode reward: 6.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.032208, mae: 3.194691, mean_q: 3.872044, mean_eps: 0.980308

1196937/1000000: episode: 2316, duration: 23.681s, episode steps: 592, steps per second: 25, episode reward: 6.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: 0.034001, mae: 3.198392, mean_q: 3.876495, mean_eps: 0.979607

1197568/1000000: episode: 2317, duration: 25.245s, episode steps: 631, steps per second: 25, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.038592, mae: 3.205916, mean_q: 3.887322, mean_eps: 0.978875

1198470/1000000: episode: 2318, duration: 36.145s, episode steps: 902, steps per second: 25, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: 0.044549, mae: 3.172741, mean_q: 3.847350, mean_eps: 0.977956

1199203/1000000: episode: 2319, duration: 29.260s, episode steps: 733, steps per second: 25, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.671 [0.000, 5.000], loss: 0.040329, mae: 3.178240, mean_q: 3.850488, mean_eps: 0.976974

1199649/1000000: episode: 2320, duration: 18.017s, episode steps: 446, steps per second: 25, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.036940, mae: 3.167322, mean_q: 3.839015, mean_eps: 0.976266

1200071/1000000: episode: 2321, duration: 16.917s, episode steps: 422, steps per second: 25, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.033223, mae: 3.238115, mean_q: 3.926864, mean_eps: 0.975746

1200682/1000000: episode: 2322, duration: 24.313s, episode steps: 611, steps per second: 25, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.037486, mae: 3.163485, mean_q: 3.835121, mean_eps: 0.975126

1201087/1000000: episode: 2323, duration: 16.073s, episode steps: 405, steps per second: 25, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.037821, mae: 3.212426, mean_q: 3.888433, mean_eps: 0.974517

1201558/1000000: episode: 2324, duration: 18.885s, episode steps: 471, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: 0.035347, mae: 3.215353, mean_q: 3.893840, mean_eps: 0.973991

1201856/1000000: episode: 2325, duration: 11.905s, episode steps: 298, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000],

mean action: 2.346 [0.000, 5.000], loss: 0.038528, mae: 3.198188, mean_q: 3.870997, mean_eps: 0.973530

1202347/1000000: episode: 2326, duration: 19.543s, episode steps: 491, steps per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: 0.041336, mae: 3.208434, mean_q: 3.888307, mean_eps: 0.973058

1202950/1000000: episode: 2327, duration: 24.130s, episode steps: 603, steps per second: 25, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: 0.037316, mae: 3.179738, mean_q: 3.854566, mean_eps: 0.972400

1203219/1000000: episode: 2328, duration: 10.731s, episode steps: 269, steps per second: 25, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.034169, mae: 3.205130, mean_q: 3.882588, mean_eps: 0.971877

1203748/1000000: episode: 2329, duration: 21.154s, episode steps: 529, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.048082, mae: 3.174771, mean_q: 3.845096, mean_eps: 0.971399

1204223/1000000: episode: 2330, duration: 18.930s, episode steps: 475, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.036893, mae: 3.175409, mean_q: 3.847626, mean_eps: 0.970797

1204591/1000000: episode: 2331, duration: 14.793s, episode steps: 368, steps per second: 25, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: 0.057107, mae: 3.173532, mean_q: 3.842516, mean_eps: 0.970290

1205421/1000000: episode: 2332, duration: 33.227s, episode steps: 830, steps per second: 25, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: 0.036400, mae: 3.206074, mean_q: 3.884537, mean_eps: 0.969570

1205687/1000000: episode: 2333, duration: 10.530s, episode steps: 266, steps per second: 25, episode reward: 5.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: 0.034800, mae: 3.174663, mean_q: 3.845546, mean_eps: 0.968913

1206310/1000000: episode: 2334, duration: 24.868s, episode steps: 623, steps per second: 25, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.031790, mae: 3.219793, mean_q: 3.900767, mean_eps: 0.968380

1206887/1000000: episode: 2335, duration: 23.054s, episode steps: 577, steps per second: 25, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.037868, mae: 3.234728, mean_q: 3.918032, mean_eps: 0.967660

1207698/1000000: episode: 2336, duration: 32.356s, episode steps: 811, steps per second: 25, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.036575, mae: 3.245357, mean_q: 3.930983, mean_eps: 0.966827

1208255/1000000: episode: 2337, duration: 22.120s, episode steps: 557, steps per second: 25, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000],

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mean action: 2.420 [0.000, 5.000], loss: 0.031056, mae: 3.221867, mean_q:
3.900925, mean_eps: 0.966006
1208829/1000000: episode: 2338, duration: 22.981s, episode steps: 574, steps
per second: 25, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.521 [0.000, 5.000], loss: 0.035840, mae: 3.235118, mean_q:
3.916047, mean_eps: 0.965327
1209291/1000000: episode: 2339, duration: 18.466s, episode steps: 462, steps
per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000],
mean action: 2.329 [0.000, 5.000], loss: 0.037905, mae: 3.208470, mean_q:
3.888250, mean_eps: 0.964706
1210022/1000000: episode: 2340, duration: 29.220s, episode steps: 731, steps
per second: 25, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.540 [0.000, 5.000], loss: 0.038034, mae: 3.274976, mean_q:
3.967313, mean_eps: 0.963990
1210614/1000000: episode: 2341, duration: 23.632s, episode steps: 592, steps
per second: 25, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.600 [0.000, 5.000], loss: 0.044298, mae: 3.318833, mean_q:
4.022338, mean_eps: 0.963196
1210915/1000000: episode: 2342, duration: 12.057s, episode steps: 301, steps
per second: 25, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000],
mean action: 2.532 [0.000, 5.000], loss: 0.035646, mae: 3.276183, mean_q:
3.964767, mean_eps: 0.962661
1211372/1000000: episode: 2343, duration: 18.398s, episode steps: 457, steps
per second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000],
mean action: 2.654 [0.000, 5.000], loss: 0.034211, mae: 3.271947, mean_q:
3.968864, mean_eps: 0.962207
1212004/1000000: episode: 2344, duration: 25.394s, episode steps: 632, steps
per second: 25, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.544 [0.000, 5.000], loss: 0.039718, mae: 3.276755, mean_q:
3.966318, mean_eps: 0.961554
1212274/1000000: episode: 2345, duration: 10.882s, episode steps: 270, steps
per second: 25, episode reward: 5.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.556 [0.000, 5.000], loss: 0.042792, mae: 3.339322, mean_q:
4.043056, mean_eps: 0.961012

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

1213114/1000000: episode: 2346, duration: 8.572s, episode steps: 840, steps per
second: 98, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.489 [0.000, 5.000], loss: 0.047027, mae: 3.328324, mean_q: 4.036859,
mean_eps: 0.960765
1213411/1000000: episode: 2347, duration: 1.256s, episode steps: 297, steps per
second: 236, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1213904/1000000: episode: 2348, duration: 2.090s, episode steps: 493, steps per
second: 236, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean
action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1214401/1000000: episode: 2349, duration: 2.093s, episode steps: 497, steps per second: 237, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1214686/1000000: episode: 2350, duration: 1.218s, episode steps: 285, steps per second: 234, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.656 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1215338/1000000: episode: 2351, duration: 2.742s, episode steps: 652, steps per second: 238, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1215823/1000000: episode: 2352, duration: 2.046s, episode steps: 485, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.365 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1216214/1000000: episode: 2353, duration: 1.652s, episode steps: 391, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1216904/1000000: episode: 2354, duration: 2.892s, episode steps: 690, steps per second: 239, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1217245/1000000: episode: 2355, duration: 1.440s, episode steps: 341, steps per second: 237, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1217727/1000000: episode: 2356, duration: 2.048s, episode steps: 482, steps per second: 235, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1218428/1000000: episode: 2357, duration: 2.953s, episode steps: 701, steps per second: 237, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1218921/1000000: episode: 2358, duration: 2.060s, episode steps: 493, steps per second: 239, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1219313/1000000: episode: 2359, duration: 1.656s, episode steps: 392, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1219793/1000000: episode: 2360, duration: 2.016s, episode steps: 480, steps per second: 238, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.356 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1220518/1000000: episode: 2361, duration: 3.046s, episode steps: 725, steps per second: 238, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1220982/1000000: episode: 2362, duration: 1.999s, episode steps: 464, steps per second: 232, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1221686/1000000: episode: 2363, duration: 2.947s, episode steps: 704, steps per second: 239, episode reward: 19.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1222096/1000000: episode: 2364, duration: 1.717s, episode steps: 410, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1222586/1000000: episode: 2365, duration: 2.049s, episode steps: 490, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1222951/1000000: episode: 2366, duration: 1.530s, episode steps: 365, steps per second: 239, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1223517/1000000: episode: 2367, duration: 2.389s, episode steps: 566, steps per second: 237, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1223824/1000000: episode: 2368, duration: 1.320s, episode steps: 307, steps per second: 233, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1224525/1000000: episode: 2369, duration: 2.936s, episode steps: 701, steps per second: 239, episode reward: 19.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1224901/1000000: episode: 2370, duration: 1.588s, episode steps: 376, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1225295/1000000: episode: 2371, duration: 1.676s, episode steps: 394, steps per second: 235, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1226265/1000000: episode: 2372, duration: 4.052s, episode steps: 970, steps per second: 239, episode reward: 17.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1226780/1000000: episode: 2373, duration: 2.191s, episode steps: 515, steps per second: 235, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1227133/1000000: episode: 2374, duration: 1.485s, episode steps: 353, steps per second: 238, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1227956/1000000: episode: 2375, duration: 3.437s, episode steps: 823, steps per second: 239, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1228443/1000000: episode: 2376, duration: 2.051s, episode steps: 487, steps per second: 237, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1228731/1000000: episode: 2377, duration: 1.214s, episode steps: 288, steps per second: 237, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1229242/1000000: episode: 2378, duration: 2.151s, episode steps: 511, steps per second: 238, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1229691/1000000: episode: 2379, duration: 1.910s, episode steps: 449, steps per second: 235, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1230255/1000000: episode: 2380, duration: 2.376s, episode steps: 564, steps per second: 237, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1230757/1000000: episode: 2381, duration: 2.090s, episode steps: 502, steps per second: 240, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1231359/1000000: episode: 2382, duration: 2.534s, episode steps: 602, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1232034/1000000: episode: 2383, duration: 2.849s, episode steps: 675, steps per second: 237, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1232507/1000000: episode: 2384, duration: 2.025s, episode steps: 473, steps per second: 234, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1232992/1000000: episode: 2385, duration: 2.043s, episode steps: 485, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.625 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1233395/1000000: episode: 2386, duration: 1.686s, episode steps: 403, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1234231/1000000: episode: 2387, duration: 3.469s, episode steps: 836, steps per second: 241, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1234755/1000000: episode: 2388, duration: 2.196s, episode steps: 524, steps per second: 239, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1235128/1000000: episode: 2389, duration: 1.574s, episode steps: 373, steps per second: 237, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1235632/1000000: episode: 2390, duration: 2.156s, episode steps: 504, steps per second: 234, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1236223/1000000: episode: 2391, duration: 2.499s, episode steps: 591, steps per second: 236, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1237008/1000000: episode: 2392, duration: 3.281s, episode steps: 785, steps per second: 239, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1237478/1000000: episode: 2393, duration: 1.971s, episode steps: 470, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1238254/1000000: episode: 2394, duration: 3.254s, episode steps: 776, steps per second: 238, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1238968/1000000: episode: 2395, duration: 2.995s, episode steps: 714, steps per second: 238, episode reward: 19.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1239642/1000000: episode: 2396, duration: 2.815s, episode steps: 674, steps per second: 239, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1240199/1000000: episode: 2397, duration: 2.345s, episode steps: 557, steps per second: 238, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1240808/1000000: episode: 2398, duration: 2.539s, episode steps: 609, steps per second: 240, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1241643/1000000: episode: 2399, duration: 3.492s, episode steps: 835, steps per second: 239, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1242073/1000000: episode: 2400, duration: 1.810s, episode steps: 430, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1242645/1000000: episode: 2401, duration: 2.501s, episode steps: 572, steps per second: 229, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1243605/1000000: episode: 2402, duration: 3.996s, episode steps: 960, steps per second: 240, episode reward: 19.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1244427/1000000: episode: 2403, duration: 3.446s, episode steps: 822, steps per second: 239, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1244838/1000000: episode: 2404, duration: 1.727s, episode steps: 411, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1245249/1000000: episode: 2405, duration: 1.739s, episode steps: 411, steps per second: 236, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1245757/1000000: episode: 2406, duration: 2.122s, episode steps: 508, steps per second: 239, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1246358/1000000: episode: 2407, duration: 2.534s, episode steps: 601, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1246702/1000000: episode: 2408, duration: 1.448s, episode steps: 344, steps per second: 237, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1247208/1000000: episode: 2409, duration: 2.133s, episode steps: 506, steps per second: 237, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1247690/1000000: episode: 2410, duration: 2.055s, episode steps: 482, steps per second: 235, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1248276/1000000: episode: 2411, duration: 2.460s, episode steps: 586, steps per second: 238, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean

action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1248804/1000000: episode: 2412, duration: 2.220s, episode steps: 528, steps per second: 238, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.269 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1249426/1000000: episode: 2413, duration: 2.606s, episode steps: 622, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1250033/1000000: episode: 2414, duration: 2.554s, episode steps: 607, steps per second: 238, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1250845/1000000: episode: 2415, duration: 3.441s, episode steps: 812, steps per second: 236, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1251135/1000000: episode: 2416, duration: 1.221s, episode steps: 290, steps per second: 237, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1251641/1000000: episode: 2417, duration: 2.116s, episode steps: 506, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1252296/1000000: episode: 2418, duration: 2.742s, episode steps: 655, steps per second: 239, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1252844/1000000: episode: 2419, duration: 2.291s, episode steps: 548, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1253299/1000000: episode: 2420, duration: 1.932s, episode steps: 455, steps per second: 235, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1253768/1000000: episode: 2421, duration: 1.973s, episode steps: 469, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1254168/1000000: episode: 2422, duration: 1.682s, episode steps: 400, steps per second: 238, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1254761/1000000: episode: 2423, duration: 2.480s, episode steps: 593, steps per second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1255552/1000000: episode: 2424, duration: 16.417s, episode steps: 791, steps per second: 48, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.039970, mae: 3.245781, mean_q: 3.928630, mean_eps: 0.987779
 1255847/1000000: episode: 2425, duration: 11.832s, episode steps: 295, steps per second: 25, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.038602, mae: 3.330646, mean_q: 4.033419, mean_eps: 0.987383
 1256347/1000000: episode: 2426, duration: 20.127s, episode steps: 500, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: 0.042308, mae: 3.267421, mean_q:

3.955352, mean_eps: 0.986906
1257207/1000000: episode: 2427, duration: 34.630s, episode steps: 860, steps per second: 25, episode reward: 19.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: 0.037907, mae: 3.263094, mean_q: 3.950167, mean_eps: 0.986090
1257803/1000000: episode: 2428, duration: 23.997s, episode steps: 596, steps per second: 25, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.032916, mae: 3.279215, mean_q: 3.971258, mean_eps: 0.985216
1258379/1000000: episode: 2429, duration: 22.993s, episode steps: 576, steps per second: 25, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.038987, mae: 3.312182, mean_q: 4.012142, mean_eps: 0.984513
1258963/1000000: episode: 2430, duration: 23.459s, episode steps: 584, steps per second: 25, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.034910, mae: 3.340738, mean_q: 4.044683, mean_eps: 0.983817
1259389/1000000: episode: 2431, duration: 17.063s, episode steps: 426, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.039629, mae: 3.276127, mean_q: 3.967002, mean_eps: 0.983210
1260102/1000000: episode: 2432, duration: 28.350s, episode steps: 713, steps per second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.040466, mae: 3.298838, mean_q: 3.993000, mean_eps: 0.982526
1260528/1000000: episode: 2433, duration: 16.978s, episode steps: 426, steps per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.048685, mae: 3.303783, mean_q: 3.997822, mean_eps: 0.981844
1260818/1000000: episode: 2434, duration: 11.655s, episode steps: 290, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.036893, mae: 3.369420, mean_q: 4.078264, mean_eps: 0.981414
1261530/1000000: episode: 2435, duration: 28.490s, episode steps: 712, steps per second: 25, episode reward: 9.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.035038, mae: 3.316474, mean_q: 4.018816, mean_eps: 0.980812
1262024/1000000: episode: 2436, duration: 19.754s, episode steps: 494, steps per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.037382, mae: 3.359632, mean_q: 4.071043, mean_eps: 0.980090
1262440/1000000: episode: 2437, duration: 16.732s, episode steps: 416, steps per second: 25, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: 0.046558, mae: 3.431010, mean_q: 4.156131, mean_eps: 0.979545
1263057/1000000: episode: 2438, duration: 24.773s, episode steps: 617, steps per second: 25, episode reward: 17.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.036765, mae: 3.356960, mean_q:

4.072978, mean_eps: 0.978923
1263571/1000000: episode: 2439, duration: 20.506s, episode steps: 514, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: 0.038258, mae: 3.318721, mean_q: 4.020717, mean_eps: 0.978244
1263985/1000000: episode: 2440, duration: 16.639s, episode steps: 414, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.036416, mae: 3.352337, mean_q: 4.064295, mean_eps: 0.977687
1264647/1000000: episode: 2441, duration: 26.335s, episode steps: 662, steps per second: 25, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: 0.039249, mae: 3.374311, mean_q: 4.086865, mean_eps: 0.977042
1265058/1000000: episode: 2442, duration: 16.368s, episode steps: 411, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.041220, mae: 3.399974, mean_q: 4.122330, mean_eps: 0.976398
1265512/1000000: episode: 2443, duration: 18.215s, episode steps: 454, steps per second: 25, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.352 [0.000, 5.000], loss: 0.040690, mae: 3.352319, mean_q: 4.067244, mean_eps: 0.975880
1265807/1000000: episode: 2444, duration: 11.857s, episode steps: 295, steps per second: 25, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: 0.036211, mae: 3.309523, mean_q: 4.015200, mean_eps: 0.975431
1266345/1000000: episode: 2445, duration: 21.484s, episode steps: 538, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.038141, mae: 3.346886, mean_q: 4.059322, mean_eps: 0.974930
1266987/1000000: episode: 2446, duration: 25.666s, episode steps: 642, steps per second: 25, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.684 [0.000, 5.000], loss: 0.040556, mae: 3.340273, mean_q: 4.050259, mean_eps: 0.974222
1267470/1000000: episode: 2447, duration: 19.416s, episode steps: 483, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: 0.037290, mae: 3.375976, mean_q: 4.089873, mean_eps: 0.973547
1267952/1000000: episode: 2448, duration: 19.343s, episode steps: 482, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.042954, mae: 3.333941, mean_q: 4.043464, mean_eps: 0.972969
1268229/1000000: episode: 2449, duration: 11.149s, episode steps: 277, steps per second: 25, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.037055, mae: 3.332269, mean_q: 4.037386, mean_eps: 0.972513
1268656/1000000: episode: 2450, duration: 17.127s, episode steps: 427, steps per second: 25, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.036682, mae: 3.362741, mean_q:

4.076570, mean_eps: 0.972090
1269050/1000000: episode: 2451, duration: 15.841s, episode steps: 394, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.292 [0.000, 5.000], loss: 0.043094, mae: 3.321751, mean_q: 4.019904, mean_eps: 0.971598
1269408/1000000: episode: 2452, duration: 14.368s, episode steps: 358, steps per second: 25, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: 0.040677, mae: 3.330489, mean_q: 4.035769, mean_eps: 0.971147
1269795/1000000: episode: 2453, duration: 15.323s, episode steps: 387, steps per second: 25, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: 0.035179, mae: 3.348005, mean_q: 4.056751, mean_eps: 0.970701
1270303/1000000: episode: 2454, duration: 20.340s, episode steps: 508, steps per second: 25, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.673 [0.000, 5.000], loss: 0.035450, mae: 3.345610, mean_q: 4.051972, mean_eps: 0.970163
1270650/1000000: episode: 2455, duration: 13.930s, episode steps: 347, steps per second: 25, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.038206, mae: 3.329246, mean_q: 4.038631, mean_eps: 0.969650
1271219/1000000: episode: 2456, duration: 22.860s, episode steps: 569, steps per second: 25, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.035798, mae: 3.351439, mean_q: 4.059841, mean_eps: 0.969100
1271510/1000000: episode: 2457, duration: 11.686s, episode steps: 291, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.749 [0.000, 5.000], loss: 0.028994, mae: 3.329112, mean_q: 4.037559, mean_eps: 0.968584
1271974/1000000: episode: 2458, duration: 18.372s, episode steps: 464, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.040091, mae: 3.368596, mean_q: 4.090682, mean_eps: 0.968130
1272438/1000000: episode: 2459, duration: 18.611s, episode steps: 464, steps per second: 25, episode reward: 13.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.036410, mae: 3.368516, mean_q: 4.083067, mean_eps: 0.967574
1273100/1000000: episode: 2460, duration: 26.523s, episode steps: 662, steps per second: 25, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: 0.035405, mae: 3.373466, mean_q: 4.088218, mean_eps: 0.966899
1274163/1000000: episode: 2461, duration: 42.503s, episode steps: 1063, steps per second: 25, episode reward: 22.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.041143, mae: 3.373522, mean_q: 4.087659, mean_eps: 0.965865
1274649/1000000: episode: 2462, duration: 19.432s, episode steps: 486, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.039478, mae: 3.379348, mean_q:


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4.096498, mean_eps: 0.964934
1274944/1000000: episode: 2463, duration: 11.908s, episode steps: 295, steps
per second: 25, episode reward: 5.000, mean reward: 0.017 [ 0.000, 1.000],
mean action: 2.536 [0.000, 5.000], loss: 0.035955, mae: 3.370955, mean_q:
4.085614, mean_eps: 0.964466
1275696/1000000: episode: 2464, duration: 30.056s, episode steps: 752, steps
per second: 25, episode reward: 16.000, mean reward: 0.021 [ 0.000, 1.000],
mean action: 2.527 [0.000, 5.000], loss: 0.035085, mae: 3.363466, mean_q:
4.077192, mean_eps: 0.963839
1276505/1000000: episode: 2465, duration: 32.401s, episode steps: 809, steps
per second: 25, episode reward: 16.000, mean reward: 0.020 [ 0.000, 1.000],
mean action: 2.529 [0.000, 5.000], loss: 0.043061, mae: 3.397441, mean_q:
4.120514, mean_eps: 0.962901
1276817/1000000: episode: 2466, duration: 12.536s, episode steps: 312, steps
per second: 25, episode reward: 4.000, mean reward: 0.013 [ 0.000, 1.000],
mean action: 2.516 [0.000, 5.000], loss: 0.041746, mae: 3.399570, mean_q:
4.122243, mean_eps: 0.962226
1277247/1000000: episode: 2467, duration: 17.052s, episode steps: 430, steps
per second: 25, episode reward: 10.000, mean reward: 0.023 [ 0.000, 1.000],
mean action: 2.579 [0.000, 5.000], loss: 0.035122, mae: 3.345435, mean_q:
4.061150, mean_eps: 0.961782
1277619/1000000: episode: 2468, duration: 14.829s, episode steps: 372, steps
per second: 25, episode reward: 5.000, mean reward: 0.013 [ 0.000, 1.000],
mean action: 2.503 [0.000, 5.000], loss: 0.041881, mae: 3.379272, mean_q:
4.094873, mean_eps: 0.961302

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

1278360/1000000: episode: 2469, duration: 14.844s, episode steps: 741, steps
per second: 50, episode reward: 12.000, mean reward: 0.016 [ 0.000, 1.000],
mean action: 2.606 [0.000, 5.000], loss: 0.032840, mae: 3.348285, mean_q:
4.056032, mean_eps: 0.960880
1278876/1000000: episode: 2470, duration: 2.182s, episode steps: 516, steps per
second: 237, episode reward: 9.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1279512/1000000: episode: 2471, duration: 2.658s, episode steps: 636, steps per
second: 239, episode reward: 11.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1280043/1000000: episode: 2472, duration: 2.230s, episode steps: 531, steps per
second: 238, episode reward: 12.000, mean reward: 0.023 [ 0.000, 1.000], mean
action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1280420/1000000: episode: 2473, duration: 1.586s, episode steps: 377, steps per
second: 238, episode reward: 4.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1281022/1000000: episode: 2474, duration: 2.525s, episode steps: 602, steps per
second: 238, episode reward: 13.000, mean reward: 0.022 [ 0.000, 1.000], mean
action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1281599/1000000: episode: 2475, duration: 2.438s, episode steps: 577, steps per second: 237, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1282552/1000000: episode: 2476, duration: 3.984s, episode steps: 953, steps per second: 239, episode reward: 21.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1283114/1000000: episode: 2477, duration: 2.351s, episode steps: 562, steps per second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1283620/1000000: episode: 2478, duration: 2.123s, episode steps: 506, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1284021/1000000: episode: 2479, duration: 1.689s, episode steps: 401, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1284650/1000000: episode: 2480, duration: 2.645s, episode steps: 629, steps per second: 238, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1285171/1000000: episode: 2481, duration: 2.201s, episode steps: 521, steps per second: 237, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1285695/1000000: episode: 2482, duration: 2.197s, episode steps: 524, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1286072/1000000: episode: 2483, duration: 1.596s, episode steps: 377, steps per second: 236, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.316 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1286467/1000000: episode: 2484, duration: 1.674s, episode steps: 395, steps per second: 236, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1287066/1000000: episode: 2485, duration: 2.519s, episode steps: 599, steps per second: 238, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1287599/1000000: episode: 2486, duration: 2.261s, episode steps: 533, steps per second: 236, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1288361/1000000: episode: 2487, duration: 3.175s, episode steps: 762, steps per second: 240, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1288931/1000000: episode: 2488, duration: 2.385s, episode steps: 570, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1289624/1000000: episode: 2489, duration: 2.892s, episode steps: 693, steps per second: 240, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1290724/1000000: episode: 2490, duration: 4.577s, episode steps: 1100, steps per second: 240, episode reward: 28.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1290991/1000000: episode: 2491, duration: 1.137s, episode steps: 267, steps per second: 235, episode reward: 7.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1291662/1000000: episode: 2492, duration: 2.812s, episode steps: 671, steps per second: 239, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1292122/1000000: episode: 2493, duration: 1.943s, episode steps: 460, steps per second: 237, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1292597/1000000: episode: 2494, duration: 2.016s, episode steps: 475, steps per second: 236, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293040/1000000: episode: 2495, duration: 1.864s, episode steps: 443, steps per second: 238, episode reward: 12.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293347/1000000: episode: 2496, duration: 1.314s, episode steps: 307, steps per second: 234, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.332 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293835/1000000: episode: 2497, duration: 2.077s, episode steps: 488, steps per second: 235, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1294375/1000000: episode: 2498, duration: 2.283s, episode steps: 540, steps per second: 237, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1294862/1000000: episode: 2499, duration: 2.046s, episode steps: 487, steps per second: 238, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1295388/1000000: episode: 2500, duration: 2.196s, episode steps: 526, steps per second: 240, episode reward: 15.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1295757/1000000: episode: 2501, duration: 1.557s, episode steps: 369, steps per second: 237, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1296332/1000000: episode: 2502, duration: 2.419s, episode steps: 575, steps per second: 238, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1296826/1000000: episode: 2503, duration: 2.086s, episode steps: 494, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1297410/1000000: episode: 2504, duration: 2.462s, episode steps: 584, steps per second: 237, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1297723/1000000: episode: 2505, duration: 1.333s, episode steps: 313, steps per second: 235, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1298136/1000000: episode: 2506, duration: 1.742s, episode steps: 413, steps per second: 237, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1298607/1000000: episode: 2507, duration: 1.991s, episode steps: 471, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1299104/1000000: episode: 2508, duration: 2.129s, episode steps: 497, steps per second: 233, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.346 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1299698/1000000: episode: 2509, duration: 2.523s, episode steps: 594, steps per second: 235, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1300415/1000000: episode: 2510, duration: 3.004s, episode steps: 717, steps per second: 239, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1300883/1000000: episode: 2511, duration: 1.974s, episode steps: 468, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1301261/1000000: episode: 2512, duration: 1.598s, episode steps: 378, steps per second: 236, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.349 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1301561/1000000: episode: 2513, duration: 1.269s, episode steps: 300, steps per second: 236, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1302249/1000000: episode: 2514, duration: 2.903s, episode steps: 688, steps per second: 237, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1302734/1000000: episode: 2515, duration: 2.043s, episode steps: 485, steps per second: 237, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1303154/1000000: episode: 2516, duration: 1.762s, episode steps: 420, steps per second: 238, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1303540/1000000: episode: 2517, duration: 1.617s, episode steps: 386, steps per second: 239, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1304581/1000000: episode: 2518, duration: 4.311s, episode steps: 1041, steps per second: 241, episode reward: 13.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1305184/1000000: episode: 2519, duration: 2.560s, episode steps: 603, steps per second: 236, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1305554/1000000: episode: 2520, duration: 1.569s, episode steps: 370, steps per second: 236, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1305841/1000000: episode: 2521, duration: 1.216s, episode steps: 287, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1306320/1000000: episode: 2522, duration: 2.015s, episode steps: 479, steps per second: 238, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.656 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1306606/1000000: episode: 2523, duration: 1.209s, episode steps: 286, steps per second: 237, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1307063/1000000: episode: 2524, duration: 1.916s, episode steps: 457, steps per second: 239, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1307564/1000000: episode: 2525, duration: 2.092s, episode steps: 501, steps per second: 239, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.339 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1308195/1000000: episode: 2526, duration: 2.652s, episode steps: 631, steps per second: 238, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1308718/1000000: episode: 2527, duration: 2.204s, episode steps: 523, steps per second: 237, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1309226/1000000: episode: 2528, duration: 2.139s, episode steps: 508, steps per second: 238, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1309783/1000000: episode: 2529, duration: 2.328s, episode steps: 557, steps per second: 239, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1310262/1000000: episode: 2530, duration: 2.010s, episode steps: 479, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1310882/1000000: episode: 2531, duration: 2.623s, episode steps: 620, steps per second: 236, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1311329/1000000: episode: 2532, duration: 1.901s, episode steps: 447, steps per second: 235, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1311749/1000000: episode: 2533, duration: 1.761s, episode steps: 420, steps per second: 238, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1312354/1000000: episode: 2534, duration: 2.527s, episode steps: 605, steps per second: 239, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1312977/1000000: episode: 2535, duration: 2.596s, episode steps: 623, steps per second: 240, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1313423/1000000: episode: 2536, duration: 1.864s, episode steps: 446, steps per second: 239, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1314263/1000000: episode: 2537, duration: 3.547s, episode steps: 840, steps per second: 237, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1314763/1000000: episode: 2538, duration: 2.090s, episode steps: 500, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1315125/1000000: episode: 2539, duration: 1.524s, episode steps: 362, steps per second: 237, episode reward: 11.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1315846/1000000: episode: 2540, duration: 3.022s, episode steps: 721, steps per second: 239, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1316260/1000000: episode: 2541, duration: 1.733s, episode steps: 414, steps per second: 239, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1316753/1000000: episode: 2542, duration: 2.077s, episode steps: 493, steps per second: 237, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1317286/1000000: episode: 2543, duration: 2.243s, episode steps: 533, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1317806/1000000: episode: 2544, duration: 2.172s, episode steps: 520, steps per second: 239, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1318189/1000000: episode: 2545, duration: 1.614s, episode steps: 383, steps per second: 237, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1318689/1000000: episode: 2546, duration: 2.100s, episode steps: 500, steps per second: 238, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1319331/1000000: episode: 2547, duration: 2.691s, episode steps: 642, steps per second: 239, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1320047/1000000: episode: 2548, duration: 3.013s, episode steps: 716, steps per second: 238, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1320338/1000000: episode: 2549, duration: 1.241s, episode steps: 291, steps per second: 234, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1320856/1000000: episode: 2550, duration: 7.067s, episode steps: 518, steps per second: 73, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: 0.032301, mae: 3.351371, mean_q: 4.056015, mean_eps: 0.987918

1321385/1000000: episode: 2551, duration: 21.420s, episode steps: 529, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.033023, mae: 3.410334, mean_q: 4.133704, mean_eps: 0.987520

1322006/1000000: episode: 2552, duration: 24.974s, episode steps: 621, steps per second: 25, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: 0.048770, mae: 3.396422, mean_q: 4.120058, mean_eps: 0.986829

1322424/1000000: episode: 2553, duration: 16.869s, episode steps: 418, steps per second: 25, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: 0.039432, mae: 3.391867, mean_q:

4.108741, mean_eps: 0.986207
 1322854/1000000: episode: 2554, duration: 17.373s, episode steps: 430, steps per second: 25, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: 0.042381, mae: 3.406969, mean_q: 4.128398, mean_eps: 0.985698
 1323375/1000000: episode: 2555, duration: 20.664s, episode steps: 521, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: 0.037709, mae: 3.379224, mean_q: 4.098985, mean_eps: 0.985127
 1324021/1000000: episode: 2556, duration: 25.787s, episode steps: 646, steps per second: 25, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.035452, mae: 3.388513, mean_q: 4.106057, mean_eps: 0.984426
 1324707/1000000: episode: 2557, duration: 27.385s, episode steps: 686, steps per second: 25, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.035326, mae: 3.400280, mean_q: 4.122557, mean_eps: 0.983627
 1325079/1000000: episode: 2558, duration: 14.891s, episode steps: 372, steps per second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.039057, mae: 3.385320, mean_q: 4.106106, mean_eps: 0.982994
 1325691/1000000: episode: 2559, duration: 24.546s, episode steps: 612, steps per second: 25, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: 0.034812, mae: 3.415884, mean_q: 4.143528, mean_eps: 0.982403
 1326136/1000000: episode: 2560, duration: 17.896s, episode steps: 445, steps per second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.036188, mae: 3.391869, mean_q: 4.113755, mean_eps: 0.981770

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 /content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_2560_episodes.h5f
 1326840/1000000: episode: 2561, duration: 28.537s, episode steps: 704, steps per second: 25, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: 0.037300, mae: 3.415164, mean_q: 4.139383, mean_eps: 0.981081
 1327282/1000000: episode: 2562, duration: 17.771s, episode steps: 442, steps per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.046973, mae: 3.454338, mean_q: 4.186946, mean_eps: 0.980392
 1327693/1000000: episode: 2563, duration: 16.526s, episode steps: 411, steps per second: 25, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.033711, mae: 3.417253, mean_q: 4.141538, mean_eps: 0.979878
 1328175/1000000: episode: 2564, duration: 19.248s, episode steps: 482, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: 0.040053, mae: 3.422761, mean_q:

4.149335, mean_eps: 0.979343
 1329001/1000000: episode: 2565, duration: 33.094s, episode steps: 826, steps per second: 25, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.035728, mae: 3.430483, mean_q: 4.153476, mean_eps: 0.978558
 1329510/1000000: episode: 2566, duration: 20.231s, episode steps: 509, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.036102, mae: 3.430683, mean_q: 4.152499, mean_eps: 0.977757
 1330214/1000000: episode: 2567, duration: 28.199s, episode steps: 704, steps per second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.042273, mae: 3.428049, mean_q: 4.157276, mean_eps: 0.977030
 1330795/1000000: episode: 2568, duration: 23.256s, episode steps: 581, steps per second: 25, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.037374, mae: 3.416980, mean_q: 4.140788, mean_eps: 0.976259
 1331403/1000000: episode: 2569, duration: 24.387s, episode steps: 608, steps per second: 25, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.044676, mae: 3.415131, mean_q: 4.144071, mean_eps: 0.975546
 1331947/1000000: episode: 2570, duration: 21.861s, episode steps: 544, steps per second: 25, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: 0.042508, mae: 3.459428, mean_q: 4.188649, mean_eps: 0.974855
 1332661/1000000: episode: 2571, duration: 28.636s, episode steps: 714, steps per second: 25, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.050265, mae: 3.430505, mean_q: 4.157868, mean_eps: 0.974099
 1332987/1000000: episode: 2572, duration: 13.076s, episode steps: 326, steps per second: 25, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.047194, mae: 3.483093, mean_q: 4.230059, mean_eps: 0.973475
 1333920/1000000: episode: 2573, duration: 37.441s, episode steps: 933, steps per second: 25, episode reward: 17.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.034262, mae: 3.421233, mean_q: 4.142990, mean_eps: 0.972722
 1334283/1000000: episode: 2574, duration: 14.608s, episode steps: 363, steps per second: 25, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: 0.039609, mae: 3.422328, mean_q: 4.144264, mean_eps: 0.971944
 1334683/1000000: episode: 2575, duration: 15.985s, episode steps: 400, steps per second: 25, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: 0.044658, mae: 3.467550, mean_q: 4.199501, mean_eps: 0.971486
 1335396/1000000: episode: 2576, duration: 28.586s, episode steps: 713, steps per second: 25, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.038848, mae: 3.426826, mean_q:

4.150291, mean_eps: 0.970818
1335695/1000000: episode: 2577, duration: 12.031s, episode steps: 299, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: 0.030470, mae: 3.387748, mean_q: 4.111220, mean_eps: 0.970211
1336402/1000000: episode: 2578, duration: 28.219s, episode steps: 707, steps per second: 25, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.036645, mae: 3.456096, mean_q: 4.186016, mean_eps: 0.969606
1336944/1000000: episode: 2579, duration: 21.573s, episode steps: 542, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.040774, mae: 3.457827, mean_q: 4.186401, mean_eps: 0.968858
1337936/1000000: episode: 2580, duration: 39.683s, episode steps: 992, steps per second: 25, episode reward: 27.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.039899, mae: 3.462942, mean_q: 4.195461, mean_eps: 0.967938
1338475/1000000: episode: 2581, duration: 21.571s, episode steps: 539, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.040817, mae: 3.428099, mean_q: 4.153570, mean_eps: 0.967019
1338967/1000000: episode: 2582, duration: 19.607s, episode steps: 492, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: 0.046125, mae: 3.456125, mean_q: 4.193140, mean_eps: 0.966400
1339436/1000000: episode: 2583, duration: 18.719s, episode steps: 469, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.039531, mae: 3.456378, mean_q: 4.195903, mean_eps: 0.965824
1340273/1000000: episode: 2584, duration: 33.458s, episode steps: 837, steps per second: 25, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.036866, mae: 3.471086, mean_q: 4.208947, mean_eps: 0.965039
1340753/1000000: episode: 2585, duration: 19.204s, episode steps: 480, steps per second: 25, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: 0.033287, mae: 3.463026, mean_q: 4.196804, mean_eps: 0.964247
1341009/1000000: episode: 2586, duration: 10.278s, episode steps: 256, steps per second: 25, episode reward: 5.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.041480, mae: 3.478965, mean_q: 4.215052, mean_eps: 0.963806
1341551/1000000: episode: 2587, duration: 21.470s, episode steps: 542, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.033713, mae: 3.450359, mean_q: 4.182319, mean_eps: 0.963328
1341947/1000000: episode: 2588, duration: 15.839s, episode steps: 396, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: 0.036735, mae: 3.451497, mean_q:

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4.187448, mean_eps: 0.962766
1342349/1000000: episode: 2589, duration: 16.093s, episode steps: 402, steps
per second: 25, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.505 [0.000, 5.000], loss: 0.037491, mae: 3.491238, mean_q:
4.234658, mean_eps: 0.962286
1343078/1000000: episode: 2590, duration: 29.170s, episode steps: 729, steps
per second: 25, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000],
mean action: 2.477 [0.000, 5.000], loss: 0.038511, mae: 3.477493, mean_q:
4.216932, mean_eps: 0.961607
1343354/1000000: episode: 2591, duration: 10.983s, episode steps: 276, steps
per second: 25, episode reward: 8.000, mean reward: 0.029 [0.000, 1.000],
mean action: 2.609 [0.000, 5.000], loss: 0.042068, mae: 3.433454, mean_q:
4.170231, mean_eps: 0.961005

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

1344310/1000000: episode: 2592, duration: 8.705s, episode steps: 956, steps per
second: 110, episode reward: 21.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.474 [0.000, 5.000], loss: 0.039138, mae: 3.540246, mean_q: 4.288804,
mean_eps: 0.960760
1344593/1000000: episode: 2593, duration: 1.194s, episode steps: 283, steps per
second: 237, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1345199/1000000: episode: 2594, duration: 2.553s, episode steps: 606, steps per
second: 237, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.398 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1345581/1000000: episode: 2595, duration: 1.620s, episode steps: 382, steps per
second: 236, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1346134/1000000: episode: 2596, duration: 2.314s, episode steps: 553, steps per
second: 239, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1346579/1000000: episode: 2597, duration: 1.887s, episode steps: 445, steps per
second: 236, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1346958/1000000: episode: 2598, duration: 1.590s, episode steps: 379, steps per
second: 238, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1347691/1000000: episode: 2599, duration: 3.049s, episode steps: 733, steps per
second: 240, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1348182/1000000: episode: 2600, duration: 2.087s, episode steps: 491, steps per
second: 235, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.625 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1348487/1000000: episode: 2601, duration: 1.297s, episode steps: 305, steps per
second: 235, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1348773/1000000: episode: 2602, duration: 1.211s, episode steps: 286, steps per second: 236, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1349261/1000000: episode: 2603, duration: 2.040s, episode steps: 488, steps per second: 239, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1349794/1000000: episode: 2604, duration: 2.230s, episode steps: 533, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1350561/1000000: episode: 2605, duration: 3.195s, episode steps: 767, steps per second: 240, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1351158/1000000: episode: 2606, duration: 2.515s, episode steps: 597, steps per second: 237, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1351554/1000000: episode: 2607, duration: 1.689s, episode steps: 396, steps per second: 235, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1351965/1000000: episode: 2608, duration: 1.737s, episode steps: 411, steps per second: 237, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1352567/1000000: episode: 2609, duration: 2.542s, episode steps: 602, steps per second: 237, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353033/1000000: episode: 2610, duration: 1.975s, episode steps: 466, steps per second: 236, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353472/1000000: episode: 2611, duration: 1.848s, episode steps: 439, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353768/1000000: episode: 2612, duration: 1.267s, episode steps: 296, steps per second: 234, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1354171/1000000: episode: 2613, duration: 1.706s, episode steps: 403, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1354497/1000000: episode: 2614, duration: 1.382s, episode steps: 326, steps per second: 236, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1354974/1000000: episode: 2615, duration: 2.010s, episode steps: 477, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1355346/1000000: episode: 2616, duration: 1.563s, episode steps: 372, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1355981/1000000: episode: 2617, duration: 2.654s, episode steps: 635, steps per second: 239, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356264/1000000: episode: 2618, duration: 1.205s, episode steps: 283, steps per second: 235, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356744/1000000: episode: 2619, duration: 2.029s, episode steps: 480, steps per second: 237, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.329 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1357395/1000000: episode: 2620, duration: 2.727s, episode steps: 651, steps per second: 239, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1357805/1000000: episode: 2621, duration: 1.718s, episode steps: 410, steps per second: 239, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1358198/1000000: episode: 2622, duration: 1.643s, episode steps: 393, steps per second: 239, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1358605/1000000: episode: 2623, duration: 1.694s, episode steps: 407, steps per second: 240, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1359053/1000000: episode: 2624, duration: 1.870s, episode steps: 448, steps per second: 240, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1359643/1000000: episode: 2625, duration: 2.454s, episode steps: 590, steps per second: 240, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1360016/1000000: episode: 2626, duration: 1.583s, episode steps: 373, steps per second: 236, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1360436/1000000: episode: 2627, duration: 1.778s, episode steps: 420, steps per second: 236, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1360892/1000000: episode: 2628, duration: 1.898s, episode steps: 456, steps per second: 240, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1361367/1000000: episode: 2629, duration: 2.012s, episode steps: 475, steps per second: 236, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.371 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1361782/1000000: episode: 2630, duration: 1.762s, episode steps: 415, steps per second: 236, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.680 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1362504/1000000: episode: 2631, duration: 3.020s, episode steps: 722, steps per second: 239, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1363470/1000000: episode: 2632, duration: 4.110s, episode steps: 966, steps per second: 235, episode reward: 21.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1363961/1000000: episode: 2633, duration: 2.052s, episode steps: 491, steps per second: 239, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1364436/1000000: episode: 2634, duration: 1.988s, episode steps: 475, steps per second: 239, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1365327/1000000: episode: 2635, duration: 3.715s, episode steps: 891, steps per second: 240, episode reward: 19.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1366116/1000000: episode: 2636, duration: 3.319s, episode steps: 789, steps per second: 238, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1366674/1000000: episode: 2637, duration: 2.339s, episode steps: 558, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1367358/1000000: episode: 2638, duration: 2.843s, episode steps: 684, steps per second: 241, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1367941/1000000: episode: 2639, duration: 2.436s, episode steps: 583, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1368638/1000000: episode: 2640, duration: 2.917s, episode steps: 697, steps per second: 239, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1369218/1000000: episode: 2641, duration: 2.470s, episode steps: 580, steps per second: 235, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1369850/1000000: episode: 2642, duration: 2.658s, episode steps: 632, steps per second: 238, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1370108/1000000: episode: 2643, duration: 1.087s, episode steps: 258, steps per second: 237, episode reward: 3.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1370480/1000000: episode: 2644, duration: 1.552s, episode steps: 372, steps per second: 240, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1371034/1000000: episode: 2645, duration: 2.329s, episode steps: 554, steps per second: 238, episode reward: 4.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1371601/1000000: episode: 2646, duration: 2.385s, episode steps: 567, steps per second: 238, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1372153/1000000: episode: 2647, duration: 2.366s, episode steps: 552, steps per second: 233, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1372590/1000000: episode: 2648, duration: 1.827s, episode steps: 437, steps per second: 239, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1372927/1000000: episode: 2649, duration: 1.416s, episode steps: 337, steps per second: 238, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1373314/1000000: episode: 2650, duration: 1.630s, episode steps: 387, steps per second: 237, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1373839/1000000: episode: 2651, duration: 2.192s, episode steps: 525, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1374221/1000000: episode: 2652, duration: 1.602s, episode steps: 382, steps per second: 238, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1374957/1000000: episode: 2653, duration: 3.089s, episode steps: 736, steps per second: 238, episode reward: 21.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1375419/1000000: episode: 2654, duration: 1.953s, episode steps: 462, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1375864/1000000: episode: 2655, duration: 1.869s, episode steps: 445, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1376229/1000000: episode: 2656, duration: 1.542s, episode steps: 365, steps per second: 237, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.386 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1376523/1000000: episode: 2657, duration: 1.244s, episode steps: 294, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1376964/1000000: episode: 2658, duration: 1.859s, episode steps: 441, steps per second: 237, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1377596/1000000: episode: 2659, duration: 2.656s, episode steps: 632, steps per second: 238, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1378198/1000000: episode: 2660, duration: 2.541s, episode steps: 602, steps per second: 237, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1378501/1000000: episode: 2661, duration: 1.283s, episode steps: 303, steps per second: 236, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.314 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1378844/1000000: episode: 2662, duration: 1.449s, episode steps: 343, steps per second: 237, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1379163/1000000: episode: 2663, duration: 1.353s, episode steps: 319, steps per second: 236, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1379718/1000000: episode: 2664, duration: 2.333s, episode steps: 555, steps per second: 238, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1380419/1000000: episode: 2665, duration: 2.942s, episode steps: 701, steps per second: 238, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1380817/1000000: episode: 2666, duration: 1.684s, episode steps: 398, steps per second: 236, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1381132/1000000: episode: 2667, duration: 1.337s, episode steps: 315, steps per second: 236, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1381581/1000000: episode: 2668, duration: 1.897s, episode steps: 449, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1382051/1000000: episode: 2669, duration: 1.984s, episode steps: 470, steps per second: 237, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1382582/1000000: episode: 2670, duration: 2.224s, episode steps: 531, steps per second: 239, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1383153/1000000: episode: 2671, duration: 2.380s, episode steps: 571, steps per second: 240, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1383850/1000000: episode: 2672, duration: 2.943s, episode steps: 697, steps per second: 237, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1384500/1000000: episode: 2673, duration: 2.718s, episode steps: 650, steps per second: 239, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1385004/1000000: episode: 2674, duration: 2.122s, episode steps: 504, steps per second: 237, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1385429/1000000: episode: 2675, duration: 1.792s, episode steps: 425, steps per second: 237, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1385916/1000000: episode: 2676, duration: 2.041s, episode steps: 487, steps per second: 239, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1386428/1000000: episode: 2677, duration: 8.277s, episode steps: 512, steps per second: 62, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.048370, mae: 3.462657, mean_q: 4.199237, mean_eps: 0.987897

1386923/1000000: episode: 2678, duration: 19.896s, episode steps: 495, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.035586, mae: 3.493477, mean_q: 4.235044, mean_eps: 0.987498

1387426/1000000: episode: 2679, duration: 20.072s, episode steps: 503, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.044929, mae: 3.479807, mean_q: 4.211142, mean_eps: 0.986898

1387809/1000000: episode: 2680, duration: 15.414s, episode steps: 383, steps per second: 25, episode reward: 11.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: 0.032035, mae: 3.487799, mean_q:

4.222031, mean_eps: 0.986366
1388237/1000000: episode: 2681, duration: 17.053s, episode steps: 428, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.034784, mae: 3.484510, mean_q: 4.216630, mean_eps: 0.985878
1388630/1000000: episode: 2682, duration: 15.705s, episode steps: 393, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.664 [0.000, 5.000], loss: 0.038791, mae: 3.497731, mean_q: 4.234856, mean_eps: 0.985386
1388885/1000000: episode: 2683, duration: 10.234s, episode steps: 255, steps per second: 25, episode reward: 5.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: 0.031855, mae: 3.444909, mean_q: 4.171142, mean_eps: 0.984998
1389307/1000000: episode: 2684, duration: 16.657s, episode steps: 422, steps per second: 25, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: 0.029039, mae: 3.468694, mean_q: 4.199641, mean_eps: 0.984592
1389592/1000000: episode: 2685, duration: 11.367s, episode steps: 285, steps per second: 25, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.688 [0.000, 5.000], loss: 0.041080, mae: 3.466177, mean_q: 4.201367, mean_eps: 0.984170
1390054/1000000: episode: 2686, duration: 18.551s, episode steps: 462, steps per second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.721 [0.000, 5.000], loss: 0.036931, mae: 3.497282, mean_q: 4.236843, mean_eps: 0.983721
1390585/1000000: episode: 2687, duration: 21.249s, episode steps: 531, steps per second: 25, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.042506, mae: 3.448983, mean_q: 4.169294, mean_eps: 0.983123
1390983/1000000: episode: 2688, duration: 15.744s, episode steps: 398, steps per second: 25, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: 0.035197, mae: 3.449719, mean_q: 4.178493, mean_eps: 0.982566
1391399/1000000: episode: 2689, duration: 16.503s, episode steps: 416, steps per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: 0.034601, mae: 3.415221, mean_q: 4.136275, mean_eps: 0.982079
1392234/1000000: episode: 2690, duration: 33.395s, episode steps: 835, steps per second: 25, episode reward: 24.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: 0.036959, mae: 3.490357, mean_q: 4.227769, mean_eps: 0.981328
1392575/1000000: episode: 2691, duration: 13.594s, episode steps: 341, steps per second: 25, episode reward: 10.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: 0.037738, mae: 3.523992, mean_q: 4.262082, mean_eps: 0.980622
1393088/1000000: episode: 2692, duration: 20.593s, episode steps: 513, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.038703, mae: 3.468500, mean_q:

4.198165, mean_eps: 0.980111
 1393589/1000000: episode: 2693, duration: 20.151s, episode steps: 501, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.038515, mae: 3.477068, mean_q: 4.212667, mean_eps: 0.979502
 1393882/1000000: episode: 2694, duration: 11.609s, episode steps: 293, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.024491, mae: 3.479841, mean_q: 4.219589, mean_eps: 0.979024
 1394173/1000000: episode: 2695, duration: 11.681s, episode steps: 291, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: 0.043645, mae: 3.459396, mean_q: 4.182445, mean_eps: 0.978674
 1394719/1000000: episode: 2696, duration: 21.656s, episode steps: 546, steps per second: 25, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: 0.045486, mae: 3.488361, mean_q: 4.226708, mean_eps: 0.978172
 1395411/1000000: episode: 2697, duration: 27.571s, episode steps: 692, steps per second: 25, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.030756, mae: 3.483751, mean_q: 4.219241, mean_eps: 0.977430
 1396075/1000000: episode: 2698, duration: 26.558s, episode steps: 664, steps per second: 25, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: 0.037255, mae: 3.511551, mean_q: 4.253244, mean_eps: 0.976617
 1396756/1000000: episode: 2699, duration: 27.240s, episode steps: 681, steps per second: 25, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.037819, mae: 3.534885, mean_q: 4.282999, mean_eps: 0.975810
 1397166/1000000: episode: 2700, duration: 16.379s, episode steps: 410, steps per second: 25, episode reward: 11.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.322 [0.000, 5.000], loss: 0.038125, mae: 3.480391, mean_q: 4.212754, mean_eps: 0.975155
 1397542/1000000: episode: 2701, duration: 14.969s, episode steps: 376, steps per second: 25, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: 0.034215, mae: 3.575600, mean_q: 4.338647, mean_eps: 0.974682
 1397963/1000000: episode: 2702, duration: 16.708s, episode steps: 421, steps per second: 25, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.033936, mae: 3.490334, mean_q: 4.226973, mean_eps: 0.974205
 1398250/1000000: episode: 2703, duration: 11.488s, episode steps: 287, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: 0.043680, mae: 3.550354, mean_q: 4.295121, mean_eps: 0.973780
 1398646/1000000: episode: 2704, duration: 15.724s, episode steps: 396, steps per second: 25, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.042187, mae: 3.540643, mean_q:

4.284301, mean_eps: 0.973370
 1399158/1000000: episode: 2705, duration: 20.452s, episode steps: 512, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.033065, mae: 3.536004, mean_q: 4.273916, mean_eps: 0.972825
 1399662/1000000: episode: 2706, duration: 20.152s, episode steps: 504, steps per second: 25, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.038147, mae: 3.573259, mean_q: 4.324093, mean_eps: 0.972215
 1400053/1000000: episode: 2707, duration: 15.636s, episode steps: 391, steps per second: 25, episode reward: 12.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.034553, mae: 3.542810, mean_q: 4.284470, mean_eps: 0.971678
 1400521/1000000: episode: 2708, duration: 18.666s, episode steps: 468, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.036238, mae: 3.529110, mean_q: 4.273978, mean_eps: 0.971162
 1400966/1000000: episode: 2709, duration: 17.638s, episode steps: 445, steps per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: 0.025356, mae: 3.569401, mean_q: 4.322937, mean_eps: 0.970614
 1401264/1000000: episode: 2710, duration: 11.924s, episode steps: 298, steps per second: 25, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.037931, mae: 3.518360, mean_q: 4.257282, mean_eps: 0.970170
 1401670/1000000: episode: 2711, duration: 16.323s, episode steps: 406, steps per second: 25, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: 0.043951, mae: 3.554066, mean_q: 4.305711, mean_eps: 0.969748
 1402018/1000000: episode: 2712, duration: 13.924s, episode steps: 348, steps per second: 25, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: 0.040743, mae: 3.534275, mean_q: 4.285376, mean_eps: 0.969294
 1403188/1000000: episode: 2713, duration: 46.749s, episode steps: 1170, steps per second: 25, episode reward: 12.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.041026, mae: 3.563857, mean_q: 4.314109, mean_eps: 0.968385
 1403773/1000000: episode: 2714, duration: 23.494s, episode steps: 585, steps per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.047037, mae: 3.562890, mean_q: 4.308552, mean_eps: 0.967331
 1404260/1000000: episode: 2715, duration: 19.429s, episode steps: 487, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.038636, mae: 3.537006, mean_q: 4.276198, mean_eps: 0.966688
 1404695/1000000: episode: 2716, duration: 17.465s, episode steps: 435, steps per second: 25, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.037959, mae: 3.592557, mean_q:

4.347244, mean_eps: 0.966136
 1405095/1000000: episode: 2717, duration: 16.033s, episode steps: 400, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: 0.048443, mae: 3.535719, mean_q: 4.285841, mean_eps: 0.965634
 1405603/1000000: episode: 2718, duration: 20.424s, episode steps: 508, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.047689, mae: 3.571698, mean_q: 4.327925, mean_eps: 0.965090
 1406323/1000000: episode: 2719, duration: 28.758s, episode steps: 720, steps per second: 25, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.045173, mae: 3.575245, mean_q: 4.324733, mean_eps: 0.964353
 1407075/1000000: episode: 2720, duration: 29.960s, episode steps: 752, steps per second: 25, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.037719, mae: 3.573920, mean_q: 4.327806, mean_eps: 0.963470

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_2720_episodes.h5f

1407634/1000000: episode: 2721, duration: 22.491s, episode steps: 559, steps per second: 25, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.034411, mae: 3.563193, mean_q: 4.308387, mean_eps: 0.962682
 1408100/1000000: episode: 2722, duration: 18.541s, episode steps: 466, steps per second: 25, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.038534, mae: 3.587251, mean_q: 4.341507, mean_eps: 0.962068
 1408877/1000000: episode: 2723, duration: 31.032s, episode steps: 777, steps per second: 25, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.036095, mae: 3.588186, mean_q: 4.342478, mean_eps: 0.961322

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

1409296/1000000: episode: 2724, duration: 6.843s, episode steps: 419, steps per second: 61, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.044882, mae: 3.617507, mean_q: 4.372258, mean_eps: 0.960767

1409756/1000000: episode: 2725, duration: 1.956s, episode steps: 460, steps per second: 235, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1410045/1000000: episode: 2726, duration: 1.224s, episode steps: 289, steps per second: 236, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1410536/1000000: episode: 2727, duration: 2.049s, episode steps: 491, steps per

second: 240, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1411542/1000000: episode: 2728, duration: 4.186s, episode steps: 1006, steps per second: 240, episode reward: 14.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1412115/1000000: episode: 2729, duration: 2.432s, episode steps: 573, steps per second: 236, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1412581/1000000: episode: 2730, duration: 1.961s, episode steps: 466, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1413178/1000000: episode: 2731, duration: 2.504s, episode steps: 597, steps per second: 238, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1413728/1000000: episode: 2732, duration: 2.315s, episode steps: 550, steps per second: 238, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1414269/1000000: episode: 2733, duration: 2.278s, episode steps: 541, steps per second: 237, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1414653/1000000: episode: 2734, duration: 1.618s, episode steps: 384, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1415124/1000000: episode: 2735, duration: 1.987s, episode steps: 471, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1415429/1000000: episode: 2736, duration: 1.299s, episode steps: 305, steps per second: 235, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.305 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1416065/1000000: episode: 2737, duration: 2.655s, episode steps: 636, steps per second: 240, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1416613/1000000: episode: 2738, duration: 2.303s, episode steps: 548, steps per second: 238, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1417014/1000000: episode: 2739, duration: 1.703s, episode steps: 401, steps per second: 235, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1417683/1000000: episode: 2740, duration: 2.791s, episode steps: 669, steps per second: 240, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1418077/1000000: episode: 2741, duration: 1.670s, episode steps: 394, steps per second: 236, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1418540/1000000: episode: 2742, duration: 1.975s, episode steps: 463, steps per second: 234, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1418933/1000000: episode: 2743, duration: 1.649s, episode steps: 393, steps per

second: 238, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1419376/1000000: episode: 2744, duration: 1.866s, episode steps: 443, steps per second: 237, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1419948/1000000: episode: 2745, duration: 2.409s, episode steps: 572, steps per second: 237, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.638 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1420377/1000000: episode: 2746, duration: 1.803s, episode steps: 429, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1420847/1000000: episode: 2747, duration: 1.981s, episode steps: 470, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1421298/1000000: episode: 2748, duration: 1.926s, episode steps: 451, steps per second: 234, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1421663/1000000: episode: 2749, duration: 1.535s, episode steps: 365, steps per second: 238, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1422253/1000000: episode: 2750, duration: 2.482s, episode steps: 590, steps per second: 238, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1422668/1000000: episode: 2751, duration: 1.758s, episode steps: 415, steps per second: 236, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1423130/1000000: episode: 2752, duration: 1.938s, episode steps: 462, steps per second: 238, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1423781/1000000: episode: 2753, duration: 2.725s, episode steps: 651, steps per second: 239, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1424181/1000000: episode: 2754, duration: 1.697s, episode steps: 400, steps per second: 236, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1424845/1000000: episode: 2755, duration: 2.783s, episode steps: 664, steps per second: 239, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1425332/1000000: episode: 2756, duration: 2.037s, episode steps: 487, steps per second: 239, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1425624/1000000: episode: 2757, duration: 1.233s, episode steps: 292, steps per second: 237, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1426258/1000000: episode: 2758, duration: 2.645s, episode steps: 634, steps per second: 240, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1426785/1000000: episode: 2759, duration: 2.229s, episode steps: 527, steps per

second: 236, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1427616/1000000: episode: 2760, duration: 3.533s, episode steps: 831, steps per second: 235, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1428185/1000000: episode: 2761, duration: 2.407s, episode steps: 569, steps per second: 236, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1428616/1000000: episode: 2762, duration: 1.828s, episode steps: 431, steps per second: 236, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1428914/1000000: episode: 2763, duration: 1.268s, episode steps: 298, steps per second: 235, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.339 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1429611/1000000: episode: 2764, duration: 2.927s, episode steps: 697, steps per second: 238, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1430000/1000000: episode: 2765, duration: 1.654s, episode steps: 389, steps per second: 235, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1430414/1000000: episode: 2766, duration: 1.759s, episode steps: 414, steps per second: 235, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1430927/1000000: episode: 2767, duration: 2.156s, episode steps: 513, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1431510/1000000: episode: 2768, duration: 2.446s, episode steps: 583, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1431959/1000000: episode: 2769, duration: 1.893s, episode steps: 449, steps per second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1432767/1000000: episode: 2770, duration: 3.377s, episode steps: 808, steps per second: 239, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1433172/1000000: episode: 2771, duration: 1.717s, episode steps: 405, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1433702/1000000: episode: 2772, duration: 2.222s, episode steps: 530, steps per second: 239, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1434251/1000000: episode: 2773, duration: 2.293s, episode steps: 549, steps per second: 239, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1434654/1000000: episode: 2774, duration: 1.694s, episode steps: 403, steps per second: 238, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1435028/1000000: episode: 2775, duration: 1.572s, episode steps: 374, steps per

second: 238, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1435547/1000000: episode: 2776, duration: 2.189s, episode steps: 519, steps per second: 237, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1436121/1000000: episode: 2777, duration: 2.444s, episode steps: 574, steps per second: 235, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1436866/1000000: episode: 2778, duration: 3.124s, episode steps: 745, steps per second: 238, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1437245/1000000: episode: 2779, duration: 1.593s, episode steps: 379, steps per second: 238, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1437915/1000000: episode: 2780, duration: 2.810s, episode steps: 670, steps per second: 238, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1438402/1000000: episode: 2781, duration: 2.054s, episode steps: 487, steps per second: 237, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.657 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1438983/1000000: episode: 2782, duration: 2.454s, episode steps: 581, steps per second: 237, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1439388/1000000: episode: 2783, duration: 1.713s, episode steps: 405, steps per second: 236, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.336 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1440106/1000000: episode: 2784, duration: 3.008s, episode steps: 718, steps per second: 239, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1440666/1000000: episode: 2785, duration: 2.343s, episode steps: 560, steps per second: 239, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.346 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1440933/1000000: episode: 2786, duration: 1.120s, episode steps: 267, steps per second: 238, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1441232/1000000: episode: 2787, duration: 1.259s, episode steps: 299, steps per second: 238, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1441776/1000000: episode: 2788, duration: 2.298s, episode steps: 544, steps per second: 237, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1442208/1000000: episode: 2789, duration: 1.846s, episode steps: 432, steps per second: 234, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1442665/1000000: episode: 2790, duration: 1.917s, episode steps: 457, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1443110/1000000: episode: 2791, duration: 1.870s, episode steps: 445, steps per

second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1443684/1000000: episode: 2792, duration: 2.409s, episode steps: 574, steps per second: 238, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1444122/1000000: episode: 2793, duration: 1.842s, episode steps: 438, steps per second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1444746/1000000: episode: 2794, duration: 2.632s, episode steps: 624, steps per second: 237, episode reward: 19.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1445234/1000000: episode: 2795, duration: 2.067s, episode steps: 488, steps per second: 236, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1445707/1000000: episode: 2796, duration: 1.976s, episode steps: 473, steps per second: 239, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.387 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1446180/1000000: episode: 2797, duration: 1.983s, episode steps: 473, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1447032/1000000: episode: 2798, duration: 3.571s, episode steps: 852, steps per second: 239, episode reward: 10.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1447510/1000000: episode: 2799, duration: 2.018s, episode steps: 478, steps per second: 237, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1448150/1000000: episode: 2800, duration: 2.707s, episode steps: 640, steps per second: 236, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1448800/1000000: episode: 2801, duration: 2.725s, episode steps: 650, steps per second: 239, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1449390/1000000: episode: 2802, duration: 2.475s, episode steps: 590, steps per second: 238, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1450143/1000000: episode: 2803, duration: 3.158s, episode steps: 753, steps per second: 238, episode reward: 10.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1450835/1000000: episode: 2804, duration: 2.914s, episode steps: 692, steps per second: 237, episode reward: 8.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1451625/1000000: episode: 2805, duration: 3.319s, episode steps: 790, steps per second: 238, episode reward: 11.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.638 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1451937/1000000: episode: 2806, duration: 6.406s, episode steps: 312, steps per second: 49, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.032288, mae: 3.595140, mean_q: 4.352017, mean_eps: 0.987911

1452381/1000000: episode: 2807, duration: 17.707s, episode steps: 444, steps per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.040941, mae: 3.639836, mean_q: 4.399911, mean_eps: 0.987558

1452916/1000000: episode: 2808, duration: 21.437s, episode steps: 535, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: 0.047224, mae: 3.615981, mean_q: 4.378455, mean_eps: 0.986973

1453604/1000000: episode: 2809, duration: 27.625s, episode steps: 688, steps per second: 25, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: 0.047100, mae: 3.609947, mean_q: 4.366692, mean_eps: 0.986241

1453892/1000000: episode: 2810, duration: 11.623s, episode steps: 288, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.039603, mae: 3.622280, mean_q: 4.389085, mean_eps: 0.985655

1454453/1000000: episode: 2811, duration: 22.519s, episode steps: 561, steps per second: 25, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.039782, mae: 3.649046, mean_q: 4.414857, mean_eps: 0.985144

1455154/1000000: episode: 2812, duration: 27.767s, episode steps: 701, steps per second: 25, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.035150, mae: 3.597859, mean_q: 4.355777, mean_eps: 0.984386

1455569/1000000: episode: 2813, duration: 16.352s, episode steps: 415, steps per second: 25, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: 0.033110, mae: 3.625972, mean_q: 4.393633, mean_eps: 0.983716

1455968/1000000: episode: 2814, duration: 15.866s, episode steps: 399, steps per second: 25, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: 0.036105, mae: 3.646539, mean_q: 4.416036, mean_eps: 0.983229

1456575/1000000: episode: 2815, duration: 24.275s, episode steps: 607, steps per second: 25, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.041238, mae: 3.590523, mean_q: 4.348790, mean_eps: 0.982626

1456931/1000000: episode: 2816, duration: 14.135s, episode steps: 356, steps per second: 25, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: 0.038343, mae: 3.617066, mean_q: 4.391861, mean_eps: 0.982048

1457456/1000000: episode: 2817, duration: 21.160s, episode steps: 525, steps per second: 25, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: 0.038499, mae: 3.619645, mean_q: 4.381007, mean_eps: 0.981520

1457764/1000000: episode: 2818, duration: 12.457s, episode steps: 308, steps per second: 25, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.038713, mae: 3.638863, mean_q: 4.405610, mean_eps: 0.981021

1458275/1000000: episode: 2819, duration: 20.462s, episode steps: 511, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.041697, mae: 3.622216, mean_q: 4.384097, mean_eps: 0.980529

1458749/1000000: episode: 2820, duration: 18.916s, episode steps: 474, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.043470, mae: 3.651821, mean_q: 4.421733, mean_eps: 0.979936

1459204/1000000: episode: 2821, duration: 18.188s, episode steps: 455, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.038354, mae: 3.635130, mean_q: 4.402650, mean_eps: 0.979379

1459708/1000000: episode: 2822, duration: 20.148s, episode steps: 504, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.042095, mae: 3.658467, mean_q: 4.431937, mean_eps: 0.978806

1460230/1000000: episode: 2823, duration: 20.899s, episode steps: 522, steps per second: 25, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.722 [0.000, 5.000], loss: 0.036900, mae: 3.643168, mean_q: 4.414711, mean_eps: 0.978189

1460730/1000000: episode: 2824, duration: 19.897s, episode steps: 500, steps per second: 25, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.672 [0.000, 5.000], loss: 0.038681, mae: 3.646728, mean_q: 4.415578, mean_eps: 0.977574

1461637/1000000: episode: 2825, duration: 36.038s, episode steps: 907, steps per second: 25, episode reward: 17.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.041960, mae: 3.644102, mean_q: 4.416921, mean_eps: 0.976730

1461941/1000000: episode: 2826, duration: 12.136s, episode steps: 304, steps per second: 25, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.355 [0.000, 5.000], loss: 0.042411, mae: 3.663137, mean_q: 4.437182, mean_eps: 0.976002

1462365/1000000: episode: 2827, duration: 16.873s, episode steps: 424, steps per second: 25, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: 0.047020, mae: 3.724551, mean_q: 4.512422, mean_eps: 0.975566

1462922/1000000: episode: 2828, duration: 22.135s, episode steps: 557, steps per second: 25, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: 0.031284, mae: 3.685351, mean_q: 4.467548, mean_eps: 0.974978

1463549/1000000: episode: 2829, duration: 25.172s, episode steps: 627, steps per second: 25, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.037258, mae: 3.712520, mean_q: 4.502307, mean_eps: 0.974267

1464217/1000000: episode: 2830, duration: 26.617s, episode steps: 668, steps per second: 25, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.037066, mae: 3.648459, mean_q: 4.420474, mean_eps: 0.973490

1464491/1000000: episode: 2831, duration: 10.947s, episode steps: 274, steps per second: 25, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.321 [0.000, 5.000], loss: 0.037555, mae: 3.682614, mean_q: 4.457483, mean_eps: 0.972926

1465143/1000000: episode: 2832, duration: 26.190s, episode steps: 652, steps per second: 25, episode reward: 6.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.036288, mae: 3.714400, mean_q: 4.496689, mean_eps: 0.972371

1465856/1000000: episode: 2833, duration: 28.461s, episode steps: 713, steps per second: 25, episode reward: 22.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: 0.036473, mae: 3.737013, mean_q: 4.529020, mean_eps: 0.971553

1466143/1000000: episode: 2834, duration: 11.509s, episode steps: 287, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.035949, mae: 3.633205, mean_q: 4.403732, mean_eps: 0.970953

1466658/1000000: episode: 2835, duration: 20.532s, episode steps: 515, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.041922, mae: 3.647941, mean_q: 4.416344, mean_eps: 0.970470

1467247/1000000: episode: 2836, duration: 23.440s, episode steps: 589, steps per second: 25, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.040246, mae: 3.627766, mean_q: 4.392861, mean_eps: 0.969808

1467716/1000000: episode: 2837, duration: 18.862s, episode steps: 469, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: 0.040154, mae: 3.658754, mean_q: 4.428239, mean_eps: 0.969174

1468419/1000000: episode: 2838, duration: 28.047s, episode steps: 703, steps per second: 25, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.046986, mae: 3.723582, mean_q: 4.509805, mean_eps: 0.968471

1468728/1000000: episode: 2839, duration: 12.480s, episode steps: 309, steps per second: 25, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: 0.043379, mae: 3.646619, mean_q: 4.416862, mean_eps: 0.967864

1469213/1000000: episode: 2840, duration: 19.404s, episode steps: 485, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.036811, mae: 3.686555, mean_q: 4.464747, mean_eps: 0.967386

1469788/1000000: episode: 2841, duration: 22.914s, episode steps: 575, steps per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: 0.034645, mae: 3.667853, mean_q: 4.449977, mean_eps: 0.966750

1470085/1000000: episode: 2842, duration: 11.905s, episode steps: 297, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.040484, mae: 3.638818, mean_q: 4.414148, mean_eps: 0.966227

1470467/1000000: episode: 2843, duration: 15.092s, episode steps: 382, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.040961, mae: 3.680287, mean_q: 4.457666, mean_eps: 0.965819

1471266/1000000: episode: 2844, duration: 31.935s, episode steps: 799, steps per second: 25, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: 0.041261, mae: 3.683763, mean_q: 4.463996, mean_eps: 0.965111

1471872/1000000: episode: 2845, duration: 24.378s, episode steps: 606, steps per second: 25, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.038168, mae: 3.670176, mean_q: 4.440987, mean_eps: 0.964269

1472587/1000000: episode: 2846, duration: 28.505s, episode steps: 715, steps per second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.043750, mae: 3.681631, mean_q: 4.456615, mean_eps: 0.963477

1473109/1000000: episode: 2847, duration: 20.971s, episode steps: 522, steps per second: 25, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.042918, mae: 3.693766, mean_q: 4.471265, mean_eps: 0.962733

1473943/1000000: episode: 2848, duration: 33.073s, episode steps: 834, steps per second: 25, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.038403, mae: 3.704151, mean_q: 4.491243, mean_eps: 0.961919

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

1474640/1000000: episode: 2849, duration: 25.003s, episode steps: 697, steps per second: 28, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: 0.042139, mae: 3.716267, mean_q: 4.502440, mean_eps: 0.961050

1475121/1000000: episode: 2850, duration: 2.029s, episode steps: 481, steps per second: 237, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1475418/1000000: episode: 2851, duration: 1.262s, episode steps: 297, steps per second: 235, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.370 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1476015/1000000: episode: 2852, duration: 2.515s, episode steps: 597, steps per second: 237, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1476303/1000000: episode: 2853, duration: 1.227s, episode steps: 288, steps per second: 235, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1476695/1000000: episode: 2854, duration: 1.699s, episode steps: 392, steps per second: 231, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1477136/1000000: episode: 2855, duration: 1.854s, episode steps: 441, steps per

second: 238, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1477430/1000000: episode: 2856, duration: 1.252s, episode steps: 294, steps per second: 235, episode reward: 9.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.374 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1478022/1000000: episode: 2857, duration: 2.467s, episode steps: 592, steps per second: 240, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1478504/1000000: episode: 2858, duration: 2.021s, episode steps: 482, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1479104/1000000: episode: 2859, duration: 2.527s, episode steps: 600, steps per second: 237, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1479664/1000000: episode: 2860, duration: 2.361s, episode steps: 560, steps per second: 237, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1480158/1000000: episode: 2861, duration: 2.071s, episode steps: 494, steps per second: 239, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1480558/1000000: episode: 2862, duration: 1.693s, episode steps: 400, steps per second: 236, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.328 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1481100/1000000: episode: 2863, duration: 2.287s, episode steps: 542, steps per second: 237, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1481525/1000000: episode: 2864, duration: 1.799s, episode steps: 425, steps per second: 236, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1482167/1000000: episode: 2865, duration: 2.717s, episode steps: 642, steps per second: 236, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1482622/1000000: episode: 2866, duration: 1.932s, episode steps: 455, steps per second: 236, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.723 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1483175/1000000: episode: 2867, duration: 2.304s, episode steps: 553, steps per second: 240, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1483819/1000000: episode: 2868, duration: 2.695s, episode steps: 644, steps per second: 239, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1484194/1000000: episode: 2869, duration: 1.584s, episode steps: 375, steps per second: 237, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1484617/1000000: episode: 2870, duration: 1.779s, episode steps: 423, steps per second: 238, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1484912/1000000: episode: 2871, duration: 1.241s, episode steps: 295, steps per

second: 238, episode reward: 10.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1485383/1000000: episode: 2872, duration: 1.998s, episode steps: 471, steps per second: 236, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1485783/1000000: episode: 2873, duration: 1.690s, episode steps: 400, steps per second: 237, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1486054/1000000: episode: 2874, duration: 1.137s, episode steps: 271, steps per second: 238, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1486740/1000000: episode: 2875, duration: 2.882s, episode steps: 686, steps per second: 238, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1487360/1000000: episode: 2876, duration: 2.603s, episode steps: 620, steps per second: 238, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1488008/1000000: episode: 2877, duration: 2.709s, episode steps: 648, steps per second: 239, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1488758/1000000: episode: 2878, duration: 3.175s, episode steps: 750, steps per second: 236, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1489281/1000000: episode: 2879, duration: 2.198s, episode steps: 523, steps per second: 238, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1489777/1000000: episode: 2880, duration: 2.091s, episode steps: 496, steps per second: 237, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1490637/1000000: episode: 2881, duration: 3.662s, episode steps: 860, steps per second: 235, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1490963/1000000: episode: 2882, duration: 1.376s, episode steps: 326, steps per second: 237, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1491664/1000000: episode: 2883, duration: 2.961s, episode steps: 701, steps per second: 237, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1492152/1000000: episode: 2884, duration: 2.045s, episode steps: 488, steps per second: 239, episode reward: 3.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1492755/1000000: episode: 2885, duration: 2.554s, episode steps: 603, steps per second: 236, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1493618/1000000: episode: 2886, duration: 3.618s, episode steps: 863, steps per second: 238, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1494819/1000000: episode: 2887, duration: 4.974s, episode steps: 1201, steps per second: 241, episode reward: 33.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1495303/1000000: episode: 2888, duration: 2.036s, episode steps: 484, steps per second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1495641/1000000: episode: 2889, duration: 1.433s, episode steps: 338, steps per second: 236, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1496159/1000000: episode: 2890, duration: 2.181s, episode steps: 518, steps per second: 237, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1496630/1000000: episode: 2891, duration: 1.986s, episode steps: 471, steps per second: 237, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1497135/1000000: episode: 2892, duration: 2.144s, episode steps: 505, steps per second: 236, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1497529/1000000: episode: 2893, duration: 1.715s, episode steps: 394, steps per second: 230, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1498480/1000000: episode: 2894, duration: 3.956s, episode steps: 951, steps per second: 240, episode reward: 13.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1499019/1000000: episode: 2895, duration: 2.254s, episode steps: 539, steps per second: 239, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1499653/1000000: episode: 2896, duration: 2.646s, episode steps: 634, steps per second: 240, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1500015/1000000: episode: 2897, duration: 1.542s, episode steps: 362, steps per second: 235, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1500393/1000000: episode: 2898, duration: 1.618s, episode steps: 378, steps per second: 234, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1500861/1000000: episode: 2899, duration: 1.965s, episode steps: 468, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1501368/1000000: episode: 2900, duration: 2.139s, episode steps: 507, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1501762/1000000: episode: 2901, duration: 1.666s, episode steps: 394, steps per second: 237, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1502146/1000000: episode: 2902, duration: 1.623s, episode steps: 384, steps per second: 237, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1502566/1000000: episode: 2903, duration: 1.779s, episode steps: 420, steps per second: 236, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1503090/1000000: episode: 2904, duration: 2.234s, episode steps: 524, steps per second: 235, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1503411/1000000: episode: 2905, duration: 1.388s, episode steps: 321, steps per second: 231, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1504039/1000000: episode: 2906, duration: 2.635s, episode steps: 628, steps per second: 238, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1504344/1000000: episode: 2907, duration: 1.296s, episode steps: 305, steps per second: 235, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1504676/1000000: episode: 2908, duration: 1.405s, episode steps: 332, steps per second: 236, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1505204/1000000: episode: 2909, duration: 2.232s, episode steps: 528, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1506073/1000000: episode: 2910, duration: 3.675s, episode steps: 869, steps per second: 236, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1506657/1000000: episode: 2911, duration: 2.464s, episode steps: 584, steps per second: 237, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1506929/1000000: episode: 2912, duration: 1.147s, episode steps: 272, steps per second: 237, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1507347/1000000: episode: 2913, duration: 1.760s, episode steps: 418, steps per second: 238, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1507669/1000000: episode: 2914, duration: 1.356s, episode steps: 322, steps per second: 237, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1508254/1000000: episode: 2915, duration: 2.465s, episode steps: 585, steps per second: 237, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1508682/1000000: episode: 2916, duration: 1.815s, episode steps: 428, steps per second: 236, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1509576/1000000: episode: 2917, duration: 3.768s, episode steps: 894, steps per second: 237, episode reward: 22.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1510029/1000000: episode: 2918, duration: 1.904s, episode steps: 453, steps per second: 238, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1510307/1000000: episode: 2919, duration: 1.178s, episode steps: 278, steps per second: 236, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.665 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1510597/1000000: episode: 2920, duration: 1.229s, episode steps: 290, steps per second: 236, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1511427/1000000: episode: 2921, duration: 3.463s, episode steps: 830, steps per second: 240, episode reward: 21.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.384 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1511945/1000000: episode: 2922, duration: 2.196s, episode steps: 518, steps per second: 236, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1512799/1000000: episode: 2923, duration: 3.585s, episode steps: 854, steps per second: 238, episode reward: 22.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1513827/1000000: episode: 2924, duration: 4.274s, episode steps: 1028, steps per second: 241, episode reward: 20.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1514287/1000000: episode: 2925, duration: 1.932s, episode steps: 460, steps per second: 238, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1514690/1000000: episode: 2926, duration: 1.703s, episode steps: 403, steps per second: 237, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1515368/1000000: episode: 2927, duration: 2.884s, episode steps: 678, steps per second: 235, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1515685/1000000: episode: 2928, duration: 1.347s, episode steps: 317, steps per second: 235, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.375 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1516234/1000000: episode: 2929, duration: 2.325s, episode steps: 549, steps per second: 236, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1516565/1000000: episode: 2930, duration: 1.395s, episode steps: 331, steps per second: 237, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1517162/1000000: episode: 2931, duration: 2.507s, episode steps: 597, steps per second: 238, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1517459/1000000: episode: 2932, duration: 5.882s, episode steps: 297, steps per second: 50, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.037947, mae: 3.712001, mean_q: 4.505592, mean_eps: 0.987921

1517944/1000000: episode: 2933, duration: 19.253s, episode steps: 485, steps per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000],

mean action: 2.581 [0.000, 5.000], loss: 0.041238, mae: 3.768277, mean_q: 4.563318, mean_eps: 0.987554

1518428/1000000: episode: 2934, duration: 19.496s, episode steps: 484, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.045312, mae: 3.737642, mean_q: 4.524874, mean_eps: 0.986973

1518718/1000000: episode: 2935, duration: 11.701s, episode steps: 290, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: 0.037677, mae: 3.711099, mean_q: 4.498336, mean_eps: 0.986507

1519287/1000000: episode: 2936, duration: 22.681s, episode steps: 569, steps per second: 25, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: 0.041819, mae: 3.700151, mean_q: 4.481893, mean_eps: 0.985991

1519990/1000000: episode: 2937, duration: 28.027s, episode steps: 703, steps per second: 25, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.044251, mae: 3.725698, mean_q: 4.513060, mean_eps: 0.985228

1520511/1000000: episode: 2938, duration: 20.767s, episode steps: 521, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.040814, mae: 3.758664, mean_q: 4.551978, mean_eps: 0.984494

1520975/1000000: episode: 2939, duration: 18.580s, episode steps: 464, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.046688, mae: 3.690830, mean_q: 4.470051, mean_eps: 0.983903

1521360/1000000: episode: 2940, duration: 15.510s, episode steps: 385, steps per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.356 [0.000, 5.000], loss: 0.035429, mae: 3.750235, mean_q: 4.550647, mean_eps: 0.983394

1521759/1000000: episode: 2941, duration: 15.908s, episode steps: 399, steps per second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: 0.042692, mae: 3.755874, mean_q: 4.544870, mean_eps: 0.982924

1522461/1000000: episode: 2942, duration: 27.860s, episode steps: 702, steps per second: 25, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: 0.041140, mae: 3.734835, mean_q: 4.525829, mean_eps: 0.982262

1522839/1000000: episode: 2943, duration: 14.943s, episode steps: 378, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.044952, mae: 3.688122, mean_q: 4.473348, mean_eps: 0.981614

1523384/1000000: episode: 2944, duration: 21.692s, episode steps: 545, steps per second: 25, episode reward: 15.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: 0.035528, mae: 3.756587, mean_q: 4.555365, mean_eps: 0.981062

1523869/1000000: episode: 2945, duration: 19.431s, episode steps: 485, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000],

mean action: 2.548 [0.000, 5.000], loss: 0.043639, mae: 3.738543, mean_q:
 4.529344, mean_eps: 0.980442
 1524144/1000000: episode: 2946, duration: 11.044s, episode steps: 275, steps
 per second: 25, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000],
 mean action: 2.735 [0.000, 5.000], loss: 0.040838, mae: 3.708846, mean_q:
 4.491660, mean_eps: 0.979986
 1524630/1000000: episode: 2947, duration: 19.472s, episode steps: 486, steps
 per second: 25, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000],
 mean action: 2.644 [0.000, 5.000], loss: 0.039457, mae: 3.748304, mean_q:
 4.544762, mean_eps: 0.979530
 1525073/1000000: episode: 2948, duration: 17.703s, episode steps: 443, steps
 per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000],
 mean action: 2.528 [0.000, 5.000], loss: 0.038822, mae: 3.746015, mean_q:
 4.538866, mean_eps: 0.978971
 1525585/1000000: episode: 2949, duration: 20.371s, episode steps: 512, steps
 per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000],
 mean action: 2.430 [0.000, 5.000], loss: 0.040157, mae: 3.724016, mean_q:
 4.513534, mean_eps: 0.978398
 1525973/1000000: episode: 2950, duration: 15.379s, episode steps: 388, steps
 per second: 25, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000],
 mean action: 2.490 [0.000, 5.000], loss: 0.046989, mae: 3.713780, mean_q:
 4.502831, mean_eps: 0.977858
 1526398/1000000: episode: 2951, duration: 16.879s, episode steps: 425, steps
 per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000],
 mean action: 2.504 [0.000, 5.000], loss: 0.056456, mae: 3.743708, mean_q:
 4.537774, mean_eps: 0.977370
 1526687/1000000: episode: 2952, duration: 11.503s, episode steps: 289, steps
 per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000],
 mean action: 2.474 [0.000, 5.000], loss: 0.040696, mae: 3.715669, mean_q:
 4.496955, mean_eps: 0.976943
 1527036/1000000: episode: 2953, duration: 13.940s, episode steps: 349, steps
 per second: 25, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000],
 mean action: 2.530 [0.000, 5.000], loss: 0.047952, mae: 3.751142, mean_q:
 4.548267, mean_eps: 0.976562
 1527438/1000000: episode: 2954, duration: 16.057s, episode steps: 402, steps
 per second: 25, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000],
 mean action: 2.470 [0.000, 5.000], loss: 0.034526, mae: 3.735258, mean_q:
 4.531160, mean_eps: 0.976110
 1527888/1000000: episode: 2955, duration: 17.970s, episode steps: 450, steps
 per second: 25, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000],
 mean action: 2.584 [0.000, 5.000], loss: 0.035361, mae: 3.780894, mean_q:
 4.582930, mean_eps: 0.975599
 1528150/1000000: episode: 2956, duration: 10.532s, episode steps: 262, steps
 per second: 25, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000],
 mean action: 2.599 [0.000, 5.000], loss: 0.034309, mae: 3.789483, mean_q:
 4.590951, mean_eps: 0.975172
 1528520/1000000: episode: 2957, duration: 14.733s, episode steps: 370, steps
 per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000],

mean action: 2.643 [0.000, 5.000], loss: 0.034981, mae: 3.818107, mean_q: 4.630349, mean_eps: 0.974793

1528992/1000000: episode: 2958, duration: 19.121s, episode steps: 472, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.042692, mae: 3.823487, mean_q: 4.633012, mean_eps: 0.974289

1529513/1000000: episode: 2959, duration: 20.909s, episode steps: 521, steps per second: 25, episode reward: 15.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: 0.040567, mae: 3.766663, mean_q: 4.566474, mean_eps: 0.973691

1530177/1000000: episode: 2960, duration: 26.395s, episode steps: 664, steps per second: 25, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: 0.039345, mae: 3.802647, mean_q: 4.607708, mean_eps: 0.972978

1530475/1000000: episode: 2961, duration: 11.801s, episode steps: 298, steps per second: 25, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.309 [0.000, 5.000], loss: 0.038320, mae: 3.775880, mean_q: 4.578182, mean_eps: 0.972402

1530905/1000000: episode: 2962, duration: 17.196s, episode steps: 430, steps per second: 25, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.032641, mae: 3.793815, mean_q: 4.599875, mean_eps: 0.971966

1531380/1000000: episode: 2963, duration: 18.968s, episode steps: 475, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.044085, mae: 3.778836, mean_q: 4.580965, mean_eps: 0.971423

1531835/1000000: episode: 2964, duration: 18.323s, episode steps: 455, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: 0.037602, mae: 3.777702, mean_q: 4.576247, mean_eps: 0.970866

1532260/1000000: episode: 2965, duration: 17.065s, episode steps: 425, steps per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: 0.042501, mae: 3.786804, mean_q: 4.586460, mean_eps: 0.970338

1532755/1000000: episode: 2966, duration: 19.744s, episode steps: 495, steps per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.037516, mae: 3.791082, mean_q: 4.603237, mean_eps: 0.969786

1533233/1000000: episode: 2967, duration: 19.097s, episode steps: 478, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: 0.042550, mae: 3.824372, mean_q: 4.642308, mean_eps: 0.969201

1533605/1000000: episode: 2968, duration: 14.779s, episode steps: 372, steps per second: 25, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: 0.041448, mae: 3.838955, mean_q: 4.658759, mean_eps: 0.968690

1534080/1000000: episode: 2969, duration: 19.114s, episode steps: 475, steps per second: 25, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000],

mean action: 2.398 [0.000, 5.000], loss: 0.046074, mae: 3.828533, mean_q: 4.636857, mean_eps: 0.968183

1534510/1000000: episode: 2970, duration: 17.228s, episode steps: 430, steps per second: 25, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: 0.043699, mae: 3.808538, mean_q: 4.612774, mean_eps: 0.967641

1534929/1000000: episode: 2971, duration: 16.844s, episode steps: 419, steps per second: 25, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: 0.043185, mae: 3.823501, mean_q: 4.635749, mean_eps: 0.967130

1535260/1000000: episode: 2972, duration: 13.305s, episode steps: 331, steps per second: 25, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.048999, mae: 3.843220, mean_q: 4.653483, mean_eps: 0.966681

1535980/1000000: episode: 2973, duration: 28.866s, episode steps: 720, steps per second: 25, episode reward: 20.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.042160, mae: 3.812676, mean_q: 4.620675, mean_eps: 0.966052

1536429/1000000: episode: 2974, duration: 18.062s, episode steps: 449, steps per second: 25, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.040804, mae: 3.795196, mean_q: 4.596576, mean_eps: 0.965349

1536970/1000000: episode: 2975, duration: 21.544s, episode steps: 541, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.043950, mae: 3.855787, mean_q: 4.675026, mean_eps: 0.964754

1537412/1000000: episode: 2976, duration: 17.757s, episode steps: 442, steps per second: 25, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.046422, mae: 3.821096, mean_q: 4.624239, mean_eps: 0.964166

1538025/1000000: episode: 2977, duration: 24.520s, episode steps: 613, steps per second: 25, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: 0.035597, mae: 3.843512, mean_q: 4.658332, mean_eps: 0.963532

1538443/1000000: episode: 2978, duration: 16.695s, episode steps: 418, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.041985, mae: 3.798902, mean_q: 4.612576, mean_eps: 0.962913

1538749/1000000: episode: 2979, duration: 12.320s, episode steps: 306, steps per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: 0.039588, mae: 3.761587, mean_q: 4.554802, mean_eps: 0.962478

1539263/1000000: episode: 2980, duration: 20.498s, episode steps: 514, steps per second: 25, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: 0.040734, mae: 3.840742, mean_q: 4.648493, mean_eps: 0.961986

1539798/1000000: episode: 2981, duration: 21.437s, episode steps: 535, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000],

mean action: 2.477 [0.000, 5.000], loss: 0.044068, mae: 3.808805, mean_q:
4.617929, mean_eps: 0.961358

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

1540307/1000000: episode: 2982, duration: 12.778s, episode steps: 509, steps
per second: 40, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.580 [0.000, 5.000], loss: 0.041511, mae: 3.855815, mean_q:
4.679977, mean_eps: 0.960858

1540818/1000000: episode: 2983, duration: 2.179s, episode steps: 511, steps per
second: 235, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1541069/1000000: episode: 2984, duration: 1.064s, episode steps: 251, steps per
second: 236, episode reward: 5.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1541882/1000000: episode: 2985, duration: 3.398s, episode steps: 813, steps per
second: 239, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1542379/1000000: episode: 2986, duration: 2.097s, episode steps: 497, steps per
second: 237, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1543152/1000000: episode: 2987, duration: 3.249s, episode steps: 773, steps per
second: 238, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1543667/1000000: episode: 2988, duration: 2.182s, episode steps: 515, steps per
second: 236, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1544150/1000000: episode: 2989, duration: 2.026s, episode steps: 483, steps per
second: 238, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1544543/1000000: episode: 2990, duration: 1.647s, episode steps: 393, steps per
second: 239, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1544907/1000000: episode: 2991, duration: 1.532s, episode steps: 364, steps per
second: 238, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1545481/1000000: episode: 2992, duration: 2.404s, episode steps: 574, steps per
second: 239, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1545816/1000000: episode: 2993, duration: 1.413s, episode steps: 335, steps per
second: 237, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1546112/1000000: episode: 2994, duration: 1.268s, episode steps: 296, steps per
second: 233, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1546768/1000000: episode: 2995, duration: 2.766s, episode steps: 656, steps per
second: 237, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean

action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1547145/1000000: episode: 2996, duration: 1.601s, episode steps: 377, steps per second: 235, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1547520/1000000: episode: 2997, duration: 1.585s, episode steps: 375, steps per second: 237, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1548198/1000000: episode: 2998, duration: 2.857s, episode steps: 678, steps per second: 237, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1548584/1000000: episode: 2999, duration: 1.638s, episode steps: 386, steps per second: 236, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1548998/1000000: episode: 3000, duration: 1.746s, episode steps: 414, steps per second: 237, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1549602/1000000: episode: 3001, duration: 2.565s, episode steps: 604, steps per second: 235, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1550446/1000000: episode: 3002, duration: 3.514s, episode steps: 844, steps per second: 240, episode reward: 11.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.386 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1550894/1000000: episode: 3003, duration: 1.882s, episode steps: 448, steps per second: 238, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1551379/1000000: episode: 3004, duration: 2.043s, episode steps: 485, steps per second: 237, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1551861/1000000: episode: 3005, duration: 2.026s, episode steps: 482, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.255 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1552320/1000000: episode: 3006, duration: 1.955s, episode steps: 459, steps per second: 235, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1552842/1000000: episode: 3007, duration: 2.206s, episode steps: 522, steps per second: 237, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1553359/1000000: episode: 3008, duration: 2.178s, episode steps: 517, steps per second: 237, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1553916/1000000: episode: 3009, duration: 2.334s, episode steps: 557, steps per second: 239, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1554389/1000000: episode: 3010, duration: 1.985s, episode steps: 473, steps per second: 238, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1554974/1000000: episode: 3011, duration: 2.445s, episode steps: 585, steps per second: 239, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1555282/1000000: episode: 3012, duration: 1.301s, episode steps: 308, steps per second: 237, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1556262/1000000: episode: 3013, duration: 4.082s, episode steps: 980, steps per second: 240, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1556555/1000000: episode: 3014, duration: 1.239s, episode steps: 293, steps per second: 237, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1557258/1000000: episode: 3015, duration: 2.965s, episode steps: 703, steps per second: 237, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1557749/1000000: episode: 3016, duration: 2.085s, episode steps: 491, steps per second: 235, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1558310/1000000: episode: 3017, duration: 2.406s, episode steps: 561, steps per second: 233, episode reward: 15.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1558853/1000000: episode: 3018, duration: 2.297s, episode steps: 543, steps per second: 236, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1559337/1000000: episode: 3019, duration: 2.041s, episode steps: 484, steps per second: 237, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1559795/1000000: episode: 3020, duration: 1.926s, episode steps: 458, steps per second: 238, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1560257/1000000: episode: 3021, duration: 1.939s, episode steps: 462, steps per second: 238, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1560782/1000000: episode: 3022, duration: 2.201s, episode steps: 525, steps per second: 238, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1561301/1000000: episode: 3023, duration: 2.196s, episode steps: 519, steps per second: 236, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1561928/1000000: episode: 3024, duration: 2.629s, episode steps: 627, steps per second: 239, episode reward: 17.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1562346/1000000: episode: 3025, duration: 1.756s, episode steps: 418, steps per second: 238, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1562660/1000000: episode: 3026, duration: 1.321s, episode steps: 314, steps per second: 238, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1562958/1000000: episode: 3027, duration: 1.264s, episode steps: 298, steps per second: 236, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.262 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1563360/1000000: episode: 3028, duration: 1.682s, episode steps: 402, steps per second: 239, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1564257/1000000: episode: 3029, duration: 3.744s, episode steps: 897, steps per second: 240, episode reward: 21.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1564661/1000000: episode: 3030, duration: 1.710s, episode steps: 404, steps per second: 236, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1565046/1000000: episode: 3031, duration: 1.609s, episode steps: 385, steps per second: 239, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1565967/1000000: episode: 3032, duration: 3.857s, episode steps: 921, steps per second: 239, episode reward: 14.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1566563/1000000: episode: 3033, duration: 2.507s, episode steps: 596, steps per second: 238, episode reward: 20.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1566961/1000000: episode: 3034, duration: 1.678s, episode steps: 398, steps per second: 237, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1567594/1000000: episode: 3035, duration: 2.676s, episode steps: 633, steps per second: 237, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1568112/1000000: episode: 3036, duration: 2.176s, episode steps: 518, steps per second: 238, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.347 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1568604/1000000: episode: 3037, duration: 2.070s, episode steps: 492, steps per second: 238, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1569009/1000000: episode: 3038, duration: 1.703s, episode steps: 405, steps per second: 238, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1569413/1000000: episode: 3039, duration: 1.702s, episode steps: 404, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.668 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1569747/1000000: episode: 3040, duration: 1.410s, episode steps: 334, steps per second: 237, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 spartial/version_1.0/pesos/dqn_v_1.0_weights_3040_episodes.h5f

1570020/1000000: episode: 3041, duration: 1.251s, episode steps: 273, steps per second: 218, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.718 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1570487/1000000: episode: 3042, duration: 1.966s, episode steps: 467, steps per

second: 237, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1570996/1000000: episode: 3043, duration: 2.132s, episode steps: 509, steps per second: 239, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1571709/1000000: episode: 3044, duration: 2.991s, episode steps: 713, steps per second: 238, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1572111/1000000: episode: 3045, duration: 1.690s, episode steps: 402, steps per second: 238, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1572456/1000000: episode: 3046, duration: 1.457s, episode steps: 345, steps per second: 237, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1573038/1000000: episode: 3047, duration: 2.482s, episode steps: 582, steps per second: 235, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1573590/1000000: episode: 3048, duration: 2.323s, episode steps: 552, steps per second: 238, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1574198/1000000: episode: 3049, duration: 2.549s, episode steps: 608, steps per second: 239, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1574782/1000000: episode: 3050, duration: 2.435s, episode steps: 584, steps per second: 240, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1575172/1000000: episode: 3051, duration: 1.645s, episode steps: 390, steps per second: 237, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.651 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1575986/1000000: episode: 3052, duration: 3.414s, episode steps: 814, steps per second: 238, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1576279/1000000: episode: 3053, duration: 1.250s, episode steps: 293, steps per second: 234, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1576661/1000000: episode: 3054, duration: 1.603s, episode steps: 382, steps per second: 238, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1577335/1000000: episode: 3055, duration: 2.825s, episode steps: 674, steps per second: 239, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.657 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1577693/1000000: episode: 3056, duration: 1.508s, episode steps: 358, steps per second: 237, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1578054/1000000: episode: 3057, duration: 1.522s, episode steps: 361, steps per second: 237, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1578525/1000000: episode: 3058, duration: 1.970s, episode steps: 471, steps per

second: 239, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1579135/1000000: episode: 3059, duration: 2.595s, episode steps: 610, steps per second: 235, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1579794/1000000: episode: 3060, duration: 2.763s, episode steps: 659, steps per second: 238, episode reward: 8.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1580602/1000000: episode: 3061, duration: 3.370s, episode steps: 808, steps per second: 240, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1580996/1000000: episode: 3062, duration: 1.667s, episode steps: 394, steps per second: 236, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.690 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1581444/1000000: episode: 3063, duration: 1.891s, episode steps: 448, steps per second: 237, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1582023/1000000: episode: 3064, duration: 2.471s, episode steps: 579, steps per second: 234, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1582398/1000000: episode: 3065, duration: 1.586s, episode steps: 375, steps per second: 237, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1583232/1000000: episode: 3066, duration: 16.596s, episode steps: 834, steps per second: 50, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: 0.054219, mae: 3.833996, mean_q: 4.642038, mean_eps: 0.987779

1583834/1000000: episode: 3067, duration: 24.053s, episode steps: 602, steps per second: 25, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: 0.047476, mae: 3.790378, mean_q: 4.593857, mean_eps: 0.987198

1584395/1000000: episode: 3068, duration: 22.427s, episode steps: 561, steps per second: 25, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: 0.044566, mae: 3.865229, mean_q: 4.685122, mean_eps: 0.986500

1584946/1000000: episode: 3069, duration: 22.085s, episode steps: 551, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.050786, mae: 3.862205, mean_q: 4.681857, mean_eps: 0.985833

1585356/1000000: episode: 3070, duration: 16.468s, episode steps: 410, steps per second: 25, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.042787, mae: 3.842558, mean_q: 4.652239, mean_eps: 0.985257

1585827/1000000: episode: 3071, duration: 18.703s, episode steps: 471, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: 0.041482, mae: 3.847557, mean_q: 4.655114, mean_eps: 0.984729

1586412/1000000: episode: 3072, duration: 23.390s, episode steps: 585, steps

per second: 25, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.561 [0.000, 5.000], loss: 0.053597, mae: 3.886834, mean_q:
4.710385, mean_eps: 0.984095

1586901/1000000: episode: 3073, duration: 19.753s, episode steps: 489, steps
per second: 25, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.354 [0.000, 5.000], loss: 0.040174, mae: 3.848967, mean_q:
4.659794, mean_eps: 0.983450

1587385/1000000: episode: 3074, duration: 19.153s, episode steps: 484, steps
per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.428 [0.000, 5.000], loss: 0.052029, mae: 3.883688, mean_q:
4.701603, mean_eps: 0.982864

1587957/1000000: episode: 3075, duration: 23.059s, episode steps: 572, steps
per second: 25, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.638 [0.000, 5.000], loss: 0.045955, mae: 3.838604, mean_q:
4.652700, mean_eps: 0.982230

1588777/1000000: episode: 3076, duration: 32.777s, episode steps: 820, steps
per second: 25, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.429 [0.000, 5.000], loss: 0.048210, mae: 3.891086, mean_q:
4.711588, mean_eps: 0.981395

1589138/1000000: episode: 3077, duration: 14.528s, episode steps: 361, steps
per second: 25, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.343 [0.000, 5.000], loss: 0.050399, mae: 3.884869, mean_q:
4.706027, mean_eps: 0.980687

1589635/1000000: episode: 3078, duration: 19.861s, episode steps: 497, steps
per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.471 [0.000, 5.000], loss: 0.042859, mae: 3.916404, mean_q:
4.754162, mean_eps: 0.980174

1590358/1000000: episode: 3079, duration: 29.016s, episode steps: 723, steps
per second: 25, episode reward: 20.000, mean reward: 0.028 [0.000, 1.000],
mean action: 2.521 [0.000, 5.000], loss: 0.044385, mae: 3.920558, mean_q:
4.755212, mean_eps: 0.979442

1591277/1000000: episode: 3080, duration: 36.599s, episode steps: 919, steps
per second: 25, episode reward: 16.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.643 [0.000, 5.000], loss: 0.044797, mae: 3.894065, mean_q:
4.717108, mean_eps: 0.978455

1591560/1000000: episode: 3081, duration: 11.339s, episode steps: 283, steps
per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.332 [0.000, 5.000], loss: 0.049445, mae: 3.823467, mean_q:
4.652599, mean_eps: 0.977735

1592104/1000000: episode: 3082, duration: 21.786s, episode steps: 544, steps
per second: 25, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.371 [0.000, 5.000], loss: 0.053783, mae: 3.832150, mean_q:
4.643732, mean_eps: 0.977241

1592833/1000000: episode: 3083, duration: 28.994s, episode steps: 729, steps
per second: 25, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.440 [0.000, 5.000], loss: 0.043291, mae: 3.886561, mean_q:
4.708264, mean_eps: 0.976475

1593278/1000000: episode: 3084, duration: 17.790s, episode steps: 445, steps

per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.535 [0.000, 5.000], loss: 0.053705, mae: 3.880748, mean_q:
4.706073, mean_eps: 0.975770

1593643/1000000: episode: 3085, duration: 14.541s, episode steps: 365, steps
per second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000],
mean action: 2.551 [0.000, 5.000], loss: 0.048910, mae: 3.946377, mean_q:
4.784426, mean_eps: 0.975285

1594459/1000000: episode: 3086, duration: 32.702s, episode steps: 816, steps
per second: 25, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.597 [0.000, 5.000], loss: 0.044868, mae: 3.928702, mean_q:
4.760799, mean_eps: 0.974577

1595078/1000000: episode: 3087, duration: 24.683s, episode steps: 619, steps
per second: 25, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.483 [0.000, 5.000], loss: 0.041082, mae: 3.963572, mean_q:
4.797419, mean_eps: 0.973715

1595782/1000000: episode: 3088, duration: 28.264s, episode steps: 704, steps
per second: 25, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.520 [0.000, 5.000], loss: 0.044873, mae: 3.898391, mean_q:
4.721115, mean_eps: 0.972921

1596248/1000000: episode: 3089, duration: 18.756s, episode steps: 466, steps
per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.539 [0.000, 5.000], loss: 0.056805, mae: 3.953122, mean_q:
4.790023, mean_eps: 0.972220

1596551/1000000: episode: 3090, duration: 12.149s, episode steps: 303, steps
per second: 25, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.472 [0.000, 5.000], loss: 0.051715, mae: 3.924187, mean_q:
4.748857, mean_eps: 0.971759

1597255/1000000: episode: 3091, duration: 28.157s, episode steps: 704, steps
per second: 25, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.530 [0.000, 5.000], loss: 0.041724, mae: 3.924989, mean_q:
4.758444, mean_eps: 0.971154

1597742/1000000: episode: 3092, duration: 19.516s, episode steps: 487, steps
per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.370 [0.000, 5.000], loss: 0.048099, mae: 3.989573, mean_q:
4.828985, mean_eps: 0.970439

1598175/1000000: episode: 3093, duration: 17.174s, episode steps: 433, steps
per second: 25, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.510 [0.000, 5.000], loss: 0.045248, mae: 3.924081, mean_q:
4.752262, mean_eps: 0.969887

1598825/1000000: episode: 3094, duration: 25.989s, episode steps: 650, steps
per second: 25, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.651 [0.000, 5.000], loss: 0.049690, mae: 3.959898, mean_q:
4.795895, mean_eps: 0.969237

1599388/1000000: episode: 3095, duration: 22.397s, episode steps: 563, steps
per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.510 [0.000, 5.000], loss: 0.046108, mae: 3.978252, mean_q:
4.822000, mean_eps: 0.968510

1599822/1000000: episode: 3096, duration: 17.444s, episode steps: 434, steps

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per second: 25, episode reward: 8.000, mean reward: 0.018 [ 0.000, 1.000],
mean action: 2.560 [0.000, 5.000], loss: 0.049309, mae: 3.939821, mean_q:
4.771248, mean_eps: 0.967912
1600240/1000000: episode: 3097, duration: 16.847s, episode steps: 418, steps
per second: 25, episode reward: 10.000, mean reward: 0.024 [ 0.000, 1.000],
mean action: 2.476 [0.000, 5.000], loss: 0.043188, mae: 3.927551, mean_q:
4.758282, mean_eps: 0.967401
1600627/1000000: episode: 3098, duration: 15.503s, episode steps: 387, steps
per second: 25, episode reward: 8.000, mean reward: 0.021 [ 0.000, 1.000],
mean action: 2.512 [0.000, 5.000], loss: 0.044533, mae: 3.932386, mean_q:
4.766177, mean_eps: 0.966918
1601025/1000000: episode: 3099, duration: 16.061s, episode steps: 398, steps
per second: 25, episode reward: 7.000, mean reward: 0.018 [ 0.000, 1.000],
mean action: 2.394 [0.000, 5.000], loss: 0.059740, mae: 3.912664, mean_q:
4.739610, mean_eps: 0.966446
1601617/1000000: episode: 3100, duration: 23.660s, episode steps: 592, steps
per second: 25, episode reward: 14.000, mean reward: 0.024 [ 0.000, 1.000],
mean action: 2.549 [0.000, 5.000], loss: 0.041680, mae: 3.977932, mean_q:
4.817883, mean_eps: 0.965850
1602006/1000000: episode: 3101, duration: 15.467s, episode steps: 389, steps
per second: 25, episode reward: 9.000, mean reward: 0.023 [ 0.000, 1.000],
mean action: 2.442 [0.000, 5.000], loss: 0.045845, mae: 3.943615, mean_q:
4.776971, mean_eps: 0.965262
done, took 25864.601 seconds

```

```

[ ]: # Evaluacion - spatial_dropout_agent
CARPETA_CHECKPOINT_SPATIAL = '/content/gdrive/MyDrive/08_MIAR/actividades/
↳proyecto practico/dqn spartial/version_1.0/mejor_modelo'
CARPETA_VIDEO_SPATIAL = '/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
↳practico/dqn spartial/version_1.0/video'

# Ejecucion del test con 100 episodios para metricas.
spatial_results = evaluar_agente(
    agent=spatial_dropout_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_SPATIAL,
    nb_episodes=100
)

# Generación de video.
spatial_video = evaluar_agente(
    agent=spatial_dropout_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_SPATIAL,
    video_dir=CARPETA_VIDEO_SPATIAL,
    nb_episodes=2
)

```

```
print_results("DQN SPATIAL", spatial_results)
```

-> Iniciando evaluación (Modo Métricas Puro)...

```
/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:
UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied
automatically.
```

```
updates=self.state_updates,
```

-> Iniciando evaluación (Modo Video Activo)...

Video exportado exitosamente a:

```
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
spartial/version_1.0/video
```

Resultados del Agente: DQN SPATIAL

Rendimiento promedio: 9.3

Rendimiento mínimo: 4.0

Rendimiento máximo: 28.0

Desviación estándar: 4.468780594300866

Implementación 3: Dueling DQN

```
[ ]: dueling_dqn_model =
    ↪ create_dueling_dqn_model(input_shape=BASE_CONFIG["INPUT_SHAPE"],
                                ↪
                                ↪ window_length=BASE_CONFIG["WINDOW_LENGTH"],
                                nb_actions = nb_actions)
print(dueling_dqn_model.summary())

dueling_agent = create_agent(model=dueling_dqn_model, nb_actions=nb_actions,
    ↪ enable_double=True)
dueling_agent = compile_agent(dueling_agent)

DQN_DUELING_VERSION_ACTUAL = "1.0"
DQN_DUELING_CARPETA = '/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
    ↪ practico/dqn dueling/'
DQN_DUELING_METRICS = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
    ↪ practico/dqn dueling/version_{DQN_DUELING_VERSION_ACTUAL}/metricas.csv'
DQN_DUELING_LOGS = f'/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
    ↪ practico/dqn dueling/version_{DQN_DUELING_VERSION_ACTUAL}/logs.json'
```

Model: "Dueling_DQN_SpaceInvaders"

```
-----
-----
Layer (type)                Output Shape                Param #   Connected to
=====
input_3 (InputLayer)        [(None, 4, 84, 84)]        0         []
```

permute_2 (Permute) ['input_3[0][0]']	(None, 84, 84, 4)	0
conv2d_6 (Conv2D) ['permute_2[0][0]']	(None, 20, 20, 32)	8224
conv2d_7 (Conv2D) ['conv2d_6[0][0]']	(None, 9, 9, 64)	32832
conv2d_8 (Conv2D) ['conv2d_7[0][0]']	(None, 7, 7, 64)	36928
flatten_2 (Flatten) ['conv2d_8[0][0]']	(None, 3136)	0
dense_4 (Dense) ['flatten_2[0][0]']	(None, 512)	1606144
dense_6 (Dense) ['flatten_2[0][0]']	(None, 512)	1606144
dense_5 (Dense) ['dense_4[0][0]']	(None, 1)	513
dense_7 (Dense) ['dense_6[0][0]']	(None, 6)	3078
dueling_output (DuelingOut ['dense_5[0][0]', putLayer) 'dense_7[0][0]']	(None, 6)	0

```
=====
=====
Total params: 3293863 (12.57 MB)
Trainable params: 3293863 (12.57 MB)
Non-trainable params: 0 (0.00 Byte)
-----
-----
None
```

```
[ ]: # Entrenamiento
      # Prepararemos los callbacks contruidos
      dqn_dueling_guardar_pesos = SaveWeightsInterval(
          agent=dueling_agent,
          version=DQN_DUELING_VERSION_ACTUAL,
```



```

        interval=160,
        base_path=DQN_DUELING_CARPETA)

dqn_dueling_guardar_mejor_modelo = SaveBestWeights(
    agent=dueling_agent,
    version=DQN_DUELING_VERSION_ACTUAL,
    base_path=DQN_DUELING_CARPETA)

dqn_dueling_metricas = MetricLogger(
    version=DQN_DUELING_VERSION_ACTUAL,
    base_path=DQN_DUELING_CARPETA)

dqn_dueling_lista_callbacks = [dqn_dueling_guardar_pesos,
                                dqn_dueling_guardar_mejor_modelo,
                                dqn_dueling_metricas,
                                FileLogger(DQN_DUELING_LOGS, interval=160),
                                ↵
                                ↪TotalStepStopper(max_steps=BASE_CONFIG['max_steps'])]

history_3 = entrenando_modelo_con_recuperacion(
    agent=dueling_agent,
    env=env,
    version=DQN_DUELING_VERSION_ACTUAL,
    max_steps=BASE_CONFIG['max_steps'],
    carpeta_guardado=DQN_DUELING_CARPETA,
    callbacks_lista=dqn_dueling_lista_callbacks,
    ruta_csv=DQN_DUELING_METRICS
)

```

No se encontraron pesos previos. Iniciando desde cero.

Retomando entrenamiento. Faltan exactamente: 1000000 steps.

Training for 1000000 steps ...

/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:

UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

updates=self.state_updates,

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_0_episodes.h5f

Nuevo récord: 22.0. Pesos guardados en

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/mejor_modelo/best_weights.h5f

920/1000000: episode: 1, duration: 5.476s, episode steps: 920, steps per

second: 168, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1599/1000000: episode: 2, duration: 2.927s, episode steps: 679, steps per second: 232, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

2136/1000000: episode: 3, duration: 2.334s, episode steps: 537, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

2751/1000000: episode: 4, duration: 2.679s, episode steps: 615, steps per second: 230, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.353 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

3233/1000000: episode: 5, duration: 2.131s, episode steps: 482, steps per second: 226, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

3638/1000000: episode: 6, duration: 1.757s, episode steps: 405, steps per second: 230, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

4011/1000000: episode: 7, duration: 1.625s, episode steps: 373, steps per second: 230, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

4582/1000000: episode: 8, duration: 2.478s, episode steps: 571, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

5301/1000000: episode: 9, duration: 3.135s, episode steps: 719, steps per second: 229, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

6103/1000000: episode: 10, duration: 3.501s, episode steps: 802, steps per second: 229, episode reward: 22.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

6658/1000000: episode: 11, duration: 2.395s, episode steps: 555, steps per second: 232, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

Nuevo récord: 24.0. Pesos guardados en

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/mejor_modelo/best_weights.h5f

7657/1000000: episode: 12, duration: 4.377s, episode steps: 999, steps per second: 228, episode reward: 24.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

8231/1000000: episode: 13, duration: 2.509s, episode steps: 574, steps per second: 229, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

8528/1000000: episode: 14, duration: 1.320s, episode steps: 297, steps per second: 225, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

9152/1000000: episode: 15, duration: 2.743s, episode steps: 624, steps per second: 227, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

9550/1000000: episode: 16, duration: 1.764s, episode steps: 398, steps per second: 226, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354: UserWarning: `Model.state\_updates` will be removed in a future version. This property should not be used in TensorFlow 2.0, as `updates` are applied automatically.

updates=self.state_updates,

10350/1000000: episode: 17, duration: 15.380s, episode steps: 800, steps per second: 52, episode reward: 9.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: 0.011793, mae: 0.055968, mean_q: 0.084157, mean_eps: 0.987789

10825/1000000: episode: 18, duration: 15.123s, episode steps: 475, steps per second: 31, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.007716, mae: 0.040529, mean_q: 0.054474, mean_eps: 0.987294

11413/1000000: episode: 19, duration: 18.227s, episode steps: 588, steps per second: 32, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: 0.010232, mae: 0.045690, mean_q: 0.061356, mean_eps: 0.986656

11981/1000000: episode: 20, duration: 18.641s, episode steps: 568, steps per second: 30, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: 0.009814, mae: 0.053815, mean_q: 0.071050, mean_eps: 0.985962

12613/1000000: episode: 21, duration: 20.772s, episode steps: 632, steps per second: 30, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.010196, mae: 0.048103, mean_q: 0.063974, mean_eps: 0.985242

13024/1000000: episode: 22, duration: 13.542s, episode steps: 411, steps per second: 30, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.009843, mae: 0.053578, mean_q: 0.078434, mean_eps: 0.984618

13498/1000000: episode: 23, duration: 15.626s, episode steps: 474, steps per second: 30, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: 0.010499, mae: 0.042688, mean_q: 0.058281, mean_eps: 0.984088

13990/1000000: episode: 24, duration: 16.190s, episode steps: 492, steps per second: 30, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.007891, mae: 0.040384, mean_q: 0.060265, mean_eps: 0.983507

14440/1000000: episode: 25, duration: 14.887s, episode steps: 450, steps per second: 30, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.010518, mae: 0.042450, mean_q: 0.062534, mean_eps: 0.982943

14875/1000000: episode: 26, duration: 14.386s, episode steps: 435, steps per second: 30, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.006596, mae: 0.040027, mean_q: 0.063480,

mean_eps: 0.982413

15409/1000000: episode: 27, duration: 16.434s, episode steps: 534, steps per second: 32, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.009164, mae: 0.049004, mean_q: 0.068889, mean_eps: 0.981830

15855/1000000: episode: 28, duration: 14.530s, episode steps: 446, steps per second: 31, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.010493, mae: 0.057614, mean_q: 0.084952, mean_eps: 0.981242

16369/1000000: episode: 29, duration: 16.997s, episode steps: 514, steps per second: 30, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: 0.007708, mae: 0.048310, mean_q: 0.073190, mean_eps: 0.980666

16768/1000000: episode: 30, duration: 13.176s, episode steps: 399, steps per second: 30, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.007803, mae: 0.044874, mean_q: 0.064077, mean_eps: 0.980118

17247/1000000: episode: 31, duration: 15.837s, episode steps: 479, steps per second: 30, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.009351, mae: 0.052181, mean_q: 0.075844, mean_eps: 0.979593

17549/1000000: episode: 32, duration: 10.049s, episode steps: 302, steps per second: 30, episode reward: 9.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.010810, mae: 0.052492, mean_q: 0.079244, mean_eps: 0.979122

18020/1000000: episode: 33, duration: 15.557s, episode steps: 471, steps per second: 30, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.008531, mae: 0.051049, mean_q: 0.078270, mean_eps: 0.978659

18427/1000000: episode: 34, duration: 13.444s, episode steps: 407, steps per second: 30, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: 0.009436, mae: 0.053744, mean_q: 0.081527, mean_eps: 0.978134

18912/1000000: episode: 35, duration: 16.082s, episode steps: 485, steps per second: 30, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.007820, mae: 0.049513, mean_q: 0.078368, mean_eps: 0.977598

19720/1000000: episode: 36, duration: 26.783s, episode steps: 808, steps per second: 30, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.008588, mae: 0.049945, mean_q: 0.078340, mean_eps: 0.976823

20244/1000000: episode: 37, duration: 16.686s, episode steps: 524, steps per second: 31, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.008210, mae: 0.059159, mean_q: 0.092517, mean_eps: 0.976024

20974/1000000: episode: 38, duration: 23.490s, episode steps: 730, steps per second: 31, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.009269, mae: 0.066095, mean_q: 0.098069,

mean_eps: 0.975270

21476/1000000: episode: 39, duration: 16.674s, episode steps: 502, steps per second: 30, episode reward: 14.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: 0.008418, mae: 0.066940, mean_q: 0.102384, mean_eps: 0.974531

21980/1000000: episode: 40, duration: 16.741s, episode steps: 504, steps per second: 30, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.008326, mae: 0.067155, mean_q: 0.097672, mean_eps: 0.973929

22653/1000000: episode: 41, duration: 22.303s, episode steps: 673, steps per second: 30, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.008744, mae: 0.065741, mean_q: 0.094388, mean_eps: 0.973221

23417/1000000: episode: 42, duration: 25.191s, episode steps: 764, steps per second: 30, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.008960, mae: 0.066076, mean_q: 0.096668, mean_eps: 0.972357

23891/1000000: episode: 43, duration: 15.604s, episode steps: 474, steps per second: 30, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.008423, mae: 0.062645, mean_q: 0.093258, mean_eps: 0.971615

24266/1000000: episode: 44, duration: 12.448s, episode steps: 375, steps per second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.008649, mae: 0.066061, mean_q: 0.100084, mean_eps: 0.971106

24727/1000000: episode: 45, duration: 15.185s, episode steps: 461, steps per second: 30, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.008351, mae: 0.066321, mean_q: 0.101159, mean_eps: 0.970605

25017/1000000: episode: 46, duration: 9.631s, episode steps: 290, steps per second: 30, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: 0.007678, mae: 0.062725, mean_q: 0.090521, mean_eps: 0.970154

25385/1000000: episode: 47, duration: 11.038s, episode steps: 368, steps per second: 33, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.009261, mae: 0.099315, mean_q: 0.138015, mean_eps: 0.969758

25924/1000000: episode: 48, duration: 17.646s, episode steps: 539, steps per second: 31, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: 0.008031, mae: 0.097890, mean_q: 0.136439, mean_eps: 0.969215

26388/1000000: episode: 49, duration: 15.485s, episode steps: 464, steps per second: 30, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.008948, mae: 0.099740, mean_q: 0.141000, mean_eps: 0.968615

26937/1000000: episode: 50, duration: 18.271s, episode steps: 549, steps per second: 30, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.007310, mae: 0.093109, mean_q: 0.126675,

```

mean_eps: 0.968006
27412/1000000: episode: 51, duration: 15.781s, episode steps: 475, steps per
second: 30, episode reward: 8.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.467 [0.000, 5.000], loss: 0.008042, mae: 0.096505, mean_q: 0.129086,
mean_eps: 0.967391
27820/1000000: episode: 52, duration: 13.612s, episode steps: 408, steps per
second: 30, episode reward: 6.000, mean reward: 0.015 [ 0.000, 1.000], mean
action: 2.527 [0.000, 5.000], loss: 0.007033, mae: 0.097671, mean_q: 0.138939,
mean_eps: 0.966863
28321/1000000: episode: 53, duration: 16.750s, episode steps: 501, steps per
second: 30, episode reward: 7.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.579 [0.000, 5.000], loss: 0.007029, mae: 0.096147, mean_q: 0.134419,
mean_eps: 0.966316
29029/1000000: episode: 54, duration: 23.438s, episode steps: 708, steps per
second: 30, episode reward: 15.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.613 [0.000, 5.000], loss: 0.006954, mae: 0.092935, mean_q: 0.128967,
mean_eps: 0.965589
29552/1000000: episode: 55, duration: 17.440s, episode steps: 523, steps per
second: 30, episode reward: 11.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.428 [0.000, 5.000], loss: 0.008109, mae: 0.096659, mean_q: 0.131954,
mean_eps: 0.964852
29913/1000000: episode: 56, duration: 12.139s, episode steps: 361, steps per
second: 30, episode reward: 7.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.355 [0.000, 5.000], loss: 0.006870, mae: 0.092584, mean_q: 0.129331,
mean_eps: 0.964322
30609/1000000: episode: 57, duration: 21.802s, episode steps: 696, steps per
second: 32, episode reward: 16.000, mean reward: 0.023 [ 0.000, 1.000], mean
action: 2.533 [0.000, 5.000], loss: 0.007831, mae: 0.129059, mean_q: 0.171498,
mean_eps: 0.963686
30906/1000000: episode: 58, duration: 9.876s, episode steps: 297, steps per
second: 30, episode reward: 4.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.710 [0.000, 5.000], loss: 0.008292, mae: 0.126904, mean_q: 0.169385,
mean_eps: 0.963090
31427/1000000: episode: 59, duration: 17.340s, episode steps: 521, steps per
second: 30, episode reward: 14.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.491 [0.000, 5.000], loss: 0.007190, mae: 0.132673, mean_q: 0.175281,
mean_eps: 0.962601
32051/1000000: episode: 60, duration: 20.747s, episode steps: 624, steps per
second: 30, episode reward: 15.000, mean reward: 0.024 [ 0.000, 1.000], mean
action: 2.553 [0.000, 5.000], loss: 0.008116, mae: 0.135810, mean_q: 0.176756,
mean_eps: 0.961914
32556/1000000: episode: 61, duration: 16.880s, episode steps: 505, steps per
second: 30, episode reward: 9.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.442 [0.000, 5.000], loss: 0.007890, mae: 0.133971, mean_q: 0.179842,
mean_eps: 0.961238

```

```

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

```

self.step += 1

33055/1000000: episode: 62, duration: 8.279s, episode steps: 499, steps per second: 60, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.008280, mae: 0.131167, mean_q: 0.176253, mean_eps: 0.960808

33445/1000000: episode: 63, duration: 1.719s, episode steps: 390, steps per second: 227, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

34229/1000000: episode: 64, duration: 3.454s, episode steps: 784, steps per second: 227, episode reward: 23.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

34674/1000000: episode: 65, duration: 1.946s, episode steps: 445, steps per second: 229, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

34970/1000000: episode: 66, duration: 1.297s, episode steps: 296, steps per second: 228, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

35498/1000000: episode: 67, duration: 2.302s, episode steps: 528, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

Nuevo récord: 25.0. Pesos guardados en

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/mejor_modelo/best_weights.h5f

36363/1000000: episode: 68, duration: 3.840s, episode steps: 865, steps per second: 225, episode reward: 25.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37331/1000000: episode: 69, duration: 4.232s, episode steps: 968, steps per second: 229, episode reward: 19.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37621/1000000: episode: 70, duration: 1.278s, episode steps: 290, steps per second: 227, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

37943/1000000: episode: 71, duration: 1.414s, episode steps: 322, steps per second: 228, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

38792/1000000: episode: 72, duration: 3.674s, episode steps: 849, steps per second: 231, episode reward: 21.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.624 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

39095/1000000: episode: 73, duration: 1.336s, episode steps: 303, steps per second: 227, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.647 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

39554/1000000: episode: 74, duration: 2.030s, episode steps: 459, steps per second: 226, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

40167/1000000: episode: 75, duration: 2.681s, episode steps: 613, steps per second: 229, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 40641/1000000: episode: 76, duration: 2.085s, episode steps: 474, steps per second: 227, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 41286/1000000: episode: 77, duration: 2.778s, episode steps: 645, steps per second: 232, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 41877/1000000: episode: 78, duration: 2.564s, episode steps: 591, steps per second: 231, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 42293/1000000: episode: 79, duration: 1.867s, episode steps: 416, steps per second: 223, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 42670/1000000: episode: 80, duration: 1.664s, episode steps: 377, steps per second: 227, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 42961/1000000: episode: 81, duration: 1.277s, episode steps: 291, steps per second: 228, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 43327/1000000: episode: 82, duration: 1.606s, episode steps: 366, steps per second: 228, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 43924/1000000: episode: 83, duration: 2.592s, episode steps: 597, steps per second: 230, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 44451/1000000: episode: 84, duration: 2.278s, episode steps: 527, steps per second: 231, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 44934/1000000: episode: 85, duration: 2.102s, episode steps: 483, steps per second: 230, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 45698/1000000: episode: 86, duration: 3.371s, episode steps: 764, steps per second: 227, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46036/1000000: episode: 87, duration: 1.482s, episode steps: 338, steps per second: 228, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46484/1000000: episode: 88, duration: 1.950s, episode steps: 448, steps per second: 230, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.279 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 46890/1000000: episode: 89, duration: 1.761s, episode steps: 406, steps per second: 230, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 47366/1000000: episode: 90, duration: 2.072s, episode steps: 476, steps per second: 230, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 47881/1000000: episode: 91, duration: 2.248s, episode steps: 515, steps per second: 229, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 48984/1000000: episode: 92, duration: 4.845s, episode steps: 1103, steps per second: 228, episode reward: 14.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 49522/1000000: episode: 93, duration: 2.352s, episode steps: 538, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 49985/1000000: episode: 94, duration: 2.030s, episode steps: 463, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 50392/1000000: episode: 95, duration: 1.784s, episode steps: 407, steps per second: 228, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 50800/1000000: episode: 96, duration: 1.801s, episode steps: 408, steps per second: 227, episode reward: 11.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 51477/1000000: episode: 97, duration: 2.977s, episode steps: 677, steps per second: 227, episode reward: 20.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 51978/1000000: episode: 98, duration: 2.182s, episode steps: 501, steps per second: 230, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 52606/1000000: episode: 99, duration: 2.753s, episode steps: 628, steps per second: 228, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 52959/1000000: episode: 100, duration: 1.558s, episode steps: 353, steps per second: 227, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 53262/1000000: episode: 101, duration: 1.339s, episode steps: 303, steps per second: 226, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 53666/1000000: episode: 102, duration: 1.764s, episode steps: 404, steps per second: 229, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 53958/1000000: episode: 103, duration: 1.309s, episode steps: 292, steps per second: 223, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 54673/1000000: episode: 104, duration: 3.119s, episode steps: 715, steps per second: 229, episode reward: 22.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 55130/1000000: episode: 105, duration: 1.985s, episode steps: 457, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 55612/1000000: episode: 106, duration: 2.109s, episode steps: 482, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 56291/1000000: episode: 107, duration: 2.937s, episode steps: 679, steps per second: 231, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean

action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 56729/1000000: episode: 108, duration: 1.918s, episode steps: 438, steps per second: 228, episode reward: 11.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 57101/1000000: episode: 109, duration: 1.651s, episode steps: 372, steps per second: 225, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 57613/1000000: episode: 110, duration: 2.248s, episode steps: 512, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 58120/1000000: episode: 111, duration: 2.198s, episode steps: 507, steps per second: 231, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 58542/1000000: episode: 112, duration: 1.841s, episode steps: 422, steps per second: 229, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 59015/1000000: episode: 113, duration: 2.065s, episode steps: 473, steps per second: 229, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 59648/1000000: episode: 114, duration: 2.757s, episode steps: 633, steps per second: 230, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 60041/1000000: episode: 115, duration: 1.766s, episode steps: 393, steps per second: 223, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 60573/1000000: episode: 116, duration: 2.334s, episode steps: 532, steps per second: 228, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 60882/1000000: episode: 117, duration: 1.351s, episode steps: 309, steps per second: 229, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 61382/1000000: episode: 118, duration: 2.170s, episode steps: 500, steps per second: 230, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 62134/1000000: episode: 119, duration: 3.256s, episode steps: 752, steps per second: 231, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 62540/1000000: episode: 120, duration: 1.803s, episode steps: 406, steps per second: 225, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.308 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 63048/1000000: episode: 121, duration: 2.262s, episode steps: 508, steps per second: 225, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 63715/1000000: episode: 122, duration: 2.894s, episode steps: 667, steps per second: 230, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 64022/1000000: episode: 123, duration: 1.345s, episode steps: 307, steps per second: 228, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.368 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 64420/1000000: episode: 124, duration: 1.740s, episode steps: 398, steps per second: 229, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.359 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 64794/1000000: episode: 125, duration: 1.623s, episode steps: 374, steps per second: 230, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 65238/1000000: episode: 126, duration: 1.931s, episode steps: 444, steps per second: 230, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 65712/1000000: episode: 127, duration: 2.114s, episode steps: 474, steps per second: 224, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 66018/1000000: episode: 128, duration: 1.350s, episode steps: 306, steps per second: 227, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 66582/1000000: episode: 129, duration: 2.430s, episode steps: 564, steps per second: 232, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 67102/1000000: episode: 130, duration: 2.257s, episode steps: 520, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 67581/1000000: episode: 131, duration: 2.074s, episode steps: 479, steps per second: 231, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 67875/1000000: episode: 132, duration: 1.293s, episode steps: 294, steps per second: 227, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.310 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 68367/1000000: episode: 133, duration: 2.167s, episode steps: 492, steps per second: 227, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 68863/1000000: episode: 134, duration: 2.182s, episode steps: 496, steps per second: 227, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 69292/1000000: episode: 135, duration: 1.864s, episode steps: 429, steps per second: 230, episode reward: 12.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 69803/1000000: episode: 136, duration: 2.225s, episode steps: 511, steps per second: 230, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

Nuevo récord: 28.0. Pesos guardados en

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn

dueling/version_1.0/mejor_modelo/best_weights.h5f

70859/1000000: episode: 137, duration: 4.643s, episode steps: 1056, steps per second: 227, episode reward: 28.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

71599/1000000: episode: 138, duration: 3.216s, episode steps: 740, steps per

second: 230, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

72322/1000000: episode: 139, duration: 3.109s, episode steps: 723, steps per second: 233, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

72841/1000000: episode: 140, duration: 2.247s, episode steps: 519, steps per second: 231, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

73356/1000000: episode: 141, duration: 2.247s, episode steps: 515, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

73917/1000000: episode: 142, duration: 2.461s, episode steps: 561, steps per second: 228, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

74491/1000000: episode: 143, duration: 2.575s, episode steps: 574, steps per second: 223, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

74762/1000000: episode: 144, duration: 1.198s, episode steps: 271, steps per second: 226, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

75123/1000000: episode: 145, duration: 1.591s, episode steps: 361, steps per second: 227, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

75846/1000000: episode: 146, duration: 12.089s, episode steps: 723, steps per second: 60, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.624 [0.000, 5.000], loss: 0.009770, mae: 0.188330, mean_q: 0.243525, mean_eps: 0.987813

76328/1000000: episode: 147, duration: 16.105s, episode steps: 482, steps per second: 30, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: 0.009012, mae: 0.178523, mean_q: 0.223007, mean_eps: 0.987340

76917/1000000: episode: 148, duration: 19.675s, episode steps: 589, steps per second: 30, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: 0.009386, mae: 0.178711, mean_q: 0.220939, mean_eps: 0.986697

77639/1000000: episode: 149, duration: 23.994s, episode steps: 722, steps per second: 30, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.009351, mae: 0.182069, mean_q: 0.226004, mean_eps: 0.985910

78130/1000000: episode: 150, duration: 16.400s, episode steps: 491, steps per second: 30, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.008975, mae: 0.177306, mean_q: 0.220036, mean_eps: 0.985182

78509/1000000: episode: 151, duration: 12.639s, episode steps: 379, steps per second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.007736, mae: 0.174630, mean_q: 0.217559, mean_eps: 0.984659

79133/1000000: episode: 152, duration: 20.779s, episode steps: 624, steps per

second: 30, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.009053, mae: 0.177906, mean_q: 0.222471, mean_eps: 0.984057

79527/1000000: episode: 153, duration: 13.081s, episode steps: 394, steps per second: 30, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.008330, mae: 0.179954, mean_q: 0.225104, mean_eps: 0.983447

80118/1000000: episode: 154, duration: 19.732s, episode steps: 591, steps per second: 30, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.008961, mae: 0.184417, mean_q: 0.227573, mean_eps: 0.982857

80675/1000000: episode: 155, duration: 18.160s, episode steps: 557, steps per second: 31, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.008406, mae: 0.187066, mean_q: 0.233060, mean_eps: 0.982168

81183/1000000: episode: 156, duration: 16.030s, episode steps: 508, steps per second: 32, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.007420, mae: 0.203524, mean_q: 0.249486, mean_eps: 0.981530

81656/1000000: episode: 157, duration: 15.874s, episode steps: 473, steps per second: 30, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: 0.008504, mae: 0.202485, mean_q: 0.247907, mean_eps: 0.980942

82211/1000000: episode: 158, duration: 18.564s, episode steps: 555, steps per second: 30, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: 0.008680, mae: 0.204074, mean_q: 0.249152, mean_eps: 0.980325

82711/1000000: episode: 159, duration: 16.709s, episode steps: 500, steps per second: 30, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.010119, mae: 0.203998, mean_q: 0.249433, mean_eps: 0.979691

83126/1000000: episode: 160, duration: 13.903s, episode steps: 415, steps per second: 30, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: 0.007790, mae: 0.195749, mean_q: 0.240499, mean_eps: 0.979142

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_160_episodes.h5f

83598/1000000: episode: 161, duration: 15.820s, episode steps: 472, steps per second: 30, episode reward: 4.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: 0.006889, mae: 0.193842, mean_q: 0.238944, mean_eps: 0.978609

84092/1000000: episode: 162, duration: 16.504s, episode steps: 494, steps per second: 30, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.007921, mae: 0.207009, mean_q: 0.253421, mean_eps: 0.978030

84703/1000000: episode: 163, duration: 20.351s, episode steps: 611, steps per

second: 30, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.007135, mae: 0.199265, mean_q: 0.245596, mean_eps: 0.977368

85553/1000000: episode: 164, duration: 28.382s, episode steps: 850, steps per second: 30, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.374 [0.000, 5.000], loss: 0.007561, mae: 0.195290, mean_q: 0.241947, mean_eps: 0.976490

85915/1000000: episode: 165, duration: 10.939s, episode steps: 362, steps per second: 33, episode reward: 12.000, mean reward: 0.033 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.010068, mae: 0.226333, mean_q: 0.277293, mean_eps: 0.975762

86520/1000000: episode: 166, duration: 20.121s, episode steps: 605, steps per second: 30, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.009363, mae: 0.238175, mean_q: 0.291963, mean_eps: 0.975184

87102/1000000: episode: 167, duration: 19.477s, episode steps: 582, steps per second: 30, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.008734, mae: 0.232550, mean_q: 0.285707, mean_eps: 0.974471

87397/1000000: episode: 168, duration: 9.849s, episode steps: 295, steps per second: 30, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: 0.007468, mae: 0.230828, mean_q: 0.284377, mean_eps: 0.973943

87695/1000000: episode: 169, duration: 9.872s, episode steps: 298, steps per second: 30, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.009120, mae: 0.227926, mean_q: 0.281461, mean_eps: 0.973588

88941/1000000: episode: 170, duration: 41.503s, episode steps: 1246, steps per second: 30, episode reward: 25.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.007675, mae: 0.228418, mean_q: 0.280689, mean_eps: 0.972662

89230/1000000: episode: 171, duration: 9.637s, episode steps: 289, steps per second: 30, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.009361, mae: 0.234500, mean_q: 0.289461, mean_eps: 0.971740

89849/1000000: episode: 172, duration: 20.689s, episode steps: 619, steps per second: 30, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.009054, mae: 0.228348, mean_q: 0.283175, mean_eps: 0.971195

90648/1000000: episode: 173, duration: 26.305s, episode steps: 799, steps per second: 30, episode reward: 23.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.007946, mae: 0.236453, mean_q: 0.292472, mean_eps: 0.970346

91201/1000000: episode: 174, duration: 17.582s, episode steps: 553, steps per second: 31, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.351 [0.000, 5.000], loss: 0.009199, mae: 0.270548, mean_q: 0.333630, mean_eps: 0.969534

91550/1000000: episode: 175, duration: 11.636s, episode steps: 349, steps per

second: 30, episode reward: 9.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.008030, mae: 0.255689, mean_q: 0.315329, mean_eps: 0.968992

92267/1000000: episode: 176, duration: 23.903s, episode steps: 717, steps per second: 30, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.386 [0.000, 5.000], loss: 0.008337, mae: 0.266422, mean_q: 0.325829, mean_eps: 0.968354

92787/1000000: episode: 177, duration: 17.379s, episode steps: 520, steps per second: 30, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.007831, mae: 0.265313, mean_q: 0.323948, mean_eps: 0.967612

93123/1000000: episode: 178, duration: 11.205s, episode steps: 336, steps per second: 30, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: 0.012245, mae: 0.272728, mean_q: 0.334147, mean_eps: 0.967098

93491/1000000: episode: 179, duration: 12.300s, episode steps: 368, steps per second: 30, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.007949, mae: 0.260940, mean_q: 0.321096, mean_eps: 0.966676

94245/1000000: episode: 180, duration: 25.279s, episode steps: 754, steps per second: 30, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.008395, mae: 0.259855, mean_q: 0.320292, mean_eps: 0.966002

95085/1000000: episode: 181, duration: 28.030s, episode steps: 840, steps per second: 30, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: 0.007372, mae: 0.262453, mean_q: 0.321737, mean_eps: 0.965044

95742/1000000: episode: 182, duration: 21.345s, episode steps: 657, steps per second: 31, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.008364, mae: 0.270773, mean_q: 0.334182, mean_eps: 0.964146

96202/1000000: episode: 183, duration: 14.684s, episode steps: 460, steps per second: 31, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.717 [0.000, 5.000], loss: 0.010829, mae: 0.295807, mean_q: 0.363625, mean_eps: 0.963477

96490/1000000: episode: 184, duration: 9.669s, episode steps: 288, steps per second: 30, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.008845, mae: 0.288195, mean_q: 0.354797, mean_eps: 0.963028

96787/1000000: episode: 185, duration: 9.902s, episode steps: 297, steps per second: 30, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.009088, mae: 0.281657, mean_q: 0.345705, mean_eps: 0.962678

97082/1000000: episode: 186, duration: 9.884s, episode steps: 295, steps per second: 30, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: 0.009454, mae: 0.277466, mean_q: 0.341012, mean_eps: 0.962322

97421/1000000: episode: 187, duration: 11.364s, episode steps: 339, steps per

second: 30, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.646 [0.000, 5.000], loss: 0.007408, mae: 0.271165, mean_q: 0.333515,
mean_eps: 0.961941

97884/1000000: episode: 188, duration: 15.502s, episode steps: 463, steps per
second: 30, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.514 [0.000, 5.000], loss: 0.008492, mae: 0.283004, mean_q: 0.348143,
mean_eps: 0.961461

98269/1000000: episode: 189, duration: 12.981s, episode steps: 385, steps per
second: 30, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.426 [0.000, 5.000], loss: 0.009355, mae: 0.281471, mean_q: 0.345036,
mean_eps: 0.960952

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

98976/1000000: episode: 190, duration: 3.998s, episode steps: 707, steps per
second: 177, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.495 [0.000, 5.000], loss: 0.010145, mae: 0.323535, mean_q: 0.396277,
mean_eps: 0.960700

99388/1000000: episode: 191, duration: 1.798s, episode steps: 412, steps per
second: 229, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

100142/1000000: episode: 192, duration: 3.262s, episode steps: 754, steps per
second: 231, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

100625/1000000: episode: 193, duration: 2.126s, episode steps: 483, steps per
second: 227, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

100923/1000000: episode: 194, duration: 1.316s, episode steps: 298, steps per
second: 227, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

101222/1000000: episode: 195, duration: 1.333s, episode steps: 299, steps per
second: 224, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean
action: 2.639 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

101660/1000000: episode: 196, duration: 1.900s, episode steps: 438, steps per
second: 231, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean
action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

102107/1000000: episode: 197, duration: 1.948s, episode steps: 447, steps per
second: 229, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

102521/1000000: episode: 198, duration: 1.800s, episode steps: 414, steps per
second: 230, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean
action: 2.263 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

102992/1000000: episode: 199, duration: 2.079s, episode steps: 471, steps per
second: 227, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.312 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

103506/1000000: episode: 200, duration: 2.253s, episode steps: 514, steps per
second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 104331/1000000: episode: 201, duration: 3.577s, episode steps: 825, steps per
 second: 231, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 104669/1000000: episode: 202, duration: 1.470s, episode steps: 338, steps per
 second: 230, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.822 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 105131/1000000: episode: 203, duration: 2.005s, episode steps: 462, steps per
 second: 230, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean
 action: 2.604 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 105766/1000000: episode: 204, duration: 2.767s, episode steps: 635, steps per
 second: 229, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 106181/1000000: episode: 205, duration: 1.864s, episode steps: 415, steps per
 second: 223, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean
 action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 106511/1000000: episode: 206, duration: 1.480s, episode steps: 330, steps per
 second: 223, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 106821/1000000: episode: 207, duration: 1.354s, episode steps: 310, steps per
 second: 229, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.329 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 107222/1000000: episode: 208, duration: 1.746s, episode steps: 401, steps per
 second: 230, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean
 action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 107767/1000000: episode: 209, duration: 2.368s, episode steps: 545, steps per
 second: 230, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 108579/1000000: episode: 210, duration: 3.526s, episode steps: 812, steps per
 second: 230, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 109231/1000000: episode: 211, duration: 2.878s, episode steps: 652, steps per
 second: 227, episode reward: 19.000, mean reward: 0.029 [0.000, 1.000], mean
 action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 109726/1000000: episode: 212, duration: 2.145s, episode steps: 495, steps per
 second: 231, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 110013/1000000: episode: 213, duration: 1.258s, episode steps: 287, steps per
 second: 228, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 110501/1000000: episode: 214, duration: 2.123s, episode steps: 488, steps per
 second: 230, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean
 action: 2.350 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 111174/1000000: episode: 215, duration: 2.936s, episode steps: 673, steps per
 second: 229, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 111860/1000000: episode: 216, duration: 3.007s, episode steps: 686, steps per
 second: 228, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 112249/1000000: episode: 217, duration: 1.712s, episode steps: 389, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 112658/1000000: episode: 218, duration: 1.784s, episode steps: 409, steps per second: 229, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 113054/1000000: episode: 219, duration: 1.725s, episode steps: 396, steps per second: 230, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 113688/1000000: episode: 220, duration: 2.735s, episode steps: 634, steps per second: 232, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 114160/1000000: episode: 221, duration: 2.061s, episode steps: 472, steps per second: 229, episode reward: 16.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 114938/1000000: episode: 222, duration: 3.416s, episode steps: 778, steps per second: 228, episode reward: 14.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 115497/1000000: episode: 223, duration: 2.435s, episode steps: 559, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 115906/1000000: episode: 224, duration: 1.773s, episode steps: 409, steps per second: 231, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 116487/1000000: episode: 225, duration: 2.530s, episode steps: 581, steps per second: 230, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 116918/1000000: episode: 226, duration: 1.901s, episode steps: 431, steps per second: 227, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 117268/1000000: episode: 227, duration: 1.544s, episode steps: 350, steps per second: 227, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 117968/1000000: episode: 228, duration: 3.078s, episode steps: 700, steps per second: 227, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 118452/1000000: episode: 229, duration: 2.115s, episode steps: 484, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 118909/1000000: episode: 230, duration: 1.988s, episode steps: 457, steps per second: 230, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 119586/1000000: episode: 231, duration: 2.955s, episode steps: 677, steps per second: 229, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 120194/1000000: episode: 232, duration: 2.659s, episode steps: 608, steps per second: 229, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean

action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 120490/1000000: episode: 233, duration: 1.314s, episode steps: 296, steps per second: 225, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 121141/1000000: episode: 234, duration: 2.865s, episode steps: 651, steps per second: 227, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 121509/1000000: episode: 235, duration: 1.594s, episode steps: 368, steps per second: 231, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 121946/1000000: episode: 236, duration: 1.916s, episode steps: 437, steps per second: 228, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 122259/1000000: episode: 237, duration: 1.364s, episode steps: 313, steps per second: 229, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 122811/1000000: episode: 238, duration: 2.390s, episode steps: 552, steps per second: 231, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 123188/1000000: episode: 239, duration: 1.664s, episode steps: 377, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.318 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 124115/1000000: episode: 240, duration: 4.027s, episode steps: 927, steps per second: 230, episode reward: 13.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 124637/1000000: episode: 241, duration: 2.270s, episode steps: 522, steps per second: 230, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 125431/1000000: episode: 242, duration: 3.432s, episode steps: 794, steps per second: 231, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 126220/1000000: episode: 243, duration: 3.448s, episode steps: 789, steps per second: 229, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 126884/1000000: episode: 244, duration: 2.896s, episode steps: 664, steps per second: 229, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 127915/1000000: episode: 245, duration: 4.455s, episode steps: 1031, steps per second: 231, episode reward: 14.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 128816/1000000: episode: 246, duration: 3.886s, episode steps: 901, steps per second: 232, episode reward: 21.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 129368/1000000: episode: 247, duration: 2.461s, episode steps: 552, steps per second: 224, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 129635/1000000: episode: 248, duration: 1.176s, episode steps: 267, steps per second: 227, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean

action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 130181/1000000: episode: 249, duration: 2.391s, episode steps: 546, steps per second: 228, episode reward: 14.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 130483/1000000: episode: 250, duration: 1.323s, episode steps: 302, steps per second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 131021/1000000: episode: 251, duration: 2.332s, episode steps: 538, steps per second: 231, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 131433/1000000: episode: 252, duration: 1.801s, episode steps: 412, steps per second: 229, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 131917/1000000: episode: 253, duration: 2.119s, episode steps: 484, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 132436/1000000: episode: 254, duration: 2.300s, episode steps: 519, steps per second: 226, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 132919/1000000: episode: 255, duration: 2.112s, episode steps: 483, steps per second: 229, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.685 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 133483/1000000: episode: 256, duration: 2.443s, episode steps: 564, steps per second: 231, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 133870/1000000: episode: 257, duration: 1.687s, episode steps: 387, steps per second: 229, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 134327/1000000: episode: 258, duration: 1.989s, episode steps: 457, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 134647/1000000: episode: 259, duration: 1.408s, episode steps: 320, steps per second: 227, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.666 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 134949/1000000: episode: 260, duration: 1.338s, episode steps: 302, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.325 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 135505/1000000: episode: 261, duration: 2.437s, episode steps: 556, steps per second: 228, episode reward: 16.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 135878/1000000: episode: 262, duration: 1.634s, episode steps: 373, steps per second: 228, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 136299/1000000: episode: 263, duration: 1.853s, episode steps: 421, steps per second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 136803/1000000: episode: 264, duration: 2.169s, episode steps: 504, steps per second: 232, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 137258/1000000: episode: 265, duration: 1.975s, episode steps: 455, steps per second: 230, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 137933/1000000: episode: 266, duration: 2.968s, episode steps: 675, steps per second: 227, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 138571/1000000: episode: 267, duration: 2.796s, episode steps: 638, steps per second: 228, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 139011/1000000: episode: 268, duration: 1.917s, episode steps: 440, steps per second: 230, episode reward: 13.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 139564/1000000: episode: 269, duration: 2.413s, episode steps: 553, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 140202/1000000: episode: 270, duration: 2.785s, episode steps: 638, steps per second: 229, episode reward: 6.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 140855/1000000: episode: 271, duration: 2.880s, episode steps: 653, steps per second: 227, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 141142/1000000: episode: 272, duration: 3.239s, episode steps: 287, steps per second: 89, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.011487, mae: 0.296674, mean_q: 0.361238, mean_eps: 0.987957
 141581/1000000: episode: 273, duration: 14.826s, episode steps: 439, steps per second: 30, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.009370, mae: 0.321372, mean_q: 0.392107, mean_eps: 0.987652
 141972/1000000: episode: 274, duration: 13.241s, episode steps: 391, steps per second: 30, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.010267, mae: 0.328570, mean_q: 0.400526, mean_eps: 0.987155
 142342/1000000: episode: 275, duration: 12.553s, episode steps: 370, steps per second: 29, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.695 [0.000, 5.000], loss: 0.008889, mae: 0.310172, mean_q: 0.377886, mean_eps: 0.986699
 142889/1000000: episode: 276, duration: 18.482s, episode steps: 547, steps per second: 30, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.009091, mae: 0.320277, mean_q: 0.391500, mean_eps: 0.986147
 143292/1000000: episode: 277, duration: 13.692s, episode steps: 403, steps per second: 29, episode reward: 1.000, mean reward: 0.002 [0.000, 1.000], mean action: 2.370 [0.000, 5.000], loss: 0.007706, mae: 0.314239, mean_q: 0.385376, mean_eps: 0.985578
 143794/1000000: episode: 278, duration: 16.974s, episode steps: 502, steps per second: 30, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.665 [0.000, 5.000], loss: 0.009541, mae: 0.313493, mean_q: 0.383998,
 mean_eps: 0.985036
 144939/1000000: episode: 279, duration: 38.557s, episode steps: 1145, steps per
 second: 30, episode reward: 19.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.495 [0.000, 5.000], loss: 0.009514, mae: 0.328737, mean_q: 0.402712,
 mean_eps: 0.984047
 145313/1000000: episode: 280, duration: 12.647s, episode steps: 374, steps per
 second: 30, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.620 [0.000, 5.000], loss: 0.008406, mae: 0.314173, mean_q: 0.386142,
 mean_eps: 0.983135
 145950/1000000: episode: 281, duration: 21.462s, episode steps: 637, steps per
 second: 30, episode reward: 17.000, mean reward: 0.027 [0.000, 1.000], mean
 action: 2.538 [0.000, 5.000], loss: 0.009144, mae: 0.325312, mean_q: 0.398877,
 mean_eps: 0.982528
 146278/1000000: episode: 282, duration: 10.496s, episode steps: 328, steps per
 second: 31, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean
 action: 2.701 [0.000, 5.000], loss: 0.011444, mae: 0.331951, mean_q: 0.405036,
 mean_eps: 0.981950
 147073/1000000: episode: 283, duration: 26.189s, episode steps: 795, steps per
 second: 30, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.434 [0.000, 5.000], loss: 0.010490, mae: 0.329187, mean_q: 0.402196,
 mean_eps: 0.981275
 147712/1000000: episode: 284, duration: 21.605s, episode steps: 639, steps per
 second: 30, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.624 [0.000, 5.000], loss: 0.010911, mae: 0.330631, mean_q: 0.407209,
 mean_eps: 0.980416
 148001/1000000: episode: 285, duration: 9.853s, episode steps: 289, steps per
 second: 29, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.581 [0.000, 5.000], loss: 0.008689, mae: 0.320530, mean_q: 0.395254,
 mean_eps: 0.979859
 148420/1000000: episode: 286, duration: 14.205s, episode steps: 419, steps per
 second: 29, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.382 [0.000, 5.000], loss: 0.009166, mae: 0.328243, mean_q: 0.405285,
 mean_eps: 0.979434
 148922/1000000: episode: 287, duration: 17.115s, episode steps: 502, steps per
 second: 29, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.622 [0.000, 5.000], loss: 0.012005, mae: 0.333542, mean_q: 0.412671,
 mean_eps: 0.978882
 149351/1000000: episode: 288, duration: 14.476s, episode steps: 429, steps per
 second: 30, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.483 [0.000, 5.000], loss: 0.009790, mae: 0.323854, mean_q: 0.396768,
 mean_eps: 0.978323
 150283/1000000: episode: 289, duration: 31.574s, episode steps: 932, steps per
 second: 30, episode reward: 24.000, mean reward: 0.026 [0.000, 1.000], mean
 action: 2.476 [0.000, 5.000], loss: 0.009768, mae: 0.329672, mean_q: 0.406824,
 mean_eps: 0.977507
 150829/1000000: episode: 290, duration: 18.568s, episode steps: 546, steps per
 second: 29, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.412 [0.000, 5.000], loss: 0.008907, mae: 0.323151, mean_q: 0.397787, mean_eps: 0.976619

151498/1000000: episode: 291, duration: 21.357s, episode steps: 669, steps per second: 31, episode reward: 7.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: 0.010447, mae: 0.354533, mean_q: 0.435396, mean_eps: 0.975890

151921/1000000: episode: 292, duration: 14.297s, episode steps: 423, steps per second: 30, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.011431, mae: 0.365947, mean_q: 0.447781, mean_eps: 0.975234

152255/1000000: episode: 293, duration: 11.263s, episode steps: 334, steps per second: 30, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.293 [0.000, 5.000], loss: 0.008804, mae: 0.352243, mean_q: 0.433090, mean_eps: 0.974781

152666/1000000: episode: 294, duration: 13.928s, episode steps: 411, steps per second: 30, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.009530, mae: 0.365120, mean_q: 0.449122, mean_eps: 0.974334

153039/1000000: episode: 295, duration: 12.601s, episode steps: 373, steps per second: 30, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: 0.010624, mae: 0.350108, mean_q: 0.434618, mean_eps: 0.973864

153594/1000000: episode: 296, duration: 18.860s, episode steps: 555, steps per second: 29, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.009053, mae: 0.351886, mean_q: 0.437921, mean_eps: 0.973307

154128/1000000: episode: 297, duration: 18.156s, episode steps: 534, steps per second: 29, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.010682, mae: 0.358123, mean_q: 0.443173, mean_eps: 0.972654

154850/1000000: episode: 298, duration: 24.486s, episode steps: 722, steps per second: 29, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: 0.010262, mae: 0.361334, mean_q: 0.446432, mean_eps: 0.971901

155350/1000000: episode: 299, duration: 16.993s, episode steps: 500, steps per second: 29, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.008664, mae: 0.361675, mean_q: 0.446264, mean_eps: 0.971166

155654/1000000: episode: 300, duration: 10.341s, episode steps: 304, steps per second: 29, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.365 [0.000, 5.000], loss: 0.008418, mae: 0.358949, mean_q: 0.445342, mean_eps: 0.970684

156150/1000000: episode: 301, duration: 16.588s, episode steps: 496, steps per second: 30, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.009599, mae: 0.357224, mean_q: 0.442863, mean_eps: 0.970204

156732/1000000: episode: 302, duration: 18.733s, episode steps: 582, steps per second: 31, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.577 [0.000, 5.000], loss: 0.009606, mae: 0.363566, mean_q: 0.453262, mean_eps: 0.969558

157135/1000000: episode: 303, duration: 13.722s, episode steps: 403, steps per second: 29, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.012045, mae: 0.364364, mean_q: 0.452096, mean_eps: 0.968968

157674/1000000: episode: 304, duration: 18.327s, episode steps: 539, steps per second: 29, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.008709, mae: 0.360360, mean_q: 0.444710, mean_eps: 0.968402

158030/1000000: episode: 305, duration: 12.098s, episode steps: 356, steps per second: 29, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.009187, mae: 0.367043, mean_q: 0.456185, mean_eps: 0.967864

158641/1000000: episode: 306, duration: 20.791s, episode steps: 611, steps per second: 29, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: 0.010198, mae: 0.369128, mean_q: 0.456122, mean_eps: 0.967283

158926/1000000: episode: 307, duration: 9.648s, episode steps: 285, steps per second: 30, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.009483, mae: 0.359546, mean_q: 0.445273, mean_eps: 0.966746

159691/1000000: episode: 308, duration: 26.003s, episode steps: 765, steps per second: 29, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.008930, mae: 0.365757, mean_q: 0.450678, mean_eps: 0.966117

160129/1000000: episode: 309, duration: 14.965s, episode steps: 438, steps per second: 29, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.007442, mae: 0.352122, mean_q: 0.436426, mean_eps: 0.965394

160522/1000000: episode: 310, duration: 13.374s, episode steps: 393, steps per second: 29, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.008955, mae: 0.357325, mean_q: 0.443033, mean_eps: 0.964895

160900/1000000: episode: 311, duration: 12.970s, episode steps: 378, steps per second: 29, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: 0.009733, mae: 0.365244, mean_q: 0.455904, mean_eps: 0.964434

161523/1000000: episode: 312, duration: 19.756s, episode steps: 623, steps per second: 32, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.011334, mae: 0.399834, mean_q: 0.495386, mean_eps: 0.963834

161934/1000000: episode: 313, duration: 13.898s, episode steps: 411, steps per second: 30, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.009954, mae: 0.398656, mean_q: 0.497045, mean_eps: 0.963213

162821/1000000: episode: 314, duration: 30.112s, episode steps: 887, steps per second: 29, episode reward: 22.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.521 [0.000, 5.000], loss: 0.011237, mae: 0.403287, mean_q: 0.503261, mean_eps: 0.962433

163637/1000000: episode: 315, duration: 27.711s, episode steps: 816, steps per second: 29, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.009516, mae: 0.397915, mean_q: 0.496590, mean_eps: 0.961410

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

164066/1000000: episode: 316, duration: 7.828s, episode steps: 429, steps per second: 55, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.008599, mae: 0.413557, mean_q: 0.515193, mean_eps: 0.960801

164458/1000000: episode: 317, duration: 1.735s, episode steps: 392, steps per second: 226, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

165175/1000000: episode: 318, duration: 3.121s, episode steps: 717, steps per second: 230, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

165970/1000000: episode: 319, duration: 3.450s, episode steps: 795, steps per second: 230, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

166804/1000000: episode: 320, duration: 3.689s, episode steps: 834, steps per second: 226, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_320_episodes.h5f

167140/1000000: episode: 321, duration: 1.597s, episode steps: 336, steps per second: 210, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

167557/1000000: episode: 322, duration: 1.814s, episode steps: 417, steps per second: 230, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

168007/1000000: episode: 323, duration: 1.956s, episode steps: 450, steps per second: 230, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

168482/1000000: episode: 324, duration: 2.071s, episode steps: 475, steps per second: 229, episode reward: 13.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

168936/1000000: episode: 325, duration: 1.994s, episode steps: 454, steps per second: 228, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

169464/1000000: episode: 326, duration: 2.300s, episode steps: 528, steps per second: 230, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

170024/1000000: episode: 327, duration: 2.503s, episode steps: 560, steps per second: 224, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

170318/1000000: episode: 328, duration: 1.280s, episode steps: 294, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

170608/1000000: episode: 329, duration: 1.275s, episode steps: 290, steps per second: 228, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

171079/1000000: episode: 330, duration: 2.070s, episode steps: 471, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

171595/1000000: episode: 331, duration: 2.247s, episode steps: 516, steps per second: 230, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

171919/1000000: episode: 332, duration: 1.425s, episode steps: 324, steps per second: 227, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

172866/1000000: episode: 333, duration: 4.126s, episode steps: 947, steps per second: 230, episode reward: 24.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

173166/1000000: episode: 334, duration: 1.329s, episode steps: 300, steps per second: 226, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

173667/1000000: episode: 335, duration: 2.206s, episode steps: 501, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

174359/1000000: episode: 336, duration: 3.009s, episode steps: 692, steps per second: 230, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

174646/1000000: episode: 337, duration: 1.255s, episode steps: 287, steps per second: 229, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

175189/1000000: episode: 338, duration: 2.373s, episode steps: 543, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

175473/1000000: episode: 339, duration: 1.274s, episode steps: 284, steps per second: 223, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

176202/1000000: episode: 340, duration: 3.230s, episode steps: 729, steps per second: 226, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

176680/1000000: episode: 341, duration: 2.089s, episode steps: 478, steps per second: 229, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177184/1000000: episode: 342, duration: 2.203s, episode steps: 504, steps per second: 229, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177571/1000000: episode: 343, duration: 1.684s, episode steps: 387, steps per second: 230, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

177856/1000000: episode: 344, duration: 1.248s, episode steps: 285, steps per second: 228, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

178264/1000000: episode: 345, duration: 1.783s, episode steps: 408, steps per second: 229, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

179153/1000000: episode: 346, duration: 3.909s, episode steps: 889, steps per second: 227, episode reward: 20.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

179545/1000000: episode: 347, duration: 1.694s, episode steps: 392, steps per second: 231, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.686 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

180030/1000000: episode: 348, duration: 2.115s, episode steps: 485, steps per second: 229, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

180369/1000000: episode: 349, duration: 1.478s, episode steps: 339, steps per second: 229, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

180741/1000000: episode: 350, duration: 1.624s, episode steps: 372, steps per second: 229, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.661 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

181340/1000000: episode: 351, duration: 2.643s, episode steps: 599, steps per second: 227, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

181740/1000000: episode: 352, duration: 1.772s, episode steps: 400, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

182246/1000000: episode: 353, duration: 2.192s, episode steps: 506, steps per second: 231, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

182761/1000000: episode: 354, duration: 2.232s, episode steps: 515, steps per second: 231, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

183142/1000000: episode: 355, duration: 1.645s, episode steps: 381, steps per second: 232, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

183675/1000000: episode: 356, duration: 2.320s, episode steps: 533, steps per second: 230, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

184239/1000000: episode: 357, duration: 2.505s, episode steps: 564, steps per second: 225, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

184778/1000000: episode: 358, duration: 2.374s, episode steps: 539, steps per second: 227, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

185072/1000000: episode: 359, duration: 1.291s, episode steps: 294, steps per second: 228, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

185912/1000000: episode: 360, duration: 3.636s, episode steps: 840, steps per second: 231, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

186378/1000000: episode: 361, duration: 2.043s, episode steps: 466, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

186707/1000000: episode: 362, duration: 1.473s, episode steps: 329, steps per second: 223, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

187185/1000000: episode: 363, duration: 2.111s, episode steps: 478, steps per second: 226, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

187765/1000000: episode: 364, duration: 2.534s, episode steps: 580, steps per second: 229, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

188144/1000000: episode: 365, duration: 1.653s, episode steps: 379, steps per second: 229, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

188704/1000000: episode: 366, duration: 2.431s, episode steps: 560, steps per second: 230, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

188994/1000000: episode: 367, duration: 1.270s, episode steps: 290, steps per second: 228, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

189551/1000000: episode: 368, duration: 2.418s, episode steps: 557, steps per second: 230, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

190060/1000000: episode: 369, duration: 2.248s, episode steps: 509, steps per second: 226, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

190477/1000000: episode: 370, duration: 1.833s, episode steps: 417, steps per second: 227, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

191179/1000000: episode: 371, duration: 3.030s, episode steps: 702, steps per second: 232, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

191699/1000000: episode: 372, duration: 2.283s, episode steps: 520, steps per second: 228, episode reward: 14.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

192113/1000000: episode: 373, duration: 1.801s, episode steps: 414, steps per second: 230, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

193171/1000000: episode: 374, duration: 4.609s, episode steps: 1058, steps per second: 230, episode reward: 21.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

193544/1000000: episode: 375, duration: 1.621s, episode steps: 373, steps per second: 230, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

194144/1000000: episode: 376, duration: 2.595s, episode steps: 600, steps per second: 231, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

194551/1000000: episode: 377, duration: 1.775s, episode steps: 407, steps per second: 229, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

195071/1000000: episode: 378, duration: 2.269s, episode steps: 520, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

195639/1000000: episode: 379, duration: 2.513s, episode steps: 568, steps per second: 226, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

196242/1000000: episode: 380, duration: 2.643s, episode steps: 603, steps per second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

196579/1000000: episode: 381, duration: 1.476s, episode steps: 337, steps per second: 228, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

196978/1000000: episode: 382, duration: 1.771s, episode steps: 399, steps per second: 225, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

197449/1000000: episode: 383, duration: 2.055s, episode steps: 471, steps per second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

197896/1000000: episode: 384, duration: 1.958s, episode steps: 447, steps per second: 228, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

198435/1000000: episode: 385, duration: 2.401s, episode steps: 539, steps per second: 225, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

198725/1000000: episode: 386, duration: 1.322s, episode steps: 290, steps per second: 219, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

199014/1000000: episode: 387, duration: 1.302s, episode steps: 289, steps per second: 222, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

199383/1000000: episode: 388, duration: 1.641s, episode steps: 369, steps per second: 225, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

199962/1000000: episode: 389, duration: 2.556s, episode steps: 579, steps per second: 227, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

200258/1000000: episode: 390, duration: 1.306s, episode steps: 296, steps per second: 227, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

200644/1000000: episode: 391, duration: 1.688s, episode steps: 386, steps per second: 229, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

201107/1000000: episode: 392, duration: 2.028s, episode steps: 463, steps per second: 228, episode reward: 13.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

201829/1000000: episode: 393, duration: 3.192s, episode steps: 722, steps per second: 226, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

202225/1000000: episode: 394, duration: 1.730s, episode steps: 396, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

202820/1000000: episode: 395, duration: 2.562s, episode steps: 595, steps per second: 232, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

203274/1000000: episode: 396, duration: 1.973s, episode steps: 454, steps per second: 230, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.678 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

204102/1000000: episode: 397, duration: 3.572s, episode steps: 828, steps per second: 232, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

204720/1000000: episode: 398, duration: 2.724s, episode steps: 618, steps per second: 227, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

205181/1000000: episode: 399, duration: 2.008s, episode steps: 461, steps per second: 230, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

205634/1000000: episode: 400, duration: 1.980s, episode steps: 453, steps per second: 229, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

206057/1000000: episode: 401, duration: 1.851s, episode steps: 423, steps per second: 229, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

206769/1000000: episode: 402, duration: 7.908s, episode steps: 712, steps per second: 90, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.009238, mae: 0.406021, mean_q: 0.504544, mean_eps: 0.987902

207257/1000000: episode: 403, duration: 16.725s, episode steps: 488, steps per second: 29, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.009558, mae: 0.416927, mean_q: 0.518286, mean_eps: 0.987513

207577/1000000: episode: 404, duration: 10.972s, episode steps: 320, steps per second: 29, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.011439, mae: 0.424539, mean_q: 0.527791, mean_eps: 0.987028

208350/1000000: episode: 405, duration: 26.512s, episode steps: 773, steps per second: 29, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.011157, mae: 0.420435, mean_q: 0.522150,

mean_eps: 0.986373

208770/1000000: episode: 406, duration: 14.388s, episode steps: 420, steps per second: 29, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.010999, mae: 0.415614, mean_q: 0.517221, mean_eps: 0.985658

209591/1000000: episode: 407, duration: 28.144s, episode steps: 821, steps per second: 29, episode reward: 23.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.010211, mae: 0.416743, mean_q: 0.518614, mean_eps: 0.984914

209991/1000000: episode: 408, duration: 13.785s, episode steps: 400, steps per second: 29, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.010982, mae: 0.412231, mean_q: 0.512310, mean_eps: 0.984182

210511/1000000: episode: 409, duration: 17.876s, episode steps: 520, steps per second: 29, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.692 [0.000, 5.000], loss: 0.009191, mae: 0.416316, mean_q: 0.519005, mean_eps: 0.983630

211029/1000000: episode: 410, duration: 17.856s, episode steps: 518, steps per second: 29, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.008301, mae: 0.410991, mean_q: 0.513449, mean_eps: 0.983006

211760/1000000: episode: 411, duration: 24.754s, episode steps: 731, steps per second: 30, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.008971, mae: 0.424084, mean_q: 0.528613, mean_eps: 0.982257

212055/1000000: episode: 412, duration: 9.323s, episode steps: 295, steps per second: 32, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.644 [0.000, 5.000], loss: 0.008916, mae: 0.452788, mean_q: 0.565381, mean_eps: 0.981642

212501/1000000: episode: 413, duration: 15.437s, episode steps: 446, steps per second: 29, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: 0.009072, mae: 0.453261, mean_q: 0.567392, mean_eps: 0.981196

212863/1000000: episode: 414, duration: 12.395s, episode steps: 362, steps per second: 29, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.009654, mae: 0.441774, mean_q: 0.549574, mean_eps: 0.980711

213314/1000000: episode: 415, duration: 15.562s, episode steps: 451, steps per second: 29, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.010235, mae: 0.450276, mean_q: 0.562608, mean_eps: 0.980224

213901/1000000: episode: 416, duration: 20.253s, episode steps: 587, steps per second: 29, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.011221, mae: 0.452010, mean_q: 0.565099, mean_eps: 0.979600

Nuevo récord: 35.0. Pesos guardados en
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn

dueling/version_1.0/mejor_modelo/best_weights.h5f

215234/1000000: episode: 417, duration: 45.950s, episode steps: 1333, steps per second: 29, episode reward: 35.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.010108, mae: 0.460845, mean_q: 0.571976, mean_eps: 0.978448

215838/1000000: episode: 418, duration: 20.746s, episode steps: 604, steps per second: 29, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: 0.010622, mae: 0.452011, mean_q: 0.565977, mean_eps: 0.977286

216685/1000000: episode: 419, duration: 29.189s, episode steps: 847, steps per second: 29, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.009336, mae: 0.450455, mean_q: 0.562215, mean_eps: 0.976415

217077/1000000: episode: 420, duration: 13.502s, episode steps: 392, steps per second: 29, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.010817, mae: 0.478912, mean_q: 0.599998, mean_eps: 0.975671

217546/1000000: episode: 421, duration: 16.136s, episode steps: 469, steps per second: 29, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.009419, mae: 0.479472, mean_q: 0.598930, mean_eps: 0.975155

217839/1000000: episode: 422, duration: 10.119s, episode steps: 293, steps per second: 29, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.009540, mae: 0.477952, mean_q: 0.596560, mean_eps: 0.974699

218288/1000000: episode: 423, duration: 15.570s, episode steps: 449, steps per second: 29, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.010064, mae: 0.473839, mean_q: 0.592922, mean_eps: 0.974255

219043/1000000: episode: 424, duration: 26.050s, episode steps: 755, steps per second: 29, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.009511, mae: 0.472777, mean_q: 0.589674, mean_eps: 0.973533

219460/1000000: episode: 425, duration: 14.487s, episode steps: 417, steps per second: 29, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.012698, mae: 0.475352, mean_q: 0.592444, mean_eps: 0.972830

220174/1000000: episode: 426, duration: 24.680s, episode steps: 714, steps per second: 29, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.009515, mae: 0.465044, mean_q: 0.581130, mean_eps: 0.972150

220847/1000000: episode: 427, duration: 23.262s, episode steps: 673, steps per second: 29, episode reward: 9.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.009229, mae: 0.473185, mean_q: 0.589737, mean_eps: 0.971318

221323/1000000: episode: 428, duration: 16.483s, episode steps: 476, steps per second: 29, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: 0.010204, mae: 0.465775, mean_q: 0.580493,

mean_eps: 0.970629

221840/1000000: episode: 429, duration: 17.300s, episode steps: 517, steps per second: 30, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.009573, mae: 0.483435, mean_q: 0.601378, mean_eps: 0.970034

222251/1000000: episode: 430, duration: 13.662s, episode steps: 411, steps per second: 30, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: 0.008652, mae: 0.524422, mean_q: 0.653217, mean_eps: 0.969477

222672/1000000: episode: 431, duration: 14.658s, episode steps: 421, steps per second: 29, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.009823, mae: 0.527813, mean_q: 0.661309, mean_eps: 0.968978

223354/1000000: episode: 432, duration: 23.599s, episode steps: 682, steps per second: 29, episode reward: 21.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.009146, mae: 0.512213, mean_q: 0.640815, mean_eps: 0.968315

223950/1000000: episode: 433, duration: 20.643s, episode steps: 596, steps per second: 29, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.009300, mae: 0.532674, mean_q: 0.664890, mean_eps: 0.967547

224665/1000000: episode: 434, duration: 24.813s, episode steps: 715, steps per second: 29, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.009641, mae: 0.514412, mean_q: 0.640839, mean_eps: 0.966760

225164/1000000: episode: 435, duration: 17.279s, episode steps: 499, steps per second: 29, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: 0.008896, mae: 0.523678, mean_q: 0.650101, mean_eps: 0.966033

225547/1000000: episode: 436, duration: 13.329s, episode steps: 383, steps per second: 29, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.008706, mae: 0.509614, mean_q: 0.633529, mean_eps: 0.965505

225920/1000000: episode: 437, duration: 13.022s, episode steps: 373, steps per second: 29, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: 0.011066, mae: 0.520453, mean_q: 0.644224, mean_eps: 0.965051

226450/1000000: episode: 438, duration: 18.438s, episode steps: 530, steps per second: 29, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.009536, mae: 0.525408, mean_q: 0.654083, mean_eps: 0.964509

227059/1000000: episode: 439, duration: 19.754s, episode steps: 609, steps per second: 31, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.008945, mae: 0.532719, mean_q: 0.666567, mean_eps: 0.963825

227442/1000000: episode: 440, duration: 13.122s, episode steps: 383, steps per second: 29, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.009179, mae: 0.530242, mean_q: 0.664095,

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mean_eps: 0.963230
227801/1000000: episode: 441, duration: 12.439s, episode steps: 359, steps per
second: 29, episode reward: 4.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.546 [0.000, 5.000], loss: 0.010463, mae: 0.535473, mean_q: 0.669355,
mean_eps: 0.962783
228382/1000000: episode: 442, duration: 20.212s, episode steps: 581, steps per
second: 29, episode reward: 15.000, mean reward: 0.026 [ 0.000, 1.000], mean
action: 2.482 [0.000, 5.000], loss: 0.008315, mae: 0.534810, mean_q: 0.666729,
mean_eps: 0.962219
229070/1000000: episode: 443, duration: 23.938s, episode steps: 688, steps per
second: 29, episode reward: 12.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.515 [0.000, 5.000], loss: 0.007868, mae: 0.543190, mean_q: 0.675326,
mean_eps: 0.961458

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

229801/1000000: episode: 444, duration: 12.489s, episode steps: 731, steps per
second: 59, episode reward: 14.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.419 [0.000, 5.000], loss: 0.008684, mae: 0.535079, mean_q: 0.665592,
mean_eps: 0.960863
230529/1000000: episode: 445, duration: 3.206s, episode steps: 728, steps per
second: 227, episode reward: 12.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
230912/1000000: episode: 446, duration: 1.682s, episode steps: 383, steps per
second: 228, episode reward: 8.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
231316/1000000: episode: 447, duration: 1.778s, episode steps: 404, steps per
second: 227, episode reward: 3.000, mean reward: 0.007 [ 0.000, 1.000], mean
action: 2.663 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
231873/1000000: episode: 448, duration: 2.455s, episode steps: 557, steps per
second: 227, episode reward: 15.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
232404/1000000: episode: 449, duration: 2.371s, episode steps: 531, steps per
second: 224, episode reward: 8.000, mean reward: 0.015 [ 0.000, 1.000], mean
action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
232917/1000000: episode: 450, duration: 2.285s, episode steps: 513, steps per
second: 224, episode reward: 10.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
233638/1000000: episode: 451, duration: 3.130s, episode steps: 721, steps per
second: 230, episode reward: 15.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
234191/1000000: episode: 452, duration: 2.426s, episode steps: 553, steps per
second: 228, episode reward: 14.000, mean reward: 0.025 [ 0.000, 1.000], mean
action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
234658/1000000: episode: 453, duration: 2.045s, episode steps: 467, steps per
second: 228, episode reward: 12.000, mean reward: 0.026 [ 0.000, 1.000], mean
action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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235302/1000000: episode: 454, duration: 2.818s, episode steps: 644, steps per second: 229, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

235672/1000000: episode: 455, duration: 1.644s, episode steps: 370, steps per second: 225, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

236154/1000000: episode: 456, duration: 2.135s, episode steps: 482, steps per second: 226, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

236461/1000000: episode: 457, duration: 1.356s, episode steps: 307, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

237766/1000000: episode: 458, duration: 5.611s, episode steps: 1305, steps per second: 233, episode reward: 20.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

238092/1000000: episode: 459, duration: 1.427s, episode steps: 326, steps per second: 228, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

238618/1000000: episode: 460, duration: 2.327s, episode steps: 526, steps per second: 226, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

239217/1000000: episode: 461, duration: 2.617s, episode steps: 599, steps per second: 229, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

239990/1000000: episode: 462, duration: 3.358s, episode steps: 773, steps per second: 230, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

240649/1000000: episode: 463, duration: 2.878s, episode steps: 659, steps per second: 229, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

241140/1000000: episode: 464, duration: 2.152s, episode steps: 491, steps per second: 228, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

241715/1000000: episode: 465, duration: 2.533s, episode steps: 575, steps per second: 227, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

242260/1000000: episode: 466, duration: 2.408s, episode steps: 545, steps per second: 226, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

242814/1000000: episode: 467, duration: 2.397s, episode steps: 554, steps per second: 231, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

243285/1000000: episode: 468, duration: 2.049s, episode steps: 471, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

244084/1000000: episode: 469, duration: 3.502s, episode steps: 799, steps per second: 228, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

244606/1000000: episode: 470, duration: 2.348s, episode steps: 522, steps per second: 222, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

244979/1000000: episode: 471, duration: 1.640s, episode steps: 373, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

245508/1000000: episode: 472, duration: 2.300s, episode steps: 529, steps per second: 230, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

245805/1000000: episode: 473, duration: 1.297s, episode steps: 297, steps per second: 229, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

246187/1000000: episode: 474, duration: 1.657s, episode steps: 382, steps per second: 231, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

246686/1000000: episode: 475, duration: 2.172s, episode steps: 499, steps per second: 230, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

247158/1000000: episode: 476, duration: 2.099s, episode steps: 472, steps per second: 225, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.288 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

247663/1000000: episode: 477, duration: 2.224s, episode steps: 505, steps per second: 227, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

248151/1000000: episode: 478, duration: 2.116s, episode steps: 488, steps per second: 231, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

249018/1000000: episode: 479, duration: 3.761s, episode steps: 867, steps per second: 231, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

249563/1000000: episode: 480, duration: 2.362s, episode steps: 545, steps per second: 231, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_480_episodes.h5f

250074/1000000: episode: 481, duration: 2.355s, episode steps: 511, steps per second: 217, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

250329/1000000: episode: 482, duration: 1.161s, episode steps: 255, steps per second: 220, episode reward: 3.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

250862/1000000: episode: 483, duration: 2.329s, episode steps: 533, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

251292/1000000: episode: 484, duration: 1.877s, episode steps: 430, steps per second: 229, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean

action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 251817/1000000: episode: 485, duration: 2.293s, episode steps: 525, steps per second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 252123/1000000: episode: 486, duration: 1.332s, episode steps: 306, steps per second: 230, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 252418/1000000: episode: 487, duration: 1.308s, episode steps: 295, steps per second: 226, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 253222/1000000: episode: 488, duration: 3.523s, episode steps: 804, steps per second: 228, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 253526/1000000: episode: 489, duration: 1.320s, episode steps: 304, steps per second: 230, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 254134/1000000: episode: 490, duration: 2.627s, episode steps: 608, steps per second: 231, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 254480/1000000: episode: 491, duration: 1.512s, episode steps: 346, steps per second: 229, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 254849/1000000: episode: 492, duration: 1.627s, episode steps: 369, steps per second: 227, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 255942/1000000: episode: 493, duration: 4.771s, episode steps: 1093, steps per second: 229, episode reward: 14.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 256233/1000000: episode: 494, duration: 1.285s, episode steps: 291, steps per second: 227, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 256699/1000000: episode: 495, duration: 2.045s, episode steps: 466, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 257050/1000000: episode: 496, duration: 1.540s, episode steps: 351, steps per second: 228, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 257705/1000000: episode: 497, duration: 2.882s, episode steps: 655, steps per second: 227, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 258545/1000000: episode: 498, duration: 3.649s, episode steps: 840, steps per second: 230, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 258923/1000000: episode: 499, duration: 1.665s, episode steps: 378, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 259626/1000000: episode: 500, duration: 3.050s, episode steps: 703, steps per second: 231, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean

action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 260637/1000000: episode: 501, duration: 4.356s, episode steps: 1011, steps per second: 232, episode reward: 17.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261020/1000000: episode: 502, duration: 1.669s, episode steps: 383, steps per second: 229, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261316/1000000: episode: 503, duration: 1.312s, episode steps: 296, steps per second: 226, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 261715/1000000: episode: 504, duration: 1.764s, episode steps: 399, steps per second: 226, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.348 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 262312/1000000: episode: 505, duration: 2.621s, episode steps: 597, steps per second: 228, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 262759/1000000: episode: 506, duration: 1.957s, episode steps: 447, steps per second: 228, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 263140/1000000: episode: 507, duration: 1.675s, episode steps: 381, steps per second: 227, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 263512/1000000: episode: 508, duration: 1.615s, episode steps: 372, steps per second: 230, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 263997/1000000: episode: 509, duration: 2.093s, episode steps: 485, steps per second: 232, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 264755/1000000: episode: 510, duration: 3.312s, episode steps: 758, steps per second: 229, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 265875/1000000: episode: 511, duration: 4.814s, episode steps: 1120, steps per second: 233, episode reward: 17.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 266209/1000000: episode: 512, duration: 1.452s, episode steps: 334, steps per second: 230, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 266774/1000000: episode: 513, duration: 2.438s, episode steps: 565, steps per second: 232, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 267371/1000000: episode: 514, duration: 2.617s, episode steps: 597, steps per second: 228, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 267842/1000000: episode: 515, duration: 2.074s, episode steps: 471, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 268371/1000000: episode: 516, duration: 2.309s, episode steps: 529, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean

action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 268742/1000000: episode: 517, duration: 1.612s, episode steps: 371, steps per second: 230, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 269204/1000000: episode: 518, duration: 2.015s, episode steps: 462, steps per second: 229, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.647 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 269736/1000000: episode: 519, duration: 2.319s, episode steps: 532, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 270265/1000000: episode: 520, duration: 2.318s, episode steps: 529, steps per second: 228, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 270946/1000000: episode: 521, duration: 2.969s, episode steps: 681, steps per second: 229, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 271456/1000000: episode: 522, duration: 2.228s, episode steps: 510, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 271920/1000000: episode: 523, duration: 2.037s, episode steps: 464, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 272224/1000000: episode: 524, duration: 3.786s, episode steps: 304, steps per second: 80, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: 0.008044, mae: 0.542381, mean_q: 0.673765, mean_eps: 0.987952
 272558/1000000: episode: 525, duration: 11.755s, episode steps: 334, steps per second: 28, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: 0.008781, mae: 0.572973, mean_q: 0.712250, mean_eps: 0.987705
 273030/1000000: episode: 526, duration: 16.639s, episode steps: 472, steps per second: 28, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.009452, mae: 0.558785, mean_q: 0.692804, mean_eps: 0.987220
 273443/1000000: episode: 527, duration: 14.522s, episode steps: 413, steps per second: 28, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: 0.010043, mae: 0.563037, mean_q: 0.702290, mean_eps: 0.986690
 273752/1000000: episode: 528, duration: 10.955s, episode steps: 309, steps per second: 28, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.009906, mae: 0.555816, mean_q: 0.688075, mean_eps: 0.986258
 274065/1000000: episode: 529, duration: 11.129s, episode steps: 313, steps per second: 28, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.008177, mae: 0.556252, mean_q: 0.691904, mean_eps: 0.985883
 274384/1000000: episode: 530, duration: 11.243s, episode steps: 319, steps per second: 28, episode reward: 8.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.429 [0.000, 5.000], loss: 0.009441, mae: 0.573441, mean_q: 0.712294, mean_eps: 0.985504

275143/1000000: episode: 531, duration: 27.059s, episode steps: 759, steps per second: 28, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: 0.009666, mae: 0.574700, mean_q: 0.715871, mean_eps: 0.984858

275432/1000000: episode: 532, duration: 10.256s, episode steps: 289, steps per second: 28, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.009717, mae: 0.576596, mean_q: 0.721788, mean_eps: 0.984230

276002/1000000: episode: 533, duration: 20.175s, episode steps: 570, steps per second: 28, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.010024, mae: 0.568565, mean_q: 0.711311, mean_eps: 0.983714

276494/1000000: episode: 534, duration: 17.358s, episode steps: 492, steps per second: 28, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: 0.008743, mae: 0.557784, mean_q: 0.697956, mean_eps: 0.983075

276879/1000000: episode: 535, duration: 13.578s, episode steps: 385, steps per second: 28, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: 0.008597, mae: 0.567297, mean_q: 0.711165, mean_eps: 0.982550

277170/1000000: episode: 536, duration: 10.325s, episode steps: 291, steps per second: 28, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.008456, mae: 0.570104, mean_q: 0.714153, mean_eps: 0.982144

277537/1000000: episode: 537, duration: 12.943s, episode steps: 367, steps per second: 28, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.009429, mae: 0.603875, mean_q: 0.751476, mean_eps: 0.981748

278300/1000000: episode: 538, duration: 26.926s, episode steps: 763, steps per second: 28, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.009481, mae: 0.597438, mean_q: 0.745647, mean_eps: 0.981071

278852/1000000: episode: 539, duration: 19.636s, episode steps: 552, steps per second: 28, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.009934, mae: 0.597397, mean_q: 0.748193, mean_eps: 0.980284

279162/1000000: episode: 540, duration: 11.066s, episode steps: 310, steps per second: 28, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.008210, mae: 0.591048, mean_q: 0.737976, mean_eps: 0.979766

279881/1000000: episode: 541, duration: 25.365s, episode steps: 719, steps per second: 28, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.009004, mae: 0.589263, mean_q: 0.734838, mean_eps: 0.979146

280382/1000000: episode: 542, duration: 17.623s, episode steps: 501, steps per second: 28, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean

action: 2.573 [0.000, 5.000], loss: 0.009534, mae: 0.578623, mean_q: 0.720768, mean_eps: 0.978414

280953/1000000: episode: 543, duration: 20.216s, episode steps: 571, steps per second: 28, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.009316, mae: 0.594177, mean_q: 0.741104, mean_eps: 0.977771

281836/1000000: episode: 544, duration: 31.199s, episode steps: 883, steps per second: 28, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.009405, mae: 0.597713, mean_q: 0.745593, mean_eps: 0.976900

282171/1000000: episode: 545, duration: 11.894s, episode steps: 335, steps per second: 28, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: 0.010165, mae: 0.592484, mean_q: 0.737161, mean_eps: 0.976170

282448/1000000: episode: 546, duration: 9.898s, episode steps: 277, steps per second: 28, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.336 [0.000, 5.000], loss: 0.011531, mae: 0.663260, mean_q: 0.825925, mean_eps: 0.975803

283065/1000000: episode: 547, duration: 21.912s, episode steps: 617, steps per second: 28, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: 0.010854, mae: 0.680499, mean_q: 0.846068, mean_eps: 0.975266

283638/1000000: episode: 548, duration: 20.274s, episode steps: 573, steps per second: 28, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.011158, mae: 0.674244, mean_q: 0.835521, mean_eps: 0.974550

284146/1000000: episode: 549, duration: 17.990s, episode steps: 508, steps per second: 28, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.010048, mae: 0.680496, mean_q: 0.845414, mean_eps: 0.973902

284445/1000000: episode: 550, duration: 10.608s, episode steps: 299, steps per second: 28, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: 0.010649, mae: 0.671549, mean_q: 0.832963, mean_eps: 0.973418

285060/1000000: episode: 551, duration: 21.774s, episode steps: 615, steps per second: 28, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: 0.009784, mae: 0.670735, mean_q: 0.834529, mean_eps: 0.972870

285682/1000000: episode: 552, duration: 22.107s, episode steps: 622, steps per second: 28, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.012325, mae: 0.672839, mean_q: 0.834914, mean_eps: 0.972129

286050/1000000: episode: 553, duration: 13.038s, episode steps: 368, steps per second: 28, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: 0.011361, mae: 0.670303, mean_q: 0.832777, mean_eps: 0.971534

286634/1000000: episode: 554, duration: 20.641s, episode steps: 584, steps per second: 28, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.644 [0.000, 5.000], loss: 0.010991, mae: 0.668971, mean_q: 0.833414, mean_eps: 0.970962

287083/1000000: episode: 555, duration: 15.908s, episode steps: 449, steps per second: 28, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.009980, mae: 0.659920, mean_q: 0.819091, mean_eps: 0.970343

287679/1000000: episode: 556, duration: 21.160s, episode steps: 596, steps per second: 28, episode reward: 19.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.011298, mae: 0.691043, mean_q: 0.857711, mean_eps: 0.969717

288056/1000000: episode: 557, duration: 13.481s, episode steps: 377, steps per second: 28, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.009137, mae: 0.677735, mean_q: 0.840362, mean_eps: 0.969134

288423/1000000: episode: 558, duration: 13.073s, episode steps: 367, steps per second: 28, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.009837, mae: 0.684737, mean_q: 0.845772, mean_eps: 0.968687

288842/1000000: episode: 559, duration: 14.895s, episode steps: 419, steps per second: 28, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.628 [0.000, 5.000], loss: 0.009899, mae: 0.677931, mean_q: 0.840121, mean_eps: 0.968214

289202/1000000: episode: 560, duration: 12.839s, episode steps: 360, steps per second: 28, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.011067, mae: 0.689618, mean_q: 0.854632, mean_eps: 0.967746

289792/1000000: episode: 561, duration: 21.015s, episode steps: 590, steps per second: 28, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: 0.010497, mae: 0.689255, mean_q: 0.853940, mean_eps: 0.967178

290179/1000000: episode: 562, duration: 13.753s, episode steps: 387, steps per second: 28, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.009462, mae: 0.681079, mean_q: 0.844839, mean_eps: 0.966592

290875/1000000: episode: 563, duration: 24.720s, episode steps: 696, steps per second: 28, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.009630, mae: 0.677533, mean_q: 0.840038, mean_eps: 0.965942

291237/1000000: episode: 564, duration: 12.919s, episode steps: 362, steps per second: 28, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.721 [0.000, 5.000], loss: 0.009410, mae: 0.695333, mean_q: 0.858698, mean_eps: 0.965306

291719/1000000: episode: 565, duration: 17.100s, episode steps: 482, steps per second: 28, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.010907, mae: 0.675746, mean_q: 0.838216, mean_eps: 0.964799

292012/1000000: episode: 566, duration: 10.517s, episode steps: 293, steps per second: 28, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.556 [0.000, 5.000], loss: 0.011285, mae: 0.688641, mean_q: 0.852115, mean_eps: 0.964336

292296/1000000: episode: 567, duration: 9.779s, episode steps: 284, steps per second: 29, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.732 [0.000, 5.000], loss: 0.011130, mae: 0.723694, mean_q: 0.896932, mean_eps: 0.963990

292820/1000000: episode: 568, duration: 17.903s, episode steps: 524, steps per second: 29, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: 0.010353, mae: 0.735767, mean_q: 0.909339, mean_eps: 0.963506

293409/1000000: episode: 569, duration: 21.081s, episode steps: 589, steps per second: 28, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.010783, mae: 0.747027, mean_q: 0.922667, mean_eps: 0.962836

293896/1000000: episode: 570, duration: 17.405s, episode steps: 487, steps per second: 28, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.010714, mae: 0.744137, mean_q: 0.917645, mean_eps: 0.962190

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

295013/1000000: episode: 571, duration: 36.580s, episode steps: 1117, steps per second: 31, episode reward: 14.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.009972, mae: 0.748895, mean_q: 0.923273, mean_eps: 0.961290

295495/1000000: episode: 572, duration: 2.126s, episode steps: 482, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

295904/1000000: episode: 573, duration: 1.816s, episode steps: 409, steps per second: 225, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

296406/1000000: episode: 574, duration: 2.211s, episode steps: 502, steps per second: 227, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

297036/1000000: episode: 575, duration: 2.755s, episode steps: 630, steps per second: 229, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

297298/1000000: episode: 576, duration: 1.158s, episode steps: 262, steps per second: 226, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.305 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

297836/1000000: episode: 577, duration: 2.357s, episode steps: 538, steps per second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

298364/1000000: episode: 578, duration: 2.298s, episode steps: 528, steps per second: 230, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

298868/1000000: episode: 579, duration: 2.219s, episode steps: 504, steps per

second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

299347/1000000: episode: 580, duration: 2.112s, episode steps: 479, steps per second: 227, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

300214/1000000: episode: 581, duration: 3.775s, episode steps: 867, steps per second: 230, episode reward: 17.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

300871/1000000: episode: 582, duration: 2.866s, episode steps: 657, steps per second: 229, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

301135/1000000: episode: 583, duration: 1.165s, episode steps: 264, steps per second: 227, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

301791/1000000: episode: 584, duration: 2.900s, episode steps: 656, steps per second: 226, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

302291/1000000: episode: 585, duration: 2.189s, episode steps: 500, steps per second: 228, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.674 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

302746/1000000: episode: 586, duration: 2.016s, episode steps: 455, steps per second: 226, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

303546/1000000: episode: 587, duration: 3.475s, episode steps: 800, steps per second: 230, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

304077/1000000: episode: 588, duration: 2.316s, episode steps: 531, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

304685/1000000: episode: 589, duration: 2.722s, episode steps: 608, steps per second: 223, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

305468/1000000: episode: 590, duration: 3.410s, episode steps: 783, steps per second: 230, episode reward: 21.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

305847/1000000: episode: 591, duration: 1.667s, episode steps: 379, steps per second: 227, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

306348/1000000: episode: 592, duration: 2.221s, episode steps: 501, steps per second: 226, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

307040/1000000: episode: 593, duration: 3.044s, episode steps: 692, steps per second: 227, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

307523/1000000: episode: 594, duration: 2.167s, episode steps: 483, steps per second: 223, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

307963/1000000: episode: 595, duration: 1.975s, episode steps: 440, steps per

second: 223, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

308791/1000000: episode: 596, duration: 3.602s, episode steps: 828, steps per second: 230, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

309243/1000000: episode: 597, duration: 1.987s, episode steps: 452, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

309553/1000000: episode: 598, duration: 1.363s, episode steps: 310, steps per second: 227, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

309994/1000000: episode: 599, duration: 1.932s, episode steps: 441, steps per second: 228, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

310292/1000000: episode: 600, duration: 1.321s, episode steps: 298, steps per second: 226, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

310858/1000000: episode: 601, duration: 2.508s, episode steps: 566, steps per second: 226, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

311417/1000000: episode: 602, duration: 2.422s, episode steps: 559, steps per second: 231, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

312101/1000000: episode: 603, duration: 2.953s, episode steps: 684, steps per second: 232, episode reward: 21.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

312508/1000000: episode: 604, duration: 1.778s, episode steps: 407, steps per second: 229, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

312909/1000000: episode: 605, duration: 1.787s, episode steps: 401, steps per second: 224, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

313407/1000000: episode: 606, duration: 2.206s, episode steps: 498, steps per second: 226, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

314097/1000000: episode: 607, duration: 2.985s, episode steps: 690, steps per second: 231, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

314656/1000000: episode: 608, duration: 2.424s, episode steps: 559, steps per second: 231, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

315293/1000000: episode: 609, duration: 2.786s, episode steps: 637, steps per second: 229, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

315678/1000000: episode: 610, duration: 1.706s, episode steps: 385, steps per second: 226, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

316140/1000000: episode: 611, duration: 2.052s, episode steps: 462, steps per

second: 225, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

316854/1000000: episode: 612, duration: 3.104s, episode steps: 714, steps per second: 230, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

317408/1000000: episode: 613, duration: 2.409s, episode steps: 554, steps per second: 230, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

317934/1000000: episode: 614, duration: 2.313s, episode steps: 526, steps per second: 227, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

318430/1000000: episode: 615, duration: 2.157s, episode steps: 496, steps per second: 230, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

318723/1000000: episode: 616, duration: 1.294s, episode steps: 293, steps per second: 226, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

319237/1000000: episode: 617, duration: 2.267s, episode steps: 514, steps per second: 227, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

319708/1000000: episode: 618, duration: 2.054s, episode steps: 471, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

320367/1000000: episode: 619, duration: 2.867s, episode steps: 659, steps per second: 230, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

320857/1000000: episode: 620, duration: 2.143s, episode steps: 490, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

321667/1000000: episode: 621, duration: 3.512s, episode steps: 810, steps per second: 231, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

321961/1000000: episode: 622, duration: 1.319s, episode steps: 294, steps per second: 223, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

323216/1000000: episode: 623, duration: 5.436s, episode steps: 1255, steps per second: 231, episode reward: 28.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

323687/1000000: episode: 624, duration: 2.069s, episode steps: 471, steps per second: 228, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

324293/1000000: episode: 625, duration: 2.638s, episode steps: 606, steps per second: 230, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

324904/1000000: episode: 626, duration: 2.721s, episode steps: 611, steps per second: 225, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

325318/1000000: episode: 627, duration: 1.822s, episode steps: 414, steps per

second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 325886/1000000: episode: 628, duration: 2.479s, episode steps: 568, steps per second: 229, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 326282/1000000: episode: 629, duration: 1.750s, episode steps: 396, steps per second: 226, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 326852/1000000: episode: 630, duration: 2.479s, episode steps: 570, steps per second: 230, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 327235/1000000: episode: 631, duration: 1.694s, episode steps: 383, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 328046/1000000: episode: 632, duration: 3.567s, episode steps: 811, steps per second: 227, episode reward: 13.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 328357/1000000: episode: 633, duration: 1.368s, episode steps: 311, steps per second: 227, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 329117/1000000: episode: 634, duration: 3.294s, episode steps: 760, steps per second: 231, episode reward: 21.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 329426/1000000: episode: 635, duration: 1.350s, episode steps: 309, steps per second: 229, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 329920/1000000: episode: 636, duration: 2.154s, episode steps: 494, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 330612/1000000: episode: 637, duration: 3.048s, episode steps: 692, steps per second: 227, episode reward: 8.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 331202/1000000: episode: 638, duration: 2.589s, episode steps: 590, steps per second: 228, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 331771/1000000: episode: 639, duration: 2.482s, episode steps: 569, steps per second: 229, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 332362/1000000: episode: 640, duration: 2.570s, episode steps: 591, steps per second: 230, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

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 dueling/version_1.0/pesos/dqn_v_1.0_weights_640_episodes.h5f

333170/1000000: episode: 641, duration: 3.626s, episode steps: 808, steps per second: 223, episode reward: 20.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

333621/1000000: episode: 642, duration: 2.019s, episode steps: 451, steps per second: 223, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

334171/1000000: episode: 643, duration: 2.397s, episode steps: 550, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

334471/1000000: episode: 644, duration: 1.313s, episode steps: 300, steps per second: 228, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

335021/1000000: episode: 645, duration: 2.401s, episode steps: 550, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

335463/1000000: episode: 646, duration: 1.930s, episode steps: 442, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

335854/1000000: episode: 647, duration: 1.719s, episode steps: 391, steps per second: 228, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

336311/1000000: episode: 648, duration: 2.037s, episode steps: 457, steps per second: 224, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

336745/1000000: episode: 649, duration: 1.923s, episode steps: 434, steps per second: 226, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

337271/1000000: episode: 650, duration: 2.297s, episode steps: 526, steps per second: 229, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

338056/1000000: episode: 651, duration: 15.310s, episode steps: 785, steps per second: 51, episode reward: 8.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.011138, mae: 0.775432, mean_q: 0.950425, mean_eps: 0.987774

338671/1000000: episode: 652, duration: 22.193s, episode steps: 615, steps per second: 28, episode reward: 18.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.011098, mae: 0.767837, mean_q: 0.942956, mean_eps: 0.987182

339005/1000000: episode: 653, duration: 12.136s, episode steps: 334, steps per second: 28, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.012236, mae: 0.776212, mean_q: 0.952996, mean_eps: 0.986610

339344/1000000: episode: 654, duration: 12.287s, episode steps: 339, steps per second: 28, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.010166, mae: 0.749580, mean_q: 0.922268, mean_eps: 0.986207

339743/1000000: episode: 655, duration: 14.453s, episode steps: 399, steps per second: 28, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.010258, mae: 0.770269, mean_q: 0.947631, mean_eps: 0.985766

340290/1000000: episode: 656, duration: 19.854s, episode steps: 547, steps per

second: 28, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.010732, mae: 0.772854, mean_q: 0.950255, mean_eps: 0.985197

340663/1000000: episode: 657, duration: 13.521s, episode steps: 373, steps per second: 28, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: 0.010958, mae: 0.778230, mean_q: 0.955615, mean_eps: 0.984645

340959/1000000: episode: 658, duration: 10.722s, episode steps: 296, steps per second: 28, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: 0.011430, mae: 0.770475, mean_q: 0.947627, mean_eps: 0.984244

341442/1000000: episode: 659, duration: 17.555s, episode steps: 483, steps per second: 28, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: 0.011001, mae: 0.768628, mean_q: 0.946070, mean_eps: 0.983776

342020/1000000: episode: 660, duration: 20.959s, episode steps: 578, steps per second: 28, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: 0.010187, mae: 0.767861, mean_q: 0.943970, mean_eps: 0.983140

342483/1000000: episode: 661, duration: 16.817s, episode steps: 463, steps per second: 28, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.011269, mae: 0.765244, mean_q: 0.939714, mean_eps: 0.982516

343004/1000000: episode: 662, duration: 19.045s, episode steps: 521, steps per second: 27, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.009929, mae: 0.785135, mean_q: 0.961985, mean_eps: 0.981926

343536/1000000: episode: 663, duration: 19.451s, episode steps: 532, steps per second: 27, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.010642, mae: 0.806120, mean_q: 0.983028, mean_eps: 0.981294

344513/1000000: episode: 664, duration: 35.515s, episode steps: 977, steps per second: 28, episode reward: 12.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.010891, mae: 0.793031, mean_q: 0.967201, mean_eps: 0.980387

345182/1000000: episode: 665, duration: 24.270s, episode steps: 669, steps per second: 28, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: 0.011052, mae: 0.786773, mean_q: 0.960916, mean_eps: 0.979398

345700/1000000: episode: 666, duration: 18.879s, episode steps: 518, steps per second: 27, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.011646, mae: 0.765750, mean_q: 0.935864, mean_eps: 0.978688

346331/1000000: episode: 667, duration: 22.892s, episode steps: 631, steps per second: 28, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.010745, mae: 0.790050, mean_q: 0.967455, mean_eps: 0.977999

346853/1000000: episode: 668, duration: 19.047s, episode steps: 522, steps per

second: 27, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.010696, mae: 0.781211, mean_q: 0.954601, mean_eps: 0.977306

347349/1000000: episode: 669, duration: 18.083s, episode steps: 496, steps per second: 27, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.009752, mae: 0.787164, mean_q: 0.960431, mean_eps: 0.976694

347972/1000000: episode: 670, duration: 22.716s, episode steps: 623, steps per second: 27, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.011171, mae: 0.812341, mean_q: 0.992558, mean_eps: 0.976024

348263/1000000: episode: 671, duration: 10.620s, episode steps: 291, steps per second: 27, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.009708, mae: 0.817527, mean_q: 0.999518, mean_eps: 0.975477

348797/1000000: episode: 672, duration: 19.460s, episode steps: 534, steps per second: 27, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: 0.010300, mae: 0.822166, mean_q: 1.004629, mean_eps: 0.974980

349154/1000000: episode: 673, duration: 12.954s, episode steps: 357, steps per second: 28, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: 0.011194, mae: 0.834207, mean_q: 1.019929, mean_eps: 0.974445

349729/1000000: episode: 674, duration: 20.984s, episode steps: 575, steps per second: 27, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: 0.010905, mae: 0.826087, mean_q: 1.011340, mean_eps: 0.973886

350308/1000000: episode: 675, duration: 21.103s, episode steps: 579, steps per second: 27, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.009730, mae: 0.826349, mean_q: 1.011378, mean_eps: 0.973194

350696/1000000: episode: 676, duration: 14.313s, episode steps: 388, steps per second: 27, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.011192, mae: 0.845488, mean_q: 1.033436, mean_eps: 0.972616

351071/1000000: episode: 677, duration: 13.734s, episode steps: 375, steps per second: 27, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.011713, mae: 0.843713, mean_q: 1.032214, mean_eps: 0.972158

351851/1000000: episode: 678, duration: 28.379s, episode steps: 780, steps per second: 27, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: 0.010005, mae: 0.828961, mean_q: 1.013870, mean_eps: 0.971464

352262/1000000: episode: 679, duration: 15.003s, episode steps: 411, steps per second: 27, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.009630, mae: 0.823677, mean_q: 1.007744, mean_eps: 0.970749

352737/1000000: episode: 680, duration: 17.191s, episode steps: 475, steps per

second: 28, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.011748, mae: 0.814417, mean_q: 0.994479, mean_eps: 0.970216

353212/1000000: episode: 681, duration: 16.165s, episode steps: 475, steps per second: 29, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.011752, mae: 0.866737, mean_q: 1.059186, mean_eps: 0.969647

353895/1000000: episode: 682, duration: 24.889s, episode steps: 683, steps per second: 27, episode reward: 21.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: 0.010268, mae: 0.865836, mean_q: 1.061177, mean_eps: 0.968954

354425/1000000: episode: 683, duration: 19.359s, episode steps: 530, steps per second: 27, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: 0.012548, mae: 0.855693, mean_q: 1.046784, mean_eps: 0.968224

355266/1000000: episode: 684, duration: 30.631s, episode steps: 841, steps per second: 27, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: 0.012077, mae: 0.872317, mean_q: 1.067596, mean_eps: 0.967401

355674/1000000: episode: 685, duration: 14.981s, episode steps: 408, steps per second: 27, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.647 [0.000, 5.000], loss: 0.011853, mae: 0.858208, mean_q: 1.051654, mean_eps: 0.966652

356069/1000000: episode: 686, duration: 14.434s, episode steps: 395, steps per second: 27, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.011092, mae: 0.862969, mean_q: 1.054393, mean_eps: 0.966170

356550/1000000: episode: 687, duration: 17.486s, episode steps: 481, steps per second: 28, episode reward: 13.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.010255, mae: 0.869293, mean_q: 1.065083, mean_eps: 0.965644

357022/1000000: episode: 688, duration: 17.213s, episode steps: 472, steps per second: 27, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.011542, mae: 0.854124, mean_q: 1.044888, mean_eps: 0.965073

357498/1000000: episode: 689, duration: 17.355s, episode steps: 476, steps per second: 27, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.010652, mae: 0.860504, mean_q: 1.056958, mean_eps: 0.964504

358116/1000000: episode: 690, duration: 21.296s, episode steps: 618, steps per second: 29, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.011266, mae: 0.878455, mean_q: 1.075053, mean_eps: 0.963849

358704/1000000: episode: 691, duration: 21.536s, episode steps: 588, steps per second: 27, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.010971, mae: 0.880237, mean_q: 1.077153, mean_eps: 0.963126

359746/1000000: episode: 692, duration: 38.095s, episode steps: 1042, steps per

second: 27, episode reward: 19.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.567 [0.000, 5.000], loss: 0.010581, mae: 0.880706, mean_q: 1.077678,
mean_eps: 0.962147

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

360660/1000000: episode: 693, duration: 26.529s, episode steps: 914, steps per
second: 34, episode reward: 14.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.477 [0.000, 5.000], loss: 0.011386, mae: 0.898287, mean_q: 1.098466,
mean_eps: 0.961101

361127/1000000: episode: 694, duration: 2.057s, episode steps: 467, steps per
second: 227, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

361422/1000000: episode: 695, duration: 1.315s, episode steps: 295, steps per
second: 224, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

361709/1000000: episode: 696, duration: 1.247s, episode steps: 287, steps per
second: 230, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

362015/1000000: episode: 697, duration: 1.342s, episode steps: 306, steps per
second: 228, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

362799/1000000: episode: 698, duration: 3.412s, episode steps: 784, steps per
second: 230, episode reward: 22.000, mean reward: 0.028 [0.000, 1.000], mean
action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

363358/1000000: episode: 699, duration: 2.497s, episode steps: 559, steps per
second: 224, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

363810/1000000: episode: 700, duration: 1.970s, episode steps: 452, steps per
second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

364110/1000000: episode: 701, duration: 1.310s, episode steps: 300, steps per
second: 229, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

364633/1000000: episode: 702, duration: 2.277s, episode steps: 523, steps per
second: 230, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

365157/1000000: episode: 703, duration: 2.276s, episode steps: 524, steps per
second: 230, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

365722/1000000: episode: 704, duration: 2.471s, episode steps: 565, steps per
second: 229, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

366508/1000000: episode: 705, duration: 3.427s, episode steps: 786, steps per
second: 229, episode reward: 21.000, mean reward: 0.027 [0.000, 1.000], mean
action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

367030/1000000: episode: 706, duration: 2.275s, episode steps: 522, steps per

second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

367319/1000000: episode: 707, duration: 1.262s, episode steps: 289, steps per second: 229, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.727 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

367643/1000000: episode: 708, duration: 1.419s, episode steps: 324, steps per second: 228, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

368161/1000000: episode: 709, duration: 2.261s, episode steps: 518, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

368451/1000000: episode: 710, duration: 1.298s, episode steps: 290, steps per second: 223, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

369490/1000000: episode: 711, duration: 4.535s, episode steps: 1039, steps per second: 229, episode reward: 17.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

370080/1000000: episode: 712, duration: 2.575s, episode steps: 590, steps per second: 229, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

370576/1000000: episode: 713, duration: 2.168s, episode steps: 496, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

370880/1000000: episode: 714, duration: 1.337s, episode steps: 304, steps per second: 227, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

371459/1000000: episode: 715, duration: 2.543s, episode steps: 579, steps per second: 228, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

371959/1000000: episode: 716, duration: 2.213s, episode steps: 500, steps per second: 226, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

372626/1000000: episode: 717, duration: 2.894s, episode steps: 667, steps per second: 230, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

373229/1000000: episode: 718, duration: 2.626s, episode steps: 603, steps per second: 230, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

373812/1000000: episode: 719, duration: 2.534s, episode steps: 583, steps per second: 230, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

374123/1000000: episode: 720, duration: 1.374s, episode steps: 311, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

374639/1000000: episode: 721, duration: 2.315s, episode steps: 516, steps per second: 223, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

375124/1000000: episode: 722, duration: 2.147s, episode steps: 485, steps per

second: 226, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

375545/1000000: episode: 723, duration: 1.834s, episode steps: 421, steps per second: 230, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.292 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

375869/1000000: episode: 724, duration: 1.422s, episode steps: 324, steps per second: 228, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

376553/1000000: episode: 725, duration: 2.967s, episode steps: 684, steps per second: 231, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

377152/1000000: episode: 726, duration: 2.615s, episode steps: 599, steps per second: 229, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

377631/1000000: episode: 727, duration: 2.103s, episode steps: 479, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

378062/1000000: episode: 728, duration: 1.901s, episode steps: 431, steps per second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

379108/1000000: episode: 729, duration: 4.524s, episode steps: 1046, steps per second: 231, episode reward: 12.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

379691/1000000: episode: 730, duration: 2.535s, episode steps: 583, steps per second: 230, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

380198/1000000: episode: 731, duration: 2.234s, episode steps: 507, steps per second: 227, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

380497/1000000: episode: 732, duration: 1.331s, episode steps: 299, steps per second: 225, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

380904/1000000: episode: 733, duration: 1.776s, episode steps: 407, steps per second: 229, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

381561/1000000: episode: 734, duration: 2.851s, episode steps: 657, steps per second: 230, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

381979/1000000: episode: 735, duration: 1.838s, episode steps: 418, steps per second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

382861/1000000: episode: 736, duration: 3.816s, episode steps: 882, steps per second: 231, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

383324/1000000: episode: 737, duration: 2.062s, episode steps: 463, steps per second: 225, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

383912/1000000: episode: 738, duration: 2.601s, episode steps: 588, steps per

second: 226, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.699 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

384214/1000000: episode: 739, duration: 1.325s, episode steps: 302, steps per second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

384767/1000000: episode: 740, duration: 2.414s, episode steps: 553, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

385335/1000000: episode: 741, duration: 2.474s, episode steps: 568, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

385673/1000000: episode: 742, duration: 1.465s, episode steps: 338, steps per second: 231, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.340 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

385954/1000000: episode: 743, duration: 1.253s, episode steps: 281, steps per second: 224, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

386476/1000000: episode: 744, duration: 2.293s, episode steps: 522, steps per second: 228, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

386920/1000000: episode: 745, duration: 1.946s, episode steps: 444, steps per second: 228, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

387458/1000000: episode: 746, duration: 2.331s, episode steps: 538, steps per second: 231, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

387825/1000000: episode: 747, duration: 1.605s, episode steps: 367, steps per second: 229, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

388146/1000000: episode: 748, duration: 1.397s, episode steps: 321, steps per second: 230, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

388442/1000000: episode: 749, duration: 1.312s, episode steps: 296, steps per second: 226, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.277 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

389383/1000000: episode: 750, duration: 4.102s, episode steps: 941, steps per second: 229, episode reward: 20.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

389695/1000000: episode: 751, duration: 1.365s, episode steps: 312, steps per second: 229, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

390166/1000000: episode: 752, duration: 2.055s, episode steps: 471, steps per second: 229, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

390558/1000000: episode: 753, duration: 1.704s, episode steps: 392, steps per second: 230, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

391089/1000000: episode: 754, duration: 2.307s, episode steps: 531, steps per

second: 230, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

391645/1000000: episode: 755, duration: 2.435s, episode steps: 556, steps per second: 228, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

391978/1000000: episode: 756, duration: 1.464s, episode steps: 333, steps per second: 227, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

392602/1000000: episode: 757, duration: 2.770s, episode steps: 624, steps per second: 225, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

392900/1000000: episode: 758, duration: 1.305s, episode steps: 298, steps per second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.326 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

393414/1000000: episode: 759, duration: 2.250s, episode steps: 514, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

394035/1000000: episode: 760, duration: 2.693s, episode steps: 621, steps per second: 231, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

394510/1000000: episode: 761, duration: 2.068s, episode steps: 475, steps per second: 230, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

395142/1000000: episode: 762, duration: 2.806s, episode steps: 632, steps per second: 225, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

395845/1000000: episode: 763, duration: 3.054s, episode steps: 703, steps per second: 230, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

396409/1000000: episode: 764, duration: 2.466s, episode steps: 564, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

396786/1000000: episode: 765, duration: 1.650s, episode steps: 377, steps per second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

397174/1000000: episode: 766, duration: 1.692s, episode steps: 388, steps per second: 229, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

397578/1000000: episode: 767, duration: 1.787s, episode steps: 404, steps per second: 226, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

398055/1000000: episode: 768, duration: 2.105s, episode steps: 477, steps per second: 227, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

398561/1000000: episode: 769, duration: 2.216s, episode steps: 506, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

398916/1000000: episode: 770, duration: 1.545s, episode steps: 355, steps per

second: 230, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

399499/1000000: episode: 771, duration: 2.529s, episode steps: 583, steps per second: 231, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.628 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

399823/1000000: episode: 772, duration: 1.406s, episode steps: 324, steps per second: 230, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

400304/1000000: episode: 773, duration: 2.122s, episode steps: 481, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

400787/1000000: episode: 774, duration: 2.146s, episode steps: 483, steps per second: 225, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

401233/1000000: episode: 775, duration: 1.956s, episode steps: 446, steps per second: 228, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

401608/1000000: episode: 776, duration: 1.643s, episode steps: 375, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

402126/1000000: episode: 777, duration: 2.265s, episode steps: 518, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

402813/1000000: episode: 778, duration: 2.998s, episode steps: 687, steps per second: 229, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

403157/1000000: episode: 779, duration: 1.520s, episode steps: 344, steps per second: 226, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

403620/1000000: episode: 780, duration: 15.299s, episode steps: 463, steps per second: 30, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.009331, mae: 0.915244, mean_q: 1.120498, mean_eps: 0.987758

404054/1000000: episode: 781, duration: 16.186s, episode steps: 434, steps per second: 27, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.010950, mae: 0.913688, mean_q: 1.117527, mean_eps: 0.987256

404628/1000000: episode: 782, duration: 21.391s, episode steps: 574, steps per second: 27, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: 0.012482, mae: 0.910200, mean_q: 1.112204, mean_eps: 0.986651

405085/1000000: episode: 783, duration: 17.077s, episode steps: 457, steps per second: 27, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.011029, mae: 0.905697, mean_q: 1.106585, mean_eps: 0.986032

405474/1000000: episode: 784, duration: 14.457s, episode steps: 389, steps per second: 27, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.010545, mae: 0.900344, mean_q: 1.100325,

mean_eps: 0.985523

406042/1000000: episode: 785, duration: 21.282s, episode steps: 568, steps per second: 27, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.010463, mae: 0.907857, mean_q: 1.109295, mean_eps: 0.984950

406500/1000000: episode: 786, duration: 17.143s, episode steps: 458, steps per second: 27, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.010898, mae: 0.909627, mean_q: 1.108807, mean_eps: 0.984335

407185/1000000: episode: 787, duration: 25.633s, episode steps: 685, steps per second: 27, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.653 [0.000, 5.000], loss: 0.011682, mae: 0.910383, mean_q: 1.111349, mean_eps: 0.983649

408261/1000000: episode: 788, duration: 39.849s, episode steps: 1076, steps per second: 27, episode reward: 14.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.010891, mae: 0.911191, mean_q: 1.113285, mean_eps: 0.982590

408679/1000000: episode: 789, duration: 14.361s, episode steps: 418, steps per second: 29, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.012137, mae: 0.972036, mean_q: 1.186180, mean_eps: 0.981695

409092/1000000: episode: 790, duration: 15.539s, episode steps: 413, steps per second: 27, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.011836, mae: 0.968347, mean_q: 1.179120, mean_eps: 0.981198

409509/1000000: episode: 791, duration: 15.687s, episode steps: 417, steps per second: 27, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: 0.011893, mae: 0.976433, mean_q: 1.191002, mean_eps: 0.980699

410282/1000000: episode: 792, duration: 28.754s, episode steps: 773, steps per second: 27, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.011864, mae: 0.970619, mean_q: 1.185476, mean_eps: 0.979984

410829/1000000: episode: 793, duration: 20.425s, episode steps: 547, steps per second: 27, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.011042, mae: 0.959434, mean_q: 1.171252, mean_eps: 0.979192

411129/1000000: episode: 794, duration: 11.188s, episode steps: 300, steps per second: 27, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.012575, mae: 0.978134, mean_q: 1.194586, mean_eps: 0.978683

411835/1000000: episode: 795, duration: 26.242s, episode steps: 706, steps per second: 27, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.011234, mae: 0.963339, mean_q: 1.175803, mean_eps: 0.978081

412201/1000000: episode: 796, duration: 13.763s, episode steps: 366, steps per second: 27, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: 0.011965, mae: 0.959101, mean_q: 1.170966,

mean_eps: 0.977438

412676/1000000: episode: 797, duration: 17.754s, episode steps: 475, steps per second: 27, episode reward: 13.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.010008, mae: 0.966499, mean_q: 1.179192, mean_eps: 0.976934

413200/1000000: episode: 798, duration: 19.772s, episode steps: 524, steps per second: 27, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.011398, mae: 0.959318, mean_q: 1.170023, mean_eps: 0.976336

413856/1000000: episode: 799, duration: 24.678s, episode steps: 656, steps per second: 27, episode reward: 18.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.010589, mae: 1.021362, mean_q: 1.246658, mean_eps: 0.975628

414372/1000000: episode: 800, duration: 19.479s, episode steps: 516, steps per second: 26, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: 0.012288, mae: 1.024402, mean_q: 1.248827, mean_eps: 0.974925

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dueling/version_1.0/pesos/dqn_v_1.0_weights_800_episodes.h5f

414961/1000000: episode: 801, duration: 22.202s, episode steps: 589, steps per second: 27, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: 0.012528, mae: 1.021350, mean_q: 1.245775, mean_eps: 0.974260

415890/1000000: episode: 802, duration: 34.631s, episode steps: 929, steps per second: 27, episode reward: 23.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.011418, mae: 1.013313, mean_q: 1.237858, mean_eps: 0.973348

416252/1000000: episode: 803, duration: 13.568s, episode steps: 362, steps per second: 27, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.012929, mae: 1.014275, mean_q: 1.236566, mean_eps: 0.972575

417103/1000000: episode: 804, duration: 31.845s, episode steps: 851, steps per second: 27, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.011918, mae: 1.005603, mean_q: 1.225441, mean_eps: 0.971848

417612/1000000: episode: 805, duration: 19.096s, episode steps: 509, steps per second: 27, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: 0.010746, mae: 0.996585, mean_q: 1.212427, mean_eps: 0.971032

418078/1000000: episode: 806, duration: 17.478s, episode steps: 466, steps per second: 27, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.010092, mae: 1.015362, mean_q: 1.236270, mean_eps: 0.970446

418544/1000000: episode: 807, duration: 17.541s, episode steps: 466, steps per second: 27, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: 0.011749, mae: 1.033188, mean_q: 1.258646,

mean_eps: 0.969887

418917/1000000: episode: 808, duration: 14.075s, episode steps: 373, steps per second: 27, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.010595, mae: 1.054709, mean_q: 1.288879, mean_eps: 0.969383

419352/1000000: episode: 809, duration: 16.314s, episode steps: 435, steps per second: 27, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.010594, mae: 1.049022, mean_q: 1.280920, mean_eps: 0.968898

419661/1000000: episode: 810, duration: 11.712s, episode steps: 309, steps per second: 26, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.011637, mae: 1.028520, mean_q: 1.258235, mean_eps: 0.968452

420430/1000000: episode: 811, duration: 28.719s, episode steps: 769, steps per second: 27, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.011622, mae: 1.049550, mean_q: 1.279677, mean_eps: 0.967804

420840/1000000: episode: 812, duration: 15.469s, episode steps: 410, steps per second: 27, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.693 [0.000, 5.000], loss: 0.010698, mae: 1.058293, mean_q: 1.293340, mean_eps: 0.967098

421257/1000000: episode: 813, duration: 15.734s, episode steps: 417, steps per second: 27, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: 0.012012, mae: 1.051690, mean_q: 1.282769, mean_eps: 0.966602

421826/1000000: episode: 814, duration: 21.240s, episode steps: 569, steps per second: 27, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.012364, mae: 1.057608, mean_q: 1.289204, mean_eps: 0.966009

422591/1000000: episode: 815, duration: 28.782s, episode steps: 765, steps per second: 27, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: 0.009911, mae: 1.030848, mean_q: 1.259229, mean_eps: 0.965210

422887/1000000: episode: 816, duration: 11.155s, episode steps: 296, steps per second: 27, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.348 [0.000, 5.000], loss: 0.011693, mae: 1.047633, mean_q: 1.276189, mean_eps: 0.964574

423447/1000000: episode: 817, duration: 20.297s, episode steps: 560, steps per second: 28, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: 0.010186, mae: 1.062950, mean_q: 1.297838, mean_eps: 0.964060

424476/1000000: episode: 818, duration: 38.150s, episode steps: 1029, steps per second: 27, episode reward: 17.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: 0.010858, mae: 1.079461, mean_q: 1.319302, mean_eps: 0.963107

424956/1000000: episode: 819, duration: 18.191s, episode steps: 480, steps per second: 26, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.352 [0.000, 5.000], loss: 0.011576, mae: 1.085105, mean_q: 1.328559,

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mean_eps: 0.962202
425662/1000000: episode: 820, duration: 26.621s, episode steps: 706, steps per
second: 27, episode reward: 15.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.588 [0.000, 5.000], loss: 0.010670, mae: 1.061816, mean_q: 1.299045,
mean_eps: 0.961490
425942/1000000: episode: 821, duration: 10.512s, episode steps: 280, steps per
second: 27, episode reward: 5.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.446 [0.000, 5.000], loss: 0.011682, mae: 1.065112, mean_q: 1.304203,
mean_eps: 0.960897

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

426239/1000000: episode: 822, duration: 2.635s, episode steps: 297, steps per
second: 113, episode reward: 4.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.448 [0.000, 5.000], loss: 0.011186, mae: 1.045861, mean_q: 1.282706,
mean_eps: 0.960705
426584/1000000: episode: 823, duration: 1.511s, episode steps: 345, steps per
second: 228, episode reward: 6.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
427113/1000000: episode: 824, duration: 2.319s, episode steps: 529, steps per
second: 228, episode reward: 10.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
427684/1000000: episode: 825, duration: 2.504s, episode steps: 571, steps per
second: 228, episode reward: 15.000, mean reward: 0.026 [ 0.000, 1.000], mean
action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
428412/1000000: episode: 826, duration: 3.191s, episode steps: 728, steps per
second: 228, episode reward: 20.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
428821/1000000: episode: 827, duration: 1.826s, episode steps: 409, steps per
second: 224, episode reward: 5.000, mean reward: 0.012 [ 0.000, 1.000], mean
action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
429445/1000000: episode: 828, duration: 2.692s, episode steps: 624, steps per
second: 232, episode reward: 14.000, mean reward: 0.022 [ 0.000, 1.000], mean
action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
430204/1000000: episode: 829, duration: 3.291s, episode steps: 759, steps per
second: 231, episode reward: 22.000, mean reward: 0.029 [ 0.000, 1.000], mean
action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
430705/1000000: episode: 830, duration: 2.185s, episode steps: 501, steps per
second: 229, episode reward: 6.000, mean reward: 0.012 [ 0.000, 1.000], mean
action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
431056/1000000: episode: 831, duration: 1.550s, episode steps: 351, steps per
second: 226, episode reward: 7.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
431452/1000000: episode: 832, duration: 1.766s, episode steps: 396, steps per
second: 224, episode reward: 7.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
432027/1000000: episode: 833, duration: 2.518s, episode steps: 575, steps per

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second: 228, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

432638/1000000: episode: 834, duration: 2.673s, episode steps: 611, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

433075/1000000: episode: 835, duration: 1.921s, episode steps: 437, steps per second: 228, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

433476/1000000: episode: 836, duration: 1.753s, episode steps: 401, steps per second: 229, episode reward: 11.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

434053/1000000: episode: 837, duration: 2.560s, episode steps: 577, steps per second: 225, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

434529/1000000: episode: 838, duration: 2.117s, episode steps: 476, steps per second: 225, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

435267/1000000: episode: 839, duration: 3.213s, episode steps: 738, steps per second: 230, episode reward: 22.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

435844/1000000: episode: 840, duration: 2.504s, episode steps: 577, steps per second: 230, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

436422/1000000: episode: 841, duration: 2.517s, episode steps: 578, steps per second: 230, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

436843/1000000: episode: 842, duration: 1.858s, episode steps: 421, steps per second: 227, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

437670/1000000: episode: 843, duration: 3.651s, episode steps: 827, steps per second: 226, episode reward: 14.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

438135/1000000: episode: 844, duration: 2.025s, episode steps: 465, steps per second: 230, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

438779/1000000: episode: 845, duration: 2.808s, episode steps: 644, steps per second: 229, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

439295/1000000: episode: 846, duration: 2.244s, episode steps: 516, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

439791/1000000: episode: 847, duration: 2.180s, episode steps: 496, steps per second: 227, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

440082/1000000: episode: 848, duration: 1.300s, episode steps: 291, steps per second: 224, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

440738/1000000: episode: 849, duration: 2.870s, episode steps: 656, steps per

second: 229, episode reward: 20.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

441060/1000000: episode: 850, duration: 1.413s, episode steps: 322, steps per second: 228, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

441601/1000000: episode: 851, duration: 2.363s, episode steps: 541, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

441988/1000000: episode: 852, duration: 1.711s, episode steps: 387, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

442477/1000000: episode: 853, duration: 2.143s, episode steps: 489, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

442850/1000000: episode: 854, duration: 1.672s, episode steps: 373, steps per second: 223, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

443881/1000000: episode: 855, duration: 4.495s, episode steps: 1031, steps per second: 229, episode reward: 11.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

444426/1000000: episode: 856, duration: 2.372s, episode steps: 545, steps per second: 230, episode reward: 14.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

445154/1000000: episode: 857, duration: 3.177s, episode steps: 728, steps per second: 229, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

445606/1000000: episode: 858, duration: 1.998s, episode steps: 452, steps per second: 226, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

445894/1000000: episode: 859, duration: 1.286s, episode steps: 288, steps per second: 224, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

446413/1000000: episode: 860, duration: 2.293s, episode steps: 519, steps per second: 226, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

446747/1000000: episode: 861, duration: 1.473s, episode steps: 334, steps per second: 227, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

447200/1000000: episode: 862, duration: 1.981s, episode steps: 453, steps per second: 229, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

447876/1000000: episode: 863, duration: 2.957s, episode steps: 676, steps per second: 229, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

448371/1000000: episode: 864, duration: 2.185s, episode steps: 495, steps per second: 227, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

448757/1000000: episode: 865, duration: 1.739s, episode steps: 386, steps per

second: 222, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.329 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
449223/1000000: episode: 866, duration: 2.046s, episode steps: 466, steps per
second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
449658/1000000: episode: 867, duration: 1.901s, episode steps: 435, steps per
second: 229, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.331 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
450090/1000000: episode: 868, duration: 1.889s, episode steps: 432, steps per
second: 229, episode reward: 11.000, mean reward: 0.025 [0.000, 1.000], mean
action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
450675/1000000: episode: 869, duration: 2.558s, episode steps: 585, steps per
second: 229, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
451345/1000000: episode: 870, duration: 2.965s, episode steps: 670, steps per
second: 226, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
451795/1000000: episode: 871, duration: 2.004s, episode steps: 450, steps per
second: 225, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
452372/1000000: episode: 872, duration: 2.512s, episode steps: 577, steps per
second: 230, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
452835/1000000: episode: 873, duration: 2.024s, episode steps: 463, steps per
second: 229, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
453312/1000000: episode: 874, duration: 2.121s, episode steps: 477, steps per
second: 225, episode reward: 3.000, mean reward: 0.006 [0.000, 1.000], mean
action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
453834/1000000: episode: 875, duration: 2.316s, episode steps: 522, steps per
second: 225, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
454332/1000000: episode: 876, duration: 2.210s, episode steps: 498, steps per
second: 225, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
454826/1000000: episode: 877, duration: 2.183s, episode steps: 494, steps per
second: 226, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
455327/1000000: episode: 878, duration: 2.170s, episode steps: 501, steps per
second: 231, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
455721/1000000: episode: 879, duration: 1.724s, episode steps: 394, steps per
second: 229, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.398 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
456235/1000000: episode: 880, duration: 2.247s, episode steps: 514, steps per
second: 229, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
456959/1000000: episode: 881, duration: 3.144s, episode steps: 724, steps per

second: 230, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

457389/1000000: episode: 882, duration: 1.905s, episode steps: 430, steps per second: 226, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

458359/1000000: episode: 883, duration: 4.239s, episode steps: 970, steps per second: 229, episode reward: 19.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

458881/1000000: episode: 884, duration: 2.296s, episode steps: 522, steps per second: 227, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

459242/1000000: episode: 885, duration: 1.573s, episode steps: 361, steps per second: 230, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

459697/1000000: episode: 886, duration: 1.971s, episode steps: 455, steps per second: 231, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

460052/1000000: episode: 887, duration: 1.564s, episode steps: 355, steps per second: 227, episode reward: 9.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

460474/1000000: episode: 888, duration: 1.878s, episode steps: 422, steps per second: 225, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

460776/1000000: episode: 889, duration: 1.319s, episode steps: 302, steps per second: 229, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.666 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

461242/1000000: episode: 890, duration: 2.052s, episode steps: 466, steps per second: 227, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

461990/1000000: episode: 891, duration: 3.248s, episode steps: 748, steps per second: 230, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

462529/1000000: episode: 892, duration: 2.357s, episode steps: 539, steps per second: 229, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

462892/1000000: episode: 893, duration: 1.599s, episode steps: 363, steps per second: 227, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.722 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

463519/1000000: episode: 894, duration: 2.795s, episode steps: 627, steps per second: 224, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

464041/1000000: episode: 895, duration: 2.301s, episode steps: 522, steps per second: 227, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

464615/1000000: episode: 896, duration: 2.508s, episode steps: 574, steps per second: 229, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

465091/1000000: episode: 897, duration: 2.090s, episode steps: 476, steps per

second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

465497/1000000: episode: 898, duration: 1.774s, episode steps: 406, steps per second: 229, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

465915/1000000: episode: 899, duration: 1.882s, episode steps: 418, steps per second: 222, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

466311/1000000: episode: 900, duration: 1.800s, episode steps: 396, steps per second: 220, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

466681/1000000: episode: 901, duration: 1.643s, episode steps: 370, steps per second: 225, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

467262/1000000: episode: 902, duration: 2.542s, episode steps: 581, steps per second: 229, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

467753/1000000: episode: 903, duration: 2.157s, episode steps: 491, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

468177/1000000: episode: 904, duration: 1.864s, episode steps: 424, steps per second: 227, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

468621/1000000: episode: 905, duration: 1.958s, episode steps: 444, steps per second: 227, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

469083/1000000: episode: 906, duration: 13.153s, episode steps: 462, steps per second: 35, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.011136, mae: 1.096458, mean_q: 1.342366, mean_eps: 0.987801

469630/1000000: episode: 907, duration: 20.875s, episode steps: 547, steps per second: 26, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.013227, mae: 1.082641, mean_q: 1.322845, mean_eps: 0.987275

470110/1000000: episode: 908, duration: 18.425s, episode steps: 480, steps per second: 26, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.012606, mae: 1.077545, mean_q: 1.319629, mean_eps: 0.986658

470561/1000000: episode: 909, duration: 17.299s, episode steps: 451, steps per second: 26, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.012251, mae: 1.079333, mean_q: 1.320014, mean_eps: 0.986099

471143/1000000: episode: 910, duration: 22.233s, episode steps: 582, steps per second: 26, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: 0.012124, mae: 1.082407, mean_q: 1.324085, mean_eps: 0.985480

471518/1000000: episode: 911, duration: 14.470s, episode steps: 375, steps per second: 26, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.621 [0.000, 5.000], loss: 0.010779, mae: 1.055894, mean_q: 1.290317, mean_eps: 0.984906

472123/1000000: episode: 912, duration: 23.185s, episode steps: 605, steps per second: 26, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.011704, mae: 1.064007, mean_q: 1.302352, mean_eps: 0.984318

472726/1000000: episode: 913, duration: 23.175s, episode steps: 603, steps per second: 26, episode reward: 6.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.012255, mae: 1.071617, mean_q: 1.310501, mean_eps: 0.983594

473246/1000000: episode: 914, duration: 19.993s, episode steps: 520, steps per second: 26, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: 0.011503, mae: 1.090204, mean_q: 1.333776, mean_eps: 0.982919

473717/1000000: episode: 915, duration: 18.177s, episode steps: 471, steps per second: 26, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.012044, mae: 1.087374, mean_q: 1.331667, mean_eps: 0.982324

473978/1000000: episode: 916, duration: 9.321s, episode steps: 261, steps per second: 28, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: 0.014495, mae: 1.114498, mean_q: 1.363595, mean_eps: 0.981885

474823/1000000: episode: 917, duration: 31.724s, episode steps: 845, steps per second: 27, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.011822, mae: 1.112476, mean_q: 1.362958, mean_eps: 0.981222

475297/1000000: episode: 918, duration: 18.312s, episode steps: 474, steps per second: 26, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.011392, mae: 1.092525, mean_q: 1.336576, mean_eps: 0.980430

476276/1000000: episode: 919, duration: 37.559s, episode steps: 979, steps per second: 26, episode reward: 19.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.012515, mae: 1.104965, mean_q: 1.350131, mean_eps: 0.979559

476662/1000000: episode: 920, duration: 14.900s, episode steps: 386, steps per second: 26, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.013149, mae: 1.117392, mean_q: 1.364713, mean_eps: 0.978741

477038/1000000: episode: 921, duration: 14.495s, episode steps: 376, steps per second: 26, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.012491, mae: 1.110275, mean_q: 1.358460, mean_eps: 0.978282

477601/1000000: episode: 922, duration: 21.732s, episode steps: 563, steps per second: 26, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.011816, mae: 1.112413, mean_q: 1.358730, mean_eps: 0.977718

477968/1000000: episode: 923, duration: 14.166s, episode steps: 367, steps per second: 26, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean

action: 2.507 [0.000, 5.000], loss: 0.012082, mae: 1.115611, mean_q: 1.362474, mean_eps: 0.977162

478569/1000000: episode: 924, duration: 23.249s, episode steps: 601, steps per second: 26, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.011672, mae: 1.113423, mean_q: 1.363062, mean_eps: 0.976581

479022/1000000: episode: 925, duration: 17.490s, episode steps: 453, steps per second: 26, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: 0.011707, mae: 1.117765, mean_q: 1.367918, mean_eps: 0.975947

479298/1000000: episode: 926, duration: 10.621s, episode steps: 276, steps per second: 26, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.772 [0.000, 5.000], loss: 0.012563, mae: 1.142537, mean_q: 1.395587, mean_eps: 0.975510

479670/1000000: episode: 927, duration: 14.320s, episode steps: 372, steps per second: 26, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.012859, mae: 1.127341, mean_q: 1.376080, mean_eps: 0.975122

480561/1000000: episode: 928, duration: 34.247s, episode steps: 891, steps per second: 26, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.012671, mae: 1.143034, mean_q: 1.395970, mean_eps: 0.974363

481239/1000000: episode: 929, duration: 26.053s, episode steps: 678, steps per second: 26, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: 0.013663, mae: 1.135465, mean_q: 1.385497, mean_eps: 0.973422

481574/1000000: episode: 930, duration: 12.952s, episode steps: 335, steps per second: 26, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: 0.014700, mae: 1.147046, mean_q: 1.401758, mean_eps: 0.972815

481867/1000000: episode: 931, duration: 11.252s, episode steps: 293, steps per second: 26, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.012101, mae: 1.139269, mean_q: 1.394110, mean_eps: 0.972438

482317/1000000: episode: 932, duration: 17.444s, episode steps: 450, steps per second: 26, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: 0.013446, mae: 1.144119, mean_q: 1.397752, mean_eps: 0.971992

482792/1000000: episode: 933, duration: 18.356s, episode steps: 475, steps per second: 26, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.013291, mae: 1.144277, mean_q: 1.399026, mean_eps: 0.971438

483611/1000000: episode: 934, duration: 31.479s, episode steps: 819, steps per second: 26, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: 0.013413, mae: 1.140956, mean_q: 1.395535, mean_eps: 0.970662

483898/1000000: episode: 935, duration: 11.167s, episode steps: 287, steps per second: 26, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean

action: 2.526 [0.000, 5.000], loss: 0.013232, mae: 1.169394, mean_q: 1.430231, mean_eps: 0.969998

484308/1000000: episode: 936, duration: 15.964s, episode steps: 410, steps per second: 26, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.011801, mae: 1.197891, mean_q: 1.462281, mean_eps: 0.969580

484891/1000000: episode: 937, duration: 22.560s, episode steps: 583, steps per second: 26, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.012069, mae: 1.183608, mean_q: 1.445542, mean_eps: 0.968985

485352/1000000: episode: 938, duration: 17.904s, episode steps: 461, steps per second: 26, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.013642, mae: 1.186293, mean_q: 1.446066, mean_eps: 0.968358

486126/1000000: episode: 939, duration: 29.966s, episode steps: 774, steps per second: 26, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.012689, mae: 1.191285, mean_q: 1.452132, mean_eps: 0.967617

486543/1000000: episode: 940, duration: 16.060s, episode steps: 417, steps per second: 26, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: 0.012015, mae: 1.171704, mean_q: 1.425833, mean_eps: 0.966902

487015/1000000: episode: 941, duration: 18.270s, episode steps: 472, steps per second: 26, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: 0.012410, mae: 1.179532, mean_q: 1.436146, mean_eps: 0.966369

487645/1000000: episode: 942, duration: 24.419s, episode steps: 630, steps per second: 26, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.344 [0.000, 5.000], loss: 0.013110, mae: 1.179127, mean_q: 1.437084, mean_eps: 0.965706

488116/1000000: episode: 943, duration: 18.313s, episode steps: 471, steps per second: 26, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.013250, mae: 1.175722, mean_q: 1.435888, mean_eps: 0.965046

488670/1000000: episode: 944, duration: 21.509s, episode steps: 554, steps per second: 26, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: 0.012874, mae: 1.180485, mean_q: 1.441807, mean_eps: 0.964432

489116/1000000: episode: 945, duration: 16.275s, episode steps: 446, steps per second: 27, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.013741, mae: 1.213499, mean_q: 1.482410, mean_eps: 0.963832

489432/1000000: episode: 946, duration: 12.159s, episode steps: 316, steps per second: 26, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.012372, mae: 1.224847, mean_q: 1.493956, mean_eps: 0.963376

489756/1000000: episode: 947, duration: 12.673s, episode steps: 324, steps per second: 26, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean

action: 2.472 [0.000, 5.000], loss: 0.011950, mae: 1.230570, mean_q: 1.499394, mean_eps: 0.962992

490277/1000000: episode: 948, duration: 20.348s, episode steps: 521, steps per second: 26, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.013488, mae: 1.216112, mean_q: 1.480368, mean_eps: 0.962483

490574/1000000: episode: 949, duration: 11.485s, episode steps: 297, steps per second: 26, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.013754, mae: 1.232505, mean_q: 1.501421, mean_eps: 0.961991

490954/1000000: episode: 950, duration: 14.788s, episode steps: 380, steps per second: 26, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: 0.014032, mae: 1.225231, mean_q: 1.493248, mean_eps: 0.961586

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

491527/1000000: episode: 951, duration: 21.965s, episode steps: 573, steps per second: 26, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: 0.012992, mae: 1.210314, mean_q: 1.472488, mean_eps: 0.961019

492018/1000000: episode: 952, duration: 2.143s, episode steps: 491, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

492310/1000000: episode: 953, duration: 1.277s, episode steps: 292, steps per second: 229, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

492775/1000000: episode: 954, duration: 2.045s, episode steps: 465, steps per second: 227, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

493216/1000000: episode: 955, duration: 1.930s, episode steps: 441, steps per second: 228, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

493612/1000000: episode: 956, duration: 1.751s, episode steps: 396, steps per second: 226, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

494368/1000000: episode: 957, duration: 3.392s, episode steps: 756, steps per second: 223, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

494875/1000000: episode: 958, duration: 2.208s, episode steps: 507, steps per second: 230, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

495435/1000000: episode: 959, duration: 2.458s, episode steps: 560, steps per second: 228, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

496193/1000000: episode: 960, duration: 3.275s, episode steps: 758, steps per second: 231, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean

action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn

dueling/version_1.0/pesos/dqn_v_1.0_weights_960_episodes.h5f

496663/1000000: episode: 961, duration: 2.184s, episode steps: 470, steps per second: 215, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

496949/1000000: episode: 962, duration: 1.297s, episode steps: 286, steps per second: 221, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

497325/1000000: episode: 963, duration: 1.658s, episode steps: 376, steps per second: 227, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

497837/1000000: episode: 964, duration: 2.249s, episode steps: 512, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

498585/1000000: episode: 965, duration: 3.262s, episode steps: 748, steps per second: 229, episode reward: 24.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

499068/1000000: episode: 966, duration: 2.133s, episode steps: 483, steps per second: 226, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

499578/1000000: episode: 967, duration: 2.287s, episode steps: 510, steps per second: 223, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.692 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

500082/1000000: episode: 968, duration: 2.237s, episode steps: 504, steps per second: 225, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

500631/1000000: episode: 969, duration: 2.425s, episode steps: 549, steps per second: 226, episode reward: 15.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

500914/1000000: episode: 970, duration: 1.247s, episode steps: 283, steps per second: 227, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.322 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

501287/1000000: episode: 971, duration: 1.649s, episode steps: 373, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

501842/1000000: episode: 972, duration: 2.442s, episode steps: 555, steps per second: 227, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.625 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

502409/1000000: episode: 973, duration: 2.521s, episode steps: 567, steps per second: 225, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

502994/1000000: episode: 974, duration: 2.625s, episode steps: 585, steps per second: 223, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

503448/1000000: episode: 975, duration: 1.996s, episode steps: 454, steps per

second: 227, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
503970/1000000: episode: 976, duration: 2.288s, episode steps: 522, steps per second: 228, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
504291/1000000: episode: 977, duration: 1.414s, episode steps: 321, steps per second: 227, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
504695/1000000: episode: 978, duration: 1.766s, episode steps: 404, steps per second: 229, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
505404/1000000: episode: 979, duration: 3.139s, episode steps: 709, steps per second: 226, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
505788/1000000: episode: 980, duration: 1.711s, episode steps: 384, steps per second: 224, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
506092/1000000: episode: 981, duration: 1.323s, episode steps: 304, steps per second: 230, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
506455/1000000: episode: 982, duration: 1.591s, episode steps: 363, steps per second: 228, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
506741/1000000: episode: 983, duration: 1.249s, episode steps: 286, steps per second: 229, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
507419/1000000: episode: 984, duration: 2.946s, episode steps: 678, steps per second: 230, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
507902/1000000: episode: 985, duration: 2.127s, episode steps: 483, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
508568/1000000: episode: 986, duration: 2.966s, episode steps: 666, steps per second: 225, episode reward: 21.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
509074/1000000: episode: 987, duration: 2.212s, episode steps: 506, steps per second: 229, episode reward: 16.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
509524/1000000: episode: 988, duration: 1.985s, episode steps: 450, steps per second: 227, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
510111/1000000: episode: 989, duration: 2.556s, episode steps: 587, steps per second: 230, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
510642/1000000: episode: 990, duration: 2.311s, episode steps: 531, steps per second: 230, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
511154/1000000: episode: 991, duration: 2.255s, episode steps: 512, steps per

second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

511457/1000000: episode: 992, duration: 1.345s, episode steps: 303, steps per second: 225, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

512250/1000000: episode: 993, duration: 3.453s, episode steps: 793, steps per second: 230, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

512631/1000000: episode: 994, duration: 1.674s, episode steps: 381, steps per second: 228, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

512969/1000000: episode: 995, duration: 1.485s, episode steps: 338, steps per second: 228, episode reward: 10.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

513470/1000000: episode: 996, duration: 2.196s, episode steps: 501, steps per second: 228, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

514028/1000000: episode: 997, duration: 2.466s, episode steps: 558, steps per second: 226, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

514370/1000000: episode: 998, duration: 1.532s, episode steps: 342, steps per second: 223, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

514727/1000000: episode: 999, duration: 1.575s, episode steps: 357, steps per second: 227, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

515157/1000000: episode: 1000, duration: 1.878s, episode steps: 430, steps per second: 229, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.333 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

515447/1000000: episode: 1001, duration: 1.263s, episode steps: 290, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

515851/1000000: episode: 1002, duration: 1.768s, episode steps: 404, steps per second: 228, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

516387/1000000: episode: 1003, duration: 2.329s, episode steps: 536, steps per second: 230, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

516934/1000000: episode: 1004, duration: 2.402s, episode steps: 547, steps per second: 228, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

517313/1000000: episode: 1005, duration: 1.694s, episode steps: 379, steps per second: 224, episode reward: 12.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

517746/1000000: episode: 1006, duration: 1.909s, episode steps: 433, steps per second: 227, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

518631/1000000: episode: 1007, duration: 3.840s, episode steps: 885, steps per

second: 230, episode reward: 22.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

519074/1000000: episode: 1008, duration: 1.937s, episode steps: 443, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

519451/1000000: episode: 1009, duration: 1.664s, episode steps: 377, steps per second: 227, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

520138/1000000: episode: 1010, duration: 3.023s, episode steps: 687, steps per second: 227, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

520866/1000000: episode: 1011, duration: 3.146s, episode steps: 728, steps per second: 231, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

521405/1000000: episode: 1012, duration: 2.363s, episode steps: 539, steps per second: 228, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

521954/1000000: episode: 1013, duration: 2.399s, episode steps: 549, steps per second: 229, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

522545/1000000: episode: 1014, duration: 2.584s, episode steps: 591, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

522829/1000000: episode: 1015, duration: 1.257s, episode steps: 284, steps per second: 226, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

523339/1000000: episode: 1016, duration: 2.237s, episode steps: 510, steps per second: 228, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

523804/1000000: episode: 1017, duration: 2.055s, episode steps: 465, steps per second: 226, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

524257/1000000: episode: 1018, duration: 2.001s, episode steps: 453, steps per second: 226, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

524892/1000000: episode: 1019, duration: 2.771s, episode steps: 635, steps per second: 229, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

525464/1000000: episode: 1020, duration: 2.501s, episode steps: 572, steps per second: 229, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

526051/1000000: episode: 1021, duration: 2.593s, episode steps: 587, steps per second: 226, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

526731/1000000: episode: 1022, duration: 2.974s, episode steps: 680, steps per second: 229, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

527023/1000000: episode: 1023, duration: 1.282s, episode steps: 292, steps per

second: 228, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

527659/1000000: episode: 1024, duration: 2.769s, episode steps: 636, steps per second: 230, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

528170/1000000: episode: 1025, duration: 2.235s, episode steps: 511, steps per second: 229, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

529257/1000000: episode: 1026, duration: 4.763s, episode steps: 1087, steps per second: 228, episode reward: 12.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

529593/1000000: episode: 1027, duration: 1.476s, episode steps: 336, steps per second: 228, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.330 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

529951/1000000: episode: 1028, duration: 1.560s, episode steps: 358, steps per second: 230, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

530392/1000000: episode: 1029, duration: 1.925s, episode steps: 441, steps per second: 229, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

530783/1000000: episode: 1030, duration: 1.706s, episode steps: 391, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

531374/1000000: episode: 1031, duration: 2.615s, episode steps: 591, steps per second: 226, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

531848/1000000: episode: 1032, duration: 2.102s, episode steps: 474, steps per second: 225, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

532305/1000000: episode: 1033, duration: 1.992s, episode steps: 457, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

532681/1000000: episode: 1034, duration: 1.640s, episode steps: 376, steps per second: 229, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

532964/1000000: episode: 1035, duration: 1.243s, episode steps: 283, steps per second: 228, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

533749/1000000: episode: 1036, duration: 3.393s, episode steps: 785, steps per second: 231, episode reward: 21.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

534263/1000000: episode: 1037, duration: 2.276s, episode steps: 514, steps per second: 226, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

535057/1000000: episode: 1038, duration: 30.349s, episode steps: 794, steps per second: 26, episode reward: 20.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: 0.013501, mae: 1.263987, mean_q: 1.540182, mean_eps: 0.987537

535472/1000000: episode: 1039, duration: 16.453s, episode steps: 415, steps per second: 25, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.014800, mae: 1.256732, mean_q: 1.533162, mean_eps: 0.986829

535971/1000000: episode: 1040, duration: 19.840s, episode steps: 499, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.014238, mae: 1.253114, mean_q: 1.531215, mean_eps: 0.986282

536438/1000000: episode: 1041, duration: 18.453s, episode steps: 467, steps per second: 25, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.012674, mae: 1.251747, mean_q: 1.526599, mean_eps: 0.985701

536923/1000000: episode: 1042, duration: 19.209s, episode steps: 485, steps per second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: 0.013903, mae: 1.260761, mean_q: 1.536834, mean_eps: 0.985130

537349/1000000: episode: 1043, duration: 16.935s, episode steps: 426, steps per second: 25, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.012356, mae: 1.239036, mean_q: 1.511361, mean_eps: 0.984582

537634/1000000: episode: 1044, duration: 11.274s, episode steps: 285, steps per second: 25, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: 0.013979, mae: 1.269915, mean_q: 1.552409, mean_eps: 0.984155

538189/1000000: episode: 1045, duration: 22.041s, episode steps: 555, steps per second: 25, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.012300, mae: 1.242173, mean_q: 1.519115, mean_eps: 0.983651

538706/1000000: episode: 1046, duration: 20.443s, episode steps: 517, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.012431, mae: 1.242090, mean_q: 1.519995, mean_eps: 0.983008

539148/1000000: episode: 1047, duration: 17.606s, episode steps: 442, steps per second: 25, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.012657, mae: 1.239725, mean_q: 1.514200, mean_eps: 0.982434

539864/1000000: episode: 1048, duration: 27.154s, episode steps: 716, steps per second: 26, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.012487, mae: 1.281098, mean_q: 1.565274, mean_eps: 0.981741

540465/1000000: episode: 1049, duration: 23.908s, episode steps: 601, steps per second: 25, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.013523, mae: 1.306400, mean_q: 1.594194, mean_eps: 0.980949

540765/1000000: episode: 1050, duration: 11.879s, episode steps: 300, steps per second: 25, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.013297, mae: 1.304832, mean_q: 1.592165, mean_eps: 0.980406

541769/1000000: episode: 1051, duration: 39.702s, episode steps: 1004, steps per second: 25, episode reward: 16.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.013508, mae: 1.313700, mean_q: 1.603341, mean_eps: 0.979624

542485/1000000: episode: 1052, duration: 28.308s, episode steps: 716, steps per second: 25, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.014469, mae: 1.315633, mean_q: 1.605685, mean_eps: 0.978592

542778/1000000: episode: 1053, duration: 11.570s, episode steps: 293, steps per second: 25, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.012120, mae: 1.288345, mean_q: 1.572915, mean_eps: 0.977987

543479/1000000: episode: 1054, duration: 27.773s, episode steps: 701, steps per second: 25, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.013274, mae: 1.293514, mean_q: 1.577227, mean_eps: 0.977392

543939/1000000: episode: 1055, duration: 18.304s, episode steps: 460, steps per second: 25, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: 0.012284, mae: 1.289845, mean_q: 1.575039, mean_eps: 0.976696

544534/1000000: episode: 1056, duration: 23.589s, episode steps: 595, steps per second: 25, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: 0.014531, mae: 1.337575, mean_q: 1.631688, mean_eps: 0.976062

545089/1000000: episode: 1057, duration: 22.066s, episode steps: 555, steps per second: 25, episode reward: 15.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: 0.015688, mae: 1.347072, mean_q: 1.639973, mean_eps: 0.975371

545565/1000000: episode: 1058, duration: 18.800s, episode steps: 476, steps per second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.012053, mae: 1.355235, mean_q: 1.651846, mean_eps: 0.974752

546463/1000000: episode: 1059, duration: 35.495s, episode steps: 898, steps per second: 25, episode reward: 25.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: 0.013163, mae: 1.340424, mean_q: 1.636296, mean_eps: 0.973929

546919/1000000: episode: 1060, duration: 18.136s, episode steps: 456, steps per second: 25, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.014181, mae: 1.349760, mean_q: 1.648543, mean_eps: 0.973118

547214/1000000: episode: 1061, duration: 11.776s, episode steps: 295, steps per second: 25, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.012538, mae: 1.349816, mean_q: 1.652969, mean_eps: 0.972666

547730/1000000: episode: 1062, duration: 20.609s, episode steps: 516, steps per second: 25, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.014635, mae: 1.361996, mean_q: 1.663137, mean_eps: 0.972179

548027/1000000: episode: 1063, duration: 11.897s, episode steps: 297, steps per second: 25, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: 0.013834, mae: 1.316614, mean_q: 1.610231, mean_eps: 0.971692

548620/1000000: episode: 1064, duration: 23.879s, episode steps: 593, steps per second: 25, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.013669, mae: 1.333912, mean_q: 1.631609, mean_eps: 0.971159

549106/1000000: episode: 1065, duration: 19.406s, episode steps: 486, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.014575, mae: 1.336662, mean_q: 1.633981, mean_eps: 0.970511

549476/1000000: episode: 1066, duration: 14.720s, episode steps: 370, steps per second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: 0.013846, mae: 1.324504, mean_q: 1.616444, mean_eps: 0.969998

549849/1000000: episode: 1067, duration: 14.986s, episode steps: 373, steps per second: 25, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.386 [0.000, 5.000], loss: 0.011713, mae: 1.334230, mean_q: 1.627274, mean_eps: 0.969551

550342/1000000: episode: 1068, duration: 19.619s, episode steps: 493, steps per second: 25, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: 0.011761, mae: 1.337660, mean_q: 1.631912, mean_eps: 0.969030

550993/1000000: episode: 1069, duration: 26.144s, episode steps: 651, steps per second: 25, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.011536, mae: 1.324360, mean_q: 1.615582, mean_eps: 0.968344

551475/1000000: episode: 1070, duration: 19.118s, episode steps: 482, steps per second: 25, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.011335, mae: 1.338489, mean_q: 1.631422, mean_eps: 0.967665

551881/1000000: episode: 1071, duration: 16.330s, episode steps: 406, steps per second: 25, episode reward: 11.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.012113, mae: 1.337097, mean_q: 1.632470, mean_eps: 0.967132

552425/1000000: episode: 1072, duration: 21.671s, episode steps: 544, steps per second: 25, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.015096, mae: 1.346422, mean_q: 1.642789, mean_eps: 0.966561

552701/1000000: episode: 1073, duration: 11.085s, episode steps: 276, steps per second: 25, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.011233, mae: 1.322319, mean_q: 1.615103, mean_eps: 0.966069

553213/1000000: episode: 1074, duration: 20.404s, episode steps: 512, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.011689, mae: 1.362223, mean_q: 1.664763, mean_eps: 0.965596

553688/1000000: episode: 1075, duration: 18.991s, episode steps: 475, steps per second: 25, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: 0.012756, mae: 1.346663, mean_q: 1.647538, mean_eps: 0.965006

554172/1000000: episode: 1076, duration: 19.445s, episode steps: 484, steps per second: 25, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.013772, mae: 1.329743, mean_q: 1.625010, mean_eps: 0.964432

554426/1000000: episode: 1077, duration: 9.828s, episode steps: 254, steps per second: 26, episode reward: 4.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.012539, mae: 1.378858, mean_q: 1.685124, mean_eps: 0.963988

555034/1000000: episode: 1078, duration: 23.421s, episode steps: 608, steps per second: 26, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.013859, mae: 1.374164, mean_q: 1.676598, mean_eps: 0.963470

555580/1000000: episode: 1079, duration: 21.782s, episode steps: 546, steps per second: 25, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.013672, mae: 1.370191, mean_q: 1.672344, mean_eps: 0.962778

556156/1000000: episode: 1080, duration: 23.196s, episode steps: 576, steps per second: 25, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: 0.012902, mae: 1.397167, mean_q: 1.702711, mean_eps: 0.962106

556658/1000000: episode: 1081, duration: 20.163s, episode steps: 502, steps per second: 25, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.012798, mae: 1.378598, mean_q: 1.677904, mean_eps: 0.961458

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

557075/1000000: episode: 1082, duration: 15.958s, episode steps: 417, steps per second: 26, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: 0.012266, mae: 1.346164, mean_q: 1.641337, mean_eps: 0.960918

557587/1000000: episode: 1083, duration: 2.256s, episode steps: 512, steps per second: 227, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

558113/1000000: episode: 1084, duration: 2.312s, episode steps: 526, steps per second: 228, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

558636/1000000: episode: 1085, duration: 2.316s, episode steps: 523, steps per second: 226, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.683 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

559078/1000000: episode: 1086, duration: 1.957s, episode steps: 442, steps per second: 226, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

559568/1000000: episode: 1087, duration: 2.161s, episode steps: 490, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

560068/1000000: episode: 1088, duration: 2.218s, episode steps: 500, steps per second: 225, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

560515/1000000: episode: 1089, duration: 1.964s, episode steps: 447, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

560954/1000000: episode: 1090, duration: 1.907s, episode steps: 439, steps per second: 230, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

561475/1000000: episode: 1091, duration: 2.275s, episode steps: 521, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

561965/1000000: episode: 1092, duration: 2.189s, episode steps: 490, steps per second: 224, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

562438/1000000: episode: 1093, duration: 2.093s, episode steps: 473, steps per second: 226, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

562876/1000000: episode: 1094, duration: 1.922s, episode steps: 438, steps per second: 228, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

563627/1000000: episode: 1095, duration: 3.260s, episode steps: 751, steps per second: 230, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

564081/1000000: episode: 1096, duration: 1.991s, episode steps: 454, steps per second: 228, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

564388/1000000: episode: 1097, duration: 1.361s, episode steps: 307, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.661 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

564811/1000000: episode: 1098, duration: 1.930s, episode steps: 423, steps per second: 219, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

565857/1000000: episode: 1099, duration: 4.547s, episode steps: 1046, steps per second: 230, episode reward: 24.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

566148/1000000: episode: 1100, duration: 1.268s, episode steps: 291, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.333 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

566676/1000000: episode: 1101, duration: 2.310s, episode steps: 528, steps per second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

567154/1000000: episode: 1102, duration: 2.091s, episode steps: 478, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

567671/1000000: episode: 1103, duration: 2.331s, episode steps: 517, steps per second: 222, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

568430/1000000: episode: 1104, duration: 3.312s, episode steps: 759, steps per second: 229, episode reward: 20.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

568931/1000000: episode: 1105, duration: 2.198s, episode steps: 501, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

569508/1000000: episode: 1106, duration: 2.521s, episode steps: 577, steps per second: 229, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.362 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

569977/1000000: episode: 1107, duration: 2.074s, episode steps: 469, steps per second: 226, episode reward: 3.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

571128/1000000: episode: 1108, duration: 5.008s, episode steps: 1151, steps per second: 230, episode reward: 14.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

571490/1000000: episode: 1109, duration: 1.587s, episode steps: 362, steps per second: 228, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

572034/1000000: episode: 1110, duration: 2.366s, episode steps: 544, steps per second: 230, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

572476/1000000: episode: 1111, duration: 1.930s, episode steps: 442, steps per second: 229, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

572792/1000000: episode: 1112, duration: 1.388s, episode steps: 316, steps per second: 228, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

573221/1000000: episode: 1113, duration: 1.891s, episode steps: 429, steps per second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

573783/1000000: episode: 1114, duration: 2.459s, episode steps: 562, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

574318/1000000: episode: 1115, duration: 2.337s, episode steps: 535, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

575040/1000000: episode: 1116, duration: 3.152s, episode steps: 722, steps per second: 229, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

575540/1000000: episode: 1117, duration: 2.191s, episode steps: 500, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

576616/1000000: episode: 1118, duration: 4.691s, episode steps: 1076, steps per second: 229, episode reward: 22.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

577028/1000000: episode: 1119, duration: 1.802s, episode steps: 412, steps per second: 229, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
577587/1000000: episode: 1120, duration: 2.436s, episode steps: 559, steps per second: 229, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_1120_episodes.h5f

578058/1000000: episode: 1121, duration: 2.162s, episode steps: 471, steps per second: 218, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
578725/1000000: episode: 1122, duration: 2.936s, episode steps: 667, steps per second: 227, episode reward: 19.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
579110/1000000: episode: 1123, duration: 1.715s, episode steps: 385, steps per second: 224, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
579618/1000000: episode: 1124, duration: 2.243s, episode steps: 508, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
580651/1000000: episode: 1125, duration: 4.512s, episode steps: 1033, steps per second: 229, episode reward: 22.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
581054/1000000: episode: 1126, duration: 1.762s, episode steps: 403, steps per second: 229, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
581716/1000000: episode: 1127, duration: 2.881s, episode steps: 662, steps per second: 230, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
582372/1000000: episode: 1128, duration: 2.917s, episode steps: 656, steps per second: 225, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
582785/1000000: episode: 1129, duration: 1.813s, episode steps: 413, steps per second: 228, episode reward: 11.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
583208/1000000: episode: 1130, duration: 1.881s, episode steps: 423, steps per second: 225, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
583918/1000000: episode: 1131, duration: 3.088s, episode steps: 710, steps per second: 230, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
584393/1000000: episode: 1132, duration: 2.067s, episode steps: 475, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
585002/1000000: episode: 1133, duration: 2.710s, episode steps: 609, steps per second: 225, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean

action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 585285/1000000: episode: 1134, duration: 1.256s, episode steps: 283, steps per second: 225, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 585675/1000000: episode: 1135, duration: 1.722s, episode steps: 390, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 586115/1000000: episode: 1136, duration: 1.929s, episode steps: 440, steps per second: 228, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 587016/1000000: episode: 1137, duration: 3.899s, episode steps: 901, steps per second: 231, episode reward: 25.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 587897/1000000: episode: 1138, duration: 3.865s, episode steps: 881, steps per second: 228, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 588746/1000000: episode: 1139, duration: 3.697s, episode steps: 849, steps per second: 230, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 589346/1000000: episode: 1140, duration: 2.613s, episode steps: 600, steps per second: 230, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 589970/1000000: episode: 1141, duration: 2.738s, episode steps: 624, steps per second: 228, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 590468/1000000: episode: 1142, duration: 2.166s, episode steps: 498, steps per second: 230, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 590889/1000000: episode: 1143, duration: 1.887s, episode steps: 421, steps per second: 223, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 591295/1000000: episode: 1144, duration: 1.777s, episode steps: 406, steps per second: 228, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 591627/1000000: episode: 1145, duration: 1.457s, episode steps: 332, steps per second: 228, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 592316/1000000: episode: 1146, duration: 2.998s, episode steps: 689, steps per second: 230, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 592893/1000000: episode: 1147, duration: 2.505s, episode steps: 577, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 593311/1000000: episode: 1148, duration: 1.848s, episode steps: 418, steps per second: 226, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 593721/1000000: episode: 1149, duration: 1.823s, episode steps: 410, steps per second: 225, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 594148/1000000: episode: 1150, duration: 1.875s, episode steps: 427, steps per second: 228, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 594429/1000000: episode: 1151, duration: 1.235s, episode steps: 281, steps per second: 228, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 594720/1000000: episode: 1152, duration: 1.272s, episode steps: 291, steps per second: 229, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 595208/1000000: episode: 1153, duration: 2.141s, episode steps: 488, steps per second: 228, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 595546/1000000: episode: 1154, duration: 1.476s, episode steps: 338, steps per second: 229, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 596135/1000000: episode: 1155, duration: 2.598s, episode steps: 589, steps per second: 227, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 596523/1000000: episode: 1156, duration: 1.728s, episode steps: 388, steps per second: 224, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 597128/1000000: episode: 1157, duration: 2.651s, episode steps: 605, steps per second: 228, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 597584/1000000: episode: 1158, duration: 1.992s, episode steps: 456, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 598057/1000000: episode: 1159, duration: 2.059s, episode steps: 473, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 598651/1000000: episode: 1160, duration: 2.597s, episode steps: 594, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 599187/1000000: episode: 1161, duration: 2.349s, episode steps: 536, steps per second: 228, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 599759/1000000: episode: 1162, duration: 2.546s, episode steps: 572, steps per second: 225, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 600079/1000000: episode: 1163, duration: 10.574s, episode steps: 320, steps per second: 30, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.011234, mae: 1.388194, mean_q: 1.692930, mean_eps: 0.987846
 600607/1000000: episode: 1164, duration: 21.437s, episode steps: 528, steps per second: 25, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: 0.012888, mae: 1.400403, mean_q: 1.705825, mean_eps: 0.987378

600971/1000000: episode: 1165, duration: 14.741s, episode steps: 364, steps per second: 25, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: 0.013031, mae: 1.394032, mean_q: 1.696156, mean_eps: 0.986843

601467/1000000: episode: 1166, duration: 20.153s, episode steps: 496, steps per second: 25, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: 0.013480, mae: 1.416139, mean_q: 1.724485, mean_eps: 0.986327

601913/1000000: episode: 1167, duration: 18.177s, episode steps: 446, steps per second: 25, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.011623, mae: 1.414313, mean_q: 1.722608, mean_eps: 0.985761

602746/1000000: episode: 1168, duration: 34.007s, episode steps: 833, steps per second: 24, episode reward: 19.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.013811, mae: 1.401842, mean_q: 1.710660, mean_eps: 0.984993

603370/1000000: episode: 1169, duration: 25.452s, episode steps: 624, steps per second: 25, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.012371, mae: 1.397886, mean_q: 1.701895, mean_eps: 0.984119

603755/1000000: episode: 1170, duration: 15.712s, episode steps: 385, steps per second: 25, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.010539, mae: 1.403432, mean_q: 1.710671, mean_eps: 0.983514

604233/1000000: episode: 1171, duration: 19.617s, episode steps: 478, steps per second: 24, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.012640, mae: 1.399625, mean_q: 1.705806, mean_eps: 0.982996

604692/1000000: episode: 1172, duration: 18.763s, episode steps: 459, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: 0.012940, mae: 1.373655, mean_q: 1.672771, mean_eps: 0.982434

605009/1000000: episode: 1173, duration: 12.435s, episode steps: 317, steps per second: 25, episode reward: 10.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: 0.012591, mae: 1.401493, mean_q: 1.705833, mean_eps: 0.981969

605598/1000000: episode: 1174, duration: 23.101s, episode steps: 589, steps per second: 25, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.012321, mae: 1.411697, mean_q: 1.719514, mean_eps: 0.981424

605860/1000000: episode: 1175, duration: 10.738s, episode steps: 262, steps per second: 24, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: 0.014940, mae: 1.442391, mean_q: 1.758733, mean_eps: 0.980915

606809/1000000: episode: 1176, duration: 38.797s, episode steps: 949, steps per second: 24, episode reward: 22.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.012645, mae: 1.417614, mean_q: 1.726733, mean_eps: 0.980188

607456/1000000: episode: 1177, duration: 26.542s, episode steps: 647, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.012521, mae: 1.417732, mean_q: 1.726737, mean_eps: 0.979230

607846/1000000: episode: 1178, duration: 16.093s, episode steps: 390, steps per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: 0.013398, mae: 1.433900, mean_q: 1.745730, mean_eps: 0.978609

608140/1000000: episode: 1179, duration: 12.085s, episode steps: 294, steps per second: 24, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.013815, mae: 1.431184, mean_q: 1.741552, mean_eps: 0.978198

608583/1000000: episode: 1180, duration: 18.087s, episode steps: 443, steps per second: 24, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: 0.013437, mae: 1.402680, mean_q: 1.704624, mean_eps: 0.977757

609064/1000000: episode: 1181, duration: 19.799s, episode steps: 481, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.012665, mae: 1.423772, mean_q: 1.733890, mean_eps: 0.977202

609737/1000000: episode: 1182, duration: 27.697s, episode steps: 673, steps per second: 24, episode reward: 6.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: 0.011095, mae: 1.404616, mean_q: 1.711230, mean_eps: 0.976509

610324/1000000: episode: 1183, duration: 24.080s, episode steps: 587, steps per second: 24, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: 0.013655, mae: 1.454795, mean_q: 1.770122, mean_eps: 0.975753

610815/1000000: episode: 1184, duration: 20.157s, episode steps: 491, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.662 [0.000, 5.000], loss: 0.013436, mae: 1.442569, mean_q: 1.758970, mean_eps: 0.975107

611137/1000000: episode: 1185, duration: 13.282s, episode steps: 322, steps per second: 24, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.013790, mae: 1.467169, mean_q: 1.787191, mean_eps: 0.974618

611681/1000000: episode: 1186, duration: 21.529s, episode steps: 544, steps per second: 25, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.013583, mae: 1.447270, mean_q: 1.764149, mean_eps: 0.974097

612291/1000000: episode: 1187, duration: 23.887s, episode steps: 610, steps per second: 26, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: 0.011551, mae: 1.451470, mean_q: 1.772159, mean_eps: 0.973406

612925/1000000: episode: 1188, duration: 25.947s, episode steps: 634, steps per second: 24, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.013950, mae: 1.455626, mean_q: 1.774277, mean_eps: 0.972659

614009/1000000: episode: 1189, duration: 44.357s, episode steps: 1084, steps per second: 24, episode reward: 27.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: 0.013574, mae: 1.443239, mean_q: 1.758162, mean_eps: 0.971627

614488/1000000: episode: 1190, duration: 19.628s, episode steps: 479, steps per second: 24, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.013530, mae: 1.461791, mean_q: 1.778117, mean_eps: 0.970691

614788/1000000: episode: 1191, duration: 12.459s, episode steps: 300, steps per second: 24, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: 0.010921, mae: 1.450018, mean_q: 1.767835, mean_eps: 0.970226

615298/1000000: episode: 1192, duration: 21.019s, episode steps: 510, steps per second: 24, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: 0.012315, mae: 1.452087, mean_q: 1.766204, mean_eps: 0.969738

615904/1000000: episode: 1193, duration: 24.979s, episode steps: 606, steps per second: 24, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.014597, mae: 1.482115, mean_q: 1.803386, mean_eps: 0.969069

616507/1000000: episode: 1194, duration: 24.920s, episode steps: 603, steps per second: 24, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: 0.011694, mae: 1.455637, mean_q: 1.772945, mean_eps: 0.968344

616993/1000000: episode: 1195, duration: 20.081s, episode steps: 486, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: 0.012701, mae: 1.459578, mean_q: 1.777482, mean_eps: 0.967689

617455/1000000: episode: 1196, duration: 18.934s, episode steps: 462, steps per second: 24, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: 0.014084, mae: 1.479244, mean_q: 1.802918, mean_eps: 0.967120

618047/1000000: episode: 1197, duration: 24.329s, episode steps: 592, steps per second: 24, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.012614, mae: 1.449230, mean_q: 1.765408, mean_eps: 0.966489

618680/1000000: episode: 1198, duration: 26.082s, episode steps: 633, steps per second: 24, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: 0.013471, mae: 1.472574, mean_q: 1.792512, mean_eps: 0.965754

619410/1000000: episode: 1199, duration: 30.084s, episode steps: 730, steps per second: 24, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.014987, mae: 1.458152, mean_q: 1.774253, mean_eps: 0.964936

620213/1000000: episode: 1200, duration: 33.162s, episode steps: 803, steps per second: 24, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.013557, mae: 1.501558, mean_q: 1.829240, mean_eps: 0.964014

620780/1000000: episode: 1201, duration: 23.328s, episode steps: 567, steps per second: 24, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.013152, mae: 1.516747, mean_q: 1.847432, mean_eps: 0.963194

621178/1000000: episode: 1202, duration: 16.523s, episode steps: 398, steps per second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: 0.013579, mae: 1.527972, mean_q: 1.859806, mean_eps: 0.962615

621889/1000000: episode: 1203, duration: 29.243s, episode steps: 711, steps per second: 24, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.013499, mae: 1.520842, mean_q: 1.854852, mean_eps: 0.961948

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

622780/1000000: episode: 1204, duration: 29.602s, episode steps: 891, steps per second: 30, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.013204, mae: 1.529877, mean_q: 1.865390, mean_eps: 0.961101

623292/1000000: episode: 1205, duration: 2.268s, episode steps: 512, steps per second: 226, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

624015/1000000: episode: 1206, duration: 3.191s, episode steps: 723, steps per second: 227, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

624411/1000000: episode: 1207, duration: 1.744s, episode steps: 396, steps per second: 227, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

624942/1000000: episode: 1208, duration: 2.341s, episode steps: 531, steps per second: 227, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

625632/1000000: episode: 1209, duration: 3.047s, episode steps: 690, steps per second: 226, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

626199/1000000: episode: 1210, duration: 2.518s, episode steps: 567, steps per second: 225, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

626590/1000000: episode: 1211, duration: 1.734s, episode steps: 391, steps per second: 226, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

627071/1000000: episode: 1212, duration: 2.116s, episode steps: 481, steps per second: 227, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

627665/1000000: episode: 1213, duration: 2.589s, episode steps: 594, steps per second: 229, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

628260/1000000: episode: 1214, duration: 2.604s, episode steps: 595, steps per

second: 228, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

628599/1000000: episode: 1215, duration: 1.493s, episode steps: 339, steps per second: 227, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

629070/1000000: episode: 1216, duration: 2.090s, episode steps: 471, steps per second: 225, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

629790/1000000: episode: 1217, duration: 3.176s, episode steps: 720, steps per second: 227, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

630231/1000000: episode: 1218, duration: 1.931s, episode steps: 441, steps per second: 228, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

630781/1000000: episode: 1219, duration: 2.451s, episode steps: 550, steps per second: 224, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

631156/1000000: episode: 1220, duration: 1.654s, episode steps: 375, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

631448/1000000: episode: 1221, duration: 1.288s, episode steps: 292, steps per second: 227, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

631804/1000000: episode: 1222, duration: 1.566s, episode steps: 356, steps per second: 227, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

632310/1000000: episode: 1223, duration: 2.231s, episode steps: 506, steps per second: 227, episode reward: 15.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

632879/1000000: episode: 1224, duration: 2.482s, episode steps: 569, steps per second: 229, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

633406/1000000: episode: 1225, duration: 2.310s, episode steps: 527, steps per second: 228, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

633896/1000000: episode: 1226, duration: 2.147s, episode steps: 490, steps per second: 228, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

634510/1000000: episode: 1227, duration: 2.675s, episode steps: 614, steps per second: 230, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

634823/1000000: episode: 1228, duration: 1.408s, episode steps: 313, steps per second: 222, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

635295/1000000: episode: 1229, duration: 2.106s, episode steps: 472, steps per second: 224, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

635661/1000000: episode: 1230, duration: 1.622s, episode steps: 366, steps per

second: 226, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

636004/1000000: episode: 1231, duration: 1.508s, episode steps: 343, steps per second: 227, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

636428/1000000: episode: 1232, duration: 1.860s, episode steps: 424, steps per second: 228, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

636935/1000000: episode: 1233, duration: 2.213s, episode steps: 507, steps per second: 229, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

637472/1000000: episode: 1234, duration: 2.348s, episode steps: 537, steps per second: 229, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

637845/1000000: episode: 1235, duration: 1.674s, episode steps: 373, steps per second: 223, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

638458/1000000: episode: 1236, duration: 2.691s, episode steps: 613, steps per second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

639052/1000000: episode: 1237, duration: 2.614s, episode steps: 594, steps per second: 227, episode reward: 5.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

640166/1000000: episode: 1238, duration: 4.840s, episode steps: 1114, steps per second: 230, episode reward: 19.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

640684/1000000: episode: 1239, duration: 2.303s, episode steps: 518, steps per second: 225, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

641064/1000000: episode: 1240, duration: 1.685s, episode steps: 380, steps per second: 226, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

641674/1000000: episode: 1241, duration: 2.658s, episode steps: 610, steps per second: 230, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

642577/1000000: episode: 1242, duration: 3.915s, episode steps: 903, steps per second: 231, episode reward: 18.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

643033/1000000: episode: 1243, duration: 1.996s, episode steps: 456, steps per second: 228, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

643385/1000000: episode: 1244, duration: 1.563s, episode steps: 352, steps per second: 225, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.332 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

643780/1000000: episode: 1245, duration: 1.776s, episode steps: 395, steps per second: 222, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

644405/1000000: episode: 1246, duration: 2.738s, episode steps: 625, steps per

second: 228, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

644870/1000000: episode: 1247, duration: 2.021s, episode steps: 465, steps per second: 230, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

645553/1000000: episode: 1248, duration: 2.987s, episode steps: 683, steps per second: 229, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

646193/1000000: episode: 1249, duration: 2.805s, episode steps: 640, steps per second: 228, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

646623/1000000: episode: 1250, duration: 1.932s, episode steps: 430, steps per second: 223, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

647544/1000000: episode: 1251, duration: 3.983s, episode steps: 921, steps per second: 231, episode reward: 25.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

647904/1000000: episode: 1252, duration: 1.579s, episode steps: 360, steps per second: 228, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

648811/1000000: episode: 1253, duration: 3.940s, episode steps: 907, steps per second: 230, episode reward: 22.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

649104/1000000: episode: 1254, duration: 1.332s, episode steps: 293, steps per second: 220, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.345 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

649520/1000000: episode: 1255, duration: 1.858s, episode steps: 416, steps per second: 224, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

650018/1000000: episode: 1256, duration: 2.180s, episode steps: 498, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

650341/1000000: episode: 1257, duration: 1.420s, episode steps: 323, steps per second: 227, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

650603/1000000: episode: 1258, duration: 1.163s, episode steps: 262, steps per second: 225, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

651126/1000000: episode: 1259, duration: 2.276s, episode steps: 523, steps per second: 230, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

651691/1000000: episode: 1260, duration: 2.458s, episode steps: 565, steps per second: 230, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

652035/1000000: episode: 1261, duration: 1.520s, episode steps: 344, steps per second: 226, episode reward: 9.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

652717/1000000: episode: 1262, duration: 2.997s, episode steps: 682, steps per

second: 228, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

653218/1000000: episode: 1263, duration: 2.204s, episode steps: 501, steps per second: 227, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

653732/1000000: episode: 1264, duration: 2.267s, episode steps: 514, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

654146/1000000: episode: 1265, duration: 1.823s, episode steps: 414, steps per second: 227, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

654421/1000000: episode: 1266, duration: 1.221s, episode steps: 275, steps per second: 225, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

654790/1000000: episode: 1267, duration: 1.641s, episode steps: 369, steps per second: 225, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

655322/1000000: episode: 1268, duration: 2.370s, episode steps: 532, steps per second: 225, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

655933/1000000: episode: 1269, duration: 2.670s, episode steps: 611, steps per second: 229, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

656426/1000000: episode: 1270, duration: 2.154s, episode steps: 493, steps per second: 229, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

656856/1000000: episode: 1271, duration: 1.888s, episode steps: 430, steps per second: 228, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

657702/1000000: episode: 1272, duration: 3.696s, episode steps: 846, steps per second: 229, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

658223/1000000: episode: 1273, duration: 2.302s, episode steps: 521, steps per second: 226, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

658654/1000000: episode: 1274, duration: 1.879s, episode steps: 431, steps per second: 229, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

659191/1000000: episode: 1275, duration: 2.368s, episode steps: 537, steps per second: 227, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

659570/1000000: episode: 1276, duration: 1.674s, episode steps: 379, steps per second: 226, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

660018/1000000: episode: 1277, duration: 1.966s, episode steps: 448, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

660935/1000000: episode: 1278, duration: 4.022s, episode steps: 917, steps per

second: 228, episode reward: 23.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 661410/1000000: episode: 1279, duration: 2.084s, episode steps: 475, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 661867/1000000: episode: 1280, duration: 2.000s, episode steps: 457, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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662286/1000000: episode: 1281, duration: 1.931s, episode steps: 419, steps per second: 217, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 662772/1000000: episode: 1282, duration: 2.131s, episode steps: 486, steps per second: 228, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 663755/1000000: episode: 1283, duration: 4.291s, episode steps: 983, steps per second: 229, episode reward: 21.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 664491/1000000: episode: 1284, duration: 3.229s, episode steps: 736, steps per second: 228, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 664902/1000000: episode: 1285, duration: 1.804s, episode steps: 411, steps per second: 228, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 665300/1000000: episode: 1286, duration: 1.739s, episode steps: 398, steps per second: 229, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 665865/1000000: episode: 1287, duration: 21.431s, episode steps: 565, steps per second: 26, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.014389, mae: 1.587212, mean_q: 1.933876, mean_eps: 0.987695
 666358/1000000: episode: 1288, duration: 20.799s, episode steps: 493, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: 0.015133, mae: 1.586550, mean_q: 1.932049, mean_eps: 0.987098
 666672/1000000: episode: 1289, duration: 13.364s, episode steps: 314, steps per second: 23, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.369 [0.000, 5.000], loss: 0.015901, mae: 1.548871, mean_q: 1.885042, mean_eps: 0.986615
 667684/1000000: episode: 1290, duration: 42.532s, episode steps: 1012, steps per second: 24, episode reward: 15.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.015149, mae: 1.588503, mean_q: 1.936801, mean_eps: 0.985821
 668653/1000000: episode: 1291, duration: 40.846s, episode steps: 969, steps per second: 24, episode reward: 21.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.513 [0.000, 5.000], loss: 0.014416, mae: 1.565922, mean_q: 1.905577,
 mean_eps: 0.984630
 668988/1000000: episode: 1292, duration: 14.140s, episode steps: 335, steps per
 second: 24, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.639 [0.000, 5.000], loss: 0.015418, mae: 1.579032, mean_q: 1.923740,
 mean_eps: 0.983848
 669366/1000000: episode: 1293, duration: 16.037s, episode steps: 378, steps per
 second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.336 [0.000, 5.000], loss: 0.015204, mae: 1.569528, mean_q: 1.909572,
 mean_eps: 0.983421
 669832/1000000: episode: 1294, duration: 19.618s, episode steps: 466, steps per
 second: 24, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.524 [0.000, 5.000], loss: 0.015739, mae: 1.585108, mean_q: 1.928345,
 mean_eps: 0.982914
 670426/1000000: episode: 1295, duration: 24.906s, episode steps: 594, steps per
 second: 24, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.416 [0.000, 5.000], loss: 0.012517, mae: 1.569501, mean_q: 1.914346,
 mean_eps: 0.982278
 670907/1000000: episode: 1296, duration: 18.925s, episode steps: 481, steps per
 second: 25, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.457 [0.000, 5.000], loss: 0.014908, mae: 1.576013, mean_q: 1.917941,
 mean_eps: 0.981633
 671187/1000000: episode: 1297, duration: 11.731s, episode steps: 280, steps per
 second: 24, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean
 action: 2.657 [0.000, 5.000], loss: 0.013644, mae: 1.537386, mean_q: 1.871389,
 mean_eps: 0.981177
 671449/1000000: episode: 1298, duration: 10.983s, episode steps: 262, steps per
 second: 24, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean
 action: 2.267 [0.000, 5.000], loss: 0.014561, mae: 1.538871, mean_q: 1.873117,
 mean_eps: 0.980850
 672060/1000000: episode: 1299, duration: 25.681s, episode steps: 611, steps per
 second: 24, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean
 action: 2.412 [0.000, 5.000], loss: 0.011995, mae: 1.582517, mean_q: 1.926907,
 mean_eps: 0.980327
 672537/1000000: episode: 1300, duration: 20.607s, episode steps: 477, steps per
 second: 23, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean
 action: 2.436 [0.000, 5.000], loss: 0.012716, mae: 1.556067, mean_q: 1.891446,
 mean_eps: 0.979674
 672913/1000000: episode: 1301, duration: 15.936s, episode steps: 376, steps per
 second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.566 [0.000, 5.000], loss: 0.012709, mae: 1.547795, mean_q: 1.883468,
 mean_eps: 0.979161
 673471/1000000: episode: 1302, duration: 23.531s, episode steps: 558, steps per
 second: 24, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.444 [0.000, 5.000], loss: 0.014329, mae: 1.562325, mean_q: 1.900007,
 mean_eps: 0.978602
 674381/1000000: episode: 1303, duration: 38.221s, episode steps: 910, steps per
 second: 24, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.390 [0.000, 5.000], loss: 0.014049, mae: 1.555106, mean_q: 1.890598,
 mean_eps: 0.977721
 675044/1000000: episode: 1304, duration: 28.129s, episode steps: 663, steps per
 second: 24, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean
 action: 2.514 [0.000, 5.000], loss: 0.012692, mae: 1.561243, mean_q: 1.899298,
 mean_eps: 0.976778
 675335/1000000: episode: 1305, duration: 12.440s, episode steps: 291, steps per
 second: 23, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.440 [0.000, 5.000], loss: 0.012589, mae: 1.580873, mean_q: 1.922650,
 mean_eps: 0.976206
 675779/1000000: episode: 1306, duration: 18.797s, episode steps: 444, steps per
 second: 24, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.502 [0.000, 5.000], loss: 0.016587, mae: 1.607518, mean_q: 1.959587,
 mean_eps: 0.975765
 676081/1000000: episode: 1307, duration: 12.874s, episode steps: 302, steps per
 second: 23, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
 action: 2.520 [0.000, 5.000], loss: 0.014353, mae: 1.579391, mean_q: 1.927389,
 mean_eps: 0.975316
 676481/1000000: episode: 1308, duration: 16.983s, episode steps: 400, steps per
 second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
 action: 2.467 [0.000, 5.000], loss: 0.017790, mae: 1.599495, mean_q: 1.945840,
 mean_eps: 0.974894
 676955/1000000: episode: 1309, duration: 20.103s, episode steps: 474, steps per
 second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.582 [0.000, 5.000], loss: 0.014301, mae: 1.592872, mean_q: 1.938318,
 mean_eps: 0.974370
 677386/1000000: episode: 1310, duration: 18.271s, episode steps: 431, steps per
 second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean
 action: 2.580 [0.000, 5.000], loss: 0.016868, mae: 1.612875, mean_q: 1.963570,
 mean_eps: 0.973828
 677755/1000000: episode: 1311, duration: 15.646s, episode steps: 369, steps per
 second: 24, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean
 action: 2.425 [0.000, 5.000], loss: 0.017139, mae: 1.611134, mean_q: 1.962352,
 mean_eps: 0.973348
 678353/1000000: episode: 1312, duration: 25.675s, episode steps: 598, steps per
 second: 23, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.505 [0.000, 5.000], loss: 0.015019, mae: 1.594006, mean_q: 1.940930,
 mean_eps: 0.972767
 678732/1000000: episode: 1313, duration: 16.241s, episode steps: 379, steps per
 second: 23, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.501 [0.000, 5.000], loss: 0.015764, mae: 1.634380, mean_q: 1.987722,
 mean_eps: 0.972182
 679021/1000000: episode: 1314, duration: 12.384s, episode steps: 289, steps per
 second: 23, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.277 [0.000, 5.000], loss: 0.016980, mae: 1.600868, mean_q: 1.947156,
 mean_eps: 0.971781
 679648/1000000: episode: 1315, duration: 26.597s, episode steps: 627, steps per
 second: 24, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean

action: 2.648 [0.000, 5.000], loss: 0.015129, mae: 1.600086, mean_q: 1.948284,
 mean_eps: 0.971231
 679967/1000000: episode: 1316, duration: 13.648s, episode steps: 319, steps per
 second: 23, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.630 [0.000, 5.000], loss: 0.016883, mae: 1.611068, mean_q: 1.961455,
 mean_eps: 0.970665
 680399/1000000: episode: 1317, duration: 18.254s, episode steps: 432, steps per
 second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean
 action: 2.516 [0.000, 5.000], loss: 0.014280, mae: 1.623870, mean_q: 1.977178,
 mean_eps: 0.970214
 680958/1000000: episode: 1318, duration: 23.800s, episode steps: 559, steps per
 second: 23, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean
 action: 2.415 [0.000, 5.000], loss: 0.013718, mae: 1.678693, mean_q: 2.044902,
 mean_eps: 0.969618
 681619/1000000: episode: 1319, duration: 27.976s, episode steps: 661, steps per
 second: 24, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean
 action: 2.478 [0.000, 5.000], loss: 0.014488, mae: 1.675961, mean_q: 2.039134,
 mean_eps: 0.968886
 682290/1000000: episode: 1320, duration: 28.533s, episode steps: 671, steps per
 second: 24, episode reward: 23.000, mean reward: 0.034 [0.000, 1.000], mean
 action: 2.495 [0.000, 5.000], loss: 0.014066, mae: 1.657342, mean_q: 2.016974,
 mean_eps: 0.968087
 682943/1000000: episode: 1321, duration: 27.658s, episode steps: 653, steps per
 second: 24, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean
 action: 2.524 [0.000, 5.000], loss: 0.014538, mae: 1.676995, mean_q: 2.039934,
 mean_eps: 0.967293
 683414/1000000: episode: 1322, duration: 20.060s, episode steps: 471, steps per
 second: 23, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean
 action: 2.456 [0.000, 5.000], loss: 0.014365, mae: 1.651047, mean_q: 2.008625,
 mean_eps: 0.966618
 683805/1000000: episode: 1323, duration: 16.705s, episode steps: 391, steps per
 second: 23, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
 action: 2.476 [0.000, 5.000], loss: 0.017781, mae: 1.669782, mean_q: 2.031770,
 mean_eps: 0.966100
 684271/1000000: episode: 1324, duration: 19.744s, episode steps: 466, steps per
 second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean
 action: 2.401 [0.000, 5.000], loss: 0.018439, mae: 1.663849, mean_q: 2.022435,
 mean_eps: 0.965586
 684767/1000000: episode: 1325, duration: 21.050s, episode steps: 496, steps per
 second: 24, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean
 action: 2.552 [0.000, 5.000], loss: 0.015447, mae: 1.677246, mean_q: 2.042305,
 mean_eps: 0.965010
 685392/1000000: episode: 1326, duration: 26.529s, episode steps: 625, steps per
 second: 24, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean
 action: 2.642 [0.000, 5.000], loss: 0.015790, mae: 1.656214, mean_q: 2.015741,
 mean_eps: 0.964338
 685888/1000000: episode: 1327, duration: 19.972s, episode steps: 496, steps per
 second: 25, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean


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action: 2.480 [0.000, 5.000], loss: 0.014403, mae: 1.689495, mean_q: 2.053242,
mean_eps: 0.963666
686532/1000000: episode: 1328, duration: 27.553s, episode steps: 644, steps per
second: 23, episode reward: 19.000, mean reward: 0.030 [0.000, 1.000], mean
action: 2.506 [0.000, 5.000], loss: 0.016318, mae: 1.673439, mean_q: 2.033831,
mean_eps: 0.962982
687004/1000000: episode: 1329, duration: 20.323s, episode steps: 472, steps per
second: 23, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.411 [0.000, 5.000], loss: 0.017450, mae: 1.664538, mean_q: 2.023878,
mean_eps: 0.962313
687528/1000000: episode: 1330, duration: 22.411s, episode steps: 524, steps per
second: 23, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.441 [0.000, 5.000], loss: 0.016297, mae: 1.690614, mean_q: 2.057217,
mean_eps: 0.961715
688126/1000000: episode: 1331, duration: 25.705s, episode steps: 598, steps per
second: 23, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.523 [0.000, 5.000], loss: 0.017187, mae: 1.669441, mean_q: 2.029461,
mean_eps: 0.961041

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

688608/1000000: episode: 1332, duration: 2.116s, episode steps: 482, steps per
second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
688988/1000000: episode: 1333, duration: 1.681s, episode steps: 380, steps per
second: 226, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
689523/1000000: episode: 1334, duration: 2.374s, episode steps: 535, steps per
second: 225, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean
action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
689968/1000000: episode: 1335, duration: 1.955s, episode steps: 445, steps per
second: 228, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
690407/1000000: episode: 1336, duration: 1.929s, episode steps: 439, steps per
second: 228, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean
action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
691106/1000000: episode: 1337, duration: 3.071s, episode steps: 699, steps per
second: 228, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
691829/1000000: episode: 1338, duration: 3.153s, episode steps: 723, steps per
second: 229, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
692206/1000000: episode: 1339, duration: 1.699s, episode steps: 377, steps per
second: 222, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.552 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
692821/1000000: episode: 1340, duration: 2.693s, episode steps: 615, steps per
second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean

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action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 693104/1000000: episode: 1341, duration: 1.254s, episode steps: 283, steps per second: 226, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 693690/1000000: episode: 1342, duration: 2.560s, episode steps: 586, steps per second: 229, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.323 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 694060/1000000: episode: 1343, duration: 1.607s, episode steps: 370, steps per second: 230, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 694532/1000000: episode: 1344, duration: 2.078s, episode steps: 472, steps per second: 227, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 694839/1000000: episode: 1345, duration: 1.365s, episode steps: 307, steps per second: 225, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 695266/1000000: episode: 1346, duration: 1.897s, episode steps: 427, steps per second: 225, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 695712/1000000: episode: 1347, duration: 1.944s, episode steps: 446, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 696274/1000000: episode: 1348, duration: 2.453s, episode steps: 562, steps per second: 229, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 696766/1000000: episode: 1349, duration: 2.145s, episode steps: 492, steps per second: 229, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 697372/1000000: episode: 1350, duration: 2.648s, episode steps: 606, steps per second: 229, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 697708/1000000: episode: 1351, duration: 1.488s, episode steps: 336, steps per second: 226, episode reward: 10.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 698302/1000000: episode: 1352, duration: 2.629s, episode steps: 594, steps per second: 226, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 698674/1000000: episode: 1353, duration: 1.623s, episode steps: 372, steps per second: 229, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 698962/1000000: episode: 1354, duration: 1.263s, episode steps: 288, steps per second: 228, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 699646/1000000: episode: 1355, duration: 2.966s, episode steps: 684, steps per second: 231, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 700122/1000000: episode: 1356, duration: 2.088s, episode steps: 476, steps per second: 228, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 700832/1000000: episode: 1357, duration: 3.143s, episode steps: 710, steps per second: 226, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 701327/1000000: episode: 1358, duration: 2.191s, episode steps: 495, steps per second: 226, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 701610/1000000: episode: 1359, duration: 1.245s, episode steps: 283, steps per second: 227, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 702131/1000000: episode: 1360, duration: 2.264s, episode steps: 521, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 702849/1000000: episode: 1361, duration: 3.113s, episode steps: 718, steps per second: 231, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 703142/1000000: episode: 1362, duration: 1.286s, episode steps: 293, steps per second: 228, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.365 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 703665/1000000: episode: 1363, duration: 2.308s, episode steps: 523, steps per second: 227, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 703933/1000000: episode: 1364, duration: 1.185s, episode steps: 268, steps per second: 226, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 704397/1000000: episode: 1365, duration: 2.017s, episode steps: 464, steps per second: 230, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 704778/1000000: episode: 1366, duration: 1.656s, episode steps: 381, steps per second: 230, episode reward: 11.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 705233/1000000: episode: 1367, duration: 1.998s, episode steps: 455, steps per second: 228, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 705614/1000000: episode: 1368, duration: 1.664s, episode steps: 381, steps per second: 229, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 706691/1000000: episode: 1369, duration: 4.675s, episode steps: 1077, steps per second: 230, episode reward: 15.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 707054/1000000: episode: 1370, duration: 1.583s, episode steps: 363, steps per second: 229, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 707596/1000000: episode: 1371, duration: 2.359s, episode steps: 542, steps per second: 230, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 708034/1000000: episode: 1372, duration: 1.911s, episode steps: 438, steps per second: 229, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean

action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 709246/1000000: episode: 1373, duration: 5.246s, episode steps: 1212, steps per second: 231, episode reward: 20.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 709588/1000000: episode: 1374, duration: 1.515s, episode steps: 342, steps per second: 226, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 710176/1000000: episode: 1375, duration: 2.568s, episode steps: 588, steps per second: 229, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 710547/1000000: episode: 1376, duration: 1.628s, episode steps: 371, steps per second: 228, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 711073/1000000: episode: 1377, duration: 2.316s, episode steps: 526, steps per second: 227, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 712027/1000000: episode: 1378, duration: 4.134s, episode steps: 954, steps per second: 231, episode reward: 14.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 712544/1000000: episode: 1379, duration: 2.298s, episode steps: 517, steps per second: 225, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 713244/1000000: episode: 1380, duration: 3.067s, episode steps: 700, steps per second: 228, episode reward: 20.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 714006/1000000: episode: 1381, duration: 3.311s, episode steps: 762, steps per second: 230, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 714401/1000000: episode: 1382, duration: 1.714s, episode steps: 395, steps per second: 230, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 714779/1000000: episode: 1383, duration: 1.658s, episode steps: 378, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 715177/1000000: episode: 1384, duration: 1.783s, episode steps: 398, steps per second: 223, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 715878/1000000: episode: 1385, duration: 3.067s, episode steps: 701, steps per second: 229, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 716142/1000000: episode: 1386, duration: 1.173s, episode steps: 264, steps per second: 225, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 716746/1000000: episode: 1387, duration: 2.613s, episode steps: 604, steps per second: 231, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 717101/1000000: episode: 1388, duration: 1.551s, episode steps: 355, steps per second: 229, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean

action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 717577/1000000: episode: 1389, duration: 2.076s, episode steps: 476, steps per second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 718067/1000000: episode: 1390, duration: 2.166s, episode steps: 490, steps per second: 226, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 718684/1000000: episode: 1391, duration: 2.737s, episode steps: 617, steps per second: 225, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 719455/1000000: episode: 1392, duration: 3.359s, episode steps: 771, steps per second: 230, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 719870/1000000: episode: 1393, duration: 1.821s, episode steps: 415, steps per second: 228, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 720404/1000000: episode: 1394, duration: 2.360s, episode steps: 534, steps per second: 226, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 720898/1000000: episode: 1395, duration: 2.211s, episode steps: 494, steps per second: 223, episode reward: 17.000, mean reward: 0.034 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 721383/1000000: episode: 1396, duration: 2.149s, episode steps: 485, steps per second: 226, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 721919/1000000: episode: 1397, duration: 2.333s, episode steps: 536, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 722451/1000000: episode: 1398, duration: 2.300s, episode steps: 532, steps per second: 231, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 722744/1000000: episode: 1399, duration: 1.278s, episode steps: 293, steps per second: 229, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 723162/1000000: episode: 1400, duration: 1.833s, episode steps: 418, steps per second: 228, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 723743/1000000: episode: 1401, duration: 2.566s, episode steps: 581, steps per second: 226, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 724027/1000000: episode: 1402, duration: 1.277s, episode steps: 284, steps per second: 222, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 724494/1000000: episode: 1403, duration: 2.021s, episode steps: 467, steps per second: 231, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.340 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 724899/1000000: episode: 1404, duration: 1.755s, episode steps: 405, steps per second: 231, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean

action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 725315/1000000: episode: 1405, duration: 1.812s, episode steps: 416, steps per second: 230, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 725867/1000000: episode: 1406, duration: 2.406s, episode steps: 552, steps per second: 229, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 726406/1000000: episode: 1407, duration: 2.373s, episode steps: 539, steps per second: 227, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 727018/1000000: episode: 1408, duration: 2.709s, episode steps: 612, steps per second: 226, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.356 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 727882/1000000: episode: 1409, duration: 3.735s, episode steps: 864, steps per second: 231, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 728383/1000000: episode: 1410, duration: 2.209s, episode steps: 501, steps per second: 227, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 729129/1000000: episode: 1411, duration: 3.251s, episode steps: 746, steps per second: 229, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 729385/1000000: episode: 1412, duration: 1.139s, episode steps: 256, steps per second: 225, episode reward: 5.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.391 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 729947/1000000: episode: 1413, duration: 2.490s, episode steps: 562, steps per second: 226, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 730437/1000000: episode: 1414, duration: 2.151s, episode steps: 490, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 731056/1000000: episode: 1415, duration: 8.910s, episode steps: 619, steps per second: 69, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.013023, mae: 1.678660, mean_q: 2.043817, mean_eps: 0.987904
 731767/1000000: episode: 1416, duration: 30.805s, episode steps: 711, steps per second: 23, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: 0.014518, mae: 1.690209, mean_q: 2.056602, mean_eps: 0.987383
 732056/1000000: episode: 1417, duration: 12.719s, episode steps: 289, steps per second: 23, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: 0.015258, mae: 1.684975, mean_q: 2.051486, mean_eps: 0.986783
 732717/1000000: episode: 1418, duration: 28.757s, episode steps: 661, steps per second: 23, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.016065, mae: 1.689484, mean_q: 2.054629, mean_eps: 0.986212
 733242/1000000: episode: 1419, duration: 23.026s, episode steps: 525, steps per

second: 23, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.014428, mae: 1.714881, mean_q: 2.087416, mean_eps: 0.985499

733948/1000000: episode: 1420, duration: 30.833s, episode steps: 706, steps per second: 23, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.017428, mae: 1.693004, mean_q: 2.058467, mean_eps: 0.984762

734242/1000000: episode: 1421, duration: 12.897s, episode steps: 294, steps per second: 23, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: 0.015570, mae: 1.703494, mean_q: 2.072464, mean_eps: 0.984162

734907/1000000: episode: 1422, duration: 28.853s, episode steps: 665, steps per second: 23, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.014803, mae: 1.689435, mean_q: 2.055370, mean_eps: 0.983586

735306/1000000: episode: 1423, duration: 17.498s, episode steps: 399, steps per second: 23, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.015805, mae: 1.703041, mean_q: 2.071538, mean_eps: 0.982948

735888/1000000: episode: 1424, duration: 25.606s, episode steps: 582, steps per second: 23, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.017180, mae: 1.694225, mean_q: 2.056751, mean_eps: 0.982360

736504/1000000: episode: 1425, duration: 25.923s, episode steps: 616, steps per second: 24, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: 0.015696, mae: 1.735741, mean_q: 2.109018, mean_eps: 0.981642

736882/1000000: episode: 1426, duration: 16.488s, episode steps: 378, steps per second: 23, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.020232, mae: 1.743216, mean_q: 2.116256, mean_eps: 0.981045

737176/1000000: episode: 1427, duration: 13.035s, episode steps: 294, steps per second: 23, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.017340, mae: 1.714960, mean_q: 2.084031, mean_eps: 0.980642

737868/1000000: episode: 1428, duration: 30.636s, episode steps: 692, steps per second: 23, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.014960, mae: 1.721729, mean_q: 2.094323, mean_eps: 0.980051

738174/1000000: episode: 1429, duration: 13.569s, episode steps: 306, steps per second: 23, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.624 [0.000, 5.000], loss: 0.016040, mae: 1.754964, mean_q: 2.133200, mean_eps: 0.979451

738650/1000000: episode: 1430, duration: 20.818s, episode steps: 476, steps per second: 23, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: 0.015519, mae: 1.732811, mean_q: 2.106221, mean_eps: 0.978981

739143/1000000: episode: 1431, duration: 21.634s, episode steps: 493, steps per

second: 23, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.018011, mae: 1.740133, mean_q: 2.112278, mean_eps: 0.978400

739610/1000000: episode: 1432, duration: 20.524s, episode steps: 467, steps per second: 23, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.018157, mae: 1.746324, mean_q: 2.122910, mean_eps: 0.977824

739995/1000000: episode: 1433, duration: 16.748s, episode steps: 385, steps per second: 23, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.015385, mae: 1.755565, mean_q: 2.130685, mean_eps: 0.977313

740486/1000000: episode: 1434, duration: 21.547s, episode steps: 491, steps per second: 23, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: 0.015395, mae: 1.755045, mean_q: 2.128580, mean_eps: 0.976787

741144/1000000: episode: 1435, duration: 28.733s, episode steps: 658, steps per second: 23, episode reward: 19.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.696 [0.000, 5.000], loss: 0.015825, mae: 1.741470, mean_q: 2.114854, mean_eps: 0.976098

742027/1000000: episode: 1436, duration: 38.721s, episode steps: 883, steps per second: 23, episode reward: 22.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: 0.015008, mae: 1.784825, mean_q: 2.168504, mean_eps: 0.975174

742404/1000000: episode: 1437, duration: 16.699s, episode steps: 377, steps per second: 23, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.015677, mae: 1.789583, mean_q: 2.175439, mean_eps: 0.974418

743136/1000000: episode: 1438, duration: 31.854s, episode steps: 732, steps per second: 23, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.016845, mae: 1.786632, mean_q: 2.172341, mean_eps: 0.973754

743617/1000000: episode: 1439, duration: 21.215s, episode steps: 481, steps per second: 23, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.020391, mae: 1.778388, mean_q: 2.159906, mean_eps: 0.973024

744381/1000000: episode: 1440, duration: 33.224s, episode steps: 764, steps per second: 23, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.016025, mae: 1.770633, mean_q: 2.152585, mean_eps: 0.972275

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_1440_episodes.h5f

744884/1000000: episode: 1441, duration: 22.101s, episode steps: 503, steps per second: 23, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.753 [0.000, 5.000], loss: 0.016430, mae: 1.779165, mean_q: 2.161809, mean_eps: 0.971517

745174/1000000: episode: 1442, duration: 12.698s, episode steps: 290, steps per

second: 23, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: 0.020414, mae: 1.826734, mean_q: 2.219068, mean_eps: 0.971042

745724/1000000: episode: 1443, duration: 24.014s, episode steps: 550, steps per second: 23, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.017954, mae: 1.799315, mean_q: 2.185906, mean_eps: 0.970538

746308/1000000: episode: 1444, duration: 24.389s, episode steps: 584, steps per second: 24, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.016310, mae: 1.772856, mean_q: 2.156010, mean_eps: 0.969858

747185/1000000: episode: 1445, duration: 38.274s, episode steps: 877, steps per second: 23, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: 0.015645, mae: 1.808544, mean_q: 2.200367, mean_eps: 0.968980

747569/1000000: episode: 1446, duration: 16.892s, episode steps: 384, steps per second: 23, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: 0.014120, mae: 1.772372, mean_q: 2.154464, mean_eps: 0.968222

748055/1000000: episode: 1447, duration: 21.215s, episode steps: 486, steps per second: 23, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.014364, mae: 1.804955, mean_q: 2.195118, mean_eps: 0.967701

748575/1000000: episode: 1448, duration: 22.849s, episode steps: 520, steps per second: 23, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.017820, mae: 1.819966, mean_q: 2.210149, mean_eps: 0.967098

749460/1000000: episode: 1449, duration: 38.748s, episode steps: 885, steps per second: 23, episode reward: 22.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: 0.016404, mae: 1.787631, mean_q: 2.171349, mean_eps: 0.966256

749932/1000000: episode: 1450, duration: 20.831s, episode steps: 472, steps per second: 23, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.016387, mae: 1.796326, mean_q: 2.184106, mean_eps: 0.965442

750222/1000000: episode: 1451, duration: 12.833s, episode steps: 290, steps per second: 23, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.693 [0.000, 5.000], loss: 0.015493, mae: 1.813951, mean_q: 2.201525, mean_eps: 0.964984

750654/1000000: episode: 1452, duration: 18.841s, episode steps: 432, steps per second: 23, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.013955, mae: 1.806372, mean_q: 2.193813, mean_eps: 0.964550

751175/1000000: episode: 1453, duration: 21.903s, episode steps: 521, steps per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: 0.017251, mae: 1.808124, mean_q: 2.198151, mean_eps: 0.963978

751538/1000000: episode: 1454, duration: 15.485s, episode steps: 363, steps per

second: 23, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: 0.015692, mae: 1.864791, mean_q: 2.269862, mean_eps: 0.963448

752130/1000000: episode: 1455, duration: 25.976s, episode steps: 592, steps per second: 23, episode reward: 16.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: 0.016132, mae: 1.841719, mean_q: 2.241620, mean_eps: 0.962874

752707/1000000: episode: 1456, duration: 25.389s, episode steps: 577, steps per second: 23, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.016421, mae: 1.830119, mean_q: 2.224572, mean_eps: 0.962174

753355/1000000: episode: 1457, duration: 28.462s, episode steps: 648, steps per second: 23, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.018370, mae: 1.837020, mean_q: 2.229504, mean_eps: 0.961439

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

753746/1000000: episode: 1458, duration: 13.898s, episode steps: 391, steps per second: 28, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.014155, mae: 1.860876, mean_q: 2.263099, mean_eps: 0.960866

754065/1000000: episode: 1459, duration: 1.424s, episode steps: 319, steps per second: 224, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

754476/1000000: episode: 1460, duration: 1.825s, episode steps: 411, steps per second: 225, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

754863/1000000: episode: 1461, duration: 1.704s, episode steps: 387, steps per second: 227, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

755238/1000000: episode: 1462, duration: 1.633s, episode steps: 375, steps per second: 230, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

755812/1000000: episode: 1463, duration: 2.498s, episode steps: 574, steps per second: 230, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.629 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

756582/1000000: episode: 1464, duration: 3.400s, episode steps: 770, steps per second: 226, episode reward: 22.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

757624/1000000: episode: 1465, duration: 4.516s, episode steps: 1042, steps per second: 231, episode reward: 29.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

758349/1000000: episode: 1466, duration: 3.146s, episode steps: 725, steps per second: 230, episode reward: 8.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

758896/1000000: episode: 1467, duration: 2.371s, episode steps: 547, steps per

second: 231, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.329 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

759509/1000000: episode: 1468, duration: 2.706s, episode steps: 613, steps per second: 227, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

760057/1000000: episode: 1469, duration: 2.408s, episode steps: 548, steps per second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

760426/1000000: episode: 1470, duration: 1.615s, episode steps: 369, steps per second: 228, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

760705/1000000: episode: 1471, duration: 1.221s, episode steps: 279, steps per second: 228, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

761217/1000000: episode: 1472, duration: 2.236s, episode steps: 512, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.346 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

761591/1000000: episode: 1473, duration: 1.645s, episode steps: 374, steps per second: 227, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.324 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

762219/1000000: episode: 1474, duration: 2.743s, episode steps: 628, steps per second: 229, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

762701/1000000: episode: 1475, duration: 2.145s, episode steps: 482, steps per second: 225, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

763092/1000000: episode: 1476, duration: 1.703s, episode steps: 391, steps per second: 230, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

763699/1000000: episode: 1477, duration: 2.638s, episode steps: 607, steps per second: 230, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

764070/1000000: episode: 1478, duration: 1.614s, episode steps: 371, steps per second: 230, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

764436/1000000: episode: 1479, duration: 1.593s, episode steps: 366, steps per second: 230, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

764730/1000000: episode: 1480, duration: 1.285s, episode steps: 294, steps per second: 229, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

765179/1000000: episode: 1481, duration: 1.974s, episode steps: 449, steps per second: 227, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

765694/1000000: episode: 1482, duration: 2.280s, episode steps: 515, steps per second: 226, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

766140/1000000: episode: 1483, duration: 1.946s, episode steps: 446, steps per

second: 229, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

766586/1000000: episode: 1484, duration: 1.963s, episode steps: 446, steps per second: 227, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

767052/1000000: episode: 1485, duration: 2.023s, episode steps: 466, steps per second: 230, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

767563/1000000: episode: 1486, duration: 2.240s, episode steps: 511, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

768078/1000000: episode: 1487, duration: 2.272s, episode steps: 515, steps per second: 227, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

768466/1000000: episode: 1488, duration: 1.729s, episode steps: 388, steps per second: 224, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

769085/1000000: episode: 1489, duration: 2.691s, episode steps: 619, steps per second: 230, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

769715/1000000: episode: 1490, duration: 2.731s, episode steps: 630, steps per second: 231, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

770092/1000000: episode: 1491, duration: 1.659s, episode steps: 377, steps per second: 227, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

770597/1000000: episode: 1492, duration: 2.217s, episode steps: 505, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

770984/1000000: episode: 1493, duration: 1.725s, episode steps: 387, steps per second: 224, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

771692/1000000: episode: 1494, duration: 3.142s, episode steps: 708, steps per second: 225, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

772351/1000000: episode: 1495, duration: 2.846s, episode steps: 659, steps per second: 232, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

772801/1000000: episode: 1496, duration: 1.958s, episode steps: 450, steps per second: 230, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

773642/1000000: episode: 1497, duration: 3.636s, episode steps: 841, steps per second: 231, episode reward: 27.000, mean reward: 0.032 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

774127/1000000: episode: 1498, duration: 2.173s, episode steps: 485, steps per second: 223, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

774735/1000000: episode: 1499, duration: 2.656s, episode steps: 608, steps per

second: 229, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

775040/1000000: episode: 1500, duration: 1.356s, episode steps: 305, steps per second: 225, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.679 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

775512/1000000: episode: 1501, duration: 2.089s, episode steps: 472, steps per second: 226, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

776213/1000000: episode: 1502, duration: 3.065s, episode steps: 701, steps per second: 229, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

777008/1000000: episode: 1503, duration: 3.491s, episode steps: 795, steps per second: 228, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

777484/1000000: episode: 1504, duration: 2.076s, episode steps: 476, steps per second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.393 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

777993/1000000: episode: 1505, duration: 2.225s, episode steps: 509, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

778553/1000000: episode: 1506, duration: 2.447s, episode steps: 560, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

779298/1000000: episode: 1507, duration: 3.232s, episode steps: 745, steps per second: 230, episode reward: 19.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

779747/1000000: episode: 1508, duration: 2.011s, episode steps: 449, steps per second: 223, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

780326/1000000: episode: 1509, duration: 2.560s, episode steps: 579, steps per second: 226, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

780826/1000000: episode: 1510, duration: 2.205s, episode steps: 500, steps per second: 227, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

781205/1000000: episode: 1511, duration: 1.666s, episode steps: 379, steps per second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

782122/1000000: episode: 1512, duration: 3.991s, episode steps: 917, steps per second: 230, episode reward: 19.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

782879/1000000: episode: 1513, duration: 3.355s, episode steps: 757, steps per second: 226, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

783337/1000000: episode: 1514, duration: 1.996s, episode steps: 458, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

783873/1000000: episode: 1515, duration: 2.329s, episode steps: 536, steps per

second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

784249/1000000: episode: 1516, duration: 1.643s, episode steps: 376, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

784592/1000000: episode: 1517, duration: 1.496s, episode steps: 343, steps per second: 229, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

784895/1000000: episode: 1518, duration: 1.324s, episode steps: 303, steps per second: 229, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.637 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

785448/1000000: episode: 1519, duration: 2.442s, episode steps: 553, steps per second: 226, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

785976/1000000: episode: 1520, duration: 2.328s, episode steps: 528, steps per second: 227, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

786439/1000000: episode: 1521, duration: 2.042s, episode steps: 463, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

787005/1000000: episode: 1522, duration: 2.462s, episode steps: 566, steps per second: 230, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

787423/1000000: episode: 1523, duration: 1.824s, episode steps: 418, steps per second: 229, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

787709/1000000: episode: 1524, duration: 1.264s, episode steps: 286, steps per second: 226, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

788122/1000000: episode: 1525, duration: 1.823s, episode steps: 413, steps per second: 227, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

788761/1000000: episode: 1526, duration: 2.824s, episode steps: 639, steps per second: 226, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

789241/1000000: episode: 1527, duration: 2.107s, episode steps: 480, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

789772/1000000: episode: 1528, duration: 2.316s, episode steps: 531, steps per second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

790218/1000000: episode: 1529, duration: 1.966s, episode steps: 446, steps per second: 227, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

790600/1000000: episode: 1530, duration: 1.700s, episode steps: 382, steps per second: 225, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

791123/1000000: episode: 1531, duration: 2.356s, episode steps: 523, steps per

second: 222, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

791724/1000000: episode: 1532, duration: 2.695s, episode steps: 601, steps per second: 223, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

792764/1000000: episode: 1533, duration: 4.482s, episode steps: 1040, steps per second: 232, episode reward: 28.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

793280/1000000: episode: 1534, duration: 2.254s, episode steps: 516, steps per second: 229, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

793617/1000000: episode: 1535, duration: 1.480s, episode steps: 337, steps per second: 228, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

794150/1000000: episode: 1536, duration: 2.345s, episode steps: 533, steps per second: 227, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

794756/1000000: episode: 1537, duration: 2.655s, episode steps: 606, steps per second: 228, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

795236/1000000: episode: 1538, duration: 2.080s, episode steps: 480, steps per second: 231, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

795782/1000000: episode: 1539, duration: 2.388s, episode steps: 546, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

796309/1000000: episode: 1540, duration: 2.306s, episode steps: 527, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

796943/1000000: episode: 1541, duration: 23.115s, episode steps: 634, steps per second: 27, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.015310, mae: 1.906432, mean_q: 2.317098, mean_eps: 0.987693

797406/1000000: episode: 1542, duration: 20.800s, episode steps: 463, steps per second: 22, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.620 [0.000, 5.000], loss: 0.020173, mae: 1.912412, mean_q: 2.319389, mean_eps: 0.987110

797960/1000000: episode: 1543, duration: 24.723s, episode steps: 554, steps per second: 22, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: 0.017949, mae: 1.861069, mean_q: 2.258477, mean_eps: 0.986500

798607/1000000: episode: 1544, duration: 28.960s, episode steps: 647, steps per second: 22, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.020020, mae: 1.889307, mean_q: 2.292850, mean_eps: 0.985780

799705/1000000: episode: 1545, duration: 49.069s, episode steps: 1098, steps per second: 22, episode reward: 19.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.017085, mae: 1.894373, mean_q:

2.298075, mean_eps: 0.984731

800201/1000000: episode: 1546, duration: 22.049s, episode steps: 496, steps per second: 22, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.611 [0.000, 5.000], loss: 0.019383, mae: 1.861568, mean_q: 2.259765, mean_eps: 0.983774

800555/1000000: episode: 1547, duration: 15.739s, episode steps: 354, steps per second: 22, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: 0.015515, mae: 1.882093, mean_q: 2.286412, mean_eps: 0.983265

801085/1000000: episode: 1548, duration: 23.639s, episode steps: 530, steps per second: 22, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.015847, mae: 1.900621, mean_q: 2.308177, mean_eps: 0.982734

801827/1000000: episode: 1549, duration: 33.054s, episode steps: 742, steps per second: 22, episode reward: 21.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.014304, mae: 1.904288, mean_q: 2.311649, mean_eps: 0.981971

802123/1000000: episode: 1550, duration: 13.255s, episode steps: 296, steps per second: 22, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: 0.013983, mae: 1.902053, mean_q: 2.313228, mean_eps: 0.981350

802425/1000000: episode: 1551, duration: 13.623s, episode steps: 302, steps per second: 22, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: 0.022882, mae: 1.928009, mean_q: 2.337939, mean_eps: 0.980990

803184/1000000: episode: 1552, duration: 33.843s, episode steps: 759, steps per second: 22, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.017314, mae: 1.944288, mean_q: 2.360401, mean_eps: 0.980354

803607/1000000: episode: 1553, duration: 18.887s, episode steps: 423, steps per second: 22, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: 0.014877, mae: 1.907160, mean_q: 2.319753, mean_eps: 0.979646

804096/1000000: episode: 1554, duration: 21.910s, episode steps: 489, steps per second: 22, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.317 [0.000, 5.000], loss: 0.020467, mae: 1.935400, mean_q: 2.353159, mean_eps: 0.979098

804517/1000000: episode: 1555, duration: 18.936s, episode steps: 421, steps per second: 22, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: 0.018743, mae: 1.931333, mean_q: 2.346708, mean_eps: 0.978551

804932/1000000: episode: 1556, duration: 18.423s, episode steps: 415, steps per second: 23, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: 0.016691, mae: 1.913538, mean_q: 2.324349, mean_eps: 0.978050

805496/1000000: episode: 1557, duration: 25.422s, episode steps: 564, steps per second: 22, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: 0.020706, mae: 1.946017, mean_q: 2.362634,

mean_eps: 0.977464

805874/1000000: episode: 1558, duration: 16.976s, episode steps: 378, steps per second: 22, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.017838, mae: 1.907161, mean_q: 2.316010, mean_eps: 0.976898

806474/1000000: episode: 1559, duration: 26.685s, episode steps: 600, steps per second: 22, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.017882, mae: 1.925475, mean_q: 2.338496, mean_eps: 0.976310

807167/1000000: episode: 1560, duration: 30.816s, episode steps: 693, steps per second: 22, episode reward: 21.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.017580, mae: 1.905406, mean_q: 2.314700, mean_eps: 0.975534

807782/1000000: episode: 1561, duration: 27.347s, episode steps: 615, steps per second: 22, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.020079, mae: 1.913990, mean_q: 2.323107, mean_eps: 0.974750

808519/1000000: episode: 1562, duration: 32.754s, episode steps: 737, steps per second: 23, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.016146, mae: 1.921848, mean_q: 2.332572, mean_eps: 0.973938

809224/1000000: episode: 1563, duration: 31.380s, episode steps: 705, steps per second: 22, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: 0.018157, mae: 1.920906, mean_q: 2.332161, mean_eps: 0.973074

809903/1000000: episode: 1564, duration: 30.309s, episode steps: 679, steps per second: 22, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: 0.018590, mae: 1.920096, mean_q: 2.330207, mean_eps: 0.972244

810272/1000000: episode: 1565, duration: 16.687s, episode steps: 369, steps per second: 22, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.016559, mae: 1.923510, mean_q: 2.337406, mean_eps: 0.971615

810777/1000000: episode: 1566, duration: 22.680s, episode steps: 505, steps per second: 22, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: 0.018389, mae: 1.918642, mean_q: 2.333003, mean_eps: 0.971090

811270/1000000: episode: 1567, duration: 21.900s, episode steps: 493, steps per second: 23, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.018233, mae: 1.919781, mean_q: 2.329635, mean_eps: 0.970490

811645/1000000: episode: 1568, duration: 16.223s, episode steps: 375, steps per second: 23, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: 0.018776, mae: 1.938404, mean_q: 2.354982, mean_eps: 0.969969

812362/1000000: episode: 1569, duration: 29.787s, episode steps: 717, steps per second: 24, episode reward: 22.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.016983, mae: 1.963888, mean_q: 2.383975,

mean_eps: 0.969314

812659/1000000: episode: 1570, duration: 12.416s, episode steps: 297, steps per second: 24, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.667 [0.000, 5.000], loss: 0.018680, mae: 1.956478, mean_q: 2.371756, mean_eps: 0.968706

813214/1000000: episode: 1571, duration: 23.128s, episode steps: 555, steps per second: 24, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: 0.018436, mae: 1.966646, mean_q: 2.386472, mean_eps: 0.968195

813953/1000000: episode: 1572, duration: 30.919s, episode steps: 739, steps per second: 24, episode reward: 19.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.020334, mae: 1.951125, mean_q: 2.366804, mean_eps: 0.967418

814364/1000000: episode: 1573, duration: 17.236s, episode steps: 411, steps per second: 24, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.015555, mae: 1.965918, mean_q: 2.386671, mean_eps: 0.966729

814813/1000000: episode: 1574, duration: 18.873s, episode steps: 449, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: 0.015481, mae: 1.941035, mean_q: 2.356385, mean_eps: 0.966213

815519/1000000: episode: 1575, duration: 29.291s, episode steps: 706, steps per second: 24, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.020607, mae: 1.963324, mean_q: 2.381957, mean_eps: 0.965519

816059/1000000: episode: 1576, duration: 22.533s, episode steps: 540, steps per second: 24, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.016196, mae: 1.964587, mean_q: 2.383802, mean_eps: 0.964773

816357/1000000: episode: 1577, duration: 12.504s, episode steps: 298, steps per second: 24, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: 0.020065, mae: 1.962711, mean_q: 2.383634, mean_eps: 0.964269

816879/1000000: episode: 1578, duration: 21.726s, episode steps: 522, steps per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.017959, mae: 1.979853, mean_q: 2.404765, mean_eps: 0.963777

817467/1000000: episode: 1579, duration: 24.595s, episode steps: 588, steps per second: 24, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.018837, mae: 1.981284, mean_q: 2.401831, mean_eps: 0.963112

817863/1000000: episode: 1580, duration: 16.522s, episode steps: 396, steps per second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.016950, mae: 1.977249, mean_q: 2.398354, mean_eps: 0.962522

818343/1000000: episode: 1581, duration: 20.054s, episode steps: 480, steps per second: 24, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.660 [0.000, 5.000], loss: 0.017988, mae: 1.968897, mean_q: 2.389864,

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mean_eps: 0.961996
818941/1000000: episode: 1582, duration: 25.038s, episode steps: 598, steps per
second: 24, episode reward: 22.000, mean reward: 0.037 [ 0.000, 1.000], mean
action: 2.602 [0.000, 5.000], loss: 0.021030, mae: 1.980758, mean_q: 2.401322,
mean_eps: 0.961348

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

819404/1000000: episode: 1583, duration: 11.639s, episode steps: 463, steps per
second: 40, episode reward: 11.000, mean reward: 0.024 [ 0.000, 1.000], mean
action: 2.443 [0.000, 5.000], loss: 0.016531, mae: 1.991800, mean_q: 2.412806,
mean_eps: 0.960834
819928/1000000: episode: 1584, duration: 2.285s, episode steps: 524, steps per
second: 229, episode reward: 14.000, mean reward: 0.027 [ 0.000, 1.000], mean
action: 2.586 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
820573/1000000: episode: 1585, duration: 2.833s, episode steps: 645, steps per
second: 228, episode reward: 14.000, mean reward: 0.022 [ 0.000, 1.000], mean
action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
821090/1000000: episode: 1586, duration: 2.297s, episode steps: 517, steps per
second: 225, episode reward: 10.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
821378/1000000: episode: 1587, duration: 1.269s, episode steps: 288, steps per
second: 227, episode reward: 6.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
821739/1000000: episode: 1588, duration: 1.595s, episode steps: 361, steps per
second: 226, episode reward: 7.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
822134/1000000: episode: 1589, duration: 1.743s, episode steps: 395, steps per
second: 227, episode reward: 9.000, mean reward: 0.023 [ 0.000, 1.000], mean
action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
822631/1000000: episode: 1590, duration: 2.174s, episode steps: 497, steps per
second: 229, episode reward: 10.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
822971/1000000: episode: 1591, duration: 1.488s, episode steps: 340, steps per
second: 229, episode reward: 6.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
823551/1000000: episode: 1592, duration: 2.535s, episode steps: 580, steps per
second: 229, episode reward: 8.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
823942/1000000: episode: 1593, duration: 1.757s, episode steps: 391, steps per
second: 223, episode reward: 7.000, mean reward: 0.018 [ 0.000, 1.000], mean
action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
824209/1000000: episode: 1594, duration: 1.181s, episode steps: 267, steps per
second: 226, episode reward: 3.000, mean reward: 0.011 [ 0.000, 1.000], mean
action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
824732/1000000: episode: 1595, duration: 2.283s, episode steps: 523, steps per
second: 229, episode reward: 9.000, mean reward: 0.017 [ 0.000, 1.000], mean

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action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 825317/1000000: episode: 1596, duration: 2.545s, episode steps: 585, steps per second: 230, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 825777/1000000: episode: 1597, duration: 2.011s, episode steps: 460, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 826400/1000000: episode: 1598, duration: 2.735s, episode steps: 623, steps per second: 228, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 826829/1000000: episode: 1599, duration: 1.925s, episode steps: 429, steps per second: 223, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 827433/1000000: episode: 1600, duration: 2.644s, episode steps: 604, steps per second: 228, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 dueling/version_1.0/pesos/dqn_v_1.0_weights_1600_episodes.h5f

827950/1000000: episode: 1601, duration: 2.362s, episode steps: 517, steps per second: 219, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 828564/1000000: episode: 1602, duration: 2.661s, episode steps: 614, steps per second: 231, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 829092/1000000: episode: 1603, duration: 2.310s, episode steps: 528, steps per second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.350 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 829728/1000000: episode: 1604, duration: 2.832s, episode steps: 636, steps per second: 225, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 830176/1000000: episode: 1605, duration: 1.959s, episode steps: 448, steps per second: 229, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 830689/1000000: episode: 1606, duration: 2.250s, episode steps: 513, steps per second: 228, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 831197/1000000: episode: 1607, duration: 2.211s, episode steps: 508, steps per second: 230, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.384 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 831623/1000000: episode: 1608, duration: 1.854s, episode steps: 426, steps per second: 230, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 832354/1000000: episode: 1609, duration: 3.218s, episode steps: 731, steps per second: 227, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 832876/1000000: episode: 1610, duration: 2.294s, episode steps: 522, steps per

second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.467 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

833437/1000000: episode: 1611, duration: 2.445s, episode steps: 561, steps per second: 229, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

833863/1000000: episode: 1612, duration: 1.849s, episode steps: 426, steps per second: 230, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

834197/1000000: episode: 1613, duration: 1.469s, episode steps: 334, steps per second: 227, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

834576/1000000: episode: 1614, duration: 1.675s, episode steps: 379, steps per second: 226, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

835288/1000000: episode: 1615, duration: 3.128s, episode steps: 712, steps per second: 228, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

835899/1000000: episode: 1616, duration: 2.701s, episode steps: 611, steps per second: 226, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

836644/1000000: episode: 1617, duration: 3.227s, episode steps: 745, steps per second: 231, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.389 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

837007/1000000: episode: 1618, duration: 1.604s, episode steps: 363, steps per second: 226, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

837501/1000000: episode: 1619, duration: 2.179s, episode steps: 494, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

838044/1000000: episode: 1620, duration: 2.383s, episode steps: 543, steps per second: 228, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

838329/1000000: episode: 1621, duration: 1.277s, episode steps: 285, steps per second: 223, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

838922/1000000: episode: 1622, duration: 2.570s, episode steps: 593, steps per second: 231, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

839257/1000000: episode: 1623, duration: 1.463s, episode steps: 335, steps per second: 229, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

839791/1000000: episode: 1624, duration: 2.330s, episode steps: 534, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

840323/1000000: episode: 1625, duration: 2.324s, episode steps: 532, steps per second: 229, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

840812/1000000: episode: 1626, duration: 2.144s, episode steps: 489, steps per

second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

841251/1000000: episode: 1627, duration: 1.959s, episode steps: 439, steps per second: 224, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

841534/1000000: episode: 1628, duration: 1.252s, episode steps: 283, steps per second: 226, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

842124/1000000: episode: 1629, duration: 2.589s, episode steps: 590, steps per second: 228, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

842584/1000000: episode: 1630, duration: 2.010s, episode steps: 460, steps per second: 229, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

843428/1000000: episode: 1631, duration: 3.646s, episode steps: 844, steps per second: 231, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

844178/1000000: episode: 1632, duration: 3.299s, episode steps: 750, steps per second: 227, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

844724/1000000: episode: 1633, duration: 2.401s, episode steps: 546, steps per second: 227, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

845254/1000000: episode: 1634, duration: 2.322s, episode steps: 530, steps per second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

845626/1000000: episode: 1635, duration: 1.632s, episode steps: 372, steps per second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

845936/1000000: episode: 1636, duration: 1.348s, episode steps: 310, steps per second: 230, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

846610/1000000: episode: 1637, duration: 2.959s, episode steps: 674, steps per second: 228, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

846955/1000000: episode: 1638, duration: 1.534s, episode steps: 345, steps per second: 225, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

847450/1000000: episode: 1639, duration: 2.168s, episode steps: 495, steps per second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

847843/1000000: episode: 1640, duration: 1.718s, episode steps: 393, steps per second: 229, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

848327/1000000: episode: 1641, duration: 2.108s, episode steps: 484, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

848972/1000000: episode: 1642, duration: 2.797s, episode steps: 645, steps per

second: 231, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

849423/1000000: episode: 1643, duration: 1.995s, episode steps: 451, steps per second: 226, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

849744/1000000: episode: 1644, duration: 1.425s, episode steps: 321, steps per second: 225, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

850480/1000000: episode: 1645, duration: 3.212s, episode steps: 736, steps per second: 229, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

850959/1000000: episode: 1646, duration: 2.086s, episode steps: 479, steps per second: 230, episode reward: 14.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.722 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

851349/1000000: episode: 1647, duration: 1.702s, episode steps: 390, steps per second: 229, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

851872/1000000: episode: 1648, duration: 2.274s, episode steps: 523, steps per second: 230, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

852355/1000000: episode: 1649, duration: 2.136s, episode steps: 483, steps per second: 226, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

852854/1000000: episode: 1650, duration: 2.230s, episode steps: 499, steps per second: 224, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

853145/1000000: episode: 1651, duration: 1.284s, episode steps: 291, steps per second: 227, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

853692/1000000: episode: 1652, duration: 2.387s, episode steps: 547, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

854637/1000000: episode: 1653, duration: 4.079s, episode steps: 945, steps per second: 232, episode reward: 12.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

855099/1000000: episode: 1654, duration: 2.018s, episode steps: 462, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

855816/1000000: episode: 1655, duration: 3.158s, episode steps: 717, steps per second: 227, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

856248/1000000: episode: 1656, duration: 1.888s, episode steps: 432, steps per second: 229, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

856775/1000000: episode: 1657, duration: 2.289s, episode steps: 527, steps per second: 230, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

857279/1000000: episode: 1658, duration: 2.213s, episode steps: 504, steps per

second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

857723/1000000: episode: 1659, duration: 1.942s, episode steps: 444, steps per second: 229, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

858149/1000000: episode: 1660, duration: 1.877s, episode steps: 426, steps per second: 227, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

858483/1000000: episode: 1661, duration: 1.481s, episode steps: 334, steps per second: 225, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

858945/1000000: episode: 1662, duration: 2.052s, episode steps: 462, steps per second: 225, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

859548/1000000: episode: 1663, duration: 2.627s, episode steps: 603, steps per second: 230, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

860036/1000000: episode: 1664, duration: 2.127s, episode steps: 488, steps per second: 229, episode reward: 14.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

860415/1000000: episode: 1665, duration: 1.673s, episode steps: 379, steps per second: 227, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

861013/1000000: episode: 1666, duration: 2.628s, episode steps: 598, steps per second: 228, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

861542/1000000: episode: 1667, duration: 2.331s, episode steps: 529, steps per second: 227, episode reward: 5.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

862116/1000000: episode: 1668, duration: 8.071s, episode steps: 574, steps per second: 71, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: 0.018343, mae: 1.967492, mean_q: 2.381306, mean_eps: 0.987911

862833/1000000: episode: 1669, duration: 30.009s, episode steps: 717, steps per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.019467, mae: 2.026292, mean_q: 2.454592, mean_eps: 0.987393

863197/1000000: episode: 1670, duration: 15.246s, episode steps: 364, steps per second: 24, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.018804, mae: 2.001396, mean_q: 2.423243, mean_eps: 0.986742

863495/1000000: episode: 1671, duration: 12.447s, episode steps: 298, steps per second: 24, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.025017, mae: 2.029159, mean_q: 2.455178, mean_eps: 0.986346

864155/1000000: episode: 1672, duration: 27.521s, episode steps: 660, steps per second: 24, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.019839, mae: 2.009279, mean_q: 2.432572,

mean_eps: 0.985773

864668/1000000: episode: 1673, duration: 21.597s, episode steps: 513, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.022937, mae: 2.025327, mean_q: 2.453197, mean_eps: 0.985070

864961/1000000: episode: 1674, duration: 12.496s, episode steps: 293, steps per second: 23, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.019141, mae: 2.031096, mean_q: 2.463026, mean_eps: 0.984585

865550/1000000: episode: 1675, duration: 24.745s, episode steps: 589, steps per second: 24, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.447 [0.000, 5.000], loss: 0.022471, mae: 2.009722, mean_q: 2.437263, mean_eps: 0.984054

866077/1000000: episode: 1676, duration: 22.162s, episode steps: 527, steps per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.018879, mae: 2.016556, mean_q: 2.445668, mean_eps: 0.983385

866505/1000000: episode: 1677, duration: 17.737s, episode steps: 428, steps per second: 24, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: 0.020493, mae: 2.013337, mean_q: 2.437107, mean_eps: 0.982811

867095/1000000: episode: 1678, duration: 24.581s, episode steps: 590, steps per second: 24, episode reward: 14.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.020166, mae: 2.036663, mean_q: 2.470830, mean_eps: 0.982202

867697/1000000: episode: 1679, duration: 25.204s, episode steps: 602, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.021751, mae: 2.068312, mean_q: 2.509497, mean_eps: 0.981486

868230/1000000: episode: 1680, duration: 22.193s, episode steps: 533, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.021141, mae: 2.069252, mean_q: 2.511358, mean_eps: 0.980805

868852/1000000: episode: 1681, duration: 26.103s, episode steps: 622, steps per second: 24, episode reward: 16.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: 0.019669, mae: 2.040626, mean_q: 2.474694, mean_eps: 0.980114

869339/1000000: episode: 1682, duration: 20.297s, episode steps: 487, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.626 [0.000, 5.000], loss: 0.019014, mae: 2.023216, mean_q: 2.456319, mean_eps: 0.979449

869738/1000000: episode: 1683, duration: 16.666s, episode steps: 399, steps per second: 24, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.017878, mae: 2.065126, mean_q: 2.503444, mean_eps: 0.978916

870248/1000000: episode: 1684, duration: 21.404s, episode steps: 510, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: 0.020387, mae: 2.055836, mean_q: 2.493459,

mean_eps: 0.978371

870920/1000000: episode: 1685, duration: 28.373s, episode steps: 672, steps per second: 24, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: 0.025387, mae: 2.034632, mean_q: 2.470345, mean_eps: 0.977663

871427/1000000: episode: 1686, duration: 21.245s, episode steps: 507, steps per second: 24, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.023372, mae: 2.054880, mean_q: 2.494999, mean_eps: 0.976955

872344/1000000: episode: 1687, duration: 38.281s, episode steps: 917, steps per second: 24, episode reward: 27.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.020935, mae: 2.068686, mean_q: 2.512146, mean_eps: 0.976101

873030/1000000: episode: 1688, duration: 28.567s, episode steps: 686, steps per second: 24, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.021077, mae: 2.082217, mean_q: 2.525115, mean_eps: 0.975138

873429/1000000: episode: 1689, duration: 16.670s, episode steps: 399, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.022665, mae: 2.086098, mean_q: 2.530157, mean_eps: 0.974486

873707/1000000: episode: 1690, duration: 11.489s, episode steps: 278, steps per second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.024975, mae: 2.101306, mean_q: 2.542838, mean_eps: 0.974080

874083/1000000: episode: 1691, duration: 15.828s, episode steps: 376, steps per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.023578, mae: 2.078938, mean_q: 2.519081, mean_eps: 0.973689

874594/1000000: episode: 1692, duration: 21.507s, episode steps: 511, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: 0.022759, mae: 2.075915, mean_q: 2.515035, mean_eps: 0.973156

875187/1000000: episode: 1693, duration: 24.730s, episode steps: 593, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.023931, mae: 2.083441, mean_q: 2.529805, mean_eps: 0.972494

875453/1000000: episode: 1694, duration: 11.106s, episode steps: 266, steps per second: 24, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: 0.018077, mae: 2.067562, mean_q: 2.506803, mean_eps: 0.971978

875780/1000000: episode: 1695, duration: 13.619s, episode steps: 327, steps per second: 24, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.364 [0.000, 5.000], loss: 0.019201, mae: 2.080396, mean_q: 2.521255, mean_eps: 0.971622

876587/1000000: episode: 1696, duration: 33.474s, episode steps: 807, steps per second: 24, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.022783, mae: 2.073530, mean_q: 2.512042,

mean_eps: 0.970943

877144/1000000: episode: 1697, duration: 23.503s, episode steps: 557, steps per second: 24, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: 0.020674, mae: 2.081003, mean_q: 2.523843, mean_eps: 0.970125

877493/1000000: episode: 1698, duration: 14.680s, episode steps: 349, steps per second: 24, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: 0.023089, mae: 2.099412, mean_q: 2.545430, mean_eps: 0.969580

878016/1000000: episode: 1699, duration: 21.867s, episode steps: 523, steps per second: 24, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: 0.022379, mae: 2.117030, mean_q: 2.567388, mean_eps: 0.969057

878418/1000000: episode: 1700, duration: 16.955s, episode steps: 402, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.020612, mae: 2.100805, mean_q: 2.548569, mean_eps: 0.968502

878795/1000000: episode: 1701, duration: 15.698s, episode steps: 377, steps per second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.026446, mae: 2.109414, mean_q: 2.556733, mean_eps: 0.968034

879294/1000000: episode: 1702, duration: 20.908s, episode steps: 499, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.017252, mae: 2.128887, mean_q: 2.580565, mean_eps: 0.967509

879811/1000000: episode: 1703, duration: 21.436s, episode steps: 517, steps per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: 0.020349, mae: 2.098607, mean_q: 2.544242, mean_eps: 0.966899

880332/1000000: episode: 1704, duration: 21.807s, episode steps: 521, steps per second: 24, episode reward: 15.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.019507, mae: 2.098256, mean_q: 2.543071, mean_eps: 0.966278

881041/1000000: episode: 1705, duration: 29.745s, episode steps: 709, steps per second: 24, episode reward: 21.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: 0.019252, mae: 2.114583, mean_q: 2.563256, mean_eps: 0.965538

881688/1000000: episode: 1706, duration: 27.070s, episode steps: 647, steps per second: 24, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.023680, mae: 2.108859, mean_q: 2.555503, mean_eps: 0.964725

882077/1000000: episode: 1707, duration: 16.332s, episode steps: 389, steps per second: 24, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: 0.023394, mae: 2.093233, mean_q: 2.536147, mean_eps: 0.964103

882356/1000000: episode: 1708, duration: 11.616s, episode steps: 279, steps per second: 24, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.731 [0.000, 5.000], loss: 0.020612, mae: 2.103078, mean_q: 2.551520,

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mean_eps: 0.963702
882825/1000000: episode: 1709, duration: 19.763s, episode steps: 469, steps per
second: 24, episode reward: 10.000, mean reward: 0.021 [ 0.000, 1.000], mean
action: 2.546 [0.000, 5.000], loss: 0.020469, mae: 2.103374, mean_q: 2.551188,
mean_eps: 0.963254
883461/1000000: episode: 1710, duration: 26.427s, episode steps: 636, steps per
second: 24, episode reward: 10.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.638 [0.000, 5.000], loss: 0.027949, mae: 2.117774, mean_q: 2.567070,
mean_eps: 0.962589
884206/1000000: episode: 1711, duration: 30.826s, episode steps: 745, steps per
second: 24, episode reward: 12.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.463 [0.000, 5.000], loss: 0.025220, mae: 2.144778, mean_q: 2.599884,
mean_eps: 0.961761
884516/1000000: episode: 1712, duration: 13.015s, episode steps: 310, steps per
second: 24, episode reward: 5.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.568 [0.000, 5.000], loss: 0.020971, mae: 2.103731, mean_q: 2.550661,
mean_eps: 0.961130

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

885034/1000000: episode: 1713, duration: 10.529s, episode steps: 518, steps per
second: 49, episode reward: 9.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.384 [0.000, 5.000], loss: 0.026935, mae: 2.132937, mean_q: 2.585646,
mean_eps: 0.960813
885525/1000000: episode: 1714, duration: 2.147s, episode steps: 491, steps per
second: 229, episode reward: 7.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
885973/1000000: episode: 1715, duration: 1.967s, episode steps: 448, steps per
second: 228, episode reward: 7.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
886279/1000000: episode: 1716, duration: 1.357s, episode steps: 306, steps per
second: 226, episode reward: 4.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
886658/1000000: episode: 1717, duration: 1.677s, episode steps: 379, steps per
second: 226, episode reward: 6.000, mean reward: 0.016 [ 0.000, 1.000], mean
action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
887136/1000000: episode: 1718, duration: 2.103s, episode steps: 478, steps per
second: 227, episode reward: 9.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
887513/1000000: episode: 1719, duration: 1.681s, episode steps: 377, steps per
second: 224, episode reward: 7.000, mean reward: 0.019 [ 0.000, 1.000], mean
action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
888163/1000000: episode: 1720, duration: 2.846s, episode steps: 650, steps per
second: 228, episode reward: 16.000, mean reward: 0.025 [ 0.000, 1.000], mean
action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
888497/1000000: episode: 1721, duration: 1.487s, episode steps: 334, steps per
second: 225, episode reward: 5.000, mean reward: 0.015 [ 0.000, 1.000], mean

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action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 889390/1000000: episode: 1722, duration: 3.988s, episode steps: 893, steps per second: 224, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 890281/1000000: episode: 1723, duration: 3.911s, episode steps: 891, steps per second: 228, episode reward: 20.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 890994/1000000: episode: 1724, duration: 3.093s, episode steps: 713, steps per second: 231, episode reward: 22.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 891465/1000000: episode: 1725, duration: 2.046s, episode steps: 471, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 892033/1000000: episode: 1726, duration: 2.509s, episode steps: 568, steps per second: 226, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 892628/1000000: episode: 1727, duration: 2.631s, episode steps: 595, steps per second: 226, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 892932/1000000: episode: 1728, duration: 1.332s, episode steps: 304, steps per second: 228, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 893316/1000000: episode: 1729, duration: 1.688s, episode steps: 384, steps per second: 227, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.654 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 894103/1000000: episode: 1730, duration: 3.418s, episode steps: 787, steps per second: 230, episode reward: 19.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 894676/1000000: episode: 1731, duration: 2.516s, episode steps: 573, steps per second: 228, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 895279/1000000: episode: 1732, duration: 2.658s, episode steps: 603, steps per second: 227, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 895709/1000000: episode: 1733, duration: 1.883s, episode steps: 430, steps per second: 228, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 896106/1000000: episode: 1734, duration: 1.732s, episode steps: 397, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.390 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 896502/1000000: episode: 1735, duration: 1.735s, episode steps: 396, steps per second: 228, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 897104/1000000: episode: 1736, duration: 2.621s, episode steps: 602, steps per second: 230, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 897837/1000000: episode: 1737, duration: 3.244s, episode steps: 733, steps per second: 226, episode reward: 6.000, mean reward: 0.008 [0.000, 1.000], mean

action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 898097/1000000: episode: 1738, duration: 1.161s, episode steps: 260, steps per second: 224, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 898469/1000000: episode: 1739, duration: 1.628s, episode steps: 372, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 899015/1000000: episode: 1740, duration: 2.380s, episode steps: 546, steps per second: 229, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 899478/1000000: episode: 1741, duration: 2.010s, episode steps: 463, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 900080/1000000: episode: 1742, duration: 2.642s, episode steps: 602, steps per second: 228, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 900786/1000000: episode: 1743, duration: 3.112s, episode steps: 706, steps per second: 227, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.336 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 901436/1000000: episode: 1744, duration: 2.825s, episode steps: 650, steps per second: 230, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 901947/1000000: episode: 1745, duration: 2.215s, episode steps: 511, steps per second: 231, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 902325/1000000: episode: 1746, duration: 1.648s, episode steps: 378, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 902920/1000000: episode: 1747, duration: 2.610s, episode steps: 595, steps per second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 903561/1000000: episode: 1748, duration: 2.826s, episode steps: 641, steps per second: 227, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 904093/1000000: episode: 1749, duration: 2.344s, episode steps: 532, steps per second: 227, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 904668/1000000: episode: 1750, duration: 2.497s, episode steps: 575, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 905282/1000000: episode: 1751, duration: 2.653s, episode steps: 614, steps per second: 231, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 905744/1000000: episode: 1752, duration: 2.003s, episode steps: 462, steps per second: 231, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.714 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 906259/1000000: episode: 1753, duration: 2.261s, episode steps: 515, steps per second: 228, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 906677/1000000: episode: 1754, duration: 1.873s, episode steps: 418, steps per second: 223, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 907292/1000000: episode: 1755, duration: 2.683s, episode steps: 615, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 907787/1000000: episode: 1756, duration: 2.177s, episode steps: 495, steps per second: 227, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.616 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 908156/1000000: episode: 1757, duration: 1.623s, episode steps: 369, steps per second: 227, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 908586/1000000: episode: 1758, duration: 1.867s, episode steps: 430, steps per second: 230, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 909187/1000000: episode: 1759, duration: 2.630s, episode steps: 601, steps per second: 228, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 909586/1000000: episode: 1760, duration: 1.769s, episode steps: 399, steps per second: 226, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

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 dueling/version_1.0/pesos/dqn_v_1.0_weights_1760_episodes.h5f

910172/1000000: episode: 1761, duration: 2.664s, episode steps: 586, steps per second: 220, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 910961/1000000: episode: 1762, duration: 3.433s, episode steps: 789, steps per second: 230, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.577 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 911694/1000000: episode: 1763, duration: 3.186s, episode steps: 733, steps per second: 230, episode reward: 19.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 912171/1000000: episode: 1764, duration: 2.095s, episode steps: 477, steps per second: 228, episode reward: 15.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 912881/1000000: episode: 1765, duration: 3.145s, episode steps: 710, steps per second: 226, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 913335/1000000: episode: 1766, duration: 2.005s, episode steps: 454, steps per second: 226, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 913898/1000000: episode: 1767, duration: 2.460s, episode steps: 563, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 914444/1000000: episode: 1768, duration: 2.383s, episode steps: 546, steps per

second: 229, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.429 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

914967/1000000: episode: 1769, duration: 2.300s, episode steps: 523, steps per second: 227, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

915546/1000000: episode: 1770, duration: 2.577s, episode steps: 579, steps per second: 225, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

916038/1000000: episode: 1771, duration: 2.157s, episode steps: 492, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

916994/1000000: episode: 1772, duration: 4.131s, episode steps: 956, steps per second: 231, episode reward: 18.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

917508/1000000: episode: 1773, duration: 2.243s, episode steps: 514, steps per second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

917886/1000000: episode: 1774, duration: 1.677s, episode steps: 378, steps per second: 225, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.677 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

918468/1000000: episode: 1775, duration: 2.583s, episode steps: 582, steps per second: 225, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

918844/1000000: episode: 1776, duration: 1.650s, episode steps: 376, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

919786/1000000: episode: 1777, duration: 4.084s, episode steps: 942, steps per second: 231, episode reward: 14.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

920211/1000000: episode: 1778, duration: 1.855s, episode steps: 425, steps per second: 229, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

920770/1000000: episode: 1779, duration: 2.461s, episode steps: 559, steps per second: 227, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

921063/1000000: episode: 1780, duration: 1.328s, episode steps: 293, steps per second: 221, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.300 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

921467/1000000: episode: 1781, duration: 1.776s, episode steps: 404, steps per second: 227, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

921870/1000000: episode: 1782, duration: 1.772s, episode steps: 403, steps per second: 227, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

922646/1000000: episode: 1783, duration: 3.405s, episode steps: 776, steps per second: 228, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

923175/1000000: episode: 1784, duration: 2.322s, episode steps: 529, steps per

second: 228, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

924250/1000000: episode: 1785, duration: 4.717s, episode steps: 1075, steps per second: 228, episode reward: 9.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

924644/1000000: episode: 1786, duration: 1.729s, episode steps: 394, steps per second: 228, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

925037/1000000: episode: 1787, duration: 1.720s, episode steps: 393, steps per second: 229, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.746 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

925523/1000000: episode: 1788, duration: 2.128s, episode steps: 486, steps per second: 228, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

926158/1000000: episode: 1789, duration: 2.794s, episode steps: 635, steps per second: 227, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

926648/1000000: episode: 1790, duration: 2.183s, episode steps: 490, steps per second: 225, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.600 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

926953/1000000: episode: 1791, duration: 1.356s, episode steps: 305, steps per second: 225, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

927509/1000000: episode: 1792, duration: 2.601s, episode steps: 556, steps per second: 214, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.018822, mae: 2.053979, mean_q: 2.487641, mean_eps: 0.987995

927915/1000000: episode: 1793, duration: 16.891s, episode steps: 406, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.022992, mae: 2.173783, mean_q: 2.631231, mean_eps: 0.987750

928474/1000000: episode: 1794, duration: 23.308s, episode steps: 559, steps per second: 24, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: 0.025092, mae: 2.191417, mean_q: 2.656228, mean_eps: 0.987172

928969/1000000: episode: 1795, duration: 20.756s, episode steps: 495, steps per second: 24, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.024824, mae: 2.170847, mean_q: 2.628741, mean_eps: 0.986538

929291/1000000: episode: 1796, duration: 13.391s, episode steps: 322, steps per second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: 0.020572, mae: 2.165316, mean_q: 2.625751, mean_eps: 0.986049

929571/1000000: episode: 1797, duration: 11.640s, episode steps: 280, steps per second: 24, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.264 [0.000, 5.000], loss: 0.021671, mae: 2.187549, mean_q: 2.650841, mean_eps: 0.985689

930367/1000000: episode: 1798, duration: 33.253s, episode steps: 796, steps per

second: 24, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: 0.019715, mae: 2.159348, mean_q: 2.616161, mean_eps: 0.985043

930887/1000000: episode: 1799, duration: 21.764s, episode steps: 520, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.020913, mae: 2.182469, mean_q: 2.644885, mean_eps: 0.984254

931667/1000000: episode: 1800, duration: 32.746s, episode steps: 780, steps per second: 24, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: 0.022819, mae: 2.137648, mean_q: 2.592362, mean_eps: 0.983474

932032/1000000: episode: 1801, duration: 15.419s, episode steps: 365, steps per second: 24, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: 0.021274, mae: 2.149517, mean_q: 2.609077, mean_eps: 0.982787

932411/1000000: episode: 1802, duration: 15.800s, episode steps: 379, steps per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: 0.020140, mae: 2.187346, mean_q: 2.654661, mean_eps: 0.982341

932788/1000000: episode: 1803, duration: 15.887s, episode steps: 377, steps per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.024817, mae: 2.154730, mean_q: 2.614235, mean_eps: 0.981887

933363/1000000: episode: 1804, duration: 24.089s, episode steps: 575, steps per second: 24, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: 0.026409, mae: 2.171081, mean_q: 2.634723, mean_eps: 0.981316

933742/1000000: episode: 1805, duration: 15.894s, episode steps: 379, steps per second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: 0.023286, mae: 2.170660, mean_q: 2.633645, mean_eps: 0.980742

934294/1000000: episode: 1806, duration: 23.061s, episode steps: 552, steps per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: 0.021955, mae: 2.156835, mean_q: 2.618242, mean_eps: 0.980183

934771/1000000: episode: 1807, duration: 19.906s, episode steps: 477, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.020566, mae: 2.164145, mean_q: 2.625071, mean_eps: 0.979566

935122/1000000: episode: 1808, duration: 14.578s, episode steps: 351, steps per second: 24, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: 0.024092, mae: 2.140336, mean_q: 2.597005, mean_eps: 0.979070

935817/1000000: episode: 1809, duration: 29.123s, episode steps: 695, steps per second: 24, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: 0.024199, mae: 2.183556, mean_q: 2.650503, mean_eps: 0.978441

936299/1000000: episode: 1810, duration: 20.066s, episode steps: 482, steps per

second: 24, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.685 [0.000, 5.000], loss: 0.025733, mae: 2.148501, mean_q: 2.607658, mean_eps: 0.977735

937161/1000000: episode: 1811, duration: 36.021s, episode steps: 862, steps per second: 24, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.022452, mae: 2.160265, mean_q: 2.623927, mean_eps: 0.976929

937565/1000000: episode: 1812, duration: 16.886s, episode steps: 404, steps per second: 24, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: 0.019779, mae: 2.150432, mean_q: 2.613532, mean_eps: 0.976168

937980/1000000: episode: 1813, duration: 17.265s, episode steps: 415, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: 0.024200, mae: 2.203815, mean_q: 2.676427, mean_eps: 0.975678

938926/1000000: episode: 1814, duration: 39.623s, episode steps: 946, steps per second: 24, episode reward: 21.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.026447, mae: 2.177648, mean_q: 2.645561, mean_eps: 0.974862

939486/1000000: episode: 1815, duration: 23.439s, episode steps: 560, steps per second: 24, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.024516, mae: 2.180597, mean_q: 2.649157, mean_eps: 0.973958

940083/1000000: episode: 1816, duration: 24.950s, episode steps: 597, steps per second: 24, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: 0.027289, mae: 2.210393, mean_q: 2.685134, mean_eps: 0.973264

940475/1000000: episode: 1817, duration: 16.327s, episode steps: 392, steps per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.332 [0.000, 5.000], loss: 0.022461, mae: 2.195125, mean_q: 2.666547, mean_eps: 0.972671

941035/1000000: episode: 1818, duration: 23.374s, episode steps: 560, steps per second: 24, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.025393, mae: 2.178154, mean_q: 2.645662, mean_eps: 0.972100

941601/1000000: episode: 1819, duration: 23.716s, episode steps: 566, steps per second: 24, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: 0.024495, mae: 2.204312, mean_q: 2.677901, mean_eps: 0.971423

942008/1000000: episode: 1820, duration: 17.208s, episode steps: 407, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: 0.024515, mae: 2.199551, mean_q: 2.673635, mean_eps: 0.970840

942516/1000000: episode: 1821, duration: 21.533s, episode steps: 508, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.661 [0.000, 5.000], loss: 0.030285, mae: 2.164295, mean_q: 2.628573, mean_eps: 0.970293

942811/1000000: episode: 1822, duration: 12.372s, episode steps: 295, steps per

second: 24, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.026467, mae: 2.197391, mean_q: 2.672453, mean_eps: 0.969810

943383/1000000: episode: 1823, duration: 23.809s, episode steps: 572, steps per second: 24, episode reward: 15.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.023268, mae: 2.194521, mean_q: 2.669624, mean_eps: 0.969290

943687/1000000: episode: 1824, duration: 12.647s, episode steps: 304, steps per second: 24, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: 0.027277, mae: 2.213811, mean_q: 2.689342, mean_eps: 0.968764

944261/1000000: episode: 1825, duration: 23.995s, episode steps: 574, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.025078, mae: 2.202210, mean_q: 2.675510, mean_eps: 0.968236

944692/1000000: episode: 1826, duration: 18.025s, episode steps: 431, steps per second: 24, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.631 [0.000, 5.000], loss: 0.025071, mae: 2.203304, mean_q: 2.674159, mean_eps: 0.967634

945002/1000000: episode: 1827, duration: 12.940s, episode steps: 310, steps per second: 24, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.025644, mae: 2.194612, mean_q: 2.662947, mean_eps: 0.967190

945572/1000000: episode: 1828, duration: 23.874s, episode steps: 570, steps per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: 0.024503, mae: 2.191431, mean_q: 2.659759, mean_eps: 0.966662

946045/1000000: episode: 1829, duration: 19.838s, episode steps: 473, steps per second: 24, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: 0.026042, mae: 2.203881, mean_q: 2.672757, mean_eps: 0.966035

946576/1000000: episode: 1830, duration: 22.288s, episode steps: 531, steps per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.019899, mae: 2.199435, mean_q: 2.668016, mean_eps: 0.965433

947171/1000000: episode: 1831, duration: 24.919s, episode steps: 595, steps per second: 24, episode reward: 16.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: 0.023309, mae: 2.206605, mean_q: 2.679952, mean_eps: 0.964758

947501/1000000: episode: 1832, duration: 13.939s, episode steps: 330, steps per second: 24, episode reward: 9.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.679 [0.000, 5.000], loss: 0.020076, mae: 2.187483, mean_q: 2.659294, mean_eps: 0.964202

947877/1000000: episode: 1833, duration: 15.748s, episode steps: 376, steps per second: 24, episode reward: 9.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.023969, mae: 2.218953, mean_q: 2.696949, mean_eps: 0.963777

948744/1000000: episode: 1834, duration: 36.113s, episode steps: 867, steps per

second: 24, episode reward: 23.000, mean reward: 0.027 [0.000, 1.000], mean
action: 2.569 [0.000, 5.000], loss: 0.025706, mae: 2.237556, mean_q: 2.715985,
mean_eps: 0.963033

949134/1000000: episode: 1835, duration: 16.398s, episode steps: 390, steps per
second: 24, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.356 [0.000, 5.000], loss: 0.023091, mae: 2.228171, mean_q: 2.705263,
mean_eps: 0.962279

949518/1000000: episode: 1836, duration: 16.060s, episode steps: 384, steps per
second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.474 [0.000, 5.000], loss: 0.027485, mae: 2.242878, mean_q: 2.723765,
mean_eps: 0.961814

950102/1000000: episode: 1837, duration: 24.374s, episode steps: 584, steps per
second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.469 [0.000, 5.000], loss: 0.028336, mae: 2.209857, mean_q: 2.681189,
mean_eps: 0.961233

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

950836/1000000: episode: 1838, duration: 9.592s, episode steps: 734, steps per
second: 77, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.560 [0.000, 5.000], loss: 0.022721, mae: 2.241572, mean_q: 2.718016,
mean_eps: 0.960782

951463/1000000: episode: 1839, duration: 2.780s, episode steps: 627, steps per
second: 226, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

952187/1000000: episode: 1840, duration: 3.182s, episode steps: 724, steps per
second: 228, episode reward: 17.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

952612/1000000: episode: 1841, duration: 1.877s, episode steps: 425, steps per
second: 226, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean
action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

953080/1000000: episode: 1842, duration: 2.092s, episode steps: 468, steps per
second: 224, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

953467/1000000: episode: 1843, duration: 1.715s, episode steps: 387, steps per
second: 226, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

954004/1000000: episode: 1844, duration: 2.400s, episode steps: 537, steps per
second: 224, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

954310/1000000: episode: 1845, duration: 1.379s, episode steps: 306, steps per
second: 222, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

954862/1000000: episode: 1846, duration: 2.432s, episode steps: 552, steps per
second: 227, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean
action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

955239/1000000: episode: 1847, duration: 1.655s, episode steps: 377, steps per

second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

956152/1000000: episode: 1848, duration: 3.971s, episode steps: 913, steps per second: 230, episode reward: 17.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

956744/1000000: episode: 1849, duration: 2.616s, episode steps: 592, steps per second: 226, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

957007/1000000: episode: 1850, duration: 1.185s, episode steps: 263, steps per second: 222, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

957429/1000000: episode: 1851, duration: 1.892s, episode steps: 422, steps per second: 223, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

957898/1000000: episode: 1852, duration: 2.062s, episode steps: 469, steps per second: 227, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

958493/1000000: episode: 1853, duration: 2.621s, episode steps: 595, steps per second: 227, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.368 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

959330/1000000: episode: 1854, duration: 3.658s, episode steps: 837, steps per second: 229, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

959893/1000000: episode: 1855, duration: 2.507s, episode steps: 563, steps per second: 225, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

960280/1000000: episode: 1856, duration: 1.721s, episode steps: 387, steps per second: 225, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.380 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

960759/1000000: episode: 1857, duration: 2.113s, episode steps: 479, steps per second: 227, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

961032/1000000: episode: 1858, duration: 1.205s, episode steps: 273, steps per second: 227, episode reward: 10.000, mean reward: 0.037 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

961574/1000000: episode: 1859, duration: 2.383s, episode steps: 542, steps per second: 227, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

962136/1000000: episode: 1860, duration: 2.485s, episode steps: 562, steps per second: 226, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

962834/1000000: episode: 1861, duration: 3.111s, episode steps: 698, steps per second: 224, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

963244/1000000: episode: 1862, duration: 1.813s, episode steps: 410, steps per second: 226, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

963804/1000000: episode: 1863, duration: 2.444s, episode steps: 560, steps per

second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

964487/1000000: episode: 1864, duration: 2.968s, episode steps: 683, steps per second: 230, episode reward: 9.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

965105/1000000: episode: 1865, duration: 2.717s, episode steps: 618, steps per second: 227, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

965638/1000000: episode: 1866, duration: 2.372s, episode steps: 533, steps per second: 225, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.642 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

966166/1000000: episode: 1867, duration: 2.319s, episode steps: 528, steps per second: 228, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

967137/1000000: episode: 1868, duration: 4.175s, episode steps: 971, steps per second: 233, episode reward: 27.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

967663/1000000: episode: 1869, duration: 2.304s, episode steps: 526, steps per second: 228, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

968363/1000000: episode: 1870, duration: 3.093s, episode steps: 700, steps per second: 226, episode reward: 19.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

969044/1000000: episode: 1871, duration: 2.983s, episode steps: 681, steps per second: 228, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

969369/1000000: episode: 1872, duration: 1.433s, episode steps: 325, steps per second: 227, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

969683/1000000: episode: 1873, duration: 1.382s, episode steps: 314, steps per second: 227, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

970195/1000000: episode: 1874, duration: 2.235s, episode steps: 512, steps per second: 229, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

970992/1000000: episode: 1875, duration: 3.505s, episode steps: 797, steps per second: 227, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

971425/1000000: episode: 1876, duration: 1.953s, episode steps: 433, steps per second: 222, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

971834/1000000: episode: 1877, duration: 1.788s, episode steps: 409, steps per second: 229, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

972427/1000000: episode: 1878, duration: 2.585s, episode steps: 593, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.629 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

972957/1000000: episode: 1879, duration: 2.311s, episode steps: 530, steps per

second: 229, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

973680/1000000: episode: 1880, duration: 3.183s, episode steps: 723, steps per second: 227, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

974096/1000000: episode: 1881, duration: 1.877s, episode steps: 416, steps per second: 222, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

974628/1000000: episode: 1882, duration: 2.353s, episode steps: 532, steps per second: 226, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

975014/1000000: episode: 1883, duration: 1.687s, episode steps: 386, steps per second: 229, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

975518/1000000: episode: 1884, duration: 2.201s, episode steps: 504, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

976003/1000000: episode: 1885, duration: 2.138s, episode steps: 485, steps per second: 227, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

976423/1000000: episode: 1886, duration: 1.881s, episode steps: 420, steps per second: 223, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

976918/1000000: episode: 1887, duration: 2.241s, episode steps: 495, steps per second: 221, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

977182/1000000: episode: 1888, duration: 1.201s, episode steps: 264, steps per second: 220, episode reward: 6.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

977596/1000000: episode: 1889, duration: 1.825s, episode steps: 414, steps per second: 227, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

977999/1000000: episode: 1890, duration: 1.791s, episode steps: 403, steps per second: 225, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

978839/1000000: episode: 1891, duration: 3.675s, episode steps: 840, steps per second: 229, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

979407/1000000: episode: 1892, duration: 2.513s, episode steps: 568, steps per second: 226, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.384 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

979908/1000000: episode: 1893, duration: 2.244s, episode steps: 501, steps per second: 223, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

981078/1000000: episode: 1894, duration: 5.054s, episode steps: 1170, steps per second: 231, episode reward: 32.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

981454/1000000: episode: 1895, duration: 1.648s, episode steps: 376, steps per

second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

982501/1000000: episode: 1896, duration: 4.559s, episode steps: 1047, steps per second: 230, episode reward: 21.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

982933/1000000: episode: 1897, duration: 1.938s, episode steps: 432, steps per second: 223, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

983331/1000000: episode: 1898, duration: 1.760s, episode steps: 398, steps per second: 226, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

983764/1000000: episode: 1899, duration: 1.886s, episode steps: 433, steps per second: 230, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

984461/1000000: episode: 1900, duration: 3.027s, episode steps: 697, steps per second: 230, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

985341/1000000: episode: 1901, duration: 3.873s, episode steps: 880, steps per second: 227, episode reward: 12.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

985723/1000000: episode: 1902, duration: 1.707s, episode steps: 382, steps per second: 224, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

986029/1000000: episode: 1903, duration: 1.377s, episode steps: 306, steps per second: 222, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

986665/1000000: episode: 1904, duration: 2.808s, episode steps: 636, steps per second: 226, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

987232/1000000: episode: 1905, duration: 2.510s, episode steps: 567, steps per second: 226, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

987525/1000000: episode: 1906, duration: 1.301s, episode steps: 293, steps per second: 225, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

987903/1000000: episode: 1907, duration: 1.680s, episode steps: 378, steps per second: 225, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

988392/1000000: episode: 1908, duration: 2.211s, episode steps: 489, steps per second: 221, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.648 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

989173/1000000: episode: 1909, duration: 3.450s, episode steps: 781, steps per second: 226, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

989716/1000000: episode: 1910, duration: 2.381s, episode steps: 543, steps per second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

990345/1000000: episode: 1911, duration: 2.784s, episode steps: 629, steps per

second: 226, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

990757/1000000: episode: 1912, duration: 1.836s, episode steps: 412, steps per second: 224, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.692 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

991439/1000000: episode: 1913, duration: 3.035s, episode steps: 682, steps per second: 225, episode reward: 21.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

991936/1000000: episode: 1914, duration: 2.185s, episode steps: 497, steps per second: 227, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

992222/1000000: episode: 1915, duration: 1.263s, episode steps: 286, steps per second: 227, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

992637/1000000: episode: 1916, duration: 1.831s, episode steps: 415, steps per second: 227, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

992931/1000000: episode: 1917, duration: 1.313s, episode steps: 294, steps per second: 224, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

993593/1000000: episode: 1918, duration: 23.644s, episode steps: 662, steps per second: 28, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.024452, mae: 2.232304, mean_q: 2.706871, mean_eps: 0.987666

994011/1000000: episode: 1919, duration: 17.436s, episode steps: 418, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.376 [0.000, 5.000], loss: 0.021785, mae: 2.222641, mean_q: 2.699877, mean_eps: 0.987086

994390/1000000: episode: 1920, duration: 15.937s, episode steps: 379, steps per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.022280, mae: 2.208452, mean_q: 2.683349, mean_eps: 0.986608

[Callback] Pesos guardados en:
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_1920_episodes.h5f

995254/1000000: episode: 1921, duration: 36.007s, episode steps: 864, steps per second: 24, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.029020, mae: 2.236760, mean_q: 2.715256, mean_eps: 0.985862

995908/1000000: episode: 1922, duration: 27.483s, episode steps: 654, steps per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.028658, mae: 2.213184, mean_q: 2.684506, mean_eps: 0.984952

996592/1000000: episode: 1923, duration: 28.729s, episode steps: 684, steps per second: 24, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.023108, mae: 2.231672, mean_q: 2.708263, mean_eps: 0.984150

997227/1000000: episode: 1924, duration: 26.543s, episode steps: 635, steps per second: 24, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: 0.024573, mae: 2.223806, mean_q: 2.702890, mean_eps: 0.983358

997906/1000000: episode: 1925, duration: 28.404s, episode steps: 679, steps per second: 24, episode reward: 17.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.021818, mae: 2.210910, mean_q: 2.684801, mean_eps: 0.982569

998330/1000000: episode: 1926, duration: 17.697s, episode steps: 424, steps per second: 24, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: 0.032006, mae: 2.221715, mean_q: 2.693232, mean_eps: 0.981906

998830/1000000: episode: 1927, duration: 20.917s, episode steps: 500, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.586 [0.000, 5.000], loss: 0.026699, mae: 2.261123, mean_q: 2.738871, mean_eps: 0.981352

999318/1000000: episode: 1928, duration: 20.437s, episode steps: 488, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: 0.028050, mae: 2.269151, mean_q: 2.752285, mean_eps: 0.980759

999627/1000000: episode: 1929, duration: 13.021s, episode steps: 309, steps per second: 24, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: 0.024676, mae: 2.264883, mean_q: 2.745473, mean_eps: 0.980282

1000165/1000000: episode: 1930, duration: 22.618s, episode steps: 538, steps per second: 24, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.027658, mae: 2.270851, mean_q: 2.752921, mean_eps: 0.979773

1000545/1000000: episode: 1931, duration: 15.827s, episode steps: 380, steps per second: 24, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: 0.027669, mae: 2.280087, mean_q: 2.764504, mean_eps: 0.979221

1001080/1000000: episode: 1932, duration: 22.245s, episode steps: 535, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.026526, mae: 2.239781, mean_q: 2.716207, mean_eps: 0.978674

1001432/1000000: episode: 1933, duration: 14.720s, episode steps: 352, steps per second: 24, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: 0.025332, mae: 2.286551, mean_q: 2.767045, mean_eps: 0.978143

1001909/1000000: episode: 1934, duration: 19.824s, episode steps: 477, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: 0.022993, mae: 2.258426, mean_q: 2.736408, mean_eps: 0.977644

1002234/1000000: episode: 1935, duration: 13.494s, episode steps: 325, steps per second: 24, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: 0.023294, mae: 2.254057, mean_q: 2.727490, mean_eps: 0.977162

1002961/1000000: episode: 1936, duration: 30.338s, episode steps: 727, steps per second: 24, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: 0.018103, mae: 2.263401, mean_q: 2.744903, mean_eps: 0.976530

1003346/1000000: episode: 1937, duration: 15.804s, episode steps: 385, steps per second: 24, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.023846, mae: 2.279249, mean_q: 2.760575, mean_eps: 0.975863

1003901/1000000: episode: 1938, duration: 23.138s, episode steps: 555, steps per second: 24, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: 0.022986, mae: 2.281724, mean_q: 2.764947, mean_eps: 0.975299

1004213/1000000: episode: 1939, duration: 12.864s, episode steps: 312, steps per second: 24, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: 0.028215, mae: 2.248876, mean_q: 2.721646, mean_eps: 0.974778

1004563/1000000: episode: 1940, duration: 14.354s, episode steps: 350, steps per second: 24, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.022446, mae: 2.281473, mean_q: 2.763350, mean_eps: 0.974382

1005181/1000000: episode: 1941, duration: 25.607s, episode steps: 618, steps per second: 24, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.025583, mae: 2.255538, mean_q: 2.732989, mean_eps: 0.973802

1005875/1000000: episode: 1942, duration: 28.782s, episode steps: 694, steps per second: 24, episode reward: 17.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: 0.025136, mae: 2.241205, mean_q: 2.716108, mean_eps: 0.973014

1006371/1000000: episode: 1943, duration: 20.471s, episode steps: 496, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.026974, mae: 2.261691, mean_q: 2.745036, mean_eps: 0.972302

1006869/1000000: episode: 1944, duration: 20.739s, episode steps: 498, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.030787, mae: 2.275648, mean_q: 2.759733, mean_eps: 0.971704

1007224/1000000: episode: 1945, duration: 14.816s, episode steps: 355, steps per second: 24, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.025195, mae: 2.245024, mean_q: 2.725414, mean_eps: 0.971193

1007735/1000000: episode: 1946, duration: 21.166s, episode steps: 511, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: 0.025321, mae: 2.265894, mean_q: 2.746695, mean_eps: 0.970674

1008144/1000000: episode: 1947, duration: 16.927s, episode steps: 409, steps per second: 24, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: 0.024943, mae: 2.245196, mean_q: 2.721717, mean_eps: 0.970122

1008450/1000000: episode: 1948, duration: 12.851s, episode steps: 306, steps per second: 24, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: 0.019599, mae: 2.310550, mean_q: 2.804246, mean_eps: 0.969693

1009089/1000000: episode: 1949, duration: 26.500s, episode steps: 639, steps per second: 24, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.026942, mae: 2.291936, mean_q: 2.778421, mean_eps: 0.969124

1009524/1000000: episode: 1950, duration: 18.172s, episode steps: 435, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.329 [0.000, 5.000], loss: 0.024574, mae: 2.299241, mean_q: 2.792022, mean_eps: 0.968481

1009854/1000000: episode: 1951, duration: 13.815s, episode steps: 330, steps per second: 24, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.029022, mae: 2.314851, mean_q: 2.806256, mean_eps: 0.968022

1010401/1000000: episode: 1952, duration: 22.697s, episode steps: 547, steps per second: 24, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.026538, mae: 2.287726, mean_q: 2.770217, mean_eps: 0.967494

1010825/1000000: episode: 1953, duration: 17.791s, episode steps: 424, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.033476, mae: 2.291236, mean_q: 2.770888, mean_eps: 0.966911

1011361/1000000: episode: 1954, duration: 22.508s, episode steps: 536, steps per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: 0.025005, mae: 2.278694, mean_q: 2.761907, mean_eps: 0.966335

1011911/1000000: episode: 1955, duration: 22.903s, episode steps: 550, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: 0.023918, mae: 2.239097, mean_q: 2.711386, mean_eps: 0.965685

1012337/1000000: episode: 1956, duration: 17.733s, episode steps: 426, steps per second: 24, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: 0.028441, mae: 2.246209, mean_q: 2.722737, mean_eps: 0.965099

1012845/1000000: episode: 1957, duration: 21.121s, episode steps: 508, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.023560, mae: 2.294785, mean_q: 2.783836, mean_eps: 0.964538

1013659/1000000: episode: 1958, duration: 33.677s, episode steps: 814, steps per second: 24, episode reward: 21.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.024085, mae: 2.302502, mean_q: 2.791649, mean_eps: 0.963746

1013965/1000000: episode: 1959, duration: 12.807s, episode steps: 306, steps per second: 24, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.025176, mae: 2.275357, mean_q: 2.753098, mean_eps: 0.963074

1014476/1000000: episode: 1960, duration: 21.169s, episode steps: 511, steps per second: 24, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.024027, mae: 2.289111, mean_q: 2.772373, mean_eps: 0.962584

1014870/1000000: episode: 1961, duration: 16.501s, episode steps: 394, steps per second: 24, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.028247, mae: 2.316430, mean_q: 2.802810, mean_eps: 0.962042

1015699/1000000: episode: 1962, duration: 34.476s, episode steps: 829, steps per second: 24, episode reward: 21.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.026524, mae: 2.297193, mean_q: 2.783904, mean_eps: 0.961307

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

1015979/1000000: episode: 1963, duration: 5.214s, episode steps: 280, steps per second: 54, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.025099, mae: 2.242069, mean_q: 2.720811, mean_eps: 0.960746

1016439/1000000: episode: 1964, duration: 2.020s, episode steps: 460, steps per second: 228, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1017142/1000000: episode: 1965, duration: 3.071s, episode steps: 703, steps per second: 229, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1017512/1000000: episode: 1966, duration: 1.614s, episode steps: 370, steps per second: 229, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1018114/1000000: episode: 1967, duration: 2.609s, episode steps: 602, steps per second: 231, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1018435/1000000: episode: 1968, duration: 1.396s, episode steps: 321, steps per second: 230, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1018903/1000000: episode: 1969, duration: 2.051s, episode steps: 468, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1019463/1000000: episode: 1970, duration: 2.447s, episode steps: 560, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.364 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1020165/1000000: episode: 1971, duration: 3.062s, episode steps: 702, steps per second: 229, episode reward: 21.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1021015/1000000: episode: 1972, duration: 3.671s, episode steps: 850, steps per second: 232, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1021784/1000000: episode: 1973, duration: 3.314s, episode steps: 769, steps per

second: 232, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.663 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1022375/1000000: episode: 1974, duration: 2.578s, episode steps: 591, steps per second: 229, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1022766/1000000: episode: 1975, duration: 1.714s, episode steps: 391, steps per second: 228, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1023681/1000000: episode: 1976, duration: 3.971s, episode steps: 915, steps per second: 230, episode reward: 9.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1024282/1000000: episode: 1977, duration: 2.598s, episode steps: 601, steps per second: 231, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1025082/1000000: episode: 1978, duration: 3.458s, episode steps: 800, steps per second: 231, episode reward: 10.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1025603/1000000: episode: 1979, duration: 2.281s, episode steps: 521, steps per second: 228, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1025991/1000000: episode: 1980, duration: 1.690s, episode steps: 388, steps per second: 230, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1026365/1000000: episode: 1981, duration: 1.638s, episode steps: 374, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1027037/1000000: episode: 1982, duration: 2.914s, episode steps: 672, steps per second: 231, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.522 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1027497/1000000: episode: 1983, duration: 2.008s, episode steps: 460, steps per second: 229, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1028011/1000000: episode: 1984, duration: 2.239s, episode steps: 514, steps per second: 230, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1028493/1000000: episode: 1985, duration: 2.144s, episode steps: 482, steps per second: 225, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1029151/1000000: episode: 1986, duration: 2.861s, episode steps: 658, steps per second: 230, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1030001/1000000: episode: 1987, duration: 3.666s, episode steps: 850, steps per second: 232, episode reward: 14.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1030459/1000000: episode: 1988, duration: 1.978s, episode steps: 458, steps per second: 232, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1031145/1000000: episode: 1989, duration: 2.975s, episode steps: 686, steps per

second: 231, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1031696/1000000: episode: 1990, duration: 2.411s, episode steps: 551, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1032113/1000000: episode: 1991, duration: 1.810s, episode steps: 417, steps per second: 230, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1032586/1000000: episode: 1992, duration: 2.062s, episode steps: 473, steps per second: 229, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1033125/1000000: episode: 1993, duration: 2.331s, episode steps: 539, steps per second: 231, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1033602/1000000: episode: 1994, duration: 2.080s, episode steps: 477, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1033908/1000000: episode: 1995, duration: 1.359s, episode steps: 306, steps per second: 225, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1034443/1000000: episode: 1996, duration: 2.348s, episode steps: 535, steps per second: 228, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1034766/1000000: episode: 1997, duration: 1.408s, episode steps: 323, steps per second: 229, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1035131/1000000: episode: 1998, duration: 1.604s, episode steps: 365, steps per second: 228, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1035434/1000000: episode: 1999, duration: 1.325s, episode steps: 303, steps per second: 229, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1036177/1000000: episode: 2000, duration: 3.203s, episode steps: 743, steps per second: 232, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1036685/1000000: episode: 2001, duration: 2.205s, episode steps: 508, steps per second: 230, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1037238/1000000: episode: 2002, duration: 2.417s, episode steps: 553, steps per second: 229, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1037624/1000000: episode: 2003, duration: 1.682s, episode steps: 386, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.635 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1038186/1000000: episode: 2004, duration: 2.437s, episode steps: 562, steps per second: 231, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1038924/1000000: episode: 2005, duration: 3.200s, episode steps: 738, steps per

second: 231, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1039554/1000000: episode: 2006, duration: 2.719s, episode steps: 630, steps per second: 232, episode reward: 18.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1040009/1000000: episode: 2007, duration: 1.988s, episode steps: 455, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1040302/1000000: episode: 2008, duration: 1.298s, episode steps: 293, steps per second: 226, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1040935/1000000: episode: 2009, duration: 2.742s, episode steps: 633, steps per second: 231, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1041447/1000000: episode: 2010, duration: 2.203s, episode steps: 512, steps per second: 232, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1041905/1000000: episode: 2011, duration: 1.987s, episode steps: 458, steps per second: 231, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1042391/1000000: episode: 2012, duration: 2.107s, episode steps: 486, steps per second: 231, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1042837/1000000: episode: 2013, duration: 1.952s, episode steps: 446, steps per second: 228, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1043221/1000000: episode: 2014, duration: 1.695s, episode steps: 384, steps per second: 227, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1043623/1000000: episode: 2015, duration: 1.767s, episode steps: 402, steps per second: 228, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1044126/1000000: episode: 2016, duration: 2.166s, episode steps: 503, steps per second: 232, episode reward: 13.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1044516/1000000: episode: 2017, duration: 1.707s, episode steps: 390, steps per second: 228, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.410 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1044829/1000000: episode: 2018, duration: 1.374s, episode steps: 313, steps per second: 228, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1045277/1000000: episode: 2019, duration: 1.949s, episode steps: 448, steps per second: 230, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1045728/1000000: episode: 2020, duration: 1.999s, episode steps: 451, steps per second: 226, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1046029/1000000: episode: 2021, duration: 1.334s, episode steps: 301, steps per

second: 226, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.229 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1046331/1000000: episode: 2022, duration: 1.314s, episode steps: 302, steps per second: 230, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.338 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1046935/1000000: episode: 2023, duration: 2.621s, episode steps: 604, steps per second: 230, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.348 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1047521/1000000: episode: 2024, duration: 2.526s, episode steps: 586, steps per second: 232, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1048028/1000000: episode: 2025, duration: 2.190s, episode steps: 507, steps per second: 232, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1048510/1000000: episode: 2026, duration: 2.103s, episode steps: 482, steps per second: 229, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049027/1000000: episode: 2027, duration: 2.275s, episode steps: 517, steps per second: 227, episode reward: 13.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049317/1000000: episode: 2028, duration: 1.270s, episode steps: 290, steps per second: 228, episode reward: 9.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049651/1000000: episode: 2029, duration: 1.460s, episode steps: 334, steps per second: 229, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.374 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1049951/1000000: episode: 2030, duration: 1.303s, episode steps: 300, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.583 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1050479/1000000: episode: 2031, duration: 2.278s, episode steps: 528, steps per second: 232, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1050951/1000000: episode: 2032, duration: 2.041s, episode steps: 472, steps per second: 231, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1051615/1000000: episode: 2033, duration: 2.894s, episode steps: 664, steps per second: 229, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.449 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052421/1000000: episode: 2034, duration: 3.499s, episode steps: 806, steps per second: 230, episode reward: 18.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.397 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052710/1000000: episode: 2035, duration: 1.253s, episode steps: 289, steps per second: 231, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.398 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1052988/1000000: episode: 2036, duration: 1.208s, episode steps: 278, steps per second: 230, episode reward: 6.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1053388/1000000: episode: 2037, duration: 1.740s, episode steps: 400, steps per

second: 230, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1053760/1000000: episode: 2038, duration: 1.630s, episode steps: 372, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1054381/1000000: episode: 2039, duration: 2.735s, episode steps: 621, steps per second: 227, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1054685/1000000: episode: 2040, duration: 1.346s, episode steps: 304, steps per second: 226, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1055218/1000000: episode: 2041, duration: 2.316s, episode steps: 533, steps per second: 230, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1055619/1000000: episode: 2042, duration: 1.739s, episode steps: 401, steps per second: 231, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1056001/1000000: episode: 2043, duration: 1.670s, episode steps: 382, steps per second: 229, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1056531/1000000: episode: 2044, duration: 2.289s, episode steps: 530, steps per second: 232, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1056976/1000000: episode: 2045, duration: 1.923s, episode steps: 445, steps per second: 231, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.339 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1057413/1000000: episode: 2046, duration: 1.911s, episode steps: 437, steps per second: 229, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1058300/1000000: episode: 2047, duration: 3.829s, episode steps: 887, steps per second: 232, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1058819/1000000: episode: 2048, duration: 11.179s, episode steps: 519, steps per second: 46, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.020911, mae: 2.326382, mean_q: 2.822717, mean_eps: 0.987854

1059394/1000000: episode: 2049, duration: 23.965s, episode steps: 575, steps per second: 24, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: 0.027453, mae: 2.333778, mean_q: 2.829149, mean_eps: 0.987364

1059798/1000000: episode: 2050, duration: 16.931s, episode steps: 404, steps per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.027461, mae: 2.361473, mean_q: 2.857546, mean_eps: 0.986776

1060253/1000000: episode: 2051, duration: 19.035s, episode steps: 455, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.024813, mae: 2.321525, mean_q: 2.811288, mean_eps: 0.986260

1060660/1000000: episode: 2052, duration: 16.990s, episode steps: 407, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: 0.023604, mae: 2.330747, mean_q: 2.821306, mean_eps: 0.985744

1060956/1000000: episode: 2053, duration: 12.468s, episode steps: 296, steps per second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.025876, mae: 2.302130, mean_q: 2.785468, mean_eps: 0.985324

1061275/1000000: episode: 2054, duration: 13.347s, episode steps: 319, steps per second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.023138, mae: 2.341290, mean_q: 2.837513, mean_eps: 0.984954

1061667/1000000: episode: 2055, duration: 16.254s, episode steps: 392, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.025310, mae: 2.330395, mean_q: 2.824996, mean_eps: 0.984527

1062181/1000000: episode: 2056, duration: 21.404s, episode steps: 514, steps per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.029610, mae: 2.336899, mean_q: 2.830483, mean_eps: 0.983982

1062654/1000000: episode: 2057, duration: 19.564s, episode steps: 473, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: 0.027710, mae: 2.337328, mean_q: 2.833942, mean_eps: 0.983390

1063176/1000000: episode: 2058, duration: 21.745s, episode steps: 522, steps per second: 24, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: 0.034932, mae: 2.331469, mean_q: 2.824986, mean_eps: 0.982794

1063459/1000000: episode: 2059, duration: 11.764s, episode steps: 283, steps per second: 24, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.026906, mae: 2.353523, mean_q: 2.853191, mean_eps: 0.982312

1063941/1000000: episode: 2060, duration: 20.197s, episode steps: 482, steps per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.028878, mae: 2.354095, mean_q: 2.849146, mean_eps: 0.981851

1064314/1000000: episode: 2061, duration: 15.511s, episode steps: 373, steps per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.340 [0.000, 5.000], loss: 0.031096, mae: 2.367046, mean_q: 2.866279, mean_eps: 0.981338

1064837/1000000: episode: 2062, duration: 21.696s, episode steps: 523, steps per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: 0.033675, mae: 2.370217, mean_q: 2.864618, mean_eps: 0.980800

1065751/1000000: episode: 2063, duration: 37.742s, episode steps: 914, steps per second: 24, episode reward: 15.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: 0.027365, mae: 2.387867, mean_q: 2.887283, mean_eps: 0.979938

1066498/1000000: episode: 2064, duration: 30.903s, episode steps: 747, steps per second: 24, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.030188, mae: 2.360189, mean_q: 2.852299, mean_eps: 0.978942

1066796/1000000: episode: 2065, duration: 12.485s, episode steps: 298, steps per second: 24, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.721 [0.000, 5.000], loss: 0.025541, mae: 2.343359, mean_q: 2.835041, mean_eps: 0.978316

1067214/1000000: episode: 2066, duration: 17.614s, episode steps: 418, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: 0.021706, mae: 2.362133, mean_q: 2.860000, mean_eps: 0.977886

1067562/1000000: episode: 2067, duration: 14.456s, episode steps: 348, steps per second: 24, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.632 [0.000, 5.000], loss: 0.024831, mae: 2.354736, mean_q: 2.851459, mean_eps: 0.977426

1068095/1000000: episode: 2068, duration: 22.022s, episode steps: 533, steps per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: 0.033520, mae: 2.384769, mean_q: 2.882440, mean_eps: 0.976898

1068718/1000000: episode: 2069, duration: 25.806s, episode steps: 623, steps per second: 24, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.400 [0.000, 5.000], loss: 0.025653, mae: 2.370572, mean_q: 2.872263, mean_eps: 0.976204

1069250/1000000: episode: 2070, duration: 22.016s, episode steps: 532, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: 0.027034, mae: 2.418450, mean_q: 2.927846, mean_eps: 0.975510

1069677/1000000: episode: 2071, duration: 17.911s, episode steps: 427, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.031951, mae: 2.429287, mean_q: 2.939309, mean_eps: 0.974934

1070231/1000000: episode: 2072, duration: 22.909s, episode steps: 554, steps per second: 24, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: 0.031531, mae: 2.395196, mean_q: 2.899833, mean_eps: 0.974346

1070708/1000000: episode: 2073, duration: 19.924s, episode steps: 477, steps per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.027646, mae: 2.431867, mean_q: 2.944306, mean_eps: 0.973730

1071256/1000000: episode: 2074, duration: 22.909s, episode steps: 548, steps per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: 0.027777, mae: 2.413503, mean_q: 2.924369, mean_eps: 0.973115

1071881/1000000: episode: 2075, duration: 26.076s, episode steps: 625, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: 0.027204, mae: 2.404061, mean_q: 2.912680, mean_eps: 0.972410

1072198/1000000: episode: 2076, duration: 13.120s, episode steps: 317, steps per second: 24, episode reward: 8.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.023732, mae: 2.420937, mean_q: 2.932746, mean_eps: 0.971843

1072887/1000000: episode: 2077, duration: 28.473s, episode steps: 689, steps per second: 24, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.026947, mae: 2.406227, mean_q: 2.912822, mean_eps: 0.971241

1073241/1000000: episode: 2078, duration: 14.783s, episode steps: 354, steps per second: 24, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: 0.025504, mae: 2.409152, mean_q: 2.916697, mean_eps: 0.970614

1073632/1000000: episode: 2079, duration: 16.328s, episode steps: 391, steps per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.024652, mae: 2.421386, mean_q: 2.933914, mean_eps: 0.970168

1073937/1000000: episode: 2080, duration: 12.760s, episode steps: 305, steps per second: 24, episode reward: 7.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: 0.025695, mae: 2.425102, mean_q: 2.935096, mean_eps: 0.969750

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_2080_episodes.h5f

1074339/1000000: episode: 2081, duration: 16.747s, episode steps: 402, steps per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.413 [0.000, 5.000], loss: 0.039084, mae: 2.416314, mean_q: 2.919353, mean_eps: 0.969326

1074977/1000000: episode: 2082, duration: 26.642s, episode steps: 638, steps per second: 24, episode reward: 16.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.033739, mae: 2.434694, mean_q: 2.942609, mean_eps: 0.968702

1075429/1000000: episode: 2083, duration: 18.640s, episode steps: 452, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: 0.033809, mae: 2.439684, mean_q: 2.949496, mean_eps: 0.968046

1075901/1000000: episode: 2084, duration: 19.728s, episode steps: 472, steps per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.581 [0.000, 5.000], loss: 0.034220, mae: 2.420867, mean_q: 2.926289, mean_eps: 0.967492

1076655/1000000: episode: 2085, duration: 31.376s, episode steps: 754, steps per second: 24, episode reward: 20.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.030605, mae: 2.433908, mean_q: 2.944767, mean_eps: 0.966758

1077060/1000000: episode: 2086, duration: 17.100s, episode steps: 405, steps per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: 0.031506, mae: 2.410650, mean_q: 2.917235, mean_eps: 0.966064

1077594/1000000: episode: 2087, duration: 22.334s, episode steps: 534, steps per second: 24, episode reward: 12.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: 0.032273, mae: 2.440424, mean_q: 2.954708, mean_eps: 0.965500

1077946/1000000: episode: 2088, duration: 14.667s, episode steps: 352, steps per second: 24, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.543 [0.000, 5.000], loss: 0.026811, mae: 2.432617, mean_q: 2.942690, mean_eps: 0.964967

1078357/1000000: episode: 2089, duration: 17.276s, episode steps: 411, steps per second: 24, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.664 [0.000, 5.000], loss: 0.030444, mae: 2.436193, mean_q: 2.944631, mean_eps: 0.964509

1078876/1000000: episode: 2090, duration: 21.566s, episode steps: 519, steps per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.029643, mae: 2.441702, mean_q: 2.952711, mean_eps: 0.963952

1079359/1000000: episode: 2091, duration: 20.068s, episode steps: 483, steps per second: 24, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.034166, mae: 2.479624, mean_q: 2.999753, mean_eps: 0.963352

1080143/1000000: episode: 2092, duration: 32.603s, episode steps: 784, steps per second: 24, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.032093, mae: 2.459888, mean_q: 2.975067, mean_eps: 0.962591

1080428/1000000: episode: 2093, duration: 12.045s, episode steps: 285, steps per second: 24, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: 0.033394, mae: 2.466775, mean_q: 2.981807, mean_eps: 0.961950

1081112/1000000: episode: 2094, duration: 28.773s, episode steps: 684, steps per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.030206, mae: 2.470686, mean_q: 2.986722, mean_eps: 0.961370

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

1081794/1000000: episode: 2095, duration: 11.595s, episode steps: 682, steps per second: 59, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: 0.031936, mae: 2.452125, mean_q: 2.967518, mean_eps: 0.960820

1082299/1000000: episode: 2096, duration: 2.198s, episode steps: 505, steps per second: 230, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1082572/1000000: episode: 2097, duration: 1.194s, episode steps: 273, steps per second: 229, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.326 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1083088/1000000: episode: 2098, duration: 2.284s, episode steps: 516, steps per second: 226, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean

action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1083451/1000000: episode: 2099, duration: 1.605s, episode steps: 363, steps per second: 226, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.361 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1083739/1000000: episode: 2100, duration: 1.260s, episode steps: 288, steps per second: 229, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.486 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1084381/1000000: episode: 2101, duration: 2.778s, episode steps: 642, steps per second: 231, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1084760/1000000: episode: 2102, duration: 1.643s, episode steps: 379, steps per second: 231, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.343 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1085234/1000000: episode: 2103, duration: 2.062s, episode steps: 474, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1085718/1000000: episode: 2104, duration: 2.123s, episode steps: 484, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1086301/1000000: episode: 2105, duration: 2.661s, episode steps: 583, steps per second: 219, episode reward: 15.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1086952/1000000: episode: 2106, duration: 2.926s, episode steps: 651, steps per second: 222, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1087730/1000000: episode: 2107, duration: 3.442s, episode steps: 778, steps per second: 226, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1088215/1000000: episode: 2108, duration: 2.127s, episode steps: 485, steps per second: 228, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1088834/1000000: episode: 2109, duration: 2.704s, episode steps: 619, steps per second: 229, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1089198/1000000: episode: 2110, duration: 1.624s, episode steps: 364, steps per second: 224, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1089933/1000000: episode: 2111, duration: 3.197s, episode steps: 735, steps per second: 230, episode reward: 18.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1090364/1000000: episode: 2112, duration: 1.866s, episode steps: 431, steps per second: 231, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1090835/1000000: episode: 2113, duration: 2.046s, episode steps: 471, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1091131/1000000: episode: 2114, duration: 1.293s, episode steps: 296, steps per second: 229, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean

action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1091879/1000000: episode: 2115, duration: 3.280s, episode steps: 748, steps per second: 228, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.406 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1092347/1000000: episode: 2116, duration: 2.039s, episode steps: 468, steps per second: 230, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1092923/1000000: episode: 2117, duration: 2.481s, episode steps: 576, steps per second: 232, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1093716/1000000: episode: 2118, duration: 3.422s, episode steps: 793, steps per second: 232, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1094190/1000000: episode: 2119, duration: 2.082s, episode steps: 474, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1094448/1000000: episode: 2120, duration: 1.142s, episode steps: 258, steps per second: 226, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1094916/1000000: episode: 2121, duration: 2.055s, episode steps: 468, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1095667/1000000: episode: 2122, duration: 3.241s, episode steps: 751, steps per second: 232, episode reward: 15.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1096281/1000000: episode: 2123, duration: 2.647s, episode steps: 614, steps per second: 232, episode reward: 19.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.366 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1096907/1000000: episode: 2124, duration: 2.716s, episode steps: 626, steps per second: 230, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1097365/1000000: episode: 2125, duration: 1.989s, episode steps: 458, steps per second: 230, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1097801/1000000: episode: 2126, duration: 1.913s, episode steps: 436, steps per second: 228, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1098267/1000000: episode: 2127, duration: 2.031s, episode steps: 466, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1098570/1000000: episode: 2128, duration: 1.322s, episode steps: 303, steps per second: 229, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.680 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1099299/1000000: episode: 2129, duration: 3.165s, episode steps: 729, steps per second: 230, episode reward: 21.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1099621/1000000: episode: 2130, duration: 1.402s, episode steps: 322, steps per second: 230, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean

action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1100019/1000000: episode: 2131, duration: 1.717s, episode steps: 398, steps per second: 232, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1100710/1000000: episode: 2132, duration: 3.043s, episode steps: 691, steps per second: 227, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1101100/1000000: episode: 2133, duration: 1.689s, episode steps: 390, steps per second: 231, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1101403/1000000: episode: 2134, duration: 1.305s, episode steps: 303, steps per second: 232, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1102005/1000000: episode: 2135, duration: 2.589s, episode steps: 602, steps per second: 233, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1102384/1000000: episode: 2136, duration: 1.648s, episode steps: 379, steps per second: 230, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1102721/1000000: episode: 2137, duration: 1.462s, episode steps: 337, steps per second: 230, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1103018/1000000: episode: 2138, duration: 1.290s, episode steps: 297, steps per second: 230, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1103408/1000000: episode: 2139, duration: 1.712s, episode steps: 390, steps per second: 228, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1103743/1000000: episode: 2140, duration: 1.476s, episode steps: 335, steps per second: 227, episode reward: 8.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1104232/1000000: episode: 2141, duration: 2.149s, episode steps: 489, steps per second: 228, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1104694/1000000: episode: 2142, duration: 2.017s, episode steps: 462, steps per second: 229, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1105111/1000000: episode: 2143, duration: 1.831s, episode steps: 417, steps per second: 228, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1105582/1000000: episode: 2144, duration: 2.057s, episode steps: 471, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1106207/1000000: episode: 2145, duration: 2.731s, episode steps: 625, steps per second: 229, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.683 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1106803/1000000: episode: 2146, duration: 2.607s, episode steps: 596, steps per second: 229, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean

action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1107290/1000000: episode: 2147, duration: 2.140s, episode steps: 487, steps per second: 228, episode reward: 14.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1107790/1000000: episode: 2148, duration: 2.211s, episode steps: 500, steps per second: 226, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1108079/1000000: episode: 2149, duration: 1.279s, episode steps: 289, steps per second: 226, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1109062/1000000: episode: 2150, duration: 4.279s, episode steps: 983, steps per second: 230, episode reward: 20.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.607 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1109493/1000000: episode: 2151, duration: 1.937s, episode steps: 431, steps per second: 223, episode reward: 8.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1110364/1000000: episode: 2152, duration: 3.783s, episode steps: 871, steps per second: 230, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1110687/1000000: episode: 2153, duration: 1.413s, episode steps: 323, steps per second: 229, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.260 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1111045/1000000: episode: 2154, duration: 1.571s, episode steps: 358, steps per second: 228, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1111433/1000000: episode: 2155, duration: 1.693s, episode steps: 388, steps per second: 229, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1111736/1000000: episode: 2156, duration: 1.331s, episode steps: 303, steps per second: 228, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.320 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1112216/1000000: episode: 2157, duration: 2.171s, episode steps: 480, steps per second: 221, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1112981/1000000: episode: 2158, duration: 3.346s, episode steps: 765, steps per second: 229, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1113396/1000000: episode: 2159, duration: 1.831s, episode steps: 415, steps per second: 227, episode reward: 1.000, mean reward: 0.002 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1113983/1000000: episode: 2160, duration: 2.571s, episode steps: 587, steps per second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1114601/1000000: episode: 2161, duration: 2.713s, episode steps: 618, steps per second: 228, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1114896/1000000: episode: 2162, duration: 1.309s, episode steps: 295, steps per second: 225, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean

action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1115347/1000000: episode: 2163, duration: 2.003s, episode steps: 451, steps per second: 225, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1115939/1000000: episode: 2164, duration: 2.579s, episode steps: 592, steps per second: 230, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1116302/1000000: episode: 2165, duration: 1.597s, episode steps: 363, steps per second: 227, episode reward: 6.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1116713/1000000: episode: 2166, duration: 1.806s, episode steps: 411, steps per second: 228, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1117394/1000000: episode: 2167, duration: 2.943s, episode steps: 681, steps per second: 231, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.272 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1118000/1000000: episode: 2168, duration: 2.678s, episode steps: 606, steps per second: 226, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1118385/1000000: episode: 2169, duration: 1.696s, episode steps: 385, steps per second: 227, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1118942/1000000: episode: 2170, duration: 2.405s, episode steps: 557, steps per second: 232, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1119556/1000000: episode: 2171, duration: 2.688s, episode steps: 614, steps per second: 228, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1119974/1000000: episode: 2172, duration: 1.820s, episode steps: 418, steps per second: 230, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1120570/1000000: episode: 2173, duration: 2.598s, episode steps: 596, steps per second: 229, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.639 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1121175/1000000: episode: 2174, duration: 2.659s, episode steps: 605, steps per second: 228, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1121473/1000000: episode: 2175, duration: 1.291s, episode steps: 298, steps per second: 231, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.638 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1121985/1000000: episode: 2176, duration: 2.218s, episode steps: 512, steps per second: 231, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.523 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1122549/1000000: episode: 2177, duration: 2.446s, episode steps: 564, steps per second: 231, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1123094/1000000: episode: 2178, duration: 2.355s, episode steps: 545, steps per second: 231, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean

action: 2.450 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1123395/1000000: episode: 2179, duration: 1.328s, episode steps: 301, steps per second: 227, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.664 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1123711/1000000: episode: 2180, duration: 1.394s, episode steps: 316, steps per second: 227, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1124165/1000000: episode: 2181, duration: 3.924s, episode steps: 454, steps per second: 116, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.037673, mae: 2.457495, mean_q: 2.973614, mean_eps: 0.987966
 1124546/1000000: episode: 2182, duration: 15.739s, episode steps: 381, steps per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.031266, mae: 2.534226, mean_q: 3.069675, mean_eps: 0.987707
 1125044/1000000: episode: 2183, duration: 20.875s, episode steps: 498, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.030341, mae: 2.498640, mean_q: 3.027340, mean_eps: 0.987182
 1125531/1000000: episode: 2184, duration: 20.392s, episode steps: 487, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: 0.033208, mae: 2.504197, mean_q: 3.036760, mean_eps: 0.986591
 1126261/1000000: episode: 2185, duration: 30.314s, episode steps: 730, steps per second: 24, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: 0.026952, mae: 2.495556, mean_q: 3.023180, mean_eps: 0.985859
 1126889/1000000: episode: 2186, duration: 26.080s, episode steps: 628, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.588 [0.000, 5.000], loss: 0.032587, mae: 2.489374, mean_q: 3.014051, mean_eps: 0.985043
 1127256/1000000: episode: 2187, duration: 15.289s, episode steps: 367, steps per second: 24, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.032866, mae: 2.500820, mean_q: 3.029570, mean_eps: 0.984448
 1127858/1000000: episode: 2188, duration: 25.054s, episode steps: 602, steps per second: 24, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.032684, mae: 2.496857, mean_q: 3.025825, mean_eps: 0.983867
 1128557/1000000: episode: 2189, duration: 29.105s, episode steps: 699, steps per second: 24, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: 0.036531, mae: 2.507789, mean_q: 3.039418, mean_eps: 0.983085
 1128989/1000000: episode: 2190, duration: 17.990s, episode steps: 432, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.465 [0.000, 5.000], loss: 0.036889, mae: 2.529235, mean_q: 3.065534, mean_eps: 0.982406
 1129681/1000000: episode: 2191, duration: 28.710s, episode steps: 692, steps

per second: 24, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.454 [0.000, 5.000], loss: 0.028449, mae: 2.494664, mean_q:
3.023789, mean_eps: 0.981731

1130097/1000000: episode: 2192, duration: 17.146s, episode steps: 416, steps
per second: 24, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.502 [0.000, 5.000], loss: 0.026595, mae: 2.523240, mean_q:
3.056103, mean_eps: 0.981066

1130605/1000000: episode: 2193, duration: 21.098s, episode steps: 508, steps
per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.642 [0.000, 5.000], loss: 0.029524, mae: 2.516344, mean_q:
3.051788, mean_eps: 0.980512

1131028/1000000: episode: 2194, duration: 17.716s, episode steps: 423, steps
per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.468 [0.000, 5.000], loss: 0.028383, mae: 2.508136, mean_q:
3.042383, mean_eps: 0.979955

1131322/1000000: episode: 2195, duration: 12.338s, episode steps: 294, steps
per second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.408 [0.000, 5.000], loss: 0.032844, mae: 2.536854, mean_q:
3.078221, mean_eps: 0.979526

1131940/1000000: episode: 2196, duration: 25.832s, episode steps: 618, steps
per second: 24, episode reward: 7.000, mean reward: 0.011 [0.000, 1.000],
mean action: 2.471 [0.000, 5.000], loss: 0.032379, mae: 2.518502, mean_q:
3.053873, mean_eps: 0.978978

1132350/1000000: episode: 2197, duration: 17.196s, episode steps: 410, steps
per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.502 [0.000, 5.000], loss: 0.032129, mae: 2.528700, mean_q:
3.065673, mean_eps: 0.978362

1132957/1000000: episode: 2198, duration: 25.185s, episode steps: 607, steps
per second: 24, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.473 [0.000, 5.000], loss: 0.029469, mae: 2.523094, mean_q:
3.059081, mean_eps: 0.977750

1133586/1000000: episode: 2199, duration: 26.216s, episode steps: 629, steps
per second: 24, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.591 [0.000, 5.000], loss: 0.024788, mae: 2.507378, mean_q:
3.044677, mean_eps: 0.977008

1134155/1000000: episode: 2200, duration: 23.637s, episode steps: 569, steps
per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.517 [0.000, 5.000], loss: 0.029837, mae: 2.488194, mean_q:
3.018464, mean_eps: 0.976290

1134633/1000000: episode: 2201, duration: 19.868s, episode steps: 478, steps
per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.586 [0.000, 5.000], loss: 0.029408, mae: 2.506227, mean_q:
3.039823, mean_eps: 0.975662

1135091/1000000: episode: 2202, duration: 19.020s, episode steps: 458, steps
per second: 24, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000],
mean action: 2.338 [0.000, 5.000], loss: 0.032525, mae: 2.510162, mean_q:
3.041026, mean_eps: 0.975100

1135474/1000000: episode: 2203, duration: 16.067s, episode steps: 383, steps

per second: 24, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.384 [0.000, 5.000], loss: 0.026673, mae: 2.536530, mean_q:
3.073980, mean_eps: 0.974596

1136260/1000000: episode: 2204, duration: 32.617s, episode steps: 786, steps
per second: 24, episode reward: 13.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.457 [0.000, 5.000], loss: 0.033091, mae: 2.531077, mean_q:
3.068715, mean_eps: 0.973895

1136665/1000000: episode: 2205, duration: 16.962s, episode steps: 405, steps
per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.420 [0.000, 5.000], loss: 0.031163, mae: 2.524994, mean_q:
3.057664, mean_eps: 0.973180

1136951/1000000: episode: 2206, duration: 11.986s, episode steps: 286, steps
per second: 24, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.434 [0.000, 5.000], loss: 0.035705, mae: 2.552153, mean_q:
3.089506, mean_eps: 0.972765

1137678/1000000: episode: 2207, duration: 30.433s, episode steps: 727, steps
per second: 24, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.545 [0.000, 5.000], loss: 0.032039, mae: 2.504839, mean_q:
3.036158, mean_eps: 0.972158

1138112/1000000: episode: 2208, duration: 18.262s, episode steps: 434, steps
per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.583 [0.000, 5.000], loss: 0.025900, mae: 2.530239, mean_q:
3.066422, mean_eps: 0.971462

1138986/1000000: episode: 2209, duration: 36.652s, episode steps: 874, steps
per second: 24, episode reward: 18.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.517 [0.000, 5.000], loss: 0.029018, mae: 2.536020, mean_q:
3.075044, mean_eps: 0.970677

1139300/1000000: episode: 2210, duration: 13.266s, episode steps: 314, steps
per second: 24, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.392 [0.000, 5.000], loss: 0.030857, mae: 2.536943, mean_q:
3.080021, mean_eps: 0.969964

1139835/1000000: episode: 2211, duration: 22.420s, episode steps: 535, steps
per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.323 [0.000, 5.000], loss: 0.028636, mae: 2.562443, mean_q:
3.112162, mean_eps: 0.969455

1140435/1000000: episode: 2212, duration: 25.157s, episode steps: 600, steps
per second: 24, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.550 [0.000, 5.000], loss: 0.025046, mae: 2.524238, mean_q:
3.061583, mean_eps: 0.968774

1141101/1000000: episode: 2213, duration: 28.031s, episode steps: 666, steps
per second: 24, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.444 [0.000, 5.000], loss: 0.029158, mae: 2.553317, mean_q:
3.094395, mean_eps: 0.968013

1141612/1000000: episode: 2214, duration: 21.208s, episode steps: 511, steps
per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.675 [0.000, 5.000], loss: 0.031902, mae: 2.512623, mean_q:
3.042642, mean_eps: 0.967307

1142075/1000000: episode: 2215, duration: 19.341s, episode steps: 463, steps

per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000],
mean action: 2.505 [0.000, 5.000], loss: 0.025444, mae: 2.542568, mean_q:
3.083832, mean_eps: 0.966724

1142772/1000000: episode: 2216, duration: 29.318s, episode steps: 697, steps
per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.458 [0.000, 5.000], loss: 0.030062, mae: 2.543535, mean_q:
3.085245, mean_eps: 0.966028

1143757/1000000: episode: 2217, duration: 41.260s, episode steps: 985, steps
per second: 24, episode reward: 19.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.525 [0.000, 5.000], loss: 0.029737, mae: 2.551660, mean_q:
3.093768, mean_eps: 0.965018

1144135/1000000: episode: 2218, duration: 15.804s, episode steps: 378, steps
per second: 24, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.640 [0.000, 5.000], loss: 0.034266, mae: 2.539443, mean_q:
3.078782, mean_eps: 0.964199

1144417/1000000: episode: 2219, duration: 11.836s, episode steps: 282, steps
per second: 24, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000],
mean action: 2.528 [0.000, 5.000], loss: 0.029243, mae: 2.568301, mean_q:
3.115521, mean_eps: 0.963803

1144827/1000000: episode: 2220, duration: 17.060s, episode steps: 410, steps
per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000],
mean action: 2.439 [0.000, 5.000], loss: 0.028716, mae: 2.559204, mean_q:
3.103273, mean_eps: 0.963388

1145349/1000000: episode: 2221, duration: 21.976s, episode steps: 522, steps
per second: 24, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000],
mean action: 2.519 [0.000, 5.000], loss: 0.037249, mae: 2.557902, mean_q:
3.100551, mean_eps: 0.962829

1145846/1000000: episode: 2222, duration: 20.721s, episode steps: 497, steps
per second: 24, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.487 [0.000, 5.000], loss: 0.031088, mae: 2.551402, mean_q:
3.088642, mean_eps: 0.962217

1146287/1000000: episode: 2223, duration: 18.384s, episode steps: 441, steps
per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.590 [0.000, 5.000], loss: 0.027258, mae: 2.551500, mean_q:
3.091764, mean_eps: 0.961655

1146757/1000000: episode: 2224, duration: 19.760s, episode steps: 470, steps
per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.496 [0.000, 5.000], loss: 0.033118, mae: 2.537186, mean_q:
3.071375, mean_eps: 0.961108

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

1147021/1000000: episode: 2225, duration: 5.704s, episode steps: 264, steps per
second: 46, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.458 [0.000, 5.000], loss: 0.035620, mae: 2.566470, mean_q: 3.104123,
mean_eps: 0.960753

1147979/1000000: episode: 2226, duration: 4.170s, episode steps: 958, steps per

second: 230, episode reward: 19.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1148795/1000000: episode: 2227, duration: 3.587s, episode steps: 816, steps per second: 227, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.547 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1149088/1000000: episode: 2228, duration: 1.308s, episode steps: 293, steps per second: 224, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.375 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1149665/1000000: episode: 2229, duration: 2.551s, episode steps: 577, steps per second: 226, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1150048/1000000: episode: 2230, duration: 1.683s, episode steps: 383, steps per second: 228, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1150493/1000000: episode: 2231, duration: 1.970s, episode steps: 445, steps per second: 226, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1150995/1000000: episode: 2232, duration: 2.204s, episode steps: 502, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1151604/1000000: episode: 2233, duration: 2.667s, episode steps: 609, steps per second: 228, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1152085/1000000: episode: 2234, duration: 2.151s, episode steps: 481, steps per second: 224, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1152561/1000000: episode: 2235, duration: 2.089s, episode steps: 476, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1152859/1000000: episode: 2236, duration: 1.316s, episode steps: 298, steps per second: 226, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1153244/1000000: episode: 2237, duration: 1.696s, episode steps: 385, steps per second: 227, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1153890/1000000: episode: 2238, duration: 2.827s, episode steps: 646, steps per second: 228, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1154350/1000000: episode: 2239, duration: 2.027s, episode steps: 460, steps per second: 227, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1154649/1000000: episode: 2240, duration: 1.356s, episode steps: 299, steps per second: 221, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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 dueling/version_1.0/pesos/dqn_v_1.0_weights_2240_episodes.h5f

1155141/1000000: episode: 2241, duration: 2.310s, episode steps: 492, steps per second: 213, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1155977/1000000: episode: 2242, duration: 3.671s, episode steps: 836, steps per second: 228, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1156775/1000000: episode: 2243, duration: 3.497s, episode steps: 798, steps per second: 228, episode reward: 11.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1157485/1000000: episode: 2244, duration: 3.153s, episode steps: 710, steps per second: 225, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1157843/1000000: episode: 2245, duration: 1.635s, episode steps: 358, steps per second: 219, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1158344/1000000: episode: 2246, duration: 2.185s, episode steps: 501, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1159112/1000000: episode: 2247, duration: 3.344s, episode steps: 768, steps per second: 230, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1159606/1000000: episode: 2248, duration: 2.171s, episode steps: 494, steps per second: 228, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.421 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1160085/1000000: episode: 2249, duration: 2.147s, episode steps: 479, steps per second: 223, episode reward: 12.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1160444/1000000: episode: 2250, duration: 1.635s, episode steps: 359, steps per second: 220, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1160836/1000000: episode: 2251, duration: 1.757s, episode steps: 392, steps per second: 223, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.444 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1161398/1000000: episode: 2252, duration: 2.469s, episode steps: 562, steps per second: 228, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1161721/1000000: episode: 2253, duration: 1.426s, episode steps: 323, steps per second: 227, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1162138/1000000: episode: 2254, duration: 1.827s, episode steps: 417, steps per second: 228, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1162434/1000000: episode: 2255, duration: 1.301s, episode steps: 296, steps per second: 228, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1162909/1000000: episode: 2256, duration: 2.091s, episode steps: 475, steps per second: 227, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1163497/1000000: episode: 2257, duration: 2.607s, episode steps: 588, steps per second: 226, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1164012/1000000: episode: 2258, duration: 2.262s, episode steps: 515, steps per second: 228, episode reward: 14.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1164682/1000000: episode: 2259, duration: 2.945s, episode steps: 670, steps per second: 228, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1165108/1000000: episode: 2260, duration: 1.872s, episode steps: 426, steps per second: 228, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1165610/1000000: episode: 2261, duration: 2.206s, episode steps: 502, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1166629/1000000: episode: 2262, duration: 4.452s, episode steps: 1019, steps per second: 229, episode reward: 17.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1167217/1000000: episode: 2263, duration: 2.559s, episode steps: 588, steps per second: 230, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1167775/1000000: episode: 2264, duration: 2.445s, episode steps: 558, steps per second: 228, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1168076/1000000: episode: 2265, duration: 1.323s, episode steps: 301, steps per second: 228, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1168487/1000000: episode: 2266, duration: 1.823s, episode steps: 411, steps per second: 225, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1168784/1000000: episode: 2267, duration: 1.319s, episode steps: 297, steps per second: 225, episode reward: 2.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.670 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1169127/1000000: episode: 2268, duration: 1.559s, episode steps: 343, steps per second: 220, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.426 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1169543/1000000: episode: 2269, duration: 1.860s, episode steps: 416, steps per second: 224, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1170110/1000000: episode: 2270, duration: 2.494s, episode steps: 567, steps per second: 227, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1170574/1000000: episode: 2271, duration: 2.039s, episode steps: 464, steps per second: 228, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1171094/1000000: episode: 2272, duration: 2.293s, episode steps: 520, steps per second: 227, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.542 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1171656/1000000: episode: 2273, duration: 2.495s, episode steps: 562, steps per second: 225, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1172073/1000000: episode: 2274, duration: 1.857s, episode steps: 417, steps per second: 225, episode reward: 12.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1172473/1000000: episode: 2275, duration: 1.771s, episode steps: 400, steps per second: 226, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.388 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1172858/1000000: episode: 2276, duration: 1.697s, episode steps: 385, steps per second: 227, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.582 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1173210/1000000: episode: 2277, duration: 1.554s, episode steps: 352, steps per second: 227, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.330 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1173823/1000000: episode: 2278, duration: 2.686s, episode steps: 613, steps per second: 228, episode reward: 18.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1174305/1000000: episode: 2279, duration: 2.159s, episode steps: 482, steps per second: 223, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1174632/1000000: episode: 2280, duration: 1.464s, episode steps: 327, steps per second: 223, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1175289/1000000: episode: 2281, duration: 2.932s, episode steps: 657, steps per second: 224, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1176189/1000000: episode: 2282, duration: 3.958s, episode steps: 900, steps per second: 227, episode reward: 17.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1176704/1000000: episode: 2283, duration: 2.277s, episode steps: 515, steps per second: 226, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1177315/1000000: episode: 2284, duration: 2.696s, episode steps: 611, steps per second: 227, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1177844/1000000: episode: 2285, duration: 2.352s, episode steps: 529, steps per second: 225, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1178164/1000000: episode: 2286, duration: 1.418s, episode steps: 320, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1178480/1000000: episode: 2287, duration: 1.393s, episode steps: 316, steps per second: 227, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1178981/1000000: episode: 2288, duration: 2.195s, episode steps: 501, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1179269/1000000: episode: 2289, duration: 1.281s, episode steps: 288, steps per second: 225, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1179694/1000000: episode: 2290, duration: 1.879s, episode steps: 425, steps per second: 226, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1180000/1000000: episode: 2291, duration: 1.352s, episode steps: 306, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1180829/1000000: episode: 2292, duration: 3.671s, episode steps: 829, steps per second: 226, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1181319/1000000: episode: 2293, duration: 2.177s, episode steps: 490, steps per second: 225, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1181810/1000000: episode: 2294, duration: 2.168s, episode steps: 491, steps per second: 226, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.316 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1182324/1000000: episode: 2295, duration: 2.290s, episode steps: 514, steps per second: 224, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1183021/1000000: episode: 2296, duration: 3.104s, episode steps: 697, steps per second: 225, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.463 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1183783/1000000: episode: 2297, duration: 3.422s, episode steps: 762, steps per second: 223, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1184389/1000000: episode: 2298, duration: 2.654s, episode steps: 606, steps per second: 228, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1185250/1000000: episode: 2299, duration: 3.787s, episode steps: 861, steps per second: 227, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1185850/1000000: episode: 2300, duration: 2.666s, episode steps: 600, steps per second: 225, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.578 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1186267/1000000: episode: 2301, duration: 1.941s, episode steps: 417, steps per second: 215, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.571 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1186881/1000000: episode: 2302, duration: 2.750s, episode steps: 614, steps per second: 223, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1187350/1000000: episode: 2303, duration: 2.078s, episode steps: 469, steps per second: 226, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1187719/1000000: episode: 2304, duration: 1.649s, episode steps: 369, steps per second: 224, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1188023/1000000: episode: 2305, duration: 1.367s, episode steps: 304, steps per second: 222, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.431 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1188436/1000000: episode: 2306, duration: 1.836s, episode steps: 413, steps per second: 225, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1189130/1000000: episode: 2307, duration: 3.078s, episode steps: 694, steps per second: 225, episode reward: 16.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.585 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1189538/1000000: episode: 2308, duration: 1.814s, episode steps: 408, steps per second: 225, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1189997/1000000: episode: 2309, duration: 15.182s, episode steps: 459, steps per second: 30, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: 0.021507, mae: 2.559409, mean_q: 3.099545, mean_eps: 0.987789

1190622/1000000: episode: 2310, duration: 26.005s, episode steps: 625, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: 0.027009, mae: 2.543602, mean_q: 3.078993, mean_eps: 0.987206

1191069/1000000: episode: 2311, duration: 18.762s, episode steps: 447, steps per second: 24, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.029380, mae: 2.543342, mean_q: 3.079451, mean_eps: 0.986562

1191695/1000000: episode: 2312, duration: 25.992s, episode steps: 626, steps per second: 24, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.623 [0.000, 5.000], loss: 0.028309, mae: 2.571739, mean_q: 3.114590, mean_eps: 0.985919

1192104/1000000: episode: 2313, duration: 17.189s, episode steps: 409, steps per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.597 [0.000, 5.000], loss: 0.024947, mae: 2.563329, mean_q: 3.103018, mean_eps: 0.985300

1192691/1000000: episode: 2314, duration: 24.597s, episode steps: 587, steps per second: 24, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: 0.029203, mae: 2.537129, mean_q: 3.073060, mean_eps: 0.984702

1193167/1000000: episode: 2315, duration: 19.757s, episode steps: 476, steps per second: 24, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: 0.027123, mae: 2.565541, mean_q: 3.109483, mean_eps: 0.984064

1193684/1000000: episode: 2316, duration: 21.881s, episode steps: 517, steps per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: 0.031155, mae: 2.551064, mean_q: 3.089053, mean_eps: 0.983469

1194200/1000000: episode: 2317, duration: 21.675s, episode steps: 516, steps per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.430 [0.000, 5.000], loss: 0.030473, mae: 2.561131, mean_q: 3.100445, mean_eps: 0.982850

1194785/1000000: episode: 2318, duration: 24.628s, episode steps: 585, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.030323, mae: 2.559894, mean_q: 3.097667, mean_eps: 0.982187

1195238/1000000: episode: 2319, duration: 18.934s, episode steps: 453, steps per second: 24, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.700 [0.000, 5.000], loss: 0.030042, mae: 2.538548, mean_q: 3.075218, mean_eps: 0.981563

1195833/1000000: episode: 2320, duration: 25.082s, episode steps: 595, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.587 [0.000, 5.000], loss: 0.037060, mae: 2.555228, mean_q: 3.088375, mean_eps: 0.980934

1196237/1000000: episode: 2321, duration: 17.043s, episode steps: 404, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.624 [0.000, 5.000], loss: 0.023527, mae: 2.562555, mean_q: 3.102211, mean_eps: 0.980334

1196810/1000000: episode: 2322, duration: 24.007s, episode steps: 573, steps per second: 24, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: 0.035688, mae: 2.563180, mean_q: 3.102622, mean_eps: 0.979749

1197318/1000000: episode: 2323, duration: 21.246s, episode steps: 508, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.031261, mae: 2.561829, mean_q: 3.101260, mean_eps: 0.979101

1197718/1000000: episode: 2324, duration: 16.676s, episode steps: 400, steps per second: 24, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.029198, mae: 2.537547, mean_q: 3.071857, mean_eps: 0.978556

1198101/1000000: episode: 2325, duration: 16.101s, episode steps: 383, steps per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.394 [0.000, 5.000], loss: 0.033453, mae: 2.564225, mean_q: 3.100055, mean_eps: 0.978086

1198830/1000000: episode: 2326, duration: 30.546s, episode steps: 729, steps per second: 24, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.030510, mae: 2.563147, mean_q: 3.099771, mean_eps: 0.977418

1199367/1000000: episode: 2327, duration: 22.492s, episode steps: 537, steps per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.676 [0.000, 5.000], loss: 0.028523, mae: 2.556248, mean_q: 3.092415, mean_eps: 0.976660

1200332/1000000: episode: 2328, duration: 40.519s, episode steps: 965, steps per second: 24, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.029054, mae: 2.564924, mean_q: 3.105093, mean_eps: 0.975760

1200675/1000000: episode: 2329, duration: 14.431s, episode steps: 343, steps per second: 24, episode reward: 10.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.032519, mae: 2.565954, mean_q: 3.104466, mean_eps: 0.974975

1200953/1000000: episode: 2330, duration: 11.614s, episode steps: 278, steps per second: 24, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: 0.031818, mae: 2.576679, mean_q: 3.117532, mean_eps: 0.974601

1201654/1000000: episode: 2331, duration: 29.329s, episode steps: 701, steps per second: 24, episode reward: 19.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: 0.025143, mae: 2.575185, mean_q: 3.115778, mean_eps: 0.974013

1202337/1000000: episode: 2332, duration: 28.640s, episode steps: 683, steps per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.363 [0.000, 5.000], loss: 0.029435, mae: 2.570450, mean_q: 3.111149, mean_eps: 0.973182

1202904/1000000: episode: 2333, duration: 23.755s, episode steps: 567, steps per second: 24, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: 0.029853, mae: 2.547985, mean_q: 3.084678, mean_eps: 0.972434

1203380/1000000: episode: 2334, duration: 20.016s, episode steps: 476, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.031793, mae: 2.555632, mean_q: 3.094614, mean_eps: 0.971810

1203831/1000000: episode: 2335, duration: 18.946s, episode steps: 451, steps per second: 24, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: 0.031618, mae: 2.543175, mean_q: 3.079748, mean_eps: 0.971253

1204312/1000000: episode: 2336, duration: 20.216s, episode steps: 481, steps per second: 24, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.026060, mae: 2.560297, mean_q: 3.100675, mean_eps: 0.970694

1204872/1000000: episode: 2337, duration: 23.601s, episode steps: 560, steps per second: 24, episode reward: 14.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: 0.032432, mae: 2.577241, mean_q: 3.118535, mean_eps: 0.970070

1205473/1000000: episode: 2338, duration: 25.563s, episode steps: 601, steps per second: 24, episode reward: 7.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.471 [0.000, 5.000], loss: 0.026763, mae: 2.570107, mean_q: 3.115904, mean_eps: 0.969371

1206042/1000000: episode: 2339, duration: 23.829s, episode steps: 569, steps per second: 24, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.462 [0.000, 5.000], loss: 0.026737, mae: 2.593974, mean_q: 3.143862, mean_eps: 0.968668

1206903/1000000: episode: 2340, duration: 36.104s, episode steps: 861, steps per second: 24, episode reward: 21.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: 0.039188, mae: 2.605268, mean_q: 3.154781, mean_eps: 0.967811

1207291/1000000: episode: 2341, duration: 16.397s, episode steps: 388, steps per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: 0.036196, mae: 2.610236, mean_q: 3.164983, mean_eps: 0.967062

1207987/1000000: episode: 2342, duration: 29.017s, episode steps: 696, steps per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: 0.030771, mae: 2.594436, mean_q: 3.140780, mean_eps: 0.966412

1208980/1000000: episode: 2343, duration: 41.757s, episode steps: 993, steps per second: 24, episode reward: 29.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: 0.029273, mae: 2.603683, mean_q: 3.152418, mean_eps: 0.965399

1209521/1000000: episode: 2344, duration: 22.834s, episode steps: 541, steps per second: 24, episode reward: 14.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: 0.034476, mae: 2.579537, mean_q: 3.123066, mean_eps: 0.964478

1210163/1000000: episode: 2345, duration: 26.926s, episode steps: 642, steps per second: 24, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: 0.030309, mae: 2.599258, mean_q: 3.148222, mean_eps: 0.963767

1210546/1000000: episode: 2346, duration: 16.182s, episode steps: 383, steps per second: 24, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.601 [0.000, 5.000], loss: 0.030818, mae: 2.641864, mean_q: 3.200339, mean_eps: 0.963153

1210921/1000000: episode: 2347, duration: 15.862s, episode steps: 375, steps per second: 24, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: 0.029356, mae: 2.646468, mean_q: 3.203467, mean_eps: 0.962697

1212065/1000000: episode: 2348, duration: 47.953s, episode steps: 1144, steps per second: 24, episode reward: 17.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: 0.033893, mae: 2.637287, mean_q: 3.190013, mean_eps: 0.961785

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow encountered in scalar add

self.step += 1

1212781/1000000: episode: 2349, duration: 16.175s, episode steps: 716, steps per second: 44, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.023471, mae: 2.600437, mean_q: 3.144005, mean_eps: 0.960890

1213153/1000000: episode: 2350, duration: 1.665s, episode steps: 372, steps per second: 223, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1213850/1000000: episode: 2351, duration: 3.081s, episode steps: 697, steps per second: 226, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.551 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1214417/1000000: episode: 2352, duration: 2.502s, episode steps: 567, steps per second: 227, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1215032/1000000: episode: 2353, duration: 2.711s, episode steps: 615, steps per second: 227, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1215701/1000000: episode: 2354, duration: 2.949s, episode steps: 669, steps per second: 227, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1216396/1000000: episode: 2355, duration: 3.125s, episode steps: 695, steps per second: 222, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1216787/1000000: episode: 2356, duration: 1.727s, episode steps: 391, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1217266/1000000: episode: 2357, duration: 2.107s, episode steps: 479, steps per second: 227, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1217806/1000000: episode: 2358, duration: 2.390s, episode steps: 540, steps per second: 226, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1218137/1000000: episode: 2359, duration: 1.476s, episode steps: 331, steps per second: 224, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1218397/1000000: episode: 2360, duration: 1.159s, episode steps: 260, steps per second: 224, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1218813/1000000: episode: 2361, duration: 1.846s, episode steps: 416, steps per second: 225, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1219419/1000000: episode: 2362, duration: 2.694s, episode steps: 606, steps per second: 225, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1220090/1000000: episode: 2363, duration: 2.945s, episode steps: 671, steps per second: 228, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1220735/1000000: episode: 2364, duration: 2.863s, episode steps: 645, steps per second: 225, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1221323/1000000: episode: 2365, duration: 2.586s, episode steps: 588, steps per second: 227, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1221720/1000000: episode: 2366, duration: 1.764s, episode steps: 397, steps per second: 225, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.496 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1222118/1000000: episode: 2367, duration: 1.793s, episode steps: 398, steps per second: 222, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1222616/1000000: episode: 2368, duration: 2.202s, episode steps: 498, steps per second: 226, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1223255/1000000: episode: 2369, duration: 2.825s, episode steps: 639, steps per second: 226, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1223656/1000000: episode: 2370, duration: 1.766s, episode steps: 401, steps per second: 227, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1224253/1000000: episode: 2371, duration: 2.607s, episode steps: 597, steps per second: 229, episode reward: 16.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1224772/1000000: episode: 2372, duration: 2.324s, episode steps: 519, steps per second: 223, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1225392/1000000: episode: 2373, duration: 2.796s, episode steps: 620, steps per second: 222, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1225833/1000000: episode: 2374, duration: 1.961s, episode steps: 441, steps per second: 225, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1226306/1000000: episode: 2375, duration: 2.097s, episode steps: 473, steps per second: 226, episode reward: 13.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1226776/1000000: episode: 2376, duration: 2.074s, episode steps: 470, steps per second: 227, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1227265/1000000: episode: 2377, duration: 2.163s, episode steps: 489, steps per second: 226, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1227961/1000000: episode: 2378, duration: 3.069s, episode steps: 696, steps per second: 227, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1228281/1000000: episode: 2379, duration: 1.396s, episode steps: 320, steps per second: 229, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1228671/1000000: episode: 2380, duration: 1.701s, episode steps: 390, steps per second: 229, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1229228/1000000: episode: 2381, duration: 2.420s, episode steps: 557, steps per second: 230, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1229738/1000000: episode: 2382, duration: 2.195s, episode steps: 510, steps per second: 232, episode reward: 14.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1230390/1000000: episode: 2383, duration: 2.849s, episode steps: 652, steps per second: 229, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1231496/1000000: episode: 2384, duration: 4.795s, episode steps: 1106, steps per second: 231, episode reward: 12.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1232518/1000000: episode: 2385, duration: 4.403s, episode steps: 1022, steps per second: 232, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.563 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1232963/1000000: episode: 2386, duration: 1.932s, episode steps: 445, steps per second: 230, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1233344/1000000: episode: 2387, duration: 1.685s, episode steps: 381, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.572 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1234025/1000000: episode: 2388, duration: 2.968s, episode steps: 681, steps per second: 229, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1234552/1000000: episode: 2389, duration: 2.311s, episode steps: 527, steps per second: 228, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.338 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1235021/1000000: episode: 2390, duration: 2.036s, episode steps: 469, steps per second: 230, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.424 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1235312/1000000: episode: 2391, duration: 1.276s, episode steps: 291, steps per second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.460 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1235832/1000000: episode: 2392, duration: 2.253s, episode steps: 520, steps per second: 231, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.385 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1236326/1000000: episode: 2393, duration: 2.190s, episode steps: 494, steps per second: 226, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1237393/1000000: episode: 2394, duration: 4.625s, episode steps: 1067, steps per second: 231, episode reward: 25.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.358 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1237762/1000000: episode: 2395, duration: 1.613s, episode steps: 369, steps per second: 229, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1238235/1000000: episode: 2396, duration: 2.068s, episode steps: 473, steps per second: 229, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1238744/1000000: episode: 2397, duration: 2.222s, episode steps: 509, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1239646/1000000: episode: 2398, duration: 3.954s, episode steps: 902, steps per second: 228, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1240194/1000000: episode: 2399, duration: 2.379s, episode steps: 548, steps per second: 230, episode reward: 14.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1240597/1000000: episode: 2400, duration: 1.762s, episode steps: 403, steps per second: 229, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.643 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn

dueling/version_1.0/pesos/dqn_v_1.0_weights_2400_episodes.h5f

1241236/1000000: episode: 2401, duration: 2.910s, episode steps: 639, steps per second: 220, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1242082/1000000: episode: 2402, duration: 3.704s, episode steps: 846, steps per second: 228, episode reward: 19.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1242458/1000000: episode: 2403, duration: 1.672s, episode steps: 376, steps per second: 225, episode reward: 10.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1243261/1000000: episode: 2404, duration: 3.536s, episode steps: 803, steps per second: 227, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1243752/1000000: episode: 2405, duration: 2.159s, episode steps: 491, steps per second: 227, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1244326/1000000: episode: 2406, duration: 2.510s, episode steps: 574, steps per second: 229, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1245094/1000000: episode: 2407, duration: 3.374s, episode steps: 768, steps per second: 228, episode reward: 19.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1245575/1000000: episode: 2408, duration: 2.147s, episode steps: 481, steps per second: 224, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.649 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1246105/1000000: episode: 2409, duration: 2.331s, episode steps: 530, steps per second: 227, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1246842/1000000: episode: 2410, duration: 3.207s, episode steps: 737, steps per second: 230, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.494 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1247495/1000000: episode: 2411, duration: 2.846s, episode steps: 653, steps per second: 229, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1247985/1000000: episode: 2412, duration: 2.184s, episode steps: 490, steps per second: 224, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.461 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1248276/1000000: episode: 2413, duration: 1.288s, episode steps: 291, steps per second: 226, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1248769/1000000: episode: 2414, duration: 2.169s, episode steps: 493, steps per second: 227, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.304 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1249382/1000000: episode: 2415, duration: 2.684s, episode steps: 613, steps per second: 228, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1249862/1000000: episode: 2416, duration: 2.097s, episode steps: 480, steps per second: 229, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean

action: 2.519 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1250278/1000000: episode: 2417, duration: 1.823s, episode steps: 416, steps per second: 228, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1250752/1000000: episode: 2418, duration: 2.095s, episode steps: 474, steps per second: 226, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1251337/1000000: episode: 2419, duration: 2.548s, episode steps: 585, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1251764/1000000: episode: 2420, duration: 1.856s, episode steps: 427, steps per second: 230, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.539 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1252085/1000000: episode: 2421, duration: 1.408s, episode steps: 321, steps per second: 228, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1252613/1000000: episode: 2422, duration: 2.284s, episode steps: 528, steps per second: 231, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1253223/1000000: episode: 2423, duration: 2.642s, episode steps: 610, steps per second: 231, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.526 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1253885/1000000: episode: 2424, duration: 2.929s, episode steps: 662, steps per second: 226, episode reward: 17.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1254460/1000000: episode: 2425, duration: 2.490s, episode steps: 575, steps per second: 231, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1254933/1000000: episode: 2426, duration: 2.064s, episode steps: 473, steps per second: 229, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1255385/1000000: episode: 2427, duration: 9.493s, episode steps: 452, steps per second: 48, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.029171, mae: 2.646569, mean_q: 3.204946, mean_eps: 0.987878
 1255913/1000000: episode: 2428, duration: 21.931s, episode steps: 528, steps per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.026109, mae: 2.624985, mean_q: 3.178456, mean_eps: 0.987441
 1256314/1000000: episode: 2429, duration: 16.830s, episode steps: 401, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.387 [0.000, 5.000], loss: 0.027018, mae: 2.624230, mean_q: 3.175291, mean_eps: 0.986884
 1256878/1000000: episode: 2430, duration: 23.671s, episode steps: 564, steps per second: 24, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: 0.033595, mae: 2.631023, mean_q: 3.185603, mean_eps: 0.986306
 1257381/1000000: episode: 2431, duration: 21.000s, episode steps: 503, steps

per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.537 [0.000, 5.000], loss: 0.037572, mae: 2.613840, mean_q:
3.165108, mean_eps: 0.985665

1257893/1000000: episode: 2432, duration: 21.403s, episode steps: 512, steps
per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.645 [0.000, 5.000], loss: 0.028581, mae: 2.634567, mean_q:
3.190436, mean_eps: 0.985055

1258261/1000000: episode: 2433, duration: 15.447s, episode steps: 368, steps
per second: 24, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.470 [0.000, 5.000], loss: 0.033540, mae: 2.642552, mean_q:
3.197970, mean_eps: 0.984527

1258814/1000000: episode: 2434, duration: 23.197s, episode steps: 553, steps
per second: 24, episode reward: 13.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.456 [0.000, 5.000], loss: 0.023404, mae: 2.629327, mean_q:
3.181698, mean_eps: 0.983975

1259764/1000000: episode: 2435, duration: 39.746s, episode steps: 950, steps
per second: 24, episode reward: 22.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.489 [0.000, 5.000], loss: 0.026266, mae: 2.640420, mean_q:
3.194294, mean_eps: 0.983075

1260420/1000000: episode: 2436, duration: 27.622s, episode steps: 656, steps
per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.562 [0.000, 5.000], loss: 0.029504, mae: 2.636035, mean_q:
3.189149, mean_eps: 0.982113

1260738/1000000: episode: 2437, duration: 13.384s, episode steps: 318, steps
per second: 24, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.459 [0.000, 5.000], loss: 0.025990, mae: 2.618902, mean_q:
3.170551, mean_eps: 0.981527

1261130/1000000: episode: 2438, duration: 16.490s, episode steps: 392, steps
per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.569 [0.000, 5.000], loss: 0.030440, mae: 2.642380, mean_q:
3.201283, mean_eps: 0.981100

1261564/1000000: episode: 2439, duration: 18.338s, episode steps: 434, steps
per second: 24, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.488 [0.000, 5.000], loss: 0.030601, mae: 2.641824, mean_q:
3.196678, mean_eps: 0.980606

1262078/1000000: episode: 2440, duration: 21.629s, episode steps: 514, steps
per second: 24, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.539 [0.000, 5.000], loss: 0.027863, mae: 2.636261, mean_q:
3.189685, mean_eps: 0.980037

1262598/1000000: episode: 2441, duration: 21.888s, episode steps: 520, steps
per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.419 [0.000, 5.000], loss: 0.031465, mae: 2.635723, mean_q:
3.186087, mean_eps: 0.979415

1263133/1000000: episode: 2442, duration: 22.361s, episode steps: 535, steps
per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.581 [0.000, 5.000], loss: 0.027967, mae: 2.650538, mean_q:
3.209180, mean_eps: 0.978782

1263486/1000000: episode: 2443, duration: 14.736s, episode steps: 353, steps

per second: 24, episode reward: 11.000, mean reward: 0.031 [0.000, 1.000],
mean action: 2.357 [0.000, 5.000], loss: 0.027627, mae: 2.650475, mean_q:
3.206382, mean_eps: 0.978249

1263816/1000000: episode: 2444, duration: 13.887s, episode steps: 330, steps
per second: 24, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000],
mean action: 2.609 [0.000, 5.000], loss: 0.024940, mae: 2.618499, mean_q:
3.169481, mean_eps: 0.977841

1264242/1000000: episode: 2445, duration: 17.832s, episode steps: 426, steps
per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.505 [0.000, 5.000], loss: 0.029382, mae: 2.649697, mean_q:
3.210617, mean_eps: 0.977387

1264761/1000000: episode: 2446, duration: 21.886s, episode steps: 519, steps
per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.561 [0.000, 5.000], loss: 0.027078, mae: 2.627409, mean_q:
3.183287, mean_eps: 0.976818

1265149/1000000: episode: 2447, duration: 16.123s, episode steps: 388, steps
per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000],
mean action: 2.410 [0.000, 5.000], loss: 0.028716, mae: 2.631758, mean_q:
3.186084, mean_eps: 0.976274

1265638/1000000: episode: 2448, duration: 20.386s, episode steps: 489, steps
per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.409 [0.000, 5.000], loss: 0.027328, mae: 2.690252, mean_q:
3.261010, mean_eps: 0.975748

1266002/1000000: episode: 2449, duration: 15.381s, episode steps: 364, steps
per second: 24, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.602 [0.000, 5.000], loss: 0.034929, mae: 2.670982, mean_q:
3.242133, mean_eps: 0.975237

1266519/1000000: episode: 2450, duration: 21.717s, episode steps: 517, steps
per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.435 [0.000, 5.000], loss: 0.028299, mae: 2.658179, mean_q:
3.227468, mean_eps: 0.974709

1267140/1000000: episode: 2451, duration: 26.157s, episode steps: 621, steps
per second: 24, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.697 [0.000, 5.000], loss: 0.029649, mae: 2.664898, mean_q:
3.233759, mean_eps: 0.974027

1268448/1000000: episode: 2452, duration: 54.765s, episode steps: 1308, steps
per second: 24, episode reward: 34.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.640 [0.000, 5.000], loss: 0.028168, mae: 2.657493, mean_q:
3.225529, mean_eps: 0.972870

1269094/1000000: episode: 2453, duration: 27.005s, episode steps: 646, steps
per second: 24, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.308 [0.000, 5.000], loss: 0.027770, mae: 2.698787, mean_q:
3.276186, mean_eps: 0.971697

1269411/1000000: episode: 2454, duration: 13.267s, episode steps: 317, steps
per second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.470 [0.000, 5.000], loss: 0.029599, mae: 2.601578, mean_q:
3.158585, mean_eps: 0.971118

1270076/1000000: episode: 2455, duration: 27.919s, episode steps: 665, steps

per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000],
mean action: 2.606 [0.000, 5.000], loss: 0.030429, mae: 2.635991, mean_q:
3.195082, mean_eps: 0.970530

1270493/1000000: episode: 2456, duration: 17.618s, episode steps: 417, steps
per second: 24, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.621 [0.000, 5.000], loss: 0.028340, mae: 2.696492, mean_q:
3.271132, mean_eps: 0.969880

1271127/1000000: episode: 2457, duration: 26.637s, episode steps: 634, steps
per second: 24, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.454 [0.000, 5.000], loss: 0.036367, mae: 2.698761, mean_q:
3.273635, mean_eps: 0.969249

1271885/1000000: episode: 2458, duration: 31.698s, episode steps: 758, steps
per second: 24, episode reward: 23.000, mean reward: 0.030 [0.000, 1.000],
mean action: 2.566 [0.000, 5.000], loss: 0.029932, mae: 2.678554, mean_q:
3.250490, mean_eps: 0.968414

1272554/1000000: episode: 2459, duration: 28.083s, episode steps: 669, steps
per second: 24, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.489 [0.000, 5.000], loss: 0.029408, mae: 2.688533, mean_q:
3.263062, mean_eps: 0.967557

1273065/1000000: episode: 2460, duration: 21.490s, episode steps: 511, steps
per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000],
mean action: 2.607 [0.000, 5.000], loss: 0.025873, mae: 2.701998, mean_q:
3.280374, mean_eps: 0.966849

1273407/1000000: episode: 2461, duration: 14.227s, episode steps: 342, steps
per second: 24, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.430 [0.000, 5.000], loss: 0.029327, mae: 2.735996, mean_q:
3.322231, mean_eps: 0.966338

1273865/1000000: episode: 2462, duration: 19.118s, episode steps: 458, steps
per second: 24, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000],
mean action: 2.587 [0.000, 5.000], loss: 0.035718, mae: 2.715277, mean_q:
3.292494, mean_eps: 0.965858

1274345/1000000: episode: 2463, duration: 20.221s, episode steps: 480, steps
per second: 24, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.665 [0.000, 5.000], loss: 0.034984, mae: 2.728673, mean_q:
3.313656, mean_eps: 0.965294

1274793/1000000: episode: 2464, duration: 18.746s, episode steps: 448, steps
per second: 24, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000],
mean action: 2.424 [0.000, 5.000], loss: 0.028944, mae: 2.717092, mean_q:
3.296856, mean_eps: 0.964737

1275153/1000000: episode: 2465, duration: 15.154s, episode steps: 360, steps
per second: 24, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000],
mean action: 2.492 [0.000, 5.000], loss: 0.033398, mae: 2.676459, mean_q:
3.246585, mean_eps: 0.964252

1275597/1000000: episode: 2466, duration: 18.688s, episode steps: 444, steps
per second: 24, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000],
mean action: 2.559 [0.000, 5.000], loss: 0.025617, mae: 2.772634, mean_q:
3.369197, mean_eps: 0.963770

1275985/1000000: episode: 2467, duration: 16.289s, episode steps: 388, steps

per second: 24, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000],
mean action: 2.590 [0.000, 5.000], loss: 0.029998, mae: 2.781533, mean_q:
3.381831, mean_eps: 0.963270

1276530/1000000: episode: 2468, duration: 22.870s, episode steps: 545, steps
per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.505 [0.000, 5.000], loss: 0.035065, mae: 2.741039, mean_q:
3.329421, mean_eps: 0.962711

1276950/1000000: episode: 2469, duration: 17.598s, episode steps: 420, steps
per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.498 [0.000, 5.000], loss: 0.032199, mae: 2.768662, mean_q:
3.365396, mean_eps: 0.962133

1277748/1000000: episode: 2470, duration: 33.480s, episode steps: 798, steps
per second: 24, episode reward: 18.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.515 [0.000, 5.000], loss: 0.030342, mae: 2.766130, mean_q:
3.359615, mean_eps: 0.961403

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add

self.step += 1

1278078/1000000: episode: 2471, duration: 9.079s, episode steps: 330, steps per
second: 36, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean
action: 2.615 [0.000, 5.000], loss: 0.036639, mae: 2.751758, mean_q: 3.338624,
mean_eps: 0.960803

1278569/1000000: episode: 2472, duration: 2.153s, episode steps: 491, steps per
second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.540 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1279058/1000000: episode: 2473, duration: 2.143s, episode steps: 489, steps per
second: 228, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1279451/1000000: episode: 2474, duration: 1.718s, episode steps: 393, steps per
second: 229, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1279951/1000000: episode: 2475, duration: 2.207s, episode steps: 500, steps per
second: 227, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean
action: 2.594 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1280446/1000000: episode: 2476, duration: 2.196s, episode steps: 495, steps per
second: 225, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean
action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1280793/1000000: episode: 2477, duration: 1.574s, episode steps: 347, steps per
second: 221, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.476 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1281312/1000000: episode: 2478, duration: 2.321s, episode steps: 519, steps per
second: 224, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1281838/1000000: episode: 2479, duration: 2.312s, episode steps: 526, steps per
second: 227, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.654 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1282222/1000000: episode: 2480, duration: 1.681s, episode steps: 384, steps per

second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1282691/1000000: episode: 2481, duration: 2.059s, episode steps: 469, steps per second: 228, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1283079/1000000: episode: 2482, duration: 1.720s, episode steps: 388, steps per second: 226, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.603 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1283708/1000000: episode: 2483, duration: 2.787s, episode steps: 629, steps per second: 226, episode reward: 14.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1284179/1000000: episode: 2484, duration: 2.064s, episode steps: 471, steps per second: 228, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1284476/1000000: episode: 2485, duration: 1.301s, episode steps: 297, steps per second: 228, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1284985/1000000: episode: 2486, duration: 2.220s, episode steps: 509, steps per second: 229, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1285482/1000000: episode: 2487, duration: 2.167s, episode steps: 497, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1286117/1000000: episode: 2488, duration: 2.770s, episode steps: 635, steps per second: 229, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1286615/1000000: episode: 2489, duration: 2.213s, episode steps: 498, steps per second: 225, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1287105/1000000: episode: 2490, duration: 2.157s, episode steps: 490, steps per second: 227, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1287397/1000000: episode: 2491, duration: 1.280s, episode steps: 292, steps per second: 228, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.353 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1287800/1000000: episode: 2492, duration: 1.750s, episode steps: 403, steps per second: 230, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.481 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1288498/1000000: episode: 2493, duration: 3.024s, episode steps: 698, steps per second: 231, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1289256/1000000: episode: 2494, duration: 3.309s, episode steps: 758, steps per second: 229, episode reward: 21.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1289789/1000000: episode: 2495, duration: 2.355s, episode steps: 533, steps per second: 226, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1290125/1000000: episode: 2496, duration: 1.470s, episode steps: 336, steps per

second: 228, episode reward: 7.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1290564/1000000: episode: 2497, duration: 1.918s, episode steps: 439, steps per second: 229, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.355 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1290878/1000000: episode: 2498, duration: 1.379s, episode steps: 314, steps per second: 228, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1291502/1000000: episode: 2499, duration: 2.708s, episode steps: 624, steps per second: 230, episode reward: 8.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1292095/1000000: episode: 2500, duration: 2.609s, episode steps: 593, steps per second: 227, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.437 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1292383/1000000: episode: 2501, duration: 1.281s, episode steps: 288, steps per second: 225, episode reward: 9.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1292673/1000000: episode: 2502, duration: 1.278s, episode steps: 290, steps per second: 227, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293206/1000000: episode: 2503, duration: 2.315s, episode steps: 533, steps per second: 230, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293491/1000000: episode: 2504, duration: 1.243s, episode steps: 285, steps per second: 229, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1293862/1000000: episode: 2505, duration: 1.605s, episode steps: 371, steps per second: 231, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1294441/1000000: episode: 2506, duration: 2.502s, episode steps: 579, steps per second: 231, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.401 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1295223/1000000: episode: 2507, duration: 3.422s, episode steps: 782, steps per second: 229, episode reward: 24.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.457 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1295496/1000000: episode: 2508, duration: 1.213s, episode steps: 273, steps per second: 225, episode reward: 4.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.681 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1296001/1000000: episode: 2509, duration: 2.204s, episode steps: 505, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1296510/1000000: episode: 2510, duration: 2.195s, episode steps: 509, steps per second: 232, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.527 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1297611/1000000: episode: 2511, duration: 4.720s, episode steps: 1101, steps per second: 233, episode reward: 28.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1297968/1000000: episode: 2512, duration: 1.596s, episode steps: 357, steps per

second: 224, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1298476/1000000: episode: 2513, duration: 2.233s, episode steps: 508, steps per second: 228, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1298876/1000000: episode: 2514, duration: 1.733s, episode steps: 400, steps per second: 231, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1299321/1000000: episode: 2515, duration: 1.928s, episode steps: 445, steps per second: 231, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1299729/1000000: episode: 2516, duration: 1.766s, episode steps: 408, steps per second: 231, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.574 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1300201/1000000: episode: 2517, duration: 2.037s, episode steps: 472, steps per second: 232, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.492 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1300533/1000000: episode: 2518, duration: 1.448s, episode steps: 332, steps per second: 229, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1301084/1000000: episode: 2519, duration: 2.433s, episode steps: 551, steps per second: 226, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1301683/1000000: episode: 2520, duration: 2.601s, episode steps: 599, steps per second: 230, episode reward: 13.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.464 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1302219/1000000: episode: 2521, duration: 2.325s, episode steps: 536, steps per second: 231, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.360 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1302686/1000000: episode: 2522, duration: 2.014s, episode steps: 467, steps per second: 232, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1303267/1000000: episode: 2523, duration: 2.528s, episode steps: 581, steps per second: 230, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1303743/1000000: episode: 2524, duration: 2.097s, episode steps: 476, steps per second: 227, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.414 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1304209/1000000: episode: 2525, duration: 2.067s, episode steps: 466, steps per second: 225, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.687 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1304642/1000000: episode: 2526, duration: 1.880s, episode steps: 433, steps per second: 230, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.619 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1305016/1000000: episode: 2527, duration: 1.633s, episode steps: 374, steps per second: 229, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.422 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1305519/1000000: episode: 2528, duration: 2.185s, episode steps: 503, steps per

second: 230, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1306117/1000000: episode: 2529, duration: 2.597s, episode steps: 598, steps per second: 230, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1306432/1000000: episode: 2530, duration: 1.366s, episode steps: 315, steps per second: 231, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1307208/1000000: episode: 2531, duration: 3.414s, episode steps: 776, steps per second: 227, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.602 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1307523/1000000: episode: 2532, duration: 1.369s, episode steps: 315, steps per second: 230, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.283 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1308128/1000000: episode: 2533, duration: 2.616s, episode steps: 605, steps per second: 231, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1308596/1000000: episode: 2534, duration: 2.029s, episode steps: 468, steps per second: 231, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1309164/1000000: episode: 2535, duration: 2.450s, episode steps: 568, steps per second: 232, episode reward: 13.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1309711/1000000: episode: 2536, duration: 2.386s, episode steps: 547, steps per second: 229, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1310158/1000000: episode: 2537, duration: 1.953s, episode steps: 447, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1310765/1000000: episode: 2538, duration: 2.648s, episode steps: 607, steps per second: 229, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.512 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1311083/1000000: episode: 2539, duration: 1.379s, episode steps: 318, steps per second: 231, episode reward: 2.000, mean reward: 0.006 [0.000, 1.000], mean action: 2.626 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1311673/1000000: episode: 2540, duration: 2.542s, episode steps: 590, steps per second: 232, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.517 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1312249/1000000: episode: 2541, duration: 2.477s, episode steps: 576, steps per second: 233, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1312600/1000000: episode: 2542, duration: 1.565s, episode steps: 351, steps per second: 224, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1312974/1000000: episode: 2543, duration: 1.632s, episode steps: 374, steps per second: 229, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1313543/1000000: episode: 2544, duration: 2.466s, episode steps: 569, steps per

second: 231, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.487 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1313840/1000000: episode: 2545, duration: 1.295s, episode steps: 297, steps per second: 229, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.525 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1314638/1000000: episode: 2546, duration: 3.454s, episode steps: 798, steps per second: 231, episode reward: 20.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1315283/1000000: episode: 2547, duration: 2.797s, episode steps: 645, steps per second: 231, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1315866/1000000: episode: 2548, duration: 2.595s, episode steps: 583, steps per second: 225, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1316315/1000000: episode: 2549, duration: 1.947s, episode steps: 449, steps per second: 231, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.379 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1316614/1000000: episode: 2550, duration: 1.304s, episode steps: 299, steps per second: 229, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1317483/1000000: episode: 2551, duration: 3.725s, episode steps: 869, steps per second: 233, episode reward: 19.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1317886/1000000: episode: 2552, duration: 1.754s, episode steps: 403, steps per second: 230, episode reward: 10.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1318358/1000000: episode: 2553, duration: 2.078s, episode steps: 472, steps per second: 227, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1318929/1000000: episode: 2554, duration: 2.510s, episode steps: 571, steps per second: 228, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1319219/1000000: episode: 2555, duration: 1.263s, episode steps: 290, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1319804/1000000: episode: 2556, duration: 2.540s, episode steps: 585, steps per second: 230, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.533 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1320376/1000000: episode: 2557, duration: 2.477s, episode steps: 572, steps per second: 231, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1320843/1000000: episode: 2558, duration: 6.512s, episode steps: 467, steps per second: 72, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: 0.026178, mae: 2.802491, mean_q: 3.401636, mean_eps: 0.987926

1321132/1000000: episode: 2559, duration: 12.141s, episode steps: 289, steps per second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.612 [0.000, 5.000], loss: 0.034309, mae: 2.791220, mean_q:

3.387075, mean_eps: 0.987681
 1321558/1000000: episode: 2560, duration: 17.826s, episode steps: 426, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.641 [0.000, 5.000], loss: 0.033392, mae: 2.766539, mean_q: 3.358438, mean_eps: 0.987251

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1322429/1000000: episode: 2561, duration: 36.298s, episode steps: 871, steps per second: 24, episode reward: 11.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.035580, mae: 2.798629, mean_q: 3.402173, mean_eps: 0.986471

1322786/1000000: episode: 2562, duration: 14.813s, episode steps: 357, steps per second: 24, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: 0.033202, mae: 2.745857, mean_q: 3.334792, mean_eps: 0.985734

1323335/1000000: episode: 2563, duration: 22.815s, episode steps: 549, steps per second: 24, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: 0.034895, mae: 2.783714, mean_q: 3.377455, mean_eps: 0.985192

1323647/1000000: episode: 2564, duration: 12.864s, episode steps: 312, steps per second: 24, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.040033, mae: 2.770851, mean_q: 3.362437, mean_eps: 0.984676

1324136/1000000: episode: 2565, duration: 20.305s, episode steps: 489, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.415 [0.000, 5.000], loss: 0.039107, mae: 2.797028, mean_q: 3.393706, mean_eps: 0.984196

1325130/1000000: episode: 2566, duration: 41.276s, episode steps: 994, steps per second: 24, episode reward: 16.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.035504, mae: 2.760519, mean_q: 3.348834, mean_eps: 0.983306

1325764/1000000: episode: 2567, duration: 26.305s, episode steps: 634, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.549 [0.000, 5.000], loss: 0.028828, mae: 2.761112, mean_q: 3.356856, mean_eps: 0.982329

1326232/1000000: episode: 2568, duration: 19.592s, episode steps: 468, steps per second: 24, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.605 [0.000, 5.000], loss: 0.033244, mae: 2.800249, mean_q: 3.403771, mean_eps: 0.981669

1327036/1000000: episode: 2569, duration: 33.257s, episode steps: 804, steps per second: 24, episode reward: 11.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: 0.034674, mae: 2.800104, mean_q: 3.397016, mean_eps: 0.980906

1327759/1000000: episode: 2570, duration: 30.057s, episode steps: 723, steps per second: 24, episode reward: 18.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: 0.047955, mae: 2.819099, mean_q:

3.420787, mean_eps: 0.979989
 1328329/1000000: episode: 2571, duration: 23.812s, episode steps: 570, steps per second: 24, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.584 [0.000, 5.000], loss: 0.030555, mae: 2.827641, mean_q: 3.440049, mean_eps: 0.979211
 1328811/1000000: episode: 2572, duration: 20.039s, episode steps: 482, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.030003, mae: 2.814234, mean_q: 3.419311, mean_eps: 0.978580
 1329107/1000000: episode: 2573, duration: 12.314s, episode steps: 296, steps per second: 24, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.476 [0.000, 5.000], loss: 0.035315, mae: 2.842017, mean_q: 3.453577, mean_eps: 0.978114
 1329736/1000000: episode: 2574, duration: 26.178s, episode steps: 629, steps per second: 24, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.552 [0.000, 5.000], loss: 0.036002, mae: 2.830483, mean_q: 3.442922, mean_eps: 0.977560
 1330201/1000000: episode: 2575, duration: 19.415s, episode steps: 465, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.557 [0.000, 5.000], loss: 0.036879, mae: 2.805911, mean_q: 3.415077, mean_eps: 0.976902
 1330860/1000000: episode: 2576, duration: 27.237s, episode steps: 659, steps per second: 24, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: 0.038453, mae: 2.817182, mean_q: 3.426136, mean_eps: 0.976228
 1331288/1000000: episode: 2577, duration: 17.789s, episode steps: 428, steps per second: 24, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.537 [0.000, 5.000], loss: 0.030056, mae: 2.855463, mean_q: 3.471189, mean_eps: 0.975578
 1332081/1000000: episode: 2578, duration: 33.219s, episode steps: 793, steps per second: 24, episode reward: 24.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: 0.030878, mae: 2.809595, mean_q: 3.418173, mean_eps: 0.974843
 1332675/1000000: episode: 2579, duration: 24.600s, episode steps: 594, steps per second: 24, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.040317, mae: 2.853057, mean_q: 3.464474, mean_eps: 0.974010
 1333059/1000000: episode: 2580, duration: 15.955s, episode steps: 384, steps per second: 24, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: 0.028047, mae: 2.843777, mean_q: 3.461715, mean_eps: 0.973425
 1333584/1000000: episode: 2581, duration: 22.174s, episode steps: 525, steps per second: 24, episode reward: 12.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: 0.033242, mae: 2.841666, mean_q: 3.453114, mean_eps: 0.972880
 1334050/1000000: episode: 2582, duration: 19.469s, episode steps: 466, steps per second: 24, episode reward: 12.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: 0.035039, mae: 2.794163, mean_q:

3.392034, mean_eps: 0.972285
1334585/1000000: episode: 2583, duration: 22.377s, episode steps: 535, steps per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: 0.030176, mae: 2.853320, mean_q: 3.469440, mean_eps: 0.971682
1335006/1000000: episode: 2584, duration: 17.530s, episode steps: 421, steps per second: 24, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.423 [0.000, 5.000], loss: 0.031318, mae: 2.818278, mean_q: 3.427632, mean_eps: 0.971109
1335684/1000000: episode: 2585, duration: 28.176s, episode steps: 678, steps per second: 24, episode reward: 20.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.040556, mae: 2.802422, mean_q: 3.402901, mean_eps: 0.970451
1336140/1000000: episode: 2586, duration: 19.141s, episode steps: 456, steps per second: 24, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.036462, mae: 2.901269, mean_q: 3.527534, mean_eps: 0.969772
1336703/1000000: episode: 2587, duration: 23.552s, episode steps: 563, steps per second: 24, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: 0.032153, mae: 2.891689, mean_q: 3.514466, mean_eps: 0.969160
1337195/1000000: episode: 2588, duration: 20.534s, episode steps: 492, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.038115, mae: 2.882567, mean_q: 3.506225, mean_eps: 0.968526
1337660/1000000: episode: 2589, duration: 19.585s, episode steps: 465, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.036146, mae: 2.869911, mean_q: 3.488315, mean_eps: 0.967953
1338100/1000000: episode: 2590, duration: 18.587s, episode steps: 440, steps per second: 24, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: 0.035058, mae: 2.874502, mean_q: 3.498013, mean_eps: 0.967410
1338385/1000000: episode: 2591, duration: 12.054s, episode steps: 285, steps per second: 24, episode reward: 7.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.488 [0.000, 5.000], loss: 0.031312, mae: 2.925513, mean_q: 3.556280, mean_eps: 0.966974
1339027/1000000: episode: 2592, duration: 26.618s, episode steps: 642, steps per second: 24, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: 0.038237, mae: 2.877096, mean_q: 3.498638, mean_eps: 0.966417
1339311/1000000: episode: 2593, duration: 11.938s, episode steps: 284, steps per second: 24, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: 0.040177, mae: 2.890603, mean_q: 3.518801, mean_eps: 0.965862
1339811/1000000: episode: 2594, duration: 20.957s, episode steps: 500, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.458 [0.000, 5.000], loss: 0.030912, mae: 2.912055, mean_q:

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3.541830, mean_eps: 0.965392
1340358/1000000: episode: 2595, duration: 22.959s, episode steps: 547, steps
per second: 24, episode reward: 12.000, mean reward: 0.022 [ 0.000, 1.000],
mean action: 2.567 [0.000, 5.000], loss: 0.050580, mae: 2.877295, mean_q:
3.491223, mean_eps: 0.964763
1340790/1000000: episode: 2596, duration: 18.043s, episode steps: 432, steps
per second: 24, episode reward: 5.000, mean reward: 0.012 [ 0.000, 1.000],
mean action: 2.495 [0.000, 5.000], loss: 0.037160, mae: 2.895144, mean_q:
3.518624, mean_eps: 0.964175
1341199/1000000: episode: 2597, duration: 16.930s, episode steps: 409, steps
per second: 24, episode reward: 8.000, mean reward: 0.020 [ 0.000, 1.000],
mean action: 2.521 [0.000, 5.000], loss: 0.027869, mae: 2.919958, mean_q:
3.554444, mean_eps: 0.963671
1341796/1000000: episode: 2598, duration: 24.862s, episode steps: 597, steps
per second: 24, episode reward: 15.000, mean reward: 0.025 [ 0.000, 1.000],
mean action: 2.459 [0.000, 5.000], loss: 0.034807, mae: 2.902499, mean_q:
3.528838, mean_eps: 0.963069
1342478/1000000: episode: 2599, duration: 28.577s, episode steps: 682, steps
per second: 24, episode reward: 14.000, mean reward: 0.021 [ 0.000, 1.000],
mean action: 2.491 [0.000, 5.000], loss: 0.035959, mae: 2.937675, mean_q:
3.575685, mean_eps: 0.962301
1343105/1000000: episode: 2600, duration: 26.096s, episode steps: 627, steps
per second: 24, episode reward: 11.000, mean reward: 0.018 [ 0.000, 1.000],
mean action: 2.491 [0.000, 5.000], loss: 0.040210, mae: 2.935059, mean_q:
3.568613, mean_eps: 0.961514

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

1343823/1000000: episode: 2601, duration: 17.268s, episode steps: 718, steps
per second: 42, episode reward: 23.000, mean reward: 0.032 [ 0.000, 1.000],
mean action: 2.570 [0.000, 5.000], loss: 0.036074, mae: 2.912880, mean_q:
3.542285, mean_eps: 0.960909
1344112/1000000: episode: 2602, duration: 1.299s, episode steps: 289, steps per
second: 222, episode reward: 5.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1344971/1000000: episode: 2603, duration: 3.725s, episode steps: 859, steps per
second: 231, episode reward: 17.000, mean reward: 0.020 [ 0.000, 1.000], mean
action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1345578/1000000: episode: 2604, duration: 2.663s, episode steps: 607, steps per
second: 228, episode reward: 8.000, mean reward: 0.013 [ 0.000, 1.000], mean
action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1345998/1000000: episode: 2605, duration: 1.844s, episode steps: 420, steps per
second: 228, episode reward: 7.000, mean reward: 0.017 [ 0.000, 1.000], mean
action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1346492/1000000: episode: 2606, duration: 2.201s, episode steps: 494, steps per
second: 224, episode reward: 7.000, mean reward: 0.014 [ 0.000, 1.000], mean
action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

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1347252/1000000: episode: 2607, duration: 3.353s, episode steps: 760, steps per second: 227, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1347548/1000000: episode: 2608, duration: 1.300s, episode steps: 296, steps per second: 228, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1348034/1000000: episode: 2609, duration: 2.143s, episode steps: 486, steps per second: 227, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.617 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1348580/1000000: episode: 2610, duration: 2.401s, episode steps: 546, steps per second: 227, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1348866/1000000: episode: 2611, duration: 1.264s, episode steps: 286, steps per second: 226, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.692 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1349983/1000000: episode: 2612, duration: 4.905s, episode steps: 1117, steps per second: 228, episode reward: 17.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1350717/1000000: episode: 2613, duration: 3.200s, episode steps: 734, steps per second: 229, episode reward: 12.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1351221/1000000: episode: 2614, duration: 2.198s, episode steps: 504, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.498 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1351715/1000000: episode: 2615, duration: 2.194s, episode steps: 494, steps per second: 225, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.490 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1352155/1000000: episode: 2616, duration: 1.938s, episode steps: 440, steps per second: 227, episode reward: 3.000, mean reward: 0.007 [0.000, 1.000], mean action: 2.595 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1352671/1000000: episode: 2617, duration: 2.291s, episode steps: 516, steps per second: 225, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353067/1000000: episode: 2618, duration: 1.751s, episode steps: 396, steps per second: 226, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.434 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353374/1000000: episode: 2619, duration: 1.354s, episode steps: 307, steps per second: 227, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1353960/1000000: episode: 2620, duration: 2.557s, episode steps: 586, steps per second: 229, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1354301/1000000: episode: 2621, duration: 1.505s, episode steps: 341, steps per second: 227, episode reward: 5.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1354650/1000000: episode: 2622, duration: 1.538s, episode steps: 349, steps per second: 227, episode reward: 8.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1355136/1000000: episode: 2623, duration: 2.136s, episode steps: 486, steps per second: 228, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356037/1000000: episode: 2624, duration: 3.941s, episode steps: 901, steps per second: 229, episode reward: 20.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.576 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356401/1000000: episode: 2625, duration: 1.602s, episode steps: 364, steps per second: 227, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356683/1000000: episode: 2626, duration: 1.248s, episode steps: 282, steps per second: 226, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.223 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1356977/1000000: episode: 2627, duration: 1.287s, episode steps: 294, steps per second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1357523/1000000: episode: 2628, duration: 2.404s, episode steps: 546, steps per second: 227, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1357770/1000000: episode: 2629, duration: 1.089s, episode steps: 247, steps per second: 227, episode reward: 2.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.425 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1358130/1000000: episode: 2630, duration: 1.595s, episode steps: 360, steps per second: 226, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1358542/1000000: episode: 2631, duration: 1.829s, episode steps: 412, steps per second: 225, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.689 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1359155/1000000: episode: 2632, duration: 2.669s, episode steps: 613, steps per second: 230, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.511 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1359741/1000000: episode: 2633, duration: 2.540s, episode steps: 586, steps per second: 231, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1360472/1000000: episode: 2634, duration: 3.192s, episode steps: 731, steps per second: 229, episode reward: 10.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1360960/1000000: episode: 2635, duration: 2.139s, episode steps: 488, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.592 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1361857/1000000: episode: 2636, duration: 3.966s, episode steps: 897, steps per second: 226, episode reward: 16.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1362524/1000000: episode: 2637, duration: 2.903s, episode steps: 667, steps per second: 230, episode reward: 22.000, mean reward: 0.033 [0.000, 1.000], mean action: 2.610 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1362825/1000000: episode: 2638, duration: 1.329s, episode steps: 301, steps per second: 226, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.645 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1363152/1000000: episode: 2639, duration: 1.439s, episode steps: 327, steps per second: 227, episode reward: 9.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1363836/1000000: episode: 2640, duration: 2.989s, episode steps: 684, steps per second: 229, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1364317/1000000: episode: 2641, duration: 2.173s, episode steps: 481, steps per second: 221, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1364937/1000000: episode: 2642, duration: 2.716s, episode steps: 620, steps per second: 228, episode reward: 7.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.505 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1365487/1000000: episode: 2643, duration: 2.425s, episode steps: 550, steps per second: 227, episode reward: 11.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1366185/1000000: episode: 2644, duration: 3.039s, episode steps: 698, steps per second: 230, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1366573/1000000: episode: 2645, duration: 1.710s, episode steps: 388, steps per second: 227, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1367276/1000000: episode: 2646, duration: 3.097s, episode steps: 703, steps per second: 227, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1368126/1000000: episode: 2647, duration: 3.699s, episode steps: 850, steps per second: 230, episode reward: 16.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1368506/1000000: episode: 2648, duration: 1.682s, episode steps: 380, steps per second: 226, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1369016/1000000: episode: 2649, duration: 2.225s, episode steps: 510, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1369409/1000000: episode: 2650, duration: 1.732s, episode steps: 393, steps per second: 227, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.377 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1369778/1000000: episode: 2651, duration: 1.657s, episode steps: 369, steps per second: 223, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1370188/1000000: episode: 2652, duration: 1.833s, episode steps: 410, steps per second: 224, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.441 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1370586/1000000: episode: 2653, duration: 1.768s, episode steps: 398, steps per second: 225, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.503 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1371005/1000000: episode: 2654, duration: 1.866s, episode steps: 419, steps per second: 225, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.325 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1371480/1000000: episode: 2655, duration: 2.106s, episode steps: 475, steps per second: 226, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1372424/1000000: episode: 2656, duration: 4.101s, episode steps: 944, steps per second: 230, episode reward: 31.000, mean reward: 0.033 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1373182/1000000: episode: 2657, duration: 3.336s, episode steps: 758, steps per second: 227, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1373884/1000000: episode: 2658, duration: 3.070s, episode steps: 702, steps per second: 229, episode reward: 11.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.504 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1374161/1000000: episode: 2659, duration: 1.216s, episode steps: 277, steps per second: 228, episode reward: 8.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.292 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1374686/1000000: episode: 2660, duration: 2.288s, episode steps: 525, steps per second: 229, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1375084/1000000: episode: 2661, duration: 1.747s, episode steps: 398, steps per second: 228, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1375593/1000000: episode: 2662, duration: 2.239s, episode steps: 509, steps per second: 227, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.564 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1375888/1000000: episode: 2663, duration: 1.314s, episode steps: 295, steps per second: 224, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1376395/1000000: episode: 2664, duration: 2.231s, episode steps: 507, steps per second: 227, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.383 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1376919/1000000: episode: 2665, duration: 2.305s, episode steps: 524, steps per second: 227, episode reward: 10.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1377304/1000000: episode: 2666, duration: 1.700s, episode steps: 385, steps per second: 226, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1377925/1000000: episode: 2667, duration: 2.708s, episode steps: 621, steps per second: 229, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.472 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1378403/1000000: episode: 2668, duration: 2.114s, episode steps: 478, steps per second: 226, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.508 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1379018/1000000: episode: 2669, duration: 2.696s, episode steps: 615, steps per second: 228, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1379423/1000000: episode: 2670, duration: 1.788s, episode steps: 405, steps per second: 227, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.402 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1379887/1000000: episode: 2671, duration: 2.038s, episode steps: 464, steps per second: 228, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1380277/1000000: episode: 2672, duration: 1.710s, episode steps: 390, steps per second: 228, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.562 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1380652/1000000: episode: 2673, duration: 1.631s, episode steps: 375, steps per second: 230, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.536 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1381128/1000000: episode: 2674, duration: 2.082s, episode steps: 476, steps per second: 229, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1381504/1000000: episode: 2675, duration: 1.689s, episode steps: 376, steps per second: 223, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1381787/1000000: episode: 2676, duration: 1.252s, episode steps: 283, steps per second: 226, episode reward: 8.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.693 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1382564/1000000: episode: 2677, duration: 3.373s, episode steps: 777, steps per second: 230, episode reward: 23.000, mean reward: 0.030 [0.000, 1.000], mean action: 2.475 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1382984/1000000: episode: 2678, duration: 1.842s, episode steps: 420, steps per second: 228, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1383306/1000000: episode: 2679, duration: 1.415s, episode steps: 322, steps per second: 228, episode reward: 7.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1383822/1000000: episode: 2680, duration: 2.253s, episode steps: 516, steps per second: 229, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1384463/1000000: episode: 2681, duration: 2.831s, episode steps: 641, steps per second: 226, episode reward: 15.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1385415/1000000: episode: 2682, duration: 4.174s, episode steps: 952, steps per second: 228, episode reward: 16.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1385698/1000000: episode: 2683, duration: 1.271s, episode steps: 283, steps per second: 223, episode reward: 5.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.633 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1386195/1000000: episode: 2684, duration: 2.187s, episode steps: 497, steps per second: 227, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1386593/1000000: episode: 2685, duration: 14.184s, episode steps: 398, steps per second: 28, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.528 [0.000, 5.000], loss: 0.037412, mae: 2.948241, mean_q: 3.595719, mean_eps: 0.987796

1387058/1000000: episode: 2686, duration: 19.422s, episode steps: 465, steps per second: 24, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000],

mean action: 2.606 [0.000, 5.000], loss: 0.046522, mae: 2.960562, mean_q: 3.600851, mean_eps: 0.987316

1387503/1000000: episode: 2687, duration: 18.600s, episode steps: 445, steps per second: 24, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: 0.042618, mae: 2.964483, mean_q: 3.612035, mean_eps: 0.986771

1388233/1000000: episode: 2688, duration: 30.397s, episode steps: 730, steps per second: 24, episode reward: 21.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: 0.030524, mae: 2.961844, mean_q: 3.611236, mean_eps: 0.986066

1388711/1000000: episode: 2689, duration: 19.682s, episode steps: 478, steps per second: 24, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: 0.035571, mae: 2.963169, mean_q: 3.612280, mean_eps: 0.985341

1389252/1000000: episode: 2690, duration: 22.685s, episode steps: 541, steps per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: 0.039775, mae: 2.955103, mean_q: 3.598050, mean_eps: 0.984731

1389688/1000000: episode: 2691, duration: 18.123s, episode steps: 436, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.599 [0.000, 5.000], loss: 0.037042, mae: 2.967373, mean_q: 3.614319, mean_eps: 0.984146

1390111/1000000: episode: 2692, duration: 17.630s, episode steps: 423, steps per second: 24, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.723 [0.000, 5.000], loss: 0.037037, mae: 2.965562, mean_q: 3.612199, mean_eps: 0.983630

1390622/1000000: episode: 2693, duration: 21.146s, episode steps: 511, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.519 [0.000, 5.000], loss: 0.033648, mae: 3.008148, mean_q: 3.664766, mean_eps: 0.983068

1391108/1000000: episode: 2694, duration: 20.113s, episode steps: 486, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.046534, mae: 2.958771, mean_q: 3.606015, mean_eps: 0.982470

1391717/1000000: episode: 2695, duration: 25.104s, episode steps: 609, steps per second: 24, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.506 [0.000, 5.000], loss: 0.030734, mae: 2.991968, mean_q: 3.652768, mean_eps: 0.981813

1392636/1000000: episode: 2696, duration: 38.109s, episode steps: 919, steps per second: 24, episode reward: 24.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.427 [0.000, 5.000], loss: 0.031403, mae: 3.021920, mean_q: 3.681999, mean_eps: 0.980896

1393081/1000000: episode: 2697, duration: 18.587s, episode steps: 445, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: 0.035126, mae: 2.964026, mean_q: 3.605986, mean_eps: 0.980078

1393387/1000000: episode: 2698, duration: 12.601s, episode steps: 306, steps per second: 24, episode reward: 6.000, mean reward: 0.020 [0.000, 1.000],

mean action: 2.271 [0.000, 5.000], loss: 0.035975, mae: 3.048578, mean_q: 3.712387, mean_eps: 0.979626

1393900/1000000: episode: 2699, duration: 21.283s, episode steps: 513, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.575 [0.000, 5.000], loss: 0.042145, mae: 3.025552, mean_q: 3.682584, mean_eps: 0.979137

1394192/1000000: episode: 2700, duration: 12.250s, episode steps: 292, steps per second: 24, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.408 [0.000, 5.000], loss: 0.035173, mae: 2.986569, mean_q: 3.637976, mean_eps: 0.978654

1394617/1000000: episode: 2701, duration: 17.627s, episode steps: 425, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.435 [0.000, 5.000], loss: 0.040738, mae: 3.002080, mean_q: 3.649581, mean_eps: 0.978222

1395441/1000000: episode: 2702, duration: 33.915s, episode steps: 824, steps per second: 24, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: 0.036303, mae: 3.012744, mean_q: 3.669649, mean_eps: 0.977471

1395791/1000000: episode: 2703, duration: 14.536s, episode steps: 350, steps per second: 24, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.706 [0.000, 5.000], loss: 0.043576, mae: 3.010775, mean_q: 3.658519, mean_eps: 0.976768

1396298/1000000: episode: 2704, duration: 21.036s, episode steps: 507, steps per second: 24, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.034812, mae: 3.000819, mean_q: 3.648585, mean_eps: 0.976254

1396794/1000000: episode: 2705, duration: 20.597s, episode steps: 496, steps per second: 24, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.419 [0.000, 5.000], loss: 0.032800, mae: 3.057997, mean_q: 3.725975, mean_eps: 0.975652

1397632/1000000: episode: 2706, duration: 34.740s, episode steps: 838, steps per second: 24, episode reward: 15.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: 0.039542, mae: 3.068417, mean_q: 3.735054, mean_eps: 0.974853

1398445/1000000: episode: 2707, duration: 33.600s, episode steps: 813, steps per second: 24, episode reward: 17.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: 0.041965, mae: 3.057598, mean_q: 3.722524, mean_eps: 0.973862

1399364/1000000: episode: 2708, duration: 38.069s, episode steps: 919, steps per second: 24, episode reward: 17.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.035879, mae: 3.062723, mean_q: 3.731237, mean_eps: 0.972822

1399888/1000000: episode: 2709, duration: 21.783s, episode steps: 524, steps per second: 24, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: 0.042153, mae: 3.050076, mean_q: 3.709752, mean_eps: 0.971958

1400234/1000000: episode: 2710, duration: 14.404s, episode steps: 346, steps per second: 24, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000],

mean action: 2.494 [0.000, 5.000], loss: 0.041629, mae: 3.021224, mean_q: 3.680001, mean_eps: 0.971435

1400526/1000000: episode: 2711, duration: 11.903s, episode steps: 292, steps per second: 25, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: 0.036427, mae: 3.068535, mean_q: 3.735264, mean_eps: 0.971051

1400951/1000000: episode: 2712, duration: 17.477s, episode steps: 425, steps per second: 24, episode reward: 11.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.367 [0.000, 5.000], loss: 0.030492, mae: 3.057224, mean_q: 3.716870, mean_eps: 0.970622

1401460/1000000: episode: 2713, duration: 21.211s, episode steps: 509, steps per second: 24, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.640 [0.000, 5.000], loss: 0.039436, mae: 3.068665, mean_q: 3.735030, mean_eps: 0.970062

1401745/1000000: episode: 2714, duration: 11.920s, episode steps: 285, steps per second: 24, episode reward: 4.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.439 [0.000, 5.000], loss: 0.039150, mae: 3.063655, mean_q: 3.723815, mean_eps: 0.969585

1402146/1000000: episode: 2715, duration: 16.483s, episode steps: 401, steps per second: 24, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: 0.032228, mae: 3.126983, mean_q: 3.801943, mean_eps: 0.969172

1402504/1000000: episode: 2716, duration: 14.927s, episode steps: 358, steps per second: 24, episode reward: 7.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: 0.027691, mae: 3.134667, mean_q: 3.819179, mean_eps: 0.968718

1403154/1000000: episode: 2717, duration: 26.853s, episode steps: 650, steps per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: 0.031558, mae: 3.105166, mean_q: 3.780356, mean_eps: 0.968114

1404221/1000000: episode: 2718, duration: 44.026s, episode steps: 1067, steps per second: 24, episode reward: 16.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.546 [0.000, 5.000], loss: 0.034610, mae: 3.105200, mean_q: 3.780081, mean_eps: 0.967082

1404817/1000000: episode: 2719, duration: 24.593s, episode steps: 596, steps per second: 24, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.589 [0.000, 5.000], loss: 0.040593, mae: 3.132040, mean_q: 3.813976, mean_eps: 0.966083

1405305/1000000: episode: 2720, duration: 20.264s, episode steps: 488, steps per second: 24, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.387 [0.000, 5.000], loss: 0.040051, mae: 3.116121, mean_q: 3.791149, mean_eps: 0.965433

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/pesos/dqn_v_1.0_weights_2720_episodes.h5f

1406043/1000000: episode: 2721, duration: 30.560s, episode steps: 738, steps per second: 24, episode reward: 21.000, mean reward: 0.028 [0.000, 1.000],

```

mean action: 2.610 [0.000, 5.000], loss: 0.036702, mae: 3.128111, mean_q:
3.809946, mean_eps: 0.964698
1406690/1000000: episode: 2722, duration: 26.723s, episode steps: 647, steps
per second: 24, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.368 [0.000, 5.000], loss: 0.034505, mae: 3.141016, mean_q:
3.824801, mean_eps: 0.963868
1407365/1000000: episode: 2723, duration: 28.196s, episode steps: 675, steps
per second: 24, episode reward: 14.000, mean reward: 0.021 [0.000, 1.000],
mean action: 2.459 [0.000, 5.000], loss: 0.038263, mae: 3.172719, mean_q:
3.857173, mean_eps: 0.963074
1408257/1000000: episode: 2724, duration: 37.032s, episode steps: 892, steps
per second: 24, episode reward: 12.000, mean reward: 0.013 [0.000, 1.000],
mean action: 2.577 [0.000, 5.000], loss: 0.038788, mae: 3.210251, mean_q:
3.908034, mean_eps: 0.962133
1408729/1000000: episode: 2725, duration: 19.683s, episode steps: 472, steps
per second: 24, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000],
mean action: 2.485 [0.000, 5.000], loss: 0.036083, mae: 3.133430, mean_q:
3.811701, mean_eps: 0.961314

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
    self.step += 1

1409214/1000000: episode: 2726, duration: 12.993s, episode steps: 485, steps
per second: 37, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000],
mean action: 2.470 [0.000, 5.000], loss: 0.040550, mae: 3.180732, mean_q:
3.867449, mean_eps: 0.960856
1410209/1000000: episode: 2727, duration: 4.326s, episode steps: 995, steps per
second: 230, episode reward: 13.000, mean reward: 0.013 [0.000, 1.000], mean
action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1410591/1000000: episode: 2728, duration: 1.690s, episode steps: 382, steps per
second: 226, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1410849/1000000: episode: 2729, duration: 1.165s, episode steps: 258, steps per
second: 221, episode reward: 5.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.430 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1411425/1000000: episode: 2730, duration: 2.524s, episode steps: 576, steps per
second: 228, episode reward: 10.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.557 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1411725/1000000: episode: 2731, duration: 1.318s, episode steps: 300, steps per
second: 228, episode reward: 9.000, mean reward: 0.030 [0.000, 1.000], mean
action: 2.370 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1412130/1000000: episode: 2732, duration: 1.793s, episode steps: 405, steps per
second: 226, episode reward: 8.000, mean reward: 0.020 [0.000, 1.000], mean
action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1412902/1000000: episode: 2733, duration: 3.347s, episode steps: 772, steps per
second: 231, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean
action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1413725/1000000: episode: 2734, duration: 3.623s, episode steps: 823, steps per

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second: 227, episode reward: 20.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1414206/1000000: episode: 2735, duration: 2.119s, episode steps: 481, steps per second: 227, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1414536/1000000: episode: 2736, duration: 1.449s, episode steps: 330, steps per second: 228, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.609 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1414977/1000000: episode: 2737, duration: 1.926s, episode steps: 441, steps per second: 229, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.510 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1415924/1000000: episode: 2738, duration: 4.116s, episode steps: 947, steps per second: 230, episode reward: 17.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1416306/1000000: episode: 2739, duration: 1.705s, episode steps: 382, steps per second: 224, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1416985/1000000: episode: 2740, duration: 2.997s, episode steps: 679, steps per second: 227, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.501 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1417442/1000000: episode: 2741, duration: 2.003s, episode steps: 457, steps per second: 228, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.567 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1417816/1000000: episode: 2742, duration: 1.641s, episode steps: 374, steps per second: 228, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.545 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1418385/1000000: episode: 2743, duration: 2.503s, episode steps: 569, steps per second: 227, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.541 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1418999/1000000: episode: 2744, duration: 2.680s, episode steps: 614, steps per second: 229, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1419304/1000000: episode: 2745, duration: 1.359s, episode steps: 305, steps per second: 224, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1419866/1000000: episode: 2746, duration: 2.491s, episode steps: 562, steps per second: 226, episode reward: 10.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1420156/1000000: episode: 2747, duration: 1.285s, episode steps: 290, steps per second: 226, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1420645/1000000: episode: 2748, duration: 2.144s, episode steps: 489, steps per second: 228, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1421042/1000000: episode: 2749, duration: 1.750s, episode steps: 397, steps per second: 227, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1421533/1000000: episode: 2750, duration: 2.142s, episode steps: 491, steps per

second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1421895/1000000: episode: 2751, duration: 1.605s, episode steps: 362, steps per second: 226, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1422765/1000000: episode: 2752, duration: 3.837s, episode steps: 870, steps per second: 227, episode reward: 15.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.440 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1423186/1000000: episode: 2753, duration: 1.844s, episode steps: 421, steps per second: 228, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.553 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1424144/1000000: episode: 2754, duration: 4.179s, episode steps: 958, steps per second: 229, episode reward: 17.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1424463/1000000: episode: 2755, duration: 1.403s, episode steps: 319, steps per second: 227, episode reward: 6.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1425115/1000000: episode: 2756, duration: 2.871s, episode steps: 652, steps per second: 227, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.615 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1425624/1000000: episode: 2757, duration: 2.254s, episode steps: 509, steps per second: 226, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.438 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1426062/1000000: episode: 2758, duration: 1.909s, episode steps: 438, steps per second: 229, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.432 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1426431/1000000: episode: 2759, duration: 1.624s, episode steps: 369, steps per second: 227, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.515 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1427075/1000000: episode: 2760, duration: 2.812s, episode steps: 644, steps per second: 229, episode reward: 13.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.470 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1427578/1000000: episode: 2761, duration: 2.211s, episode steps: 503, steps per second: 227, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1428276/1000000: episode: 2762, duration: 3.089s, episode steps: 698, steps per second: 226, episode reward: 12.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.580 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1428786/1000000: episode: 2763, duration: 2.234s, episode steps: 510, steps per second: 228, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.339 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1429174/1000000: episode: 2764, duration: 1.726s, episode steps: 388, steps per second: 225, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.479 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1430094/1000000: episode: 2765, duration: 3.983s, episode steps: 920, steps per second: 231, episode reward: 19.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.493 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1430489/1000000: episode: 2766, duration: 1.741s, episode steps: 395, steps per

second: 227, episode reward: 9.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1431710/1000000: episode: 2767, duration: 5.349s, episode steps: 1221, steps per second: 228, episode reward: 13.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1432220/1000000: episode: 2768, duration: 2.243s, episode steps: 510, steps per second: 227, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.559 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1432693/1000000: episode: 2769, duration: 2.060s, episode steps: 473, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1433101/1000000: episode: 2770, duration: 1.776s, episode steps: 408, steps per second: 230, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1433554/1000000: episode: 2771, duration: 1.996s, episode steps: 453, steps per second: 227, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.448 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1433886/1000000: episode: 2772, duration: 1.477s, episode steps: 332, steps per second: 225, episode reward: 3.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1434363/1000000: episode: 2773, duration: 2.111s, episode steps: 477, steps per second: 226, episode reward: 5.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.407 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1434778/1000000: episode: 2774, duration: 1.830s, episode steps: 415, steps per second: 227, episode reward: 9.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1435046/1000000: episode: 2775, duration: 1.181s, episode steps: 268, steps per second: 227, episode reward: 3.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1435530/1000000: episode: 2776, duration: 2.098s, episode steps: 484, steps per second: 231, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1435820/1000000: episode: 2777, duration: 1.260s, episode steps: 290, steps per second: 230, episode reward: 6.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.634 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1436429/1000000: episode: 2778, duration: 2.659s, episode steps: 609, steps per second: 229, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1437239/1000000: episode: 2779, duration: 3.539s, episode steps: 810, steps per second: 229, episode reward: 12.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.516 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1437709/1000000: episode: 2780, duration: 2.052s, episode steps: 470, steps per second: 229, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.317 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1438192/1000000: episode: 2781, duration: 2.099s, episode steps: 483, steps per second: 230, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1438907/1000000: episode: 2782, duration: 3.081s, episode steps: 715, steps per

second: 232, episode reward: 13.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1439556/1000000: episode: 2783, duration: 2.855s, episode steps: 649, steps per second: 227, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.455 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1439979/1000000: episode: 2784, duration: 1.871s, episode steps: 423, steps per second: 226, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1440566/1000000: episode: 2785, duration: 2.546s, episode steps: 587, steps per second: 231, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1441018/1000000: episode: 2786, duration: 1.962s, episode steps: 452, steps per second: 230, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1441461/1000000: episode: 2787, duration: 1.920s, episode steps: 443, steps per second: 231, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.573 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1442084/1000000: episode: 2788, duration: 2.712s, episode steps: 623, steps per second: 230, episode reward: 13.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.456 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1442779/1000000: episode: 2789, duration: 3.064s, episode steps: 695, steps per second: 227, episode reward: 13.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1443668/1000000: episode: 2790, duration: 3.856s, episode steps: 889, steps per second: 231, episode reward: 14.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1443989/1000000: episode: 2791, duration: 1.421s, episode steps: 321, steps per second: 226, episode reward: 4.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1444670/1000000: episode: 2792, duration: 2.987s, episode steps: 681, steps per second: 228, episode reward: 18.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1444978/1000000: episode: 2793, duration: 1.376s, episode steps: 308, steps per second: 224, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.412 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1445473/1000000: episode: 2794, duration: 2.235s, episode steps: 495, steps per second: 221, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.416 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1445843/1000000: episode: 2795, duration: 1.630s, episode steps: 370, steps per second: 227, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.403 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1446354/1000000: episode: 2796, duration: 2.235s, episode steps: 511, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.497 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1446814/1000000: episode: 2797, duration: 2.026s, episode steps: 460, steps per second: 227, episode reward: 9.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1447199/1000000: episode: 2798, duration: 1.675s, episode steps: 385, steps per

second: 230, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.532 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1447690/1000000: episode: 2799, duration: 2.145s, episode steps: 491, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.342 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1448097/1000000: episode: 2800, duration: 1.809s, episode steps: 407, steps per second: 225, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1448660/1000000: episode: 2801, duration: 2.502s, episode steps: 563, steps per second: 225, episode reward: 8.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1449023/1000000: episode: 2802, duration: 1.589s, episode steps: 363, steps per second: 228, episode reward: 8.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.399 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1449519/1000000: episode: 2803, duration: 2.168s, episode steps: 496, steps per second: 229, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1450196/1000000: episode: 2804, duration: 2.939s, episode steps: 677, steps per second: 230, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.378 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1450671/1000000: episode: 2805, duration: 2.079s, episode steps: 475, steps per second: 228, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1451611/1000000: episode: 2806, duration: 4.149s, episode steps: 940, steps per second: 227, episode reward: 15.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.627 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1452002/1000000: episode: 2807, duration: 9.396s, episode steps: 391, steps per second: 42, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.445 [0.000, 5.000], loss: 0.051360, mae: 3.194606, mean_q: 3.884316, mean_eps: 0.987873

1452369/1000000: episode: 2808, duration: 15.325s, episode steps: 367, steps per second: 24, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.613 [0.000, 5.000], loss: 0.036472, mae: 3.218015, mean_q: 3.913698, mean_eps: 0.987527

1452860/1000000: episode: 2809, duration: 20.481s, episode steps: 491, steps per second: 24, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.473 [0.000, 5.000], loss: 0.042241, mae: 3.234634, mean_q: 3.934563, mean_eps: 0.987014

1453494/1000000: episode: 2810, duration: 26.493s, episode steps: 634, steps per second: 24, episode reward: 12.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.513 [0.000, 5.000], loss: 0.043026, mae: 3.177545, mean_q: 3.856291, mean_eps: 0.986339

1453885/1000000: episode: 2811, duration: 16.268s, episode steps: 391, steps per second: 24, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.039173, mae: 3.166797, mean_q: 3.845259, mean_eps: 0.985722

1454365/1000000: episode: 2812, duration: 19.903s, episode steps: 480, steps per second: 24, episode reward: 7.000, mean reward: 0.015 [0.000, 1.000],

mean action: 2.446 [0.000, 5.000], loss: 0.038825, mae: 3.195007, mean_q: 3.884731, mean_eps: 0.985199

1455052/1000000: episode: 2813, duration: 28.552s, episode steps: 687, steps per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.040537, mae: 3.182053, mean_q: 3.866265, mean_eps: 0.984501

1455545/1000000: episode: 2814, duration: 20.473s, episode steps: 493, steps per second: 24, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.566 [0.000, 5.000], loss: 0.047209, mae: 3.163537, mean_q: 3.835963, mean_eps: 0.983793

1455855/1000000: episode: 2815, duration: 12.779s, episode steps: 310, steps per second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: 0.038493, mae: 3.211678, mean_q: 3.903173, mean_eps: 0.983310

1456271/1000000: episode: 2816, duration: 17.213s, episode steps: 416, steps per second: 24, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: 0.045716, mae: 3.182211, mean_q: 3.862471, mean_eps: 0.982876

1456659/1000000: episode: 2817, duration: 16.134s, episode steps: 388, steps per second: 24, episode reward: 10.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.037290, mae: 3.225999, mean_q: 3.915807, mean_eps: 0.982394

1457095/1000000: episode: 2818, duration: 18.099s, episode steps: 436, steps per second: 24, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.436 [0.000, 5.000], loss: 0.037164, mae: 3.217495, mean_q: 3.904863, mean_eps: 0.981899

1457576/1000000: episode: 2819, duration: 20.135s, episode steps: 481, steps per second: 24, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.507 [0.000, 5.000], loss: 0.037179, mae: 3.194186, mean_q: 3.871070, mean_eps: 0.981350

1458059/1000000: episode: 2820, duration: 20.141s, episode steps: 483, steps per second: 24, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: 0.041544, mae: 3.146628, mean_q: 3.813567, mean_eps: 0.980771

1458348/1000000: episode: 2821, duration: 12.150s, episode steps: 289, steps per second: 24, episode reward: 9.000, mean reward: 0.031 [0.000, 1.000], mean action: 2.550 [0.000, 5.000], loss: 0.035230, mae: 3.148601, mean_q: 3.814218, mean_eps: 0.980308

1458962/1000000: episode: 2822, duration: 25.468s, episode steps: 614, steps per second: 24, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: 0.042732, mae: 3.160747, mean_q: 3.831703, mean_eps: 0.979766

1459687/1000000: episode: 2823, duration: 30.087s, episode steps: 725, steps per second: 24, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: 0.041098, mae: 3.165806, mean_q: 3.837922, mean_eps: 0.978962

1460104/1000000: episode: 2824, duration: 17.456s, episode steps: 417, steps per second: 24, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000],

mean action: 2.643 [0.000, 5.000], loss: 0.040762, mae: 3.176601, mean_q: 3.853490, mean_eps: 0.978278
 1460788/1000000: episode: 2825, duration: 28.634s, episode steps: 684, steps per second: 24, episode reward: 10.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.618 [0.000, 5.000], loss: 0.040581, mae: 3.187212, mean_q: 3.862597, mean_eps: 0.977618
 1461324/1000000: episode: 2826, duration: 22.341s, episode steps: 536, steps per second: 24, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: 0.039583, mae: 3.164809, mean_q: 3.839153, mean_eps: 0.976886
 1461936/1000000: episode: 2827, duration: 25.535s, episode steps: 612, steps per second: 24, episode reward: 15.000, mean reward: 0.025 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: 0.042800, mae: 3.187645, mean_q: 3.864040, mean_eps: 0.976197
 1462614/1000000: episode: 2828, duration: 28.189s, episode steps: 678, steps per second: 24, episode reward: 12.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.042260, mae: 3.181840, mean_q: 3.858675, mean_eps: 0.975422
 1463205/1000000: episode: 2829, duration: 24.709s, episode steps: 591, steps per second: 24, episode reward: 12.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.646 [0.000, 5.000], loss: 0.038257, mae: 3.192202, mean_q: 3.871716, mean_eps: 0.974658
 1464005/1000000: episode: 2830, duration: 33.007s, episode steps: 800, steps per second: 24, episode reward: 16.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: 0.035841, mae: 3.198493, mean_q: 3.883273, mean_eps: 0.973823
 1464326/1000000: episode: 2831, duration: 13.114s, episode steps: 321, steps per second: 24, episode reward: 5.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: 0.027668, mae: 3.188312, mean_q: 3.867487, mean_eps: 0.973151
 1464767/1000000: episode: 2832, duration: 18.195s, episode steps: 441, steps per second: 24, episode reward: 10.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.474 [0.000, 5.000], loss: 0.041406, mae: 3.220144, mean_q: 3.898885, mean_eps: 0.972695
 1465196/1000000: episode: 2833, duration: 17.822s, episode steps: 429, steps per second: 24, episode reward: 7.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.590 [0.000, 5.000], loss: 0.033989, mae: 3.204887, mean_q: 3.885404, mean_eps: 0.972174
 1465761/1000000: episode: 2834, duration: 23.531s, episode steps: 565, steps per second: 24, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.596 [0.000, 5.000], loss: 0.038866, mae: 3.168218, mean_q: 3.843354, mean_eps: 0.971577
 1466139/1000000: episode: 2835, duration: 15.488s, episode steps: 378, steps per second: 24, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: 0.037517, mae: 3.179276, mean_q: 3.850429, mean_eps: 0.971010
 1466541/1000000: episode: 2836, duration: 16.678s, episode steps: 402, steps per second: 24, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000],

mean action: 2.570 [0.000, 5.000], loss: 0.047167, mae: 3.193434, mean_q: 3.872978, mean_eps: 0.970542

1467281/1000000: episode: 2837, duration: 30.336s, episode steps: 740, steps per second: 24, episode reward: 14.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.451 [0.000, 5.000], loss: 0.049612, mae: 3.229659, mean_q: 3.918427, mean_eps: 0.969856

1467843/1000000: episode: 2838, duration: 23.008s, episode steps: 562, steps per second: 24, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: 0.036690, mae: 3.262391, mean_q: 3.952836, mean_eps: 0.969076

1468315/1000000: episode: 2839, duration: 19.366s, episode steps: 472, steps per second: 24, episode reward: 13.000, mean reward: 0.028 [0.000, 1.000], mean action: 2.381 [0.000, 5.000], loss: 0.033805, mae: 3.231374, mean_q: 3.919549, mean_eps: 0.968457

1468859/1000000: episode: 2840, duration: 22.397s, episode steps: 544, steps per second: 24, episode reward: 7.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.351 [0.000, 5.000], loss: 0.037111, mae: 3.225222, mean_q: 3.911906, mean_eps: 0.967847

1469387/1000000: episode: 2841, duration: 21.865s, episode steps: 528, steps per second: 24, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.608 [0.000, 5.000], loss: 0.046032, mae: 3.241961, mean_q: 3.933842, mean_eps: 0.967204

1470172/1000000: episode: 2842, duration: 32.465s, episode steps: 785, steps per second: 24, episode reward: 9.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.606 [0.000, 5.000], loss: 0.037670, mae: 3.241672, mean_q: 3.928067, mean_eps: 0.966417

1470574/1000000: episode: 2843, duration: 16.675s, episode steps: 402, steps per second: 24, episode reward: 2.000, mean reward: 0.005 [0.000, 1.000], mean action: 2.396 [0.000, 5.000], loss: 0.040391, mae: 3.236359, mean_q: 3.921757, mean_eps: 0.965704

1471147/1000000: episode: 2844, duration: 23.574s, episode steps: 573, steps per second: 24, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.354 [0.000, 5.000], loss: 0.049023, mae: 3.239863, mean_q: 3.921259, mean_eps: 0.965118

1471438/1000000: episode: 2845, duration: 12.036s, episode steps: 291, steps per second: 24, episode reward: 8.000, mean reward: 0.027 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: 0.041836, mae: 3.250939, mean_q: 3.932115, mean_eps: 0.964600

1471765/1000000: episode: 2846, duration: 13.419s, episode steps: 327, steps per second: 24, episode reward: 6.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.569 [0.000, 5.000], loss: 0.036424, mae: 3.243939, mean_q: 3.930352, mean_eps: 0.964228

1472218/1000000: episode: 2847, duration: 18.764s, episode steps: 453, steps per second: 24, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.499 [0.000, 5.000], loss: 0.048047, mae: 3.256311, mean_q: 3.944088, mean_eps: 0.963760

1472681/1000000: episode: 2848, duration: 19.031s, episode steps: 463, steps per second: 24, episode reward: 5.000, mean reward: 0.011 [0.000, 1.000],

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mean action: 2.458 [0.000, 5.000], loss: 0.037768, mae: 3.263967, mean_q:
3.956833, mean_eps: 0.963210
1472974/1000000: episode: 2849, duration: 12.042s, episode steps: 293, steps
per second: 24, episode reward: 9.000, mean reward: 0.031 [0.000, 1.000],
mean action: 2.543 [0.000, 5.000], loss: 0.036741, mae: 3.235253, mean_q:
3.917978, mean_eps: 0.962757
1473480/1000000: episode: 2850, duration: 20.900s, episode steps: 506, steps
per second: 24, episode reward: 12.000, mean reward: 0.024 [0.000, 1.000],
mean action: 2.573 [0.000, 5.000], loss: 0.037730, mae: 3.255699, mean_q:
3.938220, mean_eps: 0.962279
1474184/1000000: episode: 2851, duration: 29.057s, episode steps: 704, steps
per second: 24, episode reward: 14.000, mean reward: 0.020 [0.000, 1.000],
mean action: 2.639 [0.000, 5.000], loss: 0.040847, mae: 3.208186, mean_q:
3.886347, mean_eps: 0.961554

/usr/local/lib/python3.12/dist-packages/rl/core.py:206: RuntimeWarning: overflow
encountered in scalar add
  self.step += 1

1474813/1000000: episode: 2852, duration: 16.550s, episode steps: 629, steps
per second: 38, episode reward: 9.000, mean reward: 0.014 [0.000, 1.000],
mean action: 2.533 [0.000, 5.000], loss: 0.038699, mae: 3.252468, mean_q:
3.938258, mean_eps: 0.960906
1475191/1000000: episode: 2853, duration: 1.682s, episode steps: 378, steps per
second: 225, episode reward: 6.000, mean reward: 0.016 [0.000, 1.000], mean
action: 2.373 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1475584/1000000: episode: 2854, duration: 1.727s, episode steps: 393, steps per
second: 228, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.417 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1476301/1000000: episode: 2855, duration: 3.099s, episode steps: 717, steps per
second: 231, episode reward: 15.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.484 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1476588/1000000: episode: 2856, duration: 1.258s, episode steps: 287, steps per
second: 228, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean
action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1477065/1000000: episode: 2857, duration: 2.089s, episode steps: 477, steps per
second: 228, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean
action: 2.524 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1477454/1000000: episode: 2858, duration: 1.735s, episode steps: 389, steps per
second: 224, episode reward: 6.000, mean reward: 0.015 [0.000, 1.000], mean
action: 2.332 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1477898/1000000: episode: 2859, duration: 1.950s, episode steps: 444, steps per
second: 228, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean
action: 2.507 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1478514/1000000: episode: 2860, duration: 2.656s, episode steps: 616, steps per
second: 232, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean
action: 2.453 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
1478992/1000000: episode: 2861, duration: 2.060s, episode steps: 478, steps per
second: 232, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean

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action: 2.531 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1479382/1000000: episode: 2862, duration: 1.692s, episode steps: 390, steps per second: 230, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.328 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1479743/1000000: episode: 2863, duration: 1.568s, episode steps: 361, steps per second: 230, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.529 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1480355/1000000: episode: 2864, duration: 2.645s, episode steps: 612, steps per second: 231, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.343 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1480839/1000000: episode: 2865, duration: 2.116s, episode steps: 484, steps per second: 229, episode reward: 11.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.411 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1481451/1000000: episode: 2866, duration: 2.660s, episode steps: 612, steps per second: 230, episode reward: 14.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.418 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1481826/1000000: episode: 2867, duration: 1.633s, episode steps: 375, steps per second: 230, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.392 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1482420/1000000: episode: 2868, duration: 2.600s, episode steps: 594, steps per second: 229, episode reward: 9.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.655 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1482956/1000000: episode: 2869, duration: 2.327s, episode steps: 536, steps per second: 230, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.636 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1483328/1000000: episode: 2870, duration: 1.621s, episode steps: 372, steps per second: 229, episode reward: 7.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.446 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1483820/1000000: episode: 2871, duration: 2.162s, episode steps: 492, steps per second: 228, episode reward: 10.000, mean reward: 0.020 [0.000, 1.000], mean action: 2.382 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1484603/1000000: episode: 2872, duration: 3.351s, episode steps: 783, steps per second: 234, episode reward: 15.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.480 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1484902/1000000: episode: 2873, duration: 1.307s, episode steps: 299, steps per second: 229, episode reward: 4.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.452 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1485376/1000000: episode: 2874, duration: 2.060s, episode steps: 474, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.477 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1485861/1000000: episode: 2875, duration: 2.102s, episode steps: 485, steps per second: 231, episode reward: 10.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.561 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1486494/1000000: episode: 2876, duration: 2.744s, episode steps: 633, steps per second: 231, episode reward: 11.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.502 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1486870/1000000: episode: 2877, duration: 1.656s, episode steps: 376, steps per second: 227, episode reward: 5.000, mean reward: 0.013 [0.000, 1.000], mean

action: 2.521 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1487277/1000000: episode: 2878, duration: 1.764s, episode steps: 407, steps per second: 231, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1487766/1000000: episode: 2879, duration: 2.139s, episode steps: 489, steps per second: 229, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.409 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1488066/1000000: episode: 2880, duration: 1.302s, episode steps: 300, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.337 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

[Callback] Pesos guardados en:

/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
 dueling/version_1.0/pesos/dqn_v_1.0_weights_2880_episodes.h5f

1488757/1000000: episode: 2881, duration: 3.109s, episode steps: 691, steps per second: 222, episode reward: 15.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.630 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1489213/1000000: episode: 2882, duration: 1.990s, episode steps: 456, steps per second: 229, episode reward: 2.000, mean reward: 0.004 [0.000, 1.000], mean action: 2.518 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1489958/1000000: episode: 2883, duration: 3.259s, episode steps: 745, steps per second: 229, episode reward: 16.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.454 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1490472/1000000: episode: 2884, duration: 2.234s, episode steps: 514, steps per second: 230, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1491030/1000000: episode: 2885, duration: 2.420s, episode steps: 558, steps per second: 231, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1491446/1000000: episode: 2886, duration: 1.808s, episode steps: 416, steps per second: 230, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.500 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1491941/1000000: episode: 2887, duration: 2.145s, episode steps: 495, steps per second: 231, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.622 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1492436/1000000: episode: 2888, duration: 2.190s, episode steps: 495, steps per second: 226, episode reward: 11.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.372 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1493109/1000000: episode: 2889, duration: 2.898s, episode steps: 673, steps per second: 232, episode reward: 16.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.560 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1493897/1000000: episode: 2890, duration: 3.407s, episode steps: 788, steps per second: 231, episode reward: 17.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.555 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1494530/1000000: episode: 2891, duration: 2.731s, episode steps: 633, steps per second: 232, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.548 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --
 1494985/1000000: episode: 2892, duration: 1.985s, episode steps: 455, steps per

second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.556 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1495485/1000000: episode: 2893, duration: 2.186s, episode steps: 500, steps per second: 229, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.420 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1495912/1000000: episode: 2894, duration: 1.840s, episode steps: 427, steps per second: 232, episode reward: 5.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1496483/1000000: episode: 2895, duration: 2.477s, episode steps: 571, steps per second: 230, episode reward: 9.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.489 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1496893/1000000: episode: 2896, duration: 1.786s, episode steps: 410, steps per second: 230, episode reward: 4.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.534 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1497357/1000000: episode: 2897, duration: 2.013s, episode steps: 464, steps per second: 230, episode reward: 11.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1497736/1000000: episode: 2898, duration: 1.642s, episode steps: 379, steps per second: 231, episode reward: 3.000, mean reward: 0.008 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1498260/1000000: episode: 2899, duration: 2.278s, episode steps: 524, steps per second: 230, episode reward: 6.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.433 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1498774/1000000: episode: 2900, duration: 2.258s, episode steps: 514, steps per second: 228, episode reward: 11.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.621 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1499419/1000000: episode: 2901, duration: 2.778s, episode steps: 645, steps per second: 232, episode reward: 10.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.558 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1499786/1000000: episode: 2902, duration: 1.593s, episode steps: 367, steps per second: 230, episode reward: 1.000, mean reward: 0.003 [0.000, 1.000], mean action: 2.428 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1500091/1000000: episode: 2903, duration: 1.327s, episode steps: 305, steps per second: 230, episode reward: 3.000, mean reward: 0.010 [0.000, 1.000], mean action: 2.318 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1500582/1000000: episode: 2904, duration: 2.136s, episode steps: 491, steps per second: 230, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.554 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1501070/1000000: episode: 2905, duration: 2.138s, episode steps: 488, steps per second: 228, episode reward: 8.000, mean reward: 0.016 [0.000, 1.000], mean action: 2.568 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1501463/1000000: episode: 2906, duration: 1.723s, episode steps: 393, steps per second: 228, episode reward: 7.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.466 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1501919/1000000: episode: 2907, duration: 1.973s, episode steps: 456, steps per second: 231, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.482 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1502356/1000000: episode: 2908, duration: 1.913s, episode steps: 437, steps per

second: 228, episode reward: 9.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.535 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1502809/1000000: episode: 2909, duration: 1.980s, episode steps: 453, steps per second: 229, episode reward: 6.000, mean reward: 0.013 [0.000, 1.000], mean action: 2.483 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1503173/1000000: episode: 2910, duration: 1.580s, episode steps: 364, steps per second: 230, episode reward: 5.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.591 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1503467/1000000: episode: 2911, duration: 1.281s, episode steps: 294, steps per second: 229, episode reward: 7.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.459 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1504083/1000000: episode: 2912, duration: 2.680s, episode steps: 616, steps per second: 230, episode reward: 15.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.614 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1504955/1000000: episode: 2913, duration: 3.757s, episode steps: 872, steps per second: 232, episode reward: 20.000, mean reward: 0.023 [0.000, 1.000], mean action: 2.570 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1505408/1000000: episode: 2914, duration: 1.993s, episode steps: 453, steps per second: 227, episode reward: 8.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1505832/1000000: episode: 2915, duration: 1.829s, episode steps: 424, steps per second: 232, episode reward: 10.000, mean reward: 0.024 [0.000, 1.000], mean action: 2.509 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1506126/1000000: episode: 2916, duration: 1.277s, episode steps: 294, steps per second: 230, episode reward: 5.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.565 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1506625/1000000: episode: 2917, duration: 2.159s, episode steps: 499, steps per second: 231, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.395 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1507144/1000000: episode: 2918, duration: 2.293s, episode steps: 519, steps per second: 226, episode reward: 6.000, mean reward: 0.012 [0.000, 1.000], mean action: 2.355 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1507665/1000000: episode: 2919, duration: 2.262s, episode steps: 521, steps per second: 230, episode reward: 8.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.593 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1508093/1000000: episode: 2920, duration: 1.852s, episode steps: 428, steps per second: 231, episode reward: 6.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.544 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1508677/1000000: episode: 2921, duration: 2.530s, episode steps: 584, steps per second: 231, episode reward: 11.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.491 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1509409/1000000: episode: 2922, duration: 3.126s, episode steps: 732, steps per second: 234, episode reward: 21.000, mean reward: 0.029 [0.000, 1.000], mean action: 2.579 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1509869/1000000: episode: 2923, duration: 2.013s, episode steps: 460, steps per second: 228, episode reward: 10.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.520 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1510234/1000000: episode: 2924, duration: 1.605s, episode steps: 365, steps per

second: 227, episode reward: 4.000, mean reward: 0.011 [0.000, 1.000], mean action: 2.663 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1510727/1000000: episode: 2925, duration: 2.128s, episode steps: 493, steps per second: 232, episode reward: 7.000, mean reward: 0.014 [0.000, 1.000], mean action: 2.442 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1511220/1000000: episode: 2926, duration: 2.144s, episode steps: 493, steps per second: 230, episode reward: 9.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.469 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1511816/1000000: episode: 2927, duration: 2.586s, episode steps: 596, steps per second: 230, episode reward: 11.000, mean reward: 0.018 [0.000, 1.000], mean action: 2.357 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1512262/1000000: episode: 2928, duration: 1.939s, episode steps: 446, steps per second: 230, episode reward: 4.000, mean reward: 0.009 [0.000, 1.000], mean action: 2.478 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1513123/1000000: episode: 2929, duration: 3.781s, episode steps: 861, steps per second: 228, episode reward: 13.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.495 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1513847/1000000: episode: 2930, duration: 3.114s, episode steps: 724, steps per second: 233, episode reward: 11.000, mean reward: 0.015 [0.000, 1.000], mean action: 2.468 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1514268/1000000: episode: 2931, duration: 1.828s, episode steps: 421, steps per second: 230, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.404 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1514727/1000000: episode: 2932, duration: 1.997s, episode steps: 459, steps per second: 230, episode reward: 8.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.514 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1515242/1000000: episode: 2933, duration: 2.217s, episode steps: 515, steps per second: 232, episode reward: 9.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.538 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1515825/1000000: episode: 2934, duration: 2.548s, episode steps: 583, steps per second: 229, episode reward: 12.000, mean reward: 0.021 [0.000, 1.000], mean action: 2.405 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1516295/1000000: episode: 2935, duration: 2.043s, episode steps: 470, steps per second: 230, episode reward: 9.000, mean reward: 0.019 [0.000, 1.000], mean action: 2.598 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1517037/1000000: episode: 2936, duration: 3.189s, episode steps: 742, steps per second: 233, episode reward: 16.000, mean reward: 0.022 [0.000, 1.000], mean action: 2.530 [0.000, 5.000], loss: --, mae: --, mean_q: --, mean_eps: --

1517443/1000000: episode: 2937, duration: 6.069s, episode steps: 406, steps per second: 67, episode reward: 7.000, mean reward: 0.017 [0.000, 1.000], mean action: 2.485 [0.000, 5.000], loss: 0.033988, mae: 3.340425, mean_q: 4.053717, mean_eps: 0.987930

1517752/1000000: episode: 2938, duration: 12.777s, episode steps: 309, steps per second: 24, episode reward: 8.000, mean reward: 0.026 [0.000, 1.000], mean action: 2.443 [0.000, 5.000], loss: 0.031845, mae: 3.291150, mean_q: 3.988489, mean_eps: 0.987678

1518138/1000000: episode: 2939, duration: 16.102s, episode steps: 386, steps per second: 24, episode reward: 8.000, mean reward: 0.021 [0.000, 1.000],

```
mean action: 2.637 [0.000, 5.000], loss: 0.034803, mae: 3.269838, mean_q:
3.961744, mean_eps: 0.987261
1518591/1000000: episode: 2940, duration: 18.736s, episode steps: 453, steps
per second: 24, episode reward: 5.000, mean reward: 0.011 [ 0.000, 1.000],
mean action: 2.578 [0.000, 5.000], loss: 0.040188, mae: 3.314761, mean_q:
4.011873, mean_eps: 0.986757
```

```
[ ]: CARPETA_CHECKPOINT_DUELING = '/content/gdrive/MyDrive/08_MIAR/actividades/
↳proyecto practico/dqn dueling/version_1.0/mejor_modelo'
CARPETA_VIDEO_DUELING = '/content/gdrive/MyDrive/08_MIAR/actividades/proyecto_
↳practico/dqn dueling/version_1.0/video'

# Ejecucion del test con 100 episodios para metricas.
dueling_results = evaluar_agente(
    agent=dueling_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_DUELING,
    nb_episodes=100
)

# Generación de video.
dueling_video = evaluar_agente(
    agent=dueling_agent,
    env=env,
    folder_path=CARPETA_CHECKPOINT_DUELING,
    video_dir=CARPETA_VIDEO_DUELING,
    nb_episodes=2
)

print_results("DQN DUELING", dueling_results)
```

-> Iniciando evaluación (Modo Métricas Puro)...

```
/usr/local/lib/python3.12/dist-packages/tf_keras/src/engine/training_v1.py:2354:
UserWarning: `Model.state_updates` will be removed in a future version. This
property should not be used in TensorFlow 2.0, as `updates` are applied
automatically.
```

```
updates=self.state_updates,
```

-> Iniciando evaluación (Modo Video Activo)...

Video exportado exitosamente a:

```
/content/gdrive/MyDrive/08_MIAR/actividades/proyecto practico/dqn
dueling/version_1.0/video
```

Resultados del Agente: DQN DUELING

Rendimiento promedio: 20.4

Rendimiento mínimo: 9.0

Rendimiento máximo: 36.0

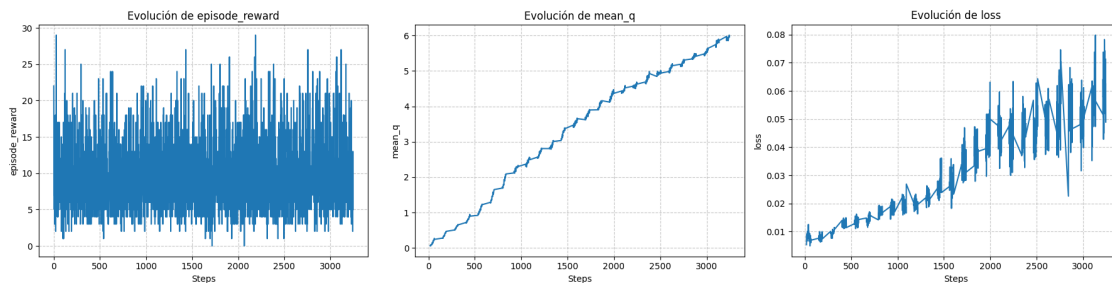
Desviación estándar: 4.8207883172775805

3. Justificación de los parámetros seleccionados y de los resultados obtenidos

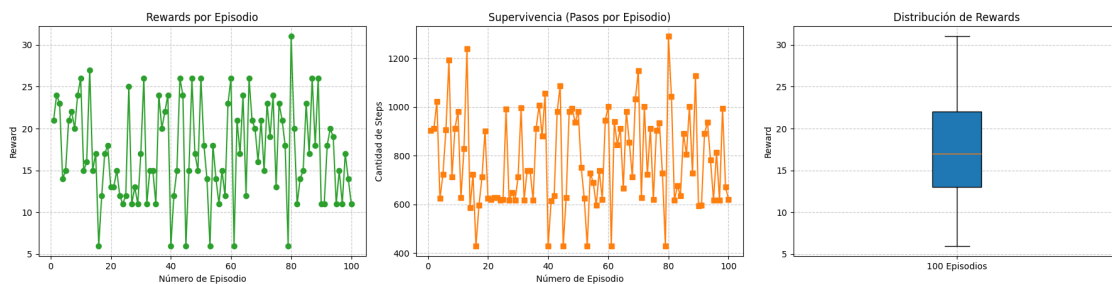
DQN Clasica

```
[ ]: graph_training(DQN_CLASICO_LOGS,model_name= "DQN_Clasico",  
    ↪variables=["episode_reward", "mean_q", "loss"])  
graph_test_results(clasico_results.history, model_name= "DQN_Clasico")  
print("")  
print_results("DQN Clásico", clasico_results)
```

Rendimiento de Entrenamiento: DQN_Clasico



Resultados de Evaluación (Test): DQN_Clasico



Resultados del Agente: DQN Clásico

Rendimiento promedio: 17.24

Rendimiento mínimo: 6.0

Rendimiento máximo: 31.0

Desviación estándar: 5.729083696368906

Conclusiones

Luego de trabajar con el primer modelo (DQN Clásico) en el entorno SpaceInvaders-v0 podemos decir que la arquitectura de tres capas convolucionales, logro establecer un punto referencia en torno a la puntuacion, a pesar de que el agente no alcanzo a llegar al reto propuesto de recompensa media de 20 puntos en 100 episodios, quedo cerca con 17.24 puntos y una desviacion estandar de 5.72, nuestro agente logro picos de puntajes de 31.0 puntos en la fase de pruebas.

Pudimos notar, debido a la desviación estándar, que nuestro agente presenta un poco de inestabilidad por lo que depende mucho de las configuraciones iniciales que utilizamos.

Estuvimos investigando un poco y vimos que mantener la red neuronal con el parámetro **with_spatial_dropout = False** resultó ser un factor clave (Ya lo veremos en el modelo 2). En los entornos visuales de Atari, donde entidades críticas (como los misiles enemigos) ocupan apenas unos pocos píxeles en la pantalla, evitar el “apagado” aleatorio de los mapas de características convolucionales hizo que el agente mantuviera una visión completa del entorno, permitiéndole reaccionar a tiempo y asegurar el rendimiento observado.

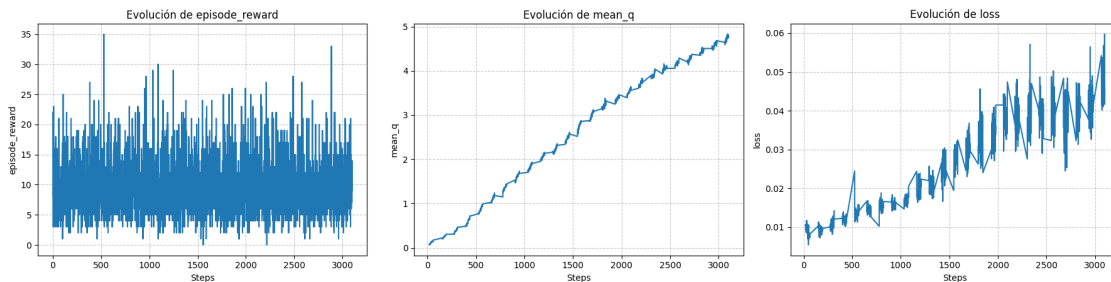
Si observamos las gráficas de entrenamiento, podremos ver la inestabilidad del agente en la puntuación, así mismo, en la función de pérdidas a medida que iban aumentando los steps, la pérdida subía, lo que nos hace concluir dicha inestabilidad, tal vez por falta de entrenamiento.

En las gráficas de test, podemos ver el diagrama de bigotes, donde si ejecutáramos y ejecutáramos muchas veces el agente con los 100 episodios probablemente en alguna ocasión obtendríamos una media de más de 20 puntos.

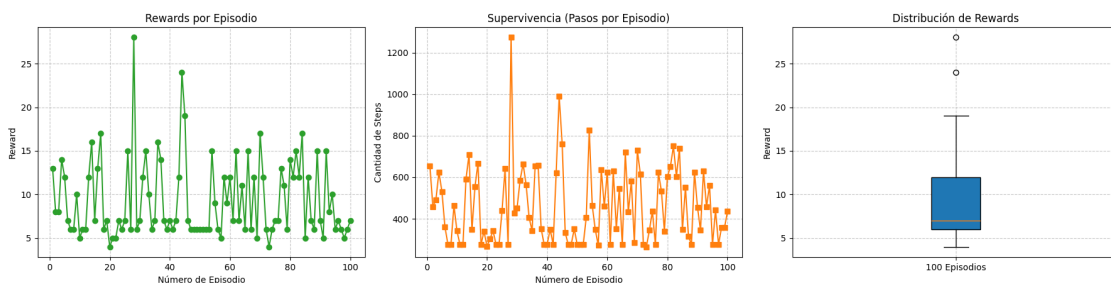
DQN Spatial

```
[ ]: graph_training(DQN_SPATIAL_LOGS,model_name= "DQN_Spatial",
    ↪variables=["episode_reward", "mean_q", "loss"])
graph_test_results(spatial_results.history, model_name= "DQN_Spatial")
print("")
print_results("DQN Spatial", spatial_results)
```

Rendimiento de Entrenamiento: DQN_Spatial



Resultados de Evaluación (Test): DQN_Spatial



Resultados del Agente: DQN Spatial
Rendimiento promedio: 9.3
Rendimiento mínimo: 4.0
Rendimiento máximo: 28.0
Desviación estándar: 4.468780594300866

Conclusiones

Luego de realizar la ejecución del segundo modelo implementado con **with_spatial_dropout = True**, podemos observar una degradación en el rendimiento, obteniendo una recompensa media de 9.3 puntos (Por debajo del objetivo de 20) con una desviación estándar (std_reward=4.46) con un máximo puntaje 28.0, esto nos confirma lo que mencionamos en el primer modelo, al tener en True el parámetro de dropout, la inestabilidad aumenta, si hemos entrenado el modelo con pocos steps.

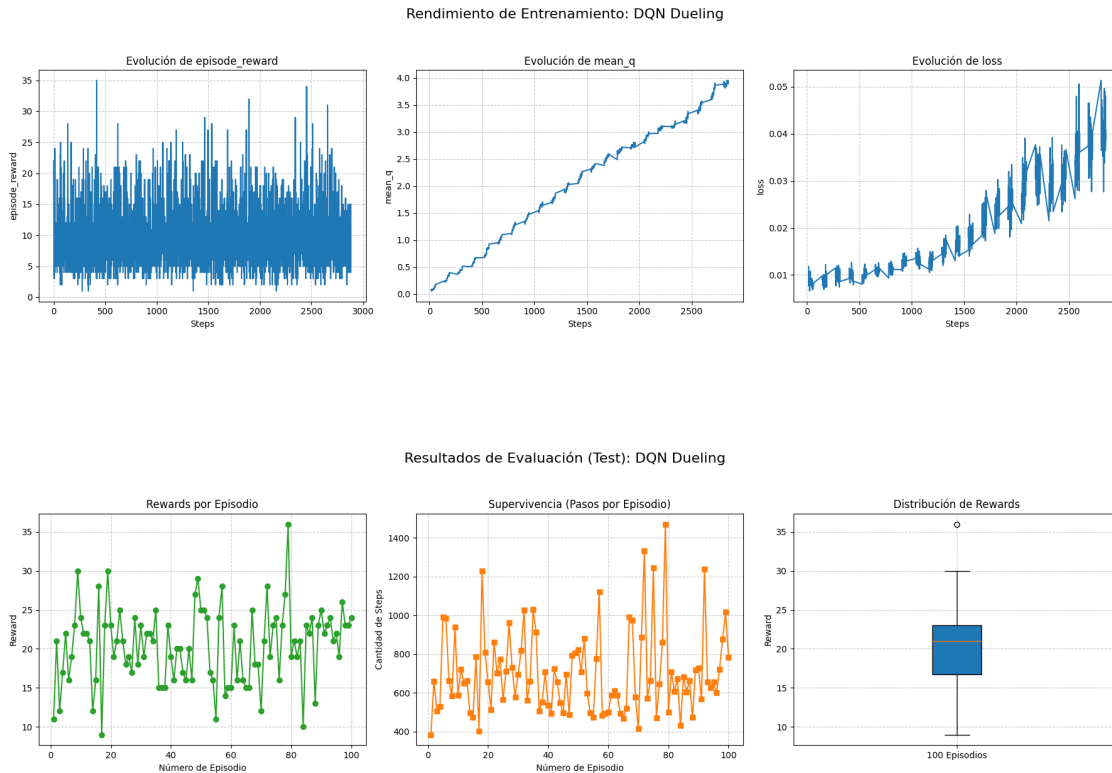
Luego de realizar investigación y viendo el entorno de Space Invaders, esta caída se justifica teóricamente por la incompatibilidad de esta técnica (el dropout = True) de regularización con entornos espacialmente sensibles:

1. **Ceguera inducida:** El Spatial Dropout desactiva mapas de características completos. En nuestro entorno, las amenazas críticas como los misiles ocupan muy pocos píxeles por lo que apagar mapas enteros durante el entrenamiento deja a la red neuronal literalmente “ciega” ante estos peligros.
2. **Corrupción del valor Q:** Al no poder percibir las amenazas de forma constante, el agente sufre muertes que considera impredecibles. Esto corrompe la estimación matemática de la función de valor (Q) y genera una política errática, impidiendo que el algoritmo converja hacia una estrategia de evasión estable.

Si analizamos las graficas de entrenamiento vemos la alta variabilidad de recompensas, al tener el dropout en true, hace que nuestro agente no pueda obtener toda la informacion del entorno en la cantidad de steps de entrenamiento seleccionada, lo que afecta de gran forma la evaluacion con 100 episodios en etapa de evaluacion, en el diagrama de bigotes podemos ver que la recompensa varia por debajo de 20 puntos, que fue el reto propuesto. Para mejorar esto luego de una investigacion realizada, es necesario entrenar por muchos mas steps el modelo para lograr una recompensa media decente.

DQN Dueling

```
[ ]: graph_training(DQN_DUELING_LOGS, model_name= "DQN Dueling",  
    ↪variables=["episode_reward", "mean_q", "loss"])  
graph_test_results(dueling_results.history, model_name= "DQN Dueling")  
print("")  
print_results("DQN Dueling", dueling_results)
```



Resultados del Agente: DQN Dueling

Rendimiento promedio: 20.4

Rendimiento mínimo: 9.0

Rendimiento máximo: 36.0

Desviación estándar: 4.8207883172775805

Conclusiones

En la evaluación del tercer modelo (Dueling DQN) superó el umbral exigido obteniendo una recompensa media de (20.4 puntos). Su característica más notable es la alta varianza: presenta una desviación estándar elevada (4.82) y logró un pico máximo de 36.0 puntos, superando las rachas del modelo clásico.

Este comportamiento bimodal se explica por la separación interna de la red en dos flujos $V(s)$ y $A(s,a)$:

Picos de rendimiento máximo: Al separar el cálculo de “qué tan bueno es el estado” de “qué tan buena es la acción”, la red puede identificar de forma general los momentos de peligro sin tener que aprender el valor de cada movimiento individualmente. Esto le permitió alcanzar episodios con un rendimiento táctico (36 puntos).

Alta dispersión (Inestabilidad): En Space Invaders, casi todos los estados requieren una acción muy específica e inmediata para sobrevivir (esquivar un misil). La separación de las capas V y A introduce una mayor complejidad de optimización matemática que, con 1.6 millones de steps,

no logró estabilizarse al 100%. Esto provocó que el agente alternara entre partidas magistrales y episodios con errores tempranos.

Justificacion & Conclusiones Generales *Justificacion:*

Cabe mencionar que el entrenamiento de los tres modelos se realizó con aproximadamente 1.600.000 steps, cada uno de ellos demora aproximadamente 7 horas entrenando, queríamos realizar dicho entrenamiento igual para todos los modelos con el fin de realizar una comparación.

Nos encontramos con dificultades a la hora de la configuración del ambiente por las incompatibilidades de versiones que se habían mencionado en clase, sin embargo, realizando investigación pudimos sortear dichos inconvenientes, además, tuvimos algunos retos en la elección de los hiperparametros porque al principio nuestros modelos no estaban entrenando correctamente, nos dimos cuenta de esto porque las métricas de perdida (loss) y media de Q (mean_q) no se estaban actualizando luego de muchos steps (perdimos dias de entrenamiento, debido a este problema), cuando aprendimos de estas experiencias tomamos la decisión de entrenar con 1.600.000 a pesar de que en algunos artículos que vimos la recomendación era intentar hacerlo con más steps, a pesar de esto, fue gratificante ver que nuestros modelos pudieron cumplir con nuestras expectativas.

La configuracion base para los tres modelos es la siguiente:

```
BASE_CONFIG = {
    # Entorno y preprocesamiento
    'ENV_NAME': 'SpaceInvaders-v0',
    'WINDOW_LENGTH': 4,
    'INPUT_SHAPE': (84, 84),
    'SEED': 123,

    # Hiperparámetros del agente
    'learning_rate': 0.00025,
    'discount_factor': 0.99,
    'batch_size': 32,
    'target_update_freq': 5000,
    'train_frequency': 4,
    'learning_starts': 10000,

    # Exploración
    'epsilon_start': 1.0,
    'epsilon_end': 0.1,
    'epsilon_decay_steps': 750000,
    'epsilon_test': 0.05,

    # Memoria
    'memory_size': 800000,

    # Entrenamiento
    'max_steps': 1000000,
    'optimizer': 'adam',
    'gradient_clip': 1.0,
```


}

A pesar de que la visualización del entorno de Atari es en sí un rectángulo, realizamos una investigación para ajustar la dimensión de las imágenes que tendría el modelo como entrada, a principio habíamos decidido mantener una configuración rectangular, sin embargo, realizando investigación decidimos optar por un cuadrado que era una configuración más estándar del entorno (`INPUT_SHAPE: (84, 84)` y `WINDOW_LENGTH: 4`). Así mismo, tuvimos en cuenta que al juntar cuatro frames consecutivos en escala de grises, íbamos a garantizar que el agente no solo perciba posiciones estáticas, sino que pueda inferir la velocidad y la trayectoria de los elementos clave (como el movimiento de los alienígenas y los proyectiles).

Para garantizar que el modelo no converja prematuramente hacia políticas subóptimas locales se decidió que el parámetro `epsilon_decay_steps` decaiga al 75% del entrenamiento total, además, el agente mantuvo un 10% por exploración en la fase final asegurando que siga descubriendo estados importantes en fases avanzadas del juego.

Intentamos que la combinación de la frecuencia de actualización de la red objetivo (`target_update_freq: 5000`) y el recorte de gradientes (`gradient_clip: 1.0`) constituyera un mecanismo de defensa contra la inestabilidad al Q-Learning profundo, a pesar de esto, vimos que aun así tuvimos gran inestabilidad en nuestros modelos.

En este punto, tuvimos un gran problema con el tema de memoria, en la primera tanda de ejecución de entrenamientos, la memoria estaba alrededor de 200.000, haciendo que nuestros modelos no entrenaran, por lo que tomamos la decisión de aumentarla a 800.000 transiciones (`memory_size`) que nos proporcione un balance entre la diversidad de datos y consumo de memoria RAM. En el periodo de calentamiento de 10.000 pasos (`learning_starts`), aseguramos que el optimizador Adam (`learning_rate: 0.00025`) comience a trabajar sobre lotes (`batch_size: 32`) de datos completamente decorrelacionados, rompiendo la dependencia temporal de las secuencias de juego y estabilizando la función de pérdida.

En la parte de entrenamiento decidimos establecer el límite en 1.000.000 de pasos (`max_steps`), sin embargo, nuestros modelos entrenaron aproximadamente 1.600.000 steps lo cual permitió un marco de ejecución predecible que evitara ciclos de entrenamiento infinitos o desgastes innecesarios, permitiendo una comparación justa y equitativa en el mismo marco de tiempo computacional para todas las modelos evaluadas.

Conclusiones Generales:

Luego de ahondar en todo el trabajo realizado podemos decir que cualquier elección errada en los hiperparámetros puede ocasionar pérdida de tiempo y de recursos computacionales (como nos pasó en esta ocasión). Aprendimos con este taller que el entrenamiento de los modelos requiere gran precisión, para que los modelos puedan converger a óptimos globales, la conclusión que sacamos de nuestro trabajo es que el entrenamiento con 1.600.000 steps se hizo insuficiente para lograr estabilizar por completo a los tres modelos, el que más se acercó a esto fue el modelo de Dueling, demostrando su versatilidad y adaptación con pocos steps.

Mientras que el DQN Clásico tendió a sobreajustarse a trayectorias de éxito específicas, la regularización introducida por el Spatial Dropout exige un mayor esfuerzo de entrenamiento inicial (necesitaba más steps para estabilizarse) pero recompensa con políticas más resistentes a las variaciones espaciales del entorno. Esto no quiere decir que sea peor que el DQN Clásico, solo que necesita más tiempo de entrenamiento para estabilizar las recompensas.

A diferencia del DQN estándar o con dropout, el enfoque Dueling descompone el problema. Esta bifurcación del flujo de datos en la red neuronal justifica su uso al estabilizar las estimaciones de Q , mitigando la sobreestimación que a menudo limita el rendimiento a largo plazo en los agentes DQN clásicos.

Para alcanzar y mantener las métricas por encima de los 20 puntos de forma sostenida durante 100 episodios, es necesario intentar entrenar los modelos con más steps. Los picos que observamos en las versiones mejoradas (Spatial y Dueling) nos confirman que modificar la arquitectura de la red neuronal impacta directamente en la capacidad del agente para superar la barrera de rendimiento del modelo clásico.

Por último, podemos decir que el objetivo principal (reto) del proyecto fue cumplido con el modelo de Dueling debido a que mantuvo una media sobre 20 puntos.
